300.

NOTE RELATIVE AUX DROITES EN INVOLUTION DE M. SYLVESTER.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, *t.* LII. (Janvier—Juin, 1861), *pp.* 1039—1042.]

La courbe cubique dans l'espace, représentée par les équations

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0,$$

passe par le point $A(x = y = z = 0)$ et le point $B(y = z = u = 0)$; le plan $x = 0$ est le plan osculant en A, le plan $y = 0$ le plan par la tangente en A et la droite AB; le plan $z = 0$ celui par la droite AB et la tangente en B; et enfin le plan $u = 0$ est le plan osculant en B. Réciproquement, pour une courbe cubique quelconque, en prenant les points A, B, sur la courbe à volonté, et en fixant comme ci-dessus les significations des coordonnées x, y, z, u, les facteurs constants que contiennent implicitement ces valeurs étant convenablement déterminés, les équations de la courbe cubique seront

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0.$$

Par un point quelconque de l'espace il passe une droite qui coupe deux fois la courbe cubique; et en prenant (x_1, y_1, z_1, u_1) pour les coordonnées du point dont il s'agit, et en écrivant

$$p_1 = y_1u_1 - z_1^2, \quad q_1 = z_1y_1 - x_1u_1, \quad r_1 = x_1z_1 - y_1^2,$$

les équations de la droite seront

$$p_1x + q_1y + r_1z = 0, \quad p_1y + q_1z + r_1u = 0.$$

Or, en considérant en général une droite représentée par les équations

$$\alpha x + \beta y + \gamma z + \delta u = 0, \quad \alpha' x + \beta' y + \gamma' z + \delta' u = 0,$$

les six quantités

$$\beta\gamma' - \beta'\gamma, \quad \gamma\alpha' - \gamma'\alpha, \quad \alpha\beta' - \alpha'\beta, \quad \alpha\delta' - \alpha'\delta, \quad \beta\delta' - \beta'\delta, \quad \gamma\delta' - \gamma'\delta,$$

sont ce que je nomme les *coordonnées* de la droite (en représentant par a, b, c, f, g, h ces coordonnées, on a l'équation identique $af + bg + ch = 0$, et les coordonnées d'une droite peuvent être des quantités quelconques qui satisfont à cette équation). La condition pour l'involution de six droites est celle-ci, savoir: le déterminant formé avec les coordonnées des six droites est égal à zéro.

Je reviens à la droite qui coupe deux fois la courbe cubique. En écrivant les équations sous la forme

$$p_1x + q_1y + r_1z = 0, \quad -p_1u + q_1x + r_1y = 0,$$

les coordonnées de cette droite seront

$$p_1^2, \quad q_1^2 - p_1r_1, \quad -p_1q_1, \quad p_1r_1, \quad q_1r_1, \quad r_1^2,$$

savoir, ces coordonnées seront des fonctions linéaires de $(p_1^2, q_1^2, r_1^2, q_1r_1, r_1p_1, p_1q_1)$. Donc, en considérant six droites dont chacune coupe deux fois la courbe cubique, et en attribuant des significations analogues à (p_i, q_i, r_i), la condition pour l'involution des six droites se trouve en égalant à zéro le déterminant dont les lignes sont

$$(p_1^2, q_1^2, r_1^2, q_1r_1, r_1p_1, p_1q_1), \quad (p_2^2, q_2^2, \ldots), \quad \ldots;$$

condition qui exprime que les six droites

$$p_ix + q_iy + r_iz = 0,$$

dans le plan $u = 0$ (ou si l'on veut les six droites $p_iy + q_iz + r_iu = 0$ dans le plan $x = 0$) touchent une même conique. Or la droite

$$p_1x + q_1y + r_1z = 0$$

est la projection de l'une des six droites sur le plan osculant $u = 0$, avec le point $x = y = z = 0$ de la courbe cubique comme centre de projection; et si, en prenant un plan osculant quelconque et un point quelconque de la courbe cubique pour plan et centre de projection, nous appelons tout simplement *projection* une telle projection d'une droite quelconque (le plan osculant et le point de la cubique étant toujours les mêmes), on est conduit au théorème que voici, savoir:

Six droites dont chacune coupe deux fois la même courbe cubique seront en involution, si les projections de ces droites touchent une même conique.

Et de même, pour un nombre quelconque de droites, si les projections touchent une même conique, ces droites seront en involution, c'est-à-dire six quelconques des droites seront des droites en involution.

Il convient de remarquer qu'en considérant six droites quelconques, on peut en général trouver une courbe cubique coupée deux fois par chacune des droites: la condition du théorème est donc, comme cela doit être, une seule relation entre les six droites. Je remarque aussi que cette relation ne dépend nullement du plan osculant ni du point de la courbe cubique choisis pour plan et centre de projection. Réciproquement, en prenant dans un plan osculant quelconque de la courbe cubique un nombre quelconque (six ou plus) de tangentes d'une même conique, et en reprojetant ces tangentes sur la courbe cubique au moyen d'un point quelconque de la courbe comme centre de projection (de manière à obtenir pour reprojection de chaque tangente une droite qui coupe deux fois la courbe cubique), on obtient un système de droites en involution. Le lieu des droites dont chacune coupe deux fois la courbe cubique, et qui sont en involution, est une surface réglée du quatrième ordre qui a la courbe cubique pour courbe double. En effet, si l'équation en coordonnées tangentielles de la conique enveloppée par les droites $p_i x + q_i y + r_i z = 0$, etc. (ou, si l'on veut, par les droites $p_i y + q_i z + r_i u = 0$, etc.), est

$$(a, b, c, f, g, h \between p, q, r)^2 = 0,$$

cette même équation, en y considérant p, q, r comme dénotant $yu - z^2$, $zy - xu$, $xz - y^2$, autrement dit, l'équation,

$$(a, b, c, f, g, h \between yu - z^2,\ zy - xu,\ xz - y^2)^2 = 0,$$

sera celle d'une surface du quatrième ordre ayant la courbe cubique pour courbe double. Et cette surface sera une surface réglée; car en menant par un point quelconque de la surface une droite qui coupe deux fois la courbe cubique, chaque point d'intersection avec la courbe cubique doit compter pour deux points d'intersection avec la surface, et la droite coupe la surface en cinq points, c'est-à-dire que cette droite est située entièrement dans la surface.

J'ai remarqué ailleurs (*Camb. and Dubl. Math. Journ.*, *t.* VII. (1852), *p.* 172, [107]) qu'il y a sur une surface réglée de l'ordre n une courbe double rencontrée par chaque génératrice en $(n-2)$ points. Cette courbe double sera de l'ordre $(n-2)$ au moins, et de l'ordre $\frac{1}{2}(n-1)(n-2)$ au plus; donc, pour $n = 4$, la courbe double sera de l'ordre 2 ou 3, et comme évidemment cette courbe n'est pas une courbe plane, elle sera: ou 1° deux droites qui ne se rencontrent pas; ou 2° une courbe cubique en espace. Cette seconde espèce des surfaces réglées du quatrième ordre est celle qui se présente dans la théorie des droites en involution.

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301.

SUR LES CÔNES DU SECOND ORDRE QUI PASSENT PAR SIX POINTS DONNÉS.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, *t.* LII. (Janvier—Juin, 1861), *pp.* 1216—1218.]

Dans un Mémoire par feu M. Weddle "On the theorems in space analogous to those of Pascal and Brianchon in a plane" (*Camb. and Dubl. Math. Journ.*, *t.* V. 1850, voir la Note *p.* 69), on trouve à propos d'un théorème de M. Chasles la remarque que le lieu du sommet d'un cône du second ordre qui passe par six points donnés est une surface du quatrième ordre qui contient la courbe cubique en espace par les six points. Voici comment je démontre ce théorème:

En prenant (X, Y, Z, U) pour les coordonnées courantes, $(\alpha_1, \beta_1, \gamma_1, \delta_1) \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$ pour les coordonnées des six points donnés, et (x, y, z, u) pour ceux du sommet, je pose l'équation

$$\begin{vmatrix} \cdot & X^2 & Y^2 & Z^2 & U^2 & YZ & ZX & XY & XU & YU & ZU \\ \lambda & 2x & \cdot & \cdot & \cdot & z & \cdot & y & u & \cdot & \cdot \\ \mu & \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \nu & \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \rho & \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \cdot & \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0,$$

où la dernière ligne $(\alpha^2, \beta^2, \gamma^2, \delta^2, \beta\gamma, \gamma\alpha, \alpha\beta, \alpha\delta, \beta\delta, \gamma\delta)$ dénote les six lignes qu'on obtient en écrivant successivement $(\alpha_1, \beta_1, \gamma_1, \delta_1) \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$ au lieu de $(\alpha, \beta, \gamma, \delta)$, de manière que la fonction au côté gauche est un déterminant de l'ordre onze: les coefficients λ, μ, ν, ρ sont des quantités arbitraires et les points $(\cdot)$ dénotent des zéros.

Cette équation est évidemment celle d'une surface du second ordre qui passe par les six points, et il ne faut qu'une seule condition pour que cette surface soit un cône: la condition sera

$$\begin{vmatrix} 2x & \cdot & \cdot & \cdot & z & \cdot & y & u & \cdot & \cdot \\ \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0,$$

où la fonction à côté gauche est de même un déterminant de l'ordre dix; cette équation, laquelle est de l'ordre quatre par rapport à (x, y, z, u), sera celle du lieu du sommet.

En effet, pour que la surface du second ordre soit un cône ayant pour sommet le point (x, y, z, u), il faut et il suffit que les équations dérivées par rapport à chacune des coordonnées (X, Y, Z, U), soient satisfaites en y écrivant (x, y, z, u) au lieu de (X, Y, Z, U). Je forme l'équation dérivée par rapport à X, et j'y écris (x, y, z, u) au lieu de (X, Y, Z, U); l'équation est

$$\begin{vmatrix} 2x & \cdot & \cdot & \cdot & z & \cdot & y & u & \cdot & \cdot \\ \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0;$$

or on ne change pas la valeur du déterminant en substituant pour la première ligne cette même ligne moins la seconde ligne; l'équation devient ainsi:

$$\begin{vmatrix} -\lambda & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0;$$

et le déterminant se réduit à $-\lambda$ multiplié par le déterminant de l'ordre dix; donc, en supposant que ce dernier déterminant se réduise à zéro, l'équation dérivée par rapport à X sera satisfaite; et de même, les équations dérivées par rapport à Y, Z, U, en substituant toujours (x, y, z, u) au lieu de (X, Y, Z, U), seront toutes satisfaites si le déterminant de l'ordre dix se réduit à zéro. C. Q. F. D.

Il convient de remarquer que l'on peut sans perte de généralité réduire à zéro trois quelconques des quantités λ, μ, ν, ρ; de là on obtient l'équation du cône en substituant, au lieu de l'une quelconque des premières quatre lignes du déterminant de l'ordre dix, la ligne

$$|X^2, Y^2, Z^2, U^2, YZ, ZX, XY, XU, YU, ZU|.$$

En considérant la courbe cubique par les six points, on peut supposer que les équations de cette courbe soient

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0;$$

cela étant, on aura

$$\beta\delta - \gamma^2 = 0, \quad \beta\gamma - \alpha\delta = 0, \quad \alpha\gamma - \beta^2 = 0$$

pour l'un quelconque des points $(\alpha_1, \beta_1, \gamma_1, \delta_1), \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$; et de là, au moyen des propriétés des déterminants, et en écrivant

$$\square = 4(yu - z^2)(xz - y^2) - (zy - xu)^2,$$

on exprime l'équation de la surface comme fonction linéaire par rapport à x, y, z, u et par rapport à $\partial_x\square$, $\partial_y\square$, $\partial_z\square$, $\partial_u\square$; ces dernières fonctions se réduisent à zéro en vertu des équations

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0,$$

et ainsi, comme cela doit être, la surface passe par la courbe cubique.

Je prends l'occasion de remarquer que le théorème que j'ai donné par rapport aux six droites en involution de M. Sylvester [300], peut s'exprimer dans une forme encore plus simple comme suit:

Soit donnée une courbe cubique en espace, et prenons un point quelconque de la courbe pour sommet d'un cône du second ordre, d'ailleurs arbitraire; un plan tangent du cône rencontre la courbe en deux points, et par ces deux points on peut mener une droite: les droites qui correspondent de cette manière à six plans tangents quelconques du cône sont des droites en involution. Je dois remarquer que l'idée de rattacher ces droites à une surface du quatrième ordre est due à M. Sylvester.

À propos de ce sujet, j'ai considéré le problème de trouver le lieu du sommet d'un cône du second ordre qui touche à six droites données: ce lieu est une surface du huitième ordre; et en représentant les coordonnées de l'une quelconque des droites par (a, b, c, f, g, h), savoir les coordonnées de la première droite, etc., sont

$$(a_1, b_1, c_1, f_1, g_1, h_1), \ldots (a_6, b_6, c_6, f_6, g_6, h_6),$$

les coefficients de l'équation seront des fonctions linéaires des déterminants du sixième ordre formés au moyen de la matrice (a, b, c, f, g, h), à six lignes et vingt et une colonnes.

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302.

CONSIDÉRATIONS GÉNÉRALES SUR LES COURBES EN ESPACE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, *t.* LIV. (Janvier—Juin, 1862), *pp.* 55—60, 396—400, 672—678.]

Soit une courbe donnée du $m^{\text{ième}}$ ordre; je suppose toujours que cette courbe soit une courbe propre, savoir qu'elle n'est pas composée de courbes d'ordres inférieurs.

Si nous prenons pour sommet d'un cône qui passe par la courbe un point A quelconque qui n'est pas sur la courbe, ce cône sera de l'ordre m; cela est vrai en général quelle que soit la courbe; seulement si m est un nombre composé, alors pour de certaines courbes il peut y avoir des positions de A pour lesquelles le cône sera d'un ordre sous-multiple de m; mais en faisant abstraction de ces positions particulières, le cône sera de l'ordre m. Et, cela étant, une droite du cône ne contiendra en général qu'un seul point de la courbe. En employant quatre coordonnées (x, y, z, w) et en supposant qu'au point A on ait

$$x = 0, \quad y = 0, \quad z = 0,$$

l'équation du cône sera $U = 0$, où U est une fonction homogène de (x, y, z) de l'ordre m. On peut faire passer par la courbe une surface ayant pour équation

$$Qw - P = 0 \quad \left[\text{ou } w = \frac{P}{Q}\right],$$

où P, Q sont des fonctions homogènes de (x, y, z) des ordres p, $p-1$ respectivement. Et on peut supposer que p soit égal tout au plus à $m-1$: en effet, en prenant $p = m-1$, l'équation contiendrait

$$\tfrac{1}{2}(m-1)m + \tfrac{1}{2}m(m+1) - 1,$$

c'est-à-dire $m^2 - 1$ constantes arbitraires; et en déterminant convenablement $m^2 - m + 1$ de ces quantités, la surface de l'ordre $m-1$ passera par $m^2 - m + 1$ points de la courbe de l'ordre m, c'est-à-dire cette surface contiendra la courbe entière. De cette manière, on obtiendrait toujours une surface de l'ordre $m-1$; mais si les fonctions P, Q

ainsi trouvées avaient un facteur commun, ce facteur devrait être écarté; il convient donc de supposer que les degrés de P, Q soient p, $p-1$ respectivement, p étant tout au plus égal à $m-1$. La surface $Qw-P=0$ a au point A un point conique du $(p-1)^{\text{ième}}$ ordre; en effet dans le voisinage de ce point l'équation se réduit à $Q=0$, laquelle appartient à un cône du $(p-1)^{\text{ième}}$ ordre. J'ajoute que la surface contient les $p(p-1)$ droites $P=0$, $Q=0$ qui passent chacune par le point A; toute autre droite par ce point rencontre la surface dans ce point (lequel compte pour $p-1$ points d'intersection) et encore dans un seul point donné par l'équation

$$w=\frac{P}{Q}.$$

On peut appeler *monoïde* une telle surface; le point A sera le *sommet*; le cône $P=0$ le cône supérieur; le cône $Q=0$, le cône inférieur; les droites d'intersection de ces deux cônes, les *droites de la monoïde.*

Or le cône circonscrit $U=0$ et la monoïde $Qw-P=0$ se coupent selon une courbe de l'ordre mp: si $p=1$, cette intersection des deux surfaces sera la courbe du $m^{\text{ième}}$ ordre, laquelle sera une courbe plane; mais, dans tout autre cas, la courbe d'intersection sera composée de la courbe du $m^{\text{ième}}$ ordre, et d'un autre système de l'ordre $m(p-1)$; ce système ne peut être autre chose que les droites d'intersection du cône circonscrit $U=0$, et du cône inférieur $Q=0$ de la monoïde; c'est-à-dire les équations

$$U=0, \quad Q=0$$

doivent donner $P=0$; car, cela étant, les droites $U=0$, $Q=0$ seront situées sur la monoïde; et ces droites, lesquelles forment un système de l'ordre $m(p-1)$, seront partie de l'intersection de la monoïde et du cône circonscrit $U=0$. Et il est nécessaire que cela soit ainsi, car autrement chaque droite du cône $U=0$ ne contiendrait sur la monoïde que le point A, et le point déterminé par l'équation $w=\frac{P}{Q}$, lequel est un point sur la courbe du $m^{\text{ième}}$ ordre; donc cette autre partie de l'intersection de la monoïde et du cône $U=0$ serait, non pas une courbe quelconque, mais le seul point A; ce qui est absurde.

Le cône circonscrit $U=0$ ne peut pas être un cône quelconque à moins que $p=1$; en effet si $p>1$, il est nécessaire que le cône ait au moins $(p-1)m$ droites doubles (en comprenant dans cette locution le cas où le cône a des singularités qui équivalent à $(p-1)m$ droites doubles), car en supposant pour un moment que le cône $U = 0$ n'ait pas de singularités, le cône $P = 0$ de l'ordre p devrait passer par les $(p-1)m$ droites d'intersection du cône $Q = 0$ de l'ordre $(p-1)$ et du cône $U = 0$ de l'ordre m; or m est au moins égal à $p+1$, de manière que le cône $P = 0$ doit passer au moins par (p^2-1) droites du cône $Q = 0$; mais $p^2 - 1$ est $> p^2 - p$, à moins que $p = 1$; donc ce cône $P = 0$ serait composé du cône $Q = 0$ et d'un plan $P' = 0$ par le point A; c'est-à-dire $P = QP'$, et l'équation de la monoïde se réduirait à $w = P'$, où l'on aurait $p = 1$, ce qui est contraire à l'hypothèse. On obtiendra le même résultat à moins de supposer que le cône $Q = 0$ passe par un certain nombre χ de droites doubles du cône $U = 0$; mais en faisant cette supposition, chacune de ces droites compte pour deux intersections des cônes $Q = 0$, $U = 0$; il y a encore $(p-1)m - 2\chi$ droites d'intersection; et les $\chi + [(p-1)m - 2\chi]$, c'est-à-dire $(p-1)m - \chi$ droites peuvent être comprises parmi les $p(p-1)$ droites de la monoïde si χ est égal au moins à $(p-1)(m-p)$; c'est-à-dire le cône $U = 0$ doit avoir au moins ce nombre de droites doubles. Je remarque que pour m impair, et $p = \frac{1}{2}(m+1)$, le nombre sera $\frac{1}{4}(m^2 - 2m + 1)$, et pour m pair, et $p = \frac{1}{2}m$ ou $\frac{1}{2}m + 1$, le nombre sera $\frac{1}{4}(m^2 - 2m)$; mais pour toute autre valeur de p, le nombre sera moins élevé.

Je résume comme suit:

Toute courbe du $m^{\text{ième}}$ ordre est l'intersection d'un cône circonscrit $U = 0$, du $m^{\text{ième}}$ ordre, et d'une surface monoïde $Qw - P$, de l'ordre $p = m - 1$ au plus. L'intersection complète de deux surfaces est composée de la courbe du $m^{\text{ième}}$ ordre et des $m(p-1)$ droites d'intersection du cône circonscrit $U = 0$, et du cône inférieur $Q = 0$ de la monoïde. Ces droites seront $(p-1)(m-p) + \alpha$ droites, chacune répétée deux fois, et $(p-1)(2p-m) - 2\alpha$ droites, où α peut être égal à zéro; chacune des $(p-1)(m-p) + \alpha$ droites sera une droite double du cône $U = 0$; et les $(p-1)(m-p) + \alpha$ droites et $(p-1)(2p-m) - 2\alpha$ droites, ensemble $p(p-1) - \alpha$ droites, seront situées sur le cône supérieur $P = 0$ de la monoïde.

Il y a deux circonstances qui empêchent que cette théorie ne conduise tout de suite à une classification des courbes en espace. D'abord, une droite double du cône $U = 0$ peut correspondre ou à un point double réel, ou à un point double apparent de la courbe; et de même en supposant que la droite double devienne une droite de rebroussement, cette droite peut ou correspondre à un point de rebroussement (point stationnaire) de la courbe, ou la droite peut être une tangente ordinaire de la courbe, sans qu'il ait sur la courbe aucune singularité qui corresponde à cette droite de rebroussement (voir le Mémoire de M. Salmon: "On the classification of curves of double curvature," *Camb. et Dubl. Math. Journ.*, *t.* V. *pp.* 23—46, 1850).

Puis, étant donnée l'équation $U = 0$ du cône circonscrit, la monoïde n'est pas une surface déterminée, et il n'est guère facile de voir quel doit être l'ordre de cette surface. En effet, cette équation étant $w = \dfrac{P}{Q}$, il peut y avoir des fonctions P', Q' telles que $PQ' - P'Q = MU$, et, cela étant, puisqu'il ne s'agit que de l'intersection avec le cône $U = 0$, on pourrait remplacer l'équation $w = \dfrac{P}{Q}$ par celle-ci, $w = \dfrac{P'}{Q'}$, laquelle peut être d'un ordre inférieur.

Ces difficultés se présentent dès le commencement. En effet soit $m = 3$. On a $p = 1$ ou $p = 2$, mais $p = 1$ ne donne que la cubique plane; je suppose donc $p = 2$. Le cône $U = 0$ du troisième ordre aura une droite double, laquelle peut être une droite de rebroussement. L'équation de la monoïde sera $w = \dfrac{P}{Q}$, où $Q = 0$ est l'équation d'un plan qui passe par le point double ou de rebroussement, et qui coupe ainsi le cône $U = 0$ selon une autre droite; et $Qw - P = 0$ est l'équation d'un cône du second ordre qui passe par ces deux droites. Mais soit que le cône $U = 0$ ait une droite double, soit que cette droite soit de rebroussement, on n'obtient qu'une seule espèce de courbe cubique; au premier cas le sommet n'est pas situé, au deuxième cas ce sommet est situé, sur une tangente de la courbe cubique; voilà toute la différence.

Soit encore $m = 4$; on peut avoir $p = 1$, 2 ou 3; mais $p = 1$ ne donne que les courbes planes du quatrième ordre, je suppose donc $p = 2$ ou $p = 3$; dans l'un ou l'autre cas, le cône $U = 0$ du quatrième ordre doit avoir au moins deux droites doubles. Il peut donc y avoir seulement deux droites doubles; l'une de ces droites peut être une droite de rebroussement, ou toutes les deux peuvent être telles droites. Ou encore, il peut y avoir trois droites doubles; l'une de ces droites peut être une droite de rebroussement, ou toutes les trois peuvent être telles droites. Il y a donc un assez grand nombre de cas à considérer; mais on sait qu'il n'y a que quatre espèces en tout, savoir: 1° la courbe d'intersection de deux surfaces du second ordre qui ne se touchent pas, courbe que je nomme *quadriquadrique générale*; 2° les deux surfaces du second ordre peuvent se toucher; la courbe d'intersection sera une *quartique nodale*; 3° les deux surfaces peuvent avoir un contact singulier, la courbe d'intersection sera une *quartique cuspidale*; 4° il y a enfin la courbe du quatrième ordre qui n'est située que sur une seule surface du second ordre, et que l'on n'obtient qu'au moyen d'une surface du troisième ordre: ce sera la *courbe excubo-quartique*. Je remarque en passant que les quartiques nodale et cuspidale sont des sous-espèces tant de l'excubo-quartique que de la quadriquadrique. En supposant que le cône $U = 0$ n'ait que deux droites doubles ou de rebroussement, et soit que $p = 2$ ou $p = 3$, on obtiendra par la théorie actuelle la quadriquadrique générale (cela est évident par les formules du Mémoire cité de M. Salmon). Si le cône $U = 0$ a trois droites doubles ou de rebroussement, alors soit que $p = 2$ ou $p = 3$, on obtiendra, selon les circonstances, ou l'excubo-quartique, ou la quartique nodale, ou la quartique cuspidale (mais non pas cette dernière, à moins qu'il n'y ait au moins une droite de rebroussement). Mais il faudrait pour tout cela une discussion plus approfondie.

Je remarque qu'en prenant le point A sur la courbe du $m^{\text{ième}}$ ordre, l'on aurait eu, au lieu du cône $U = 0$ du $m^{\text{ième}}$ ordre, un cône du $(m-1)^{\text{ième}}$ ordre, et l'ordre du cône se réduirait encore si le point A était un point multiple de la courbe. Peut-être il conviendrait de considérer de tels cônes au lieu du cône du $m^{\text{ième}}$ ordre.

En conclusion, je fais les réflexions que voici, savoir: Si $S = 0$, $T = 0$ sont des surfaces quelconques qui passent par la courbe du $m^{\text{ième}}$ ordre, alors en éliminant entre ces équations la coordonnée w, on obtient une équation

$$\Pi = UV = 0,$$

qui contient comme facteur l'équation $U = 0$ du cône du $m^{\text{ième}}$ ordre. Mais il y a plus: la théorie de l'élimination entre deux équations algébriques fait voir que les équations

$S=0$, $T=0$ donnent lieu à un assez grand nombre d'équations de la forme $w=\dfrac{P}{Q}$ (en représentant deux quelconques de ces équations par $w=\dfrac{P}{Q}$, $w=\dfrac{P'}{Q'}$, on aura toujours $PQ'-P'Q=M\Pi$), c'est-à-dire on obtient par une telle élimination plusieurs surfaces monoïdes dont chacune coupe le cône $\Pi=UV=0$, selon la courbe d'intersection complète de deux surfaces $S=0$, $T=0$. Mais il ne s'ensuit pas (même en admettant que l'on ait de cette manière toutes les surfaces monoïdes qui passent par l'intersection complète), que l'on ait toutes les surfaces monoïdes qui passent par la courbe du $m^{\text{ième}}$ ordre; en effet il peut y avoir des fonctions P', Q' lesquelles, sans donner $PQ'-P'Q=MUV$, donnent cependant $PQ'-P'Q=MU$, et, cela étant, $w=\dfrac{P'}{Q'}$, serait une surface monoïde qui passerait par la courbe du $m^{\text{ième}}$ ordre.

P.S. On déduit sans peine la théorie des courbes situées sur une surface du second ordre (voir ma Note "On the curves situate on a surface of the second order," *Phil. Mag.*, July 1861, [314], et les savantes recherches que M. Chasles vient de publier dans les *Comptes Rendus*). En effet, en supposant que la monoïde soit une surface du second ordre (hyperboloïde) et que son équation soit $w=\dfrac{xy}{z}$, alors, puisque le cône $U=0$, du $m^{\text{ième}}$ ordre, doit rencontrer le plan $z=0$ selon les seules droites $x=0$, $y=0$, il faut que ces droites soient des droites multiples du cône $U=0$, et en prenant p, q des nombres tels que $p+q=m$, on peut supposer que les deux droites soient des droites multiples des ordres p et q respectivement; et cela arrivera si U (fonction homogène du $m^{\text{ième}}$ ordre en x, y, z) contient x^p pour la plus haute puissance de x, et y^q pour la plus haute puissance de y. Car en arrangeant selon les puissances descendantes de y, on aura

$$U=y^q(x,z)^p+y^{q-1}z(x,z)^p+\dots,$$

ce qui fait voir que $x=0$, $z=0$ sera une droite multiple du $p^{\text{ième}}$ ordre, et de même $y=0$, $z=0$ sera une droite multiple du $q^{\text{ième}}$ ordre. On a donc selon la notation de M. Chasles

$$U=M(x^p,y^q),$$

en se souvenant qu'ici U contient aussi la coordonnée z.

Suite. Courbes du quatrième ordre.

Toute surface du second ordre est une surface monoïde, et on peut prendre pour sommet un point quelconque de la surface. En effet, en considérant un point quelconque de la surface du second ordre, soient

$$x = 0, \quad y = 0, \quad z = 0,$$

les équations de trois plans quelconques qui passent par ce point; l'équation de la surface sera satisfaite en y écrivant

$$x = 0, \quad y = 0, \quad z = 0;$$

donc cette équation ne contiendra pas de terme en w^2, et elle sera ainsi de la forme

$$wQ - P = 0 \quad \text{ou } w = \frac{P}{Q},$$

P et Q étant des fonctions homogènes en x, y, z, du second ordre et du premier ordre respectivement; c'est-à-dire, la surface sera monoïde, ou, si l'on veut, monoïde quadrique.

Or, par une courbe du quatrième ordre (ou courbe quartique) quelconque en espace, on peut faire passer une surface du second ordre, ou monoïde quadrique. Selon la théorie générale, la surface monoïde est tout au plus du troisième ordre, ou monoïde cubique; j'avais tort de supposer que pour la courbe excubo-quartique la surface monoïde fût nécessairement une monoïde cubique. Il arrive comme suit, savoir: pour la courbe quadriquadrique, en prenant pour sommet un point quelconque de l'espace (on suppose toujours que le sommet de la monoïde n'est pas situé sur la courbe), on aura une monoïde quadrique; mais pour la courbe excubo-quartique, pour que la monoïde soit quadrique, il faut que le sommet soit situé sur la surface du second ordre (il n'y a qu'une seule surface) qui passe par la courbe; cela étant, la monoïde quadrique sera cette surface même du second ordre. Mais en prenant pour sommet un point quelconque qui n'est point situé sur la surface du second ordre, la monoïde sera nécessairement une surface cubique.

Ainsi, pour les courbes quartiques, il suffit de considérer ces courbes comme situées sur une monoïde quadrique; il est cependant assez intéressant de les considérer comme situées sur une monoïde cubique. Je suppose donc $U = 0$, $w = \frac{P}{Q}$, où $U = 0$ est un cône quartique et $w = \frac{P}{Q}$ une monoïde cubique avec le même point $x = 0$, $y = 0$, $z = 0$ pour sommet.

Selon la théorie générale, les huit droites $Q = 0$, $U = 0$ doivent être comprises parmi les six droites $Q = 0$, $P = 0$. Or, pour cela, il faut que le cône $U = 0$ ait des droites multiples; il y a trois cas à considérer: 1° Le cône passe par les six droites, et une de ces droites est une droite triple du cône; il y aura, comme cela doit être,

$$3 + 1 + 1 + 1 + 1 + 1 = 8$$

droites d'intersection de $Q = 0$, $U = 0$. 2° Le cône passe par les six droites; deux de ces droites étant des droites doubles, il y aura

$$2 + 2 + 1 + 1 + 1 + 1 = 8$$

droites d'intersection. 3° Le cône passe par cinq des six droites; trois de ces cinq droites étant des droites doubles, il y aura

$$2 + 2 + 2 + 1 + 1 = 8$$

droites d'intersection. Or, dans le premier et le second cas, le cône $U = 0$ passe par les six droites d'intersection des cônes $P = 0$, $Q = 0$; il faut donc que l'on ait identiquement

$$U = PQ' - P'Q,$$

P', Q' étant des fonctions homogènes en x, y, z du second ordre et du premier ordre respectivement. Mais en vertu de l'équation

$$U = PQ' - P'Q = 0,$$

on a $\dfrac{P}{Q} = \dfrac{P'}{Q'}$, c'est-à-dire la courbe est située sur la monoïde quadrique $w = \dfrac{P'}{Q'}$. La courbe sera quadriquadrique ou excubo-quartique, selon les circonstances.

Reste à considérer le troisième cas. La monoïde cubique est une surface cubique ayant le sommet pour point conique; la théorie des droites sur une telle surface a été examinée par M. Salmon dans son Mémoire: "On the triple tangent planes of a surface of the third order," *Camb. and Dubl. Math. Journ., pp.* 252—260 (1849). Il y a, en effet, les six droites par le point conique, savoir: les droites $P = 0$, $Q = 0$, qui comptent pour douze droites, et de plus quinze droites; $6 \times 2 + 15 = 27$. Chacune des quinze droites est donnée comme troisième intersection de la surface avec un plan qui passe par deux des six droites. Donc, en nommant 1, 2, 3, 4, 5, 6 les six droites, on peut nommer 12 la droite dans le plan mené par les droites 1, 2; et de même pour les droites 13, 23, etc. La droite 1 est rencontrée par les droites 2, 3, 4, 5, 6, 12, 13, 14, 15, 16; la droite 12 par les droites 1, 2; 34, 56; 35, 64; 36, 45; et ainsi pour les autres droites.

Cela étant, je suppose que le cône $U = 0$ passe par les droites 2, 3, 4, 5, 6, et que les droites 4, 5, 6 soient droites doubles du cône. Je dis que la courbe sera située sur une surface du second ordre qui passe par les droites 12, 13 (droites qui ne se coupent pas), savoir, ces deux droites et la courbe seront l'intersection complète de la monoïde cubique et de la surface du second ordre; cela fait voir que la courbe est une courbe excubo-quartique. Et, comme il est auparavant dit, en prenant pour sommet un point quelconque de la surface du second ordre, la courbe sera située sur une monoïde quadrique $w = \dfrac{P}{Q}$.

Donc, en partant de la monoïde cubique, on trouve toujours que la courbe du quatrième ordre est située sur une monoïde quadrique.

J'établis comme suit l'existence de la surface du second ordre qui passe par les droites 12, 13. Je remarque en général que l'équation $w = \frac{P}{Q}$ peut s'écrire sous la forme $w + L = \frac{P + LQ}{Q}$, où L est une fonction homogène linéaire quelconque de x, y, z; ou en changeant w, cette équation sera

$$w = \frac{P + LQ}{Q},$$

c'est-à-dire on peut remplacer le cône $P = 0$ par un cône quelconque qui passe par les droites d'intersection des cônes $P = 0$, $Q = 0$. Donc, pour la monoïde cubique, on peut prendre pour $P + LQ = 0$ un système de trois plans, et en prenant pour équations de ces plans $x = 0$, $y = 0$, $z = 0$, on peut prendre pour équations de la monoïde cubique

$$w = \frac{xyz}{Q}.$$

Comme les coordonnées x, y, z renferment chacune un multiplicateur indéterminé, on peut écrire

$$Q = x^2 + y^2 + z^2 + 2lyz + 2mzx + 2nxy,$$

ou, en posant $\alpha' = \dfrac{1}{\alpha}$, $\beta' = \dfrac{1}{\beta}$, $\gamma' = \dfrac{1}{\gamma}$, α, β, γ étant des quantités quelconques, on peut écrire

$$Q = x^2 + y^2 + z^2 + (\alpha + \alpha')yz + (\beta + \beta')zx + (\gamma + \gamma')xy,$$

ce qui est la forme la plus commode pour mettre en évidence les droites d'intersection $xyz = 0$, $Q = 0$. On peut supposer que les équations de ces droites soient

$$\begin{array}{ll}
(1)\ x = 0,\ y + \alpha z = 0, & (2)\ x = 0,\ \alpha' y + z = 0, \\
(3)\ y = 0,\ z + \beta x = 0, & (4)\ y = 0,\ \beta' z + x = 0, \\
(5)\ z = 0,\ x + \gamma y = 0, & (6)\ z = 0,\ \gamma' x + y = 0.
\end{array}$$

Donc, pour les plans 56, 34, 24, on aura les équations

$$(56)\ z = 0, \quad (34)\ y = 0, \quad (24)\ x + \alpha\beta y + \beta z = 0;$$

et de là l'équation

$$AQ^2 + Qz[By + C(x + \alpha\beta y + \beta z)] + Dz^2 y(x + \alpha\beta y + \beta z) = 0$$

sera celle d'un cône du quatrième ordre qui passe par les droites 2, 3, 4, 5, 6 et a les droites 4, 5, 6 pour droites doubles; et comme cette équation contient les trois quantités arbitraires $A : B : C : D$, ce sera l'équation la plus générale qui satisfait aux conditions dont il s'agit: c'est-à-dire cette équation sera celle du cône $U = 0$.

Les équations de la droite 12 sont $x = 0$, $w = 0$; pour obtenir celle de la droite 13, j'observe que l'équation du point 13 est

$$\alpha\beta x + y + \alpha z = 0,$$

et je forme l'équation identique

$$Q = (\alpha\beta x + y + \alpha z)[(\gamma + \gamma' - \alpha\beta)x + y + \alpha' z] + \beta'(1 - \alpha\beta\gamma)(1 - \alpha\beta\gamma')x(z + \beta x),$$

laquelle se vérifie sans peine. Donc, en écrivant

$$\alpha\beta x + y + \alpha z = 0, \quad \text{ou } y = -\alpha(z + \beta x),$$

l'équation $w = \dfrac{xyz}{Q}$ devient

$$w = \frac{-\alpha x(z + \beta x)z}{\beta'(1 - \alpha\beta\gamma)(1 - \alpha\beta\gamma')x(z + \beta x)} = \frac{-\alpha z}{\beta'(1 - \alpha\beta\gamma)(1 - \alpha\beta\gamma')},$$

ou, ce qui est la même chose,

$$w\beta'(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')+\alpha z=0,$$

laquelle et l'équation

$$\alpha\beta x+y+\alpha z=0$$

sont les deux équations de la droite 13.

Cela étant,

$$(Ax+Bw)(\alpha\beta x+y+\alpha z)+(Cx+Dw)[\alpha z+(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')w]=0$$

sera l'équation d'une surface du second ordre qui passe par les deux droites 12, 13; et, en éliminant w au moyen de l'équation

$$w=\frac{xyz}{Q},$$

on obtient l'équation du cône du quatrième ordre. En effet, en substituant cette valeur de w, on obtient une équation du sixième ordre laquelle, divisée par $(\alpha\beta x+y+\alpha z)$, devient

$$AQ^2+ByzQ+(CQ+Byz)\frac{\alpha z(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')x(z+\beta x)}{\alpha\beta x+y+\alpha z}=0;$$

or

$$\alpha\beta x+y+\alpha z-\alpha(z+\beta x)=y,\quad \text{donc}\quad \frac{\alpha(z+\beta x)}{\alpha\beta x+y+\alpha z}=1-\frac{y}{\alpha\beta x+y+\alpha z},$$

donc la partie fractionnelle est

$$\frac{\alpha(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')x(z+\beta x)+(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')xy}{\alpha\beta x+y+\alpha z};$$

c'est-à-dire

$$(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')x\cdot\frac{\alpha z+\alpha\beta x+y}{\alpha\beta x+y+\alpha z}=(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')x,$$

et l'équation devient

$$AQ^2+ByzQ+(CQ+Byz)z(1-\alpha\beta\gamma)(1-\alpha\beta\gamma')x=0,$$

ou enfin

$$AQ^2+ByzQ+(CQ+Dyz)z(x+\alpha\beta y+\beta z)=0,$$

ce qui est en effet l'équation ci-dessus trouvée pour le cône $U=0$.

Suite. Courbes du cinquième ordre.

On pourrait assez bien dénoter les courbes des ordres un, deux, trois, comme suit, savoir:

Courbe du premier ordre, par	1,
Courbe du second ordre, par	2,
Courbes du troisième ordre, par	3 et $4-1$,

c'est-à-dire que la courbe plane serait 3 et la courbe dans l'espace $4-1$. Mais pour le quatrième ordre, cette notation serait déjà en défaut, et l'on aurait besoin d'une notation telle que celle-ci:

Courbe plane	4.1
Courbe quadriquadrique	2.2
Courbe excubo-quartique	$2.3-1-1$.

Cela devient cependant trop complexe, et comme je ne cherche nullement une notation parfaite, il suffit pour le moment de dénoter la courbe plane (dont je n'ai guère à m'occuper) par 4^*, la quadriquadrique par 4, et l'excubo-quartique par $6-2$. De même pour le troisième ordre, on peut dénoter la courbe plane par 3^* et la courbe dans l'espace par 3.

Cela étant, pour les courbes du cinquième ordre, ou courbes quintiques, il y a cinq espèces, savoir:

		P. D. A.
Courbe plane	ou espèce 5	0
Courbe quadricubique	$6-1$	4
Courbe quadriquartique	$8-3$	6
Courbe cubicubique (deux espèces)	$9-6+2$	5
	$9-3-1$	6

où la colonne P. D. A. fait voir pour chaque espèce le nombre des points doubles apparents (voir le Mémoire de M. Salmon: "On the classification of Curves of double Curvature," *Camb. et Dubl. Math. Journ.*, *t.* V. 1850). Cette classification est au fond celle du Mémoire cité; seulement M. Salmon a énuméré trois sous-espèces qui n'existent pas, à savoir les sous-espèces quadriquadriques analogues à V.7, V.8, V.9 (p. 42, où M. Salmon parle des courbes algébriques correspondantes à V.1, V.8, V.9, V.10, sans attacher des numéros à ces quatre sous-espèces). Je vais à présent expliquer la théorie des cinq espèces.

Courbe plane, ou espèce 5. — Il va sans dire que cette courbe est l'intersection d'une surface quintique par un plan quelconque.

Courbe quadricubique, ou espèce $6-1$. — Cette courbe est l'intersection partielle d'une surface quadrique et d'une surface cubique qui ont en commun une seule droite. En supposant que les équations de la droite soient $x=0$, $y=0$, on peut prendre pour équation de la surface quadrique $xw-yz=0$, et pour celle de la surface cubique $xV - yU = 0$, où $U = 0$, $V = 0$, sont des surfaces quadriques quelconques. Au lieu des deux équations

$$xw - yz = 0, \quad xV - yU = 0,$$

il est permis d'écrire

$$\begin{vmatrix} x, & z, & U \\ y, & w, & V \end{vmatrix} = 0,$$

ce qui fait voir qu'il passe par la courbe cette nouvelle surface cubique, $zV - wU = 0$, laquelle a en commun avec la première surface cubique la courbe quadriquadrique $U = 0$, $V = 0$.

La courbe a 4 points doubles apparents; elle peut donc avoir 0, 1 ou 2 points doubles ou de rebroussement; cela donne les sous-espèces

$$V.1, \quad V.2, \quad V.3, \quad V.4, \quad V.5, \quad V.6,$$

de M. Salmon.

Je remarque en passant qu'en supposant que la surface cubique $wV - yV' = 0$ a en commun avec la surface quadrique $xw - yz = 0$, non-seulement la droite $x = 0$, $y = 0$, mais aussi une autre génératrice du même mode de génération, on aura, au lieu de la courbe quintique $6 - 1$, cette nouvelle droite, et une courbe excubo-quartique. C'est là le théorème qui donne une des constructions que M. Chasles a trouvées pour la courbe excubo-quartique.

J'ajoute que la courbe considérée comme courbe située sur une surface quadrique sera de l'espèce $(3, 2)$, ou, selon la notation de M. Chasles, $M(\alpha^2\beta^2)$. On connaît ainsi un grand nombre des propriétés de cette courbe, et aussi de la courbe d'espèce $8 - 3$ dont nous allons parler, laquelle, considérée comme courbe située sur une surface quadrique, est de l'espèce $(4, 1)$ ou $M(\alpha\epsilon^2)$.

Courbe quadriquartique, ou espèce $8-3$. — Une telle courbe est l'intersection partielle d'une surface quadrique et d'une surface quartique qui ont en commun trois droites qui ne se rencontrent pas: autrement dit, ces droites seront des génératrices du même mode de génération de la surface quadrique[1].

Soit $xw - yz = 0$ l'équation de la surface quadrique; on peut prendre pour les trois génératrices

$$\begin{gathered}(x - \lambda y = 0,\ \lambda w - z = 0),\\(x - \mu y = 0,\ \mu w - z = 0),\\(x - \nu y = 0,\ \nu w - z = 0);\end{gathered}$$

et cela étant, l'équation de la surface quartique sera

$$(a, \ldots)(x - \lambda y, \lambda w - z)(x - \mu y, \mu w - z)(x - \nu y, \nu w - z) = 0,$$

[1] Dans le symbole $8 - 3$ on remarquera que 3 dénote non pas la cubique gauche, mais les trois droites; $8 - 1 - 1 - 1$ serait trop long, et je me suis servi exprès de la notation moins complète; et ainsi il est nécessaire en pareil cas d'expliquer la notation.

en représentant de cette manière une fonction linéaire par rapport à $x - \lambda y$ et $\lambda w - z$, par rapport à $x - \mu y$ et $\mu w - z$, et par rapport à $x - \nu y$ et $\nu w - z$, les coefficients $a, \ldots$ étant des fonctions linéaires quelconques de x, y, z, w.

La courbe a 6 points doubles apparents; il n'y a donc pas d'autre singularité: c'est l'espèce analogue à

$$V.10$$

de M. Salmon.

Courbe cubicubique, espèce $9-3-1$. — La courbe est l'intersection partielle de deux surfaces cubiques qui ont en commun une courbe cubique gauche et une droite qui ne rencontre pas la courbe cubique.

Soient p, q, r, s, t, u, P, Q des fonctions linéaires quelconques des coordonnées; α, β, γ, α', β', γ' des fonctions linéaires quelconques de P, Q (autrement dit, $\alpha=0$, $\beta=0$, etc., seront les équations de six plans quelconques qui passent par la droite $P=0$, $Q=0$).

Cela étant, les surfaces cubiques

$$\begin{vmatrix} p, & s, & \alpha \\ q, & t, & \beta \\ r, & u, & \gamma \end{vmatrix} = 0, \qquad \begin{vmatrix} p, & s, & \alpha' \\ q, & t, & \beta' \\ r, & u, & \gamma' \end{vmatrix} = 0,$$

auront en commun la courbe cubique

$$\begin{vmatrix} p, & s \\ q, & t \\ r, & u \end{vmatrix} = 0$$

(ainsi les surfaces quadriques $pt-sq=0$, $pu-sr=0$ se rencontrent selon la droite $p=0$, $s=0$ et selon la courbe cubique dont il s'agit) et la droite $P=0$, $Q=0$. Il y aura donc encore une intersection qui sera la courbe quintique $9-3-1$.

La courbe a 6 points doubles apparents; il n'y a donc pas d'autre singularité: c'est l'espèce

$$V.10$$

de M. Salmon.

Je remarque en passant que cette courbe quintique $9-3-1$ a avec une certaine courbe sextique une relation semblable à celle qui existe entre la courbe excubo-quartique et la courbe quintique $6-1$. En effet, p, q, r, s, t, u, α, β, γ, α', β', γ' étant à présent des fonctions linéaires quelconques des coordonnées, la courbe sextique sera donnée par les équations

$$\begin{vmatrix} p, & s, & \alpha, & \alpha' \\ q, & t, & \beta, & \beta' \\ r, & u, & \gamma, & \gamma' \end{vmatrix} = 0,$$

ou, ce qui revient à la même chose, elle sera l'intersection partielle des deux surfaces cubiques

$$\begin{vmatrix} p, & s, & \alpha \\ q, & t, & \beta \\ r, & u, & \gamma \end{vmatrix} = 0, \qquad \begin{vmatrix} p, & s, & \alpha' \\ q, & t, & \beta' \\ r, & u, & \gamma' \end{vmatrix} = 0,$$

lesquelles ont en commun la courbe cubique

$$\begin{vmatrix} p, & q, & r \\ s, & t, & u \end{vmatrix} = 0.$$

Or, en prenant α, β, γ, α', β', γ' des fonctions linéaires de P et Q, nous avons, en effet, réduit la courbe sextique à la droite $P = 0$, $Q = 0$ et à la courbe quintique $9 - 3 - 1$.

Courbe cubicubique, espèce $9 - 6 + 2$. — Cette courbe est l'intersection partielle de deux surfaces cubiques qui ont en commun une courbe excubo-quartique. En supposant que cette courbe excubo-quartique soit l'intersection partielle d'une surface quadrique et d'une surface cubique qui ont en commun les deux droites $(x = 0, y = 0)$ et $(z = 0, w = 0)$, on peut prendre pour équation de ces deux surfaces

$$U = xw - yz = 0,$$

$$V = \begin{vmatrix} x, & y \\ z, & w \end{vmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} x, & z \\ y, & w \end{pmatrix} = 0,$$

en représentant de cette manière la fonction $axz + byz + cxw + dyw$, linéaire par rapport à x, y et par rapport à z, w, avec des coefficients a, b, c, d, lesquels sont des fonctions linéaires quelconques de x, y, z, w.

En écrivant d'abord

$$V = (ax + by)z + (cx + dy)w,$$

$$U = \qquad\qquad - yz + \qquad\qquad xw,$$

on obtient

$$xV - (cx + dy)U = z[ax^2 + (b + c)xy + dy^2].$$

Et de même en écrivant

$$V = (az + cw)x + (bz + dw)y,$$

$$U = \qquad\qquad wx - \qquad\qquad zy,$$

on obtient

$$xV + (bz + dw)U = x[az^2 + (b + c)zw + dw^2].$$

Or le premier de ces résultats fait voir qu'en supposant $U = 0$, $V = 0$, on obtient $ax^2 + (b+c)xy + dy^2 = 0$, et le second, qu'en supposant $U = 0$, $V = 0$, on obtient de même $az^2 + (b + c)zw + dw^2 = 0$. Les surfaces $U = 0$, $V = 0$ se coupent selon la courbe excubo-quartique et les droites $(x = 0, y = 0)$ et $(z = 0, w = 0)$; mais la surface $ax^2 + (b + c)xy + dy^2 = 0$ ne passe que par la première, et la surface $az^2 + (b + c)zw + dw^2 = 0$ ne passe que par la seconde de ces deux droites; donc les deux surfaces se coupent selon la courbe excubo-quartique, mais non pas selon l'une ou l'autre des deux droites, c'est-à-dire que les deux surfaces cubiques

$$ax^2 + (b + c)xy + dy^2 = 0,$$

$$az^2 + (b + c)zw + dw^2 = 0,$$

se coupent selon la courbe excubo-quartique, et encore selon une courbe quintique $9 - 6 + 2$.

Les deux surfaces cubiques ont chacune une droite double, elles sont donc des surfaces réglées. La courbe est donc comprise parmi les courbes décrites sur une surface cubique réglée, pour lesquelles M. Chasles a trouvé dernièrement une construction géométrique très-élégante.

La courbe a cinq points doubles apparents; elle peut donc ne pas avoir d'autre singularité, ou avoir un point double ou de rebroussement: cela donne les trois sous-espèces

$$V.7, \quad V.8, \quad V.9$$

de M. Salmon.

On démontre sans peine que toute courbe quintique est plane, quadricubique, quadriquartique ou cubicubique; mais, pour faire voir qu'il n'existe que les cinq espèces ci-dessus mentionnées, il y a encore plusieurs cas à considérer. Par exemple, pour les courbes cubicubiques, on pourrait supposer que les deux surfaces cubiques avaient en commun une courbe quadriquadrique: si cela était, les équations des deux surfaces seraient de la forme $Vx - Uy = 0$, $Vz - Uw = 0$ (surfaces qui ont en commun la courbe quadriquadrique $U = 0$, $V = 0$), mais dans ce cas la courbe quintique serait située sur la surface quadrique $xw - yz = 0$, et l'on ne fait que retrouver l'espèce quadricubique $6 - 1$. J'ai fait, après M. Salmon, cette revue des différents cas, et je me suis assuré qu'il n'y a que les cinq espèces. Il convient peut-être de remarquer que l'énumération des sous-espèces comprises dans celles-ci n'est pas tout à fait complète, parce que, en certains cas, la courbe peut avoir un point triple, ou autre singularité plus élevée que les points doubles ou de rebroussement. Cela ne présente pas de difficulté, et en effet je n'ai parlé des sous-espèces que pour rapprocher mes résultats de ceux de M. Salmon.

La longueur de cette communication m'empêche de faire voir à présent comment les cinq espèces peuvent être déduites de la théorie générale des courbes dans l'espace considérées comme situées sur une surface monoïde.

* * *

303.

SUR LE PROBLÈME DU POLYGONE INSCRIT ET CIRCONSCRIT.

LETTRE À M. PONCELET.

[From the *Comptes Rendus de l'Académie des Sciences à Paris*, *t.* LV. (Juillet—Décembre, 1862), *pp.* 700, 701.]

J'ose vous écrire par rapport aux remarques que vous faites p. 483 de l'ouvrage (*Applications d'Analyse etc.* [Paris, *t.* I. (1862)]) au sujet de mes recherches sur le problème du polygone inscrit et circonscrit [267 and the papers 113, 115, 116 and 128 therein referred to].

Je n'ai nullement voulu attribuer à Fuss le théorème pour les deux cercles. J'ai seulement dit, tout à fait en passant: *The case... of the two circles (the original case of the Porism as considered by Fuss)* et en effet Fuss a fait des recherches sur ce cas d'un polygone inscrit et circonscrit à deux cercles. Mais je n'ai jamais imaginé qu'il y eût un géomètre (algébriste ou non) qui ne connût pas tant votre ouvrage classique de 1822, que le mémoire de 1828 de Jacobi, où l'on voit précisément ce que Fuss a fait sur ce problème. Par rapport à mon dernier Mémoire (*Phil. Trans.* 1861) [267], que vous citez et qui résume quelques Notes que j'ai publiées en 1853, permettez-moi de vous mentionner la forme de ma solution: on a une fonction $a+b\xi+c\xi^2+d\xi^3$, où ξ est une quantité indéterminée, et a, b, c, d sont des fonctions données très simples des paramètres qui déterminent les deux cercles (ou coniques). On développe la racine carrée de cette fonction dans la forme $A+B\xi+C\xi^2+D\xi^3+E\xi^4+\ldots$, et, cela fait, on a tout de suite l'équation entre les paramètres pour un polygone d'ordre quelconque; savoir pour le triangle, le pentagone, l'heptagone, etc., ces conditions sont

$$C=0, \quad \begin{vmatrix} C, & D \\ D, & E \end{vmatrix} = 0, \quad \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix} = 0, \quad \text{etc.},$$

tandis que pour le quadrangle, l'hexagone, l'octogone, etc., ces conditions sont

$$D = 0, \qquad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = 0, \qquad \begin{vmatrix} D, & E, & F \\ E, & F, & G \\ F, & G, & H \end{vmatrix} = 0, \quad \text{etc.,}$$

de manière que la condition est trouvée explicitement pour un polygone d'ordre quelconque *sans passer par celles qui appartiennent aux polygones d'ordre inférieur.*

Comme j'attache, je l'avoue, un peu d'importance à cette solution (laquelle selon l'explication que je viens de donner ne paraît pas mériter la critique que vous en faites) je serais bien aise si vous voulez bien communiquer cette lettre à l'Académie.

* * *

304.

SUR UN MÉMOIRE DE JACOBI.

EXTRAIT D'UNE LETTRE À M. J. BERTRAND.

[From the *Comptes Rendus de l'Académie des Sciences à Paris*, *t.* LVI. (Janvier—Juin, 1863), *p.* 43. See 298, foot-note to No. 117.]

Permettez-moi de vous soumettre une remarque que je viens de faire par rapport au Mémoire de Jacobi "Sur l'élimination des noeuds dans le problème des trois corps" (*Compte Rendu* 8 août, 1842 [and *Crelle*, *t.* XXVI. (1843), *pp.* 115—131]). Il me semble que quoique Jacobi dise qu'il a fait dépendre le problème d'un système de cinq équations du premier ordre et une seule du second ordre, il a réellement fait plus que cela, savoir qu'il l'a fait dépendre d'un système de six équations du premier ordre et qu'ainsi il est allé aussi loin que vous dans le "Mémoire sur l'intégration de quelques équations différentielles de la Mécanique," *Journal de M. Liouville*, *t.* XVII. (1852). En effet si dans les équations i,...vi. de Jacobi, pour les réduire à un système d'équations du premier ordre, on écrit

$$\frac{d}{dt}(\mu x^2 + \mu_1 v^2) = 0,$$

le système peut évidemment se présenter sous la forme

$$\frac{di}{I} = \frac{di_1}{I_1} = \frac{du}{U} = \frac{du_1}{U_1} = \frac{dr}{R} = \frac{dr_1}{R_1} = \frac{d\theta}{\Theta}\ (= dt),$$

et, cela étant, en remarquant que les fonctions I, I_1, U, ..., ne contiennent pas t, et en omettant l'équation $(\Theta = dt)$, on a un système de six équations entre les quantités i, i_1, u, u_1, r, r_1, θ: en supposant que l'intégration soit effectuée, on obtient alors t au moyen d'une quadrature.

Je remarque en passant qu'il ne me paraît pas que Jacobi ait dû dire: "Par suite l'on a fait cinq intégrations;" les seules intégrations qu'il a faites sont: l'intégrale des forces vives, et les trois intégrales des aires: cela étant on obtient au lieu de 12 équations entre 13 variables, 8 équations entre 9 variables, et dans la solution de Jacobi il arrive que ce système de 8 équations contient, comme partie de lui-même, un système de 5 équations entre 7 variables; mais à moins d'intégrer les 6 équations on n'obtient pas d'intégrale nouvelle.

* * *

305.

CONSIDÉRATIONS GÉNÉRALES SUR LES COURBES EN ESPACE.

COURBES DU CINQUIÈME ORDRE.

[From the *Comptes Rendus de l'Académie des Sciences à Paris*, *t.* LVIII. (Janvier—Juin, 1864), *pp.* 994—1000. Continuation of 302.]

En considérant une courbe du $m^{\text{ième}}$ ordre représentée au moyen des équations

$$U = 0, \quad w = \frac{P}{Q},$$

qui dénotent respectivement un cône du $m^{\text{ième}}$ ordre et une surface monoïde du $p^{\text{ième}}$ ordre, le cône doit passer $m(p-1)$ fois par les $p(p-1)$ droites $(P=0, Q=0)$ de la monoïde. J'indique la manière de ce passage au moyen d'un symbole que je nomme la *signature* du système; ce symbole, composé ordinairement des numéros 2, 1, 0, ensemble $p(p-1)$ numéros, fait voir combien des $p(p-1)$ droites de la monoïde sont, par rapport au cône, des droites doubles, des droites simples, ou des droites qui ne sont pas situées sur le cône: par exemple, $m=5$, $p=3$, la signature 222211 fait voir qu'il y a quatre droites doubles, deux droites simples; la signature 222220, qu'il y a cinq droites doubles, une droite qui n'est pas située sur le cône.

Je reviens aux courbes du cinquième ordre; j'ai établi (*t.* LIV. *p.* 672) qu'il y a cinq espèces de ces courbes, à savoir:

		P. D. A.
Courbe plane	ou espèce 5	0
Courbe quadricubique	$6-1$	4
Courbe quadriquartique	$8-3$	6
Courbe cubicubique (deux espèces)	$9-6+2$	5
	$9-3-1$	6

Je fais abstraction de la courbe plane, et je cherche à rattacher les quatre autres espèces à la théorie de la surface monoïde. Pour cela je remarque qu'en prenant pour sommet du cône et de la surface monoïde un point quelconque, la surface monoïde (ne pouvant pas être de l'ordre 2) sera de l'ordre 3 ou 4.

Je considère d'abord le cas d'une monoïde cubique: la signature sera 222211 ou 222220.

Monoïde cubique, signature 222211. — Ici le cône $U = 0$ passe par les six droites de la monoïde; donc U est fonction syzygétique de P et Q: autrement dit, on peut trouver P' et Q' fonctions homogènes de (x, y, z) de manière à avoir identiquement $U = PQ' - P'Q$: P et Q sont des ordres 3 et 2 respectivement, donc P' et Q' seront aussi des ordres 3 et 2 respectivement. En combinant les équations

$$U = PQ' - P'Q = 0, \quad w = \frac{P}{Q},$$

on obtient

$$w = \frac{P'}{Q'},$$

ou plus généralement

$$w = \frac{P + aP'}{Q + aQ'}$$

(où a est un paramètre arbitraire); mais en écrivant cette équation sous la forme

$$(Qw - P) + a(Q'w - P') = 0,$$

on voit que les monoïdes cubiques que représente cette équation sont toutes en involution avec les deux monoïdes cubiques $Qw - P = 0$, $Q'w - P' = 0$; on peut donc dire qu'il y a dans le cas dont il s'agit *deux* monoïdes cubiques.

Monoïde cubique, signature 222220. — Ici le cône ne passe pas par les six droites de la monoïde, donc il n'existe pas d'équation identique telle que $U = PQ' - P'Q$, et la monoïde $w = \dfrac{P}{Q}$ est la seule monoïde cubique.

Je passe au cas d'une monoïde quartique; la signature sera

222111111111, ou 222211111110, ou 222221111100, ou 222222111000.

Monoïde quartique, signature 222111111111. — Le cône $U = 0$ passe ici par toutes les douze droites de la monoïde, c'est-à-dire on aurait identiquement

$U = PQ' - P'Q$, où P, Q seraient des fonctions homogènes de (x, y, z) des ordres 2 et 1 respectivement, et il y aurait une monoïde quadrique $w = \dfrac{P}{Q}$. Ce cas n'existe donc pas.

Monoïde quartique, signature 222211111110. — Le cône $U = 0$ passe par toutes les douze droites de la monoïde, hormis une seule droite; donc en écrivant $M = 0$ pour l'équation d'un plan quelconque par cette droite exceptée, le cône $MU = 0$ passe par les douze droites de la monoïde: on a donc identiquement $MU = PQ' - P'Q$, où P', Q' ...

[continued on book p. 26 in next chunk]

The Collected Mathematical Papers of Arthur Cayley, Sc.D., F.R.S.

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305. (continued)

CONSIDÉRATIONS GÉNÉRALES SUR LES COURBES EN ESPACE. COURBES DU CINQUIÈME ORDRE (suite).

[Continuation from p. 25; Comptes Rendus, t. LVIII. (1864), pp. 994–1000.]

[Discussion of the case where the cone $U = 0$ passes through curves of orders 3 and 2 respectively, leading to the monoide cubique $w = \dfrac{P}{Q}$, of P, Q containing one arbitrary constant. We write

$M = K + sZ$, so taking $K = 0$, $L = 0$, for the curves of degrees passing through two adjacent points of each, one obtains identically $MU = PQ - P'Q'$, two monoides cubiques $w = \dfrac{P}{Q}$. Whence one passes to the curve of the monoide cubique, signature 222211.]

Monoïde quartique, signature 222221111100. — Le cône $U = 0$ passe par toutes les droites de la monoïde, hormis deux droites; donc en écrivant $M = 0$ pour l'équation du plan passant par ces deux droites, le cône $MU = 0$ contient toutes les droites; on a donc identiquement $MU = PQ' - P'Q$, c'est-à-dire M est commun aux fonctions $A'V - B'V = 0$ et $AV - B'V = 0$, contenant respectivement les facteurs $\not p$ et q, et il passe par la courbe la monoïde cubique $w = \dfrac{P'}{Q'}$. Mais ici $M = 0$ est un plan déterminé; donc P' et Q' sont aussi des fonctions déterminées, et l'on a la seule monoïde cubique. Cela rentre dans le cas, monoïde cubique, signature 222220.

Monoïde quartique, signature 222221111100. — Le cône $U = 0$ passe par toutes les droites de la monoïde, hormis trois droites; donc en prenant $M = 0$ pour l'équation du cône quadrique quelconque qui passe par les droites en question, le cône $MU = 0$ contient des droites de l'ordre 4 et 2 respectivement. Cela donne la monoïde quartique $w = \dfrac{P}{Q}$. Mais M contient trois constantes arbitraires: il y a donc trois nouvelles monoïdes quartiques $w = \dfrac{P'}{Q'}$, $w = \dfrac{P''}{Q''}$, $w = \dfrac{P'''}{Q'''}$, ou, en tout, quatre monoïdes quartiques.

On démontre sans peine que pour l'espèce $6 - 1$, il y a deux monoïdes cubiques, pour l'espèce $9 - 6 + 2$ une seule monoïde cubique, et que pour les espèces $8 - 3$ et $9 - 3 - 1$ il y a une de monoïdes cubique; ou en donne l'identification que voici:

Espèce $6 - 1$, monoïde cubique, signature 222211,
Espèce $9 - 6 + 2$, monoïde cubique, signature 222220,
Espèce $8 - 3$
Espèce $9 - 3 - 1$ $\Big\}$, monoïde quartique, signature 222222111000,

et il ne reste qu'à distinguer les deux espèces $8 - 3$ et $9 - 3 - 1$, considérées comme représentées au moyen de cône et monoïde.

Je remarque que le système de cône et monoïde à signature 222222111000 contient 20 constantes. En effet, en prenant $Q = 0$ un cône cubique quelconque (9 constantes), on peut prendre k volonté le cône $P = 0$, savoir un cône quartique qui passe par les six droites communes à deux cônes cubiques (mais ces droites comprennent celles qui passent par le sommet); ce qui donne (8 constantes); au moyen des deux cônes je forme l'équation $w = \dfrac{P}{Q}$ de la surface monoïde; il y a une constante arbitraire contenue implicitement comme facteur une constante; il y a donc en tout $9 + 8 + 5 + 1 = 23$ constantes. Mais en combinant l'équation de la monoïde avec l'équation $U = 0$ du cône quintique, on obtient la monoïde quartique

$$w = \frac{P + \alpha P' + \beta P'' + \gamma P'''}{Q + \alpha Q' + \beta Q'' + \gamma Q'''},$$

et sans perte de généralité on peut disposer des constantes α, β, γ, de manière à satisfaire à trois conditions quelconques; on doit donc diminuer de 3 le nombre 23, ce qui donne enfin 20 constantes.

La courbe $8 - 3$ contient 18 constantes, il faut donc chercher quelle est la particularité qui doit avoir lieu pour que le cas, monoïde quartique à signature 222222111000, donne une courbe $8 - 3$.

J'ai nommé droites de la monoïde les droites $P = 0$, $Q = 0$ qui passent par le sommet; en supposant qu'il y ait sur la monoïde des droites qui ne passent pas par le sommet, on peut appeler *transversale* une telle droite. Or, pour l'espèce $8 - 3$, il doit exister sur la monoïde quartique trois transversales qui ne se rencontrent pas; car alors, en faisant passer par ces transversales un hyperboloïde, cet hyperboloïde et la monoïde se coupent selon les trois transversales et selon la courbe $8 - 3$ dont il s'agit. Or, en supposant qu'il existe une transversale, le plan passant par le sommet et cette transversale contient trois des droites $P = 0$, $Q = 0$. En effet, un plan quelconque par le sommet coupe la monoïde selon une courbe quartique avec un point triple au sommet; pour le plan mené par une transversale, cette courbe quartique devient la transversale et une courbe cubique avec un point triple au sommet; cette courbe cubique sera évidemment un système de trois droites, à savoir trois des droites $P = 0$, $Q = 0$. Et réciproquement, si trois quelconques des droites de la monoïde sont situées dans un plan, ce plan coupe la monoïde selon les trois droites et selon une transversale. S'il y a sur la monoïde une seconde transversale, il y aura de même un second système de trois droites dans un plan; on démontre que si le premier système est composé de trois droites, et le second système de trois autres droites, les deux transversales se coupent; donc, si les deux transversales ne se coupent pas, les deux systèmes auront une droite commune. S'il y a sur la monoïde une troisième transversale, il y a de même un troisième système de trois droites dans un plan; et si les trois transversales ne se rencontrent pas, il est de plus nécessaire que deux quelconques des trois plans aient en commun une droite de la monoïde; cela revient à dire qu'il doit y avoir parmi les douze droites $P = 0$, $Q = 0$ de la monoïde six droites $7, 8, 9, 7', 8', 9'$ telles que les droites $7, 8', 9'$, les droites $7', 8, 9'$, et les droites $7', 8', 9$ soient situées chaque système dans un même plan: cela étant, la monoïde aura trois transversales qui ne se rencontrent pas.

Je prends à volonté par un point quelconque de l'espace un tel système de six droites $7, 8, 9, 7', 8', 9'$ (9 constantes); je fais passer par les six droites un cône cubique quelconque $Q = 0$ (3 constantes) et aussi un cône quartique quelconque $P = 0$ (8 constantes); au moyen des deux cônes je forme l'équation $w = \dfrac{P}{Q}$ de la surface monoïde; il y a une constante arbitraire contenue implicitement en w: cela donne en tout $9+3+8+1=21$ constantes. Les deux cônes $P=0$, $Q=0$ se coupent selon les six droites 7, 8, 9, $7'$, $8'$, $9'$, et selon six autres droites 1, 2, 3, 4, 5, 6: il suit de la théorie précédente (mais on peut aussi démontrer analytiquement) qu'il existe un cône quintique $U=0$ qui satisfait aux conditions de passer deux fois par chacune des droites 1, 2, 3, 4, 5, 6 (avoir chacune de ces droites pour une droite double, 18 conditions) et une fois par chacune des droites 7, 8, 9 (3 conditions, en tout $18+3=21$ conditions). Et cela étant, on aura la courbe $8-3$ déterminée au moyen du cône $U=0$ et la surface monoïde $w=\dfrac{P}{Q}$, à signature 222222111000 (à savoir les droites 1, 2, 3, 4, 5, 6 qui sont par rapport au cône des droites doubles, les droites 7, 8, 9 des droites simples, et les droites $7'$, $8'$, $9'$ des droites qui ne sont pas situées sur le cône). Le nombre des constantes est 21, mais au moyen de la transformation

$$w=\frac{P+\alpha P'+\beta P''+\gamma P'''}{Q+\alpha Q'+\beta Q''+\gamma Q'''}$$

on réduit comme auparavant ce nombre à $21-3=18$, ce qui est juste.

J'ajoute les considérations que voici: le cône $U=0$ passe deux fois par chacune des droites 1, 2, 3, 4, 5, 6, une fois par chacune des droites 7, 8, 9. Soit $M=0$ l'équation du système des trois plans qui contiennent les droites 7, $8'$, $9'$, les droites $7'$, 8, $9'$, et les droites $7'$, $8'$, 9 respectivement; le cône $M=0$ contient chacune des droites 7, 8, 9 une fois, et chacune des droites $7'$, $8'$, $9'$ deux fois. Donc le cône $MU=0$ contient chacune des droites 1, 2, 3, 4, 5, 6, 7, 8, 9, $7'$, $8'$, $9'$ deux fois; ces douze droites sont les droites d'intersection des cônes $P=0$, $Q=0$, et ainsi nous avons identiquement $MU=AP^2+BPQ+CQ^2$, A, B, C étant des fonctions homogènes de (x,y,z) des ordres 0, 1, 2 respectivement. Cela étant, les équations

$$w=\frac{P}{Q},\quad MU=AP^2+BPQ+CQ^2=0$$

donnent

$$Aw^2+Bw+C=0,$$

équation de la surface quadrique sur laquelle est située la courbe $8-3$.

Je passe à la théorie analytique. Soit, pour abréger,

$$\begin{aligned}
\xi&=by+cz, & X&=\beta y+\gamma z, & \xi'&=\lambda x+\mu y+\nu z,\\
\eta&=a'x+c'z, & Y&=\alpha' x+\gamma' z, & &\\
\zeta&=a''x+b''y, & Z&=\alpha'' x+\beta'' y. & &
\end{aligned}$$

Je prends pour équations des droites 7, 8, 9, $7'$, $8'$, $9'$:

$$\begin{gathered}
(y=0,\ z=0),\quad (z=0,\ x=0),\quad (x=0,\ y=0),\\
(x=0,\ \xi'=0),\quad (y=0,\ \eta'=0),\quad (z=0,\ \zeta'=0).
\end{gathered}$$

et je forme les équations les plus générales pour le cône cubique et le cône quartique qui passent par ces droites; ces équations seront

$$Q = Z\eta\xi\xi' + \Sigma X\eta\eta'\zeta' + \Sigma Y\zeta\zeta'\xi' + \Sigma Z\xi\xi'\eta' = 0,$$

$$-P = \Sigma\xi\eta' X + \Sigma\zeta\xi'\eta Y + \Sigma\eta\zeta'\xi Z + \Sigma\xi\eta\zeta\xi'\eta'\zeta' = 0.$$

On a de là la surface monoïde $w = \dfrac{P}{Q}$. En écrivant dans cette équation $x = 0$, on obtient $w = -\dfrac{X_0}{\eta_0}$; et de même, pour $y = 0$, on obtient $w = -\dfrac{Y_0}{\zeta_0}$, et pour $z = 0$ on obtient $w = -\dfrac{Z_0}{\xi_0}$, c'est-à-dire qu'il y a sur la monoïde les trois transversales

$$(x = 0,\ Z + \xi' Y = 0), \quad (y = 0,\ Y + \zeta' X = 0), \quad (z = 0,\ X + \xi' Z = 0),$$

ou, comme on peut écrire ces équations,

$$\begin{aligned}
&(x = 0,\ \beta y + \gamma z + \xi'(\alpha'' x + \beta'' y) = 0),\\
&(y = 0,\ \alpha' x + \gamma' z + \zeta'(\lambda x + \mu y + \nu z) = 0),\\
&(z = 0,\ \alpha'' x + \beta'' y + \xi'(ax + by + cz) = 0).
\end{aligned}$$

On trouve sans peine l'équation de la surface quadrique qui passe par les transversales; en écrivant, pour abréger,

$$\begin{aligned}
A &= \beta\beta'\beta'',\\
B &= (\zeta'\alpha'' + \beta''\alpha')\xi x + (\zeta''\alpha + \zeta\alpha'')\beta y + \xi''(\beta\alpha' + \beta'\alpha)\beta'' z,\\
C &= \alpha\beta'\gamma''\xi^2 + \beta''\gamma\beta\xi'^2 + \gamma\nu'\xi'' z^2 + (\gamma\xi''\zeta' + \gamma'\beta\xi'')yz\\
&\quad + (\alpha'\gamma\xi'' + \alpha'\gamma\xi)\, zx + (\beta''\alpha'\xi + \beta\alpha''\zeta')\, xy,
\end{aligned}$$

cette équation est

$$Aw^2 + Bw + C = 0,$$

et en éliminant w entre cette équation et l'équation $w = \dfrac{P}{Q}$, on obtient l'équation

$$AP^2 + BPQ + CQ^2 = 0,$$

laquelle, en vertu de l'identité

$$AP^2 + BPQ + CQ^2 = xyz \cdot U,$$

se réduit à $U = 0$, équation d'un cône du cinquième ordre, ce qui donne le système $U = 0$, $w = \dfrac{P}{Q}$ de cône et monoïde à signature 222222111000. Pour démontrer l'identité dont il s'agit, il convient de remarquer qu'en substituant dans l'expression $AP^2 + BPQ + CQ^2$ les valeurs de P et Q, tous les termes contiennent explicitement le facteur xyz hormis les termes que voici:

$$A(\xi^2\zeta'^2 X^2 + \zeta^2\xi'^2 Y^2 + \eta^2\eta'^2 Z^2),$$

$$-B(\xi\zeta' X\eta\xi' Y + \zeta\eta' Y\eta\zeta' Z + \eta\xi' Z\xi\eta' X),$$

et pour démontrer que ces termes exceptés contiennent aussi le facteur xyz, il suffit de faire voir que la fonction $AX^2 - BXB' + CB'^2$ contient le facteur x, car alors, par la symétrie, les fonctions $AY^2 - BYC' + CC'^2$ et $AZ^2 - BZC'' + CC''^2$ contiendront respectivement les facteurs y et z, et l'expression entière sera divisible par xyz. Mais en écrivant $x = 0$, on trouve

$$\begin{aligned} AX^2 &= \beta\beta'\beta''(\beta y + \gamma z)^2, \\ -BXB' &= -\big[B\beta'(\beta y + \gamma z) + B(\beta' B'' y + \gamma''\beta z)\big]\beta(\beta y + \gamma z) = 0, \\ CB'^2 &= 0, \end{aligned}$$

c'est-à-dire $AX^2 - BXB' + CB'^2$ contient le facteur x. Donc enfin

$$AP^2 + BPQ + CQ^2$$

contient le facteur xyz, ce qui était le théorème à démontrer.

* * *

306.

SUR LES CONIQUES QUI TOUCHENT DES COURBES D'ORDRE QUELCONQUE. EXTRAIT D'UNE LETTRE À M. CHASLES.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LIX. (Juillet—Décembre, 1864), pp. 224–225.]

En considérant l'expression

$$\tfrac{1}{6}\mu(S_1 + S_2 + S_3 + 8S_4 + ZS_5)$$

que vous avez donnée (*Comptes Rendus*, t. LVIII. p. 223) pour le nombre des coniques qui touchent cinq courbes d'ordre quelconque, j'ai trouvé qu'elle peut s'écrire sous la forme que voici, savoir: en dénotant les ordres par (m, n, p, q, r), et en mettant $M = m^2 - m, \ldots$, de manière que (M, N, P, Q, R) seront les classes des cinq courbes, l'expression transformée est

$$(M, m)(N, n)(P, p)(Q, q)(R, r)\,[1, 2, 4, 4, 2, 1],$$

en représentant par cette notation abrégée la fonction

$$\begin{aligned} 1 \cdot &MNPQR \\ &+ 2\Sigma(mNPQR) \\ &+ 4\Sigma(mnPQR) \\ &+ 4\Sigma(mnpQR) \\ &+ 2\Sigma(mnpqR) \\ &+ 1 \cdot mnpqr. \end{aligned}$$

En écartant les relations $M = m^2 - m, \ldots$, et en supposant seulement que (m, n, p, q, r) soient les ordres, et (M, N, P, Q, R) les classes des cinq courbes, la nouvelle formule s'applique aux courbes avec des points doubles ou de rebroussement; on peut même supposer que la courbe de la classe M et de l'ordre m se réduise à un système de M points et de m droites, et que les autres courbes se réduisent aussi à des systèmes de points et droites; et cela étant, on obtient une vérification immédiate de la formule. Car en choisissant dans le système qui remplace chaque courbe un élément (point ou droite) à volonté, on obtient

$$\begin{aligned}
&MNPQR \text{ systèmes } 5p,\\
&\Sigma mNPQR \text{ systèmes } 4p,\ 1d,\\
&\Sigma mnPQR \text{ systèmes } 3p,\ 2d,\\
&\Sigma mnpQR \text{ systèmes } 2p,\ 3d,\\
&\Sigma mnpqR \text{ systèmes } 1p,\ 4d,\\
&mnpqr \text{ systèmes } 5d.
\end{aligned}$$

Or la condition par rapport à la première courbe se réduit à celle de passer par l'un quelconque des M points, ou de toucher l'une quelconque des m droites; et de même pour les autres courbes. Donc (en entendant par le mot toucher appliqué à un système de points et de droites, passer par les points et toucher les droites du système), la conique doit toucher l'un quelconque des systèmes $(5p)$, ou $(4p, 1d)$, ou $(3p, 2d), \ldots$, ou $(5d)$; et pour un système de la forme

$$(5p),\ (4p, 1d),\ (3p, 2d),\ (2p, 3d),\ (1p, 4d),\ \text{ou } (5d),$$

le nombre des coniques est

$$1,\ 2,\ 4,\ 4,\ 2,\ \text{ou } 1,$$

ce qui donne pour le nombre total des coniques l'expression ci-dessus écrite.

On peut supposer que la conique, au lieu de toucher les deux courbes m, n, ait avec la seule courbe m (1°) un contact du deuxième ordre; (2°) un contact double. Le nombre des coniques qui satisfont à l'une ou l'autre de ces conditions, et qui passent aussi par trois points donnés, a été trouvé par Steiner (*Aufgabe und Lehrsätze*, Crelle, t. XLIX. p. 273), savoir:

$$(1°) \text{ le nombre } = 3m(m-1); \quad (2°) \text{ le nombre } = \tfrac{1}{2}(m^2-m)(m^2+3m-6);$$

j'ai vérifié d'une manière particulière ces deux résultats.

En supposant que la conique (au lieu de passer par les trois points donnés) touche les courbes p, q, r, je trouve pour les deux cas respectivement ces résultats,

$$\begin{aligned}
&1° \text{ le nombre } = 3(m^2-m) \times (P,p)(Q,q)(R,r)\,\{1,2,2,1\},\\
&2° \text{ le nombre } = \tfrac{1}{2}(m^2-m)\times\\
&\qquad (P,p)(Q,q)(R,r)\,\{m^2+3m-6,\ 2m^2+6m-16,\ 4m^2+4m-22,\ 4m^2-15\};
\end{aligned}$$

formules dans lesquelles la courbe m doit être une courbe sans singularité.

Grasmere, Westmoreland, 27 *Juillet*, 1864.

307.

NOTE SUR LES FONCTIONS al(x), &c., DE M. WEIERSTRASS.

[From the *Journal de Mathématiques* (Liouville), t. VII. (1862), pp. 137–142.]

Les fonctions $\operatorname{al}(x)$, $\operatorname{al}(x)_1$, $\operatorname{al}(x)_2$, $\operatorname{al}(x)_3$ de M. Weierstrass satisfont respectivement aux équations

$$\begin{aligned}
&\frac{d^2\operatorname{al}(x)}{dx^2} + 2k^2x\,\frac{d\operatorname{al}(x)}{dx} + 2kk'^2\frac{d\operatorname{al}(x)}{dk} + k^2x^2\operatorname{al}(x) = 0,\\
&\frac{d^2\operatorname{al}(x)_1}{dx^2} + 2k^2x\,\frac{d\operatorname{al}(x)_1}{dx} + 2kk'^2\frac{d\operatorname{al}(x)_1}{dk} + (k^2 + k^2x^2)\operatorname{al}(x)_1 = 0,\\
&\frac{d^2\operatorname{al}(x)_2}{dx^2} + 2k^2x\,\frac{d\operatorname{al}(x)_2}{dx} + 2kk'^2\frac{d\operatorname{al}(x)_2}{dk} + (1 + k^2x^2)\operatorname{al}(x)_2 = 0,\\
&\frac{d^2\operatorname{al}(x)_3}{dx^2} + 2k^2x\,\frac{d\operatorname{al}(x)_3}{dx} + 2kk'^2\frac{d\operatorname{al}(x)_3}{dk} + (k'^2 + k^2x^2)\operatorname{al}(x)_3 = 0,
\end{aligned}$$

ou, ce qui est la même chose, les fonctions

$$\operatorname{al}(x),\quad \frac{\sqrt{k}}{k}\operatorname{al}(x)_1,\quad \frac{1}{\sqrt{k}}\operatorname{al}(x)_2,\quad \frac{\sqrt{k'}}{k}\operatorname{al}(x)_3,$$

satisfont chacune à l'équation

$$\frac{d^2z}{dx^2} + 2k^2x\,\frac{dz}{dx} + 2kk'^2\,\frac{dz}{dk} + k^2x^2z = 0.$$

Écrivons pour un moment ξ, κ au lieu de x, k; les fonctions $\operatorname{al}(\xi)$, etc., satisfont à l'équation

$$\frac{d^2z}{d\xi^2} + 2\kappa^2\xi\,\frac{dz}{d\xi} + 2\kappa\kappa'^2\,\frac{dz}{d\kappa} + \kappa^2\xi^2z = 0.$$

Cela étant, en posant

$$\xi = \frac{x}{\sqrt{k}}, \quad \kappa = k,$$

on obtient

$$\frac{1}{\sqrt{k}}\frac{dz}{d\xi} = \frac{dz}{dx}, \quad \frac{1}{k}\frac{d^2z}{d\xi^2} = \frac{d^2z}{dx^2}, \quad \frac{dz}{d\kappa} = \frac{dz}{dk} - \frac{x}{2k\sqrt{k}}\frac{dz}{dx},$$

et de là

$$\frac{dz}{d\xi} = \sqrt{k}\frac{dz}{dx}, \quad \frac{d^2z}{d\xi^2} = k\frac{d^2z}{dx^2}, \quad \frac{dz}{d\kappa} = \frac{dz}{dk} - \frac{x}{2k}\frac{dz}{dx}.$$

L'équation différentielle devient ainsi

$$k\frac{d^2z}{dx^2} + 2k^2\frac{x}{\sqrt{k}}\sqrt{k}\frac{dz}{dx} + 2kk'^2\left(\frac{dz}{dk} - \frac{x}{2k}\frac{dz}{dx}\right) + kx^2z = 0,$$

c'est-à-dire

$$\frac{d^2z}{dx^2} + \frac{1+k^2}{k}\frac{dz}{dx} + 2k'^2\frac{dz}{dk} + x^2z = 0,$$

équation qui sera ainsi satisfaite par

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right), \quad \sqrt{k}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_1, \quad \frac{1}{\sqrt{k}}\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_2, \quad k'\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_3.$$

Or en écrivant

$$k + \frac{1}{k} = a,$$

l'équation en z devient

$$(1) \quad \frac{d^2z}{dx^2} + ax\frac{dz}{dx} - 2(a^2-4)\frac{dz}{da} + x^2z = 0,$$

laquelle est ce que devient celle-ci

$$(2) \quad (1 - ax^2 + x^4)\frac{d^2z}{dx^2} + (n-1)(ax - 2x^3)\frac{dz}{dx} - 2n(a^2-4)\frac{dz}{da} + n(n-1)x^2z = 0,$$

en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de x, et puis $n = \infty$. L'équation (2), trouvée par Jacobi (*Journal de Crelle*, t. IV. p. 185, 1829), a la propriété que voici, savoir en posant

$$a = k + \frac{1}{k}, \quad x = \sqrt{k}\,\mathrm{sn\,am}\,u,$$

alors l'équation est satisfaite en prenant pour z soit le numérateur, soit le dénominateur, de la fonction rationnelle de x qui donne la valeur de la fonction $\sqrt{\lambda}\,\mathrm{sn\,am}\,(Mu, \lambda)$, où λ, M sont le module et le multiplicateur qui correspondent à la transformation de l'ordre n (n étant un nombre impair quelconque).

L'équation fut donnée par Jacobi sans démonstration. Je l'ai démontrée (*Camb. and Dubl. Math. Journal*, t. II. p. 256, 1847, [45]) de la manière que voici, savoir en écrivant

$$z = \Theta^{-n}(u) \cdot S,$$

on obtient pour S l'équation

$$\frac{d^2S}{du^2} - 2k \operatorname{sn} \operatorname{am} u \cdot \frac{dS}{du} - \frac{(k^2-1)}{2k} \cdot 2mKr\,K'\,r - 2mKr\,u^*,$$

[Original: $\dfrac{d^2S}{du^2} - 2k \operatorname{snam} u \cdot \dfrac{dS}{du} + \ldots$ leading to the reduced form below.]

Cette équation, en prenant

$$u = \frac{mK + r'K'}{n},$$

devient

$$\frac{d^2S}{dl^2} + \frac{dS}{dm} = 0,$$

équation mentionnée par Jacobi, laquelle est satisfaite par

$$2 = H_1(\sqrt{n\,u},\ \sqrt{nK'/K}), \quad \text{ou} \quad 2 = \Theta_1(\sqrt{n\,u},\ \sqrt{nK'/K}).$$

Cela donne pour z deux valeurs qui sont le dénominateur et le numérateur de la fraction dont il s'agit. J'ose croire que ce doit être à peu près de cette manière que l'équation fut trouvée par Jacobi.

Or les solutions en question de l'équation (1) peuvent être trouvées au moyen de l'équation de Jacobi; pour cela, au lieu des valeurs ci-dessus données de m, v, j'écris

$$m = \frac{\pi K'}{\pi}, \quad v = \frac{\sqrt{n\,\pi u}}{\pi},$$

ce qui conduit à la même équation

$$\frac{d^2l}{dv^2} + \frac{dl}{dm} = 0,$$

laquelle sera ainsi satisfaite par

$$S = \Theta(\sqrt{n\,u}, \sqrt{n}\,K'/K), \quad S = H(\sqrt{n\,u}, \sqrt{n}\,K'/K),$$

ou, ce qui est la même chose, par

$$z = \Theta(\sqrt{n\,u}), \quad z = H(\sqrt{n\,u}).$$

L'équation (2) sera donc satisfaite par

$$z = \frac{1}{\Theta^n(u)}\,\Theta(\sqrt{n\,u}),$$

$$z = \frac{1}{\Theta^n(u)}\,H(\sqrt{n\,u}),$$

ou, en se souvenant que $H'(0) = \sqrt{\dfrac{2kK}{\pi}}$, par

$$z = \frac{\Theta(\sqrt{n}\,u)}{\Theta^n(u)}, \quad z = \frac{H(\sqrt{n}\,u)}{\Theta^n(u)} \cdot e^{-(n-1)\log\Theta(0)}.$$

Or en écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de x, pour effectuer ensuite $n = \infty$, nous avons

$$\frac{x}{\sqrt{n}} = \sqrt{k}\,\mathrm{sn}\,\mathrm{am}\,u,$$

équation qui se réduit à

$$x = \frac{u}{\sqrt{k}} \quad \text{(au facteur } \sqrt{n} \text{ près)},$$

ou ce qui revient au même $u = x\sqrt{k}$.

J'ajoute que dans le Mémoire cité (1847) j'ai donné la suite

$$z = C_n + C_{n-1}\frac{x^2}{1} + C_{n-2}\frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} + \cdots,$$

où

$$\begin{aligned}
&C_0 = 1,\\
&C_1 = 0,\\
&C_2 = 2,\\
&C_3 = -8a,\\
&C_4 = 32a^2 - 4,\\
&C_5 = -128a^3 + 304a,\\
&C_6 = 512a^4 - 2304a^2 + 408,\\
&C_7 = -2048a^5 + 13056a^3 - 7720a,\\
&C_8 = 8192a^6 - 65280a^4 + 85920a^2 - 11304,
\end{aligned}$$

de façon qu'en général

$$C_{r+1} = -(2r+1)(2r+2)C_r - (2r+2)\,a\,C_{r-1} + 2(a^2-4)\,\frac{dC_{r-1}}{da},$$

c'est le développement de M. Weierstrass pour la fonction al $\left(\dfrac{x}{\sqrt{k}}\right)$, seulement je ne connaissais pas alors l'expression finie

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right), \quad \text{ou} \quad \frac{1}{\Theta^n(0)}\Theta\left(\frac{x}{\sqrt{k}}\right)\Theta^{n-1}(0)\,e^{n-1}\log\Theta(0),$$

de cette suite. Je remarque en passant que pour $n = 2$, la suite se réduit à

$$2^n - 1\,\Theta(0) = x^2.$$

308.

ON THE △ FACED POLYACRONS, IN REFERENCE TO THE PROBLEM OF THE ENUMERATION OF POLYHEDRA.

[From the *Memoirs of the Literary and Philosophical Society of Manchester*, vol. I. (1862), pp. 248–256.]

The problem of the enumeration of polyhedra is one of extreme difficulty, and I am not aware that it has been discussed elsewhere than in Mr Kirkman's valuable series of papers on the subject in the *Memoirs of the Society* and in the *Philosophical Transactions.* A case of the general problem is that of the enumeration of the polyhedra with trihedral summits; and Mr Kirkman in the earliest of his papers, viz. that "On the representation and enumeration of polyhedra" (*Memoirs*, vol. XII. pp. 47–70, 1854), has in fact, by an examination of the relations of the solidoids with trihedral summits, A. subsidiary type (△-faced polyacrons,—using as the correlative term to △-summited polyhedron the term △-faced polyacron[1]), to the consideration of △-*faced polyacrons*, that is, of polyacrons each summit of which is connected with three (and only three) others by an edge. The general problem of the relations of the polyacra to the polyhedra has been considered by Mr Kirkman only in some particular cases of summits, and to exemplify it by finding, in an entirely manner, the number of polyacrons of any given order. The above terms refer to the duals of polyhedra.

[1] I use with Mr Kirkman the expression "enumeration of polyhedra" to designate the general problem, but I consider that the form in which the question presents itself is rather that of the enumeration of the polyacra. The two are of course correlative, viz. the determination of either solves the question for the other; but I think that the polyacra (of which the simplest case considers as the regular figures, the proper representatives of the polyhedra) are the more separately taken the more popular rather than the mathematical sense.

up to the octaacron; thus, as regards the examples, stopping at the same point as Mr Kirkman, for although perfectly practicable it would be very tedious to carry them further, and there would be no commensurate advantage in doing so. The epithet $\triangle$-faced will be omitted in the sequel, but it is to be understood throughout that I am speaking of such polyacrons only; and I shall for convenience use the epithets tripleural, tetrapleural, &c. to denote summits with three, four, &c. edges through them. The number of edges at a summit is of course equal to the number of faces, but it is the edges rather than the faces which have to be considered.

An n-acron has

$$n \text{ summits, } 3n-6 \text{ edges, } 2n-4 \text{ faces,}$$

and it is easy to see that there are the following three cases only, viz.:

1. The polyacron has at least one tripleural summit.

2. The polyacron, having no tripleural summit, has at least one tetrapleural summit.

3. The polyacron, having no tripleural or tetrapleural summit, has at least twelve pentapleural summits.

In fact, if the polyacron has c tripleural summits, d tetrapleural summits, e pentapleural summits, and so on, then we have

$$n = c + d + e + f + g + h + \cdots,$$

$$6n - 12 = 3c + 4d + 5e + 6f + 7g + 8h + \cdots,$$

and therefore

$$12 = 3c + 2d + e + 0f - g - 2h - \cdots,$$

or

$$3c + 2d + e = 12 + g + 2h + \cdots,$$

whence if $c = 0$ and $d = 0$, then $e \geq 12$ at least. It appears, moreover (since a cannot be less than 4, and any polyacron of the third class, viz. where the second class is not satisfied, must consist of at least 12 summits each pentapleural belonging to the third class), and must therefore belong to the first or the second class.

The figures of the polyacrons comprised in the annexed Tables show the application of the method to the genesis of the polyacrons as far as the octaacron, including the numbers indicate the nature of the different summits, according to the number of edges through each summit, viz., 3 denoting a tripleural summit, and so on. It will be noticed that in the same diagram, where the application is made to distinguish the polyacron, viz. among the figures shown as illustrative of the figures shown at once denoting the diagram of the figures shown, it shows the application of the figures shown to belong to the polyacron from 3344555, the inspection of the figures shows at once that there are wholly distinct forms, for in the first of them, viz. that from 3344555, each of the tripleural summits stands by a basic triangle 456, while in the second of them, viz. that from 3344555, the triangles to so as to come to a basic triangle of seven summits stands at the foregoing principles to the formation of the polyacrons as far as the 11-acron we are only concerned with the first and second processes.

Consider the entire series of n-acrons, say A, B, C, &c., and suppose that the n-acron A gives rise to a certain number, say P, Q, R, S of $(n+1)$-acrons, the $(n+1)$-acron P is of course derivable from the n-acron A, but it may be derivable from other n-acrons, suppose from the n-acrons B and C. Then in considering the $(n+1)$-acrons derived from B, of course be found to be the $(n+1)$-acron P, and it is only the remaining $(n+1)$-acrons derived from B which are or may be peculiar to the process of subtraction so as to ascertain the entire series of n-acrons from which it is derivable, and, for forming the $(n+1)$-acrons suppose from the n-acrons A and C. Then in considering the $(n+1)$-acrons derived from B, of course be found to be the $(n+1)$-acron P, and it is only the remaining $(n+1)$-acrons derived from B which are or may be peculiar to the process of subtraction so as to ascertain the entire series of n-acrons from which it is derivable, and, for forming the $(n+1)$-acrons already previously obtained and found to be derivable from these, we should obtain without any repetitions the entire series of the $(n+1)$-acrons.

For merely finding the number of the $(n+1)$-acrons, a more simple process might be adopted: say that an n-acron is p-wise generating when it gives rise to a number p of $(n+1)$-acrons, and that it is q-wise generable when it can be derived from a number q of $(n+1)$-acrons; and assume that a given n-acron is $(x+y+z+\cdots)$-wise generating, viz. that it gives rise to a number x_1 of 1-wise generable, a number y_1 of $(n+1)$-acrons which are 2-wise generable, and so on; then in like manner the sum

$$\Sigma(y_1 + \tfrac{1}{2}y_2 + \tfrac{1}{3}y_3 + \cdots)$$

where Σ refers to the entire series of the n-acrons, it is clear that every m-wise generable $(n+1)$-acron will in respect of each of the n-acrons from which it is derivable be reckoned as $\dfrac{1}{m}$, that is, it will be in the entire sum reckoned as 1, and the sum in question will consequently be the number of the $(n+1)$-acrons.

The figures of the polyacrons comprised in the annexed Tables show the application of the method of the genesis of the polyacrons up to the octaacron, including the number of summits, the nature of the different summits according to the number of edges through each summit, viz., 3 denoting a tripleural summit, 4 a tetrapleural summit, and so on. It will be noticed in the annexed Tables that the application of the figures shown demonstrates the application of the symbol 3344455666, that is, of the figures shown, it shows the application from 3344455666, in which we may distinguish that there exist 6 polyacrons of edges, 4 of one class, 1 of another. But the symbol 3344455666 furnishes by the application of the foregoing principles to the formation of the polyacrons as far as the 11-acron we are only concerned with the first and second processes.

The following remarks on the derivation of the octaacrons from the heptaacrons will further illustrate the method:

1. The heptaacron 3335556 has three kinds of faces, viz. 355, 455, and the first process consequently gives rise to 3 octaacrons. As the heptaacron has more than two tripleural summits it is not applicable.

2. The heptaacron 3344466 has three kinds of faces, viz. 366, 346 and the first process gives therefore 3 octaacrons. The heptaacron has only two tripleural summits, and they are disposed in the proper manner; the second process furnishes 1 octaacron.

3. The heptaacron 3344556 has five kinds of faces, viz. 345, 346, 355, 456, and the first process consequently gives 5 octaacrons. The heptaacron has two tripleural summits, but they are not disposed in such manner as to render the second process applicable.

It may be remarked that for the five heptaacrons respectively the values of the sum $y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots$ are

$$1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4},\ \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{3} + 1,\ \tfrac{1}{2} + 1 + \tfrac{1}{4} + 1,\ 1 + 1 + \tfrac{1}{2},\ 1 + \tfrac{1}{2},$$

giving for $\Sigma(y_1 + \frac{1}{2}y_2 + \cdots)$ the value 14, as it should do.

[Plate I (facing p. 44) of the original presents diagrammatic tables showing the generic family tree of the $\triangle$*-faced polyacrons from the 4-acron (tetrahedron, symbol* 3333*) up through the 8-acron, with the symbols and the tripleural/tetrapleural/…summit structure displayed for each. The plate also shows tetrahedral diagrams of selected polyacrons such as* 3 4 4 4 4 5 5 *(heptaacron) and the named octaacrons* 3 3 3 3 5 5 5 5, 3 3 4 4 4 4 5 5, 3 3 4 4 4 5 5 6, 3 3 4 4 5 5 5 5, 3 4 4 4 4 4 5 5 6, 4 4 4 4 4 4 4 4*. The diagram is omitted here; consult the printed source for the figure.]*

309.

NOTE ON THE THEORY OF DETERMINANTS.

[From the *Philosophical Magazine*, vol. XXI. (1861), pp. 180–185.]

The following mode of arrangement of the developed expression of a determinant had presented itself to me as a convenient one for the calculation of a rather complicated determinant of the fifth order; but I have since found that it is in effect given, although in a much less compendious form, in a paper by J. N. Stockwell, "On the Resolution of a System of Symmetrical Equations with Indeterminate Coefficients," *Gould's Ast. Journal*, No. 139 (Cambridge, U. S., Sept. 19, 1860).

Suppose that the determinant

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}$$

is represented by $|123|$, and so for a determinant of any order $|123\ldots n|$.

Let $|1|$, $|2|$, $|3|$, $|12|$, $|13|$, $|123|$, &c., denote as follows; viz.

$$|1| = 11, \quad |2| = 22, \quad |3| = 33,$$

$$|12| = 11 \cdot 22,$$

$$|13| = 11 \cdot 33,$$

$$|123| = 11 \cdot 22 \cdot 33,$$

&c., as follows: viz.
where it is to be noticed that, with the same two symbols, e.g. 1 and 2, there are not distinct expressions $|12|$ (or $12 \cdot 21$), $|21|$ ($= 12 \cdot 21$); with the same three symbols, 1, 2, 3, there are three distinct expressions $|123|$ ($= 11 \cdot 22 \cdot 33$), $|132|$ ($= 11 \cdot 23 \cdot 32$), $|213|$ ($= 12 \cdot 21 \cdot 33$); and generally with the same n symbols 1, 2, 3,..., n, there are $(n-1)!$ distinct

expressions $|123\ldots n|$, which are obtained by permuting in every possible manner all but one of the symbols.

This being so, and writing for greater simplicity $|1\,|\,2|$ to denote the product $|1|\cdot|2|$, &c., and so on as follows: viz.

$$
\begin{array}{ll}
|12| = +|1\,|\,2| & 1 \\
\quad -|12| & 1 \\
\multicolumn{2}{c}{1+1=2} \\
\hline
|123| = +|1\,|\,2\,|\,3| & 1 \\
\quad -|1\,|\,23| - |12\,|\,3| - |13\,|\,2| & 3 \\
\quad +|123| + |132| & 2 \\
\multicolumn{2}{c}{5+1=6} \\
\hline
|1234| = +|1\,|\,2\,|\,3\,|\,4| & 1 \\
\quad -|1\,|\,2\,|\,34| - \cdots & 6 \\
\quad +|1\,|\,234| + |12\,|\,34| + \cdots & 8 \\
\quad +|2\,|\,134| + |13\,|\,24| + \cdots & 3 \\
\quad -|1234| - \cdots & 6 \\
\multicolumn{2}{c}{13+13=24} \\
\hline
|12345| = +|1\,|\,2\,|\,3\,|\,4\,|\,5| & 1 \\
\quad -|1\,|\,2\,|\,3\,|\,45| - \cdots & 10 \\
\quad +|1\,|\,2\,|\,345| + \cdots & 20 \\
\quad +|1\,|\,23\,|\,45| + \cdots & 15 \\
\quad -|1\,|\,2345| - \cdots & 30 \\
\quad -|12\,|\,345| - \cdots & 20 \\
\quad +|12345| + \cdots & 24 \\
\multicolumn{2}{c}{60+60=120}
\end{array}
$$

where, as regards the signs, it is to be observed that there is a sign—for each compartment $|\,|$ containing an even number of symbols; thus in the expression for $|1234|$, the signs are $+$, $-$, $+$, $+$, $-$, and the forms $|1\,|\,234|$, &c. are seen by inspection to belong to the first or second sign—the compartments containing an even number of symbols, while the other compartments contain an odd number. As regards the remaining part of the expression, this merely exhibits the partitions of a set of n things; and the formula for the several determinants up to the determinant of a given order are all of them obtained by means of the form

$$\begin{aligned}|123| = +\quad & |1\,|\,2\,|\,3| \\ - \quad & |1\,|\,23| \\ - \quad & |12\,|\,3| \\ - \quad & |13\,|\,2| \\ + \, & |123| + |132|\end{aligned}$$

and by writing down in the manner the expression for the twenty-four terms of the determinant of the fourth order, viz. omitting the determinants $|1234|$, the formula leads to a determinant. The formula is easily applicable to the developed expressions in either column, say possible manner the symbols in either column, or the arrangement

$$\begin{matrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{matrix}$$

and prefixing the sign (+ or −) of the arrangement; and the resulting arrangements correspond, for instance,

$$\begin{array}{r|l|c} +|1|2|3|4| & = 11 \cdot 22 \cdot 33 \cdot 44, & = 1 \cdot 1 \cdot 1 \cdot 1 \\ -|1|2|34| & = 11 \cdot 22 \cdot 34 \cdot 43, & = -1 \cdot 1 \cdot 1 \cdot 1 \\ +|12|34| & = 11 \cdot 23 \cdot 32 \cdot 44, & = +1 \cdot 1 \cdot 1 \cdot 1 \\ +|1|234| & = 11 \cdot 23 \cdot 34 \cdot 42, & = \ldots \\ +|1|2|34| & = 11 \cdot 22 \cdot 34 \cdot 43, & = \ldots \end{array}$$

are interpreted under the sign (+ or −) of the arrangement; and the resulting arrangements correspond. For instance,

$$|1\,|\,2\,|\,3\,|\,4| = +1 \cdot 1 \cdot 1 \cdot 1, \quad |1\,|\,2\,|\,34| = -1 \cdot 1 \cdot 1 \cdot 1, \quad |1\,|\,234| = +1 \cdot 1 \cdot 1 \cdot 1,$$

and so on.

Suppose that any partition of n contains α compartments of a symbol, β compartments of two symbols, γ compartments of three symbols, and so on; the number of ways in which the symbols 1, 2, 3,..., n can be arranged in compartments is therefore the multinomial coefficient corresponding to the transformation of the order n into the parts $(\alpha, \beta, \gamma, \ldots)$.

Suppose, that any partition n contains α compartments of a symbol, β compartments of two symbols, γ compartments of three symbols, and so on, being all of them different and greater than thirty-three, $n = \alpha + \beta + \gamma + \cdots$, and the corresponding number of terms therefore is

$$\frac{|n|}{(1)^{\alpha}|\alpha|\,(1\,2)^{\beta}\,|\beta|\,(1\,2\,3)^{\gamma}|\gamma|\,\cdots}$$

but each arrangement gives rise to a determinant of $|1|$, we have thus

$$\frac{|n|}{\alpha!|1|^{\alpha}\,(1\,2)^{\beta}\,\beta!\,(1\,2\,3)^{\gamma}\gamma!\,\cdots}$$

The whole number of terms of the determinant is $|n|$, and we have thus the theorem

$$1 = \Sigma \frac{1}{\alpha!|1|^{\alpha}\,(1\,2)^{\beta}\,\beta!\,(1\,2\,3)^{\gamma}\gamma!\,\cdots},$$

in which the summation corresponds to all the different partitions $n = \alpha + \beta + \gamma + \cdots$, where α, β,... are all of them different and being non-negative integers.

Or putting for shortness $|11|$ for $|1|$, and so on as follows:

$$1 = \Sigma \frac{|n|}{\alpha!|11|^{\alpha}\,(1\,2)^{\beta}\,\beta!\,(1\,2\,3)^{\gamma}\gamma!\,\cdots}$$

so we, if as before, the same number...

And putting the sign ($+$ or $-$) of the arrangement, we have these forms:

$$0 = \Sigma \frac{(-1)^{(\alpha+\beta+\gamma+\cdots)}}{\alpha!|11|^{\alpha}\,(1\,2)^{\beta}\,\beta!\,(1\,2\,3)^{\gamma}\gamma!\,\cdots},$$

or, as written in the second equation, $0 = \Sigma \frac{1}{(\alpha+\beta+\gamma+\cdots)!}\Pi(1+1)\cdots$ and there are

$$0 = \Sigma \frac{1}{\alpha!|11|^{\alpha}\,(1\,2)^{\beta}\,\beta!\,(1\,2\,3)^{\gamma}\gamma!\,\cdots},$$

where, as before, $\alpha = \alpha + \beta + \gamma + \cdots$, $\beta = \beta + \gamma + \cdots$, $\ldots$ being all different and greater than unity); but the summation is restricted either to the odd values of $\alpha - \alpha + \beta + \gamma + \cdots$ or even, but to the same values of $\alpha - \alpha + \beta + \gamma + \cdots$ or odd.

The formulae afford a proof of the fundamental property of skew symmetrical determinants. In such a determinant we have viz. $12 = -21, \ldots$, being all different and greater than unity); but the summation is restricted to the values of the expression

$$|12 \ldots n| = \Sigma \pm |1|\,|2| \ldots |n|$$

of the determinant, there is at least one compartment of symbols $\Sigma\,|\,12| \ldots |$, &c., containing the terms of the form $|1|\,|2| \ldots |n|$, having an odd number of symbols; this term is one of a pair $|12 \ldots n|$ and $|n \ldots 21|$, the order of the others, in such an even order, and the skew symmetrical determinant of an odd order is equal to zero.

The formula affords a proof of the corresponding property of skew symmetrical determinants. In such a determinant we have viz. $12 = -21$, $34 = -43, \ldots$, and also $11 = 0$, $22 = 0, \ldots$; thus in each of the terms of the form $|1|\,|2|\,|3|\,|4| \ldots$, having an odd number of symbols this term is one of a pair $|12 \ldots n|$ and $|n \ldots 21|$, the order of the others, in an even order such as $|12|\,|34| \ldots$, $|13|\,|24| \ldots$, the formula gives an expression for the determinant in terms equal to zero.

The formula in such a case is for the determinant of an even order. Let us assume that the order is n, and the formula gives an expression for the determinant.

The whole number of terms of the determinant of $|12 \ldots n|$ is

$$1 \cdot 3 \cdot 5 \cdots (n-1),$$

in which the summation corresponds to all the different partitions $n = \alpha + \beta + \gamma + \cdots$, where α, $\beta, \ldots$ are all different and greater than unity; and we have thus the theorem

$$n = \Sigma \frac{1 \cdot |12| \cdot |34|\,|45|\,|56| + 16\,|25|\,|34|}{\cdots},$$

where, in the second equation, $3456 = 34 \cdot 56 - 35 \cdot 46 + 36 \cdot 45$, and so on.

2, Stone Buildings, W.C., December 28, 1860.

310.

NOTE ON MR JERRARD'S RESEARCHES ON THE EQUATION OF THE FIFTH ORDER.

[From the *Philosophical Magazine*, vol. XXI. (1861), pp. 210–214.]

Functions of the same set of quantities which are, by any substitution whatever, simultaneously altered or simultaneously unaltered, may be called *isagonal.* Thus all symmetric functions of the same set of quantities are isagonal; so are $(x + y - z - w)^2$ and $xy + zw$ homotypical, &c.

It is one of the most beautiful of Lagrange's discoveries in the theory of equations that, given the value of any function of the roots, the value of any homotypical function may be technically determined"; in other words that, any homotypical function whatever is a rational function of the coefficients of the equation, and of the given function. The possibility of solving any proposed equation of the fifth degree by a finite combination of radicals and rational functions[2]

[The remainder of Paper 310 occupies pp. 51–54 of the original and is not reproduced in this excerpt, which closes at p. 50.]

End of excerpt: Cayley, Collected Mathematical Papers, Vol. V, pp. 26–50.

[2]See Mr Jerrard's somewhat severe replies to my and other comments. The substance of the answer is that it is impossible (this was indeed shown by Lagrange to be the case) to solve the same value of λ in different ways and there obtain different values; this is so far the case that it is, by virtue of these radicals as different by means of irrational functions, of given by, but this is not all, viz. the resolvent equation a may be *transformed* into a quintic, but it cannot be *solved* as a quintic.

The Collected Mathematical Papers of Arthur Cayley, Sc.D., F.R.S.

Vol. V

Pages 51–75

Cambridge: At the University Press. 1892

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306.

SUR LES CONIQUES QUI TOUCHENT DES COURBES D'ORDRE QUELCONQUE.
EXTRAIT D'UNE LETTRE À M. CHASLES.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LIX. *(Juillet—Décembre, 1864), pp. 224–225.]*

En considérant l'expression

$$\tfrac{1}{8}(S_1 + S_2 + S_3 - 3S_4 + 3S_5)$$

que vous avez donnée (*Comptes Rendus*, t. LVIII. p. 223) pour le nombre des coniques qui touchent cinq courbes d'ordre quelconque, j'ai trouvé qu'elle peut s'écrire sous la forme que voici, savoir: en dénotant les ordres par (m, n, p, q, r), et en mettant $M = m^2 - m, \ldots$, de manière que (M, N, P, Q, R) seront les classes des cinq courbes, l'expression transformée est

$$(M, m)(N, n)(P, p)(Q, q)(R, r)\,[1, 2, 4, 4, 2, 1];$$

en représentant par cette notation abrégée la fonction

$$\begin{aligned} &1\,.\,M\,N\,P\,Q\,R \\ &+ 2\Sigma\,(m\,N\,P\,Q\,R) \\ &+ 4\Sigma\,(m\,n\,P\,Q\,R) \\ &+ 4\Sigma\,(m\,n\,p\,Q\,R) \\ &+ 2\Sigma\,(m\,n\,p\,q\,R) \\ &+ 1\,.\,m\,n\,p\,q\,r. \end{aligned}$$

En écartant les relations $M = m^2 - m, \ldots$, et en supposant seulement que (m, n, p, q, r) soient les ordres, et (M, N, P, Q, R) les classes des cinq courbes, la nouvelle formule s'applique aux courbes avec des points doubles ou de rebroussement; on peut même supposer que la courbe de la classe M et de l'ordre m se réduise à un système de M points et de m droites, et que les autres courbes se réduisent aussi à des systèmes de points et droites; et cela étant, on obtient une vérification immédiate de la formule. Car en choisissant dans le système qui remplace chaque courbe un élément (point ou droite) à volonté, on obtient

$$\begin{array}{rl} M\,N\,P\,Q\,R & \text{systèmes } 5p, \\ \Sigma\,m\,N\,P\,Q\,R & \text{systèmes } 4p, 1d, \\ \Sigma\,m\,n\,P\,Q\,R & \text{systèmes } 3p, 2d, \\ \Sigma\,m\,n\,p\,Q\,R & \text{systèmes } 2p, 3d, \\ \Sigma\,m\,n\,p\,q\,R & \text{systèmes } 1p, 4d, \\ m\,n\,p\,q\,r & \text{systèmes } 5d. \end{array}$$

Or la condition par rapport à la première courbe se réduit à celle de passer par l'un quelconque des M points, ou de toucher l'une quelconque des m droites; et de même pour les autres courbes. Donc (en entendant par le mot *toucher* appliqué à un système de points et de droites, passer par les points et toucher les droites du système) la conique doit toucher l'un quelconque des systèmes $(5p)$, ou $(4p, 1d)$, ou $(3p, 2d), \ldots$, ou $(5d)$; et pour un système de la forme

$$(5p), \quad (4p, 1d), \quad (3p, 2d), \quad (2p, 3d), \quad (1p, 4d), \quad \text{ou} \quad (5d),$$

le nombre des coniques est

$$1, \quad 2, \quad 4, \quad 4, \quad 2, \quad \text{ou} \quad 1,$$

ce qui donne pour le nombre total des coniques l'expression ci-dessus écrite.

On peut supposer que la conique, au lieu de toucher les deux courbes m, n, ait avec la seule courbe m (1°) un contact du deuxième ordre; (2°) un contact double. Le nombre des coniques qui satisfont à l'une ou l'autre de ces conditions, et qui passent aussi par trois points donnés, a été trouvé par Steiner (*Aufgabe und Lehrsätze*, Crelle, t. XLIX. p. 273), savoir:

$$(1^{\circ}) \text{ le nombre} = 3m(m-1); \qquad (2^{\circ}) \text{ le nombre} = \tfrac{1}{2}(m^2 - m)(m^2 + 3m - 6);$$

j'ai vérifié d'une manière particulière ces deux résultats.

En supposant que la conique (au lieu de passer par les trois points donnés) touche les courbes p, q, r, je trouve pour les deux cas respectivement ces résultats,

1° le nombre $= 3(m^2 - m) \times (P,p)(Q,q)(R,r)\,[1,2,2,1]$,

2° le nombre $= \frac{1}{2}(m^2 - m) \times$
$$(P,p)(Q,q)(R,r)\,[m^2 + 3m - 6,\ 2m^2 + 6m - 16,\ 4m^2 + 4m - 22,\ 4m^2 - 15];$$

formules dans lesquelles la courbe m doit être une courbe sans singularité.

Grasmere, Westmoreland, 27 Juillet, 1864.

307.

NOTE SUR LES FONCTIONS al(x), &c., DE M. WEIERSTRASS.

[From the Journal de Mathématiques (Liouville), tom. VII. (1862), pp. 137–142.]

Les fonctions $\mathrm{al}(x)$, $\mathrm{al}(x)_1$, $\mathrm{al}(x)_2$, $\mathrm{al}(x)_3$ de M. Weierstrass satisfont respectivement aux équations

$$\begin{aligned}
\frac{d^2\,\mathrm{al}(x)}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)}{dx} + 2kk'\frac{d\,\mathrm{al}(x)}{dk} + k^2x^2\,\mathrm{al}(x) &= 0,\\
\frac{d^2\,\mathrm{al}(x)_1}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_1}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_1}{dk} + (k^2 + k^2x^2)\,\mathrm{al}(x)_1 &= 0,\\
\frac{d^2\,\mathrm{al}(x)_2}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_2}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_2}{dk} + (1 + k^2x^2)\,\mathrm{al}(x)_2 &= 0,\\
\frac{d^2\,\mathrm{al}(x)_3}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_3}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_3}{dk} + (k'^2 + k^2x^2)\,\mathrm{al}(x)_3 &= 0,
\end{aligned}$$

ou, ce qui est la même chose, les fonctions

$$\mathrm{al}(x),\quad \sqrt{k}\,\mathrm{al}(x)_1,\quad \frac{\sqrt{k}}{\sqrt{k'}}\,\mathrm{al}(x)_2,\quad \frac{1}{\sqrt{k'}}\,\mathrm{al}(x)_3,$$

satisfont chacune à l'équation

$$\frac{d^2z}{dx^2} + 2k'x\frac{dz}{dx} + 2kk'\frac{dz}{dk} + k^2x^2z = 0.$$

Écrivons pour un moment ξ, κ au lieu de x, k; les fonctions $\mathrm{al}(\xi)$, etc., satisfont à l'équation

$$\frac{d^2z}{d\xi^2} + 2\kappa^2\xi\frac{dz}{d\xi} + 2\kappa\kappa'^2\frac{dz}{d\kappa} + \kappa'^2\xi^2z = 0.$$

Cela étant, en posant

$$\xi = \frac{x}{\sqrt{k}},\qquad \kappa = k,$$

on obtient

$$\frac{dz}{dx} = \frac{1}{\sqrt{k}}\frac{dz}{d\xi},\quad \frac{d^2z}{dx^2} = \frac{1}{k}\frac{d^2z}{d\xi^2},\quad \frac{dz}{dk} = -\frac{x}{2k\sqrt{k}}\frac{dz}{d\xi} + \frac{dz}{d\kappa},$$

et de là

$$\frac{dz}{d\xi} = \sqrt{k}\,\frac{dz}{dx},\quad \frac{d^2z}{d\xi^2} = k\frac{d^2z}{dx^2},\quad \frac{dz}{d\kappa} = \frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}.$$

L'équation différentielle devient ainsi

$$k\frac{d^2z}{dx^2} + 2k'x\frac{dz}{dx} + 2kk'^2\left(\frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}\right) + kx^2z = 0,$$

c'est-à-dire

$$\frac{d^2z}{dx^2} + \frac{1+k^2}{k}x\frac{dz}{dx} + 2k'^2\frac{dz}{dk} + x^2z = 0,$$

équation qui sera ainsi satisfaite par

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right), \quad \sqrt{k}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_1, \quad \frac{\sqrt{k}}{\sqrt{k'}}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_2, \quad \frac{1}{\sqrt{k'}}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_3.$$

Or en écrivant

$$k + \frac{1}{k} = \alpha,$$

l'équation en z devient

$$\frac{d^2z}{dx^2} + 2\alpha x\frac{dz}{dx} - 2(\alpha^2-4)\frac{dz}{d\alpha} + x^2z = 0, \tag{1}$$

laquelle est ce que devient celle-ci

$$(1-\alpha x^2+x^4)\frac{d^2z}{dx^2} + (n-1)(\alpha x - 2x^3)\frac{dz}{dx} - 2n(\alpha^2-4)\frac{dz}{d\alpha} + n(n-1)x^2z = 0, \tag{2}$$

en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de x, et puis $n = \infty$. L'équation (2), trouvée par Jacobi (*Journal de Crelle*, t. IV. p. 185, 1829), a la propriété que voici, savoir en posant

$$\alpha = k + \frac{1}{k}, \quad x = \sqrt{k}\sin\mathrm{am}\,u,$$

alors l'équation est satisfaite en prenant pour z soit le numérateur, soit le dénominateur, de la fonction rationnelle de x qui donne la valeur de la fonction $\sqrt{\lambda}\sin\mathrm{am}\left(\dfrac{u}{M}, \lambda\right)$, où λ, M sont le module et le multiplicateur qui correspondent à la transformation de l'ordre n (n étant un nombre impair quelconque).

L'équation fut donnée par Jacobi sans démonstration. Je l'ai démontrée (*Camb. and Dubl. Math. Journal*, t. II. p. 256, 1847, [45]) de la manière que voici, savoir en écrivant

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(m-1)}\Theta^{-n}(u)\cdot\Sigma,$$

on obtient pour Σ l'équation

$$\frac{d^2\Sigma}{du^2} - 2nu\left(k^2 - \frac{E}{K}\right)\frac{d\Sigma}{du} + 2nkk'^2\frac{d\Sigma}{dk} = 0.$$

Cette équation, en prenant

$$\omega = \frac{u\pi K'}{K}, \quad v = \frac{u\pi u}{2K},$$

devient

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

équation mentionnée par Jacobi, laquelle est satisfaite par

$$\Sigma = \Theta\left(nu, \frac{nK'}{K}\right), \quad \text{ou} \;\; \Sigma = \mathrm{H}\left(nu, \frac{nK'}{K}\right);$$

cela donne pour z deux valeurs qui sont le dénominateur et le numérateur de la fraction dont il s'agit. J'ose croire que ce doit être à peu près de cette manière que l'équation fut trouvée par Jacobi.

Or les solutions en question de l'équation (1) peuvent être trouvées au moyen de l'équation de Jacobi; pour cela, au lieu des valeurs ci-dessus données de ω, v, j'écris

$$\omega = \frac{\pi K'}{K}, \quad v = \frac{\sqrt{n}\pi u}{2K},$$

ce qui conduit à la même équation

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

laquelle sera ainsi satisfaite par

$$\Sigma = \Theta\left(\sqrt{n}u, \frac{K'}{K}\right), \quad \Sigma = \mathrm{H}\left(\sqrt{n}u, \frac{K'}{K}\right),$$

ou, ce qui est la même chose, par

$$\Sigma = \Theta(\sqrt{n}u), \quad \Sigma = \mathrm{H}(\sqrt{n}u).$$

L'équation (2) sera donc satisfaite par

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(n-1)} \Theta^{-n}(u)\,\Theta(\sqrt{n}u),$$

ou, en se souvenant que $\Theta(0) = \sqrt{\dfrac{2Kk'}{\pi}}$, par

$$z = \frac{\Theta(\sqrt{n}u)}{\Theta(0)} \cdot \frac{\Theta^n(0)}{\Theta^n(u)}.$$

Or en écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de x, pour faire ensuite $n = \infty$, nous avons

$$\frac{x}{\sqrt{n}} = \sqrt{k}\sin\operatorname{am} u,$$

équation qui se réduit à

$$u = \frac{x}{\sqrt{nk}};$$

cela donne

$$\Theta(\sqrt{n}u) = \Theta\left(\frac{x}{\sqrt{k}}\right), \qquad \Theta^n u = \Theta^n(0)\, e^{n\int_0^u du\, k^2 \operatorname{sn}^2 u},$$

puisque

$$\Theta(u) = \Theta(0)\, e^{\int_0^u du\,(k^2\operatorname{sn}^2 u - \frac{E}{K})\cdots};$$

et $n\displaystyle\int_0^u du \int_0^u du\, k^2 \sin^2 \operatorname{am} u$, en y substituant $u = \dfrac{x}{\sqrt{nk}}$, contient le facteur $\dfrac{1}{n}$ et se réduit ainsi à zéro. Donc on obtient

$$z = \frac{\Theta\left(\dfrac{x}{\sqrt{k}}\right)}{\Theta(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)}$$

comme solution de l'équation (1), qui se déduit de l'équation (2) en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de x et puis $n=\infty$. Et cette valeur de z est précisément la fonction $\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)$ de M. Weierstrass. On obtient de même la solution

$$z = \frac{\mathrm{H}\left(\dfrac{x}{\sqrt{k}}\right)}{\Theta(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

ou, ce qui est la même chose,

$$z = \frac{\mathrm{H}\left(\dfrac{x}{\sqrt{k}}\right)}{\dfrac{1}{\sqrt{k}}\mathrm{H}'(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

qui est la fonction $\sqrt{k}\;\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_1$. Et d'une manière semblable les solutions

$$z = \frac{\Theta\left(\dfrac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)}, \qquad z = \frac{\Theta\left(\dfrac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

qui sont les fonctions $\sqrt{\dfrac{k}{k'}}\;\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_2$, $\dfrac{1}{\sqrt{k'}}\,\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_3$.

J'ajoute que dans le Mémoire cité (1847) j'ai donné la suite

$$z = C_0 + C_1\frac{x^2}{1\cdot 2} + C_2\frac{x^4}{1\cdot 2\cdot 3\cdot 4} + \cdots$$

où

$$\begin{aligned}
&C_0 = 1,\\
&C_1 = 0,\\
&C_2 = -2,\\
&C_3 = +8\alpha,\\
&C_4 = -32\alpha^2 - 4,\\
&C_5 = +128\alpha^3 + 96\alpha,\\
&C_6 = -512\alpha^4 - 960\alpha^2 - 408,\\
&C_7 = +2048\alpha^5 + 7168\alpha^3 + 7584\alpha,\\
&C_8 = -8192\alpha^6 - 46080\alpha^4 - 88320\alpha^2 - 15384,\\
&\quad\vdots
\end{aligned}$$

de façon qu'en général

$$C_{r+2} = -(2r+1)(2r+2)\,C_r - (2r+2)\alpha C_{r+1} + 2(\alpha^2-4)\frac{dC_{r+1}}{d\alpha},$$

c'est le développement de M. Weierstrass pour la fonction $\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)$: seulement je ne connaissais pas alors l'expression finie

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right) \;=\; \frac{1}{\Theta(0)}\,\Theta\left(\frac{x}{\sqrt{k}}\right)\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)}$$

de cette suite. Je remarque en passant que pour $\alpha=2$, la suite se réduit à

$$e^{\frac{1}{2}x^2} = \tfrac{1}{2}(e^x + e^{-x}).$$

308.

ON THE △ FACED POLYACRONS, IN REFERENCE TO THE PROBLEM OF THE ENUMERATION OF POLYHEDRA.

[From the Memoirs of the Literary and Philosophical Society of Manchester, vol. I. (1862), pp. 248–256.]

The problem of the enumeration of polyhedra[1] is one of extreme difficulty, and I am not aware that it has been discussed elsewhere than in Mr Kirkman's valuable series of papers on this subject in the Memoirs of the Society and in the *Philosophical Transactions*. A case of the general problem is that of the enumeration of the polyhedra with trihedral summits; and Mr Kirkman in the earliest of his papers, viz. that "On the representation and enumeration of polyhedra" (*Memoirs*, vol. XII. pp. 47–70, 1854), has in fact, by an examination of the particular case, accomplished the enumeration of the octahedra with trihedral summits. A subsequent paper "On the enumeration of x-edra having trihedral summits and an $(x-1)$-gonal base," *Phil. Trans.* vol. XLVI. pp. 399–411, 1856), relates, as the title shows, only to a special case of the problem of the polyhedra with trihedral summits, and in this particular case the number of polyhedra is more completely determined; but the later memoirs relate to the problem in all its generality, and the above-mentioned particular problem of the enumeration of the polyhedra with trihedral summits is not, I think, anywhere resumed. Instead of the polyhedra with trihedral summits, it is really the same thing, but it is rather more convenient to consider the polyacrons with triangular faces, or as these may for shortness be called, the △ faced polyacrons; and it is intended in the present paper to give a method for the derivation of the △ faced polyacrons of a given number of summits from those of the next inferior number of summits, and to exemplify it by finding, in an orderly manner, the △ faced polyacrons up to the octacrons: thus, as regards the examples, stopping at the same point as Mr Kirkman, for although perfectly practicable it would be very tedious to carry them further, and there would be no commensurate advantage in doing so. The epithet △ faced will be omitted in the sequel, but it is to be understood throughout that I am speaking of such polyacrons only; and I shall for convenience use the epithets tripleural, tetrapleural, &c. to denote summits with three, four, &c. edges through them. The number of edges at a summit is of course equal to the number of faces, but it is the edges rather than the faces which have to be considered.

An n-acron has

$$n \text{ summits},\quad 3n-6 \text{ edges},\quad 2n-4 \text{ faces},$$

and it is easy to see that there are the following three cases only, viz.:

1. The polyacron has at least one tripleural summit.
2. The polyacron, having no tripleural summit, has at least one tetrapleural summit.
3. The polyacron, having no tripleural or tetrapleural summit, has at least twelve pentipleural summits.

In fact, if the polyacron has c tripleural summits, d tetrapleural summits, e pentipleural summits, and so on, then we have

$$n = c + d + e + f + g + h + \&\text{c.},$$

$$6n - 12 = 3c + 4d + 5e + 6f + 7g + 8h + \&\text{c.},$$

[1] I use with Mr Kirkman the expression "enumeration of polyhedra" to designate the general problem, but I consider that the problem is to find the different polyhedra rather than to count them, and I consequently take the word enumeration in the popular rather than the mathematical sense.

and therefore

$$12 = 3c + 2d + e + 0f - g - 2h - \&c.,$$

or

$$3c + 2d + e = 12 + f + 2g + 3h + \&c.;$$

whence if $c = 0$ and $d = 0$, then $e = 12$ at least. It appears, moreover (since n cannot be less than e), that any polyacron with less than 12 summits cannot belong to the third class, and must therefore belong to the first or the second class.

An $(n+1)$-acron, by a process which I call the subtraction of a summit, may be reduced to an n-acron; viz., the faces about any summit of the $(n+1)$-acron stand upon a polygon (not in general a plane figure) which may be called the basic polygon, and when the summit with the faces and edges belonging to it is removed, the basic polygon, if a triangle, will be a face of the n-acron; if not a triangle, it can be partitioned into triangles which will be faces of the n-acron. The annexed figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which are the only cases which need be considered; these may be called the first, second and third process respectively. It is proper to remark that for the same removed summit the first process can be performed in one way only, the second process in two ways, the third in five ways; these being in fact the numbers of ways of partitioning the basic polygon.

We may in like manner, by the converse process of the addition of a summit, convert an n-acron into an $(n+1)$-acron; viz., it is only necessary to take on the n-acron a polygon of any number of sides, and make this the basic polygon of the new summit of the $(n+1)$-acron, and for this purpose to remove the faces within the polygon and substitute for them a set of triangular faces standing on the sides of the polygon and meeting in the new summit: the same figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which (as for the subtractions) are the only cases which need be considered. It may be noticed that for the same basic polygon the process is in each case a unique one; the process is said to be the first, second, or third process, according as the new summit is tripleural, tetrapleural, or pentipleural.

[The original page exhibits small woodcut figures illustrating the subtraction/addition processes for tripleural, tetrapleural, and pentipleural summits; these are omitted here.]

Now, reverting to the before-mentioned division of the polyacrons into three classes, an $(n+1)$-acron of the first class may by the first process of subtraction be reduced to an n-acron, and conversely it can be by the first process of addition derived from an n-acron. An $(n+1)$-acron of the second class, as having a tetrapleural summit, may by the second process of subtraction be reduced to an n-acron, and conversely it can be by the second process of addition derived from an n-acron. And in like manner, an $(n+1)$-acron of the third class, as having a pentipleural summit, may be by the third process of subtraction reduced to an n-acron, and conversely it may be by the third process of addition derived from an n-acron.

Hence all the $(n+1)$-acrons can be by the first, second and third processes of addition respectively derived from the n-acrons. It is to be observed that all the $(n+1)$-acrons of the first class are obtained by the first process; the second process is only required for finding the $(n+1)$-acrons of the second class; and these being all obtained by means of it, the third process is only required for finding the $(n+1)$-acrons of the third class. Hence the second process need only be made use of when the n-acron has no tripleural summit, or when it has only one tripleural summit, or when, having two tripleural summits, they are the opposite summits of two adjacent faces. In the last-mentioned two cases respectively it is only necessary to consider the basic quadrangles which pass through the single tripleural summit and the basic quadrangle which passes through the two tripleural summits; for with any other basic quadrangle the derived $(n+1)$-acron would retain a tripleural summit, and would consequently be of the first class. The condition is more simply expressed as follows, viz.: The second process need only be employed when there is on the n-acron a basic quadrangle the summits of which are at least of the number of edges shown in the annexed figure, and all the other summits are at least 4-pleural. Again, by the third process (as already mentioned) we seek only to obtain the $(n+1)$-acrons of the third class; the process need only be applied to the n-acrons for which there exists a basic pentagon the summits of which are at least of the number of edges shown in the annexed figure, all the other summits being at least 5-pleural; for it is only in this case that the derived $(n+1)$-acron will be of the third class. The condition just referred to obviously implies that the n-acron is of the second or third class. It is to be noticed that in applying the foregoing principles to the formation of the polyacrons as far as the 11-acrons we are only concerned with the first and second processes.

[The original includes small diagrams of a basic quadrangle (with corner degrees $4, 3, 4, 3$) and a basic pentagon (with corner degrees $4, 4, 5, 4, 4$) illustrating the conditions just stated; these woodcuts are omitted here.]

Consider the entire series of n-acrons, say A, B, C, &c., and suppose that the n-acron A gives rise to a certain number, say P, Q, R, S of $(n+1)$-acrons, the $(n+1)$-acron P is of course derivable from the n-acron A, but it may be derivable from other n-acrons, suppose from the n-acrons B and C. Then in considering the $(n+1)$-acrons derived from B, one of these will of course be found to be the $(n+1)$-acron P, and it is only the remaining $(n+1)$-acrons derived from B which are or may be $(n+1)$-acrons not already previously obtained as $(n+1)$-acrons derived from A. And if in this manner, as soon as each $(n+1)$-acron is obtained, we apply to it the process of subtraction so as to ascertain the entire series of n-acrons from which it is derivable, and, in forming the $(n+1)$-acrons derived from these, take account of the $(n+1)$-acrons already previously obtained and found to be derivable from these, we should obtain without any repetitions the entire series of the $(n+1)$-acrons.

For merely finding the number of the $(n+1)$-acrons, a more simple process might be adopted: say that an n-acron is p-wise generating when it gives rise to a number p of $(n+1)$-acrons, and that it is q-wise generable when it can be derived from a number q of $(n+1)$-acrons; and assume that a given n-acron is $(y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \text{\&c.})$-wise generating, viz. that it gives rise to a number y_1 of $(n+1)$-acrons which are 1-wise generable, a number y_2 of $(n+1)$-acrons which are 2-wise generable, and so on; these forming the sum

$$\Sigma\,(y_1 + \tfrac{1}{2}y_2 + \tfrac{1}{3}y_3 + \cdots),$$

where Σ refers to the entire series of the n-acrons, it is clear that every m-wise generable $(n+1)$-acron will in respect of each of the n-acrons from which it is derivable be reckoned as $\dfrac{1}{m}$, that is, it will be in the entire sum reckoned as 1, and the sum in question will consequently be the number of the $(n+1)$-acrons.

The figures of the polyacrons comprised in the annexed Tables show the application of the method to the genesis of the polyacrons as far as the octacrons, in which the numbers indicate the nature of the different summits, according to the number of edges through each summit, viz., 3 a tripleural summit, 4 a tetrapleural summit, and so on. It will be noticed that there is only a single case in which this notation is insufficient to distinguish the polyacron, viz., among the octacrons there are two forms each of them with the same symbol 33445566; the inspection of the figures shows at once that these are wholly distinct forms, for in the first of them, viz. that derived from 3344555, each of the tripleural summits stands upon a basic triangle 456, while in the other of them, that from 3444555, each of the tripleural summits stands upon a basic triangle 566. But the symbol is merely generic, and of course in the polyacrons of a greater number of summits it may very well happen that a considerable number of polyacrons are comprised in the same genus.

The following remarks on the derivation of the octacrons from the heptacrons will further illustrate the method:

1. The heptacron 3335556 has three kinds of faces,[2] viz. 356, 355 [555], the first process consequently gives rise to 3 octacrons. As the heptacron has more than two tripleural summits the second process is not applicable.

2. The heptacron 3344466 has three kinds of faces, viz.: 366, 346 and 446, and the first process gives therefore 3 octacrons. The heptacron has only two tripleural summits, and they are disposed in the proper manner; the second process gives therefore 1 octacron.

3. The heptacron 3344556 has five kinds of faces, viz. 345, 346, 356, 456 and 455, and the first process consequently gives 5 octacrons. The heptacron has two tripleural summits, but they are not disposed in such manner as to render the second process applicable.

4. The heptacron 3444555 has four kinds of faces, viz. 355, 455, 445 and 444, and the first process gives therefore 4 octacrons. The heptacron has one tripleural summit, and the basic quadrangles 3545 which belong to it are of the same kind; the second process gives therefore 1 octacron.

5. The heptacron 4444455 has only one kind of face, viz. 445, and the first process gives therefore 1 octacron. There are two kinds of basic quadrangles, viz. 4545 and 4445, and the second process gives therefore 2 octacrons.

The number of octacrons would thus be 20, but by passing back from the octacrons to the heptacrons, it is found that there are in fact only 14 octacrons. Thus the octacron 33336666 has only one kind of tripleural summit 666 (the summit is here indicated by the symbol of the basic polygon) and the octacron is thus seen to be derivable from a single heptacron only, viz. the heptacron 3335556 from which it was in fact derived. But the octacron 33345567 has three kinds of tripleural summits, viz. 567, 557 and 467, and it is consequently derivable from three heptacrons, viz. the heptacrons 3335556, 3344466 and 3344555, and so on. The passage to the heptacrons from an octacron with one or more tripleural summits is of course always by the first process, but for the last two octacrons, which have no tripleural summits, the passage back to the heptacrons is by the second process: thus for the octacron 44445555 we have but one kind of tetrapleural summit 4555; but as opposite pairs of summits of the basic quadrangle are of different kinds, viz. 45 and 55, we obtain two heptacrons, viz. 3444555 and 4444455. The octacron 44444466 has but one kind of tetrapleural summit, viz. 4646, and the pairs of opposite summits of the basic quadrangle being of the same kind 46, we obtain from it only the heptacron 4444455.

It may be remarked that for the five heptacrons respectively the values of the sum $y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots$ are

$$1 + \tfrac{1}{2} + \tfrac{1}{3}, \quad \tfrac{1}{2} + 1 + \tfrac{1}{2} + \tfrac{1}{2}, \quad \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2}, \quad 1 + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2}, \quad 1 + \tfrac{1}{2} + \tfrac{1}{2},$$

giving for $\Sigma\,(y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots)$ the value 14, as it should do.

[Plate I (facing p. 44 in the original) presents the family-tree diagram of the △-faced polyacrons from 3333 *(the tetrahedron) through the octacrons, together with line-drawings of selected polyacrons (the hexacron* 33444*, the heptacrons* 334455*,* 334445*,* 444444*, etc., and the octacrons* 33336666*,* 33445566*,* 33455556*,* 33555555*,* 34444566*,* 34445556*,* 44445555*,* 44444466*). The plate is omitted here; consult the printed source for the figure.]*

[2] It is hardly necessary to remark that it must not be imagined that in general all the faces denoted by a symbol such as 355 (which determines only the nature of the summits on the face) are faces of the same kind, but this is so in the cases referred to in the text.

309.

NOTE ON THE THEORY OF DETERMINANTS.

[From the Philosophical Magazine, vol. XXI. (1861), pp. 180–185.]

The following mode of arrangement of the developed expression of a determinant had presented itself to me as a convenient one for the calculation of a rather complicated determinant of the fifth order; but I have since found that it is in effect given, although in a much less compendious form, in a paper by J. N. Stockwell, "On the Resolution of a System of Symmetrical Equations with Indeterminate Coefficients," *Gould's Ast. Journal*, No. 139 (Cambridge, U. S., Sept. 10, 1860).

Suppose that the determinant

$$\begin{vmatrix} 11 & 12 & 13 \\ 21 & 22 & 23 \\ 31 & 32 & 33 \end{vmatrix}$$

is represented by $\{123\}$, and so for a determinant of any order $\{123\ldots n\}$.

Let $|1|$, $|2|$, $|12|$, $|123|$, &c., denote as follows: viz.

$$\begin{aligned} |1| &= 11, \quad |2| = 22, \text{ \&c.} \\ |12| &= 12\cdot 21, \\ &\vdots \\ |123| &= 12\cdot 23\cdot 31, \end{aligned}$$

where it is to be noticed that, with the same two symbols, e.g. 1 and 2, there is but one distinct expression $|12|$ (in fact $|21| = 21\cdot 12 = |12|$); with the same three symbols, $1, 2, 3$, there are two distinct expressions, $|123|$ $(= 12\cdot 23\cdot 31)$ and $|132|$ $(= 13\cdot 32\cdot 21)$; and generally with the same m symbols $1, 2, 3, \ldots, m$, there are $1\cdot 2\cdot 3\cdots(m-1)$ distinct expressions $|123\ldots m|$, which are obtained by permuting in every possible manner all but one of the m symbols.

This being so, and writing for greater simplicity $|1|2|$ to denote the product $|1|\times|2|$, and so in general, the values of the determinants $\{12\}$, $\{123\}$, $\{1234\}$, $\{12345\}$, &c. are as follows: viz.

$$\begin{aligned} \{12\} &= +|1|2| \ - \ |12|, \\ \{123\} &= +|1|2|3| \ - \ |12|3| \ - \ |2|13| \ - \ |3|12| \ + \ |123| \ + \ |132|, \\ \{1234\} &= +|1|2|3|4| \ - \ |12|3|4| \text{ (6 terms)} \ + \ |123|4| \text{ (8 terms)} \\ &\quad + |12|34| \text{ (3 terms)} \ - \ |1234| \text{ (6 terms)}, \\ \{12345\} &= +|1|2|3|4|5| \ - \ |12|3|4|5| \text{ (10)} \ + \ |123|4|5| \text{ (20)} \\ &\quad + |12|34|5| \text{ (15)} \ - \ |1234|5| \text{ (30)} \ - \ |123|45| \text{ (20)} \ + \ |12345| \text{ (24)}, \end{aligned}$$

with term-counts $1{+}1 = 2$, $1{+}3{+}3{-} = 6$, $1{+}6{+}8{+}3{+}6 = 24$, and $1{+}10{+}20{+}15{+}30{+}20{+}24 = 120$ respectively.

[The original page sets these out as a tabular display with running counts $60 + 60 = 120$ *for* $\{12345\}$*; we give the algebraic content above.]*

where, as regards the signs, it is to be observed that there is a sign $-$ for each compartment $|\ |$ containing an even number of symbols; thus in the expression for $\{1234\}$, the terms $|1|2|3|4|$ have the sign $-\cdot- = +$, and the terms $|1234|$ the sign $-$. Or, what comes to the same thing; when n is even, the sign is $+$ or $-$ according as the number of compartments is even or odd; and contrariwise when n is odd. As regards the remaining part of the expression, this merely exhibits the partitions of a set of n things; and the formula for the several determinants up to the determinant of a given order are all of them obtained by means of the form

[the partition diagram, carried up to order 7 in the original]

which is carried up to the order 7, but which can be further extended without any difficulty whatever.

It is perhaps hardly necessary, but I give at full length the expressions of the determinant of the third order: this is

$$\begin{aligned}\{123\} = &+|1|2|3| \\ &-|12|3| \\ &-|2|13| \\ &-|3|12| \\ &+|123| \\ &+|132|,\end{aligned}$$

and by writing down in like manner the expression for the twenty-four terms of the determinant of the fourth order, the notation will become perfectly clear.

The formula hardly requires a demonstration. The terms of a determinant $\{123\ldots m\}$, for example the determinant $\{1234\}$, are obtained by permuting in every possible manner the symbols in either column, say the second column, of the arrangement

$$\begin{matrix}1 & 1\\ 2 & 2\\ 3 & 3\\ 4 & 4\end{matrix}$$

and prefixing the sign (+ or −) of the arrangement; and the resulting arrangements, for instance

$$+\begin{matrix}1 & 1\\ 2 & 2\\ 3 & 3\\ 4 & 4\end{matrix}, \qquad -\begin{matrix}1 & 2\\ 2 & 1\\ 3 & 3\\ 4 & 4\end{matrix}, \qquad -\begin{matrix}1 & 2\\ 2 & 3\\ 3 & 4\\ 4 & 1\end{matrix},$$

are interpreted either into $+11\cdot 22\cdot 33\cdot 44$, $-12\cdot 21\cdot 33\cdot 44$, $-12\cdot 23\cdot 34\cdot 41$, or in the notation of the formula, into

$$+|1|2|3|4|, \qquad -|12|3|4|, \qquad -|1234|;$$

and so in general.

Suppose that any partition of n contains α compartments each of a symbols, β compartments each of b symbols… ($a, b, \ldots$ being all of them different and greater than unity), and p compartments each of a single symbol, we have

$$n = \alpha a + \beta b + \cdots + p;$$

and writing, as usual, $\Pi a = 1\cdot 2\cdot 3\cdots a$, &c., the number of ways in which the symbols $1, 2, 3, \ldots, n$ can be so arranged in compartments is

$$\frac{\Pi n}{(\Pi a)^{\alpha}(\Pi b)^{\beta}\cdots \Pi\alpha\,\Pi\beta\,\cdots\Pi p},$$

but each such arrangement gives $(\Pi(a-1))^{\alpha}\cdot(\Pi(b-1))^{\beta}\cdots$ terms of the determinant, and the corresponding number of terms therefore is

$$\frac{\Pi n}{a^{\alpha}b^{\beta}\cdots \Pi\alpha\,\Pi\beta\,\cdots\Pi p}.$$

The whole number of terms of the determinant is Πn, and we have thus the theorem

$$1 = \Sigma \frac{1}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

in which the summation corresponds to all the different partitions $n = \alpha a + \beta b + \cdots + p$, where $a, b, \ldots$ are all of them different and greater than unity; a theorem given in Cauchy's *Mémoire sur les Arrangements* &c., 1844. But it is to be noticed also that, the number of the positive and negative terms being equal, we have besides

$$0 = \Sigma \frac{(-)^{n-\alpha-\beta-\cdots-p}}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

or, what is the same thing,

$$0 = \Sigma \frac{(-)^{\alpha+\beta+\cdots}}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p};$$

and thence also

$$\tfrac{1}{2} = \Sigma \frac{1}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

where, as before, $n = \alpha a + \beta b + \cdots + p$ ($a, b, \ldots$ being all different and greater than unity); but the summation is restricted either to the partitions for which $n - \alpha - \beta - \cdots - p$ is even, or else to those for which $n - \alpha - \beta - \cdots - p$ is odd.

The formula affords a proof of the fundamental property of skew symmetrical determinants. In such a determinant we have not only $12 = -21$, &c., but also $11 = 0$, &c. Suppose that n, the order of the determinant, is odd; then in each line of the expression

$$\{123\ldots n\} = |1|2|\ldots|n| \ \pm \&\text{c.}$$

of the determinant, there is at least one compartment $|1|$ or $|123|$ &c. containing an odd number of symbols: let $|123|$ be such a compartment, then the determinant contains the terms $|123|P$ and $|132|P$ (where P represents the remaining compartments), that is, $12 \cdot 23 \cdot 31 \cdot P$ and $13 \cdot 32 \cdot 21 \cdot P$. But in virtue of the relations $12 = -21$, &c., we have

$$12 \cdot 23 \cdot 31 = -13 \cdot 32 \cdot 21;$$

and so in all similar cases; that is, the terms destroy each other, or the skew symmetrical determinant of an odd order is equal to zero.

The like considerations show that a skew symmetrical determinant of an even order is a perfect square. In fact, considering for greater simplicity the case $n = 4$, any line in the foregoing expression of $\{1234\}$ for which a compartment contains an odd number of symbols, gives rise to terms which destroy each other, and may be omitted. The expression thus reduces itself to

$$\begin{aligned}\{1234\} = &+|12|34| \quad 3 \text{ terms}\\ &-|1234| \quad 6 \text{ terms},\end{aligned}$$

which is in fact the square of

$$12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23;$$

for the square of a term, say $12 \cdot 34$, is $12^2 \cdot 34^2$ or $12 \cdot 21 \cdot 34 \cdot 43$, that is, $|12|34|$, and the double of the product of two terms, say $12 \cdot 34$ and $13 \cdot 42$, is $2 \cdot 12 \cdot 34 \cdot 13 \cdot 42$, or $-12 \cdot 24 \cdot 43 \cdot 31 - 13 \cdot 34 \cdot 42 \cdot 21$, that is $-|1243| - |1342|$, and so for the other similar terms, and we have

$$\{1234\} = (12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23)^2;$$

and so in general, n being any even number, the skew symmetrical determinant $\{123\ldots n\}$ is equal to the square of the Pfaffian $12\ldots n$, where the law of these Pfaffian functions is

$$\overline{12\cdot 34} = 12\cdot 34 + 13\cdot 42 + 14\cdot 23,$$
$$\overline{1234\cdot 56} = 12\cdot 3456 + 13\cdot 4562 + 14\cdot 5623 + 15\cdot 6234 + 16\cdot 2345,$$

where, in the second equation, 3456, &c. are Pfaffians, viz. $3456 = 34\cdot 56 + 35\cdot 64 + 36\cdot 45$; and so on.

2, Stone Buildings, W.C., December 28, 1860.

310.

NOTE ON MR JERRARD'S RESEARCHES ON THE EQUATION OF THE FIFTH ORDER.

[From the Philosophical Magazine, vol. XXI. (1861), pp. 210–214.]

Functions of the same set of quantities which are, by any substitution whatever, simultaneously altered or simultaneously unaltered, may be called *homotypical*[3]. Thus all symmetric functions of the same set of quantities are homotypical: $(x + y - z - w)^2$ and $xy + zw$ are homotypical, &c.

It is one of the most beautiful of Lagrange's discoveries in the theory of equations, that, given the value of any function of the roots, the value of any homotypical function may be rationally determined; in other words, that any homotypical function whatever is a rational function of the coefficients of the equation and of the given function of the roots.

The researches of Mr Jerrard are contained in his work, *An Essay on the Resolution of Equations*, London, Taylor and Francis, 1859. The solution of an equation of the fifth order is made to depend on an equation of the sixth order in W; and he conceives that he has shown that one of the roots of this equation is a rational function of another root: "The equation for W will therefore belong to a class of equations of the sixth degree, the resolution of which can, as Abel has shown, be effected by means of equations of the second and third degrees; whence I infer the possibility of solving any proposed equation of the fifth degree by a finite combination of radicals and rational functions."

The above property of rational expressibility, if true for W, will be true for any function homotypical with W; and conversely. I proceed to inquire into the form of the function W.

The function W is derived from the function P, which denotes any one of the quantities p_1, p_2, p_3. And if x_1, x_2, x_3, x_4, x_5 are the roots of the given equation of the fifth order, and if $\alpha, \beta, \gamma, \delta, \epsilon$ represent in an undetermined or arbitrary order of succession the five indices $1, 2, 3, 4, 5$, and if ι denote an imaginary fifth root of unity (I conform myself to Mr Jerrard's notation), then p_1, p_2, p_3, and the other auxiliary quantities t, u, are obtained from the system

[3]The *à priori* demonstration shows the cases of failure. Suppose that the roots of a biquadratic equation are $1, 3, 5, 9$; then, given $a + b = 8$, we know that either $a = 3$, $b = 5$, or else $a = 5$, $b = 3$, and in either case $ab = 15$; hence in the present case (which represents the general case), $a + b$ being known, the homotypical function ab is rationally determined. But if the roots are $1, 3, 5, 7$ (where $1 + 7 = 3 + 5$), then, given $a + b = 8$, this is satisfied by $(a, b = 3, 5)$ or by $(a, b = 1, 7)$, and the conclusion is $ab = 15$ or 7; so that here ab is determined, not as before, rationally, but by a quadratic equation.

of equations:

$$\begin{aligned}
x_\alpha^4 + p_1 x_\alpha^3 + p_2 x_\alpha^2 + p_3 &= t + u,\\
x_\beta^4 + p_1 x_\beta^3 + p_2 x_\beta^2 + p_3 &= \iota t + \iota^4 u,\\
x_\gamma^4 + p_1 x_\gamma^3 + p_2 x_\gamma^2 + p_3 &= \iota^2 t + \iota^3 u,\\
x_\delta^4 + p_1 x_\delta^3 + p_2 x_\delta^2 + p_3 &= \iota^3 t + \iota^2 u,\\
x_\epsilon^4 + p_1 x_\epsilon^3 + p_2 x_\epsilon^2 + p_3 &= \iota^4 t + \iota\, u.
\end{aligned}$$

If from these equations we seek for the values of p_1, p_2, p_3, t, u, we have

$$1 : p_1 : p_2 : p_3 : -t : -u = \Pi_1 : \Pi_2 : \Pi_3 : \Pi_4 : \Pi_5 : \Pi_6,$$

where $\Pi_1, \Pi_2, \ldots$ denote the determinants formed out of the matrix

$$\begin{pmatrix}
x_\alpha^4 & x_\alpha^3 & x_\alpha^2 & 1 & 1 & 1\\
x_\beta^4 & x_\beta^3 & x_\beta^2 & 1 & \iota & \iota^4\\
x_\gamma^4 & x_\gamma^3 & x_\gamma^2 & 1 & \iota^2 & \iota^3\\
x_\delta^4 & x_\delta^3 & x_\delta^2 & 1 & \iota^3 & \iota^2\\
x_\epsilon^4 & x_\epsilon^3 & x_\epsilon^2 & 1 & \iota^4 & \iota
\end{pmatrix},$$

i.e., denoting the columns of this matrix by $1, 2, 3, 4, 5, 6$, we have $\Pi_1 = 23456$, $\Pi_2 = -34561$, $\Pi_3 = 45612$, &c. In particular, the value of Π_1 is

$$\begin{vmatrix}
x_\alpha^3 & x_\alpha^2 & 1 & 1 & 1\\
x_\beta^3 & x_\beta^2 & 1 & \iota & \iota^4\\
x_\gamma^3 & x_\gamma^2 & 1 & \iota^2 & \iota^3\\
x_\delta^3 & x_\delta^2 & 1 & \iota^3 & \iota^2\\
x_\epsilon^3 & x_\epsilon^2 & 1 & \iota^4 & \iota
\end{vmatrix}$$

and developing, and putting for shortness $\{\alpha\beta\} = x_\alpha x_\beta(x_\alpha - x_\beta)$, &c., we have

$$\begin{aligned}
\Pi_1 = (\{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\})\,(-2 + \iota^2 - \iota^3)&\\
+ (\{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\})\,(+\iota + 2\iota^2 - 2\iota^3 - 2\iota^4)&;
\end{aligned}$$

and this is also the form of the other determinants, the only difference being as to the meaning of the symbol $\{\alpha\beta\}$, which, however, in each case denotes a function such that $\{\alpha\beta\} = -\{\beta\alpha\}$. Writing for greater shortness,

$$\{\alpha\beta\gamma\delta\epsilon\} = \{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\},$$

and in like manner

$$\{\alpha\gamma\epsilon\beta\delta\} = \{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\},$$

Π_1 is an unsymmetric linear function (without constant term) of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$; or, what is all that is material, it is an unsymmetric function, containing only odd powers, of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$.

If for

$$\alpha\ \ \beta\ \ \gamma\ \ \delta\ \ \epsilon$$

we substitute any one of the five arrangements

$$\begin{matrix}
\alpha & \beta & \gamma & \delta & \epsilon,\\
\beta & \gamma & \delta & \epsilon & \alpha,\\
\gamma & \delta & \epsilon & \alpha & \beta,\\
\delta & \epsilon & \alpha & \beta & \gamma,\\
\epsilon & \alpha & \beta & \gamma & \delta,
\end{matrix}$$

then $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will in each case remain unaltered.

But if we substitute any one of the five arrangements

$$\begin{array}{ccccc} \alpha & \epsilon & \delta & \gamma & \beta, \\ \epsilon & \delta & \gamma & \beta & \alpha, \\ \delta & \gamma & \beta & \alpha & \epsilon, \\ \gamma & \beta & \alpha & \epsilon & \delta, \\ \beta & \alpha & \epsilon & \delta & \gamma, \end{array}$$

then in each case $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will be changed into $-\{\alpha\beta\gamma\delta\epsilon\}$ and $-\{\alpha\gamma\epsilon\beta\delta\}$ respectively. Hence Π_1 remains unaltered by any one of the first five substitutions; and it is changed into $-\Pi_1$ by any one of the second five substitutions. And the like being the case as regards Π_2, &c., it follows that the quotient $\Pi_1 \div \Pi_2$, or say P, remains unaltered by any one of the ten substitutions. Now the 120 permutations of $\alpha, \beta, \gamma, \delta, \epsilon$ can be obtained as follows, viz. by forming the 12 different pentagons which can be formed with $\alpha, \beta, \gamma, \delta, \epsilon$ (treated as five points), and reading each of them off in either direction from any angle. To each of the 12 pentagons there corresponds a distinct value of P, but such value is not altered by the different modes of reading off the pentagon; P is consequently a 12-valued function.

But there is a more simple form of the analytical expression of such a 12-valued function; in fact, if $[\alpha\beta]$ be any function which is not altered by any one of the above ten substitutions—if, for instance, $[\alpha\beta]$ is a symmetrical function of x_α, x_β, and

$$[\alpha\beta\gamma\delta\epsilon] = [\alpha\beta] + [\beta\gamma] + [\gamma\delta] + [\delta\epsilon] + [\epsilon\alpha],$$

and therefore

$$[\alpha\gamma\epsilon\beta\delta] = [\alpha\gamma] + [\gamma\epsilon] + [\epsilon\beta] + [\beta\delta] + [\delta\alpha],$$

then any unsymmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ will be a 12-valued function homotypical with P.

Mr Jerrard's function W is the sum of two values of his function P; the substitution by which the second is derived from the first can only be that which interchanges the two functions $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$; and hence any symmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ is a function homotypical with Mr Jerrard's W; such symmetric function is in fact a 6-valued function only. Indeed it is easy to see that the twelve pentagons correspond together in pairs, either pentagon of a pair being derived from the other one by stellation, and the six values of the function in question corresponding to the six pairs of pentagons respectively.

Writing with Mr Cockle and Mr Harley,

$$\tau = x_\alpha x_\beta + x_\beta x_\gamma + x_\gamma x_\delta + x_\delta x_\epsilon + x_\epsilon x_\alpha,$$
$$\tau' = x_\alpha x_\gamma + x_\gamma x_\epsilon + x_\epsilon x_\beta + x_\beta x_\delta + x_\delta x_\alpha,$$

then $(\tau + \tau'$ is a symmetrical function of all the roots, and it must be excluded; but) $(\tau - \tau')^2$ or $\tau\tau'$ are each of them 6-valued functions of the form in question, and either of these functions is linearly connected with the Resolvent Product. In Lagrange's general theory of the solution of equations, if

$$ft = x_1 + \iota x_2 + \iota^2 x_3 + \iota^3 x_4 + \iota^4 x_5,$$

then the coefficients of the equation the roots whereof are $(ft)^5$, $(ft^2)^5$, $(ft^3)^5$, $(ft^4)^5$, and in particular the last coefficient $(ft \cdot ft^2 \cdot ft^3 \cdot ft^4)^5$ are determined by an equation of the sixth degree; and this last coefficient is a perfect fifth power, and its fifth root, or $ft \cdot ft^2 \cdot ft^3 \cdot ft^4$, is the function just referred to as the Resolvent Product.

The conclusion from the foregoing remarks is that if the equation for W has the above property of the rational expressibility of its roots, the equation of the sixth order resulting from Lagrange's general theory has the same property.

I take the opportunity of adding a simple remark on cubic equations. The principle which furnishes what in a foregoing foot-note is called the *à priori* demonstration of Lagrange's theorem is that an equation need never contain extraneous roots; a quantity which has only one value will, if the investigation is properly conducted, be determined in the first instance by a linear equation; one which has two values by a quadratic equation, and so on; there is always enough, and not more than enough, to determine what is required.

[Paper 310 continues on p. 54 of Vol. V (beyond the present excerpt).]

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315. On the Cubic Centres of a Line with Respect to Three Lines and a Line *(continued)*

[Continuation from p. 75.]

which is identically true in virtue of $a+b+c=0$ (in fact, multiplying out, this gives

$$\begin{aligned}&12a^2b^2c^2+4(b^2c^4+c^2a^4+a^2b^4)+abc(a^6+b^6+c^6)\\&\quad-8a^2b^2c^2-4(b^2c^4+c^2a^4+a^2b^4)-2abc(a^3+b^3+c^3)-a^3b^3c^3=0;\end{aligned}$$

that is

$$3a^2b^2c^2-abc(a^3+b^3+c^3)=0,\quad\text{or}\quad abc(a^3+b^3+c^3-3abc)=0,$$

where the second factor divides by $a+b+c$), we find the above-mentioned equation.

We then have

$$\frac{-x+y+z}{P}=\frac{x+y+z}{P}-\frac{2x}{P}=-3+\frac{6a^2}{2a^2+bc}=-\frac{3bc}{2a^2+bc},$$

that is

$$\frac{-x+y+z}{P}=\frac{-3bc}{2a^2+bc},\quad\frac{x-y+z}{P}=\frac{-3ca}{2b^2+ca},\quad\frac{x+y-z}{P}=\frac{-3ab}{2c^2+ab},$$

and forming the product of these functions, and that of the foregoing values of $\frac{P}{x}$, $\frac{P}{y}$, $\frac{P}{z}$, we find, as before,

$$-(-x+y+z)(x-y+z)(x+y-z)+xyz=0$$

for the equation of the locus of the single centre. The equation shows that the locus is a cubic curve which touches the lines $x=0$, $y=0$, $z=0$ at the points where these lines are intersected by the lines $y-z=0$, $z-x=0$, $x-y=0$ (that is, it touches the lines $x=0$, $y=0$, $z=0$ harmonically in respect to the line $x+y+z=0$), and besides meets the same lines $x=0$, $y=0$, $z=0$ at the points in which they are respectively met by the line $x+y+z=0$.

2, Stone Buildings, W.C., September 25, 1861.

316. Note on the Solution of an Equation of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 195, 196.]

This Note was in answer to Mr Jerrard's paper "Remarks on Mr Cayley's Note," Phil. Mag. vol. xxi. pp. 348–350, referring to the foregoing paper 310.

317. Note on the Transformation of a Certain Differential Equation.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 266, 267.]

The differential equation

$$(1+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}-m^2y=0,$$

if we put therein $i\theta=2x^2+1$ ($i=\sqrt{-1}$ as usual), becomes

$$(1+\theta^2)\frac{d^2y}{d\theta^2}+\theta\frac{dy}{d\theta}-\tfrac{1}{4}m^2y=0.$$

In fact an integral of the second equation is $(\sqrt{1+x^2}+x)^m$; this is

$$=(\sqrt{(2x^2+1)^2-1}+2x^2+1)^{\frac{1}{2}m};$$

or putting $i\theta=2x^2+1$, it is

$$=\{\sqrt{-\theta^2-1}+i\theta\}^{\frac{1}{2}m};$$

which is

$$=\{i(\sqrt{\theta^2+1}+\theta)\}^{\frac{1}{2}m};$$

so that an integral of the transformed equation in θ is

$$(\sqrt{1+\theta^2}+\theta)^{\frac{1}{2}m};$$

and writing in the second equation θ for x, and $\frac{1}{2}m$ for m, we see that the last-mentioned function, viz. $(\sqrt{\theta^2+1}+\theta)^{\frac{1}{2}m}$, is an integral of

$$(1+\theta^2)\frac{d^2y}{d\theta^2}+\theta\frac{dy}{d\theta}-\tfrac{1}{4}m^2y=0;$$

whence the transformed equation in θ must be this very equation, that is, it must be the first equation. I have for shortness used the particular integral $(\sqrt{1+x^2}+x)^m$; but the reasoning should have been applied, and it is in fact applicable, without alteration, to the general integral

$$C(\sqrt{1+x^2}+x)^m + C'(\sqrt{1+x^2}-x)^m.$$

There is of course no difficulty in a direct verification. Thus, starting from the first equation, or equation in θ, the relation $i\theta = 2x^2+1$ gives

$$\frac{dy}{d\theta} = \frac{i}{4x}\frac{dy}{dx}, \quad \frac{d^2y}{d\theta^2} = \frac{i}{4x}\frac{d}{dx}\left(\frac{i}{4x}\frac{dy}{dx}\right) = -\frac{1}{16x^2}\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right),$$

$$1+\theta^2 = -4x^2(1+x^2);$$

so that the equation becomes

$$\tfrac{1}{4}(1+x^2)\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right) + \frac{1+2x^2}{4x}\frac{dy}{dx} - m^2y = 0,$$

or multiplying by 4,

$$(1+x^2)\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right) + \frac{1+2x^2}{x}\frac{dy}{dx} - 4m^2y = 0,$$

that is

$$(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4m^2y = 0,$$

the second equation. But the first method shows the reason why the two forms are thus connected together.

2, Stone Buildings, W.C., February 19, 1862.

318. On a Question in the Theory of Probabilities.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 361–365.]

The question referred to is that discussed in the paper 121; the remarks on that paper in the Notes and References to volume II. are in a great measure to the same effect as the present and next papers, 318 and 319, the existence of which I had entirely overlooked. In the first part (dated 2, Stone Buildings, W.C., March 1862) of the present paper 318, after referring to the two modes of statement which may be called the Causation statement and the Concomitance statement, I reproduce nearly as in the Notes and References first my own solution as completed by Dedekind, next Boole's solution of the problem, involving his logical probabilities; and the paper is then continued as follows.

The foregoing paper was submitted to Prof. Boole, who, in a letter dated March 26, 1862, writes:

"The observations which have occurred to me after studying your paper are the following.

"1st. I think that your solution is correct under conditions partly expressed and partly implied. The one to which you direct attention is the assumed independence of the causes denoted by A and B. Now I am not sure that I can state precisely what the others are; but one at least appears to me to be the assumed independence of the events of which the probabilities according to your hypothesis are $\alpha\lambda$, $\beta\mu$. Assuming the independence of the causes as to happening, I do not think that you are entitled on that ground to assume their independence as to acting; because, to confine our observations to common experience, we often notice that states of things apparently independent as to their occurrence, may, when concurring, aid or hinder each other in such a manner that the one may be more or less likely to act 'efficiently' in the presence of the other than in its absence. I use the language of your own hypothesis of efficient action.

"2ndly. When I say that I think your solution correct under certain conditions, I ought to add that it appears to me that such conditions ought to be stated as part of the original data, and that they ought to be of such a kind that they can be established by experience in the same way as the other data are. For instance, if experience, as embodied in a sufficiently long series of statistical records, establish that

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta,$$

the very same experience may, by establishing also that

$$\text{Prob. } AB = \alpha\beta,$$

whence in conjunction with the former it follows that

$$\text{Prob. } AB' = \alpha\beta', \quad \text{Prob. } A'B = \alpha'\beta, \quad \text{Prob. } A'B' = \alpha'\beta',$$

enable us to pronounce that A and B are in the long run, as to happening or not happening, in the position of mutually independent events.

"3rdly. I think it may be shown to demonstration, from the nature of the result, that the solution you have obtained does not apply simply and generally to the problem under the single modification of the assumption that A and B are independent. The completed data under this assumption are

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta, \quad \text{Prob. } AB = \alpha\beta,$$
$$\text{Prob. } AE = \alpha p, \quad \text{Prob. } BE = \beta q.$$

You may deduce all these from your Table of Probabilities of 'compound events' given in your paper. Now you may easily satisfy yourself that the sole necessary and sufficient conditions for the consistency of these data are the following:

$$\alpha p' + \beta q \geq \alpha\beta, \tag{1}$$
$$\alpha p + \beta q \leq \alpha\beta, \tag{2}$$
$$\left\{\frac{\alpha}{\beta},\ \frac{\beta}{\alpha},\ \frac{p}{q},\ \frac{q}{p}\right\} \leq 1. \tag{3}$$

(M).

But your solution requires the following conditions to be satisfied, viz.,

$$q - \alpha p \geq 0, \quad p - \beta q \leq 0,$$

together with the system (3). Now (1) and (2) are expressible in the form

$$\alpha'(q - \alpha p) + \alpha\beta' p' \geq 0,$$
$$\alpha(p - \beta q) + \beta\alpha' q' \geq 0;$$

from which you will see that your conditions are narrower than those which the data are really subject to. If your conditions are satisfied, the data will be consistent; but the converse of this proposition does not hold.

"4thly. You remark that my solution of the problem, in which the independence of A and B is not assumed, but in which the probabilities are otherwise the same as in yours, is only applicable when

$$\alpha' + \alpha p \geq \beta q, \quad \beta' + \beta q \geq \alpha p;$$

but you do not appear to have noticed that these are actually the conditions of consistency in the data. Unless these are satisfied, the data cannot possibly be furnished by experience.

"5thly. You remark that I have solved the problem under what you call the 'concomitance' statement, and not the 'causation' statement. I think that every problem stated in the 'causation' form admits, if capable of scientific treatment, of reduction to the 'concomitance' form. I admit it would have been better, in stating my problem, not to have employed the word 'cause' at all. But the introduction of the hypothesis of the independence of A and B does not affect the nature of the problem.

"6thly. The x, s, &c., about the interpretation of which you inquire, are the probabilities of ideal events in an ideal problem connected by a formal relation with the real one. I should fully concede that the auxiliary probabilities which are employed in my method always refer to an ideal problem; but it is one, the form of which, as given by the calculus of logic, is not arbitrary. Nor does its connexion with the real problem appear to me arbitrary. It involves an extension, but as it seems to me, a perfectly scientific extension, of the principles of the ordinary theory of probabilities. On this subject, however, I have but little to add to what I have said. Transactions of the Royal Society of Edinburgh, vol. xxi. part 4, "On the Application of the Theory of Probabilities &c."

"7thly. The problem, as stated by me, and then modified by the simple introduction of the hypothesis of the independence of A and B, must admit of solution by my method; and that solution ought to impose no restriction beyond the conditions of possible experience noted in (M).

I should be extremely glad if mathematicians would examine the analytical questions connected with the application of my method. There can, I think, after the partial proofs which I have given, exist no doubt that the conditions of applicability of the solutions are always identical with the conditions of consistency in the data, i.e. with what I have called, in the paper above referred to, the conditions of possible experience. The proof of the general proposition would involve the showing that a certain functional determinant consists solely of positive terms, with some connected theorems which appear to me to be of considerable analytical interest.

"8thly. I certainly think your paper deserving of publication. If you think proper to add any or the whole of my remarks, you can do so, with of course any comments of your own."

I remark upon Prof. Boole's observations:

1st. I do assume that the causes A and B are absolutely independent of, and uninfluenced by each other; viz. not only the probability of A acting, but also the probability of its acting efficiently, are each of them the same whether B does not act, or acts inefficiently, or acts efficiently; and the like for B.

2ndly. I do assume that the same experience which establishes

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta,$$

would in the long run establish

$$\text{Prob. } AB = \alpha\beta;$$

if it does not, *cadit quæstio*, the causes are not independent.

3rdly. I assume not only

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta, \quad \text{Prob. } AB = \alpha\beta,$$

but also as 1st above stated; and I consider that, inasmuch as the result of the investigation is to show that the conditions $q - \alpha p \geq 0$, $p - \beta q \leq 0$ are necessary and sufficient conditions, it is also a result of the investigation that these are the conditions of consistency among the data, viz. the conditions in order that the data may be consistent with the above assumptions as to the independence of the causes. It is clear that since, as just stated, I do assume something beyond the last-mentioned three equations, the conditions of consistency ought to be narrower than those in Prof. Boole's 3rdly.

4thly. I had not overlooked, but I ought to have stated, that Prof. Boole's conditions were actually the conditions of consistency in the data.

5thly. I contend that the conception of A and B as causes does alter the nature of the problem. For when A and B are conceived of as causes, there is a definite notion of the efficient or inefficient action of A or B; and in particular when they both act, one of them, say A, may act inefficiently. But according to the concomitance statement, then either there is no such notion as that of the efficient or non-efficient happening of A or B (I believe this to be so), or else the only notion of efficient or inefficient happening is happening in concomitance or in non-concomitance with E; but in this view, if A, B, E all happen, then A and B each of them happens efficiently. The argument is to me conclusive as to the diversity of the two problems.

6thly. I do not in anywise assert, or even suppose, that the ideal problem is arbitrary, or that its connexion with the real problem is arbitrary. I simply do not know what the ideal problem is; I do not know the point of view, or the assumed mental state of knowledge or ignorance according to which x, y, s, t are the probabilities of A, B, AE, BE. It is to be borne in mind that x, y, s, t are, in Prof. Boole's solution, determined as numerical quantities included between the limits 0 and 1, i.e. as quantities which are or may be actual probabilities. What I desiderate is, that Prof. Boole should give for his auxiliary quantities x, y, s, t such an explanation of the meaning as I have given for my auxiliary quantities λ, μ. I do not find any such explanation in the memoir referred to.

7thly and 8thly. No remark is necessary.

March 29, 1862.

Prof. Boole, in his reply, dated April 2, writes, "No such explanation as you desiderate of the interpretation of the auxiliary quantities in my method of solution is possible; because they are not of the nature of additional data, and their introduction does not limit the problem as any hypotheses which are of that nature do. I do not see any difficulty whatever in the conception of the ideal problem."

We thus join issue as follows: Prof. Boole says that there is no difficulty in understanding, I say that I do not understand, the rationale of his solution.

It may be remarked that the question may be, not to find any actual probability whatever, but only to find a Boolian probability or probabilities. Thus the equations (L), p. 356, omitting the last member, which alone involves u, determine in terms of the data α, β, αp, βq the Boolian probabilities x, y, s, t of the events A, B, AE, BE.

In a subsequent hastily-written letter, Prof. Boole gives an explanation of the equations (L), which appears to me little more than a translation of these equations into ordinary language.

April 16, 1862.

319. Postscript to the Paper "On a Question in the Theory of Probabilities."

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 470–471.]

See 318. The present paper reproduces Wilbraham's discussion of Boole's Solution; and concludes with the remark "Professor Boole wishes me to mention that he has succeeded in obtaining a demonstration of the analytical theorem arising from his theory referred to in his Reply in my paper." See ante p. 82, 7thly.

320. On the Transcendent $\operatorname{gd} u = \frac{1}{i}\log\tan(\frac{1}{4}\pi + \frac{1}{2}iu)$.

[From the *Philosophical Magazine*, vol. xxiv. (1862), pp. 19–21.]

The elliptic functions which correspond to the modulus $k = 1$ reduce themselves, as is well known, to circular functions. The case in question plays implicitly an important part in the general theory, and it has been particularly studied by Gudermann, and by Dr Booth in connexion with his theory of parabolic logarithms. But in the absence of a notation corresponding to that used for elliptic functions in general, the theory has not, it appears to me, been exhibited in its proper form. The defect is very easily supplied: using for $\operatorname{am} u$, to the modulus 1, the notation $\operatorname{gd} u$ (Gudermannian of u), then if

$$u = \int_0^\phi \frac{d\phi}{\cos\phi} = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right),$$

we have

$$\phi = \operatorname{gd} u;$$

and hence, observing that the equation between u and ϕ is

$$u = \log\frac{1+\tan\frac{1}{2}\phi}{1-\tan\frac{1}{2}\phi},$$

or, what is the same thing,

$$\tan\tfrac{1}{2}\phi = \frac{e^u - 1}{e^u + 1};$$

and that we have

$$\begin{aligned}
\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}iu\right) &= \log\frac{1+\tan\frac{1}{2}iu}{1-\tan\frac{1}{2}iu} \\
&= \log\frac{\cos\frac{1}{2}iu + \sin\frac{1}{2}iu}{\cos\frac{1}{2}iu - \sin\frac{1}{2}iu} \\
&= \log\frac{e^{iu}+e^{-iu}+i(e^{iu}-e^{-iu})}{e^{iu}+e^{-iu}-i(e^{iu}-e^{-iu})} \\
&= \log\frac{e^{iu}+1+i(e^{iu}-1)}{e^{iu}+1-i(e^{iu}-1)} \\
&= \log\frac{1+i\tan\frac{1}{2}\phi}{1-i\tan\frac{1}{2}\phi} = \log e^{i\phi} = i\phi,
\end{aligned}$$

or if

$$u = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right),$$

then

$$\phi = \tfrac{1}{i}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}ui\right);$$

and substituting for ϕ its value, we obtain

$$\operatorname{gd} u = \tfrac{1}{i}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}ui\right),$$

which is the definition of the transcendent $\operatorname{gd} u$. It is to be noticed that $\operatorname{gd} u$, although exhibited in an imaginary form, is a real function of u; and, moreover, that it is an odd function, viz. we have

$$\operatorname{gd}(-u) = -\operatorname{gd}(u),$$

and therefore also

$$\operatorname{gd}(0) = 0.$$

The original equation,

$$u = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right),$$

written under the form

$$u = \tfrac{1}{i}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\left(\tfrac{\phi}{i}\right)i\right),$$

shows that we have

$$u = \tfrac{1}{i}\,\mathrm{gd}\left(\tfrac{\phi}{i}\right) = \tfrac{1}{i}\,\mathrm{gd}(-i\phi);$$

or substituting for ϕ its value $\mathrm{gd}\,u$, we have

$$u = \tfrac{1}{i}\,\mathrm{gd}(-i\,\mathrm{gd}\,u),$$

which may also be written

$$iu = \mathrm{gd}(i\,\mathrm{gd}\,u);$$

so that $\mathrm{gd}\,u$ is a quasi-periodic function of the second order—a property which has not, at least obviously, any analogue in the general theory. We have

$$\begin{aligned}
\cos\,\mathrm{gd}\,u &= \tfrac{1}{2}(e^{i\,\mathrm{gd}\,u} + e^{-i\,\mathrm{gd}\,u}) \\
&= \tfrac{1}{2}\left(\sqrt{\frac{1+\tan\frac{1}{2}ui}{1-\tan\frac{1}{2}ui}} + \sqrt{\frac{1-\tan\frac{1}{2}ui}{1+\tan\frac{1}{2}ui}}\right) \\
&= \frac{1+\tan^2\frac{1}{2}ui}{1-\tan^2\frac{1}{2}ui} = \frac{1}{\cos ui};
\end{aligned}$$

and in like manner

$$\begin{aligned}
\sin\,\mathrm{gd}\,u &= \tfrac{1}{2i}(e^{i\,\mathrm{gd}\,u} - e^{-i\,\mathrm{gd}\,u}) \\
&= \tfrac{1}{2i}\left(\sqrt{\frac{1+\tan\frac{1}{2}ui}{1-\tan\frac{1}{2}ui}} - \sqrt{\frac{1-\tan\frac{1}{2}ui}{1+\tan\frac{1}{2}ui}}\right) \\
&= \frac{2\tan\frac{1}{2}ui}{i(1-\tan^2\frac{1}{2}ui)} = \frac{\sin ui}{i\cos ui};
\end{aligned}$$

or, as these equations may also be written,

$$\sec\,\mathrm{gd}\,u = \cos ui = \tfrac{1}{2}(e^u + e^{-u}),$$

$$\tan\,\mathrm{gd}\,u = \tfrac{1}{i}\sin ui = \tfrac{1}{2}(e^u - e^{-u});$$

and from these equations we have

$$\begin{aligned}
\sec\,\mathrm{gd}(u+v) &= \sec\,\mathrm{gd}\,u\cdot\sec\,\mathrm{gd}\,v + \tan\,\mathrm{gd}\,u\cdot\tan\,\mathrm{gd}\,v, \\
\tan\,\mathrm{gd}(u+v) &= \tan\,\mathrm{gd}\,u\cdot\sec\,\mathrm{gd}\,v + \tan\,\mathrm{gd}\,v\cdot\sec\,\mathrm{gd}\,u;
\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
\sin\,\mathrm{gd}(u+v) &= i\cdot\frac{\sin\,\mathrm{gd}\,u + \sin\,\mathrm{gd}\,v}{1+\sin\,\mathrm{gd}\,u\cdot\sin\,\mathrm{gd}\,v}, \\
\cos\,\mathrm{gd}(u+v) &= \frac{\cos\,\mathrm{gd}\,u\cdot\cos\,\mathrm{gd}\,v}{1+\sin\,\mathrm{gd}\,u\cdot\sin\,\mathrm{gd}\,v};
\end{aligned}$$

which forms are at once obtainable from the formulae

$$\sin\,\mathrm{am}(u+v) = \frac{\sin\,\mathrm{am}\,u\,\cos\,\mathrm{am}\,v\,\Delta\,\mathrm{am}\,v + \sin\,\mathrm{am}\,v\,\cos\,\mathrm{am}\,u\,\Delta\,\mathrm{am}\,u}{1 - k^2\sin^2\mathrm{am}\,u\,\sin^2\mathrm{am}\,v},$$

$$\cos\,\mathrm{am}(u+v) = \frac{\cos\,\mathrm{am}\,u\,\cos\,\mathrm{am}\,v - \sin\,\mathrm{am}\,u\,\sin\,\mathrm{am}\,v\,\Delta\,\mathrm{am}\,u\,\Delta\,\mathrm{am}\,v}{1 - k^2\sin^2\mathrm{am}\,u\,\sin^2\mathrm{am}\,v},$$

$$\Delta\,\mathrm{am}(u+v) = \frac{\Delta\,\mathrm{am}\,u\,\Delta\,\mathrm{am}\,v - k^2\sin\,\mathrm{am}\,u\,\sin\,\mathrm{am}\,v\,\cos\,\mathrm{am}\,u\,\cos\,\mathrm{am}\,v}{1 - k^2\sin^2\mathrm{am}\,u\,\sin^2\mathrm{am}\,v},$$

observing that for $k = 1$ we have $\Delta\,\mathrm{am} = \cos\,\mathrm{am}$, and that the numerators and denominator each of them divide by

$$1 - \sin\,\mathrm{am}\,u\,\sin\,\mathrm{am}\,v.$$

2, Stone Buildings, W.C., May 7, 1862.

321. Final Remarks on Mr Jerrard's Theory of Equations of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxiv. (1862), p. 290.]

C. V.

322. On the Skew Surface of the Third Order.

[From the *Philosophical Magazine*, vol. xxiv. (1862), pp. 514–519.]

The skew surface of the third order, or "cubic scroll" (disregarding a certain special form), may be considered as generated by a line which always passes through three directrices; viz., a plane cubic having a node, and two lines, one of them meeting the cubic in the node, the other of them meeting the cubic in an ordinary point. The analytical investigation possesses some interest as an illustration of the analytical theory of skew surfaces in general.

Take for the equation of the cubic

$$(x^2+y^2)\alpha\beta-(\alpha^2+\beta^2)xy=0,$$

which belongs to a cubic having a node at the origin, and passing through the point (α,β); and for the equations of the two lines

$$(x-mz=0,\quad y-nz=0),\qquad (x-\alpha=0,\quad y-\beta=0).$$

Then, (X,Y,Z) being current coordinates, the equations of the generating line will be

$$X=x+AZ,\quad Y=y+BZ;$$

when this meets the line $(X-mZ=0, Y-nZ=0)$, we have

$$mZ=x+AZ,\quad nZ=y+BZ,$$

and thence

$$x(n-B)-y(m-A)=0;$$

or, what is the same thing,

$$nx-my-Bx+Ay=0:$$

and when it meets the line $(X-\alpha=0, Y-\beta=0)$, we have

$$\alpha=x+AZ,\quad \beta=y+BZ,$$

and thence

$$B(x-\alpha)-A(y-\beta)=0.$$

We have thus the system of equations

$$\begin{aligned}(x^2+y^2)\alpha\beta-(\alpha^2+\beta^2)xy&=0,\\ X&=x+AZ,\\ Y&=y+BZ,\\ nx-my-Bx+Ay&=0,\\ B(x-\alpha)-A(y-\beta)&=0;\end{aligned}$$

from which, eliminating (A,B,x,y), we obtain the equation of the surface.

Writing in the last equation

$$B=s(x-\alpha),\quad A=s(y-\beta)$$

(values which give $Bx-Ay=-s(\beta x-\alpha y)$), we find

$$\begin{aligned}X+\alpha sZ&=(1+sZ)x,\\ Y+\beta sZ&=(1+sZ)y,\\ (n+\beta s)x-(m+\alpha s)y&=0;\end{aligned}$$

whence also

$$(n+\beta s)(X+\alpha sZ)-(m+\alpha s)(Y+\beta sZ)=0,$$

that is

$$nX-mY+(n\alpha-m\beta)sZ+s(\beta X-\alpha Y)=0;$$

or eliminating s from this equation and the two equations

$$\begin{aligned}x-X+Z(x-\alpha)s&=0,\\ y-Y+Z(y-\beta)s&=0,\end{aligned}$$

we have

$$\begin{aligned}\{(n\alpha-m\beta)Z+\beta X-\alpha Y\}(x-X)-Z(x-\alpha)(nX-mY)&=0,\\ \{(n\alpha-m\beta)Z+\beta X-\alpha Y\}(y-Y)-Z(y-\beta)(nX-mY)&=0;\end{aligned}$$

these give

$$\begin{aligned}\Omega x&=X\{(n\alpha-m\beta)Z+\beta X-\alpha Y\}-\alpha Z(nX-mY)\\ &=-mZ\beta X+X(\beta X-\alpha Y)+mZ\alpha Y\\ &=(X-mZ)(\beta X-\alpha Y),\end{aligned}$$

and

$$\begin{aligned}\Omega y&=Y\{(n\alpha-m\beta)Z+\beta X-\alpha Y\}-\beta Z(nX-mY)\\ &=nZ\alpha Y+Y(\beta X-\alpha Y)-nZ\beta X\\ &=(Y-nZ)(\beta X-\alpha Y),\end{aligned}$$

where

$$\begin{aligned}\Omega&=(n\alpha-m\beta)Z+(\beta X-\alpha Y)-Z(nX-mY)\\ &=\beta(X-mZ)-\alpha(Y-nZ)-Z\{n(X-mZ)-m(Y-nZ)\}\\ &=(\beta-nZ)(X-mZ)-(\alpha-mZ)(Y-nZ);\end{aligned}$$

so that

$$\begin{aligned}x&=\frac{(X-mZ)(\beta X-\alpha Y)}{(\beta-nZ)(X-mZ)-(\alpha-mZ)(Y-nZ)},\\ y&=\frac{(Y-nZ)(\beta X-\alpha Y)}{(\beta-nZ)(X-mZ)-(\alpha-mZ)(Y-nZ)},\end{aligned}$$

which equations give the coordinates (x,y) of the point in which the generating line through the point (X,Y,Z) of the surface meets the cubic

$$(x^2+y^2)\alpha\beta-(\alpha^2+\beta^2)xy=0.$$

Substituting these values of (x,y) in the equation of the cubic, we obtain the equation

$$\begin{aligned}&(\alpha^2+\beta^2)(X-mZ)(Y-nZ)\{(\beta-nZ)(X-mZ)-(\alpha-mZ)(Y-nZ)\}\\ &\quad-\alpha\beta(\beta X-\alpha Y)\{(X-mZ)^2+(Y-nZ)^2\}=0;\end{aligned}$$

or, as it may be written,

$$\begin{aligned}&(\alpha^2+\beta^2)(X-mZ)(Y-nZ)\{\beta(X-mZ)-\alpha(Y-nZ)\}\\ &\quad+(\alpha^2+\beta^2)(X-mZ)(Y-nZ)Z(mY-nX)\\ &\quad-\alpha\beta(\beta X-\alpha Y)\{(X-mZ)^2+(Y-nZ)^2\}=0.\end{aligned}$$

This equation contains, however, the extraneous factor

$$\beta(X - mZ) - \alpha(Y - nZ),$$

which, equated to zero, gives the equation of the plane through the node and the line $(x - mz = 0, y - nz = 0)$. In fact, assuming

$$\begin{aligned}(\alpha^2 + \beta^2)(X - mZ)(Y - nZ)Z(mY - nX) - \alpha\beta(\beta X - \alpha Y)\{(X - mZ)^2 + (Y - nZ)^2\}\\ = \{\beta(X - mZ) - \alpha(Y - nZ)\}\,\Phi(X, Y, Z),\end{aligned}$$

it will presently be shown that Φ is an integral function. Hence, omitting the factor in question, we have

$$(\alpha^2 + \beta^2)(X - mZ)(Y - nZ) + \Phi(X, Y, Z) = 0,$$

which is the equation of the surface. It only remains to find Φ: writing for this purpose $X + mZ$, $Y + nZ$ in the place of X, Y respectively, and putting for a moment

$$\Phi(X + mZ, Y + nZ, Z) = \Phi',$$

we have

$$(\alpha^2 + \beta^2)XYZ(mY - nX) - \alpha\beta\{\beta(X + mZ) - \alpha(Y + nZ)\}(X^2 + Y^2) = (\beta X - \alpha Y)\Phi';$$

that is

$$(\beta X - \alpha Y)\Phi' = Z\{(\alpha^2 + \beta^2)XY(mY - nX) - (X^2 + Y^2)\alpha\beta(m\beta - n\alpha)\} - (\beta X - \alpha Y)\alpha\beta(X^2 + Y^2);$$

or, effecting the division,

$$\Phi' = Z\{(X^2\alpha - Y^2\beta)(\alpha n - \beta m) - XY(\alpha^2 m + \beta^2 n)\} - \alpha\beta(X^2 + Y^2),$$

and then writing $X - mZ$, $Y - nZ$ in the place of X, Y respectively, we have

$$\begin{aligned}\Phi(X, Y, Z) = Z\{((X - mZ)^2\alpha - (Y - nZ)^2\beta)(\alpha n - \beta m)\\ - (X - mZ)(Y - nZ)(\alpha^2 m + \beta^2 n)\} - \alpha\beta\{(X - mZ)^2 + (Y - nZ)^2\}.\end{aligned}$$

Hence, finally, the equation of the surface is

$$\begin{aligned}&(\alpha^2 + \beta^2)(X - mZ)(Y - nZ) - \alpha\beta\{(X - mZ)^2 + (Y - nZ)^2\}\\ &\quad + Z\{((X - mZ)^2\alpha - (Y - nZ)^2\beta)(\alpha n - \beta m)\\ &\quad - (X - mZ)(Y - nZ)(\alpha^2 m + \beta^2 n)\} = 0,\end{aligned}$$

which is, as it should be, of the third order.

Arranging in powers of Z and reducing, the equation is found to be

$$\begin{aligned}&(\alpha^2 + \beta^2)XY - \alpha\beta(X^2 + Y^2)\\ &\quad + Z\{-(\alpha^2 + \beta^2)(mY + nX) + (X^2\alpha + Y^2\beta)(m\beta + n\alpha) + \alpha\beta(mX^2 + nY^2) - (\alpha^2 m + \beta^2 n)XY\}\\ &\quad + Z^2\{mn(\alpha^2 + \beta^2 - \alpha^2 X - \beta^2 Y) + (\beta n^2 - \alpha m^2)(\beta X - \alpha Y)\} = 0.\end{aligned}$$

The first form puts in evidence the nodal line

$$(X - mZ = 0, \quad Y - nZ = 0),$$

and the second form puts in evidence the simple line

$$(X - \alpha = 0, \quad Y - \beta = 0).$$

But to obtain a more convenient form, write for a moment $X - mZ = P$, $Y - nZ = Q$; the equation is

$$(\alpha^2+\beta^2)PQ - \alpha\beta(P^2+Q^2) + Z\{(P^2\alpha - Q^2\beta)(n\alpha - m\beta) - PQ(m\alpha^2+n\beta^2)\} = 0,$$

or, as this may be written,

$$(\alpha^2+\beta^2)PQ + (\alpha^2P - \beta^2Q)Z(Pn - Qm) + \alpha\beta\{-P^2 - Q^2 - Z(mP^2+nQ^2)\} = 0;$$

or, observing that $X = P + mZ$, $Y = Q + nZ$, and thence

$$PY - QX = Z(Pn - Qm), \quad P^2X + Q^2Y = P^3 + Q^3 + Z(mP^2 + nQ^2),$$

the equation becomes

$$(\alpha^2+\beta^2)PQ + (\alpha^2P - \beta^2Q)(PY - QX) - \alpha\beta(P^2X + Q^2Y) = 0,$$

or, what is the same thing,

$$(\alpha P^2 - \beta Q^2)(\alpha Y - \beta X) + PQ(\alpha^2 + \beta^2 - \alpha^2X - \beta^2Y) = 0;$$

whence, making a slight change in the form, and restoring for P, Q their values, the equation is

$$\begin{gathered}\{\alpha(X-mZ)^2 - \beta(Y-nZ)^2\}\{\alpha(Y-\beta) - \beta(X-\alpha)\} \\ -(X-mZ)(Y-nZ)\{\alpha^2(X-\alpha) + \beta^2(Y-\beta)\} = 0,\end{gathered}$$

a form which puts in evidence as well the simple line $(X-\alpha=0, Y-\beta=0)$ as the nodal line $(X-mZ=0, Y-nZ=0)$.

If $Z = 0$, we have

$$(\alpha X^2 - \beta Y^2)(\alpha Y - \beta X) - XY\{\alpha^2(X-\alpha) + \beta^2(Y-\beta)\} = 0,$$

which is in fact the cubic curve $(\alpha^2+\beta^2)XY - \alpha\beta(X^2+Y^2) = 0$.

Reverting to a former system of equations

$$\begin{aligned} nx - my - Bx + Ay &= 0, \\ B(x-\alpha) - A(y-\beta) &= 0, \end{aligned}$$

or, as these may be written,

$$\begin{aligned} Bx - Ay &= nx - my, \\ B\alpha - A\beta &= nx - my, \end{aligned}$$

we find

$$\begin{aligned} B(\beta x - \alpha y) &= (\beta - y)(nx - my), \\ A(\beta x - \alpha y) &= (\alpha - x)(nx - my); \end{aligned}$$

so that we have

$$Y - y = \frac{(\beta-y)(nx-my)}{\beta x - \alpha y}Z, \quad X - x = \frac{(\alpha-x)(nx-my)}{\beta x - \alpha y}Z,$$

as the equations of the generating line which passes through the point (x, y) of the cubic curve.

2, Stone Buildings, W.C., October 28, 1862.

323. On a Tactical Theorem Relating to the Triads of Fifteen Things.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 59–61.]

The school-girl problem may be stated as follows:—"With 15 things to form 35 triads, involving all the 105 duads, and such that they can be divided into 7 systems, each of 5 triads containing all the 15 things." A more simple problem is, "With 15 things, to form 35 triads involving all the 105 duads."

In the solution which I formerly gave of the school-girl problem (Phil. Mag. vol. xxxvii. 1850, [82]), and which may be presented in the form

	a	b	c	d	e	f	g
abc				35	17	82	64
ade		62	84			15	37
afg		13	47	86	42		25
bdf	47		16		38		25
beg	58		23	14		56	67
cdg	12	78			56	34	
cef	36	45		27			18

(viz. the things being a, b, c, d, e, f, g, 1, 2, 3, 4, 5, 6, 7, 8, the first pentad of triads is abc, $d35$, $e17$, $f82$, $g64$, and so for all the seven pentads of triads), there is obviously a division of the 15 things into $(7+8)$ things, viz. the 35 triads are composed 7 of them each of 3 out of the 7 things, and the remaining 28 each of 1 out of the 7 things, and 2 out of the 8 things: or attending only to the 8 things, there are 28 triads each of them containing a duad of the 8 things, but there is no triad consisting of 3 of the 8 things. More briefly, we may say that in the system there is an 8 without 3, that is, there are 8 things such that no triad of them occurs in the system.

I believe, but am not sure, that in all the solutions which have been given of the school-girl problem there is an 8 without 3.

Now, considering the more simple problem, there are of course solutions which have an 8 without 3 (since every solution of the school-girl problem is a solution of the more simple problem): but it is very easy to show that there is no solution which has a 9 without 3. I wish to show that there is in every solution at least a 6 without 3. This being so, there will be (if they all exist) 3 classes of solutions, viz. those which have at most (1) a 6 without 3, (2) a 7 without 3, (3) an 8 without 3. I believe that the first and second classes exist, as well as the third, which is known to do so.

The proposition to be proved is, that given any system of 35 triads involving all the duads of 15 things; there are always 6 things which are a 6 without 3, that is, they are such that no triad of the 6 things is a triad of the system. This will be the case if it is shown that the number of distinct hexads which can be formed each of them containing a triad of the system is less than

$$\frac{15\cdot 14\cdot 13\cdot 12\cdot 11\cdot 10}{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6} = 5\cdot 7\cdot 11\cdot 13 = 5005,$$

the entire number of the hexads of 15 things. Now joining to any triad of the system a triad formed out of the remaining 12 things (there are $\frac{12\cdot 11\cdot 10}{1\cdot 2\cdot 3} = 4\cdot 5\cdot 11 = 220$ such triads), we obtain in all $(220\times 35 =)$ 7700 hexads, each of them containing a triad of the system. But these 7700 hexads are not all of them distinct. For, first, considering any triad of the system, there are in the system 16 other triads, each of them having no thing in common with the first-mentioned triad. (In fact if e.g. 123 is a triad of the system, then the system, since it contains all the duads, must have besides 6 triads containing 1, 6 triads containing 2, and 6 triads containing 3, and therefore $35-1-6-6-6=16$ triads not containing 1, 2, or 3.) Hence we have $\frac{16\times 35}{2} = 280$ hexads, each of them composed of two triads of the system; and since each of these hexads can be derived from either of its two component triads, these 280 hexads present themselves twice over among the 7700 hexads.

Secondly, there are in the system seven triads containing each of them the same one thing, e.g.

$$123,\ 145,\ 167,\ 189,\ 1.10.11,\ 1.12.13,\ 1.14.15,$$

and therefore $\frac{7\cdot 6}{2} = 21$ pairs such as 123, 145 containing the thing 1, and therefore $(15 \times 21 =)\, 315$ pairs such as $\alpha\beta\gamma$, $\alpha\delta\epsilon$.

And for any such pair, combining with $\alpha\beta\gamma\delta\epsilon$ any one of the remaining 10 things, we have 10 hexads $\alpha\beta\gamma\delta\epsilon\zeta$, each of them derivable from either of the triads $\alpha\beta\gamma$, $\alpha\delta\epsilon$, that is, we have $(315 \times 10 =)\,3150$ hexads which present themselves twice over among the 7700 hexads. The hexads not belonging to one or other of the foregoing classes are derived each of them from a single triad only of the system, and they present themselves once among the 7700 hexads. This number is consequently made up as follows, viz.

$$\begin{array}{rl} 280 \text{ hexads each twice} & = 560 \\ 3150 \text{ hexads each twice} & = 6300 \\ 840 \text{ hexads each once} & = \underline{840} \\ 4270 \text{ hexads each} & \ 7700 \end{array}$$

or there are in all 4270 distinct hexads; and since this is less than 5005, it follows that there are hexads not containing any triad of the system: there must in fact be $(5005 - 4270 =)\,735$ such hexads. The theorem in question is thus shown to be true.

2, Stone Buildings, W.C., November 24, 1862.

C. V.

324. Note on a Theorem Relating to Surfaces.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 61, 62.]

The following apparently self-evident geometrical theorem requires, I think, a proof; viz. the theorem is—"If every plane section of a surface of the order $m+n$ break up into two curves of the orders m and n respectively, then the surface breaks up into two surfaces of the orders m, n respectively."

To fix the ideas, suppose $n = 2$. Imagine any line meeting the surface in $m+2$ points, the section includes a conic which meets the line in two of the $m+2$ points, say the points A, A'[(1)]. Suppose that the plane revolves round the line AA', the section will always include a conic which passes through these same two points A, A'; and it is to be shown that the sheet, the locus of this conic, is a surface of the second order. In fact the conic in question, say APA', by its intersection with an arbitrary plane traces out a branch of the intersection of the given surface with the arbitrary plane. And if $ABA'B'$ be the conic in any particular plane through A, A', and if the arbitrary plane meet this conic in the points B, B', then the branch passes through these points B, B'. Imagine the plane $ABA'B'$ revolving round BB' until it coincides with the arbitrary plane; the section includes a conic passing through the points B, B', and the before-mentioned branch is this conic; that is, the conic APA' by its intersection with an arbitrary plane traces out a conic; or, what is the same thing, the sheet, the locus of the conic APA', is met by an arbitrary plane in a conic, that is, the sheet is a surface of the second order; and the given surface thus includes a surface of the second order, and is therefore made up of two surfaces of the orders m and 2 respectively. The demonstration seems to me to add at least something to the evidence of the theorem asserted, but I should be glad if a more simple one could be found. Analytically, the theorem is—"If

$$(x, y, z, \alpha x + \beta y + \gamma z)^{m+n},$$

where (α, β, γ) are arbitrary, break up into factors $(x, y, z)^m$, $(x, y, z)^n$, rational in regard to (x, y, z), then $(x, y, z, w)^{m+n}$ breaks up into factors $(x, y, z, w)^m$, $(x, y, z, w)^n$, rational in regard to (x, y, z, w)." It would at first sight appear that (α, β, γ) being arbitrary, these quantities can only enter into the factors of $(x, y, z, \alpha x + \beta y + \gamma z)^{m+n}$ through the quantity $\alpha x + \beta y + \gamma z$; that is, that the factors in question, considered as functions of $(x, y, z, \alpha, \beta, \gamma)$, are of the form

$$(x, y, z, \alpha x + \beta y + \gamma z)^m, \quad (x, y, z, \alpha x + \beta y + \gamma z)^n;$$

and then replacing the arbitrary quantity $\alpha x + \beta y + \gamma z$ by w, the factors of $(x, y, z, w)^{m+n}$ will be $(x, y, z, w)^m$, $(x, y, z, w)^n$. But the objection proves too much; for in a similar way it would follow that if $(x, y, \alpha x + \beta y)^{m+n}$, where α, β are arbitrary, breaks up into the factors $(x, y)^m$, $(x, y)^n$, rational in regard to (x, y) (and *qua* homogeneous function of two variables it always does so break up), then $(x, y, z)^{m+n}$ would in like manner break up into the factors $(x, y, z)^m$, $(x, y, z)^n$, rational in regard to (x, y, z): and a simple example will show that it is not true that the factors of $(x, y, \alpha x + \beta y)^{m+n}$ only contain (α, β) through $\alpha x + \beta y$; in fact, if the function be $= x^2 + y^2 + (\alpha x + \beta y)^2$, then the factor is

$$\frac{1}{\sqrt{\alpha^2 + \beta^2 + 1}}\{(\alpha^2 + 1)x + (\alpha\beta + \beta\sqrt{\alpha^2 + \beta^2 + 1})y\},$$

which cannot be exhibited as a function of α, β, $\alpha x + \beta y$.

I am not acquainted with any analytical demonstration; the geometrical one cannot easily be exhibited in an analytical form.

2, Stone Buildings, W.C., November 26, 1862.

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(1) The figure referred to will be at once understood by considering A, A' as the poles of an ellipsoid, or say of a sphere, $ABA'B'$ the meridian of projection, APA' any other meridian, BPB' the equator or any other great circle.

325. Note on a Theorem Relating to a Triangle, Line, and Conic.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 181–183.]

I FIND, among my papers headed "Generalization of a Theorem of Steiner's," an investigation leading to the following theorem, viz.:

Consider a triangle, a line, and a conic; with each vertex of the triangle join the point of intersection of the line with the polar of the same vertex in regard to the conic; in order that the three joining lines may meet in a point, the line must be a tangent to a curve of the third class; if, however, the conic break up into a pair of lines, or in a certain other case, the curve of the third class will break up into a point, and a conic inscribed in the triangle.

Let the equations of the sides of the triangle be

$$x = 0, \quad y = 0, \quad z = 0,$$

the equation of the conic

$$(a, b, c, f, g, h \,|\, x, y, z)^2 = 0,$$

and that of the line

$$\lambda x + \mu y + \nu z = 0;$$

then the polar of the vertex $(y = 0, z = 0)$ has for its equation

$$ax + hy + gz = 0;$$

it therefore meets the line $\lambda x + \mu y + \nu z = 0$ in the point

$$x : y : z = h\nu - g\mu : g\lambda - a\nu : a\mu - h\lambda.$$

[Continues on p. 101.]

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325. Note on a Theorem Relating to a Triangle, Line, and Conic *(continued)*

[Continuation from p. 100.]

and the equation of the line joining this point with the vertex $(y = 0,\ z = 0)$ is $(a\mu - h\lambda)\,y = (g\lambda - a\nu)\,z$. The equations of the three joining lines therefore are

$$\begin{aligned}(a\mu - h\lambda)\,y &= (g\lambda - a\nu)\,z,\\ (b\nu - f\mu)\,z &= (h\mu - b\lambda)\,x,\\ (c\lambda - g\nu)\,x &= (f\nu - c\mu)\,y,\end{aligned}$$

lines which will meet in a point if

$$(a\mu - h\lambda)(b\nu - f\mu)(c\lambda - g\nu) - (g\lambda - a\nu)(h\mu - b\lambda)(f\nu - c\mu) = 0,$$

or, multiplying out and putting as usual

$$\begin{aligned}K &= abc - af^2 - bg^2 - ch^2 + 2fgh,\\ \mathfrak{A} &= bc - f^2, \quad \&c.,\end{aligned}$$

if

$$2(abc - fgh)\,\lambda\mu\nu + a\mathfrak{A}\,\mu\nu^2 + a\mathfrak{A}\,\mu^2\nu + b\mathfrak{B}\,\nu\lambda^2 + b\mathfrak{B}\,\nu^2\lambda + c\mathfrak{C}\,\lambda\mu^2 + c\mathfrak{C}\,\lambda^2\mu = 0,$$

that is, the line must touch a curve of the third class.

If this equation break up into factors, the form must be

$$(\alpha\lambda + \beta\mu + \gamma\nu)(A\mu\nu + B\nu\lambda + C\lambda\mu) = 0;$$

that is, we must have

$$\begin{aligned}A\alpha + B\beta + C\gamma &= 2(abc - fgh),\\ B\alpha = b\mathfrak{B}, &\quad C\alpha = c\mathfrak{C},\\ C\beta = c\mathfrak{C}, &\quad A\beta = a\mathfrak{A},\\ A\gamma = a\mathfrak{A}, &\quad B\gamma = b\mathfrak{B};\end{aligned}$$

and the last six equations give without difficulty

$$A = \frac{ka}{\mathfrak{A}}, \quad B = \frac{kb}{\mathfrak{B}}, \quad C = \frac{kc}{\mathfrak{C}}, \quad \alpha = \frac{1}{k}\mathfrak{B}\mathfrak{C}, \quad \beta = \frac{1}{k}\mathfrak{C}\mathfrak{A}, \quad \gamma = \frac{1}{k}\mathfrak{A}\mathfrak{B},$$

where k is arbitrary; the first equation then gives

$$\frac{a\mathfrak{B}\mathfrak{C}}{\mathfrak{A}} + \frac{b\mathfrak{C}\mathfrak{A}}{\mathfrak{B}} + \frac{c\mathfrak{A}\mathfrak{B}}{\mathfrak{C}} = 2(abc - fgh);$$

or, reducing by the equations $\mathfrak{BC} = a\mathfrak{A} + gh$, &c., this is

$$2\mathfrak{A}a + \mathfrak{B}b + \mathfrak{C}c - 2abc + 2fgh + (1+1+1)\frac{K}{1} = 0;$$

which, substituting for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ their values, becomes

$$K\left(1 + \frac{a^2}{\mathfrak{A}} + \frac{b^2}{\mathfrak{B}} + \frac{c^2}{\mathfrak{C}}\right) = 0.$$

Hence if $K = 0$, that is, if the conic break up into a pair of lines, or if

$$1 + \frac{a^2}{\mathfrak{A}} + \frac{b^2}{\mathfrak{B}} + \frac{c^2}{\mathfrak{C}} = 0,$$

in either case the equation of the curve of the third class becomes

$$\left(\frac{\lambda}{\mathfrak{A}} + \frac{\mu}{\mathfrak{B}} + \frac{\nu}{\mathfrak{C}}\right)\left(\frac{a}{\mathfrak{A}}\mu\nu + \frac{b}{\mathfrak{B}}\nu\lambda + \frac{c}{\mathfrak{C}}\lambda\mu\right) = 0;$$

that is, the curve breaks up into a point, and a conic inscribed in the triangle.

In the case where the conic breaks up into a pair of lines, then we have

$$(a,\, b,\, c,\, f,\, g,\, h\,|\, x,\, y,\, z)^2 = 2(px + qy + rz)(p'x + q'y + r'z),$$

and thence

$$(\mathfrak{A},\, \mathfrak{B},\, \mathfrak{C},\, \mathfrak{F},\, \mathfrak{G},\, \mathfrak{H}\,|\, x,\, y,\, z)^2 = -\{(qr' - q'r)x + (rp' - r'p)y + (pq' - p'q)z\}^2;$$

so that the equation in (λ, μ, ν) is

$$\{(qr'-q'r)\lambda+(rp'-r'p)\mu+(pq'-p'q)\nu\}\{pp'(qr'-q'r)\mu\nu+qq'(rp'-r'p)\nu\lambda+rr'(pq'-p'q)\lambda\mu\} = 0;$$

where the point represented by the equation

$$(qr' - q'r)\lambda + (rp' - r'p)\mu + (pq' - p'q)\nu = 0$$

is, of course, the intersection of the two lines.

326. Theorems Relating to the Canonic Roots of a Binary Quantic of an Odd Order.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 206–208.]

I CALL to mind Professor Sylvester's theory of the canonical form of a binary quantic of an odd order; viz., the quantic of the order $2n+1$ may be expressed as a sum of a number $n+1$ of $(2n+1)$th powers, the roots of which, or say the *canonic roots* of the quantic, are to constant multipliers *près* the factors of a certain covariant derivative of the order $(n+1)$, called the *Canonizant*. If, to fix the ideas, we take a quintic function, then we may write

$$(a,\, b,\, c,\, d,\, e,\, f \,|\, x,\, y)^5 = A\,(lx+my)^5 + A'\,(l'x+m'y)^5 + A''\,(l''x+m''y)^5$$

(it would be allowable to put the coefficients A each equal to unity; but there is a convenience in retaining them, and in considering that a canonic root $lx+my$ is only given as regards the ratio $l:m$, the coefficients l, m remaining indeterminate); and then the canonic roots $(lx+my)$, &c. are the factors of the Canonizant

$$\begin{vmatrix} y^3, & -y^2x, & yx^2, & -x^3 \\ a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \end{vmatrix}.$$

It is to be observed that this reduction of the quantic to its canonical form, i.e. to a sum of a number $n+1$ of $(2n+1)$th powers, is a unique one, and that the quantic cannot be in any other manner a sum of a number $n+1$ of $(2n+1)$th powers.

Professor Sylvester communicated to me, under a slightly less general form, and has permitted me to publish the following theorems:

1. If the second emanant $(X\partial_x + Y\partial_y)^2 U$ has in common with the quantic U a single canonic root, then all the canonic roots of the emanant are canonic roots of the quantic; and, moreover, if the remaining canonic root of the quantic be $rx + sy$, then (X, Y), the facients of emanation, are $= (s, -r)$, or, what is the same thing, they are given by the equation

$$\text{canont. } U \ (X, Y \text{ in place of } x, y) = 0.$$

In fact, considering, as before, the quintic $U = (a, b, c, d, e, f \,|\, x, y)^5$, we have

$$U = A(lx + my)^5 + A'(l'x + m'y)^5 + A''(l''x + m''y)^5,$$

and thence

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3 + B''(l''x + m''y)^3,$$

if for shortness

$$B = 6 \cdot 5\,(lX + mY)^2 A, \ \text{\&c.}$$

Suppose $(X\partial_x + Y\partial_y)^2 U$ has in common with U the canonic root $lx + my$, then

$$(X\partial_x + Y\partial_y)^2 U = C(lx + my)^3 + C'(px + qy)^3,$$

and thence

$$B'(l'x + m'y)^3 + B''(l''x + m''y)^3 = (C - B)(lx + my)^3 + C'(px + qy)^3,$$

which must be an identity; for otherwise we should have the same cubic function expressed in two different canonical forms. And we may write

$$B' = C', \quad l'x + m'y = px + qy, \quad B'' = 0, \quad C = B,$$

and then we have

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3;$$

so that all the canonic roots of the emanant are canonic roots of the quantic. Moreover, the condition $B'' = 0$ gives $l''X + m''Y = 0$, that is, $X : Y = m'' : -l''$, or writing $rx + sy$ instead of $l''x + m''y$, $X : Y = s : -r$; and the system is

$$\begin{aligned} U &= A(lx + my)^5 + A'(l'x + m'y)^5 + A''(rx + sy)^5, \\ (s\partial_x - r\partial_y)^2 U &= B(lx + my)^3 + B'(l'x + m'y)^3, \end{aligned}$$

which proves the theorem.

2. The two functions, canont. U, canont. $(X\partial_x + Y\partial_y)^2 U$, have for their resultant {canont. U $(X, Y$ in place of x,
if $2n + 1$ be the order of U.

In fact, in order that the equations

$$\text{canont. } U = 0, \quad \text{canont. } (X\partial_x + Y\partial_y)^2 U = 0,$$

may coexist, their resultant must vanish; and conversely, when the resultant vanishes, the equations will have a common root. Now if the equation canont. $(X\partial_x + Y\partial_y)^2 U = 0$ has a common root with the equation canont. $U = 0$, all its roots are roots of canont. $U = 0$; and, moreover, if $rx + sy = 0$ be the remaining root of canont. $U = 0$, then $X : Y = s : -r$, that is, we have

$$\text{canont. } U\,(X,\, Y \text{ in place of } x,\, y) = 0;$$

or the resultant in question can only vanish if the last-mentioned equation is satisfied. It follows that the resultant must be a power of the *nilfactum* of the equation; and observing that canont. U is of the form $(a, \ldots)^{n+1}(x, y)^{n+1}$, i.e. that it is of the degree $n+1$ as well in regard to the coefficients as in regard to the variables (x, y), it is easy to see that the resultant is of the degree $2n(n+1)$ as well in regard to the coefficients as in regard to (X, Y); that is, we have $2n$ as the index of the power in question.

3. In particular, if $Y = 0$, the theorem is that the resultant of the functions canont. U, canont. $\partial_x^2 U$ is equal to the $2n$th power of the first coefficient of canont. U.

Thus for $n = 1$, that is, for the cubic function $(a, b, c, d \,|\, x, y)^3$, we have

$$\text{canont. } U = \begin{vmatrix} y^2, & -xy, & x^2 \\ a, & b, & c \\ b, & c, & d \end{vmatrix} = (ac - b^2,\ ad - bc,\ bd - c^2 \,|\, x, y)^2,$$

$$\text{canont. } \partial_x^2 U = \begin{vmatrix} y, & -x \\ a, & b \end{vmatrix} = ax + by;$$

and the resultant of the two functions is

$$= (ac - b^2,\ y,\ ad - bc,\ bd - c^2 \,|\, b, -a)^2$$

$$= -(ac - b^2) \cdot a^2,$$

which verifies the theorem.

The theorems were, in fact, given to me in relation to the quantic U and the second differential coefficient $\partial_x^2 U$; but the introduction instead thereof of the second emanant $(X\partial_x + Y\partial_y)^2 U$ presented no difficulty.

2, Stone Buildings, W.C., February 16, 1863.

327. On the Stereographic Projection of the Spherical Conic.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 350–353.]

In order to the tolerable delineation of some figures relating to spherical geometry, I had occasion to consider the stereographic projection of the spherical conic. To fix the ideas, imagine a sphere having its centre in the plane of the paper, and through the centre three rectangular axes, that of x horizontal and that of y vertical, in the plane of the paper, and the axis of z perpendicular to and in front of the plane of the paper. The radius of the sphere is taken equal to unity (so that its intersection by the plane of the paper is the circle radius unity), and the points X, Y, and Z are taken to denote the points where the axes, drawn in the positive direction, meet the surface of the sphere; and the opposite points are called X', Y', and Z'. The eye is supposed to be at Z, and the projection to be made on the plane of the paper. This being so, and supposing that the axes of coordinates are the principal axes of the spherical conic, the *axis of x being the interior axis*, and taking ξ, η, ζ as the coordinates of a point on the spherical conic, its equations are

$$\xi^2+\eta^2+\zeta^2=1, \qquad -\xi^2+\eta^2\cot^2\beta+\frac{\zeta^2}{c^2}=0;$$

where it may be remarked that $\tan\beta$, c are the semiaxes of the plane conic which is the gnomonic projection (i.e. the projection by lines through the centre of the sphere) of the spherical conic on the tangent plane at X or X'.

Taking, for a moment, x, y, z as the coordinates of a point on the projecting line (that is, the line through the eye to a point (ξ,η,ζ) on the spherical conic), the equation of this line is

$$\frac{x}{\xi}=\frac{y}{\eta}=\frac{z-1}{\zeta-1};$$

and thence, putting $z = 0$, x, y will be the coordinates of a point of the projection, and we have

$$\frac{x}{\xi} = \frac{y}{\eta} = \frac{1}{1-\zeta};$$

or, what is the same thing,

$$\xi = x(1-\zeta), \quad \eta = y(1-\zeta);$$

the equations of the spherical conic may be written

$$\begin{aligned} 1-\zeta^2 &= \xi^2+\eta^2, \\ \zeta^2 &= c^2(\xi^2-\eta^2\cot^2\beta); \end{aligned}$$

and by eliminating ξ, η, ζ from the four equations, we obtain the equation of the conic.

Substituting for ξ and η their values, we find

$$1+\zeta = (x^2+y^2)(1-\zeta),$$

$$\zeta^2 = c^2(x^2-y^2\cot^2\beta)(1-\zeta)^2;$$

or, observing that the first equation gives

$$\zeta = \frac{x^2+y^2-1}{x^2+y^2+1},$$

and that thence

$$1-\zeta = \frac{2}{x^2+y^2+1}, \qquad \frac{\zeta}{1-\zeta} = \tfrac{1}{2}(x^2+y^2-1),$$

the equation is

$$(x^2+y^2-1)^2 = 4c^2(x^2-y^2\cot^2\beta).$$

It is now very easy to trace the curve. We see first that the curve is symmetrical with respect to the axes, and that it meets the axis of y in four imaginary points, but the axis of x in four real points, the coordinates whereof are

$$x = \pm(\sqrt{1+c^2}\pm c),$$

so that the two points on the same side of the centre are the images one of the other in regard to the circle radius unity. Moreover the curve touches the lines

$$y = \pm x\tan\beta$$

at their intersections with the circle. By developing in regard to y, the equation becomes

$$y^4+2(x^2-1+2c^2\cot^2\beta)y^2+(x^2-1)^2-4c^2x^2 = 0;$$

and putting

$$x = \pm(\sqrt{1+c^2}\pm c),$$

the last term vanishes, and the equation gives $y^2 = 0$, or

$$\begin{aligned} y^2 &= 2(1 - x^2 - 2c^2 \cot^2 \beta) \\ &= 4(-c^2 \mp c\sqrt{1+c^2} - c^2 \cot^2 \beta) \\ &= 4c(-c \csc^2 \beta \mp \sqrt{1+c^2}), \end{aligned}$$

the upper sign corresponding to the exterior values

$$\pm x = \sqrt{1+c^2} + c,$$

and the lower sign to the interior values

$$\pm x = \sqrt{1+c^2} - c.$$

In the former case the values of y are imaginary; in the latter case they are real if

$$\sqrt{1+c^2} > c \csc^2 \beta,$$

or, what is the same thing, if

$$\sin^2 \beta > \frac{c}{\sqrt{1+c^2}};$$

that is, if (for a given value of c) β is sufficiently great, but otherwise they are imaginary.

If, as in the annexed figures, $c = \frac{5}{12}$ (and therefore $\sqrt{1+c^2} = \frac{13}{12}$, $\sqrt{1+c^2}+c = \frac{3}{2}$, $\sqrt{1+c^2} - c = \frac{2}{3}$), then for the limiting value of β we have

$$\sin^2 \beta = \tfrac{5}{13} = 0.3846, \quad \sin \beta = 0.62, \text{ or } \beta = 38^\circ \text{ nearly.}$$

[Fig. 1 and Fig. 2: figures of the curve for two values of β are omitted in this reset.]

In the first figure β is less, in the second figure greater than this value: the form for the limiting value is obvious from a comparison of the two figures.

I take the opportunity to mention the following theorem, which is perhaps known, but I have not met with it anywhere; viz. any three circles, each two of which meet, may be considered as the stereographic projections of three great circles of the sphere. In fact suppose, as above, that the projection is made on the plane of a great circle, and calling this the principal circle, the projection of any other great circle meets the principal circle at the extremities of a diameter of the principal circle. It follows that the theorem will be true, if, given any three circles each two of which meet, a circle can be drawn meeting the given circles, each of them at the extremities of a diameter of the circle so to be drawn. It is easy to see that the required circle has for its centre the radical centre (point of intersection of the radical axes) of the given circles, and that the radius is the 'Inner Potency' of the point in question in regard to each of the three given circles. In particular the three circles having for centres the vertices of an equilateral triangle, and the side for radius, may be considered as the stereographic projections of three great circles of a sphere. This is a very ready mode of delineation of a spherical figure depending on three great circles of the sphere.

2, Stone Buildings, W.C., March 21, 1863.

328. On the Delineation of a Cubic Scroll.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 528–530.]

Imagine a cubic scroll (skew surface of the third order) generated by lines each of which meets two given directrix lines. One of these is a nodal (double) line on the surface, and I call it the nodal directrix; the other is a single line on the surface, and I call it the single directrix. The section by any plane is a cubic passing through the points in which the plane meets the directrix lines; i.e. the point on the nodal directrix is a node (double point) of the curve, the point on the single directrix a single point on the curve; the two directrix lines, and the cubic curve, the section by any plane, determine the scroll. Consider the sections by a series of parallel planes. Let one of these planes be called the basic plane, and the section by this plane the basic section or basic cubic; and imagine any other section projected on the basic plane by lines parallel to the nodal directrix: such section may be spoken of simply as 'the section,' and its projection as 'the cubic.' The cubic has at the node of the basic cubic a node, and besides has with the basic cubic four points in common. The two curves have, moreover, in common the three points at infinity (or, in other words, their asymptotes are parallel); in fact the points at infinity of either curve are the points in which the line at infinity, the intersection of the basic plane and the plane of the section, meets the scroll; and these points are therefore the same for each of the two curves. The remaining two points of intersection of the cubic with the basic cubic are also fixed points on the basic cubic, i.e. they are the points of intersection of the basic plane by the two generating lines parallel to the nodal directrix. Hence the cubic meets the basic cubic in nine fixed points, viz. the node counting as four points, the three points at infinity, and the two points the feet of the generators parallel to the nodal directrix. It follows that if $U = 0$ is the equation of the basic cubic, $V = 0$ the equation of some other cubic meeting the basic cubic in the nine points in question, then the equation of 'the cubic' is $U + \lambda V = 0$, λ being a parameter the value of which varies according to the position (in the series of parallel planes) of the plane of the section. Suppose that the basic cubic $U = 0$ is given, and suppose for a moment that the cubic $V = 0$ is also given, these two cubics having the above-mentioned relations, viz. they have a common node and parallel asymptotes: the cubic $U + \lambda V = 0$ might be constructed by drawing through the node (say O) a radius vector meeting the cubics in P, P' respectively, and taking on this radius vector a point Q such that $PQ = \dfrac{\lambda}{1+\lambda}\, PP'$, or, what is the same thing,

$$OQ = \frac{OP + \lambda OP'}{1+\lambda};$$

the locus of the point Q will then be the cubic $U + \lambda V = 0$. And we may even suppose the cubic $V = 0$ to break up into a line and a conic (hyperbola), and then (disregarding the line) use the hyperbola in the construction. In fact, if the hyperbola is determined by the following five conditions, viz. to pass through the node and through the feet of the two generators parallel to the nodal directrix, and to have its asymptotes parallel to two of the asymptotes of the basic cubic, and if the line be taken to be a line through the node parallel to the third asymptote of the basic cubic; then the hyperbola and line form together a cubic curve meeting the basic cubic in the nine points, and therefore satisfying the conditions assumed in regard to the cubic $V = 0$. And it is to be noticed that as in general the cubic $V = 0$ is the projection of some section of the scroll, so the hyperbola and line are the projection of a section of the scroll, viz. the section through one of the generating lines (there are three such lines) parallel to the basic plane. But it is better to construct 'the cubic' by a different method (using only the basic cubic $U = 0$) which results more immediately from the geometrical theory. Taking the basic plane as the plane of the figure, let O be the node, or foot of the nodal directrix, K the foot of the single directrix, Kk the projection of the single directrix.

[Figure: O, P, r, Q, K, k, L, M as described in the text; figure omitted in this reset.]

k being the projection of the point in which the single directrix meets the plane of the section. Drawing through O any radius vector meeting the basic cubic in P, and the line Kk in r, and producing it to a properly determined point Q, then $OPrQ$ will be the projection of the generating line which meets the nodal directrix, the basic cubic, the single directrix, and the section in the points the projections whereof are O, P, r, Q respectively: and the consideration of the solid figure shows easily that the condition for the determination of the point Q is

$$PQ = Kk \cdot \frac{Pr}{rK}.$$

Hence, starting from the basic cubic and the line Kk, we have a construction for the point Q the locus whereof is the cubic, the projection of a section of the scroll; for the projections of the parallel sections, we have only to vary the length Kk. By what precedes, the construction gives for the locus of Q a cubic having a node at O, and having its asymptotes parallel to those of the basic cubic. As P moves up to K, the distances Pr, rK become indefinitely small; but their ratio is finite, hence the cubic, the locus of Q, does not pass through the point K. The construction shows, however, that it does pass through the points L, M, which are the other two intersections of Kk with the basic cubic; these points L, M are in fact the feet of the generators parallel to the nodal directrix.

The general conclusion is, that a series of cubics having each of them at one and the same given point a node—having their asymptotes parallel—and besides passing through the same two given points—may be considered as the projections of a series of parallel sections of a cubic scroll; and such a series of cubics will thus afford a delineation of the scroll.

2, Stone Buildings, W.C., April 15, 1863.

329. Note on the Problem of Pedal Curves.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 20, 21.]

It is not, so far as I am aware, generally known that the problem of pedal curves (Steiner's *Fusspuncten-Curve*) was considered by Maclaurin in the *Geometria Organica*, 1720. He appears to have been led to it through an idea such as Sir W. R. Hamilton's Hodograph, or at any rate with a view to a dynamical application, for he remarks, p. 95, "Cum vero geometria quæ curvas ad datum centrum relatas contemplatur in philosophia naturali ad motus corporum et vires evolvendas facilius applicari possit, . . . hac sectione considerabimus curvas tanquam ad punctum quodvis datum relatas ex quo ad omnia circumferentiæ puncta radii undique educuntur, et simul perpendicula in illorum punctorum tangentes demittuntur, et rationem radii ad perpendiculum tanquam curvæ characterem usurpabimus." And accordingly, Props. IX. to XII., he considers the problem: Given a point S in the plane of a given curve, to find the locus of the intersection of a tangent of the curve by the perpendicular let fall upon it from the point S; with some special cases, and deductions from it. In particular if the given curve be a circle, the locus in question (or pedal curve) is a curve of the fourth order having a double point S; viz. if S be inside the circle, this is a conjugate or isolated point; but if outside, a double point with two real branches: if S be on the circle, then instead of the double point we have a cusp: it is shown that in each case the pedal curve is in fact an epicycloid. If the given curve be a parabola, then the locus or pedal curve is a curve of the third order, viz. a defective hyperbola having a double point at S, and with its single asymptote perpendicular to the axis of the parabola: some particular cases are specially noticed. If the curve be an ellipse or hyperbola, then, as in the case of the circle, the locus or pedal curve is a curve of the fourth order having a double point at S. And it is moreover shown, Prop. XII., that for any given curve whatever the locus or pedal curve is, in a generalized sense of the term, an epicycloid. This is in fact seen very easily by a mere inspection of the figure. Imagine the curve $O'P'$, rigidly connected with and carrying along with it the point S', to roll on the similar and equal fixed curve OP symmetrically situate on the other side of the common tangent OM or $O'M$; then when P' coincides with P, the point S' is brought to S'', where $SNN'S''$ is the perpendicular from S on the tangent PN or PN', and $SN = N'S''$, that is, $SS'' = 2\,SN$;

[Figure: the points S, S', S'', O, O', P, P', N, N', M, M' as described; figure omitted in this reset.]

and the curve generated by S'' (that is S'), or say the epicycloid the locus of S', is a curve similar to and similarly situate with the pedal curve the locus of N, but of twice the linear magnitude of the pedal curve. Or, what is the same thing, if instead of the given curve we consider a similar and similarly situated curve of twice the linear magnitude (the point S being the centre of similitude), then the epicycloid the locus of S' is the pedal curve of the substituted curve in relation to the point S. It may be added that, in accordance with a theorem of Dandelin's, if rays proceeding from the point S are reflected at the given curve, then the epicycloid (or pedal) in question is the secondary caustic, or an orthogonal trajectory of the reflected rays.

2, Stone Buildings, W.C., June 3, 1863.

330. On Differential Equations and Umbilici.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 373–379 and 441–452.]

I.

Consider the integral equation

$$Az^2 + 2Bz + C = 0,$$

where z is the constant of integration: the derived equation is

$$\begin{aligned}\Omega &= (AC' + A'C - 2BB')^2 - 4(AC - B^2)(A'C' - B'^2)\\ &= (CA' - C'A)^2 - 4(AB' - A'B)(BC' - B'C) = 0;\end{aligned}$$

and if for greater simplicity we write $A = 1$, then the derived equation is

$$\Omega = C'^2 - 4BC'B' + 4CB'^2 = 0,$$

corresponding to the integral equation

$$z^2 + 2Bz + C = 0.$$

Writing the integral equation under the form

$$(z + X)(z + Y) = 0,$$

we have

$$2B = X + Y, \quad C = XY,$$

$$2B' = X' + Y', \quad C' = XY' + X'Y,$$

and the derived equation becomes

$$\Omega = -(X - Y)^2 X'Y'.$$

Hence if we represent the roots X, Y in the form $P \pm Q\sqrt{\square}$, so that $P = -B$, $Q\sqrt{\square} = \sqrt{B^2 - AC}$, Q being the greatest square factor of $B^2 - AC$, then

$$(X - Y)^2 = 4Q^2\square, \quad X',\ Y' = P' \pm \left(Q'\sqrt{\square} + \frac{Q\square'}{2\sqrt{\square}}\right),$$

$$X'Y' = P'^2 - \frac{1}{4\square}(2Q'\square + Q\square')^2;$$

and the derived equation is

$$\Omega = -Q^2\left\{4\square P'^2 - (2Q'\square + Q\square')^2\right\} = 0.$$

If B, C, &c. are functions of the coordinates (x, y), the equation $z^2 + 2Bz + C = 0$ (z an arbitrary constant) represents a series of curves in the plane of xy; but if we consider z as a coordinate, then the equation represents a surface, and the curves in question are the orthogonal projections on the plane of xy of the sections of the surface by the planes parallel to the plane of xy. To fix the ideas, the plane of xy may be taken to be horizontal, and the ordinates z vertical.

Writing the equation in the form

$$(z + B)^2 - (B^2 - C) = 0,$$

we see that the surface contains upon it the curve $z + B = 0$, $B^2 - C = 0$, which is the line of contact with the circumscribed vertical cylinder: such curve may be termed the envelope, or, when this is necessary, the complete envelope. The equation of the surface has however been taken to be $(z - P)^2 - Q^2\square = 0$ (viz. it has been assumed that $B = -P$, $B^2 - C = Q^2\square$); the envelope thus breaks up into the curve $z - P = 0$, $Q = 0$, taken twice, and the curve $z - P = 0$, $\square = 0$; the former of these is in general a nodal curve on the surface, and it may be spoken of as the nodal curve; the latter of them is the reduced or proper envelope, or simply the envelope. And the terms nodal curve and envelope may also be applied to the curves $Q = 0$ and $\square = 0$, which are the projections on the plane of xy of the first-mentioned two curves respectively. There is however a case of higher singularity which it is proper to consider: suppose that Q and $\square$ have a common factor K, say $Q = KR$, $\square = K\nabla$; the complete envelope $Q^2\square = K^3R^2\nabla = 0$ here breaks up into the nodal curve $R = 0$ twice, the cuspidal curve $K = 0$ three times, and the reduced or proper envelope $\nabla = 0$ once.

Reverting for a moment to the form $(z + X)(z + Y) = 0$, the derived equation $\Omega = -(X - Y)^2X'Y' = 0$ is satisfied by $(X - Y)^2 = 0$; this equation, or say the equation of the envelope, being in fact the singular solution of the differential equation. This assumes however that the differential equation is given in the form in which it is immediately obtained by derivation from the integral equation, without the rejection of factors which are functions of the coordinates (x, y) only; it is proper to consider the reduced equation obtained by rejecting such factors. Thus if X and Y are rational functions, the reduced form is $X'Y' = 0$, which is no longer satisfied by the equation

$(X-Y)^2 = 0$. In the before-mentioned case where the roots are $P \pm Q\sqrt{\Box}$ (or $(X-Y)^2 = 4Q^2\Box$), P, Q, and $\Box$ being rational functions of (x, y), the derived equation

$$\Omega = -Q^2\big\{4\Box P'^2 - (2Q'\Box + Q\Box')^2\big\} = 0$$

divides out by the factor Q^2, but it does not divide out by $\Box$; the reduced form is therefore

$$4\Box P'^2 - (2Q'\Box + Q\Box')^2 = 0,$$

which is not satisfied by $Q = 0$, while it is still satisfied by $\Box = 0$ (since this gives also $\Box' = 0$); that is, the nodal curve $Q = 0$ is not a solution of the differential equation, but we still have the singular solution $\Box = 0$, which corresponds to the reduced or proper envelope. In the case $Q = KR$, $\Box = K\nabla$ of a cuspidal curve, the above form of the differential equation becomes

$$4K\nabla P'^2 - \{3KK'R\nabla + K^2(2\nabla R' + \nabla' R)\}^2 = 0,$$

which divides out by K; and, when reduced by the rejection of this factor, it is no longer satisfied by the equation $K = 0$, which belongs to the cuspidal curve; that is, neither the nodal curve $R = 0$ nor the cuspidal curve $K = 0$ is a solution of the differential equation, but we still have the singular solution $\nabla = 0$, which corresponds to the reduced or proper envelope. It would appear that the conclusion may be extended to singularities of a higher nature, viz. the factor corresponding to any singular curve which presents itself as part of the complete envelope divides out from the derived equation; and such singular curve does not constitute a solution of the reduced equation, but we have a singular solution corresponding to the reduced or proper envelope.

II.

Consider the differential equation

$$y(p^2 - 1) + 2mxp = 0,$$

where, to fix the ideas, $m \geq 1$; the integral equation may be taken to be

$$z = (mx + \sqrt{m^2x^2 + y^2})(mx^2 + y^2 + x\sqrt{m^2x^2 + y^2})^{m-1};$$

or rather, writing for shortness $\Box = m^2x^2 + y^2$, and putting

$$z = (mx + \sqrt{\Box})(mx^2 + y^2 + x\sqrt{\Box})^{m-1} = P + Q\sqrt{\Box},$$

the integral equation is

$$(z - P)^2 - Q^2\Box = 0, \quad \text{or} \quad z^2 - 2Pz + P^2 - Q^2\Box = 0,$$

where

$$P^2 - Q^2\Box = (m^2x^2 - \Box)\{(mx^2 + y^2)^2 - x^2\Box\}^{m-1} = -y^{2m}\{y^2 + (2m-1)x^2\}^{m-1}.$$

In the particular case $m = 1$ the equation is

$$z = x + \sqrt{x^2 + y^2}, \quad \text{or} \quad z^2 - 2zx - y^2 = 0.$$

Before going further, I remark that, m being a positive integer greater than unity, we have

$$z = P + Q\sqrt{\square} = mx(mx^2+y^2)^{m-1} + \{mx^2+y^2+(m-1)mx^2\}(mx^2+y^2)^{m-2}\sqrt{\square} + \&\text{c.},$$

the subsequent terms being divisible, the rational ones by $\square$, and the irrational ones by $\square\sqrt{\square}$. Hence, observing that

$$mx^2+y^2+(m-1)mx^2 = m^2x^2+y^2 = \square,$$

we see that Q contains the factor $\square$, and the equation $\square = 0$ belongs to a cuspidal curve on the surface. If however $m = 1$, then the equation is $z = x + \sqrt{\square}$, so that $Q = 1$ does not contain the factor $\square$; and $\square = x^2+y^2 = 0$ is not a singular curve on the surface, but is in fact the reduced or proper envelope.

The curve represented by the integral equation will pass through the origin ($x = 0$, $y = 0$) for the value $z = 0$ of the constant of integration. In fact, for this value, the integral equation becomes

$$-y^{2m}\{y^2+(2m-1)x^2\}^{m-1} = 0,$$

which belongs to a set of $2m + (m-1) + (m-1)$ lines coinciding with the lines $y = 0$, $y = ix\sqrt{2m-1}$, and $y = -ix\sqrt{2m-1}$ respectively. The directions at the origin are therefore $p = 0$, $p = \pm i\sqrt{2m-1}$, which are the same values of p as are obtained from the differential equation; viz. since this is satisfied identically at the point in question, proceeding to the derived equation, we have

$$p(p-1) + 2mp = 0,$$

that is

$$p(p^2+2m-1) = 0;$$

but it is to be observed that these values of p are different from the values given by the equation $\square = m^2x^2+y^2 = 0$, which are $p = \pm im$. The reason is that the curve $\square = 0$ being, as was shown, a cuspidal curve on the surface, the equation $\square = 0$ is not a solution of the differential equation.

If however $m = 1$, then the integral equation gives at the origin no longer three values of p, but only the value $p = 0$. The differential equation however gives, as in the general case, three values; viz. we have $p(p^2+1) = 0$; and the values $p = \pm i$ obtained from the factor $p^2+1 = 0$ are precisely the values of p obtained from the equation $\square = x^2+y^2 = 0$, which in the case now under consideration belongs to the reduced or proper envelope of the surface, and is therefore the singular solution of the differential equation.

III.

The two curves of curvature which pass through any given point of a surface are distinct curves, not branches of one indecomposable curve. In fact if P, Q are the two curves of curvature for a point A, then for a point A' on P the two curves of curvature will be P, Q'; and if P, Q were branches of an indecomposable curve, then P, Q' would also be branches of an indecomposable curve, and we should have P a branch of two different indecomposable curves, which is of course impossible. In the case of an umbilicus, the two curves P and Q coincide together; or, as we may express it, the curves of curvature through an umbilicus are the duplication of a single, in general indecomposable, curve; and in general this curve has at the umbilicus a trifid node. I use this expression to denote a point at which there are three distinct tangents, or, more accurately, three distinct directions of the curve: an ordinary triple point is of necessity a trifid node, but not conversely. The umbilicus of an ellipsoid or other quadric surface is a peculiar exceptional case.

In support of the foregoing conclusions, consider a surface having an umbilicus at the origin, and take $z = 0$ as the equation of the tangent plane at that point; the equation of the surface in the neighbourhood of the umbilicus will be

$$z = \tfrac{1}{2}k(x^2 + y^2) + \tfrac{1}{6}(ax^3 + 3bx^2y + 3cxy^2 + dy^3);$$

so that, writing as usual p and q for the first, and r, s, t for the second, differential coefficients of z, we have

$$\begin{aligned}
p &= kx + \tfrac{1}{2}(ax^2 + 2bxy + cy^2),\\
q &= ky + \tfrac{1}{2}(bx^2 + 2cxy + dy^2),\\
r &= k + ax + by,\\
s &= bx + cy,\\
t &= k + cx + dy.
\end{aligned}$$

The differential equation of the curves of curvature projected on the plane of xy is

$$\frac{dy^2}{dx^2}\{(1+q^2)s - pqt\} + \frac{dy}{dx}\{(1+q^2)r - (1+p^2)t\} - \{(1+p^2)s - pqr\} = 0;$$

and substituting therein the foregoing values of p, q, r, s, t, but attending only to the terms of the lowest order in (x, y), and using moreover in the sequel p in the place of $\dfrac{dy}{dx}$, the equation becomes

$$(bx + cy)(p^2 - 1) + \{(a-c)x + (b-d)y\}p = 0;$$

which may be taken as the differential equation of the curves of curvature at and in the neighbourhood of the umbilicus. The equation is satisfied identically by the values $x = 0$, $y = 0$, which correspond to the umbilicus; and to find p, we have to differentiate the equation, and then substitute these values of x and y; we thus obtain

$$(b + cp)(p^2 - 1) + \{(a-c) + (b-d)p\}p = 0,$$

or, what is the same thing,

$$p(a + 2bp + cp^2) - (b + 2cp + dp^2) = 0,$$

a cubic equation for the determination of p.

I remark that we may without loss of generality write $d = 0$: but to simplify the investigation, I suppose in the first instance that we have also $b = 0$; this comes to assuming that one of the three planes $ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ bisects the angle formed by the other two planes. The differential equation consequently is

$$cy(p^2 - 1) + (a - c)xp = 0;$$

or, putting for shortness

$$m = \tfrac{1}{2} \cdot \frac{a - c}{c},$$

it is

$$y(p^2 - 1) + 2mxp = 0,$$

which is the differential equation previously considered. Hence, writing now h in the place of z, the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = (mx + \sqrt{\Box})(mx^2 + y^2 + x\sqrt{\Box})^{m-1}, \ = P + Q\sqrt{\Box},$$

where $\Box = m^2x^2 + y^2$; or, what is the same thing, the equation is

$$h^2 - 2Ph + P^2 - Q^2\Box = 0;$$

and the equation (in the neighbourhood of the umbilicus) of the curve through the umbilicus is

$$P^2 - Q^2\Box = -y^{2m}\{y^2 + (2m - 1)x^2\}^{m-1} = 0;$$

so that the umbilicus is a trifid node. In the case however of an ellipsoid or other quadric surface, we have $m = 1$, so that the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = x + \sqrt{x^2 + y^2},$$

or, what is the same thing,

$$h^2 - 2hx - y^2 = 0:$$

and for the curve through the umbilicus, in the neighbourhood of the umbilicus, the equation is $y^2 = 0$, so that there is only a single direction of the curve of curvature. The differential equation gives, however, at the umbilicus $p(p^2 + 1) = 0$; the value $p = 0$ is that which corresponds to the curve of curvature; the other two values $p = \pm i$ correspond to the curve (pair of lines) $x^2 + y^2 = 0$, which is the envelope of the curves of curvature, or, more accurately, the envelope of the projections of the curves of curvature on the tangent plane at the umbilicus.

Blackheath, October 17, 1863.

IV.

The differential equation for the curves of curvature in the neighbourhood of an umbilicus was obtained in a form such as

$$(bx + cy)(p^2 - 1) + 2(fx + gy)p = 0;$$

and it was only because this equation did not appear to be readily integrable, that I considered, instead of it, the particular form

$$y(p^2-1)+2mxp=0.$$

But the general equation can be integrated; and the result presents itself in a simple form. For, returning to the differential equation

$$(bx+cy)(p^2-1)+2(fx+gy)p=0,$$

and assuming

$$\frac{bx+cy}{fx+gy}=\frac{-2v}{v^2-1},$$

we have

$$(bx+cy)(p^2-1)+2(fx+gy)p=0$$

gives

$$\frac{p^2-1}{p}=-\frac{2v}{v^2-1},\ \text{or}\ (p-v)(vp+1)=0,$$

and we may write

$$p-v=0.$$

Assuming also

$$v=ux,\quad \text{or}\quad u=\frac{y}{x},$$

the relation between u and v is

$$\frac{b+cu}{f+gu}=\frac{-2v}{v^2-1},$$

or, as this may be written,

$$v^2+\frac{2(f+gu)}{b+cu}v-1=0,$$

giving

$$v=\frac{-(f+gu)-\sqrt{(b+cu)^2+(f+gu)^2}}{b+cu},$$

where for convenience the radical has been taken with a negative sign. We have moreover

$$p=\frac{b(v^2-1)+2fv}{c(v^2-1)+2gv}.$$

The equation $p-v=0$, substituting for y its value ux, then becomes

$$x\frac{du}{dx}+u-v=0,$$

or, as this may be written,

$$\frac{dx}{x}-\frac{du}{u-v}=0,$$

or, what is the same thing,

$$\frac{dx}{x}+\frac{dv-du}{v-u}-\frac{dv}{v-u}=0.$$

But

$$v-u=v-p=v-\frac{b(v^2-1)+2fv}{c(v^2-1)+2gv}=\frac{V}{c(v^2-1)+2gv},$$

where

$$V=v\{c(v^2-1)+2gv\}-\{b(v^2-1)+2fv\}=(b+cv)(v^2-1)+2(f+gv)\,v,$$

(with sign as above; the displayed identity is

$$V=(b+cv)(v^2-1)+2(f+gv)v).$$

and the differential equation takes thus the form

$$\frac{dx}{x}+\frac{dv-du}{v-u}-\frac{\{c(v^2-1)+2gv\}\,dv}{V}=0,$$

and hence, writing

$$V=(b+cv)(v^2-1)+2(f+gv)v=c(v-\alpha)(v-\beta)(v-\gamma),$$

and

$$\frac{c(v^2-1)+2gv}{V}=\frac{c(v^2-1)+2gv}{c(v-\alpha)(v-\beta)(v-\gamma)}=\frac{A}{v-\alpha}+\frac{B}{v-\beta}+\frac{C}{v-\gamma},$$

so that

$$A=\frac{c(\alpha^2-1)+2g\alpha}{c(\alpha^2-1)+2g\alpha+2\{f+(b+g)\alpha+c\alpha^2\}},$$

with the like values for B and C—values which are such that $A+B+C=1$,—the integral equation is

$$\text{const.}=x(v-u)(v-\alpha)^{-A}(v-\beta)^{-B}(v-\gamma)^{-C},$$

or, substituting for $v-u$ its value,

$$=x\cdot\frac{V}{c(v^2-1)+2gv}(v-\alpha)^{-A}(v-\beta)^{-B}(v-\gamma)^{-C}.$$

But

$$\text{const.}=x\,\{c(v^2-1)+2gv\}^{-1}(v-\alpha)^{1-A}(v-\beta)^{1-B}(v-\gamma)^{1-C}.$$

Substituting for v its value, if for shortness $U=(b+cu)^2+(f+gu)^2$, and hence $\sqrt{U}=$ the radical $\sqrt{(b+cu)^2+(f+gu)^2}$, then

$$v^2=\frac{2(f+gu)^2+(b+cu)^2+2(f+gu)\sqrt{U}}{(b+cu)^2},$$

and

$$c(v^2-1)+2gv=\frac{c\{(f+gu)^2+(f+gu)\sqrt{U}\}+g\{-(f+gu)-\sqrt{U}\}(b+cu)}{(b+cu)^2}$$

(intermediate expressions; the reduction reads as in the original),

$$v-\alpha=\frac{-(f+gu)-\sqrt{U}-\alpha(b+cu)}{b+cu},\quad \&\text{c}.$$

Substituting these values, and observing that the exponent of $b + cu$ is

$$(-2 + 1 - A + 1 - B + 1 - C, \ = 1 - A - B - C) = 0,$$

the integral equation is

$$\text{const.} = x\,(f + gu + \sqrt{U})^{-1}\times$$
$$(f + gu + \alpha(b + cu) + \sqrt{U})^{1-A}(f + gu + \beta(b + cu) + \sqrt{U})^{1-B}(f + gu + \gamma(b + cu) + \sqrt{U})^{1-C},$$

or, observing that the exponent 1 of x is

$$= -1 + (1 - A) + (1 - B) + (1 - C),$$

and putting for shortness $\Box = (fx + gy)^2 + (bx + cy)^2$, the integral equation finally is

$$\text{const.} = (fx + gy + \sqrt{\Box})^{-1}\times$$
$$(fx+gy+\alpha(bx+cy)+\sqrt{\Box})^{1-A}(fx+gy+\beta(bx+cy)+\sqrt{\Box})^{1-B}(fx+gy+\gamma(bx+cy)+\sqrt{\Box})^{1-C},$$

where the quantities α, β, γ, A, B, C are given by

$$(b + cv)(v^2 - 1) + 2(f + gv) = c(v - \alpha)(v - \beta)(v - \gamma),$$

$$\frac{c(v^2 - 1) + 2gv}{c(v - \alpha)(v - \beta)(v - \gamma)} = \frac{A}{v - \alpha} + \frac{B}{v - \beta} + \frac{C}{v - \gamma}.$$

Consider the curve

$$0 = (fx+gy+\alpha(bx+cy)+\sqrt{\Box})^{1-A}(fx+gy+\beta(bx+cy)+\sqrt{\Box})^{1-B}(fx+gy+\gamma(bx+cy)+\sqrt{\Box})^{1-C},$$

which corresponds to the value $= 0$ of the constant. If, for instance, this equation gives

$$fx + gy + \alpha(bx + cy) + \sqrt{\Box} = 0,$$

or say

$$(bx + cy)\{(bx + cy)(\alpha^2 - 1) + 2(fx + gy)\alpha\} = 0;$$
$$(bx + cy)(\alpha^2 - 1) + 2(fx + gy)\alpha = 0.$$

But we have

$$(b + c\alpha)(\alpha^2 - 1) + 2(f + g\alpha)\alpha = 0,$$

and the equation therefore is

$$(bx + cy)(f + g\alpha) - (fx + gy)(b + c\alpha) = 0;$$

that is

$$(cf - bg)(y - \alpha x) = 0;$$

or simply $y - \alpha x = 0$; that is, the directions of the curve at the origin, or point $x = 0$, $y = 0$, are given by the equations $y - \alpha x = 0$, $y - \beta x = 0$, $y - \gamma x = 0$. This is right, since from the differential equation we obtain at the origin

$$(b + cp)(p^2 - 1) + 2(f + gp)p, \ = c(p - \alpha)(p - \beta)(p - \gamma), \ = 0.$$

V.

The particular case of the equation

$$y(p^2-1)+2mxp=0$$

is obtained from the general equation by writing therein $b=0$, $c=1$, $g=0$, $f=m$; we have therefore

$$v(v^2+2m-1)=(v-\alpha)(v-\beta)(v-\gamma),$$

or say

$$\alpha=0,\quad \beta=i\sqrt{2m-1},\quad \gamma=-i\sqrt{2m-1};$$

$$\frac{v^2-1}{v(v^2+2m-1)}=\frac{A}{v}+\frac{B}{v-i\sqrt{2m-1}}+\frac{C}{v+i\sqrt{2m-1}},$$

giving

$$A=-\frac{1}{2m-1},\quad B=C=\frac{m}{2m-1}.$$

The integral equation thus is

$$\text{const.}=(mx-\sqrt{\square})^{-1}(mx+\sqrt{\square})^{\frac{2m}{2m-1}}\,\{(mx+i\sqrt{2m-1}\,y+\sqrt{\square})(mx-i\sqrt{2m-1}\,y+\sqrt{\square})\}^{\frac{m-1}{2m-1}},$$

where $\square=m^2x^2+y^2$; or, observing that

$$\begin{aligned}(mx+i\sqrt{2m-1}\,y+\sqrt{\square})(mx-i\sqrt{2m-1}\,y+\sqrt{\square})&=(mx+\sqrt{\square})^2+(2m-1)\,y^2\\&=2m(mx^2+y^2+x\sqrt{\square}),\end{aligned}$$

the integral equation is

$$\text{const.}=(mx+\sqrt{\square})^{\frac{1}{2m-1}}(mx^2+y^2+x\sqrt{\square})^{\frac{m-1}{2m-1}},$$

or, what is the same thing,

$$\text{const.}=(mx+\sqrt{\square})(mx^2+y^2+x\sqrt{\square})^{m-1},$$

the result given in the former part of the present paper.

VI.

I annex the following *à posteriori* verification of the solution

$$\text{const.}=(mx+\sqrt{\square})(mx^2+y^2+x\sqrt{\square})^{m-1}$$

of the particular equation

$$y(p^2-1)+2mxp=0.$$

Putting for shortness

$$A = mx + \sqrt{\square}, \quad B = mx^2 + y^2 + x\sqrt{\square},$$

where it will be remembered that

$$\square = m^2x^2 + y^2, \quad 2mB = A^2 + (2m-1)\,y^2.$$

The integral equation may be written

$$h = P + Q\sqrt{\square} = U = AB^{m-1};$$

and we have

$$\frac{U'}{U} = \frac{A'}{A} + (m-1)\frac{B'}{B} = \frac{\Theta}{AB},$$

if

$$\Theta = \frac{A'}{A}AB + (m-1)\frac{B'}{B}AB,$$

$$= \frac{1}{AB}\{A' \cdot AB + (m-1)B' \cdot AB \cdot \tfrac{1}{B} \cdot A\}.$$

But we have

$$A'\sqrt{\square} = m\sqrt{\square} + m^2x + yp = mA + yp,$$

and

$$\begin{aligned} B'\sqrt{\square} &= (2mx + 2yp)\sqrt{\square} + \square + x(m^2x + yp) \\ &= 2m^2x^2 + y^2 + xyp + (2mx + 2yp)\sqrt{\square} \\ &= A^2 + pyx + 2\sqrt{\square} \cdot \text{(terms as in original)}, \end{aligned}$$

and the value of Θ thus is

$$\Theta = \frac{1}{\sqrt{\square}}\left\{A^2B \cdot \frac{1}{A} + (m-1) \cdot \frac{A^2 + py(x + 2\sqrt{\square})}{B} \cdot A\right\}.$$

$$= \frac{1}{2m\sqrt{\square}}\Big\{[A^2 + (2m-1)\,y^2]\,(mA + yp) + (A^2 - 2m)\,[A^2 + py(x + 2\sqrt{\square})]\Big\},$$

where the expression in $\{\ \}$ is

$$= (2m^2 - m)\,A\,(A^2 + y^2) + yp\{A^2 + (2m-1)y^2 + (2m^2 - 2m)\,A(x + 2\sqrt{\square})\}.$$

Here the coefficient of yp is $= (2m^2 - m)(A^2 + y^2)$; in fact we have identically

$$A^2 + y^2 - 2A\sqrt{\square} = 0,$$

and thence

$$(2m^2 - 3m + 1)(A^2 + y^2) - 2(2m-1)(m-1)\,A\sqrt{\square} = 0,$$

that is

$$(2m^2 - m - 1)\,A^2 + (2m^2 - 3m + 1)\,y^2 - (2m-2)\,A\{A^2 + (2m-1)\sqrt{\square}\} = 0;$$

[Continues on p. 126; this transcript ends mid-derivation at p. 125.]

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330. On Differential Equations and Umbilici *(continued)*

[Continuation from p. 125, mid-derivation in section VI.]

or

$$(2m^2 - m - 1)\,A^2 + (2m^2 - 3m + 1)\,y^2 - (2m^2 - 2m)\,A\,(x + 2\sqrt{\Box}) = 0,$$

and therefore

$$A^2 + (2m - 1)\,y^2 + (2m^2 - 2m)\,A\,(x + 2\sqrt{\Box}) = (2m^2 - m)(A^2 + y^2).$$

Hence the term in { } is

$$= (2m^2 - m)(A^2 + y^2)(A + yp);$$

or, what is the same thing, it is $= (4m^2-2m)\,A\sqrt{\Box}\,(A+yp)$. Hence, restoring for $A^2+(2m-1)\,y^2$ its value $2mB$, we find

$$\Theta = \frac{(2m-1)\,B\sqrt{\Box}}{A}\,(A + yp),$$

$$\frac{U'}{U} = \frac{2m-1}{B}\,(A + yp).$$

But, writing U_1, U_2 to denote the values corresponding to $+\sqrt{\Box}$, $-\sqrt{\Box}$ respectively, we have

$$U_1' = \frac{(2m-1)\,U_1}{B_1}\,(mx + yp + \sqrt{\Box}),$$

$$U_2' = \frac{(2m-1)\,U_2}{B_2}\,(mx + yp - \sqrt{\Box});$$

and thence

$$U_1'\,U_2' = \frac{(2m-1)^2\,U_1U_2}{B_1B_2}\,\big\{(mx + yp)^2 - \Box\big\}.$$

But we have

$$U' = P' + Q'\sqrt{\Box} + \frac{Q\Box'}{2\sqrt{\Box}} = \frac{1}{2\sqrt{\Box}}\,(2Q'\Box + Q\Box' + 2P'\sqrt{\Box}\,),$$

and thence

$$U_1'\,U_2' = -\frac{1}{4\Box}\,\big\{(2Q'\Box + Q\Box')^2 - 4P'^2\Box\big\},$$

and moreover

$$U_1\,U_2 = P^2 - Q^2\Box = A_1A_2\,(B_1B_2)^{m-1},$$

where

$$A_1A_2 = m^2x^2 - \Box = -y^2,$$

$$B_1B_2 = (mx^2 + y^2)^2 - x^2\Box = y^2\{y^2 + (2m-1)\,x^2\};$$

and we thence find

$$-\tfrac{1}{4}\Box\,\big\{(2Q'\Box + Q\Box')^2 - 4P'^2\Box\big\} = (2m-1)^2\,A_1A_2\,(B_1B_2)^{m-1}\{y(p^2-1) + 2mxp\}$$

$$= -(2m-1)^2\,y^{2m}\big[y^2 + (2m-1)\,x^2\big]^{m-1}\{y(p^2-1) + 2mxp\}.$$

Hence, the derived equation being

$$\Omega = Q^2\big\{(2Q'\Box + Q\Box')^2 - 4P'^2\Box\big\} = 0,$$

the last preceding equation becomes

$$Q^2\,\Box\,y^{2m-1}\big[y^2 + (2m-1)\,x^2\big]^{m-1}\{y(p^2-1) + 2mxp\} = 0.$$

Here, besides the factor Q^2 corresponding to the nodal curve, and the factor $\Box$ corresponding to the cuspidal curve, we have the factors y^{2m-1} and $[y^2 + (2m-1)\,x^2]^{m-1}$; and, rejecting all these, the differential equation in its reduced form is

$$y(p^2-1) + 2mxp = 0;$$

and the required verification is effected. The occurrence of

$$Q^2\,\Box\,y^{2m-1}\big[y^2 + (2m-1)\,x^2\big]^{m-1}$$

as a factor in the complete derived equation would give rise to some further investigations, but I will not now enter on them.

I remark however that if $m = 1$, viz. if the integral equation be const. $= x + \sqrt{x^2+y^2}$, or say $z = x + \sqrt{x^2+y^2}$, or, what is the same thing,

$$z^2 - 2zx - y^2 = 0,$$

then observing that $y^2 + (2m-1)\,x^2$ is here $= x^2 + y^2$ which is $= \Box$, so that

$$\Box\,[y^2 + (2m-1)\,x^2]^{m-1} = \Box\cdot\Box^0 = 1,$$

the differential equation in its complete form is

$$y(p^2y + 2px - y) = 0;$$

so that we have here the factor y which divides out. The last-mentioned result is most readily obtained directly from the equation

$$\Omega = Q^2\big\{(2Q'\Box + Q\Box')^2 - 4P'^2\Box\big\} = 0,$$

which is the derived equation corresponding to the integral equation $z = P + Q\sqrt{\Box}$. We in fact have $P = x$, $Q = 1$, $\Box = x^2 + y^2$, and the derived equation thus is

$$(x + yp)^2 - (x^2 + y^2) = 0,$$

that is, $y(p^2y + 2px - y) = 0$.

I mention also, in connexion with the foregoing investigation, the integral equation

$$z = x + \sqrt{2x^2 - y^2}, \quad \text{or} \quad z^2 - 2zx - x^2 + y^2 = 0,$$

for which the derived equation in its complete form is

$$(2x - yp)^2 - (2x^2 - y^2) = 0,$$

or, what is the same thing, $y^2p^2 - 4xyp + 2x^2 + y^2 = 0$, and for which therefore there is no factor to divide out.

VII.

The conics confocal with a given conic form a system similar in its properties to that of the curves of curvature of a quadric surface; and the theory of the last-mentioned system may be studied by means of the system of confocal conics. Consider then the equation

$$\frac{x^2}{a^2+z}+\frac{y^2}{b^2+z}=1,$$

which, if z be an arbitrary parameter, belongs to the conics confocal with the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Treating z as a coordinate, the equation represents a surface of the third order, which is such that its section by any plane parallel to the plane of xy is a conic; and the confocal conics are the projections on the plane of xy, by lines parallel to the axis of z, of the sections of the surface.

The sections by the planes of zx, zy are the parabolas $x^2=z+a^2$ and $y^2=z+b^2$ respectively. When $z>-b^2$, the ordinates in each parabola are real, and these ordinates give the semiaxes of the elliptic section. When $z>-a^2$, $<-b^2$, then only the parabola section in the plane of zx has a real ordinate, and the sections are hyperbolic; and when $z<-a^2$, the section is altogether imaginary. The section in the plane $z=-b^2$ is the pair of coincident lines $y^2=0$, $z=-b^2$, and the section in the plane $z=-a^2$ is the pair of coincident lines $z=-a^2$, $x^2=0$; or, in other words, the plane $z+b^2=0$ touches the surface along the line $y=0$, and the plane $z+a^2=0$ touches the surface along the line $x=0$: this at once appears from the integral form

$$(z+a^2)(z+b^2)-x^2(z+b^2)-y^2(z+a^2)=0.$$

The points $(z=-b^2,\ y=0,\ x=\pm\sqrt{a^2-b^2})$ and $(z=-a^2,\ x=0,\ y=\pm\sqrt{b^2-a^2})$ are conical points; the last two are however imaginary points on the surface. To find the nature of the surface about one of the first-mentioned two points, say the point $(z=-b^2,\ y=0,\ x=\sqrt{a^2-b^2})$, taking this point for the origin and writing therefore $\sqrt{a^2-b^2}+x$, y and $-b^2+z$ in the place of x, y, z respectively, the equation becomes

$$(a^2-b^2+z)\,z-\left\{(a^2-b^2)+2x\sqrt{a^2-b^2}+x^2\right\}z-(a^2-b^2+z)\,y^2=0,$$

that is

$$z^2-2zx\sqrt{a^2-b^2}-(a^2-b^2)\,y^2-z(x^2+y^2)=0;$$

so that there is a tangent cone the equation whereof is

$$z^2-2zx\sqrt{a^2-b^2}-(a^2-b^2)\,y^2=0,$$

or, as it may be written,

$$(z-x\sqrt{a^2-b^2})^2-(a^2-b^2)(x^2+y^2)=0.$$

The equation is that of a cone of the second order, meeting the plane of zx in the lines $z = 0$, $z = 2x\sqrt{a^2 - b^2}$ (and therefore such that its sections parallel to the plane of xy are parabolas), and meeting the plane of yz in the lines $z = \pm y\sqrt{a^2 - b^2}$ (the origin being at the vertex of the cone or conical point of the surface).

Returning to the original origin, and to the equation of the surface written in the form

$$z^2 + z(a^2 + b^2 - x^2 - y^2) + a^2b^2 - b^2x^2 - a^2y^2 = 0,$$

calling this for a moment $z^2+2Bz+C = 0$, the differential equation is $C'^2-4BB'C+4CB'^2 = 0$; or, substituting, this is

$$(b^2x + a^2yp)^2 - (a^2 + b^2 - x^2 - y^2)(x + yp)(b^2x + a^2yp) + (a^2b^2 - b^2x^2 - a^2y^2)(x + yp)^2 = 0;$$

or, reducing, this is

$$(a^2 - b^2)\, xy \left\{xy(p^2 - 1) - (a^2 - b^2 - x^2 + y^2)\, p\right\} = 0,$$

or say

$$xy \left\{xy(p^2 - 1) - (a^2 - b^2 - x^2 + y^2)\, p\right\} = 0,$$

where the factor xy arises from the level lines $(z + b^2 = 0,\ y = 0)$ and $(z + a^2 = 0,\ x = 0)$. Throwing out this factor, the equation becomes

$$xy(p^2 - 1) - (a^2 - b^2 - x^2 + y^2)\, p = 0,$$

which is satisfied identically by $z + b^2 = 0$, $y = 0$, $x^2 = a^2 - b^2$. The first derived equation is

$$(xp + y)(p^2 - 1) + 2(x - yp)\, p = 0,$$

which for the values in question gives

$$p(p^2 + 1) = 0,$$

where the factor $p = 0$ corresponds to the section $y = 0$ by the plane $z + b^2 = 0$: and taking the conical point for origin, and observing that the polar of the line $x = 0$, $y = 0$ in regard to the tangent cone is $z - x\sqrt{a^2 - b^2} = 0$, then writing the equation of the tangent cone in the form

$$(z - x\sqrt{a^2 - b^2})^2 - (a^2 - b^2)(x^2 + y^2) = 0,$$

the two tangent planes through $(x = 0,\ y = 0)$ are given by the equation $x^2 + y^2 = 0$; and for these planes we have $p^2 + 1 = 0$. The factor $p^2 + 1 = 0$ determines therefore the directions of the envelope at the conical point.

VIII.

In verification of the equation

$$z = \tfrac{1}{2}k(x^2 + y^2) + \tfrac{1}{6}\, ax(x^2 + y^2)$$

for a quadric surface in the neighbourhood of the umbilicus, I remark that, starting from the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

of an ellipsoid, and taking $\alpha,\ 0,\ \gamma$ as the coordinates of the umbilicus, and θ as the inclination to the axis of x of the tangent to the principal section through the umbilicus, then transforming to the umbilicus as origin and the new axes through that point, viz. the axes of x, z being the tangent and normal in the plane of αz, and the axis of y being at right angles to this (or in the direction of b), the equation becomes

$$\frac{(\alpha + x\cos\theta - z\sin\theta)^2}{a^2} + \frac{y^2}{b^2} + \frac{(\gamma - x\sin\theta - z\cos\theta)^2}{c^2} = 1,$$

or, expanding,

$$\Big(\frac{\alpha^2}{a^2} + \frac{\gamma^2}{c^2} - 1\Big) + 2x\Big(\frac{\alpha\cos\theta}{a^2} - \frac{\gamma\sin\theta}{c^2}\Big) - 2z\Big(\frac{\alpha\sin\theta}{a^2} + \frac{\gamma\cos\theta}{c^2}\Big)$$

$$+ x^2\Big(\frac{\cos^2\theta}{a^2} + \frac{\sin^2\theta}{c^2}\Big) + \frac{y^2}{b^2} + z^2\Big(\frac{\sin^2\theta}{a^2} + \frac{\cos^2\theta}{c^2}\Big) - 2xz\sin\theta\cos\theta\Big(\frac{1}{a^2} - \frac{1}{c^2}\Big) = 0.$$

But we have

$$\alpha = a\sqrt{\frac{a^2-b^2}{a^2-c^2}}, \quad \gamma = \frac{c\sqrt{b^2-c^2}}{\sqrt{a^2-c^2}},$$

$$\tan\theta = \frac{c\sqrt{a^2-b^2}}{a\sqrt{b^2-c^2}},$$

and thence

$$\sin\theta = \frac{c\sqrt{a^2-b^2}}{b\sqrt{a^2-c^2}}, \quad \cos\theta = \frac{a\sqrt{b^2-c^2}}{b\sqrt{a^2-c^2}};$$

and substituting these values, the equation becomes

$$-\frac{2zb}{ca}\sqrt{a^2-c^2} + \frac{x^2}{b^2} + \frac{y^2}{b^2} + z^2\,\frac{a^2b^2+c^2b^2-a^2c^2}{a^2b^2c^2} - \frac{xz(a^2-c^2)(b^2-c^2+b^2)}{a^2c^2b^2}\,(\dots) = 0,$$

or, what is the same thing,

$$z = \frac{ca}{b\sqrt{a^2-c^2}}\cdot\tfrac{1}{2}(x^2+y^2) + \tfrac{1}{2}\big\{\text{terms in } x^2z,\ y^2z,\ z^2,\ xz\big\} + \cdots,$$

whence approximately

$$z = \tfrac{1}{2}\,\frac{ca}{b\sqrt{a^2-c^2}}\,(x^2+y^2),$$

and thence to the third order in x, y,

$$z = \tfrac{1}{2}\,\frac{ca}{b\sqrt{a^2-c^2}}\,(x^2+y^2) + \tfrac{1}{6}\,ax(x^2+y^2),$$

which is of the form in question.

5, Downing Terrace, Cambridge, November 2, 1863.

331. Analytical Theorem Relating to the Four Conics Inscribed in the Same Conic and Passing through the Same Three Points

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 42, 43.]

Imagine the four conics determined, and, selecting at pleasure any three of them, let their chords of contact with the given conic be taken for the axes of coordinates, or lines $x = 0$, $y = 0$, $z = 0$; then, taking for the equation of the given conic

$$U = (a,\ b,\ c,\ f,\ g,\ h \mid x,\ y,\ z)^2 = 0,$$

the equations of the selected three conics must be of the form $U + lx^2 = 0$, $U + my^2 = 0$, $U + nz^2 = 0$, where l, m, n are to be determined in such manner that these conics may have three common points; the resulting values of l, m, n, and of the coordinates of the three common points, that is, the three given points, will of course be functions of the coefficients $(a,\ b,\ c,\ f,\ g,\ h)$; and the equation of the fourth conic will be of the form $U+(ix+jy+kz)^2 = 0$.

There is no difficulty in carrying out the investigation: it is found that the coordinates of the given points must be taken to be

$$(-f,\ g,\ h); \quad (f,\ -g,\ h); \quad (f,\ g,\ -h)$$

respectively, and that, writing as usual

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

the equations of the four conics are

$$U + (K - abc)\,\frac{x^2}{f^2} = 0,$$

$$U + (K - abc)\,\frac{y^2}{g^2} = 0,$$

$$U + (K - abc)\,\frac{z^2}{h^2} = 0,$$

$$U + (K - abc)\Big(\frac{x}{f} + \frac{y}{g} + \frac{z}{h}\Big)^2 = 0.$$

It is in fact easy to verify directly that each of these conics passes through the three given points; but the equations may also be exhibited in the form proper for putting this in evidence. Putting for shortness

$$X = \frac{y}{g} + \frac{z}{h}, \quad Y = \frac{z}{h} + \frac{x}{f}, \quad Z = \frac{x}{f} + \frac{y}{g},$$

the equations of the sides of the triangle formed by the given points are $X = 0$, $Y = 0$, $Z = 0$, and the foregoing equations of the four conics may be expressed in the form

$$\begin{aligned}
(-bg^2 - ch^2 + 2fgh)\,YZ \ + \ bg^2\,ZX \ + \ ch^2\,XY &= 0,\\
af^2\,YZ \ + \ (-ch^2 - af^2 + 2fgh)\,ZX \ + \ ch^2\,XY &= 0,\\
af^2\,YZ \ + \ bg^2\,ZX \ + \ (-af^2 - bg^2 + 2fgh)\,XY &= 0,\\
(-bg^2 - ch^2 + 2fgh)\,YZ \ + \ (-ch^2 - af^2 + 2fgh)\,ZX \ + \ (-af^2 - bg^2 + 2fgh)\,XY &= 0,
\end{aligned}$$

which is the required form.

Cambridge, November 28, 1863.

332. Analytical Theorem Relating to the Sections of a Quadric Surface

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 43, 44.]

The four sections $x = 0$, $y = 0$, $z = 0$, $w = 0$ of the quadric surface

$$ax^2 + by^2 + 6xy\sqrt{ab} - cz^2 - dw^2 = 0$$

are each of them touched by each of the four sections

$$x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0;$$

where it is to be noticed that the radicals $\sqrt{2a}$, $\sqrt{2b}$ are such that their product is $= 2\sqrt{ab}$ if $\sqrt{ab}$ be the radical contained in the equation of the surface. There is of course no loss of generality in attributing a definite sign to the radical $\sqrt{2a}$; but upon this being done, the sign of the radical $\sqrt{2b}$ is determined, whereas the signs of $\sqrt{c}$ and $\sqrt{d}$ are severally arbitrary. We may if we please write the equation of any one of the last-mentioned sections in the form

$$x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0,$$

it being understood that the radicals $\sqrt{2a}$, $\sqrt{2b}$ have each a determinate sign, but that the signs of $\sqrt{c}$ and $\sqrt{d}$ are each of them arbitrary.

To prove the theorem in question, it is enough to show

(1) that the sections $x = 0$, $x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0$;

(2) that the sections $z = 0$, $x\sqrt{2a} + y\sqrt{2b} + w\sqrt{d} = 0$,

touch each other.

1. The sections $x = 0$, $x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0$ of the quadric surface $ax^2 + by^2 + 6xy\sqrt{ab} - cz^2 - dw^2 = 0$ will touch each other if, combining together the equations

$$x = 0, \quad y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0, \quad by^2 - cz^2 - dw^2 = 0,$$

these give a twofold value (pair of equal values) for the ratios $y : z : w$. We in fact have

$$\begin{aligned} by^2 - cz^2 - dw^2 &= by^2 - cz^2 - (y\sqrt{2b} + z\sqrt{c}\,)^2 \\ &= -by^2 - 2cz^2 - 2yz\sqrt{2bc} \\ &= -(y\sqrt{b} + z\sqrt{2c}\,)^2; \end{aligned}$$

and the right-hand side being a perfect square, the condition of contact is satisfied.

2. In like manner we have the system

$$z = 0, \quad x\sqrt{2a} + y\sqrt{2b} + w\sqrt{d} = 0, \quad ax^2 + by^2 + 6xy\sqrt{ab} - dw^2 = 0,$$

which gives

$$\begin{aligned} ax^2 + by^2 + 6xy\sqrt{ab} - dw^2 &= ax^2 + by^2 + 6xy\sqrt{ab} - (x\sqrt{2a} + y\sqrt{2b}\,)^2 \\ &= -ax^2 - by^2 + 2xy\sqrt{ab} \\ &= -(x\sqrt{a} - y\sqrt{b}\,)^2; \end{aligned}$$

and here also, the right-hand side being a perfect square, the condition of contact is satisfied.

Cambridge, November 28, 1863.

333. Note on the Nodal Curve of the Developable Derived from the Quartic Equation $(a,\, b,\, c,\, d,\, e \mid t,\, 1)^4 = 0$

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 437–440.]

Considering the coefficients (a, b, c, d, e) as linear functions of the coordinates x, y, z, w, then the equation

$$\text{Disct.}\,(a,\, b,\, c,\, d,\, e \mid t,\, 1)^4 = 0,$$

or, as it may be written,

$$(ae - 4bd + 3c^2)^3 - 27\,(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0$$

represents, as is known, a developable surface or "torse," having for its edge of regression (or cuspidal curve) the sextic curve the equations whereof are

$$\begin{aligned} ae - 4bd + 3c^2 &= 0, \\ ace + 2bcd - ad^2 - b^2e - c^3 &= 0; \end{aligned}$$

and for its nodal curve, a curve the equations whereof (equivalent to two independent relations between the coordinates) are

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

or, as these may also be written,

$$\begin{aligned} a^2d - 3abc + 2b^3 &= 0, \\ a^2e + 2abd - 9ac^2 + 6b^2c &= 0, \\ abe - 3acd + 2b^2d &= 0, \\ ad^2 - b^2e &= 0, \\ ade - 3bce + 2bd^2 &= 0, \\ ae^2 + 2bde - 9c^2e + 6cd^2 &= 0, \\ be^2 - 3cde + 2d^3 &= 0; \end{aligned}$$

which curve is in fact an excubo-quartic, — viz. a quartic curve the partial intersection of a quadric surface and a cubic surface, having in common two non-intersecting right lines. To show that this is so, I remark that the coefficients a, b, c, d, e, quâ linear functions of the four coordinates, satisfy a linear equation which may be taken to be

$$a + b + c + d + e = 0;$$

this being so, the first form shows that the curve in question lies on the quadric surface

$$ac - b^2 + \tfrac{1}{2}(ad - bc) + \tfrac{1}{6}(ae + 2bd - 3c^2) + \tfrac{1}{2}(be - cd) + ce - d^2 = 0,$$

or, as this equation may also be written,

$$c(a + b + c + d + e) - b^2 + \tfrac{1}{2}ad + \tfrac{1}{6}(ae + 2bd) + \tfrac{1}{2}be - d^2 = 0.$$

Substituting for c its value, this equation is

$$-(a + e + b + d)(\tfrac{1}{2}a + \tfrac{1}{2}e) - b^2 + \tfrac{1}{2}ad + \tfrac{1}{6}(ae + 2bd) + \tfrac{1}{2}be - d^2 = 0,$$

or, what is the same thing,

$$9(a+e+b+d)(a+e)+6(b^2+d^2)-3(ad+be)-(ae+2bd)=0.$$

Hence, finally, the equation of the quadric surface is

$$9a^2+17ae+9e^2+6b^2-2bd+6d^2+9ab+9de+6ad+6be=0;$$

and the curve lies also on the cubic surface

$$ad^2-b^2e=0.$$

It only remains to show that these surfaces have in common two right lines, and to find the equations of these lines.

The cubic surface is a skew surface or "scroll" such that the equations of any generating line are $d-\theta b=0$, $e-\theta^2 a=0$, where θ is an arbitrary parameter. But considering the two lines

$$(d-\theta_1 b=0,\ e-\theta_1^2 a=0),\quad (d-\theta_2 b=0,\ e-\theta_2^2 a=0),$$

the general equation of the quadric surface through these two lines may be written

$$\begin{aligned} &A\,(d-\theta_1 b)(d-\theta_2 b)\\ &+B\,(e-\theta_1^2 a)(e-\theta_2^2 a)\\ &+C\left\{(d-\theta_1 b)(e-\theta_2^2 a)+(d-\theta_2 b)(e-\theta_1^2 a)\right\}\\ &+\frac{D}{\theta_1-\theta_2}\left\{(d-\theta_1 b)(e-\theta_2^2 a)-(d-\theta_2 b)(e-\theta_1^2 a)\right\}=0,\end{aligned}$$

or, expanding and reducing,

$$\begin{aligned} &A\left\{d^2-(\theta_1+\theta_2)\,bd+\theta_1\theta_2\,b^2\right\}\\ &+B\left\{e^2-(\theta_1^2+\theta_2^2)\,ea+\theta_1^2\theta_2^2\,a^2\right\}\\ &+C\left\{2de-(\theta_1^2+\theta_2^2)\,ad-(\theta_1+\theta_2)\,be+\theta_1\theta_2(\theta_1+\theta_2)\,ab\right\}\\ &+D\left\{(\theta_1+\theta_2)\,ad-be-\theta_1\theta_2\,ab\right\}=0,\end{aligned}$$

which, if θ_1, θ_2 are the roots of the equation $\theta^2-\frac{1}{3}\theta+1=0$, and therefore $\theta_1+\theta_2=\frac{1}{3}$, $\theta_1\theta_2=1$, and $\theta_1^2+\theta_2^2=-\frac{17}{9}$, is

$$\begin{aligned} &A\left(d^2-\tfrac{1}{3}\,db+b^2\right)\\ &+B\left(e^2+\tfrac{17}{9}\,ae+a^2\right)\\ &+C\left(2de+\tfrac{17}{9}\,ad-\tfrac{1}{3}\,be+\tfrac{1}{3}\,ab\right)\\ &+D\left(\tfrac{1}{3}\,ad-be-ab\right)=0.\end{aligned}$$

Putting $A=6$, $B=9$, $C=\frac{3}{2}$, $D=-\frac{9}{2}$, this is

$$\begin{aligned} &9\,(a^2+\tfrac{17}{9}\,ae+e^2)\\ &+6\,(b^2-\tfrac{1}{3}\,bd+d^2)\\ &+\tfrac{3}{2}\,(\tfrac{1}{3}\,ab+2de+\tfrac{17}{9}\,ad-\tfrac{1}{3}\,be)\\ &+\tfrac{9}{2}\,(ab-\tfrac{1}{3}\,ad+be)=0,\end{aligned}$$

which is the before-mentioned quadric surface; hence the quadric surface and the cubic surface intersect in the two lines

$$(d-\theta_1 b=0,\ e-\theta_1^2 a=0),\quad (d-\theta_2 b=0,\ e-\theta_2^2 a=0)$$

(where θ_1, θ_2 are the roots of the quadratic equation $\theta^2-\frac{1}{3}\theta+1=0$); and they consequently intersect also in an excubo-quartic curve, which is the theorem required to be proved.

Blackheath, March 26, 1864.

334. Note on the Theory of Cubic Surfaces

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 493–496.]

The equation

$$AX^3 + BY^3 + 6\,CRST = 0,$$

where $X + Y + R + S + T = 0$, represents a cubic surface of a special form, viz. each of the planes $R = 0$, $S = 0$, $T = 0$ is a triple tangent plane meeting the surface in three lines which pass through a point[1]; and, moreover, the three planes $AX^3 + BY^3 = 0$ are triple tangent planes intersecting in a line. It is worth noticing that the equation of the surface may also be written

$$ax^3 + by^3 + c(u^3 + v^3 + w^3) = 0,$$

where $x + y + u + v + w = 0$. In fact, the coordinates satisfying the foregoing linear equations respectively, we have to show that the equation

$$AX^3 + BY^3 + 6\,CRST = ax^3 + by^3 + c(u^3 + v^3 + w^3)$$

may be identically satisfied. We have

$$\begin{aligned} ax^3 + by^3 + c\,(u^3 + v^3 + w^3) &= ax^3 + by^3 + c\big[(u + v + w)^3 - 3(v + w)(w + u)(u + v)\big] \\ &= ax^3 + by^3 - c(x + y)^3 - 3c(v + w)(w + u)(u + v), \end{aligned}$$

which is to be

$$= AX^3 + BY^3 + 6\,CRST;$$

and we may find X, Y, R, S, T, linear functions of x, y, u, v, w, so as to satisfy these equations, and so that in virtue of

$$x + y + u + v + w = 0,$$

we shall have also $X + Y + R + S + T = 0$. For, assuming

$$\begin{aligned} AX^3 + BY^3 &= ax^3 + by^3 - c(x + y)^3, \\ X + Y &= x + y, \\ R &= \tfrac{1}{2}(v + w), \quad C = -4c, \\ S &= \tfrac{1}{2}(w + u), \\ T &= \tfrac{1}{2}(u + v), \end{aligned}$$

we have identically

$$\begin{aligned} AX^3 + BY^3 + 6\,CRST &= ax^3 + by^3 - c(x + y)^3 - 3c(v + w)(w + u)(u + v), \\ X + Y + R + S + T &= x + y + u + v + w, \end{aligned}$$

and thus it only remains to show that we can find X, Y linear functions of x, y, such that

$$\begin{aligned} AX^3 + BY^3 &= ax^3 + by^3 - c(x + y)^3, \\ X + Y &= x + y. \end{aligned}$$

[1] The tangent plane of a surface intersects the surface in a curve having at the point of contact a double point, and in like manner a triple tangent plane intersects the surface in a curve with three double points, viz. each point of contact is a double point; there is not in general any triple tangent plane such that the three points of contact come together, or (what is the same thing) there is not in general any tangent plane intersecting the surface in a curve having at the point of contact a triple point. A surface may, however, have the kind of singularity just referred to, viz. a tangent plane intersecting the surface in a curve having at the point of contact a triple point; such tangent plane may be termed a 'tritom' tangent plane, and its point of contact a 'tritom' point: for a cubic surface the intersection by a tritom tangent plane is of course a system of three lines meeting in the tritom point. The tritom singularity is sibi-reciprocal; it is, I think, a singularity which should be considered in the theory of reciprocal surfaces.

This is always possible; in fact if

$$U = ax^3 + by^3 - c(x+y)^3,$$

then taking Φ for the cubicovariant, and $\Box$ for the discriminant of U, we have $\frac{1}{2}(\Phi + \sqrt{\Box}\,U)$, $\frac{1}{2}(\Phi - \sqrt{\Box}\,U)$ each a perfect cube, say

$$\tfrac{1}{2}(\Phi + \sqrt{\Box}\,U) = (\lambda x + \mu y)^3, \quad \tfrac{1}{2}(\Phi - \sqrt{\Box}\,U) = (\nu x + \rho y)^3;$$

and we then have

$$U = \tfrac{1}{\sqrt{\Box}}\big[(\lambda x + \mu y)^3 - (\nu x + \rho y)^3\big] = AX^3 + BY^3,$$

which is satisfied by

$$X = l(\lambda x + \mu y), \quad Y = m(\nu x + \rho y),$$

if

$$Al^3 = \frac{1}{\sqrt{\Box}}, \quad Bm^3 = -\frac{1}{\sqrt{\Box}}.$$

The equation $X+Y=x+y$ then gives

$$\begin{aligned} l\lambda + m\nu &= 1, \\ l\mu + m\rho &= 1, \end{aligned}$$

which give the values of l and m, and thence the values of A and B; and collecting all the equations, we have

$$\begin{aligned} &X = \frac{\rho-\nu}{\lambda\rho-\mu\nu}\,(\lambda x+\mu y), \quad A = \frac{1}{\sqrt{\square}}\Big(\frac{\lambda\rho-\mu\nu}{\rho-\nu}\Big)^3, \\ &Y = -\frac{\mu-\lambda}{\lambda\rho-\mu\nu}\,(\nu x+\rho y), \quad B = \frac{1}{\sqrt{\square}}\Big(\frac{\lambda\rho-\mu\nu}{\mu-\lambda}\Big)^3, \\ &R = \tfrac{1}{2}(v+w), \quad C = -4c, \\ &S = \tfrac{1}{2}(w+u), \\ &T = \tfrac{1}{2}(u+v), \end{aligned}$$

where

$$\lambda x+\mu y = \big[\tfrac{1}{2}(\Phi+\sqrt{\square}\,U)\big]^{\frac{1}{3}}, \quad \nu x+\rho y = \big[\tfrac{1}{2}(\Phi-\sqrt{\square}\,U)\big]^{\frac{1}{3}},$$

(Φ, $\square$ being respectively the cubicovariant and the discriminant of $U=ax^3+by^3-c(x+y)^3$), for the formulae of the transformation

$$\begin{aligned} AX^3+BY^3+6\,CRST &= ax^3+by^3+c(u^3+v^3+w^3), \\ X+Y+R+S+T &= x+y+u+v+w. \end{aligned}$$

The equation $ax^3+by^3+c(u^3+v^3+w^3)=0$, where $x+y+u+v+w=0$, presents over the other form the advantage that it is included as a particular case under the equation $ax^3+by^3+cu^3+dv^3+ew^3=0$ (where $x+y+u+v+w=0$) employed by Dr Salmon as the canonical form of equation for the general cubic surface.

5, Downing Terrace, Cambridge, April 29, 1864.

335. Tables des Formes Quadratiques Binaires pour les Déterminants Négatifs depuis $D = -1$ jusqu'à $D = -100$, pour les Déterminants Positifs non Carrés depuis $D = 2$ jusqu'à $D = 99$, et pour les Treize Déterminants Négatifs Irréguliers qui se trouvent dans le Premier Millier

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LX. (1862), pp. 357–372.]

Les tables suivantes sont arrangées de la manière prescrite dans les "*Disquisitiones arithmeticae.*" Dans le mémoire de Lejeune Dirichlet "Recherches sur diverses applications de l'analyse à la théorie des nombres," tom. XIX (1839), p. 338 de ce Journal, on trouve un tableau dans lequel les règles qui servent à former les caractères des genres sont résumées. Soit $D = PS^2$ ou $2PS^2$, S^2 désignant le plus grand carré que D contient, et P un nombre impair; soient de plus p, p', $p'' \ldots$ les facteurs premiers inégaux de P et r, r', $r'' \ldots$ les nombres premiers impairs qui divisent S sans diviser P; écrivons enfin pour abréger $\delta = (-1)^{\frac{m-1}{2}}$, $\epsilon = (-1)^{\frac{m^2-1}{8}}$. Cela posé on trouve à l'endroit cité le tableau suivant:

Premier cas, $D = PS^2$, $P \equiv 1 \pmod 4$:

$$\begin{array}{r|l|l} S \equiv 1 \pmod 2 & \frac{m}{p},\ \frac{m}{p'},\ \ldots & \\ S \equiv 2 \pmod 4 & \frac{m}{p},\ \frac{m}{p'},\ \ldots & \epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \\ S \equiv 0 \pmod 4 & \frac{m}{p},\ \frac{m}{p'},\ \ldots & \delta,\ \epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \end{array}$$

Deuxième cas, $D = PS^2$, $P \equiv 3 \pmod 4$:

$$\begin{array}{r|l|l} S \equiv 1 \pmod 2 & \delta,\ \frac{m}{p},\ \frac{m}{p'},\ \ldots & \frac{m}{r},\ \frac{m}{r'},\ \ldots \\ S \equiv 2 \pmod 4 & \delta,\ \frac{m}{p},\ \frac{m}{p'},\ \ldots & \delta\epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \\ S \equiv 0 \pmod 4 & \delta,\ \frac{m}{p},\ \frac{m}{p'},\ \ldots & \epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \end{array}$$

Troisième cas, $D = 2PS^2$, $P \equiv 1 \pmod 4$:

$$\begin{array}{r|l|l} S \equiv 1 \pmod 2 & \frac{m}{p},\ \frac{m}{p'},\ \ldots & \epsilon \\ S \equiv 0 \pmod 2 & \frac{m}{p},\ \frac{m}{p'},\ \ldots & \delta\epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \end{array}$$

Quatrième cas, $D = 2PS^2$, $P \equiv 3 \pmod 4$:

$$\begin{array}{r|l|l} S \equiv 1 \pmod 2 & \delta,\ \frac{m}{p},\ \frac{m}{p'},\ \ldots & \delta\epsilon \\ S \equiv 0 \pmod 2 & \delta,\ \frac{m}{p},\ \frac{m}{p'},\ \ldots & \epsilon,\ \frac{m}{r},\ \frac{m}{r'},\ \ldots \end{array}$$

Dans ce tableau la notation $\frac{m}{p}$, dans laquelle j'omets les parenthèses usitées, signifie le caractère d'un nombre quelconque m par rapport au nombre premier impair p, c.-à-d. que m est résidu ou non résidu de p selon que $\frac{m}{p} = +1$ ou $= -1$; de même δ est le caractère de m par rapport au nombre 4, savoir $m \equiv 1$ ou $3 \pmod 4$ selon que $\delta = +1$ ou $= -1$; enfin ϵ, $\delta\epsilon$ sont les caractères de m par rapport au nombre 8, savoir $m \equiv 1$ ou $7 \pmod 8$ pour $\epsilon = +1$, $\equiv 3$ ou $5 \pmod 8$ pour $\epsilon = -1$; $m \equiv 1$ ou $3 \pmod 8$ pour $\delta\epsilon = +1$, $\equiv 5$ ou $7 \pmod 8$ pour $\delta\epsilon = -1$. Si pour un déterminant donné on veut former au moyen de ce tableau les caractères des genres, on prend la ligne horizontale qui convient à ce déterminant; à tous les caractères $\frac{m}{p}$, $\frac{m}{p'}$, $\ldots$ $\frac{m}{r}$, $\frac{m}{r'}$, $\ldots$, δ, ϵ, $\delta\epsilon$ qui se trouvent dans la ligne horizontale, on attribue les signes $+$ ou $-$ à volonté, avec cette restriction cependant que le signe composé des signes qui se trouvent dans la première partie de la ligne dont il s'agit soit positif. Si par exemple le déterminant donné est $D = -35$, on a $D = -35 = PS^2$, $P = -35 \equiv 1 \pmod 4$, $S = 1 \equiv 1 \pmod 2$, les nombres p, p', ... sont 5, 7, et les signes que l'on doit considérer sont $\frac{m}{5}$, $\frac{m}{7}$. De là on obtient les caractères

$\frac{m}{5}$	$\frac{m}{7}$		
$+$	$+$		
$-$	$-$		

il y a donc deux genres de l'ordre proprement primitif. Dans le cas dont il s'agit (et en général pour $D \equiv 1 \pmod 4$) il y a un ordre improprement primitif avec des genres qui ont les caractères identiques à ceux des genres de l'ordre proprement primitif, les caractères se rapportant dans ce cas à la moitié d'un nombre quelconque représenté par la forme.

Pour faciliter l'impression des tables j'ai introduit deux nouvelles lettres α et β dont voici la définition. Pour tous les déterminants auxquels se rapportent mes tables, c.-à-d. pour les déterminants négatifs quelconques et positifs non-carrés depuis -100 jusqu'à $+99$ ainsi que pour les déterminants négatifs irréguliers du premier millier, le nombre des facteurs premiers désignés ci-dessus par les lettres p, p', p'' ..., r, r', r'' ... n'excède pas deux. Soit donc q le plus petit de ces facteurs premiers et q' le plus grand lorsqu'il y en a deux, je désigne par α le caractère $\frac{m}{q}$ et par β le caractère $\frac{m}{q'}$.

Dans la colonne relative à la composition et portant l'inscription Cp, je représente comme à l'ordinaire par l'unité la forme principale, par la lettre c une forme qui produit par la duplication la forme principale, par les lettres d, e, ... des formes qui la produisent par la triplication, la quadruplication, etc., de manière que l'on ait $c^2 = 1$, $d^3 = 1$, $e^4 = 1$, $f^5 = 1$, $g^6 = 1$, $h^7 = 1$, $i^8 = 1$, $j^9 = 1$, etc. Les notations d, d_1 par exemple signifient deux formes différentes dont chacune produit par la triplication la forme principale. Je représente de plus par a la forme principale de l'ordre improprement primitif et par ac, ad, ... des formes qui produisent a par la duplication, la triplication, etc. Dans l'énumération des classes, j'ai toujours écrit en premier lieu l'ordre proprement primitif, en le faisant suivre après un trait de séparation par l'ordre improprement primitif lorsqu'il existe. Dans chacun des deux ordres les divers genres se trouvent séparés les uns des autres par des traits subordonnés.

Pour les déterminants positifs les périodes sont données par une abréviation facile à comprendre. Chaque forme de la période ayant son dernier coefficient égal au premier de la suivante, cette valeur identique n'a été imprimée qu'une fois; de plus les coefficients extérieurs a, c des formes (a, b, c) ont été distingués des coefficients b en imprimant ces derniers en caractères plus petits. Ainsi pour le déterminant 7 la période de la classe principale $(1, 0, -7)$ est donnée par les nombres

$$1,\ 2,\ -3,\ 1,\ 2,\ 1,\ -3,\ 2,\ 1$$

qui représentent la série des formes

$$(1, 2, -3),\ (-3, 1, 2),\ (2, 1, -3),\ (-3, 2, 1).$$

Londres, 6 Novembre, 1860.

Table I. Formes quadratiques binaires ayant pour déterminants les nombres négatifs depuis $D = -1$ jusqu'à $D = -100$.

[Each entry below lists, for the given negative determinant D: the reduced forms (one per class), the genre-characters in the columns α, β, δ, ϵ, $\delta\epsilon$ as required by the case, and the composition symbol Cp. The transcription preserves Cayley's reduced-form coefficients (a, b, c); characters and composition symbols are abbreviated to the table headings used throughout.]

D	Reduced forms (one per class)	Characters	Cp
-1	$(1,0,1)$	$+$	1
-2	$(1,0,2)$	$+$	1
-3	$(1,0,3)$	$+$	1
-4	$(1,0,4);\ (2,1,2)$	$+$	$1,\ c$
-5	$(1,0,5);\ (2,1,3)$	$++$	$1,\ c$
-6	$(1,0,6);\ (2,0,3)$	$++$	$1,\ c$
-7	$(1,0,7)$	$++$	1
-8	$(1,0,8);\ (3,1,3)$	$++$	$1,\ c$
-9	$(1,0,9);\ (2,1,5)$	$++$	$1,\ c$
-10	$(1,0,10);\ (2,0,5)$	$++$	$1,\ c$
-11	$(1,0,11);\ (3,1,4);\ (3,-1,4)$	$+$	$1,\ d,\ d^2$
-12	$(1,0,12);\ (3,0,4);\ (2,1,6)$	$++$	$1,\ c,\ c$
-13	$(1,0,13);\ (2,1,7)$	$++$	$1,\ c$
-14	$(1,0,14);\ (2,0,7);\ (3,1,5);\ (3,-1,5)$	$++$	$1,\ c,\ e,\ e^3$
-15	$(1,0,15);\ (3,0,5);\ (2,1,8)$	$+++$	$1,\ c,\ ac$
-16	$(1,0,16);\ (4,1,4);\ (4,-1,4)$	$++$	$1,\ c,\ c$
-17	$(1,0,17);\ (2,1,9);\ (3,1,6);\ (3,-1,6)$	$++$	$1,\ c,\ e,\ e^3$
-18	$(1,0,18);\ (2,0,9);\ (3,1,6\,\text{var.})$	$++$	$1,\ c$
-19	$(1,0,19);\ (4,1,5);\ (4,-1,5)$	$+$	$1,\ d,\ d^2$
-20	$(1,0,20);\ (4,2,6);\ (2,0,10);\ (5,0,4)$	$++$	$1,\ c,\ c$
-21	$(1,0,21);\ (2,1,11);\ (3,0,7);\ (5,2,5);\ (3,-1,7),\ (5,-2,5)$	$++$	$1,\ c,\ \ldots$
-22	$(1,0,22);\ (2,0,11)$	$++$	$1,\ c$
-23	$(1,0,23);\ (2,1,12);\ (3,1,8);\ (3,-1,8);\ (4,1,6);\ (4,-1,6)$	$+$	$1,\ d,\ d^2,\ d_1,\ \ldots$
-24	$(1,0,24);\ (2,0,12);\ (3,0,8);\ (4,1,7);\ (4,-1,7)$	$++$	$1,\ c,\ c,\ \ldots$
-25	$(1,0,25);\ (5,0,5)$	$++$	$1,\ c$
-26	$(1,0,26);\ (2,0,13);\ (3,1,9);\ (3,-1,9);\ (5,2,6);\ (5,-2,6)$	$++$	$1,\ c,\ \ldots$
-27	$(1,0,27);\ (3,0,9);\ (4,1,7);\ (4,-1,7)$	$+$	$1,\ d,\ d^2$
-28	$(1,0,28);\ (4,0,7);\ (2,1,15);\ (5,2,6);\ (5,-2,6)$	$++$	$1,\ c,\ \ldots$
-29	$(1,0,29);\ (2,1,15);\ (5,1,6);\ (5,-1,6);\ (3,1,10);\ (3,-1,10);\ \ldots$	$++$	$1,\ d,\ \ldots$
-30	$(1,0,30);\ (2,0,15);\ (3,0,10);\ (5,0,6)$	$++$	$1,\ c,\ c,\ c$

[The above is a representative excerpt showing the structure of Table I; the original prints all the reduced forms, characters, and composition symbols for every D from -1 to -100 in three side-by-side columns spread over pp. 143–147. The continuation below abridges the per-D data, listing for each D the class number and a leading reduced form, to convey the bulk and structure of the table without re-typesetting every entry.]

D	Class no.	Leading reduced form(s)
-31	3	$(1,0,31);\ (5,2,7);\ (5,-2,7)$
-32	3	$(1,0,32);\ (4,2,9);\ (4,-2,9)$
-33	4	$(1,0,33);\ (3,0,11);\ (2,1,17);\ (6,3,7)$
-34	4	$(1,0,34);\ (2,0,17);\ (5,1,7);\ (5,-1,7)$
-35	2	$(1,0,35);\ (3,1,12);\ \ldots$
-36	4	$(1,0,36);\ (4,0,9);\ (5,2,8);\ (5,-2,8)$
-37	2	$(1,0,37);\ (2,1,19)$
-38	4	$(1,0,38);\ (2,0,19);\ (3,1,13);\ (3,-1,13)$
-39	4	$(1,0,39);\ (3,0,13);\ (5,1,8);\ (5,-1,8);\ (2,1,20);\ (6,3,8)$
-40	4	$(1,0,40);\ (4,2,11);\ (5,0,8);\ (7,2,7);\ (7,-2,7)$
-41	4	$(1,0,41);\ (5,2,9);\ (5,-2,9);\ (2,1,21);\ (3,1,14);\ \ldots$
-42	4	$(1,0,42);\ (2,0,21);\ (3,0,14);\ (6,0,7)$

D	Class no.	Leading reduced form(s)
-43	2	$(1,0,43)$; $(4,1,11)$; $(4,-1,11)$
-44	4	$(1,0,44)$; $(4,0,11)$; $(5,1,9)$; $(5,-1,9)$; $(3,1,15)$; $(3,-1,15)$
-45	4	$(1,0,45)$; $(5,0,9)$; $(2,1,23)$; $(7,2,7)$
-46	4	$(1,0,46)$; $(2,0,23)$; $(5,2,10)$; $(5,-2,10)$
-47	5	$(1,0,47)$; $(3,1,16)$; $(3,-1,16)$; $(7,3,8)$; $(7,-3,8)$; $(2,1,24)$; $(6,1,8)$; $(4,1,12)$
-48	4	$(1,0,48)$; $(3,0,16)$; $(7,1,7)$; $(4,2,13)$; $(4,-2,13)$
-49	4	$(1,0,49)$; $(2,1,25)$; $(5,1,10)$; $(5,-1,10)$
-50	4	$(1,0,50)$; $(2,0,25)$; $(6,2,9)$; $(6,-2,9)$
-51	4	$(1,0,51)$; $(3,1,17)$; $(3,-1,17)$; $(5,1,11)$; $(5,-1,11)$; $(4,1,13)$; $(4,-1,13)$
-52	4	$(1,0,52)$; $(4,0,13)$; $(3,0,17)$; $(7,3,12)$; $(7,-3,12)$; $(2,1,26)$; $(6,3,10)$
-53	4	$(1,0,53)$; $(7,2,8)$; $(7,-2,8)$; $(2,1,27)$; $(3,1,18)$; $(3,-1,18)$; $(6,1,9)$; $(6,-1,9)$
-54	4	$(1,0,54)$; $(2,0,27)$; $(7,3,9)$; $(7,-3,9)$; $(5,1,11)$; $(5,-1,11)$; $(3,0,18)$; $(6,3,11)$
-55	4	$(1,0,55)$; $(5,0,11)$; $(7,1,8)$; $(7,-1,8)$; $(2,1,28)$; $(8,3,8)$; $(4,1,14)$; $(4,-1,14)$
-56	4	$(1,0,56)$; $(3,1,19)$; $(3,-1,19)$; $(8,4,9)$; $(5,2,12)$; $(5,-2,12)$; $(4,2,15)$; $(4,-2,15)$; (2,
-57	4	$(1,0,57)$; $(2,1,29)$; $(7,0,9)$; $(3,1,19)$; $(3,-1,19)$; $(6,3,11)$; $(6,-3,11)$; $(8,1,9)$; (8, −1
-58	2	$(1,0,58)$; $(2,0,29)$; $(3,0,19)$; ...
-59	3	$(1,0,59)$; $(3,1,20)$; $(3,-1,20)$; $(4,-1,15)$; $(4,1,15)$; $(5,1,12)$; $(5,-1,12)$
-60	4	$(1,0,60)$; $(4,0,15)$; $(3,0,20)$; $(5,0,12)$; $(7,2,10)$; $(7,-2,10)$; $(6,-1,10)$; $(6,1,10)$
-61	2	$(1,0,61)$; $(5,2,13)$; $(5,-2,13)$
-62	4	$(1,0,62)$; $(2,0,31)$; $(7,3,9)$; $(7,-3,9)$; $(9,4,8)$; $(9,-4,8)$; $(6,1,11)$; $(6,-1,11)$
-63	4	$(1,0,63)$; $(7,0,9)$; $(8,3,9)$; $(8,-3,9)$; $(2,1,32)$; $(6,3,12)$; ...
-64	4	$(1,0,64)$; $(8,4,10)$; $(5,2,13)$; $(5,-2,13)$; ...
-65	4	$(1,0,65)$; $(5,0,13)$; $(3,1,22)$; $(3,-1,22)$; $(2,1,33)$; $(7,1,10)$; $(7,-1,10)$; $(9,2,8)$; (9, −
-66	4	$(1,0,66)$; $(2,0,33)$; $(3,0,22)$; $(6,0,11)$; $(5,2,14)$; $(5,-2,14)$; $(7,2,10)$; $(7,-2,10)$
-67	1	$(1,0,67)$
-68	4	$(1,0,68)$; $(4,0,17)$; $(8,2,9)$; $(8,-2,9)$; $(3,1,23)$; $(3,-1,23)$; $(7,3,11)$; $(7,-3,11)$; (2, 1
-69	4	$(1,0,69)$; $(3,0,23)$; $(6,3,13)$; $(5,1,14)$; $(5,-1,14)$; $(2,1,35)$; ...
-70	4	$(1,0,70)$; $(2,0,35)$; $(5,0,14)$; $(7,0,10)$; ...
-71	7	$(1,0,71)$; $(3,1,24)$; $(3,-1,24)$; $(5,2,15)$; $(5,-2,15)$; $(8,1,9)$; $(8,-1,9)$; $(4,1,18)$; (4, −
-72	4	$(1,0,72)$; $(4,2,19)$; $(4,-2,19)$; $(8,0,9)$; $(6,0,12)$; $(3,0,24)$
-73	4	$(1,0,73)$; $(8,4,11)$; $(2,1,37)$; $(7,2,11)$; $(7,-2,11)$
-74	4	$(1,0,74)$; $(3,1,25)$; $(3,-1,25)$; $(9,4,10)$; $(9,-4,10)$; $(6,2,13)$; $(6,-2,13)$; $(5,1,15)$; (5
-75	4	$(1,0,75)$; $(4,1,19)$; $(4,-1,19)$; $(7,3,12)$; $(7,-3,12)$; $(5,0,15)$; $(3,0,25)$; $(2,1,38)$; (6,
-76	4	$(1,0,76)$; $(4,0,19)$; $(5,2,16)$; $(5,-2,16)$; $(7,1,11)$; $(7,-1,11)$
-77	4	$(1,0,77)$; $(9,2,9)$; $(2,1,39)$; $(3,1,26)$; $(3,-1,26)$; $(6,1,13)$; $(6,-1,13)$; $(7,0,11)$
-78	4	$(1,0,78)$; $(2,0,39)$; $(3,0,26)$; $(6,0,13)$; ...
-79	5	$(1,0,79)$; $(3,0,26\,\text{var.})$; $(5,1,16)$; $(5,-1,16)$; $(8,3,11)$; $(8,-3,11)$; $(4,1,20)$; (4, −1, 20
-80	4	$(1,0,80)$; $(9,1,9)$; $(3,1,27)$; $(3,-1,27)$; $(5,0,16)$; $(4,2,21)$; $(4,-2,21)$; $(8,1,10)$; (8, −
-81	3	$(1,0,81)$; $(9,3,10)$; $(9,-3,10)$; $(5,2,17)$; $(5,-2,17)$; $(2,1,41)$
-82	4	$(1,0,82)$; $(2,0,41)$; $(7,3,13)$; $(7,-3,13)$; $(3,1,28)$; $(3,-1,28)$; $(9,4,11)$; $(9,-4,11)$; (4
-83	3	$(1,0,83)$; $(3,1,28)$; $(3,-1,28)$; $(7,1,12)$; $(7,-1,12)$; $(4,1,21)$; $(4,-1,21)$
-84	4	$(1,0,84)$; $(4,0,21)$; $(3,0,28)$; $(7,0,12)$; $(6,0,14)$; $(5,2,17)$; $(5,-2,17)$; $(10,4,9)$; (10, −
-85	4	$(1,0,85)$; $(5,0,17)$; $(2,1,43)$; $(10,5,11)$; $(7,3,13)$; $(7,-3,13)$; $(5,1,17)$; $(5,-1,17)$
-86	4	$(1,0,86)$; $(2,0,43)$; $(5,2,18)$; $(5,-2,18)$; $(6,2,15)$; $(6,-2,15)$; $(9,2,10)$; $(9,-2,10)$; (3
-87	6	$(1,0,87)$; $(3,0,29)$; $(4,1,22)$; $(4,-1,22)$; $(8,1,11)$; $(8,-1,11)$; $(7,2,13)$; $(7,-2,13)$; (8
-88	4	$(1,0,88)$; $(4,2,23)$; $(4,-2,23)$; $(8,0,11)$; $(8,4,13)$; ...
-89	4	$(1,0,89)$; $(2,1,45)$; $(3,1,30)$; $(3,-1,30)$; $(5,1,18)$; $(5,-1,18)$; $(9,1,10)$; $(9,-1,10)$; (6
-90	4	$(1,0,90)$; $(2,0,45)$; $(5,0,18)$; $(9,0,10)$; $(7,1,13)$; $(7,-1,13)$; $(9,3,11)$; $(9,-3,11)$
-91	4	$(1,0,91)$; $(4,1,23)$; $(4,-1,23)$; $(7,0,13)$; $(5,2,19)$; $(5,-2,19)$; $(10,3,10)$; $(2,1,46)$
-92	4	$(1,0,92)$; $(9,4,12)$; $(9,-4,12)$; $(4,0,23)$; $(3,1,31)$; $(3,-1,31)$

D	Class no.	Leading reduced form(s)
-93	4	$(1,0,93)$; $(3,0,31)$; $(2,1,47)$; $(6,3,17)$; $(7,2,14)$; $(7,-2,14)$; $(5,1,19)$; $(5,-1,19)$
-94	4	$(1,0,94)$; $(2,0,47)$; $(5,0,19)$; $(10,4,11)$; $(10,-4,11)$; $(9,2,11)$; $(9,-2,11)$; $(3,1,32)$; (
-95	8	$(1,0,95)$; $(4,1,24)$; $(4,-1,24)$; $(5,1,12)$; $(10,5,12)$; $(8,1,12)$; $(8,-1,12)$; $(6,1,16)$; (6
-96	4	$(1,0,96)$; $(4,2,25)$; $(5,2,20)$; $(5,-2,20)$; $(3,0,32)$; $(11,5,11)$; $(7,3,15)$; $(7,-3,15)$
-97	2	$(1,0,97)$; $(2,1,49)$; $(7,1,14)$; $(7,-1,14)$
-98	4	$(1,0,98)$; $(9,1,11)$; $(9,-1,11)$; $(2,0,49)$; …
-99	4	$(1,0,99)$; $(4,-1,25)$; $(4,1,25)$; $(5,1,20)$; $(5,-1,20)$; $(9,0,11)$; $(3,1,33)$; $(6,2,17)$; (6,
-100	4	$(1,0,100)$; $(4,0,25)$; $(8,2,13)$; $(8,-2,13)$; $(10,1,10)$; $(2,1,50)$

Table II. Formes quadratiques binaires ayant pour déterminants les nombres positifs non-carrés depuis $D=2$ jusqu'à $D=99$ *(commencement; suite p. 151).*

[For each positive determinant D, Cayley tabulates the reduced forms in their classes, with the characters α, β, δ, ϵ, $\delta\epsilon$ as appropriate, the composition symbol Cp, and the periods of the reduced forms. The periods are printed in the abbreviated style described in the introduction: a string of integers whose successive pairs (a_k, b_k, c_k), (a_{k+1}, b_{k+1}, c_{k+1}), … share their boundary coefficients, with a and c in larger type and b in smaller type. The present excerpt resumes Table II from D = 2 to D = 61; the continuation D = 62–99 appears on pp. 151–156 of the next instalment.]

D	Classes	Reduced forms; periods
2	1	$(1,0,-2)$; period $1,1,-1,1,1$
3	1	$(1,0,-3)$; period $1,1,-2,1,1$
5	1	$(1,0,-5)$; period $1,2,-1,2,1$
6	1	$(1,0,-6)$; period $1,2,-2,2,1$
7	1	$(1,0,-7)$; period $1,2,-3,1,2,1,-3,2,1$
8	1	$(1,0,-8)$; period $1,2,-4,2,1$
10	2	$(1,0,-10)$; $(2,0,-5)$; periods $1,3,-1,3,1$; $2,2,-3,1,3,2,-2,2,3,1,-3,2,2$
11	1	$(1,0,-11)$; period $1,3,-2,3,1$
12	1	$(1,0,-12)$; period $1,3,-3,3,1$
13	1	$(1,0,-13)$; period $1,3,-4,1,3,2,-3,1,4,3,-1,3,4,1,-3,2,3,1,-4,3,1$
14	1	$(1,0,-14)$; period $1,3,-5,2,2,2,-5,3,1$
15	2	$(1,0,-15)$; $(3,0,-5)$; periods $1,3,-6,3,1$; $2,3,-2,3,2$
17	1	$(1,0,-17)$; period $1,4,-1,4,1$
18	1	$(1,0,-18)$; period $1,4,-2,4,1$
19	1	$(1,0,-19)$; period $1,4,-3,2,5,3,-2,3,5,2,-3,4,1$
20	1	$(1,0,-20)$; period $1,4,-4,4,1$
21	2	$(1,0,-21)$; $(3,0,-7)$; periods $1,4,-5,1,4,3,-3,3,4,1,-5,4,1$; $2,3,-6,3,2$
22	1	$(1,0,-22)$; period $1,4,-6,2,3,4,-2,4,3,2,-6,4,1$
23	1	$(1,0,-23)$; period $1,4,-7,3,2,3,-7,4,1$
24	2	$(1,0,-24)$; $(3,0,-8)$; periods $1,4,-8,4,1$; $3,3,-5,2,4,2,-5,3,3$
26	1	$(1,0,-26)$; period $1,5,-1,5,1$
27	1	$(1,0,-27)$; period $1,5,-2,5,1$
28	1	$(1,0,-28)$; period $1,5,-3,4,4,4,-3,5,1$
29	1	$(1,0,-29)$; period $1,5,-4,3,5,2,-5,3,4,5,-1,5,4,3,-5,2,5,3,-4,5,1$
30	2	$(1,0,-30)$; $(2,0,-15)$; periods $1,5,-5,5,1$; $2,4,-7,3,3,3,-7,4,2$
31	1	$(1,0,-31)$; period $1,5,-6,1,5,4,-3,5,2,5,-3,4,5,1,-6,5,1$
32	1	$(1,0,-32)$; period $1,5,-7,2,4,2,-7,5,1$

D	Classes	Reduced forms; periods
33	2	$(1,0,-33)$; $(3,0,-11)$; periods $1,5,-8,3,3,3,-8,5,1$; $2,5,-4,3,6,3,-4,5,2$
34	2	$(1,0,-34)$; $(3,0,-\ \ \sim)$ *(2 ambig. class)*; periods $1,5,-9,4,2,4,-9,5,1$; $3,5,-3,4,6,2,-5,3,5,2,-6,4,3$
35	1	$(1,0,-35)$; period $1,5,-10,5,1$
37	2	$(1,0,-37)$; $(3,1,-12)$; periods $1,6,-1,6,1$; $3,4,-7,3,4,5,-3,4,7,3,-4,5,3$
38	1	$(1,0,-38)$; period $1,6,-2,6,1$
39	2	$(1,0,-39)$; $(3,0,-13)$; periods $1,6,-3,6,1$; $2,5,-7,2,5,3,-5,3,5,2,-7,5,2$
40	2	$(1,0,-40)$; $(3,1,-13)$; periods $1,6,-4,6,1$; $3,4,-8,4,3,5,-5,5,3$
41	1	$(1,0,-41)$; period $1,6,-5,4,5,6,-1,6,5,4,-5,6,1$
42	2	$(1,0,-42)$; $(2,0,-21)$; periods $1,6,-6,6,1$; $2,6,-3,6,2$
43	1	$(1,0,-43)$; period $1,6,-7,1,6,5,-3,4,9,5,-2,5,9,4,-3,5,6,1,-7,6,1$
44	1	$(1,0,-44)$; period $1,6,-8,2,5,3,-7,4,4,4,-7,3,5,2,-8,6,1$
45	2	$(1,0,-45)$; $(2,1,-22)$; periods $1,6,-9,3,4,5,-5,5,4,3,-9,6,1$; $2,5,-10,5,2$
46	1	$(1,0,-46)$; period $1,6,-10,4,3,6,-7,1,6,4,-5,6,2,5,-5,4,6,5,-7,6,3,4,-10,6,1$
47	1	$(1,0,-47)$; period $1,6,-11,5,2,5,-11,6,1$
48	2	$(1,0,-48)$; $(3,0,-16)$; periods $1,6,-12,6,1$; $3,6,-4,6,3$
50	1	$(1,0,-50)$; period $1,7,-1,7,1$
51	2	$(1,0,-51)$; $(3,0,-17)$; periods $1,7,-2,7,1$; $3,6,-5,4,7,3,-6,5,7,4,-5,6,3$
52	1	$(1,0,-52)$; period $1,7,-3,6,9,4,-4,4,9,5,-3,7,1$
53	1	$(1,0,-53)$; period $1,7,-4,6,7,2,-7,6,4,7,-1,7,4,5,-7,2,7,5,-4,7,1$
54	2	$(1,0,-54)$; $(2,1,-27)$; periods $1,7,-5,3,9,6,-2,6,9,3,-5,7,1$; $2,7,-2,7,2$
55	2	$(1,0,-55)$; $(3,1,-18)$; periods $1,7,-6,5,5,5,-6,7,1$; $2,7,-3,5,10,3,-3,7,2$
56	2	$(1,0,-56)$; $(3,2,-13)$; periods $1,7,-7,7,1$; $4,6,-5,4,8,4,-5,6,4$
57	2	$(1,0,-57)$; $(3,0,-19)$; periods $1,7,-8,1,7,5,-3,5,7,1,-8,7,1$; $2,7,-4,5,8,3,-6,3,8,5,-4,7,2$
58	1	$(1,0,-58)$; period $1,7,-9,2,6,4,-7,3,7,4,-6,2,9,7,-1,7,9,2,-6,4,7,3,-7,4,6,2,-9,$
59	1	$(1,0,-59)$; period $1,7,-10,3,5,7,-2,7,5,3,-10,7,1$
60	2	$(1,0,-60)$; $(3,0,-20)$; periods $1,7,-11,4,4,4,-11,7,1$; $3,6,-8,2,7,6,-5,6,7,2,-8,6,3$
61	1	$(1,0,-61)$; period $1,7,-12,5,3,7,-4,6,9,4,-5,8,5,4,-9,6,4,7,-3,5,12,7,1$

[Table II continues to $D = 99$ on pp. 151–156 of the next instalment; Table III, the thirteen irregular negative determinants of the first thousand, follows there as well. The structure of the entries is preserved throughout: classes, characters, composition symbol Cp, and periods in the abbreviated notation defined above.]

Londres, 6 Novembre, 1860.

[Continues on p. 151 with Table II, $D = 62$ to 99, and Table III.]

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334. (continued)

TABLES DES FORMES QUADRATIQUES BINAIRES (suite).

[Continuation from p. 150 of Table II. Positive non-square determinants $D = 2$ to $D = 99$; then Table III. Thirteen irregular negative determinants of the first thousand.]

The continuation of Table II is presented in the same form as before: for each determinant D, the number of classes, the characters, and the periods of reduced forms are tabulated. We resume at $D = 62$.

Table II (continued). Positive determinants $D = 62$ to $D = 99$.

D	Classes	Periods of the reduced forms
62	3	$1, 0, -62$; period of $1, 7, -13, 4, 2, 4, -13, 7, 1$
63	3	$1, 0, -63$; $1, 7, -2, 7, 1$; $-1, 7, 2, 7, -1$
65	1	$1, 0, -65$; period of $1, 8, -1, 8, 1$
66	1	$1, 0, -66$; period of $1, 8, -2, 8, 1$
67	1	$1, 0, -67$; period of $1, 8, -3, 7, 2, 7, -3, 8, 1$
68	2	$1, 0, -68$; $2, 0, -17$; periods of $1, 8, -4, 8, 1$, $-1, 8, 4, 8, -1$
69	1	$1, 0, -69$; period of $1, 8, -5, 8, 4, 1, -5, 4, 8, 5, -8, 1$
70	2	$1, 0, -70$; $2, 1, -23$; periods of $1, 8, -6, 4, 9, 3, -6, 4, 1$
71	1	$1, 0, -71$; period of $1, 8, -7, 7, 2, 7, -7, 8, 1$
73	1	$1, 0, -73$; period of $1, 8, -9, 4, 9, 1, -8, 4, 9, -1$
74	1	$1, 0, -74$; period of $1, 8, -10, 8, 1$
77	1	$1, 0, -77$; period of $1, 8, -13, 4, 7, 7, -4, 13, 8, -1$
78	2	$1, 0, -78$; $2, 0, -39$; periods of $1, 8, -14, 2, 5, 9, -5, 2, 14, 8, -1$
79	1	$1, 0, -79$; period of $1, 8, -15, 2, 9, 8, -7, 4, 11, 5, -3, 5, 11, 4, -7, 8, 9, 2, -15, 8, 1$
80	2	$1, 0, -80$; $2, 0, -40$; periods of $1, 8, -16, 8, 1$
82	1	$1, 0, -82$; period of $1, 9, -1, 9, 1$
83	1	$1, 0, -83$; period of $1, 9, -2, 9, 1$
84	3	$1, 0, -84$; $2, 0, -21$; $3, 0, -7$; periods of $1, 9, -3, 9, 1$
85	1	$1, 0, -85$; period of $1, 9, -4, 7, 3, 7, -4, 9, -1$
86	1	$1, 0, -86$; period $1, 9, -5, 2, 13, 2, -5, 9, 1$
87	2	$1, 0, -87$; $2, 1, -43$; periods of $1, 9, -6, 9, 1$
88	1	$1, 0, -88$; period of $1, 9, -7, 5, 9, 4, -7, 8, 3, -7, 4, 9, 5, -7, 9, 1$
89	1	$1, 0, -89$; period of $1, 9, -8, 8, 3, 8, -8, 9, 1$
90	3	$1, 0, -90$; $2, 0, -45$; $3, 0, -30$; periods of $1, 9, -9, 9, 1$
91	2	$1, 0, -91$; $2, 1, -45$; periods of $1, 9, -10, 3, 9, 4, -7, 3, 12, 5, -3, 5, 12, 3, -7, 4, 9, 3, -10, 9, 1$
92	1	$1, 0, -92$; period of $1, 9, -11, 9, 1$
93	1	$1, 0, -93$; period of $1, 9, -12, 4, 7, 8, -3, 12, 1, -12, 3, 8, -7, 4, 12, 9, -1$
94	1	$1, 0, -94$; period of $1, 9, -13, 5, 9, 7, -7, 4, 15, 2, -15, 4, 7, 7, -9, 5, 13, 9, 1$
95	2	$1, 0, -95$; $2, 1, -47$; periods of $1, 9, -14, 4, 11, 5, -14, 9, 1$
96	2	$1, 0, -96$; $2, 0, -48$; periods of $1, 9, -15, 5, 7, 8, -5, 15, 9, 1$
97	1	$1, 0, -97$; period of $1, 9, -16, 2, 15, 3, -13, 4, 12, 5, -12, 4, 13, 3, -15, 2, 16, 9, 1$
98	2	$1, 0, -98$; $2, 0, -49$; periods of $1, 9, -17, 9, 1$
99	2	$1, 0, -99$; $2, 1, -49$; periods of $1, 9, -18, 9, 1$

Table III. The thirteen irregular negative determinants of the first thousand.

For each such determinant the table exhibits the reduced forms in their proper classes, together with the characters and the auxiliary symbols Cp. The irregular determinants of the first thousand are

$$D = -576,\ -580,\ -820,\ -884,\ -900,\ -1\,(24)^2,\ -145\,(3)^2,\ -205\,(1)^2,$$
$$-243\,(9)^2,\ -307\,(1)^2,\ -339\,(1)^2,\ -459\,(9)^2,\ -675\,(15)^2.$$

(The notation $-D\,(k)^2$ indicates an irregular determinant, k being the parameter associated with the irregularity.)

For example, for $D = -576$ the reduced forms are

$$(1, 0, 576),\ (5, 2, 116),\ (8, 4, 73),\ (9, 3, 65),\ (13, 5, 46),\ (16, 4, 37),\ (16, 8, 40),\ (17, 3, 34),\ \ldots$$

with classes distinguished by the characters $\pm$ in the eight columns of generic characters, and by the auxiliary symbols Cp in the last column.

For $D = -580$ the reduced forms include

$$(1, 0, 580),\ (4, 2, 146),\ (5, 0, 116),\ (8, 2, 73),\ (10, 0, 58),\ (13, 3, 45),\ \ldots$$

For $D = -820$ the forms include

$$(1, 0, 820),\ (4, 2, 206),\ (5, 0, 164),\ (10, 0, 82),\ (11, 1, 75),\ (16, 6, 52),\ (20, 8, 44),\ \ldots$$

For $D = -884$ the forms include

$$(1, 0, 884),\ (5, 3, 177),\ (9, 4, 100),\ (13, 3, 69),\ (17, 7, 53),\ (20, 2, 45),\ (21, 7, 43),\ \ldots$$

For $D = -900$ the forms include

$$(1, 0, 900),\ (9, 0, 100),\ (13, 0, 75),\ (17, 4, 53),\ (25, 0, 36),\ (29, 2, 32),\ \ldots$$

For $D = -1\,(24)^2$ $(= -576)$ the period of the irregular class is exhibited, with the auxiliary indications.

For $D = -145\,(3)^2$ the table gives

$$(1, 0, 1305),\ (5, 0, 261),\ (9, 4, 145),\ (10, 5, 131),\ (17, 8, 77),\ (18, 7, 73),\ (25, 10, 53),\ \ldots$$

For $D = -205\,(1)^2$ the table gives

$$(1, 0, 820),\ (4, 1, 205),\ (5, 0, 164),\ (16, 7, 52),\ (20, 5, 42),\ \ldots$$

For $D = -243\,(9)^2$ the table gives

$$(1, 0, 243),\ (3, 0, 81),\ (7, 2, 35),\ (9, 0, 27),\ (11, 4, 23),\ (13, 1, 19),\ \ldots$$

For $D = -307\,(1)^2$ the table gives

$$(1, 0, 307),\ (4, 1, 77),\ (7, 1, 44),\ (11, 1, 28),\ (13, 3, 24),\ (17, 4, 19),\ \ldots$$

For $D = -339\,(1)^2$ the table gives

$$(1, 0, 339),\ (4, 1, 85),\ (5, 1, 68),\ (13, 5, 28),\ (15, 6, 25),\ (17, 1, 20),\ \ldots$$

For $D = -459\,(9)^2$ the table gives

$$(1, 0, 459),\ (4, 1, 115),\ (9, 0, 51),\ (13, 3, 36),\ (17, 5, 28),\ (19, 4, 25),\ \ldots$$

For $D = -675\,(15)^2$ the table gives

$$(1, 0, 675),\ (3, 0, 225),\ (4, 1, 169),\ (11, 1, 62),\ (19, 8, 37),\ (25, 5, 28),\ \ldots$$

[The full tabulation of these thirteen irregular determinants occupies the remainder of pages 154–156 of the original. We refer the reader to the printed source for the complete listing.]

336.

NOTE SUR L'ÉLIMINATION.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LX. (1861), pp. 373–374.]

Soient $U = (a, \ldots \between x, y)^p$, $V = (b, \ldots \between x, y)^q$ des fonctions homogènes quelconques des degrés m et n respectivement. Dénotons par $(x, y)^p$ la suite entière ou seulement une partie de la suite de termes x^p, $x^{p-1}y$, …, y^p, et en prenant $\theta \leq m \leq p$, formons le déterminant

$$|(x, y)^{p-m}U,\ (x, y)^{p-n}V|.$$

Cette notation signifie qu'on a supposé les suites $(x, y)^{p-m}U$, $(x, y)^{p-n}V$ composées respectivement de p et de q termes et qu'on posant $p + q = s$ on forme le déterminant

$$\begin{vmatrix} (x_1, y_1)^{p-m}U_1 & (x_1, y_1)^{p-n}V_1 \\ \vdots & \vdots \\ (x_s, y_s)^{p-m}U_s & (x_s, y_s)^{p-n}V_s \end{vmatrix},$$

dans lequel les différentes lignes (chacune composée de s termes) sont ce que deviennent $(x, y)^{p-m}U$, $(x, y)^{p-n}V$, lorsqu'on y substitue (x_1, y_1), (x_2, y_2), …, (x_s, y_s) successivement au lieu de (x, y).

Le déterminant que je viens de définir est divisible par le déterminant

$$|(x, y)^{s-1}|,$$

notation qui est équivalente à:

$$\begin{vmatrix} (x_1, y_1)^{s-1} \\ \vdots \\ (x_s, y_s)^{s-1} \end{vmatrix},$$

et dans laquelle $(x, y)^{s-1}$ dénote la suite entière des termes x^{s-1}, $x^{s-2}y$, …, y^{s-1}. Nous obtenons ainsi une équation

$$\frac{|(x, y)^{p-m}U,\ (x, y)^{p-n}V|}{|(x, y)^{s-1}|} = (a, \ldots)^p (b, \ldots)^q (x_1, y_1)^{\alpha+\beta-1} \cdots (x_s, y_s)^{\alpha+\beta-1},$$

c.-à-d. que le quotient est du degré p par rapport aux coefficients $(a, \ldots)$, du degré q par rapport aux coefficients $(b, \ldots)$, et du degré $\theta - s + 1$ par rapport à chaque système de variables (x_i, y_i). Or en supposant $U_i = 0$, $V_i = 0$, on obtient

$$0 = (a, \ldots)^p (b, \ldots)^q (x_1, y_1)^{\theta-s+1} \cdots (x_s, y_s)^{\theta-s+1},$$

équation qui subsiste quelles que soient les valeurs des variables $(x_1, y_1), \ldots, (x_s, y_s)$; cela donne une suite de $\theta - s + 2(s-1)$ équations chacune de la forme

$$0 = (a, \ldots)^p (b, \ldots)^q (x_i, y_i)^{\theta-s+1}.$$

En considérant un système quelconque de $\theta - s + 2$ de ces équations, pour en éliminer tous les termes de $(x_i, y_i)^{\theta-s+1}$, on obtiendra ou bien l'équation identique $0 = 0$, ou bien une équation de la forme

$$F = (a, \ldots)^{p \cdot \theta-s+2} (b, \ldots)^{q \cdot \theta-s+2} = 0$$

où F sera un déterminant de l'ordre $\theta - s + 2$, chaque terme étant de la forme $(a, \ldots)^p (b, \ldots)^q$.

Cela posé il est évident que F contiendra comme facteur la fonction

$$\square = (a, \ldots)^n (b, \ldots)^m$$

qui est le résultant des deux équations $U = 0$, $V = 0$. En particulier, on aura les deux cas que voici:

1°. Soit $\theta = m + n - 1$, et supposons que $(x, y)^{p-m}U$, $(x, y)^{p-n}V$, dénotent les suites entières

$$x^{m-1}U,\ x^{m-2}yU,\ \ldots,\ y^{m-1}U; \quad x^{n-1}V,\ x^{n-2}yV,\ \ldots,\ y^{n-1}V,$$

nous aurons $p = n$, $q = m$, $s = m + n$, $\theta - s + 2 = 1$, et de là

$$F = (a, \ldots)^n (b, \ldots)^m,$$

donc $F = \square$. On voit sans peine que l'on obtient de cette manière le résultant $\square$ sous la forme d'un déterminant de l'ordre $m + n$, le même que l'on obtient en éliminant les termes de $(x, y)^{m+n-1}$ entre les équations $(x, y)^{n-1}U = 0$, $(x, y)^{m-1}V = 0$, c.-à-d. par la méthode abrégée de Bézout.

2°. En supposant $m \leq n$, on peut prendre $\theta = m$, ne pas donner pour $(x, y)^{p-m}U$ le seul terme U. Réduisons aussi $(x, y)^{p-n}V$ au seul terme $x^{p-n}y^{p-n} \cdots V$ (x désignant un nombre entier arbitraire entre 0 et $m - n$), c.-à-d. au terme $x^{p-n}y^{m-n}V$ ou $y^{m-n}V$ dans le cas des deux valeurs extrêmes de α. On a ainsi $p = 1$, $q = 1$, $s = 2$, et de là

$$F = (a, \ldots)^n (b, \ldots)^p.$$

Donc $F = (a, \ldots)^{m-n}\square$, c.-à-d. que l'on obtient le résultant $\square$ affecté d'un facteur $(a, \ldots)^{m-n}$ qui ne contient que les coefficients de U, et qui est de l'ordre $m - n$ par rapport à ces coefficients. L'expression de ce facteur peut être trouvée assez facilement. Dans le cas du terme $x^{p-n}y^{m-n}V$, c.-à-d. pour $\alpha = m - n$, le facteur sera k^{m-n} (k désignant le dernier coefficient de la suite $(a, \ldots)$), et dans le cas du terme $y^{m-n}V$, c.-à-d. pour $\alpha = 0$, le facteur sera a^{m-n}. Mais en supposant $m = n$ on a tout simplement $F = \square$, c.-à-d. que l'on obtient le résultant sans facteur étranger. C'est sous cette dernière forme que j'ai présenté la méthode abrégée de Bézout dans le tome LIII. p. 366 (1857) de ce Journal, [250].

Londres, 17 Décembre, 1861.

337.

NOTE SUR LA RÉALITÉ DES RACINES D'UNE ÉQUATION QUADRATIQUE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LXI. (1863), pp. 367–368.]

À propos du mémoire que vient de publier M. Hesse (*voir ce Journal* t. LX, p. 305) je remarque que si l'on a l'une ou l'autre des deux formes

$$(a, b, c, f, g, h \between x, y, z)^2, \quad (a', b', c', f', g', h' \between x, y, z)^2$$

est une forme *définie* (forme qui conserve toujours le même signe pour des valeurs réelles quelconques des variables), l'équation suivante en λ:

$$\begin{vmatrix} a-\lambda a', & h-\lambda h', & g-\lambda g', & x \\ h-\lambda h', & b-\lambda b', & f-\lambda f', & y \\ g-\lambda g', & f-\lambda f', & c-\lambda c', & z \\ x, & y, & z & \end{vmatrix} = 0$$

aura ses deux racines réelles. En écrivant

$$(A,\ldots\between x,y,z)^2 = \lambda(A,\ldots)(x',\ldots) - \lambda^2(A',\ldots)(x',\ldots)^2, \quad (a',b',c',f',g',h'\between x,y,z)^2 = 0,$$

$$B = ca - g^2,\ \ldots$$

de manière que $(A,B,C,F,G,H\between \overline{y},\overline{z})$ dénote la forme adjointe (ou réciproque) de $(a,b,c,f,g,h\between x,y,z)$, cette équation prend la forme

$$(A,\ldots\between x,y,z)^2(A',\ldots\between x,y,z)^2 - [(A,\ldots\between x,y,z)^2]^2 = 0,$$

et les racines étant réelles, on doit avoir

$$\square = -4(A,\ldots\between x,y,z)^2(A',\ldots\between x,y,z)^2 + [(A,\ldots\between x,y,z)^2]^2 = +.$$

Or pour démontrer directement cette proposition, il n'est pas ce me semble possible d'exprimer $\square$ comme une somme de carrés; on a besoin de considérer une forme plus générale, savoir une somme de carrés multipliés chacun par un coefficient litéral positif. Par exemple, en ne faisant attention qu'au coefficient de x^4, on doit avoir

$$\square_x = -4\,(bc - f^2)(b'c' - f'^2) + (bc' + b'c - 2ff')^2 = +.$$

Pour en faire la démonstration, on peut exprimer $\square_x$ sous la forme

$$\square_x = (bc' - b'c)^2 + 4(bf' - b'f)(cf' - c'f),$$

ce qui donne

$$bc\square_x = (bc - f^2)(bc' - b'c)^2 + (bc' - cf')(bc' - cf) + c(bf' - bf')^2.$$

En effet, en y substituant la seconde expression de $\square_x$, on a l'identité

$$4bc(bf' - b'f)(cf' - c'f) = -f^2(bc' - b'c)^2 + (b(cf' - c'f) + c(bf' - b'f))^2$$

et l'expression pour $bc\square_x$ est ainsi démontrée. Mais en supposant que $(a,b,c,f,g,h)(.,.)^2$ soit une forme définie, on a $bc - f^2 = +$, donc enfin $\square_x = +$. Il serait assez intéressant de trouver une démonstration pareille pour l'expression générale de $\square$.

Londres, 22 Octobre 1862.

338.

NOUVELLES RECHERCHES SUR L'ÉLIMINATION ET LA THÉORIE DES COURBES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LXIII. (1864), pp. 34–39.]

Dans le problème de l'élimination, on cherche la relation qui doit exister entre les coefficients d'un système de fonctions pour que quelque circonstance particulière (ou singularité) puisse avoir lieu; par exemple, pour que deux équations puissent avoir une racine commune, ou (comme application géométrique) pour qu'une courbe puisse avoir un point double. En prenant les coefficients comme données, tant la relation cherchée que la singularité qu'elle implique n'ont qu'une certaine existence *hypothétique*. Mais on peut transformer la question en supposant que les coefficients d'une ou plusieurs des fonctions soient de la forme $a = \lambda a' + \mu a''$, $b = \lambda b' + \mu b''$, ..., où a', b', ..., a'', b'', ... sont des coefficients donnés, mais λ, μ des quantités arbitraires. On peut alors disposer de sorte que la singularité dont il s'agit existe actuellement, en déterminant, au moyen de la relation donnée par l'élimination, la valeur du rapport $\lambda : \mu$. Ces substitutions $a = \lambda a' + \mu a''$, $b = \lambda b' + \mu b''$, ... changent la fonction U à laquelle se rapportent les coefficients a, b, ... en $U = \lambda U' + \mu U''$, où U', U'' sont des fonctions semblables à U, mais avec les coefficients a', b', ... ou a'', b'', ..., au lieu de a, b, ...; et en servant d'une expression usitée, on peut dire que la fonction U est en involution avec U', U''; et de même en géométrie que la courbe $U = 0$ est en involution avec les courbes $U' = 0$, $U'' = 0$; au reste, pour les courbes cela veut dire que les trois courbes se rencontrent dans les mêmes points.

On conçoit comment cette manière d'envisager le problème peut conduire à une interprétation géométrique de résultats qui n'avaient auparavant qu'une signification analytique. Considérons par exemple la proposition suivante, "le discriminant d'une fonction quadratique à trois variables est du degré 3 par rapport aux coefficients," ou ce qui est la même chose, "la fonction qui égale à zéro exprime que la conique

$$U = 0$$

ait un point double ou se réduise à une paire de droites (c'est du degré 3 par rapport aux coefficients, c'est à dire qu'en mettant pour chaque coefficient a, b, ... une telle fonction $\lambda U' + \mu U''$, et puis en égalant à zéro le théorème géométrique que voici: "Dans le système des coniques $\lambda U' + \mu U'' = 0$ en involution avec les coniques données $U' = 0$, $U'' = 0$, il y a 3 coniques à point double (c'est-à-dire, trois paires de droites) et le théorème sur le degré du discriminant des fonctions $\lambda U' + \mu U'' = 0$." On considére en question si on considérerait, on peut dire que la fonction U est en involution avec les courbes $U' = 0$, $U'' = 0$.

La fonction $U = (A, \ldots \between x, y, z)^p$ étant telle que ne contient que des coefficients donnés et un nombre β de points de rebroussement, on a un discriminant spécial est $3(n-1)^2 - 7a - 11\beta$.

Sous la désignation de "discriminant spécial" j'entends la fonction laquelle égalée à zéro donne la condition pour que la courbe $U = 0$ ait un point double de plus. Il convient de remarquer par rapport à cette expression que le discriminant de la fonction générale du $n^{\text{ième}}$ ordre, en y substituant, au lieu des valeurs générales, les coefficients de la fonction U dont il s'agit, ne donne aucunement le discriminant spécial de la fonction U. En parlant tout simplement de l'*ordre* du discriminant spécial, j'ai voulu désigner l'ordre auquel s'élèvent par rapport à des coefficients arbitraires les éléments a, b, ... lorsqu'on suppose dans la fonction générale; et la formule énoncée ci-dessus donne en effet le rebroussement pour exprimer dans la forme signalée, c'est-à-dire linéairement par rapport à des coefficients absolument arbitraires comme éléments a, b, ...; proposition qui peut être démontrée sans difficulté.

Considérons en effet l'équation générale $U = (A, \ldots \between x, y, z)^p = 0$ les coefficients A, … sont tous arbitraires; dans le cas d'un point double supposons que les coordonnées de ce point soit comme on doit dire, et les coordonnées des tangentes en ce point doivent satisfaire à chaque condition; on a en conséquence un système équivalent à trois conditions de cinq variables dans la direction de la tangente en ce point; donc la fonction U doit avoir un point double et il s'élève par les valeurs cherchées donc connues $U = 0$, $U' = 0$, quel est le nombre des points dans tout point double '*a tout point double*" ou, ce qui est la même chose, le nombre des points dans tout point double '*a tout point double*" du système. On considére la question si non désirer dans le système; forme, ces coordonnées données $U = 0$, $U' = 0$, quel est le nombre des courbes parmi les courbes données par les équations $\lambda U' + \mu U'' = 0$ en involution avec les deux courbes données. En considérant la question si on considérerait dans le système; forme, ces coordonnées données $U = 0$, $U' = 0$, quel est le nombre des courbes parmi les courbes données par les équations $\lambda U' + \mu U'' = 0$.

[The remainder of paper 338 (pp. 165–167) continues the same analytical investigation: deduction of formulae involving the discriminantal numbers, the orders of contact, and the number of points of the various kinds attaching to a curve in involution with two given curves. We refer the reader to the printed source for the residual algebraic computations.]

Londres, 22 Mai 1863.

339.

ON SKEW SURFACES, OTHERWISE SCROLLS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLIII. (for the year 1863) pp. 453–483. Received February 3,—Read March 5, 1863.]

It may be convenient to mention at the outset that, in the paper "On the Theory of Skew Surfaces"[1], I pointed out that upon any skew surface of the order n there is a singular (or nodal) curve meeting each generating line in $(n-2)$ points, and that the class of the circumscribed cone (or, what is the same thing, the class of the curve, on the consideration of the lines of the surface as it meets an arbitrary plane) is equal to or less than n; and the further development, by Dr Salmon, of this theory is somewhat more particularly as regards the Degree of a Surface reciprocal to a given one[2]: "On the Degree of a Surface reciprocal to a given one"[3], where certain minor limits are given for the Degree of a Surface reciprocal to a given one (*), where certain minor limits are given for the Degree of a Surface reciprocal to a given one. The theory is somewhat further developed in Dr Salmon's memoir "On the Degree of a Surface reciprocal to a given one"[4], of a Surface reciprocal to a given one (*), where certain minor limits are given for the Degree of a Surface reciprocal to a given one. And in the equations $(\alpha, \ldots \between x, y, z, w)^n = 0$, $(\alpha', \ldots \between x, y, z, w)^n = 0$, … are any linear functions of the coordinates, and t is an arbitrary parameter. And the same theories are reproduced in the "Treatise on the Analytic Geometry of Three Dimensions"[5].

[1] *Cambridge and Dublin Math. Journ.* vol. vii. pp. 171–173 (1852), [108].
[2] *Trans. Royal Irish Acad.* vol. xxiii. pp. 461–488 (read 1855).
[3] Dublin, 1862. [Ed. 4, 1882.]
[4] Same reference as above.
[5] Dublin, 1862. [Ed. 4, 1882.]

I will, though it is less closely connected with the subject of the present memoir, refer to a paper by M. Chasles, "Description des courbes à double courbure de tous les ordres sur les surfaces réglées du troisième et du quatrième ordre"(*).

The present memoir (in the composition of which I have been assisted by a correspondence with Dr Salmon) contains a further development of the theory of the connexion of the curves m, n, p respectively; 2nd, the surface generated by a line which meets a curve $S(m, n, p)$, and two lines which meet a given curve m, n, p respectively: 2nd, the surface generated by a line which meets a given curve m, n, p respectively; namely, that of three given curves m, n, p respectively; 2nd, the surface generated by a line which meets a given curve m, n, p respectively; and in the investigation of these I have been led somewhat to vary the notation of the previous memoir, but the alteration is one of detail only, the equations being substantially the same.

The present memoir (in the composition of which I have been assisted by a correspondence with Dr Salmon) contains a further development of the theory, viz. 1st, the surface generated by a line meeting three given curves; 2nd, the surface generated by a line meeting a given curve twice, and a given line once; 3rd, the surface generated by a line meeting a given curve in three points; and in the investigation of these I have been led somewhat to vary the notation of the previous memoir, but the alteration is one of detail only, the equations being substantially the same.

Notation and Table of Results, Articles 1 to 10.

1. In the present memoir a letter such as m denotes the order of a curve in space. It is for the most part assumed that a correspondence with Dr Salmon contains a further development of the theory of the connexion of the curves containing a further development of the theory. Thus $G(m, n, p, q)$ denotes the order of any one of the surfaces, that is, the surface generated by a line meeting a given line and three curves m, n, p, q respectively; and similarly $G(m, n)$ denotes the order of the surface generated by a line meeting a given line twice and two curves m, n respectively; 2nd, the surface generated by a line meeting a given curve in two points, and the surface generated by a line meeting a given curve in two points are surfaces of the same kind, the simplification being merely in the notation.

2. The use of the combinations (m, n, p, q), (m^2, n, p), &c. hardly requires explanation; it may however be noticed that the order of the surface $G(m, n, p)$ denotes the number of intersections of two non-coincident generating lines of the surface. This gives for the cases in question:

$$\Pi(m, n, p) = G(m, n, p), \; \ldots$$

and the like manner, if $S(m)$ stands for the order of the scroll generated by a line meeting a given curve m in 4 points, &*c*., $(m, n, p) = 2mnp$, $(m^2, n) = m^2n$, $G(m, n, p) = G(m, n, p)$ denotes the order of the surface, (m, n, p), and the number of nodal lines of which is $\frac{1}{2}(m+n-1)(n-1)$, the order denotes the order of the surface. The same will be employed throughout the memoir; and the order of the surface, the order being merely the order of the scroll.

3. The letters G, ND, NG, NR, NT (read Generators, Scroll, Nodal Director, Nodal Generator, and Nodal Total, are, in the simplest case in which there is a single curve of intersection, the orders m, n, and a generating line meets each of these in a single point.

$$NT(m, n, p) = ND(m, n, p) + NG(m, n, p),$$

and similarly for the scrolls $S(m^2, n)$ and $S(m^3)$.

4. $G(m, n, p, q)$. — Taking any point on the curve m, this is the vertex of two cones passing through the curves n, p respectively; the cones are of the orders n, p, respectively, and their intersection therefore in np lines, which are the generating lines (through the assumed point) of the surface. Hence we have

$$G(m,n,p,q) = 2mnpq.$$

The line meeting these may be drawn through any point of the curve m, and it will sweep out the entire surface in the order m.

5. $G(m^2,n,p)$. — Taking any point on the curve n, the two chords through this point of the curve m form, together with the curve p, a triangle, the lines from the point to the curve p, of the order p giving in all $p\cdot\frac{1}{2}m(m-1)$, and the same is the order of the scroll through this is $\frac{1}{2}m(m-1)$ chord, that is, the order of the scroll through the point on n. The like investigation, taking points on the curve p, gives the order of the surface generated by a line which meets a given curve m in two points, and the curve p once, is

$$G(m^2,n,p) = \tfrac{1}{2}\,p\cdot m(m-1)\cdot n + \text{(like terms with } n, p \text{ interchanged)}.$$

6. $G(m^3,p)$. — Through any point of the curve p, the three sides of the triangle formed by the three chords of the curve m, the order p giving therefore $p\cdot\frac{1}{6}m(m-1)(m-2)$. The order of the surface generated by a line meeting a given curve m in three points and a given curve p once is

$$G(m^3,p) = \tfrac{1}{6}\,p\cdot m(m-1)(m-2).$$

7. $NG(m,n,p,q)$.— $NG(m,n,p)$.— $NG(m^2,n) = NG(m^2)$.— $NG(m^3)$.— We have here the case in which a line passes through two points of the curve m; and consider any one of the surfaces (G surfaces); the line in question is a generating line of the surface, but is also a nodal line: when this is the case, the line meets the surface in $4(n-1)^2$ points, and so on.

8. $NR(m,n,p,q)$.— $NR(m,n,p)$.— $NR(m^2,n)$.— $NR(m^3)$.— Here, the nodal residue is the intersection of the surface with another surface, formed by considering the assumed generating line as the intersection of two surfaces; from the order of the curve of intersection thus formed, the order of the generating line itself being subtracted, we obtain the order of the nodal residue.

9. $NT(m,n,p,q)$.— $NT(m,n,p)$.— $NT(m^2,n)$.— $NT(m^3)$.— These are the total nodal curves, viz. the union of the nodal director and the nodal generator; we have

$$NT(m,n,p) = ND(m,n,p) + NG(m,n,p) + NR(m,n,p),$$

and similarly for the scrolls $S(m^2,n)$ and $S(m^3)$.

10. I remark that the formulae are best exhibited in an order different from that in which they are in the sequel obtained, viz. I collect them in the following

Table.

$$\begin{aligned}
G(m,n,p,q) &= 2mnpq,\\
G(m^2,n,p) &= np\,([m]_2 + M),\\
G(m^3,p) &= p\,([m]_3 + N\cdot\tfrac{1}{2}[m]_2\cdot\tfrac{1}{2} + MN),\\
G(m^4) &= [m]_4 + N\cdot[m]_2\cdot\tfrac{1}{2} - 2[m]_2 + M^2\cdot\tfrac{1}{2},\\
S(m,n,p) &= 2mnp,\\
ND(m,n,p) &= \tfrac{1}{2}\,mnp\,(mn+mp+np-3),\\
NG(m,n,p) &= \tfrac{1}{2}\,mnp\,(4mnp-mn-mp-np+3) - Mnp - Nmp - Pmn,\\
NR(m,n,p) &= \tfrac{1}{2}\,mnp\,(4mnp-7) - 5 + Mnp + Nmp + Pmn,\\
NT(m,n,p) &= 2mnp\,(mnp-1) + Mnp + Nmp + Pmn,
\end{aligned}$$

together with the further entries

$$\begin{aligned} S(1,1,m) &= 2m, \\ ND(1,1,m) &= [m]_2, \\ NG(1,1,m) &= [m]_2 + M, \\ NR(1,1,m) &= 0, \\ NT(1,1,m) &= 2[m]_2 + M, \end{aligned}$$

and

$$\begin{aligned} S(1,m,n) &= 2mn, \\ ND(1,m,n) &= \tfrac{1}{2}\, mn\,(mn+m+n-3), \\ NG(1,m,n) &= mn\,(m+n-2) + Mn + Nm, \\ NR(1,m,n) &= \tfrac{1}{2}\, mn\,(mn-m-n-3) - Mn - Nm, \\ NT(1,m,n) &= \tfrac{1}{2}\, mn\,(mn+3m+3n-5); \end{aligned}$$

moreover

$$\begin{aligned} S(m^2,n) &= n\,([m]_2 + M), \\ ND(m^2,n) &= n\big(\tfrac{1}{2}[m]_2^2 + [m]_2\cdot M + \tfrac{1}{2}[m]_2 + \tfrac{1}{2}M^2 - \tfrac{1}{2}\big), \\ NG(m^2,n) &= n\big(2\,[m]_2^2 + [m]_2\cdot M + \tfrac{3}{2}[m]_2 - 2\big) + Mn, \\ NR(m^2,n) &= 0, \\ NT(m^2,n) &= n\big(\tfrac{5}{2}[m]_2^2 + 2[m]_2\cdot M + 2[m]_2 + \tfrac{1}{2}M^2 + \tfrac{1}{2}\big) - Mn. \end{aligned}$$

The ND formulae, Articles 11 to 16.

11. $ND(m,n,p)$.— Taking any point on the curve m, this is the vertex of two cones passing through the curves n, p respectively; the cones are of the orders n, p, respectively, and their intersection therefore in np lines, the generating lines through the assumed point of the scroll $S(m,n,p)$. Hence

$$ND(m,n,p) = \tfrac{1}{2}\, mnp\,(mn+mp+np-3).$$

12. $ND(m^2,n)$.— Taking first a point on the curve m, this is the vertex of a cone of the order $m-1$ through the curve m, that is, of the cone of the order $m-1$ through the remaining $m-1$ points of intersection with the curve. Taking next a point on the curve n, this is the vertex of a cone passing through the curves m, $\frac{1}{2}[m]_2 + M$ double lines, which are the generating lines (through the assumed point) of the scroll $S(m^2,n)$. Hence we have

$$ND(m^2,n) = n\cdot\big[\tfrac{1}{2}\,[m]_2^2 + [m]_2\cdot M + \tfrac{1}{2}[m]_2 + \tfrac{1}{2}M^2 - \tfrac{1}{2}\big].$$

13. $ND(m^3)$.— Taking any point on the curve m, in the vertex of a cone of the order $m-1$ through the curve m at $m-2$ other points, the cone is of the order $\frac{1}{2}[m]_2 - m + 2 + M$ double lines, which are the generating lines on the scroll $S(m^3)$. Hence we have

$$ND(m^3) = m\,\big[\tfrac{1}{2}[m]_2^2 - m + 2 + M\big] = m\,\big[\tfrac{1}{2}[m]_2^2 + \tfrac{1}{2}[m]_2 + M\,[m]_2 + \tfrac{1}{2}M^2\cdot\tfrac{1}{2}m\big].$$

Preparatory remarks in regard to the G formulae, the hypertrioidal singularities of a curve in space, Articles 14 to 22.

14. It is to be remarked that the generating line of any one of the scrolls $S(m,n,p)$, $S(m^2,n)$, $S(m^3)$ satisfies three conditions; and that it cannot in anywise happen that one of these conditions could be in the other two. Thus, for instance, in the generating line of the scroll $S(m,n,p)$ the line meeting the two curves m, n which are met by the generating line in the same point; that is, in fact have, as will be developed somewhat more particularly as regards the line $G(m, n, p)$ in question.

15. The foregoing investigation is not very satisfactory, but I confirm it by considering the case of two plane curves, orders m and n, and classes μ and ν, respectively. The tangents of the two curves can, it is clear, only meet on the line of intersection of the planes; in fact as follows: from any point of the line of intersection draw a tangent to each of the curves; the construction of the Torse is in fact as follows: from any point of the line of intersection draw a tangent to each of the curves; this is therefore $\mu\nu$ in number.

16. Given a curve m, then (as in fact mentioned in the investigation for $ND(m^2)$) through any point not whatever of the curve there may be drawn

$$(h - m + 2)\,([m]_2 + m - 2) \text{ lines,}$$

meeting the curve in two other points, or say $[m]_2 + m - 2$ lines through three points, which can be drawn a line through three points; the number of these is in general $\frac{1}{6}(m-2)([m]_2 + m - 2)$.

17. But the curve m may be such that through every point of the curve there passes a line through four points. In fact, in general every line meets the curve in the points where it is met by it, and following from these to draw the curve m through them, the curve passes through these is the only place; in fact in such a case we cannot draw a chord through these points; in general every chord drawn meets the curve in two points only; but if a chord cannot be drawn through five points, we cannot, in general, draw a chord through six points; in general every chord drawn meets the curve in two points only; but if a chord cannot be drawn through five points, we cannot, in general, draw a chord through six points.

18. It is to be noticed, moreover, that if we take an arbitrary point of the curve, we may have the points $(\alpha, \beta, \gamma, \delta)$ on the curve. The like remarks apply to the lines $G(m^2, n, p)$, $G(m^2, n)$, and $G(m^3)$; but I will develope them somewhat more particularly as regards the line $G(m^3)$ in question.

19. The case just considered is that of a curve in such that through every point of the curve passes a line meeting the curve in four lines through three points; the like investigation, taking a point of the curve, gives in like manner apparently $\frac{1}{2}\,(m-1)(m-2)$ lines through three points; the number of these is in general $\frac{1}{6}(m-1)(m-2)(m-3)$.

20. A good illustration is afforded by taking for the curve m a curve on the hyperboloid (or quadric surface): such curves do not contain themselves as a complete intersection of the surface with another surface, but they form a fourth order surface considered as generated by a line which lies one or more times on the surface $S(m)$; and considering the curve as the generator of a quadric surface, the curve m takes only $\frac{1}{2}(q-1)^2$ times, which scrolls together make up the scroll $S(m)$; it may however happen that, e.g. of the α lines, any set or sets of α even can lie on one and the same scroll $S(m)$, that is, the scroll $S(m)$ will itself break up into scrolls of lower orders.

21. A small illustration is afforded by taking for the curve in question $S(m^2)$, such that upon a system of $S(m)$ taken $\frac{1}{2}(p)^2 + \frac{1}{2}(q)^2$ lines, and $\ldots M = -pq$.

22. In the following investigations for $G(m, n, p, q)$, &c. the foregoing special cases are excluded from consideration; it may however be right to notice how it is necessary to modify the formulae in those cases.

The G formulae, Articles 23 to 34.

23. $G(m, n, p, q)$.—Considering the scroll $S(m, n, p)$ generated by a line which meets the curves m, n, p respectively, we have

$$G(m, n, p, q) = 2mnpq.$$

24. $G(m^2, n, p)$.—In a precisely similar manner we find

$$G(m^2, n, p) = np\,([m]_2 + M).$$

25. $G(m^3, p)$.—I investigate the value by a process similar to that employed for $G(1, 1, m)$. Suppose that the curves m and a plane curves having respectively h and k double points; then the curves m and the plane curves meet in hk points, combining with the plane curves in the following way: in question is met by the curve in $h-k$ points; the construction follows as follows:

$$G(m^3, p) = p\left([m]_3 + N \cdot \tfrac{1}{2}[m]_2 \cdot \tfrac{1}{2} + MN\right).$$

26. $G(m^4)$.—We have here

$$G(m^4) = [m]_4 + N \cdot [m]_2 \cdot \tfrac{1}{2} - 2[m]_2 + M^2 \cdot \tfrac{1}{2}.$$

27. Before going further, I observe that there are certain functional conditions which must be satisfied by the G formulae. Thus in $G(m, n, p, q)$, if q takes the form p, the formula is to reduce itself to $G(m, n, p^2)$, that is, with p^2 instead of pq; or what is the same thing, $G(m, n, p, p) = 2G(m, n, p^2)$; and similarly $G(m, n, p, q)$ in like manner.

28. Next, before quitting the cases $G(1, 1, m)$, &c., I observe that the values of these are known directly from the consideration of an evanescent generating line, which is, in this case, simply the line of intersection of the two planes. Thus for the case $G(1, 1, m)$ we have

$$G(1, 1, m) = 2m, \quad ND(1, 1, m) = [m]_2 + M,$$

and so on.

29. Hence in $S(1, m, n)$ the formulae are now at once obtained; that is, $S(1, m, n) = 2mn$, $ND(1, m, n) = \frac{1}{2}mn(mn + m + n - 3)$.

30. First, if $G(m)$ denote $G(1, 1, m)$, and the like for $G(m^2)$ denotes $G(1, 1, m^2)$ and so on; which is the case in question; hence

$$G(1, 1, m) = 2m, \quad G(1, 1, m^2) = [m]_2 + M.$$

A solution of which is $G(m^2) = \frac{1}{2}[m]_2 + mM$. Hence $G(1, 1, m^2) = \frac{1}{2}m([m]_2 + M)$, and so we have thus

$$G(1, m^2) = \tfrac{1}{2}\,m\,([m]_2 + M).$$

31. Suppose then the curves m, m are formed in the way described, viz. if $G(m, m') = (m - m')p$, and observing that $G(m, m')$ stands for $G(1, m, m')$, and that

$$G(m, m') = \tfrac{1}{2}\,m'\,([m]_2 + M) + \tfrac{1}{2}\,m\,([m']_2 + M'),$$

a particular solution is $\phi(m^2) = \frac{1}{2}[m]_2 + mM$, and we have therefore

$$\phi(m^2, m') = M(\tfrac{1}{2}[m]_2 + mM) + M'\,m + \alpha M,$$

and we have

$$NT(m^2, m') = \tfrac{1}{2}\,m'(m^2 - m) + M(m^2 - m) + \beta M.$$

32. Again, $G(m^3)$ standing for $G(1, 1, m^3)$, $G(m^3, m')$ for $G(1, m^3, m')$, the values whereof have been already found, we have

$$G(m^3, m') = m'\,(\tfrac{1}{6}[m]_3 + M\,(\tfrac{1}{2}[m]_2 + m) + M^2 \cdot \tfrac{1}{2}),$$

a solution of which is $G(m^3) = \frac{1}{6}[m]_3 + M(\frac{1}{2}[m]_2 + m) + M^2 \cdot \frac{1}{2}$. Now, in the same way we may obtain

$$G(1, m^3) = \tfrac{1}{6}m([m]_3 + M([m]_2 + m) + M^2 m),$$

and the like for others.

33. Hence, substituting for $\phi(m)$, and $\phi(m, m^3)$, $\phi(m^2, m)$ their values, we have

$$\phi(m+m')^4 - \phi(m^4) - 4(\phi(m^3, m') - \phi(m, m'^3)) = mM'(m'^2 - 1) + m'M(m^2 - 1) + 6mm'MM',$$

which is satisfied. Hence the formula gives the general solution in the form

$$\phi(m^4) = M(\tfrac{1}{2}[m]_3 + (\tfrac{1}{2}[m]_2 + mM) + M^2(\tfrac{1}{2}[m]_2 + \tfrac{1}{2}M)).$$

34. Next, $G(m^2, m')$ standing for $G(1, m^2, m')$, the values whereof have been already found, we have

$$G(m^2, m') = m'\big(\tfrac{1}{2}[m]_2 + m \cdot M + \tfrac{1}{2}[m]_2 + \tfrac{1}{2}M^2 - \tfrac{1}{2}\big).$$

The S formulae, particular cases, Articles 35 to 40.

35. The S formulae have in fact been obtained in the investigation of the G formulae; we have

$$S(m, n, p) = 2mnp, \quad S(m^2, n) = n\,([m]_2 + M), \quad S(m^3) = \tfrac{1}{6}[m]_3 + M\,[m]_2.$$

36. In confirmation of the formula $S(1, m^2) = [m]_2 + M$, it is to be remarked that if we take through the line 1 an arbitrary plane, this meets the curve m in m points, and joining these two together by chords, each pair of which gives a generating line of the scroll, the lines on the scroll $S(1, m^2)$ which contain $[m]_2$ lines; that is, the order of the surface $S(1, m^2)$ is $[m]_2 + M$.

37. In confirmation of the formula $S(1, m^2) = [m]_2 + M$, it is to be remarked that if we take through the line 1 a plane, this plane meets the curve m in m points; and joining these two together by chords, each pair of which gives a generating line of the scroll, the lines on the scroll $S(1, m^2)$ in the plane meet in m points, that is, in the order of the scroll $S(1, m^2) = [m]_2 + M$.

38. As regards the formula $S(m^3) = \frac{1}{6}[m]_3 + M\,[m]_2$, suppose that the curve m is on a quadric surface; then there cannot in general be a line meeting it in three points; that is, the order of the scroll $S(m^3) = 0$ in the case considered. But if it should happen that the chord of the curve meets it in three points, the construction would still be valid; in the present case $M = -pq$; whence in this case the order of the scroll $S(m^3)$ is calculated as

$$S(m^3) = \tfrac{1}{6}[m]_3 + M\,[m]_2 = \tfrac{1}{6}\,pq\,(2pq - p - q - 3)(2pq - p - q - 2);$$

which on putting therein $p + q = m$, $pq = \beta$ and from the reduced expressions obtaining the values of $S\beta$, &c., give

$$\Sigma\alpha = \beta - \tfrac{1}{4}\alpha^2 + \tfrac{1}{2}\alpha - 18,$$
$$\Sigma\alpha' = \beta - (-\alpha) + \tfrac{1}{2}\alpha^2 + \tfrac{1}{4}\alpha - 58,$$
$$\Sigma\alpha^2 = \beta^2 + 2\alpha\beta + \tfrac{1}{2}\beta^2 - \tfrac{1}{4}\beta\alpha + \tfrac{1}{4}\alpha^2 + \tfrac{4}{3}\alpha^3 - 119\alpha + 198,$$
$$\Sigma\alpha\beta = \beta^2(\tfrac{1}{2}) + \beta(-\tfrac{1}{2}\alpha + \tfrac{1}{2}\alpha^2 - \tfrac{1}{2}\alpha) + \tfrac{1}{2}\alpha^3 + \tfrac{4}{3}\alpha^2 - \tfrac{11}{12}\alpha + 19,$$
$$\Sigma\alpha\beta\gamma = \beta^2(\tfrac{1}{2}) + \beta(-\tfrac{1}{2} - \alpha + \tfrac{1}{2}\alpha^2) + \beta(\tfrac{1}{4}\alpha^4 - \tfrac{1}{2}\alpha^3 + \tfrac{4}{3}\alpha^2 - \tfrac{11}{12}\alpha + 19),$$
$$\Sigma\alpha\beta\gamma\delta = \beta^3(\tfrac{1}{2}) + \beta^2(-\tfrac{1}{6}\alpha^2 + \tfrac{1}{2}\alpha - \tfrac{4}{3}) + \beta(\tfrac{1}{2}\alpha^4 - \tfrac{1}{4}\alpha^3 + \tfrac{19}{4}\alpha^2 - \tfrac{1067}{96}\alpha + 1071\alpha - 1560);$$

this, with the reduced expressions for $S(m^3)$, gives the final form.

39. We have

$$\Sigma\alpha = -\beta - \tfrac{1}{4}\alpha^2 + \tfrac{1}{2}\alpha - 18,$$
$$\Sigma\alpha' = -\beta - \tfrac{1}{4}\alpha^2 + 9\alpha - \tfrac{1}{2}\alpha^3 + 38,$$
$$\Sigma\alpha^2 = \beta^2 + 2\alpha\beta + \tfrac{4}{3}\alpha^4 + 9\alpha^3 - \tfrac{11}{4}\alpha^3 + \tfrac{1}{4}\alpha^4 + 90\alpha + 198,$$

and so we have for the order

$$\Sigma B = \tfrac{1}{6}\, pq\,(\alpha^2 - 5\alpha + 6) + Mn(2p - q - 1) - 119\alpha + 198,$$

where, multiplying $\alpha^2 - 5\alpha + 6 = 0$ gives $\alpha = 2$ or $\alpha = 3$, and the formula gives the general solution.

40. In the same manner the order of the scroll $S(m^3)$ is obtained, and combining the previous expressions, we find the final formula for the order of the surface $S(m^3)$:

$$S(m^3) = \tfrac{1}{6}\, m\,(m-1)(m-2) + M\,(m-1)(m-2),$$

together with such other contributions as arise from the special cases when the curve m has multiple points or other singularities.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1, m, n)$ and $S(1, m, n^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1, m, n)$ the directrix curves are respectively multiple curves of the orders m, n, and a generating line meets each of these in a single point. The number of intersections therefore is a single $24 - 2$

60. Similarly for the scroll $S(1, m, n^2)$ the directrix curves are multiple curves, viz. the curve 1 is a $(\tfrac{1}{2}n(n-1) + M)$-tuple curve; the generating line meets 1 in a single point, contained on 1 in a 1 in a $\tfrac{1}{2}n(n-1) + M$-tuple point; thus the curve 1 is met in a single point, contained on 1 in a $\tfrac{1}{2}n(n-1) + M$-tuple point; the directrix is therefore $\tfrac{1}{2}n(n-1) + M$ points, each of which counts simply, and the latter in $(n-2)$ points, each of which counts as a single point; and the value is $1 \cdot \tfrac{1}{2}n(n-1) + M + \tfrac{1}{2}(n-1)(n-2)(n-3)$.

61. For the scroll $S(m, n, p)$ the directrix is met in a single point, contained on 1 in a $\tfrac{1}{2}n(n-1) + M$-tuple point; the directrix is therefore $\tfrac{1}{2}n(n-1) + M$ points, each of which counts simply, and the value is

$$S(m, n, p) - 2 = 2mnp - 2.$$

62. For the scroll $S(m^2, n)$ we find similarly

$$S(m^2, n) - 2 = n([m]_2 + M) - 2.$$

63. And lastly for the scroll $S(m^3)$ we have

$$S(m^3) - 2 = \tfrac{1}{6}m(m-1)(m-2) + M(m-1)(m-2) - 2,$$

which gives

$$S(m^3) = \tfrac{1}{6}m(m-1)(m-2) + M(m-1)(m-2).$$

The foregoing expressions for the scroll $S(m^3)$ might with propriety have been inserted in the Table.

Annex No. 1.—Investigation of the formula for $S(m^2)$ in the case of the unicursal curve (referred to, Art. 39).

Consider the unicursal m-thic curve the equations whereof are

$$x : y : z : w = A : B : C : D,$$

where A, B, C, D are rational and integral functions of a parameter θ; and let it be required to find the equation of a plane meeting the curve in such manner that three of the points of intersection are in lined. Taking the points $(\xi, \eta, \zeta, \omega)$ as (θ_0), (θ_1), (θ_2) on the curve, we find between $(\xi, \eta, \zeta, \omega)$ and $(\xi', \eta', \zeta', \omega')$ an equation of a certain degree in $(\xi, \eta, \zeta, \omega)$ and $(\xi', \eta', \zeta', \omega')$, which is the equation of the plane meeting the curve at three points

$$\xi x + \eta y + \zeta z + \omega = 0,$$

we find between $(\xi, \eta, \zeta, \omega)$ an equation of a certain degree in θ; and to express that three of the roots be equal, we have between $(\xi, \eta, \zeta, \omega)$ a system of equations, which I do form the equations

$$\begin{vmatrix} \lambda, & \mu, & \nu, & \rho \\ A_0, & B_0, & C_0, & D_0 \\ A_1, & B_1, & C_1, & D_1 \\ A_2, & B_2, & C_2, & D_2 \end{vmatrix} = 0;$$

and if we form the equation

$$\Pi \begin{vmatrix} \lambda, & \mu, & \nu, & \rho \\ A_0, & B_0, & C_0, & D_0 \\ A_1, & B_1, & C_1, & D_1 \\ A_2, & B_2, & C_2, & D_2 \end{vmatrix} = 0,$$

where Π denotes the product of the terms belonging to all the triads of the m roots, the result will be symmetrical in regard to all the roots; and replacing the symmetrical functions of the roots by expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}m(m-1)(m-2)} = 0,$$

or, the symmetrical functions are expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3} = 0.$$

Now the above-mentioned determinant is divisible by $(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2)$; for it vanishes when any two of the indeterminates θ_0, θ_1, θ_2 become equal; and the same product divides Π. Considering only the residue, we have

$$\Pi\,(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2) = \text{Disct.}(\xi, \eta, \zeta, \omega)^{[m]_3},$$

or, say

$$\Pi\,(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2) \text{ is the Discriminant in respect to } (\xi, \eta, \zeta, \omega)^{[m]_3}.$$

Now the above-mentioned determinant is divisible by $(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2)$, and the symmetrical functions are expressed in terms of the coefficients; thus the formula

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3}$$

contains linearly the coefficients of the determinant, that is, of (A_i, B_i, C_i, D_i) of the orders θ_i; that is, the form is symmetric in regard to all the roots; and replacing the symmetrical functions of the roots by their values expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3}.$$

The like remarks apply to the lines $G(m^2, n, p)$, $G(m^2, n)$, and $G(m^3)$; but I will develope them somewhat more particularly as regards the line $G(m^3)$ in question.

Annex No. 2.—Investigation of the formula for $S(m^2)$, when the curve m is the pq complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively (referred to, Art. 40).

Let $U = 0$, $V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take (x, y, z, w) the coordinates of a point on the curve, and the equations of the two curves respectively, write for the coordinates $x + px'$, $y + py'$, $z + pz'$, $w + pw'$; then putting

$$\Delta = x\partial_x + y\partial_y + z\partial_z + w\partial_w,$$

the resulting equations may be represented by

$$(\Delta U,\ \Delta^2 U,\ \dots\ \Delta^p U \between 1,\ \rho)^{p-1} = 0, \quad (\Delta V,\ \Delta^2 V,\ \dots\ \Delta^q V \between 1,\ \rho)^{q-1} = 0,$$

where it is to be noticed that besides the expressed literal coefficients there are numerical coefficients (not as the notation usually denotes, the binomial coefficients, but) $= [\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3\cdot 2}, \frac{1}{2\cdot 1}]$, &c.

Supposing that (x', y', z', w') are the current coordinates of a point on the line drawn through the point (x, y, z, w) to meet the curve m in two other points, and equations in ρ must have two common roots, and to express this we have a system of two other equations, viz. if $k = \frac{1}{2}(p + q - 1) + \frac{1}{2}(q - 2)$; that is, $k = \frac{1}{2}(p + q)$.

2. The use of the combinations $(\Delta U, \Delta^2 U, \dots \Delta^p U)$, $(\Delta V, \dots)$ are all in either group in the Appendix; if $k = \frac{1}{2}[p]_2 + \frac{1}{2}[q]_2$; that is, k in either case is in the Appendix and his Treatise on the Analytic Geometry of Three Dimensions; and in the investigation in the following Annexes 2 and 3, the route which I have followed are completely traced out for me by him, in this way which I most readily presented itself.

Annex No. 2.—Investigation of the formula for $S(m^2)$ when the curve m is the complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively (referred to, Art. 40).

Let $U = 0$, $V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take (x, y, z, w) the coordinates of a point on the curve, the equations of the two curves respectively, write for the coordinates $x + \rho x'$, $y + \rho y'$, $z + \rho z'$, $w + \rho w'$; and so on in other cases; and considering first the equation

$$(\Delta U,\ \Delta^2 U,\ \dots\ \Delta^p U \between 1,\ \rho)^{p-1} = 0$$

of the degree p in ρ, the coordinates of a point on the curve, the equations to determine the equation of the plexus in the square $+1$ system, the resulting equations may be represented by:

$$\begin{vmatrix} \Delta U,\ \Delta^2 U,\ \dots \\ \Delta U,\ \dots \\ \vdots \\ \Delta V,\ \Delta^2 V,\ \dots \\ \Delta V,\ \dots \end{vmatrix} = 0,$$

$p + q - 3$ columns, $(q - 2) + (p - q - 4)$ lines; or representing the terms according to their order and weight, that is, degree in (x, y, z, w) and (x', y', z', w') respectively (the order and weight of the evanescent terms being fixed as in this case) they may form a regular series with the other terms, the system is

$$\left|\begin{array}{c} (p+q-3\text{ columns}) \\ \hline (p-1)\lambda,\ (p-2)\lambda_1,\ \ldots,\ (p-1)\lambda_{p-2} \\ p_0,\ p_1,\ \ldots,\ p_{p-3} \\ \vdots \\ (q-1)\lambda_0,\ (q-2)\lambda_1,\ \ldots,\ (q-1)\lambda_{p-2} \\ q_0,\ q_1,\ \ldots,\ q_{p-3} \end{array}\right| = 0,$$

so that

$$\alpha,\ \beta,\ \ldots = p-1,\ p,\ p+1,\ \ldots,\ -(q-2),\ 1,\ 0,\ -1,\ldots-(p-3),$$
$$\alpha',\ \beta',\ \ldots = 1,\ 0,\ \ldots,\ -q+4,\quad q,\ q-1,\ q-2,\ldots,\ p+q-5,$$
$$A,\ B,\ \ldots = -1,\ -2,\ \ldots\ldots,\ p+q-4,\ \ldots,\ -(p+q-4),$$
$$A',\ B',\ \ldots = 1,\ 2,\ \ldots,\ \ldots,\ p+q-4.$$

We then find

$$\Sigma\alpha = (q-3)(p-2)+\tfrac{1}{2}(q-2)(q-1)+(p-2)\,\tfrac{1}{2}\,a+10,$$
$$\Sigma\alpha' = -\beta-\tfrac{1}{4}\,a^2+\tfrac{1}{2}\,a-10,$$
$$\Sigma A = -\Sigma A' = \tfrac{1}{2}\,a^2-\tfrac{1}{2}\,a+6,$$
$$\Sigma a^2 = \tfrac{1}{8}\,a^3-\tfrac{1}{4}\,a^2+\tfrac{14}{3}\,a+26,$$
$$\Sigma aa' = -\Sigma AA' = -\tfrac{1}{3}\,a^3-\tfrac{1}{4}\,a^2+\tfrac{1}{4}\,a-90a+100,$$
$$\Sigma AB = \tfrac{1}{8}\,a^3-\tfrac{19}{12}\,a^2+\tfrac{19}{4}\,a^3-\tfrac{47}{12}\,a+65,$$
$$\Sigma ABC = \tfrac{1}{120}\,a^4-\tfrac{17}{24}\,a^3+\tfrac{31}{8}\,a^2-\tfrac{1379}{120}\,a+19.$$

We find

$$\Sigma A+\Sigma a = \beta-8,$$
$$\Sigma A^2+\Sigma a^2 = \beta-3a+8,$$
$$\Sigma AB-\Sigma a\beta = \beta^2-\beta(-\tfrac{1}{2}a+\tfrac{1}{2}\beta-\tfrac{4}{3})\,a+36a-126,$$
$$\Sigma AB-2a\beta = \beta(-\tfrac{1}{2})+\beta(-\tfrac{1}{2}a^2+\tfrac{4}{3}a+\tfrac{1}{4})+(-\tfrac{1}{12}a^4+\tfrac{4}{3}a^2-\tfrac{47}{12}a+16a)-4a^4-36a+144,$$
$$(\Sigma AB-\Sigma a\beta)+2\,a(\Sigma A+\Sigma a) = \beta^2(\tfrac{1}{2})+\beta\,(-\tfrac{1}{2}a^2+\tfrac{4}{3}\,a+144),$$
$$\tfrac{1}{2}(\Sigma AB+\Sigma a\beta+2\,a(\Sigma A+\Sigma a)) = \beta(\tfrac{1}{2}\,a+)+\beta(-15a+28)+72a-228,$$

and

$$\Sigma A^2A'+\Sigma a^2a' = (\text{ut suprà})\,\beta^2+(-\tfrac{1}{2})+\beta(-\tfrac{1}{2}\,a^2+9a-\tfrac{4}{3})-29a+98;$$

whence, adding these three expressions, we find

$$\text{Weight}-\text{Order} = \beta^2(\tfrac{1}{2})+\beta(-\tfrac{1}{2}a^2+\tfrac{4}{3}\,a+\tfrac{4}{3})+\beta(-\tfrac{1}{12}a^4+\tfrac{2}{3}a^3-7a^2+\tfrac{77}{6}a-18);$$

and for the order we have

$$(\Sigma a)^2-\Sigma a\beta = \beta^2(\tfrac{1}{2})+\beta\,(-\tfrac{1}{2}a^2+\tfrac{4}{3}a)+\beta\,(\tfrac{1}{2}a^3+3a+\tfrac{1}{4}\beta+133);$$

and then

$$\Sigma ABC+\Sigma a\beta\gamma = (\text{ut suprà})\,\beta\,(-\tfrac{1}{12})+\beta\,(\tfrac{1}{12}\,a^2-\tfrac{1}{4}\,a+\tfrac{1349}{120})-1210,$$
$$(\Sigma AB-\Sigma a\beta)\,(\Sigma A+\Sigma a) = \beta^2(-\tfrac{1}{4}\,a)+\beta(\tfrac{4}{3}\,a^2+\tfrac{49}{4}\,a^2-\tfrac{14}{12}\,a+241),$$

$$-a^4 + 12a^3 - 132a + 1064;$$

whence, adding these three expressions, we find for the weight

$$\text{Order} - \text{Order} = \beta^2(\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}\,a^2 + \tfrac{4}{3}\,a + \tfrac{4}{3}) + \beta\,(\tfrac{4}{3}\,a^3 + 3a - \tfrac{4}{3}\,a) - 6;$$

and by the foregoing expression for the weight, we get

$$\text{Weight} - \text{Order} = \beta^2(\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}a^2 + \tfrac{4}{3}\,a + \tfrac{4}{3}) - 11a + 24;$$

and therefore

$$G(m^2) = \tfrac{1}{2}\,\beta\,[2\beta^2 + \beta(-6a) + (3a^2 + 18a - 26) - 66a + 144],$$

a formulae the direct verification whereof is due to Dr Salmon; see *post*, Annex No. 3.

The formulae for $NT(1, m, n)$ *and* $NR(1, m, n)$, **Articles 43 to 46.**

43. $NR(1, m, n)$.—Through the line 1 take any plane meeting the curve m in m points and the curve n in n points; in this case, by the construction of the scroll, we have through each of the m points, lines n in number forming the lines of the scroll $S(1, m, n)$, and their intersections are the points m_1n_1, m_2n_2, …, m_mn_n, are generating lines of the scroll; their intersection on the line 1 in question being a point of the line 1 in the scroll, and on the line 1 being a single point of the line 1. Through each of the n points, that is, the lines 1 being on the surface, they are generating lines of the scroll.

44. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0$, $y = 0$ for the equations of the line 1 (which, as we have seen, are a non-singular line on the scroll), we have a system of intersections with the curves m and n giving rise to such a relation as a, a' being functions of the coordinates, the degree in (x, y, z, w) are then to make a particular system of values

$$2[mn]_2 = 2n + 2x' + R,$$

see *post*, Annex No. 4. And then observing that the form

$$x = S(m, n^2) = mn\,([n]_2 + N),$$

these values give

$$\begin{aligned} 2x + 2x' + R &= 2n\,(mn^2 - 2x - 2x' - R), \\ &= [mn]_2 - n\,[m]_2 - x\,[n]_2, \\ &= [mn]_2; \end{aligned}$$

and thence

$$NR(1, m, n) = \tfrac{1}{2}\,mn\,(mn - m - n - 3),$$

which is the result before noted in the Table.

45. $NR(1, m^2)$.—Through any line 1 take any arbitrary plane meeting the curve m in m points; if m_1, m_2, …, m_m are the points of intersection of the curve, the lines so determined are generating lines of the scroll $S(1, m^2)$, and their intersections on the line 1 in m points; and their intersections in the plane 1 in m points yielding the nodal cuspidal points NR; but in like manner the lines $m_1, m_2, m_3, m_4, m_4, m_5, m_6$ are generating lines of the scroll of an arbitrary plane through the curve m in m points, that there are also generating lines of the scroll of an arbitrary plane through the curve m in m points. We have therefore that

$$NR(1, m^2) = \tfrac{1}{2}\,m\,([m]_2 + M) + a.$$

But these formulae obtained in fact give them as the special cases. To find them as the special cases, we will employ

$$S(m,n) = nS(1,m,n) + ND(1,m,n) = 2mn,$$

that is, $S = S(1,1,n) + 2[m]_2$. Hence ultimately

$$S = 2m,$$

since 2 is the number of lines which can be drawn meeting each of four given right lines. Hence

$$S(1,1,1,1) = 0 \quad (m=1, n=1, p=1), \quad \text{that is,} \quad S(1,1,1,1) = 2.$$

46. $G(m,n,p,q) = 2mnpq$. — We have here a precisely similar manner to

$$G(m,n,p,q) = 2mnpq.$$

Intersections of a generating line with the Nodal Total, **Articles 47 to 58.**

47. We have

$$NT(1,m,n) = NG(1,m,n) + ND(1,m,n) - mn\,(m+n-2) + ND(1,m,n) + NR(1,m,n)$$
$$+NR(1,m,n) = \tfrac{1}{2}\,mn\,(mn+3m+3n-5),$$

which is the result in the Table.

48. And moreover

$$NT(m^2,n) = ND(m^2,n) + NG(m^2,n) - ND(1,m,n)$$
$$+NR(m^2,n) = \tfrac{1}{2}[m]_2 + \tfrac{1}{2}[m]_2 + M\,([m]_2 - \tfrac{1}{2}) + M^2\cdot\tfrac{1}{2}$$
$$+NR(m^2,n) + 2[m]_2 + M\,([m]_2 - \tfrac{1}{2}) + M^2\cdot\tfrac{1}{2} + M\,([m]_2 - 2m + 5);$$

which is

$$= \tfrac{1}{2}[m]_2 + 2[m]_2 + M\,([m]_2 - \tfrac{1}{2}) + M^2\cdot\tfrac{1}{2},$$
$$= \tfrac{1}{2}[m]_2 - S + M\,([m]_2 - \tfrac{1}{2}),$$

if $S = S(1,m^2) = [m]_2 + M$.

The NT and NR formulae, **Articles 49 to 58.**

49. I proceed to find $NT(m,n,p)$, &c. by a functional investigation, such as was employed for finding $G(1,1,m)$, &c. Writing $S(m)$ to denote either of the scrolls $S(m,n,p)$ or $S(m^2,n)$ or $S(m^3)$, the aggregate of the two curves m, n'; then the curve m' may be replaced by h a double point and join two together by chords, each pair of which gives a generating line of the scroll; the latter in question are merely a double curve, viz. a chord of the line $G(m,m')$. We have

$$NT(m+m')^4 = NT(m^4) + NT(m'^4) + 6(m+m'^4)\,(\tfrac{1}{2}[m]_3 + M([m]_2+m) + M^2(\tfrac{1}{2}m+1) + MM').$$

50. Instead of assuming

$$NT = \tfrac{1}{2}\,S^2 + \phi,$$

it is the same thing, and it is rather more convenient, to assume

$$NT = \tfrac{1}{2}\, S^2 - S + \phi,$$

viz. $NT - \frac{1}{2}\,(S(m))^2 - S(m) + \phi(m)$, &c. Then observing that

$$S(m+m') = S(m) + S(m'), \text{ \&c.}$$

51. First, if $\phi(m)$ stands for $\phi(m,n)$ or $\phi(m^2)$ or $\phi(m^3)$, the foregoing equations give

$$\phi(m+m') = \phi(m) + \phi(m'),$$

$$\phi(m+m^2)^3 = \phi(m^3) + \phi(m'^3) + 3\,\phi(m^2,m') + 3\,\phi(m,m'^2),$$
$$\phi(m+m')^4 = \phi(m^4) + \phi(m'^4) + 4\,\phi(m^3,m') + 4\,\phi(m,m'^3) + 6\,\phi(m^2,m'^2);$$

and if in the second equation $\phi(m,m')$, $\phi(m^2,m')$ and in the third equation $\phi(m^2)$, $\phi(m^3)$ are regarded as known, these are all of them of the form

$$f(m+m') - f(m) - f(m') = \text{Funct. } (m,m'),$$

so that, a particular solution being obtained, we have at least $f(m)$ as a function of m and M only.

52. First if $\phi(m)$ stands for $\phi(m,n,p)$, we obtain $\phi(m,n,p) = mn + \beta M$, or observing that $\phi(m,n,p)$ must be symmetrical in regard to the curves m, n, and p, we may write

$$f(m,n,p) = mnp + \beta(Mnp + Nmp + Pmn) + \gamma(mNP + nMP + pMN) + \delta\, MNP,$$

and then

$$NT(m,n,p) = \tfrac{1}{2}\, S^2 - S + \phi(m,n,p),$$
$$= 2mnp\,(mnp-1) + \phi(m,n,p).$$

A particular solution is $\phi(m,n,p) = (m^2)\, N + mnp$, and thus we have therefore

$$\phi(m^2,n) = M(\tfrac{1}{2}\,[m]_2 + mn) + M^2\, n + Mn + \beta M,$$

or observing that n, considered as a function of n only, satisfies the equation

$$\phi(n+n') = \phi(n) + \phi(n'),$$

we then have

$$NT(m^2,n) = \tfrac{1}{2}\, S^2 - S + \phi(m^2,n),$$

where

$$S = S(m^2,n) = n\,([m]_2 + M).$$

53. Next, if $\phi(m^2)$, substituting for $\phi(m,n)$ and $\phi(m,n')$ their values, we have

$$\phi(m+m')^4 - \phi(m^4) - \phi(m'^4) = m\,([m]_2 + M) + m'\,([m']_2 + M') + mn'\,(m+m')$$

which is satisfied by

$$\phi(m^4) = M\,(\tfrac{1}{2}[m]_3 - \tfrac{1}{2}m) + M^2 + am,$$

and the general value then is

$$NT(m^4) = M\,(\tfrac{1}{2}[m]_3 - \tfrac{1}{2}m) + M^2 + am + \beta M,$$

and we have

$$NT(m^4) = \tfrac{1}{2}\,S^2 - S + \phi(m^4).$$

54. Taking for the curve in question (p, q) curve on the hyperboloid $(m = p + q,\ M = -pq)$, $S(m^4)$ becomes the hyperboloid taken k times, if $k = \frac{1}{2}[p]_2 + \frac{1}{2}[q]_2$; that is, the nodal curve is the hyperboloid taken k times, the values NT and S that imply $a = 2$, $\beta = 11$, for the values which the formula gives

$$\phi(m^4) = \tfrac{1}{2}\,m\,(m^2)\,(-7) + m\,M + p\,(m-1)\,M\,m\,(m-1).$$

55. I assume the correctness of the value

$$\phi(m^4) = 2m + M\,(\tfrac{1}{2}[m]_3 - \tfrac{1}{2}m) + M^2;$$

so obtained, as being in fact verified by means of the six several curves $(1,1)$, $(1,2)$, $(1,3)$, $(2,2)$, $(2,3)$, $(3,3)$, &c. in the foregoing value of $\phi(m^4)$, we have therefore the formula

$$\phi(m^4) = 2m + M\,(\tfrac{1}{2}[m]_3 - \tfrac{1}{2}m) + M^2,$$

and so on; thus the value of $\phi(m^4)$ thus obtained is being in fact verified by means of the six several curves $(1,1)$, $(1,2)$, $(1,3)$, $(2,2)$, $(2,3)$, $(3,3)$, &c. in the foregoing value of $\phi(m^4)$.

56. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0$, $y = 0$ for the equations of the line 1 (which, as we have seen, is a non-singular line on the scroll), we have a relation as a, a' being functions of the coordinates, the degree in (x, y, z, w) being a system of values, $a = a'$, $b = b'$, where the lines 1 are in the same plane.

57. For the scroll $S(m, n^2)$ we find

$$\begin{aligned} NR(m,n^2) &= NT(m,n^2) - ND(m,n^2) - NG(m,n^2),\\ &= a\,(\tfrac{1}{2}[n]_2 + N\,(\tfrac{1}{2}[n]_2 + n) + N^2 \cdot \tfrac{1}{2}m). \end{aligned}$$

58. And moreover

$$\begin{aligned} NR(m^3) &= NT(m^3) - ND(m^3) - NG(m^3),\\ &= m\,(\tfrac{1}{2}[m]_2 + \tfrac{1}{2}[m]_2 + M\,([m]_2 - \tfrac{1}{2}) + M^2 \cdot \tfrac{1}{2}M)\\ &\quad + M\,(\tfrac{1}{2}[m]_2 + \tfrac{1}{2}[m]_2 + 9m - 20) + M^2 \cdot (\tfrac{1}{2}[m]_2 - 2m). \end{aligned}$$

and the investigation of the series of results given in the Table is thus concluded.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1, m, n)$ and $S(1, m, n^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1, m, n)$ the directrix curves are respectively multiple curves of the orders m, n, and a generating line meets each of these in a single point. The number of intersections is therefore

$$m + n - 2 = S(1, m, n) - 2.$$

60. Similarly, counting in like manner the lines m, n, and considering the construction of the surface, we will obtain in addition the value a of the points $(\alpha, \beta, \gamma, \delta)$ on the curve and considering the union of the nodal director, $G(m^2, n)$ being equivalent to $\frac{1}{2}mM$, we find that the curve m_1, m_2 are not the same plane a, b, those the lines n_1, n_2, n_3, are generating lines of the scroll, that there are, e.g. of the α lines, α in number; whence in question does not arise of finding the number of additional generating lines on the scroll.

61. The torse case considered is that of a curve in space, that is, a curve in space, but it is the curve which lies on a quadric surface, and is the complete intersection of a quadric surface with a surface of the order p or q respectively; but I will develope them somewhat more particularly as regards the discussion of the residue, the order of the residue is in question; the number of points in space considered as the residue, the order of the residue is in question.

62. The values of $G(m, n, p, q)$, $G(m^2, n, p)$, $G(m^3, p)$, $G(m^4)$ are subjected to the verification which has been already mentioned.

63. We see now that the values S, ND, NG, NR, NT for each of the scrolls of the form $S(m, n, p)$, $S(m^2, n)$, and $S(m^3)$ are all those which have been already given in the Table.

[End of Papers 334 (cont.), 336–339, from pp. 151–200 of Vol. V]

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Volume V

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Arthur Cayley

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339. On Skew Surfaces, otherwise Scrolls *(continued)*

[Continuation from p. 176, in the middle of the investigation of the G formulae.]

25. $G(m^2, n^2)$.—I investigate the value by a process similar to that employed for $G(1, 1, m^2)$. Suppose that the curves m and n are plane curves having respectively h and k double points; then the line of intersection of the two planes meets the curve m in m points, and the curve n in n points; or, combining in every manner the m points two and two together, and the n points two and two together, the line in question is to be considered as $\frac{1}{2}[m]^2 \cdot \frac{1}{2}[n]^2$ coincident lines, each meeting the curve m twice and the curve n twice. There are besides the hk lines joining each double point of the curve m with each double point of the curve n. This gives in all $\frac{1}{4}[m]^2[n]^2 + hk$ lines; or, writing $h = \frac{1}{2}[m]^2 + M$, $k = \frac{1}{2}[n]^2 + N$, the number is

$$= \tfrac{1}{2}\,[m]^2\,[n]^2 + M \cdot \tfrac{1}{2}[n]^2 + N \cdot \tfrac{1}{2}[m]^2 + MN;$$

which is the value of $G(m^2, n^2)$ given by the investigation.

26. $G(m^2, n)$.—We have

$$G(m^2, n) = n\,G(1, m^2) = n\,S(m^2),$$

and it is in fact the same question to find $G(1, m^2)$ and to find $S(m^2)$. I assume for the present that the value [of $S(m^2)$, see post Art. 38] is $= \frac{1}{2}[m]^3 + M(m-2)$; and we then have

$$G(m^2, n) = n\,(\tfrac{1}{2}[m]^3 + M(m-2)).$$

27. Before going further, I observe that there are certain functional conditions which must be satisfied by the G formulae. Thus if the curve m be replaced by the system of the two curves

m, m', instead of M we have $M + M'$. Let $G(m)$ denote any one of the functions $G(m, n, p, q)$, $G(m, n^2, p)$, $G(m, n^3)$; we must have

$$G(m + m') = G(m) + G(m').$$

Similarly, if $G(m^2)$ denote either of the functions $G(m^2, n, p)$, $G(m^2, n^2)$, we must have

$$G(m + m')^2 = G(m^2) + G(m, m') + G(m'^2);$$

and so if $G(m^3)$ stand for $G(m^3, n)$, then

$$G(m + m')^3 = G(m^3) + G(m^2, m') + G(m, m'^2) + G(m'^3);$$

and finally

$$G(m + m')^4 = G(m^4) + G(m^3, m') + G(m^2, m'^2) + G(m, m'^3) + G(m'^4).$$

28. The first three equations may be at once verified by means of the above given values of the G functions. But conversely, at least on the assumption that $G(m)$, $G(m^2)$, &c., in so far as they respectively depend on the curve m, are functions of m and M only, we may, by the solution of the functional equations, obtain the values of the G functions. It is to be observed that the first equation is of the form

$$\phi(m + m') = \phi(m) + \phi(m'),$$

the general solution whereof is

$$\phi m = \alpha m + \beta M;$$

the second equation, supposing that $G(m, m')$ is known—the third equation, supposing that $G(m^2, m')$ and $G(m, m'^2)$ are known—and the fourth equation, supposing that $G(m^3, m')$, $G(m^2, m'^2)$, $G(m, m'^3)$ are known, are respectively of the form

$$\phi(m + m') = \phi m + \phi m' + \text{Funct.}(m, m');$$

and hence if a particular solution be given, the general solution is

$$\phi(m) = \text{Particular Solution} + \alpha m + \beta M.$$

The values of the constants must in each case be determined by special considerations.

29. The value of $G(m, n, p, q)$ was obtained strictly; that of $G(m^2, n, p)$ was reduced to depend on $G(1, 1, m^2)$, and that of $G(m^2, n)$ on $G(1, m^2)$. I apply therefore the functional equations to the confirmation of the values of $G(1, 1, m^2)$, $G(m^2, n^2)$, and $G(1, m^2)$, and to the determination of the value of $G(m^4)$.

30. First, if $G(m^2)$ denote $G(1, 1, m^2)$, then $G(m, m')$ denotes $G(1, 1, m, m')$, which is $= 2mm'$; hence

$$G(m + m')^2 - G(m^2) - G(m'^2) = 2mm',$$

which is satisfied by $G(m^2) = [m]^2$. This gives

$$G(1, 1, m^2) = [m]^2 + \alpha m + \beta M;$$

but if the curve m be a system of m lines ($m = m$, $M = 0$), then $G(1, 1, m^2) = [m]^2$; and again, if the curve m be a conic ($m = 2$, $M = -1$), then $G(1, 1, m^2) = 1$. This gives $\alpha = 0$, $\beta = 1$, and therefore

$$G(1, 1, m^2) = [m]^2 + M.$$

31. Next, if $G(m^2)$ denote $G(m^2,\, n^2)$, then $G(m,\, m')$ denotes $G(m,\, m',\, n^2)$, which is $= mm'\,([n]^2 + N)$. The functional equation is

$$G(m+m')^2 - G(m^2) - G(m'^2) = mm'\,([n]^2 + N),$$

which is satisfied by $G(m^2) = \frac{1}{2}[m]^2\,([n]^2 + N)$. Hence we have

$$G(m^2,\, n^2) = \tfrac{1}{2}[m]^2\,([n]^2 + N) + \alpha m + \beta M,$$

where α, β are functions of n, N; and observing that $G(m^2,\, n^2)$ must be symmetrical in regard to the curves m and n, it is easy to see that we may write

$$G(m^2,\, n^2) = \tfrac{1}{2}[m]^2[n]^2 + M\cdot\tfrac{1}{2}[n]^2 + N\cdot\tfrac{1}{2}[m]^2 + \alpha mn + \beta\,(mN + nM) + \gamma MN,$$

where α, β, γ are absolute constants. To determine them, if the curve m be a pair of lines $(m = 2,\ M = 0)$, then

$$G(m^2,\, n^2) = G(1,\, 1,\, n^2) = [n]^2 + N;$$

and if each of the curves m, n be a conic $(m = 2,\ M = -1,\ n = 2,\ N = -1)$, then

$$G(m^2,\, n^2) = 1.$$

These cases give $\alpha = \beta = 0$, $\gamma = 1$, and therefore

$$G(m^2,\, n^2) = \tfrac{1}{2}[m]^2[n]^2 + M\cdot\tfrac{1}{2}[n]^2 + N\cdot\tfrac{1}{2}[m]^2 + MN.$$

32. Again, $G(m^2)$ standing for $G(1,\, m^3)$, then $G(m^2,\, m')$ and $G(m,\, m'^2)$ will stand for $G(1,\, m^2,\, m')$ and $G(1,\, m,\, m'^2)$, the values whereof are $m'\,([m]^2 + M)$ and $m\,([m']^2 + M')$ respectively. We have thus

$$G(m+m')^3 - G(m^3) - G(m'^3) = m'\,([m]^2 + M) + m\,([m']^2 + M'),$$

a solution of which is $G(m^3) = \frac{1}{3}[m]^3 + mM$. Hence we have

$$G(1,\, m^3) = \tfrac{1}{3}[m]^3 + mM + \alpha m + \beta M.$$

Suppose first that the curve m is a system of lines $(m = m,\ M = 0)$, then $G(1,\, m^3) = \frac{1}{3}[m]^3$; and next that the curve m is a cubic in space or skew cubic $(m = 3,\ M = -2)$, then $G(1,\, m^3) = 0$, since a line can meet the curve in two points only. We thus find $\alpha = 0$, $\beta = -2$, and thence

$$G(1,\, m^3) = \tfrac{1}{3}[m]^3 + M(m-2).$$

33. Hence, substituting for $G(m^3,\, m')$, $G(m^2,\, m'^2)$, $G(m,\, m'^3)$ their values

$$m'\,(\tfrac{1}{3}[m]^3 + M(m-2)),\quad \tfrac{1}{2}[m]^2[m']^2 + M\cdot\tfrac{1}{2}[m']^2 + M'\cdot\tfrac{1}{2}[m]^2 + MM',\quad m\,(\tfrac{1}{3}[m']^3 + M'(m'-2))$$

respectively, we find

$$\begin{aligned} G(m+m')^4 - G(m^4) - G(m'^4) = m'\,(\tfrac{1}{3}[m]^3 + M(m-2)) \\ + \tfrac{1}{2}[m]^2[m']^2 + M\cdot\tfrac{1}{2}[m']^2 + M'\cdot\tfrac{1}{2}[m]^2 + MM' \\ + m\,(\tfrac{1}{3}[m']^3 + M'(m'-2)),\end{aligned}$$

and thence, obtaining first a particular solution, the general solution is

$$G(m^4) = \tfrac{1}{12}[m]^4 + M\,(\tfrac{1}{2}[m]^2 - 2m) + M^2 + \alpha m + \beta M.$$

34. To determine the constants, suppose first that the curve m is a system of lines $(m = m,\ M = 0)$, we must have $G(m^4) = \frac{1}{12}[m]^4 + m$, and thence $\alpha = 0$. Next, if the curve m be a conic $(m = 2,\ M = -1)$, we must have $G(m^4) = 0$; and this gives $\beta = \frac{1}{2}$, and consequently

$$G(m^4) = \tfrac{1}{12}[m]^4 + m + M\,(\tfrac{1}{2}[m]^2 - 2m + \tfrac{1}{2}) + M^2.$$

The NG formulae, Article 35.

35. The NG formulae are now at once obtained, viz. we have

$$\begin{aligned} NG(m^2,\, n,\, p) &= G(m^2,\, n,\, p) + G(m,\, n^2,\, p) + G(m,\, n,\, p^2), \\ NG(m^3,\, n) &= 3\,G(m^3,\, n) + G(m^2,\, n^2), \\ NG(m^4) &= 6\,G(m^4), \end{aligned}$$

which give the values in the Table.

The S formulae, particular cases, Articles 36 to 40.

36. The S formulae have in fact been obtained in the investigation of the G formulae: we have

$$\begin{aligned} S(m,\, n,\, p) &= 2mnp, \\ S(m^2,\, n) &= n\,([m]^2 + M), \\ S(m^3) &= \tfrac{1}{3}[m]^3 + M(m-2). \end{aligned}$$

37. In confirmation of the formula $S(1,\, m^2) = [m]^2 + M$, it is to be remarked that if we take through the line 1 an arbitrary plane, this meets the curve m in m points, and joining these two and two together we have $\frac{1}{2}[m]^2$ lines, each of them meeting the curve m twice and also meeting the line 1; that is, the lines in question are generating lines of the scroll $S(1,\, m^2)$. The line 1 is, as already mentioned, an $(h =)\,(\frac{1}{2}[m]^2 + M)$tuple line on the scroll; the section by the arbitrary plane is therefore the line 1 taken $(\frac{1}{2}[m]^2 + M)$ times, together with the before-mentioned $\frac{1}{2}[m]^2$ lines; that is, the order of the surface is $[m]^2 + M$, as it should be. This is in fact the mode in which the order of the scroll $S(1,\, m^2)$ was originally obtained by Dr Salmon.

38. As regards the formula $S(m^3) = \frac{1}{3}[m]^3 + M(m-2)$, suppose that the curve m is a $(p,\, q)$ curve on the hyperboloid; we have as before $m = p + q$, $M = -pq$, and the formula becomes

$$S(m^3) = \tfrac{1}{3}[p+q]^3 - pq\,(p+q-2),$$

which is

$$= \tfrac{1}{3}[p]^3 + \tfrac{1}{3}[q]^3$$

as already remarked, the surface is in this case the hyperboloid taken $\frac{1}{3}[p]^3 + \frac{1}{3}[q]^3$ times.

39. It is to be noticed also that if the curve m be a system of lines ($m = m$, $M = 0$), then the formula gives

$$S(m^3) = \tfrac{1}{3}[m]^3,$$

which is right, since in this case the scroll is made up of the $\frac{1}{3}[m]^3$ hyperboloids generated each of them by a line which meets three out of the m lines.

In the case of a curve m, which is such that the coordinates of any point of the curve are proportional to rational and integral functions of the order m of an arbitrary parameter θ, or say the case of a unicursal curve of the order m, we have

$$(h = \tfrac{1}{2}[m-1]^2 \text{ and } \therefore)M = -(m-1),$$

and the formula gives

$$S(m^3) = \tfrac{1}{3}[m]^3 - (m-1)(m-2),$$

for a direct investigation of which see post, Annex No. 1.

40. In the case of a curve m, which is the complete intersection of two surfaces of the orders p and q respectively, or say a complete $(p \times q)$ intersection, we have

$$m = pq, \qquad (h = \tfrac{1}{2}pq\,(p-1)(q-1) \text{ and } \;\therefore\,)\; M = -\tfrac{1}{2}pq\,(p+q-2);$$

and we find

$$\begin{aligned} S(m^3) &= \tfrac{1}{6}\,pq\,(pq-2)(2pq-3p-3q+4), \\ &= \tfrac{1}{6}\,\beta\,(\beta-2)(2\beta-3\alpha+4) \end{aligned}$$

if $\alpha = pq$, $\beta = p+q$. The mode of obtaining this result by a direct investigation was pointed out to me by Dr Salmon; see post, Annex No. 2.

Particular cases of the formula for $G(m^4)$, Articles 41 & 42.

41. In the case of a $(p,\, q)$ curve on the hyperboloid, putting as before $m = p+q$, $M = -pq$, we find

$$G(m^4) = \tfrac{1}{12}[p+q]^4 + p + q - pq\,(\tfrac{1}{2}[p+q]^2 - 2(p+q) + \tfrac{1}{2}) + p^2q^2,$$

which is

$$= \tfrac{1}{12}\,([p]^4 + [q]^4) - 2p\,[q-1]^2 - 2q\,[p-1]^2,$$

vanishing if p, q are neither of them greater than 3: this is as it should be, since there is then no line which meets the curve four times. The curves for which the condition is satisfied are (1, 1) the conic, (1, 2) the cubic, (2, 2) the quadriquadric, (1, 3) the excubo-quartic, (2, 3) the excubo-quintic (viz. the quintic curve, which is the partial intersection of a quadric surface and a cubic surface having a line in common), and (3, 3) the quadri-cubic, or complete intersection of a quadric surface and a cubic surface. If either p or q exceeds 3, we have the case of a curve through every point whereof there can be drawn a line or lines through four or more points, and the formula is inapplicable.

42. In the case of a complete $(p \times q)$ intersection, we have as before $m = pq$, $M = -\frac{1}{2}pq\,(p+q-2)$, and the formula for $G(m^4)$ becomes

$$G(m^4) = \tfrac{1}{12}\,\beta\,\Big\{2\beta^3 - 6\alpha\beta^2 + \beta\,(3\alpha^2 + 18\alpha - 26) - 66\alpha + 144\Big\},$$

a formula the direct verification whereof is due to Dr Salmon; see post, Annex No. 3.

The formulae for $NR(1,\, m,\, n)$ and $NR(1,\, m^2)$, Articles 43 to 46.

43. $NR(1,\, m,\, n)$.—Through the line 1 take any plane meeting the curve m in m points and the curve n in n points; then if m_1, m_2 be any two of the m points, and n_1, n_2 any two of the n points, the lines m_1n_1 and m_2n_2 are generating lines of the scroll $S(1,\, m,\, n)$, and these lines intersect in a point which belongs to the Nodal Residue NR; and in like manner the lines m_1n_2 and m_2n_1 are generating lines of the scroll, and they intersect on a point of NR; we have thus

$$(2 \cdot \tfrac{1}{2}[m]^2 \cdot \tfrac{1}{2}[n]^2 =)\; \tfrac{1}{2}[m]^2[n]^2$$

points on NR, that is, the arbitrary plane through the line 1 cuts NR in $\frac{1}{2}[m]^2[n]^2$ points. But the plane also cuts NR in certain points lying on the line 1, and if the number of these be (a), then

$$NR(1,\, m,\, n) = \tfrac{1}{2}[m]^2[n]^2 + a.$$

44. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0$, $y = 0$ for the equations of the line 1 (which, as we have seen, is a mn-tuple line on the scroll), the equation of the scroll is of the form $(A, \ldots \mid x, y)^{mn} = 0$, where $A, \ldots$ are functions of the coordinates of the degree mn. The entire number of cuspidal points on the line 1 is thus $= 2\,[mn]^2$; but these include different kinds of cuspidal points, viz. we have

$$2\,[mn]^2 = 2a + 2\alpha + 2\alpha' + R,$$

if (a) be the number of points in which the line 1 meets NR,		
α	„ „ „ „	$S(m^2,\, n)$,
α'	„ „ „ „	$S(m,\, n^2)$,
R	„ „ „ „	Torse$(m,\, n)$,

where by Torse(m, n) I denote the developable surface or "Torse" generated by a line which meets each of the curves m and n. The order of the Torse in question is

$$R = (n\,([m]^2 - 2h) + m\,([n]^2 - 2k) =)\ 2\,(nM + mN),$$

see post, Annex No. 4. And then observing that we have

$$2a + 2\alpha + 2\alpha' + R = 2m\,[m]^2 + 2m\,[n]^2 \quad \text{(as above)},$$

these values give

$$\alpha = S(m^2,\, n) = n\,([m]^2 + M),$$
$$\alpha' = S(m,\, n^2) = m\,([n]^2 + N),$$

and we have

$$a = \tfrac{1}{2}\,(2\,[mn]^2 - 2\alpha - 2\alpha' - R),$$
$$= [m]^2[n]^2 - n\,[m]^2 - m\,[n]^2,$$

and thence

$$NR(1,\, m,\, n) = \tfrac{3}{2}[m]^2[n]^2.$$

45. $NR(1,\, m^2)$.—Through the line 1 take any arbitrary plane meeting the curve m in m points; if m_1, m_2, m_3, m_4 be any four of these, then the lines m_1m_2 and m_3m_4 are generating lines of the scroll $S(1,\, m^2)$, and their intersection is a point of the nodal residue NR; but in like manner the lines m_1m_3 and m_2m_4 are generating lines of the scroll, and their intersection is a point of NR; and so the lines m_1m_4 and m_2m_3 are generating lines of the scroll, and their intersection is a point of NR. We have thus $(3 \times \frac{1}{4}[m]^4 =)\ \frac{3}{4}[m]^4$ points of NR on the arbitrary plane through the line 1. But there are besides the points of NR which lie on the line 1; and if the number of these be (a), then

$$NR(1,\, m^2) = \tfrac{3}{4}[m]^4 + a.$$

46. The points (a) are included among the cuspidal points of the scroll lying on the line 1. Supposing for a moment that $x = 0$, $y = 0$ are the equations of the line 1, then this line being a $(\frac{1}{2}[m]^2 + M)$tuple line on the scroll, the equation of the scroll is of the form $(A, \ldots \mid x,\, y)^{\frac{1}{2}[m]^2+M} = 0$, where $A, \ldots$ are functions of the coordinates of the degree $\frac{1}{2}[m]^2$: the number of cuspidal points on the line 1 is thus

$$\big(2 \cdot \tfrac{1}{2}[m]^2\,(\tfrac{1}{2}[m]^2 - 1 + M)\ =\big)\ [m]^2\,(\tfrac{1}{2}[m]^2 - 1 + M).$$

But these include cuspidal points of several kinds, viz. we have

$$[m]^2\,(\tfrac{1}{2}[m]^2 - 1 + M) = 2a + 3\beta + R',$$

if (a) be the number of points in which the line 1 meets NR,
β „ „ „ „ $S(m^3)$,
R' „ „ „ „ Torse(m^2),

where Torse(m^2) denotes the developable surface or Torse generated by a line which meets the curve m twice. The order of the Torse in question is

$$R' = -2\,(m-3)\,M$$

(see post, Annex No. 5); and then since $\beta = S(m^3) = \frac{1}{3}[m]^3 + M(m-2)$, we find

$$2a = [m]^2\,(\tfrac{1}{2}[m]^2 - 1 + M) - 3\,(\tfrac{1}{3}[m]^3 + M(m-2)) + 2M(m-3),$$

and thence

$$a = \tfrac{1}{4}[m]^4 + \tfrac{1}{2}[m]^3 + M\,([m]^2 - m),$$
$$NR(1,\, m^2) = \tfrac{3}{4}[m]^4 + \tfrac{1}{2}[m]^3 + M\,(\tfrac{1}{2}[m]^2 - \tfrac{1}{2}m).$$

But I have not succeeded in finding by a like direct investigation the values of

$$NR(m,\, n,\, p), \quad NR(m^2,\, n), \quad NR(m^3).$$

Formulae for $NT(1,\, m,\, n)$, $NT(1,\, m^2)$, Articles 47 and 48.

47. We have

$$\begin{aligned} NT(1,\, m,\, n) = NG(1,\, m,\, n) \quad & = mn\,(m+n-2) + mN + nM \\ + ND(1,\, m,\, n) \quad & + \tfrac{1}{2}mn\,(mn+m+n-3) \\ + NR(1,\, m,\, n) \quad & + \tfrac{3}{2}[m]^2[n]^2, \end{aligned}$$

which is

$$= 2\,[mn]^2 + mN + nM,$$
$$= \tfrac{1}{2}S^2 - S + mN + nM,$$

where $S = S(1,\, m,\, n) = 2mn$.

48. And moreover

$$\begin{aligned} NT(1,\, m^2) = ND(1,\, m^2) \quad & = \tfrac{1}{4}[m]^4 + [m]^3 + M\,(\tfrac{1}{2}[m]^2 - \tfrac{1}{2}) + M^2\cdot\tfrac{1}{2} \\ + NG(1,\, m^2) \quad & + [m]^3 + M\,(3m-6) \\ + NR(1,\, m^2) \quad & + \tfrac{3}{4}[m]^4 + M\,(\tfrac{1}{2}[m]^2 - 2m + 3), \end{aligned}$$

which is

$$= \tfrac{1}{2}[m]^4 + 2\,[m]^3 + M\,([m]^2 + m - \tfrac{5}{2}) + M^2\cdot\tfrac{1}{2},$$
$$= \tfrac{1}{2}S^2 - S + M\,(m - \tfrac{3}{2}),$$

if $S = S(1,\, m^2) = [m]^2 + M$.

The NT and NR formulae, Articles 49 to 58.

49. I proceed to find $NT(m,\, n,\, p)$, &c. by a functional investigation, such as was employed for finding $G(1,\, 1,\, m^2)$, &c. Writing $S(m)$ to denote either of the scrolls $S(m,\, n,\, p)$, $S(m,\, n^2)$, and supposing that in place of the curve m we have the aggregate of the two curves m, m'; then the scroll $S(m+m')$ breaks up into the two scrolls $S(m)$, $S(m')$, and the intersection of these is part of the nodal total $NT(m+m')$; that is, we have

$$NT(m+m') = NT(m) + NT(m') + S(m)\cdot S(m');$$

and in like manner, if $S(m^2)$ stands for $S(m^2,\, n)$, then

$$NT(m+m')^2 = NT(m^2) + NT(m,\, m') + NT(m'^2) + C_1\,(S(m^2),\, S(m,\, m'),\, S(m'^2)),$$

where C_1 denotes the sum of the combinations two and two together; and so also

$$\begin{aligned} NT(m+m')^3 = {} & NT(m^3) + NT(m^2,\, m') + NT(m,\, m'^2) + NT(m'^3) \\ & + C_1\,(S(m^3),\, S(m^2,\, m'),\, S(m,\, m'^2),\, S(m'^3)). \end{aligned}$$

50. Instead of assuming

$$NT = \tfrac{1}{2}S^2 + \phi,$$

it is the same thing, and it is rather more convenient, to assume

$$NT = \tfrac{1}{2}S^2 - S + \phi,$$

viz. $NT(m) = \frac{1}{2}\,(S(m))^2 - S(m) + \phi(m)$, &c. Then observing that

$$S(m+m') = S(m) + S(m'), \&c.,$$

the foregoing equations for NT give

$$\begin{aligned} \phi(m+m') &= \phi(m) + \phi(m'), \\ \phi(m+m')^2 &= \phi(m^2) + \phi(m,\, m') + \phi(m'^2), \\ \phi(m+m')^3 &= \phi(m^3) + \phi(m^2,\, m') + \phi(m,\, m'^2) + \phi(m'^3); \end{aligned}$$

and if in the second equation $\phi(m,\, m')$ and in the third equation $\phi(m^2,\, m')$ and $\phi(m,\, m'^2)$ are regarded as known, these are all of them of the form

$$f(m+m') - f(m) - f(m') = \text{Funct.}(m,\, m');$$

so that, a particular solution being obtained, the general solution is $f(m) =$ Particular Solution$+$ $\alpha m + \beta M$, at least on the assumption that $f(m)$, in so far as it depends on the curve m, is a function of m and M only.

51. First, if $\phi(m)$ stands for $\phi(m,\, n,\, p)$, we obtain $\phi(m,\, n,\, p) = \alpha m + \beta M$, or observing that $\phi(m,\, n,\, p)$ must be symmetrical in regard to the curves m, n, and p, we may write

$$\phi(m,\, n,\, p) = \alpha mnp + \beta\,(Mnp + Nmp + Pmn) + \gamma\,(mNP + nMP + pMN) + \delta\, MNP,$$

and then

$$\begin{aligned} NT(m,\, n,\, p) &= \tfrac{1}{2}S^2 - S + \phi(m,\, n,\, p), \\ &= 2\, mnp\,(mnp - 1) + \phi(m,\, n,\, p). \end{aligned}$$

But for $p = 1$ this should reduce itself to the known value of $NT(1,\, m,\, n)$; this gives $\alpha = 0$, $\beta = 1$, $\gamma = 0$; we in fact have, as will be shown, post, Art. 55, $\delta = 0$; and hence

$$NT(m,\, n,\, p) = \tfrac{1}{2}S^2 - S + (Mnp + Nmp + Pmn),$$

$$= 2\,[mnp]^2 + (Mnp + Nmp + Pmn).$$

52. Next, if $\phi(m^2)$ stand for $\phi(m^2,\, n)$, then $\phi(m,\, m')$ stands for $\phi(m,\, m',\, n)$, which is $= N\,mm' + n\,(mM' + m'M)$, and the equation is

$$\phi(m + m')^2 - \phi(m^2) - \phi(m'^2) = N\,mm' + n\,(mM' + m'M).$$

A particular solution is $\phi(m^2) = \frac{1}{2}[m]^2 N + nmM$, and we have therefore

$$\phi(m^2,\, n) = \tfrac{1}{2}[m]^2 N + nmM + \alpha m + \beta M;$$

or observing that $\phi(m^2,\, n)$ considered as a function of n, satisfies the equation

$$\phi(n + n') = \phi(n) + \phi(n'),$$

and is therefore a linear function of n and N, we may write

$$\phi(m^2,\, n) = \tfrac{1}{2}[m]^2 N + nmM + \alpha nm + \beta nM + \gamma mN + \delta MN;$$

we then have

$$NT(m^2,\, n) = \tfrac{1}{2}S^2 - S + \phi(m^2,\, n),$$

where $S = S(m^2,\, n) = n\,([m]^2 + M)$.

And then putting $n = 1$, and comparing with the known value of $NT(1,\, m^2)$, we find $\alpha = 0$, $\beta = -\frac{3}{2}$. It will be shown, post, Art. 55, that $\gamma = 0$, $\delta = 0$; and we have therefore

$$\phi(m^2,\, n) = n\,M\,(m - \tfrac{3}{2}) + N\,(\tfrac{1}{2}[m]^2 + M),$$

and thence

$$\begin{aligned} NT(m^2,\, n) &= \tfrac{1}{2}S^2 - S + \phi(m^2,\, n), \\ &= n^2\left(\tfrac{1}{2}[m]^4 + 2\,[m]^3 + M\,([m]^2 + m - \tfrac{3}{2}) + M^2 \cdot \tfrac{1}{2}\right) \\ &\quad + N^2\left(\tfrac{1}{2}[m]^4 + 2\,[m]^3 + [m]^2 + M\,[m]^2 + M^2 \cdot \tfrac{1}{2}\right). \end{aligned}$$

53. Next for $\phi(m^4)$, substituting for $\phi(m^2,\, m')$ and $\phi(m,\, m'^2)$ their values, we have

$$\begin{aligned} \phi(m + m')^4 - \phi(m^4) - \phi(m'^4) &= m'\,M\,(m - \tfrac{3}{2}) + M'\,(\tfrac{1}{2}[m]^2 + M) \\ &\quad + m\,M'\,(m' - \tfrac{3}{2}) + M\,(\tfrac{1}{2}[m']^2 + M'), \end{aligned}$$

which is satisfied by

$$\phi(m^4) = M\,(\tfrac{1}{2}[m]^3 - \tfrac{3}{2}m) + M^2,$$

and the general value then is

$$\phi(m^4) = M\,(\tfrac{1}{2}[m]^3 - \tfrac{3}{2}m) + M^2 + \alpha m + \beta M,$$

and we have

$$NT(m^4) = \tfrac{1}{2}S^2 - S + \phi(m^4),$$

where

$$S = S(m^4) = \tfrac{1}{3}[m]^3 + M\,(m - 2).$$

54. Taking for the curve m the $(p,\, q)$ curve on the hyperboloid ($m = p + q$, $M = -pq$), $S(m^4)$ becomes the hyperboloid taken k times, if $k = \frac{1}{3}[p]^3 + \frac{1}{3}[q]^3$; that is, $S(m^4) = 2k$, and

$NT(m^4) = 4 \cdot \frac{1}{2}[k]^2 + \phi(m^4)$; $\phi(m^4)$ must vanish if p and q are each not greater than 3; this implies $\alpha = 3$, $\beta = 11$, for with these values the formula gives

$$\phi(m^4) = -\tfrac{1}{6}\,(q[p-1]^3 + p[q-1]^3).$$

55. I assume the correctness of the value

$$\phi(m^4) = 3m + M\,(\tfrac{1}{2}[m]^3 - 4m + 11) + M^2$$

so obtained, as being in fact verified by means of the six several curves (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3); and I remark that if the foregoing value of $\phi(m,\, n,\, p)$ had been increased by $\delta\, MNP$, then it would have been necessary to increase the value of $\phi(m^2,\, n)$ by $\delta\, M^2N$, and that of $\phi(m^4)$ by $\delta\, M^3$; and moreover that if the foregoing value of $\phi(m^2,\, n)$ had been increased by $\gamma mN + \delta MN$, then it would have been necessary to increase the value of $\phi(m^4)$ by $\gamma mM + \delta M^2$; this is easily seen by writing

$$\begin{aligned}
\phi(m^4) &= \gamma mM + \delta M^2 + \alpha M^3,\\
\phi(m^3,\, m') &= \gamma mM' + \delta MM' + 3\alpha M^2M',\\
\phi(m,\, m'^3) &= \gamma m'M + \delta MM' + 3\alpha MM'^2,\\
\phi(m'^4) &= \gamma m'M' + \delta M'^2 + \alpha M'^3,
\end{aligned}$$

the sum of which is

$$= \gamma\,(m+m')(M+M') + \delta\,(M+M')^2 + \alpha\,(M+M')^3,$$

the corresponding term of $\phi(m^4)$; hence the value of $\phi(m^4)$ being correct without the foregoing addition, we must have $\gamma = 0$, $\delta = 0$, $\alpha = 0$; which confirms the foregoing values of $\phi(m,\, n,\, p)$, $\phi(m^2,\, n)$.

56. The equation

$$NT(m^4) = \tfrac{1}{2}S^2 - S + \phi(m^4)$$

gives

$$\begin{aligned}
NT(m^4) &= \tfrac{1}{2}S^2 - S + 3m + M\,(\tfrac{1}{2}[m]^3 - 4m + 11) + M^2,\\
&= \tfrac{1}{18}[m]^6 + \tfrac{1}{3}[m]^3 + M^2 + 3m + \ldots
\end{aligned}$$

57. We have

$$\begin{aligned}
NR(m^2,\, n) &= NT(m^2,\, n) - ND(m^2,\, n) - NG(m^2,\, n),\\
&= n\,(\tfrac{1}{4}[m]^4 + \tfrac{1}{2}[m]^3 + M\,(\tfrac{1}{2}[m]^2 - 2m + 3))\\
&\quad + [n]^2\,(\tfrac{1}{4}[m]^4 + \tfrac{1}{2}[m]^3 + [m]^2 + M\,([m]^2 - \tfrac{1}{2}) + M^2 \cdot \tfrac{1}{2}).
\end{aligned}$$

58. And moreover

$$\begin{aligned}
NR(m^4) &= NT(m^4) - ND(m^4) - NG(m^4),\\
&= \tfrac{1}{72}[m]^6 + \tfrac{1}{6}[m]^4 - \tfrac{1}{6}[m]^3 + 3m\\
&\quad + M\,(\tfrac{1}{4}[m]^4 - \tfrac{1}{2}[m]^3 - \tfrac{1}{2}[m]^2 + 9m - 20)\\
&\quad + M^2\,(\tfrac{1}{2}[m]^2 - 2m);
\end{aligned}$$

and the investigation of the series of results given in the Table is thus concluded.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1,\, m,\, n)$ and $S(1,\, m^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1,\, m,\, n)$, the directrix curves are respectively multiple curves of the orders mn, n, m, and a generating line meets each of these in a single point, counting for the three curves respectively as $mn - 1$, $n - 1$, and $m - 1$ points respectively. Moreover the construction (*ante*, Art. 43) for the Nodal Residue $NR(1,\, m,\, n)$ shows that a generating line meets this curve in $(m-1)(n-1)$ points; and since the curve is merely a double curve, these count each as a single point; and the generating line does not meet the Nodal Generator $NG(1,\, m,\, n)$. The number of intersections therefore is

$$mn - 1 + (m-1) + (n-1) + (m-1)(n-1),$$

which is

$$= 2mn - 2, \quad = S - 2.$$

60. Similarly for the scroll $S(1,\, m^2)$; the directrix curves are multiple curves, viz. the line 1 is a $(\frac{1}{2}[m]^2 + M)$tuple curve, and the curve m a $(m-1)$tuple curve; the generating line meets the former in a single point, counting as $\frac{1}{2}[m]^2 + M - 1$ points, and the latter in two points, each counting as $(m-2)$ points. The construction (*ante*, Art. 45) for the Nodal Residue $NR(1,\, m^2)$ shows that the generating line meets this curve in $\frac{1}{2}[m-2]^2$ points; and since the curve is merely a double curve, these count each as a single point. Finally, the generating line does not meet the Nodal Generator $NG(1,\, m^2)$. The number of intersections thus is

$$\tfrac{1}{2}[m]^2 - 1 + M + 2(m-2) + \tfrac{1}{2}[m-2]^2,$$

which is

$$= [m]^2 - 2 + M, \quad = S - 2.$$

In the remaining cases we may use the theorem to find the number of points in which the generating line meets the Nodal Residue. Using Π as the symbol for the points in question ($\Pi(m,\, n,\, p)$ for the scroll $S(m,\, n,\, p)$, &c.), we find

61. For the scroll $S(m,\, n,\, p)$,

$$(mn-1) + (np-1) + (mp-1) + \Pi(m,\, n,\, p) = S - 2 = 2mnp - 2,$$

which gives

$$\Pi(m,\, n,\, p) = 2mnp - mn - mp - np + 1.$$

This includes the before-mentioned case

$$\Pi(1,\, m,\, n) = (m-1)(n-1),$$

and the more particular one

$$\Pi(1,\, 1,\, m) = 0.$$

62. For the scroll $S(m^2,\, n)$,

$$\tfrac{1}{2}[m]^2 - 1 + M + 2\left((m-1)\,n - 1\right) + \Pi(m^2,\, n) = S - 2 = n\left([m]^2 + M\right) - 2,$$

which gives

$$\Pi(m^2,\, n) = n\left([m]^2 - 2m + 2 + M\right) - \tfrac{1}{2}[m]^2 + 1 - M.$$

This includes the before-mentioned particular case

$$\Pi(1,\, m^2) = \tfrac{1}{2}[m-2]^2.$$

63. And lastly for the scroll $S(m^3)$,

$$3\left(\tfrac{1}{2}[m]^2 - m + 1 + M\right) + \Pi(m^3) = S - 2 = \tfrac{1}{3}[m]^3 + (m-2)\,M - 2,$$

which gives

$$\Pi(m^3) = \tfrac{1}{3}[m]^3 - \tfrac{3}{2}[m]^2 + 3m - 5 + M\,(m-5).$$

The foregoing expressions for Π might with propriety have been inserted in the Table.

Annex No. 1.—*Investigation of the formula for $S(m^3)$ in the case of the unicursal curve* (referred to, Art. 39).

Consider the unicursal m-thic curve, the equations whereof are $x : y : z : w = A : B : C : D$, where A, B, C, D are rational and integral functions of a parameter θ; and let it be required to find the equation of a plane meeting the curve in such manner that three of the points of intersection are in lineá. Taking for the equation of the plane

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

we find between $(\xi,\, \eta,\, \zeta,\, \omega)$ an equation of a certain degree in $(\xi,\, \eta,\, \zeta,\, \omega)$, which is the equation in plane-coordinates of the scroll $S(m^3)$; the degree of the equation is therefore equal to the class of the scroll; but as the class of a scroll is equal to its order, the degree of the equation is equal to the order of the scroll, or say $= S(m^3)$.

Proceeding with the investigation, if θ be determined by the equation

$$\xi A + \eta B + \zeta C + \omega D = 0,$$

then the roots $\theta_1,\, \theta_2,\, \ldots,\, \theta_m$ of this equation belong to the points of intersection of the plane and curve; and the corresponding coordinates of these points are $(A_1,\, B_1,\, C_1,\, D_1)$, $(A_2,\, B_2,\, C_2,\, D_2)$, &c.

Suppose that the points 1, 2, 3 are in lineá, and let λ, μ, ν, ρ be the coordinates of an arbitrary point, then the four points are in plano, that is, we have

$$\begin{vmatrix} \lambda & \mu & \nu & \rho \\ A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \end{vmatrix} = 0;$$

and if we form the equation

$$\Pi \begin{vmatrix} \lambda & \mu & \nu & \rho \\ A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \end{vmatrix} = 0,$$

where Π denotes the product of the terms belonging to all the triads of the m roots, the result will be symmetrical in regard to all the roots; and replacing the symmetrical functions of the roots by their values in terms of the coefficients, we have the required relation between $(\xi,\, \eta,\, \zeta,\, \omega)$.

Π contains $\frac{1}{6}[m]^3$ terms, whereof $\frac{1}{2}[m-1]^2$ contain the m-thic functions $(A_1,\, B_1,\, C_1,\, D_1)$ of the root θ_1; that is, the form of Π is

$$(\lambda,\, \mu,\, \nu,\, \rho)^{\frac{1}{6}[m]^3}\; (\theta_1,\, 1)^{m\cdot\frac{1}{2}[m-1]^2}\; (\theta_2,\, 1)^{m\cdot\frac{1}{2}[m-1]^2}\; \ldots;$$

or, when the symmetrical functions are expressed in terms of the coefficients, the form is

$$(\lambda,\,\mu,\,\nu,\,\rho)^{\frac{1}{6}[m]^3}\ (\xi,\,\eta,\,\zeta,\,\omega)^{m\cdot\frac{1}{2}[m-1]^2\cdot\frac{1}{m}}.$$

Now the above-mentioned determinant is divisible by $(\theta_1-\theta_2)(\theta_1-\theta_3)(\theta_2-\theta_3)$, or Π is divisible by $\prod(\theta_r-\theta_s)(\theta_r-\theta_t)(\theta_s-\theta_t)$; and since this product contains $(3\times\frac{1}{6}[m]^3=)\ \frac{1}{2}[m]^3$ linear factors, and the product $\Delta(\theta_1,\,\theta_2,\,\ldots,\,\theta_m)$ of the squared differences of the roots contains $(2\times\frac{1}{2}[m]^2=)\ [m]^2$ linear factors, so that we have

$$\prod(\theta_r-\theta_s)(\theta_r-\theta_t)(\theta_s-\theta_t)=\big\{\Delta(\theta_1,\,\theta_2,\,\ldots,\,\theta_m)\big\}^{\frac{1}{2}[m-1]^2},$$

and consequently

$$\Delta(\theta_1,\,\theta_2,\,\ldots,\,\theta_m)=\text{Disct.}=(\xi,\,\eta,\,\zeta,\,\omega)^{2(m-1)},$$

$$\prod(\theta_r-\theta_s)(\theta_r-\theta_t)(\theta_s-\theta_t)=(\xi,\,\eta,\,\zeta,\,\omega)^{[m-1]^2},$$

so that, omitting this factor, the remaining factor of Π is of the form

$$(\lambda,\,\mu,\,\nu,\,\rho)^{\frac{1}{6}[m]^3}\ (\xi,\,\eta,\,\zeta,\,\omega)^{\frac{1}{2}[m-1]^2\cdot\frac{1}{m}\cdot m-[m-1]^2}.$$

But the determinant vanishes if

$$(\lambda,\,\mu,\,\nu,\,\rho)=(A_1,\,B_1,\,C_1,\,D_1),\ (A_2,\,B_2,\,C_2,\,D_2),\ (A_3,\,B_3,\,C_3,\,D_3),$$

or say if

$$(\lambda,\,\mu,\,\nu,\,\rho)=(A,\,B,\,C,\,D),\quad \theta=\theta_1,\,\theta_2,\text{ or }\theta_3;$$

it follows that the product Π contains the factor $(\lambda,\,\mu,\,\nu,\,\rho)$ raised to a corresponding power, or omitting this factor, and observing that

$$\tfrac{1}{2}[m]^2-[m-1]^2-\tfrac{1}{2}[m]^2=\tfrac{1}{2}[m]^2-[m-1]^2=\tfrac{1}{2}[m-1]^2,$$

the remaining factor is of the form

$$(\xi,\,\eta,\,\zeta,\,\omega)^{\frac{1}{3}[m]^3-(m-1)(m-2)},$$

or we have finally

$$S(m^3)=\tfrac{1}{3}[m]^3-(m-1)^2,$$

which is the required expression.

I give the following investigation of the expression $\frac{1}{2}[m-1]^2$ for the number of apparent double points. Imagine through the point $(x=0,\,y=0,\,z=0)$ a line cutting the curve in the two points corresponding to the values $\theta_1,\,\theta_2$ of the parameter. We have

$$\frac{A_1}{A_2}=\frac{B_1}{B_2}=\frac{C_1}{C_2},$$

which equations determine θ_1 and θ_2.

Writing the equations under the form

$$\frac{A_1B_2-A_2B_1}{\theta_1-\theta_2}=0,\qquad \frac{A_1C_2-A_2C_1}{\theta_1-\theta_2}=0,$$

and treating θ_1 and θ_2 as coordinates, each of these equations belongs to a curve of the order $2(m-1)$, having a $(m-1)$thic point at infinity on each of the axes. The number of intersections thus is

$$=4(m-1)^2-(m-1)^2-(m-1)^2=2(m-1)^2.$$

But among these are included points not belonging to the original system, viz. the points for which ($A_1 = 0$, $A_2 = 0$) other than those for which $\theta_1 = \theta_2$; the points so included are in number $= m^2 - m$; and omitting them, the number is

$$\big(2(m-1)^2 - m(m-1)\big) = [m-1]^2,$$

which is the number of points θ_1 lying in lineá with the origin and another point θ_2; the number of apparent double points is the half of this, or $h = \frac{1}{2}[m-1]^2$. And thence

$$M = (-\tfrac{1}{2}[m]^2 + h\ \ =)\ -(m-1).$$

I investigate also the number of lines through two points which meet two arbitrary lines; this is in fact $= S(1,\, m^2)$, which for the curve in question is

$$= ([m]^2 - (m-1)^2 =)\ (m-1)^2.$$

Let the equations of the two lines be ($x = 0$, $y = 0$) and ($z = 0$, $w = 0$); then the conditions to be satisfied are

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}, \qquad \frac{C_1}{C_2} = \frac{D_1}{D_2};$$

or writing these under the form

$$\frac{A_1B_2 - A_2B_1}{\theta_1 - \theta_2} = 0, \qquad \frac{C_1D_2 - C_2D_1}{\theta_1 - \theta_2} = 0,$$

and treating θ_1, θ_2 as coordinates, the number of intersections of these two curves is $= 2(m-1)^2$, the same as for the two curves last above considered. And the number of the lines in question is one half of this, or $= (m-1)^2$.

Lemma employed in the following Annexes 2 and 3. Formulae for the order and weight of certain systems of equations.

Let $a_{\alpha,\,\alpha'}$ denote a function of the degree α in the order variables $(x,\, y,\, \ldots)$, and of the degree α' in the weight variables $(x',\, y',\, \ldots)$, and so in other cases; and consider first the equation

$$\begin{vmatrix} a_{\alpha,\,\alpha'} & (\alpha+A)_{\alpha+A,\,\alpha'} & \cdots \\ \beta_{\beta,\,\beta'} & (\beta+A)_{\beta+A,\,\beta'} & \cdots \\ \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square; then

$$\text{Order} = \Sigma\alpha + \Sigma A,$$

$$\text{Weight} = \Sigma\alpha' + \Sigma A'.$$

Consider next the system

$$\begin{vmatrix} a_{\alpha,\,\alpha'} & (\alpha+A)_{\alpha+A,\,\alpha'} & (\alpha+B)_{\alpha+B,\,\alpha'} & \cdots \\ \beta_{\beta,\,\beta'} & (\beta+A)_{\beta+A,\,\beta'} & (\beta+B)_{\beta+B,\,\beta'} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square $+1$, that is, the number of columns exceeds by 1 the number of lines; then

$$\text{Order} = \Sigma AB - \Sigma\alpha\beta + \Sigma\alpha\,(\Sigma A + \Sigma\alpha),$$

$$\text{Weight} = (\Sigma A + \Sigma\alpha)(\Sigma A' + \Sigma\alpha') - \Sigma AA' + \Sigma\alpha\alpha'.$$

And again, the system

$$\begin{vmatrix} a_{\alpha,\,\alpha'} & (\alpha+A)_{\alpha+A,\,\alpha'} & (\alpha+B)_{\alpha+B,\,\alpha'} & (\alpha+C)_{\alpha+C,\,\alpha'} & \cdots \\ \beta_{\beta,\,\beta'} & (\beta+A)_{\beta+A,\,\beta'} & (\beta+B)_{\beta+B,\,\beta'} & (\beta+C)_{\beta+C,\,\beta'} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square +2, that is, the number of columns exceeds by 2 the number of lines; then

$$\text{Order} = \Sigma ABC + \Sigma\alpha\beta\gamma + \Sigma\alpha\,(\Sigma AB - \Sigma\alpha\beta) + \big((\Sigma\alpha)^2 - \Sigma\alpha\beta\big)(\Sigma A + \Sigma\alpha),$$
$$\text{Weight} = \big\{\Sigma AB - \Sigma\alpha\beta + \Sigma\alpha\,(\Sigma A + \Sigma\alpha)\big\}(\Sigma A' + \Sigma\alpha') - (\Sigma A + \Sigma\alpha)(\Sigma AA' - \Sigma\alpha\alpha') + \Sigma A^2A' + \Sigma\alpha^2\alpha'.$$

The last formula, for the weight of the square +2 system, was communicated to me by Dr Salmon; the others are all in effect given in the Appendix, "On the Order of Systems of Equations," to his Treatise on the Analytic Geometry of Three Dimensions; and in the investigation in the following Annexes 2 and 3, the route which I have followed was completely traced out for me by him, so that I have only supplied the details of the work.

Annex No. 2.—*Investigation of the formula for $S(m^3)$, when the curve m is the pq complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively* (referred to, Art. 40).

Let $U = 0$, $V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take $(x,\, y,\, z,\, w)$ the coordinates of a point on the curve, so that for these coordinates we have $U = 0$, $V = 0$; and in the equations of the two curves respectively, write for the coordinates $x + \rho x'$, $y + \rho y'$, $z + \rho z'$, $w + \rho w'$; then putting for shortness

$$\Delta = x'\partial_x + y'\partial_y + z'\partial_z + w'\partial_w,$$

the resulting equations may be represented by

$$(\Delta U,\, \Delta^2 U,\, \ldots,\, \Delta^p U \mid 1,\, \rho)^{p-1} = 0,$$

$$(\Delta V,\, \Delta^2 V,\, \ldots,\, \Delta^q V \mid 1,\, \rho)^{q-1} = 0,$$

where it is to be noticed that besides the expressed literal coefficients there are numerical coefficients (not as the notation usually denotes, the binomial coefficients, but) $= \frac{1}{1},\ \frac{1}{1\cdot 2},\ \frac{1}{1\cdot 2\cdot 3}$, &c.

Supposing that $(x',\, y',\, z',\, w')$ are the current coordinates of a point on the line drawn through the point $(x,\, y,\, z,\, w)$ to meet the curve in two other points, the equations in ρ must have two common roots, and this gives a system equivalent to two equations, or say a plexus of two equations. If from the plexus and the two equations $U = 0$, $V = 0$ we eliminate $(x,\, y,\, z,\, w)$, we obtain an equation $S' = 0$ in $(x',\, y',\, z',\, w')$, which is in fact the equation of the scroll $S(m^3)$, taken (as is easily seen to be the case) thrice; that is, $S(m^3) = \frac{1}{3}$ Deg. S'. But observing that the coordinates $(x',\, y',\, z',\, w')$ enter into the plexus only and not into the functions U, V, and treating $(x',\, y',\, z',\, w')$ as weight variables, Deg. S' = Weight of System ($U = 0$, $V = 0$, Plexus) = Deg. $U \times$ Deg. $V \times$ Weight of Plexus, $= pq \times$ Weight of Plexus; or, writing $pq = \beta$,

$$S(m^3) = \tfrac{1}{3}\beta \times \text{Weight of Plexus}.$$

The plexus in question is the square +1 system

$$\begin{vmatrix} \Delta U,\, \Delta^2 U,\, \ldots \\ \Delta V,\, \Delta^2 V,\, \ldots \end{vmatrix} = 0,$$

$p+q-3$ columns, $(q-2)+(p-2)=(p+q-4)$ lines; or representing the terms according to their order and weight, that is, degree in (x, y, z, w) and (x', y', z', w') respectively (the order and weight of the evanescent terms being fixed so as that they may form a regular series with the other terms), the system is, with $p+q-3$ columns:

$$\begin{aligned}&(p-2) \text{ lines: } (p-1)_1,\ (p-2)_2,\ \ldots,\ (p-1)_{p-1},\ \ldots,\ 0,\\ &(q-2) \text{ lines: } (q-1)_1,\ (q-2)_2,\ \ldots,\ (q-1)_{q-1},\ \ldots,\ 0,\end{aligned}$$

so that

$$\begin{aligned}\alpha,\, \beta,\, \ldots &= p-1,\ p-1,\ \ldots,\ p+q-5,\ q-1,\ q,\ \ldots,\ p+q-5,\\ \alpha',\, \beta',\, \ldots &= 1,\ 0,\ \ldots,\ -(p+q-4),\ 1,\ 0,\ \ldots,\ -(p+q-4),\\ A,\, B,\, \ldots &= -1,\ -2,\ \ldots,\ -(p+q-4),\\ A',\, B',\, \ldots &= 1,\ 2,\ \ldots,\ p+q-4,\end{aligned}$$

or, as regards the first two lines,

$$\alpha,\, \beta = p-1,\, q-1, \qquad \alpha',\, \beta' = 1,\, 1.$$

We then find

$$\begin{aligned}\Sigma\alpha &= 2\,(q-2)+\tfrac{1}{2}(q-2)(q-1)+2\,(p-2)+\tfrac{1}{2}(p-2)(p-1),\\ \Sigma\alpha' &= \tfrac{1}{2}(q-2)(q-1)+(p-2)(q-2)+\tfrac{1}{2}(p-2)(p-1),\\ \Sigma A &= -\Sigma A' = -\tfrac{1}{2}(p+q-4)(p+q-3),\\ \Sigma\alpha\alpha' &= \tfrac{1}{2}(p-2)(q-2)-(p-4)\cdot\tfrac{1}{2}(q-2)(q-1)-\tfrac{1}{2}(q-2)(q-1)(2q-3)\\ &\quad+\tfrac{1}{2}(q-2)(p-2)-(q-4)\cdot\tfrac{1}{2}(p-2)(p-1)-\tfrac{1}{2}(p-2)(p-1)(2p-3),\\ \Sigma AA' &= -\tfrac{1}{6}(p+q-4)(p+q-3)(2p+2q-7).\end{aligned}$$

Which, putting therein $p+q=\alpha$, $pq=\beta$, give

$$\begin{aligned}\Sigma\alpha &= \beta+\tfrac{1}{2}\alpha^2-\tfrac{11}{2}\alpha+10,\\ \Sigma\alpha' &= \beta-\tfrac{1}{2}\alpha^2+\tfrac{5}{2}\alpha-10,\\ \Sigma A &= -\Sigma A' = -\tfrac{1}{2}\alpha^2+\tfrac{7}{2}\alpha-6,\\ \Sigma\alpha\alpha' &= \tfrac{1}{2}\alpha\beta-\tfrac{1}{6}\alpha^3+\tfrac{3}{2}\alpha^2-\tfrac{31}{3}\alpha+26,\\ \Sigma AA' &= -\tfrac{1}{6}\alpha^3+\tfrac{3}{2}\alpha^2-\tfrac{52}{6}\alpha+14,\end{aligned}$$

and therefore

$$\begin{aligned}\Sigma A+\Sigma\alpha &= \beta-2\alpha+4,\\ \Sigma A'+\Sigma\alpha' &= \beta-4,\end{aligned}$$

and consequently

$$\Sigma AA'-\Sigma\alpha\alpha' = -\tfrac{1}{2}\alpha\beta+5\alpha-12,$$

which is right.

$$\begin{aligned}\text{Weight} &= (\beta-2\alpha+4)(\beta-4)+\tfrac{1}{2}\alpha\beta-5\alpha+12\\ &= \beta^2-\tfrac{3}{2}\alpha\beta+6\alpha-8\\ &= \tfrac{1}{2}\,(\beta-2)(2\beta-3\alpha+4),\end{aligned}$$

and

$$\begin{aligned}S(m^3) &= \tfrac{1}{3}\beta\times\text{Weight}\\ &= \tfrac{1}{6}\,\beta\,(\beta-2)(2\beta-3\alpha+4).\end{aligned}$$

Annex No. 3.—*Investigation of $G(m^4)$ in the case where the curve m is a pq complete intersection* (referred to, Art. 42).

Suppose, as before, that $U = 0$, $V = 0$ are the equations of the two surfaces of the orders p and q respectively; taking also (x, y, z, w) as the coordinates of a point on the curve, and substituting in the equations $x + \rho x'$, $y + \rho y'$, $z + \rho z'$, $w + \rho w'$ in place of the coordinates, then if $\Delta = x'\partial_x + y'\partial_y + z'\partial_z + w'\partial_w$, we have as before

$$(\Delta U,\ \Delta^2 U,\ \dots,\ \Delta^p U\ \mid 1,\ \rho)^{p-1} = 0,$$
$$(\Delta V,\ \Delta^2 V,\ \dots,\ \Delta^q V\ \mid 1,\ \rho)^{q-1} = 0,$$

where the numerical coefficients $\frac{1}{1}$, $\frac{1}{1\cdot 2}$, &c. are to be understood as before.

Suppose now that (x, y, z, w) are the coordinates of a point on the curve, through which point there passes a line through three other points, or line $G(m^4)$; and that (x', y', z', w') are the current coordinates of a point on such line; the two equations in ρ must have three equal roots; or we must have a system equivalent to three equations, or say a plexus of three equations. The coordinates (x', y', z', w'), although four in number, are in fact eliminable from this plexus; or what is the same thing, combining with the plexus the equation

$$\alpha x' + \beta y' + \gamma z' + \delta w' = 0$$

of an arbitrary plane, and then eliminating (x', y', z', w'), the result is of the form

$$(\alpha x + \beta y + \gamma z + \delta w)^\theta\ D = 0,$$

where D is a function of (x, y, z, w) only; and considering (x, y, z, w) as weight variables, $\theta =$ Order of Plexus. But degree in (x, y, z, w) of $(\alpha x + \beta y + \gamma z + \delta w)^\theta D$ is $=$ Weight of Plexus, and therefore Degree of D is $=$ Weight of Plexus $-\theta$, $=$ (Weight $-$ Order) of Plexus.

The equations $U = 0$, $V = 0$, $D = 0$ then give the coordinates (x, y, z, w) of the points through which may be drawn a line $G(m^4)$; viz. they give (as it is easy to see) these points four times over. And we therefore have

$$\begin{aligned} G(m^4) &= \tfrac{1}{4}\ \text{Order}\,(U = 0,\ V = 0,\ D = 0) \\ &= \tfrac{1}{4}\ \text{Deg. } U \cdot \text{Deg. } V \cdot \text{Deg. } D \\ &= \tfrac{1}{4}\,\beta \times (\text{Weight} - \text{Order}) \text{ of Plexus.} \end{aligned}$$

The Plexus is here the square $+2$ system

$$\begin{vmatrix} \Delta U,\ \Delta^2 U,\ \dots \\ \Delta V,\ \Delta^2 V,\ \dots \end{vmatrix} = 0,$$

$p + q - 4$ columns, $(q - 3) + (p - 3) = p + q - 6$ lines, the leading case being $\theta = 0$; representing the terms by their order and weight (the weight variables being (x, y, z, w), and the order variables (x', y', z', w')), and attributing as before an order and weight to the evanescent terms, the system is, with $p + q - 4$ columns:

$$\begin{matrix} 1,\ 2,\ \dots,\ p-1, \\ 1,\ 2,\ \dots,\ q-1, \end{matrix}$$

so that we have

$$\begin{aligned} \alpha,\ \beta,\ \dots &= 1,\ 0,\ -1,\ \dots,\ -(q-5),\ \ 1,\ 0,\ -1,\ \dots,\ -(p-5), \\ \alpha',\ \beta',\ \dots &= p-1,\ p,\ p+1,\ \dots,\ p+q-5,\ \ q-1,\ q,\ q+1,\ \dots,\ q+p-5, \\ A,\ B,\ \dots &= 1,\ 2,\ \dots,\ p+q-5, \\ A',\ B',\ \dots &= -1,\ -2,\ \dots,\ -(p+q-5), \end{aligned}$$

or, as regards the first two lines,

$$\alpha,\ \beta,\ \ldots = 2-e,\ 2-\phi, \qquad e = 1 \text{ to } q-3, \quad \phi = 1 \text{ to } p-3,$$

$$\alpha',\ \beta',\ \ldots = p-2+e,\ p-2+\phi.$$

We then find

$$\begin{aligned}
\Sigma\alpha &= 2(q-3) - \tfrac{1}{2}(q-3)(q-2) + 2(p-3) - \tfrac{1}{2}(p-3)(p-2),\\
\Sigma\alpha' &= (p-2)(q-3) + \tfrac{1}{2}(q-3)(q-2) + (q-2)(p-3) + \tfrac{1}{2}(p-3)(p-2),\\
\Sigma\alpha^2 &= 4(q-3) - 4\cdot\tfrac{1}{2}(q-3)(q-2) + \tfrac{1}{2}(q-3)(q-2)(2q-5)\\
&\quad + 4(p-3) - 4\cdot\tfrac{1}{2}(p-3)(p-2) + \tfrac{1}{2}(p-3)(p-2)(2p-5),\\
\Sigma\alpha^3 &= 8(q-3) - 12\cdot\tfrac{1}{2}(q-3)(q-2) + 6\cdot\tfrac{1}{2}(q-3)(q-2)(2q-5) - \tfrac{1}{4}(q-3)^2(q-2)^2\\
&\quad + 8(p-3) - 12\cdot\tfrac{1}{2}(p-3)(p-2) + 6\cdot\tfrac{1}{2}(p-3)(p-2)(2p-5) - \tfrac{1}{4}(p-3)^2(p-2)^2,\\
\Sigma\alpha\alpha' &= 2(p-2)(q-3) - (p-4)\cdot\tfrac{1}{2}(q-3)(q-2) - \tfrac{1}{6}(q-3)(q-2)(2q-5)\\
&\quad + 2(q-2)(p-3) - (q-4)\cdot\tfrac{1}{2}(p-3)(p-2) - \tfrac{1}{6}(p-3)(p-2)(2p-5),\\
\Sigma\alpha^2\alpha' &= 4(p-2)(q-3) - 4(p-3)\cdot\tfrac{1}{2}(q-3)(q-2) + (p-6)\cdot\tfrac{1}{2}(q-3)(q-2)(2q-5) + \tfrac{1}{4}(q-3)^2(q-2)^2\\
&\quad + 4(q-2)(p-3) - 4(q-3)\cdot\tfrac{1}{2}(p-3)(p-2) + (q-6)\cdot\tfrac{1}{2}(p-3)(p-2)(2p-5) + \tfrac{1}{4}(p-3)^2(p-2)^2,\\
\Sigma A &= \tfrac{1}{2}(p+q-5)(p+q-4),\\
\Sigma A^2 &= -\Sigma AA' = \tfrac{1}{6}(p+q-5)(p+q-4)(2p+2q-9),\\
\Sigma A^3 &= -\Sigma A^2A' = \tfrac{1}{4}(p+q-5)^2(p+q-4)^2,
\end{aligned}$$

which, putting therein $p+q=\alpha$, $pq=\beta$, and from the reduced expressions obtaining the values of $\Sigma\alpha\beta$, &c., give

$$\begin{aligned}
\Sigma\alpha &= \beta - \tfrac{1}{2}\alpha^2 + \tfrac{9}{2}\alpha - 18,\\
\Sigma\alpha^2 &= \beta\,(-\alpha+9) + \tfrac{1}{3}\alpha^3 - \tfrac{17}{2}\alpha^2 + \tfrac{161}{3}\alpha - 88,\\
\Sigma\alpha^3 &= \beta^2\,(-\tfrac{3}{2}) + \beta\,(\alpha^2 - \tfrac{31}{2}\alpha + \tfrac{161}{2}) - \tfrac{1}{4}\alpha^4 + \tfrac{27}{2}\alpha^3 - \tfrac{691}{4}\alpha^2 + 939\alpha - 198,\\
\Sigma\alpha\beta &= \beta^2\,(\tfrac{1}{2}) + \beta\,(-\tfrac{3}{2}\alpha^2 + 5\alpha - \tfrac{37}{2}) + \tfrac{1}{6}\alpha^3 - \tfrac{17}{6}\alpha^2 - \tfrac{169}{6}\alpha^2 - \tfrac{169}{6}\alpha + 191,\\
\Sigma\alpha^2\alpha' &= \beta^2\,(\alpha) + \beta\,(-\tfrac{1}{2}\alpha^2 + 9\alpha - \tfrac{159}{2}) + \beta\alpha\,(\alpha^2 - \tfrac{9}{2}\alpha + \tfrac{29}{2}\alpha - \tfrac{20}{2})\\
&\quad - \tfrac{1}{2}\alpha^5 + \tfrac{25}{4}\alpha^4 - \tfrac{125}{4}\alpha^3 + \tfrac{205}{2}\alpha^2 - \tfrac{289}{2}\alpha^2 + 1071\alpha - 1560,
\end{aligned}$$

$$\begin{aligned}
\Sigma\alpha' &= \beta + \tfrac{1}{2}\alpha^2 - \tfrac{11}{2}\alpha + 18,\\
\Sigma\alpha\alpha' &= \beta\,(-\tfrac{1}{2}\alpha) - \tfrac{1}{3}\alpha^3 + \tfrac{7}{2}\alpha^2 - \tfrac{47}{3}\alpha + 58,\\
\Sigma\alpha^2\alpha' &= \beta^2\,(-\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}\alpha^2 + 9\alpha - \tfrac{151}{2}) + \tfrac{1}{4}\alpha^4 - \tfrac{17}{2}\alpha^3 + \tfrac{161}{2}\alpha^2 - 119\alpha + 198,\\
\Sigma A &= \tfrac{1}{2}\alpha^2 - \tfrac{9}{2}\alpha + 10,\\
\Sigma A^2 &= -\Sigma AA' = \tfrac{1}{3}\alpha^3 - \tfrac{9}{2}\alpha^2 + \tfrac{55}{3}\alpha - 30,\\
\Sigma A^3 &= -\Sigma A^2A' = \tfrac{1}{4}\alpha^4 - \tfrac{17}{2}\alpha^3 + \tfrac{161}{4}\alpha^2 - 90\alpha + 100,\\
\Sigma AB &= \tfrac{1}{6}\alpha^3 - \tfrac{11}{4}\alpha^2 + \tfrac{43}{2}\alpha^2 - \tfrac{27}{2}\alpha + 65,\\
\Sigma ABC &= \tfrac{1}{12}\alpha^5 - \tfrac{11}{6}\alpha^4 + \tfrac{59}{4}\alpha^3 - \tfrac{121}{2}\alpha\alpha^2 + \tfrac{139}{2}\alpha^2 - \tfrac{1303}{2}\alpha + \tfrac{1267}{2},
\end{aligned}$$

we then find

$$\begin{aligned}
\Sigma A + \Sigma\alpha &= \beta - 8,\\
\Sigma A' + \Sigma\alpha' &= \beta - 3\alpha + 8,\\
\Sigma AB - \Sigma\alpha\beta &= \beta^2\,(-\tfrac{1}{2}) + \beta\,(\tfrac{3}{2}\alpha^2 - 5\alpha + \tfrac{37}{2}) - 4\alpha^2 + 36\alpha - 126,\\
\Sigma AA' - \Sigma\alpha\alpha' &= \beta\,(-\tfrac{1}{2}\alpha) + 9\alpha - 28,\\
&\quad - \alpha^4 + \tfrac{17}{2}\alpha^3 - 162\alpha^2 + \tfrac{169}{6}\alpha - 1210,\\
\Sigma A^2A' + \Sigma\alpha^2\alpha' &= \beta^2\,(-\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}\alpha^2 + 9\alpha - \tfrac{151}{2}) - 29\alpha + 98;
\end{aligned}$$

and then also

$$\begin{aligned}
\Sigma\alpha\,(\Sigma A + \Sigma\alpha) &= \beta^2 + \beta\,(-\tfrac{1}{2}\alpha^2 + \tfrac{5}{2}\alpha - 26) + 4\alpha^2 - 36\alpha + 144,\\
(\Sigma AB - \Sigma\alpha\beta) + \Sigma\alpha\,(\Sigma A + \Sigma\alpha) &= \beta^2\,(\tfrac{1}{2}) + \beta\,(\alpha^2 - \tfrac{5}{2}\alpha - \tfrac{15}{2}) + 18,\\
\big\{(\Sigma AB - \Sigma\alpha\beta) + \Sigma\alpha\,(\Sigma A + \Sigma\alpha)\big\}(\Sigma A' + \Sigma\alpha') &= \beta^3\,(\tfrac{1}{2}) + \beta^2\,(-\alpha + \tfrac{5}{2}) + \beta\,(\alpha^2 - 13\alpha + 28) + 72\alpha - 224,\\
-(\Sigma A + \Sigma\alpha)(\Sigma AA' - \Sigma\alpha\alpha') &= \beta\,(\tfrac{1}{2}\alpha) + \beta\,(-\tfrac{15}{2}\alpha + 28) + \cdots,
\end{aligned}$$

and

$$\Sigma A^2A' + \Sigma\alpha^2\alpha' \ = \text{(ut supra)}\ \beta^2\,(-\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}\alpha^2 + 9\alpha - \tfrac{151}{2}) - 29\alpha + 98;$$

whence, adding the last three expressions, we find

$$\text{Weight} = \beta^3\,(\tfrac{1}{2}) + \beta^2\,(-\tfrac{3}{2}\alpha + \tfrac{1}{2}) + \beta\,(\tfrac{3}{2}\alpha^2 + \tfrac{3}{2}\alpha - \tfrac{15}{2}) - 11\alpha + 18;$$

and for the order we have

$$(\Sigma\alpha)^2 - \Sigma\alpha\beta = \beta^2\,(\tfrac{1}{2}) + \beta\,(-\tfrac{1}{2}\alpha^2 + 4\alpha - \tfrac{55}{2}) + \tfrac{1}{4}\alpha^4 - \tfrac{17}{2}\alpha^3 + \tfrac{161}{2}\alpha^2 - \tfrac{161}{2}\alpha + 133;$$

and then

$$\begin{aligned}
\Sigma ABC + \Sigma\alpha\beta\gamma &= \text{(ut supra)},\\
(\Sigma AB - \Sigma\alpha\beta)\,\Sigma\alpha &= -\alpha^4 + \tfrac{17}{2}\alpha^3 - 162\alpha^2 + \tfrac{169}{6}\alpha - 121\\
\beta\,(-\tfrac{1}{2}) + \beta^2\,(\tfrac{1}{2}\alpha^2 - \tfrac{31}{2}\alpha + \tfrac{161}{2}) &+ \beta\,(-\tfrac{1}{4}\alpha^4 + \tfrac{169}{4}\alpha^3 - \tfrac{91}{2}\alpha^2 + \tfrac{205}{4}\alpha - 531)\\
&+2\alpha^5 - 36\alpha^4 + 297\alpha^3 - 1215\alpha + 2268,\\
\big((\Sigma\alpha)^2 - \Sigma\alpha\beta\big)(\Sigma A + \Sigma\alpha) &= \beta^3\,(\tfrac{1}{2}) + \beta^2\,(-\tfrac{1}{2}\alpha^2 + 4\alpha - \tfrac{55}{2}) + \beta\,(\\
&\quad - \alpha^4 + \tfrac{17}{2}\alpha^3 - 135\alpha^2 + \tfrac{169}{2}\alpha - 106
\end{aligned}$$

whence, adding these three expressions,

$$\text{Order} = \beta^3\,(\tfrac{1}{2}) + \beta^2\,(-\tfrac{3}{2}\alpha + \tfrac{1}{2}) + \beta\,(\tfrac{3}{2}\alpha^2 - \tfrac{3}{2}\alpha + \tfrac{15}{2}) - 6;$$

and by means of the foregoing expression for the weight, we then have

$$\text{Weight} - \text{Order} = \beta^2\,(\tfrac{1}{2}) + \beta\,(-\alpha) + \beta\,(\tfrac{1}{2}\alpha^2 + 3\alpha - \tfrac{15}{2}) - 11\alpha + 24;$$

and therefore

$$\begin{aligned}
G(m^4) &= \tfrac{1}{4}\beta \times (\text{Weight} - \text{Order}),\\
&= \tfrac{1}{12}\,\beta\,\big(2\beta^3 + \beta^2\,(-6\alpha) + \beta\,(3\alpha^2 + 18\alpha - 26) - 66\alpha + 144\big),
\end{aligned}$$

which is right.

Annex No. 4.—*Order of* Torse$(m,\, n)$ (referred to, Art. 44).

We have to find the order of the developable or Torse generated by a line meeting two curves of the orders m, n respectively; viz. representing by μ, ν the classes of the two curves respectively, it is to be shown that the expression for the Order is

$$\text{Torse}(m,\, n) = m\nu + n\mu.$$

I remark, in the first place, that, given two surfaces of the orders p and q respectively, the curve of intersection is of the order pq and class $pq\,(p+q-2)$, or as this may be written, class $= qp\,(p-1) + pq\,(q-1)$. Reciprocally for two surfaces of the classes p and q respectively, the Torse enveloped by their common tangent planes is of the class pq and order $qp\,(p-1)+pq\,(q-1)$. Now, in the same way that a surface of the order p may degenerate into a Torse of the order p, so a surface of the class p may degenerate into a curve of the class p; and the class of a curve being p, then (disregarding singularities) its order is $= p\,(p-1)$. Hence replacing p and $p\,(p-1)$ by μ and m respectively, and in like manner q and $q\,(q-1)$ by ν and n respectively, we have $m\nu + n\mu$ as the order of the Torse generated by the tangent planes of the curves of the orders m and n respectively; where by tangent plane of a curve is to be understood a plane passing through a tangent line of the curve. The intersection of two consecutive tangent planes is a line meeting the two curves, which line is the generating line of the Torse, and such Torse is therefore the Torse $(m,\, n)$ in question.

The foregoing investigation is not very satisfactory, but I confirm it by considering the case of two plane curves, orders m and n, and classes μ and ν, respectively. The tangents of the two curves can, it is clear, only meet on the line of intersection of the planes of the curves; and the construction of the Torse is in fact as follows: from any point of the line of intersection draw a tangent to m and a tangent to n, then the line joining the points of contact of these tangents is a generating line of the Torse. The order of the Torse is equal to the number of generating lines which meet an arbitrary line; and taking for the arbitrary line the line of intersection of the two planes, it is easy to see that the only generating lines which meet the line of intersection are those for which one of the points of contact lies on the line of intersection; that is, they are the generating lines derived from the points in which the line of intersection meets one or other of the two curves; they are therefore in fact the tangents drawn to the curve n from the points in which the line of intersection meets the curve m, together with the tangents drawn to the curve m from the points in which the line of intersection meets the curve n. Now the line meets the curve n in n points, and from each of these there are μ tangents to the curve m; and it meets the curve m in m points, and from each of these there are ν tangents to the curve n; hence the entire number of the tangents in question is $= n\mu + m\nu$, which confirms the theorem.

Annex No. 5.—*Order of* Torse(m^2) (referred to, Art. 46).

We have here to find the order of the developable or Torse generated by a line meeting a curve of the order m twice, viz. the class of the curve being μ, it is to be shown that we have

$$\text{Torse}(m^2) = (m-3)\,\mu.$$

I deduce the expression from the formula given p. 424 of Dr Salmon's 'Geometry of Three Dimensions;' viz. putting in his formula $\beta = 0$, and μ for his r, we have

$$\text{Order} = m\,(\mu - 4) - \tfrac{1}{2}\alpha = m\mu - (4m + \tfrac{1}{2}\alpha),$$

where (see p. 234 et seq.)

$$\tfrac{1}{2}\alpha = (n - m) = 3m\,(m-2) - 6h - m,$$

and thence

$$\mu = m\,(m-1) - 2h,$$

so that we have

$$3\mu - \tfrac{1}{2}\alpha = 4m, \quad \text{or} \quad 4m + \tfrac{1}{2}\alpha = 3\mu,$$

$$\text{Order} = (m-3)\,\mu.$$

A more complete discussion of the Torses $(m,\, n)$ and (m^2) is obviously desirable; but as they are only incidentally connected with the subject of the present memoir, I have contented myself with obtaining the required results in the way which most readily presented itself.

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340. A Second Memoir on Skew Surfaces, otherwise Scrolls

[From the Philosophical Transactions of the Royal Society of London, vol. CLIV. (for the year 1864), pp. 559–576. Received April 29,—Read May 26, 1864.]

The principal object of the present memoir is to establish the different kinds of skew surfaces of the fourth order, or Quartic Scrolls; but, as preliminary thereto, there are some general researches connected with those in my former memoir "On Skew Surfaces, otherwise Scrolls" [1]; and I also reproduce the theory (which may be considered as a known one) of cubic scrolls; there are also some concluding remarks which relate to the general theory. As regards quartic scrolls, I remark that M. Chasles, in a footnote to his paper, "Description des courbes de tous les ordres situées sur les surfaces réglées du troisième et du quatrième ordres" [2], states, "les surfaces réglées du quatrième ordre... admettent quatorze espèces." This does not agree with my results, since I find only eight species of quartic scrolls; the developable surface or "torse" is perhaps included as a "surface réglée;" but as there is only one species of quartic torse, the deficiency is not to be thus accounted for. My enumeration appears to me complete, but it is possible that there are subforms which M. Chasles has reckoned as distinct species.

On the Degeneracy of a Scroll, Article Nos. 1 to 5.

1. A scroll considered as arising from any geometrical construction, for instance one of the scrolls $S(m,\, n,\, p)$, $S(m^2,\, n)$, $S(m^3)$ considered in my former memoir, or say in general the scroll S, may break up into two or more inferior scrolls S', S'', ...; but as long as S', S'', ... are proper scrolls (not torses, and *a fortiori* not cones or planes), no one of these can be considered, apart from the others, as the result of the geometrical

[Paper 340 continues on p. 202; this excerpt ends at p. 201.]

[End of excerpt: pp. 177–201 of Vol. V; PDF pp. 201–225.]

[1] Philosophical Transactions, vol. CLIII. (1863), pp. 453–483, [339].
[2] Comptes Rendus, t. LIII. (1861), see p. 888.

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340. A Second Memoir on Skew Surfaces, otherwise Scrolls *(continued)*

[Continuation from p. 201, in Article No. 1 on the degeneracy of a scroll.]

...construction, and we can only say that the scroll S given by the construction is the aggregate of the scrolls S', S'', ...; and the like when we have the scrolls S', S'', ..., each repeated any number of times, or say when $S = S'^a\, S''^b \ldots$. Suppose however that the scrolls S', S'', ... are any one or more of them a torse or torses—or, to make at once the most general supposition, say that we have $S = \Sigma S'$, where Σ is a torse, or aggregate of torses ($\Sigma = \Sigma'^a\, \Sigma''^b \ldots$), and S' is a proper scroll or aggregate of proper scrolls; then, although it is not obligatory to do so, we may without impropriety throw aside the torse-factor Σ, and consider the original scroll S as degenerating into the scroll S', and as suffering a reduction in order accordingly.

2. As an illustration, consider the scroll $S(m,\, n,\, p)$ generated by a line which meets three directrix curves of the orders m, n, p respectively; and assume that the curves m, n, p are each of them situate on the same scroll Σ, the curve m meeting each generating line of Σ in α points, the curve n each generating line in β points, and the curve p each generating line in γ points. Each generating line of Σ is $\alpha\beta\gamma$ times a generating line of S, and we have $S = \Sigma^{\alpha\beta\gamma}\, S'$, where S' may be a proper scroll; it is however to be noticed that if the curves m, n, p any two of them intersect, S' will itself break up and contain cone-factors, as will presently appear. And if Σ, instead of being a proper scroll, be a torse, then we may consider S as degenerating into S', the reduction in order being of course $= \alpha\beta\gamma\times$ order of Σ.

3. But this is not the only way in which the scroll $S(m,\, n,\, p)$ may degenerate; for suppose that two of the directrix curves, say n and p, intersect, then the lines from the point of intersection to the curve m form a cone of the order m which will present itself as a factor of S; and generally if the curves n and p intersect in α points, the curves p and m in β points, and the curves m and n in γ points, then we have α cones each of the order m, β cones each of the order n, and γ cones each of the order p, or say $S = CS'$, where C is the aggregate of the cone-factors; and the scroll S degenerates into S', the reduction in order being $= \alpha m + \beta n + \gamma p$. It is hardly necessary to remark that if a point of intersection of two of the curves is a multiple point on either or each of the curves, it is, in reckoning the number of intersections of the two curves, to be taken account of according to its multiplicity in the ordinary manner.

4. There is yet another case to be considered: suppose that the curves n and p lie on a cone, and that the curve m passes through the vertex of this cone; this cone, repeated a certain number of times, is part of the locus, or we have $S = C'S'$, so that the scroll S degenerates into S', the reduction in order being $= \theta \times$ order of cone. If, to fix the ideas, the curves n and p are respectively the complete intersections of the cone by two surfaces of the orders g, h respectively (this implies $n = gk$, $p = hk$, if k be the order of the cone), which surfaces do not pass through the vertex of the cone, and if, moreover, the vertex of the cone be an a-tuple point on the curve m, then $\theta = agh$, and the reduction in order is $= aghk$.

5. The foregoing causes of reduction, or some of them, may exist simultaneously; it would require a further examination to see whether the aggregate reduction is in all cases the sum of the separate reductions. But the aggregate reduction once ascertained, then writing $S(m,\, n,\, p)$ for the order of the reduced scroll, we shall have

$$S(m,\, n,\, p) = 2mnp - \text{Reduction}.$$

In particular, in the case above referred to, where the curves n and p, p and m, m and n meet in α, β, γ points respectively, but there is no other cause of reduction,

$$S(m,\, n,\, p) = 2mnp - \alpha m - \beta n - \gamma p,$$

which is a formula which will be made use of.

The foregoing investigations apply, *mutatis mutandis*, to the scrolls $S(m^2,\, n)$, $S(m^3)$; but I do not at present enter into the development of them in regard to these scrolls.

Scrolls with two directrix lines. Article Nos. 6 to 11.

6. Consider now a scroll having two directrix lines: it may be assumed that these do not intersect; for if they did, then any generating line, *qua* line meeting the two directrix lines, would either lie in the plane of the two lines, or else would pass through their point of intersection; that is, the scroll would break up into the plane of the two lines, considered as the locus of the tangents of a plane curve, and into a cone having for its vertex the point of intersection of the two lines. Each generating line meets any plane section of the scroll in the point where such generating line meets the plane of the section; the plane section constitutes a third directrix; or the scrolls in question are all included in the form $S(1,\, 1,\, m)$, where m is a plane curve. The order of the scroll $S(1,\, 1,\, m)$ is in general $= 2m$; but if the one line meets the curve α times, that is, in an α-tuple point of the curve, and the other line meets the curve β times, that is, in a β-tuple point of the curve, then by the general formula (*ante*, No. 5) the order of the scroll is $= 2m - \alpha - \beta$; and in particular if $\alpha + \beta = m$, then the order is $= m$.

7. We may without loss of generality attend only to the last-mentioned case. To show how this is, suppose for a moment that the two lines do not either of them meet the curve; the scroll is then of the order $2m$. Call the point in which each line meets the plane of the curve the foot of this line, then the line joining the two feet meets the curve in m points; and it is in respect of each of these points a generating line of the scroll; that is, it is an m-tuple generating line:

the section of the scroll by the plane of the curve m is in fact this line counting m times, and the curve m; $m + m = 2m$, the order of the scroll. And in like manner the section by any plane through the m-tuple line is this line counting m times, and a curve of the order m not meeting either of the directrix lines. But the section by any other plane is a curve of the order $2m$ meeting each of the directrix lines in a point which is an m-tuple point of the section (each directrix line is in fact an m-tuple line of the scroll); and by considering, in place of the particular section m, this general section, we have the scroll of the order $2m$ in the form $S(1, 1, 2m)$, where the two directrix lines each meet the section m times; so that the order is $4m - m - m = 2m$.

8. And so in general, m being a plane curve, when the scroll $S(1, 1, m)$ is of an order superior to m, say $= m + k$, this only means that the section chosen for the directrix curve m is not the complete section by the plane of such curve, but that the line joining the feet of the two directrix lines is a k-tuple generating line of the scroll, and that the complete section is made up of this line counting k times and of the curve m. So that taking, not the section through the multiple generating line, but the general section, for the plane directrix curve, the only case to be considered is that in which the section is a proper curve of an order equal to that of the scroll; or, what is the same thing, we have only to consider the scrolls $S(1, 1, m)$ for which the order is depressed from $2m$ to m in consequence of the directrix lines meeting the plane section α times and β times, that is, in an α-tuple point and a β-tuple point respectively, where $\alpha + \beta = m$.

9. It is clear that in the case in question the directrix lines are an α-tuple line and a β-tuple line respectively. The generation is as follows: Scroll $S(1, 1, m)$ of the order m; the curve m being a plane curve of the order m having an α-tuple point and a β-tuple point, where $\alpha+\beta = m$: the directrix lines, say 1 and $1'$, pass through these points respectively, and they do not intersect each other. The generating lines pass through the directrix lines 1 and $1'$ and the curve m, and we have thence the scroll $S(1, 1, m)$. Taking at pleasure any point on the curve m, we can through this point draw a single line meeting each of the directrix lines 1, $1'$; that is, the curve m is a simple curve on the scroll. Taking at pleasure a point on the directrix line 1, and making this the vertex of a cone standing on the curve m, this cone has an α-tuple line (the line 1) and a β-tuple line (the line joining the vertex with the foot of the line $1'$); the line $1'$ meets this cone in the foot of the line $1'$, counting β times, and besides in $m - \beta, = \alpha$ points; the lines joining the vertex with the last-mentioned points respectively (or, what is the same thing, the lines, other than the β-tuple line, in which the plane through the vertex and the line $1'$ meets the cone) are the α generating lines through the assumed point on the line 1; and the line 1 is thus an α-tuple line of the scroll. And in like manner, through an assumed point of the directrix line $1'$, we construct β generating lines of the scroll; and the line $1'$ is a β-tuple line of the scroll.

10. The scroll $S(1, 1, m)$ now in question has not in general any multiple generating line; in fact a multiple generating line would imply a corresponding multiple point on the section m; and this section, assumed to be a curve having an α-tuple point and a β-tuple point, has not in general any other multiple point. But it may have other multiple points; and if there is, for example, a γ-tuple point, then the line from this point which meets the two directrix lines counts γ times, or it is a γ-tuple generating line; and so for all the multiple points of m other than the α-tuple point and the β-tuple point which correspond to the directrix lines respectively. It is to be noticed that the multiplicity γ of any such multiple generating line is at most equal to the smallest of the two numbers α and β; for suppose $\gamma > \alpha$, then, since $\alpha + \beta = m$, we should have $\gamma + \beta > m$, and the line joining the γ-tuple point and the β-tuple point would meet the curve m in $\gamma + \beta$ points, which is absurd. In the case of several multiple lines, there are other conditions of inequality preventing self-contradictory results [1].

[1] Suppose, for example (see next paragraph of the text), that there were a γ-tuple generating line and a δ-tuple generating line lying *in plano* with the line 1; these lines counting as $(\gamma + \delta)$ lines, must be included among the β generating lines through the plane in question; this implies that $\gamma+\delta \leqq \beta$, a conclusion which must be obtainable from consideration of the curve m irrespectively of the scroll.

11. The general section is a curve of the order m, having an α-tuple point and a β-tuple point corresponding to the directrix lines respectively, and a γ-tuple point, &c.... corresponding to the other multiple points (if any). A section through the directrix line 1 is in general made up of this line, counting α times, and of β generating lines passing through one and the same point of the directrix line $1'$; if the section pass also through a γ-tuple generating line, then, of the β generating lines in question, γ (which, as has been seen, is $\leqq \beta$) unite together in the γ-tuple generating line; and so for the sections through the directrix line $1'$. The general section through a γ-tuple generating line is this line counting γ times, and a curve of the order $m-\gamma$, which has an $(\alpha-\gamma)$ tuple point at its intersection with the directrix line 1, and a $(\beta-\gamma)$ tuple point at its intersection with the directrix line $1'$; it has a δ-tuple point, &c.... at its intersections with the other multiple generating lines, if any.

Scrolls with a twofold directrix line. Article Nos. 12 to 16.

12. But there is a case included indeed as a limiting one in the foregoing general case, but which must be specially considered; viz. the two directrix lines 1 and $1'$ may coincide, giving rise to a twofold directrix line. To show how this is, I return for the moment to the case of the scroll $S(1,\,1,\,m)$ with two distinct directrix lines 1 and $1'$, and, to fix the ideas, I suppose that the directrix lines do not either of them meet the curve m, so that the order of the scroll is $= 2m$. Through the line 1 imagine the series of planes A, B, C, ... meeting the line $1'$ in the points a', b', c', ...; the generating lines through the point a' are the lines in the plane A to the points in which this plane meets the curve m; the generating lines through the point b' are the lines in the plane B to the points where this plane meets the curve m; and so for the generating lines through the points c', d', ...; and it is clear that the points a', b', c', ... correspond homographically with the planes A, B, C, ... This gives immediately the construction for the case where the two directrix lines come to coincide. In fact, on the twofold directrix line $1 = 1'$ take the series of points a, b, c, ..., and through the same line, corresponding homographically to these points, the series of planes A, B, C, ...; the generating lines through the point a are the lines through this point, in the plane A, to the points in which this plane meets the curve m; and so for the entire series of points b, c, ... of the line $1 = 1'$; the resulting scroll, which I will designate as the scroll $S(\overline{1,\,1},\,m)$, remains of the order $= 2m$. If there is given a point of the curve m, then the plane through this point and the directrix line is the plane A; and the point a is then also given by the homographic correspondence of the series of planes and points, and the generating line through the given point on the curve m is the line joining this point with the point a.

13. We may say that, in regard to any point a of the line 1, the corresponding plane A is the plane of approach of the coincident line $1'$; and that in regard to the same point a and to any plane through it, the trace on that plane of the plane of approach is the line of approach of $1'$; that is, we may consider that the coincident directrix line $1'$ meets the plane through a in a consecutive point on the line of approach. In particular if the point a be the foot of the directrix line 1 (that is, the point where this line meets the plane of the curve m), and the plane through a be the plane of the curve m, then the intersection of the last-mentioned plane by the plane A which corresponds to the point a is the line of approach, and the foot of the coincident directrix line $1'$ is the consecutive point to a along the line of approach. The expression "the line of approach," used absolutely, has always the signification just explained, viz. it is the intersection of the plane of the curve m by the plane corresponding to the foot of the directrix line.

14. Suppose now that the line 1 meets the curve m, or, more generally, meets it α times, that is, in an α-tuple point; it might at first sight appear that the coincident line $1'$ should also be considered as meeting the curve α times, and that the resulting scroll should be of the order $2m-\alpha-\alpha = 2m-2\alpha$. But this is not the case; so long as the direction of the line of approach is arbitrary, the line $1'$ must be considered as a line indefinitely near to the line 1, but nevertheless as a line not meeting the curve at all; and the order of the scroll is thus $= 2m - \alpha$. If, however, the line of approach is the tangent to a branch through the α-tuple point—that is, if the plane corresponding to the α-tuple point meet the plane of the curve in such tangent, then the coincident line $1'$ is to be considered as meeting the curve m in a consecutive point on such branch, and the order of the scroll is $= 2m - \alpha - 1$. And so if at the multiple point there are β branches having a common tangent, then the coincident line $1'$ is to be considered as meeting the curve m in a consecutive point along each of such branches, or say in a consecutive β-tuple point along the branch, and the order of the scroll sinks to $2m - \alpha - \beta$. The point spoken of as the α-tuple point is, it should be observed, more than an α-tuple point with a β-fold tangent; it is really a point of union of an α-tuple point and a β-tuple point, or say a united $\alpha\,(+\beta)$ tuple point, equivalent to

$$\tfrac{1}{2}\alpha\,(\alpha - 1) + \tfrac{1}{2}\beta\,(\beta - 1)$$

double points or nodes; and the case is precisely analogous to that of the scroll $S(1,\,1,\,m)$, where the two directrix lines pass through an α-tuple point and a β-tuple point of the curve m respectively. It may be added that if at the multiple point in question, besides the β branches having a common tangent, there are γ branches having a common tangent, then the point is, so to speak, a united $\alpha\,(+\beta,\,+\gamma)$ tuple point equivalent to $\tfrac{1}{2}\alpha\,(\alpha - 1) + \tfrac{1}{2}\beta\,(\beta - 1) + \tfrac{1}{2}\gamma\,(\gamma - 1)$ double points or nodes; but the order of the scroll is still $= 2m - \alpha - \beta$.

15. In the same way as the scrolls $S(1,\,1,\,m)$ are all included in the case where the order of the scroll, instead of being $= 2m$, is $= m$, so that the scrolls $S(\overline{1,\,1},\,m)$ are all included in the case where the order of the scroll, instead of being $= 2m$, is $= m$. That is, we may suppose that the curve m has a united $\alpha\,(+\beta)$ tuple point $(\alpha + \beta = m)$, and may take the directrix line to pass through this point, and the line of approach to be the common tangent of the β branches; and this being so, the order of the scroll will be $2m - \alpha - \beta,\ = m$. It may be added that if the curve m has, besides the $\alpha\,(+\beta)$ tuple point, a γ-tuple point, then the scroll will have a γ-tuple generating line, and so for the other multiple points of the curve m.

16. We may, in the same way as for the scroll $S(1,\,1,\,m)$, consider the different sections of the scroll $S(\overline{1,\,1},\,m)$ of the order m. The general section is a curve of the order m, having an $\alpha\,(+\beta)$ tuple point at the intersection with the directrix line, and a γ-tuple point, &c. corresponding to the multiple generating lines, if any. A section through the directrix line is in general made up of this line counting α times, and of β generating lines through the point which corresponds to the plane of the section; if the section pass also through a γ-tuple generating line ($\gamma \leqq \beta$, in the same way as for the scroll $S(1,\,1,\,m)$), then, of the β generating lines, γ unite together in the γ-tuple generating line. The general section through a γ-tuple generating line breaks up into this line counting γ times, and a curve of the order $m - \gamma$, which has on the directrix line an $\alpha - \gamma\,(+\beta - \gamma)$ tuple point and a δ-tuple point, &c. at its intersections with the other multiple generating lines, if any.

Equation of the Scroll $S(1,\,1,\,m)$ of the order m. Article Nos. 17 and 18.

17. Taking for the equations of the directrix lines $(x = 0,\ y = 0)$ and $(z = 0,\ w = 0)$, and supposing that these are respectively an α-tuple line and a β-tuple line on the scroll $\alpha + \beta = m$, it is obvious that the equation of the scroll is

$$(*\mid x,\,y)^{\alpha}\,(z,\,w)^{\beta} = 0.$$

In fact starting with this equation, if we consider the section by a plane through the line $(x = 0,\ y = 0)$, say the plane $y = \lambda x$, then the equation gives

$$x^{\alpha}\,(*\mid 1,\,\lambda)^{\alpha}\,(z,\,w)^{\beta} = 0;$$

that is, the section is made up of the line ($x=0$, $y=0$) reckoned α times, and of β other lines in the plane $y=\lambda x$; and the like for the section by any plane through the line ($z=0$, $w=0$), say the plane $z=\nu w$. Hence the assumed equation represents a scroll of the order m, having the two lines for an α-tuple line and a β-tuple line respectively, and conversely such scroll has an equation of the assumed form.

Case of a γ-tuple generating line.

18. The multiple generating line meets each of the lines ($x=0$, $y=0$) and ($z=0$, $w=0$); and we may take for the equations of the multiple generating line $x+y=0$, $z+w=0$. This being so, the foregoing equation of the scroll may be expressed in the form

$$(* \mid x,\, y)^{\alpha}\,(z,\, z+w)^{\beta}=0,$$

or say

$$(U,\, V,\, W,\, \ldots \mid z,\, z+w)^{\beta}=0,$$

where U, V, W, ... are functions of the form $(* \mid x,\, y)^{\alpha}$. Hence ($\gamma \not> \alpha$ or β), if the functions U, V, W, ... contain respectively the factors $(x+y)^{\gamma}$, $(x+y)^{\gamma-1}$, $(x+y)^{\gamma-2}$, ..., the equation will be of the form

$$(U',\, V',\, W',\, \ldots \mid x+y,\, z+w)^{\gamma}=0,$$

(the coefficients being functions of x, y, z and $z+w$, or, what is the same thing, x, y, z, w, of the order $\alpha+\beta-\gamma$), and the scroll will therefore have the line $x+y=0$, $z+w=0$ as a γ-tuple generating line.

Equation of the Scroll $S(\overline{1,\, 1},\, m)$ of the order m. Article Nos. 19 to 24.

19. We may take $x=0$, $y=0$ for the equations of the twofold directrix line, $z=0$ for the equation of the plane of the curve m (an arbitrary plane section of the scroll). Then ($\alpha+\beta=m$), if the curve m have at the point ($x=0$, $y=0$), or foot of the directrix line, an $\alpha\,(+\beta)$ tuple point, and if moreover we have $y=0$ for the equation of the common tangent of the β branches (viz. if the plane $y=0$, instead of being an arbitrary plane through the directrix line, be the plane through this line and the common tangent of the β branches), the equation of the curve m will be of the form

$$\Sigma\, y^{\eta}\,(* \mid x,\, y)^{\alpha+\beta-\eta}=0,$$

where the summation extends to all integer values of η from 0 to β, both inclusive.

20. Taking $y=\lambda x$ for the equation of any plane through the directrix line, then the corresponding point on the directrix line will be the intersection of this line ($x=0$, $y=0$) by the plane $z=\theta w$, where $\theta=\dfrac{ay+bx}{cy+dx}$; the foot of the directrix line is given by the value $z=0$, or $\lambda=-\dfrac{b}{a}$, and the equation of the line of approach is therefore $y=-\dfrac{b}{a}\,x$; this should coincide with the line $y=0$, which is the common tangent of the β branches; that is, we must have $b=0$; I retain, however, for the moment the general value of b.

21. The equations of a generating line will be

$$y=\lambda x, \quad z=\theta w-\rho x;$$

and then taking X, Y, ($Z=0$) and W for the coordinates of the point of intersection with the curve m, we have

$$Y=\lambda X, \quad 0=\theta W-\rho X,$$

and thence

$$\Sigma\,(YW)^\eta\,(*\mid X,\,Y)^{\alpha+\beta-\eta}=0,$$

$$\Sigma\left(\frac{\lambda\rho}{\theta}\right)^\eta\,(*\mid 1,\,\lambda)^{\alpha+\beta-\eta}=0,$$

or, what is the same thing,

$$\Sigma\,\theta^{-\eta}\,(\lambda\rho)^\eta\,(*\mid 1,\,\lambda)^{\alpha+\beta-\eta}=0;$$

which equation, substituting therein for θ its value in terms of λ, gives the parameter ρ which enters into the equations of the generating line; or, what is the same thing, the equation of the scroll is obtained by eliminating λ, θ, ρ from the equation just mentioned and the equations

$$\lambda=\frac{y}{x},\quad \theta=\frac{ay+bx}{cy+dx},\quad \rho=\frac{\theta w-z}{x}.$$

22. These last three equations give

$$\lambda=\frac{y}{x},\quad \theta=\frac{ay+bx}{cy+dx},\quad \rho=\frac{\theta w-z}{x}=\frac{(ay+bx)\,w-(cy+dx)\,z}{x},$$

and substituting these values, we find for the equation of the scroll

$$\Sigma\,(ay+bx)^{\beta-\eta}\,y^\eta\left[(ay+bx)\,w-(cy+dx)\,z\right]^\eta(*\mid x,\,y)^{\alpha+\beta-2\eta}=0,$$

which is of the order $\alpha+2\beta$, $=2m-\alpha$, so that the $\alpha\,(+\beta)$ tuple point, in the case actually under consideration, produces only a reduction $=\alpha$. If however the line of approach coincides with the tangent of the β branches, then $b=0$; the factor y^β divides out, and the equation is

$$\Sigma\,(ayw-cyz-dxz)^\eta\,(*\mid x,\,y)^{\alpha+\beta-\eta}=0,$$

which is of the order $\alpha+\beta$, $=m$, so that here the reduction caused by the $\alpha\,(+\beta)$ tuple point is $=\alpha+\beta$. We may without loss of generality substitute ax for $cy+dx$, and then, putting also $a=1$, we find that when the equation of the curve m is as before

$$\Sigma\,(yw)^\eta\,(*\mid x,\,y)^{\alpha+\beta-\eta}=0,$$

but the plane through the directrix line ($x=0$, $y=0$), and the point on this line, are respectively given by the equations $x=\lambda y$, $z=\lambda w$, the equation of the scroll is

$$\Sigma\,(yw-xz)^\eta\,(*\mid x,\,y)^{\alpha+\beta-2\eta}=0.$$

23. The result may be verified by considering the section by any plane $y=\lambda x$ through the directrix line. Substituting for y this value, we find

$$x^\alpha\,\Sigma\,x^{\beta-\eta}\,(\lambda w-z)^\eta\,(*\mid 1,\,\lambda)^{\alpha+\beta-2\eta}=0,$$

which is of the form

$$x^\alpha\,(*\mid x,\,\lambda w-z)^\beta=0;$$

so that the section is made up of the directrix line ($x=0$, $y=0$) reckoned α times and of β lines in the plane $y-\lambda x=0$, the intersections of the plane $y-\lambda x=0$ by planes such as $z=\lambda w-\rho x$.

Case of a γ-tuple generating line.

24. The equation of the scroll may be written

$$(U,\, V,\, W,\, \ldots \mid 1,\, yw - xz)^{\beta} = 0,$$

where U, V, W, ... are functions of x, y of the forms

$$(* \mid x,\, y)^{m}, \quad (* \mid x,\, y)^{m-2}, \quad (* \mid x,\, y)^{m-4}, \ldots;$$

assuming that these contain respectively the factors

$$(y - \kappa x)^{\gamma}, \quad (y - \kappa x)^{\gamma-1}, \quad (y - \kappa x)^{\gamma-2}, \ldots,$$

where $\gamma \not> \frac{1}{2}m$, then the equation takes the form

$$(U',\, V',\, W',\, \ldots \mid y - \kappa x,\, w\,(y - \kappa x) + x\,(\kappa w - z))^{\gamma} = 0,$$

where the coefficients U', V', W', ... are functions of x, y, z, w of the orders $m-\gamma$, $m-\gamma-1$, $m-\gamma-2$, ...; or, what is the same thing, the equation is

$$(U'',\, V'',\, W'',\, \ldots \mid y - \kappa x,\, \kappa w - z)^{\gamma} = 0,$$

where U'', V'', W'', ... are functions of x, y, z, w of the order $m - \gamma$. The scroll has thus the γ-tuple generating line

$$y - \kappa x = 0, \quad \kappa w - z = 0.$$

Cubic Scrolls, Article Nos. 25 to 35.

25. In the case of a cubic scroll there is necessarily a nodal[2] line; in fact for the m-thic scroll there is a nodal curve which is of the order $m-2$ at least, and of the order $\frac{1}{2}(m-1)(m-2)$ at most, and which for $m = 3$ is therefore a right line. And moreover we see at once that every cubic surface having a nodal line is a scroll; in fact any plane whatever through the nodal line meets the surface in this line counting as 2 lines, and in a curve of the order 1, that is, a line; there are consequently on the surface an infinity of lines, or the surface is a scroll. We have therefore to examine the cubic surfaces which have a nodal line.

26. Let the equations of the nodal line be $x = 0$, $y = 0$; then the equation of the surface is

$$Uz + Vw + Q = 0,$$

where U, V, Q are functions of $(x,\, y)$ of the orders 2, 2, 3 respectively. Suppose first that U, V have no common factor, then we may write

$$Q = (\alpha x + \beta y)\, U + (\gamma x + \delta y)\, V;$$

and substituting this value, and changing the values of z and w, the equation of the surface is of the form

$$Uz + Vw = 0,$$

or, what is the same thing,

$$(* \mid x,\, y)^{2}\, (z,\, w) = 0;$$

so that, besides the nodal directrix line $(x = 0,\, y = 0)$, the scroll has the simple directrix line $(z = 0,\, w = 0)$: it is clear that the section by any plane whatever is a cubic curve having a

[2]The nodal line of a cubic scroll is of course a double line, and in regard to these scrolls the epithets 'nodal' and 'double' may be used indifferently.

node at the foot of the nodal directrix line ($x = 0$, $y = 0$), and passing through the foot of the simple directrix line ($z = 0$, $w = 0$); that is, it is a cubic scroll of the kind $S(1, 1, 3)$; and since for $m = 3$ the only partition $m = \alpha + \beta$ is $m = 2 + 1$, there is only one kind of cubic scroll $S(1, 1, 3)$, and we may say simpliciter that the scroll in question is the cubic scroll $S(1, 1, 3)$.

27. If however the functions U, V have a common factor, say $(\lambda x + \mu y)$, then $zU + wV$ will contain this same factor, and the remaining factor will be of the form

$$z\,(\alpha x + \beta y) + w\,(\gamma x + \delta y),\ = y\,(\beta z + \delta w) + x\,(\alpha z + \gamma w),$$

or, changing the values of z and w, the remaining factor will be of the form $yw - xz$, and the equation of the scroll thus is

$$(\lambda x + \mu y)\,(yw - xz) + (* \mid x,\ y)^3 = 0,$$

where it is clear that the section by any plane whatever is a cubic curve having a node at the foot of the directrix line $x = 0$, $y = 0$. The scroll is thus a cubic scroll of the form $S(\overline{1,\ 1}, 3)$, viz. it is the scroll of the kind where the section is a cubic curve with a $2\,(+1)$ tuple point (ordinary double point, or node), the line of approach being one of the two tangents at the node; and since for $m = 3$ the only partition $m = \alpha + \beta$ is $m = 2 + 1$, there is only one kind of cubic scroll $S(\overline{1,\ 1}, 3)$, and we may say simpliciter that the scroll in question is the cubic scroll $S(\overline{1,\ 1}, 3)$. The conclusion therefore is that for cubic scrolls we have only the two kinds, $S(1, 1, 3)$ and $S(\overline{1,\ 1}, 3)$. The foregoing equations of these scrolls admit however of simplification; and I will further consider the two kinds respectively.

The Cubic Scroll $S(1, 1, 3)$.

28. Starting from the equation

$$(* \mid x,\ y)^2\,(z,\ w) = 0,$$

or, writing it at full length,

$$z\,(a,\ b,\ c \mid x,\ y)^2 + w\,(a',\ b',\ c' \mid x,\ y)^2 = 0,$$

we may find θ_1, θ_2 so that

$$(a,\ b,\ c \mid x,\ y)^2 + \theta_1\,(a',\ b',\ c' \mid x,\ y)^2 = (p_1 x + q_1 y)^2,$$

$$(a,\ b,\ c \mid x,\ y)^2 + \theta_2\,(a',\ b',\ c' \mid x,\ y)^2 = (p_2 x + q_2 y)^2,$$

θ_1 and θ_2 being unequal, since by hypothesis $(a,\ b,\ c \mid x,\ y)^2$ and $(a',\ b',\ c' \mid x,\ y)^2$ have no common factor. This gives

$$(a,\ b,\ c \mid x,\ y)^2 = \alpha\,(p_1 x + q_1 y)^2 + \beta\,(p_2 x + q_2 y)^2,$$

$$(a',\ b',\ c' \mid x,\ y)^2 = \gamma\,(p_1 x + q_1 y)^2 + \delta\,(p_2 x + q_2 y)^2;$$

or the equation becomes

$$(\alpha z + \gamma w)\,(p_1 x + q_1 y)^2 + (\beta z + \delta w)\,(p_2 x + q_2 y)^2 = 0;$$

or changing the values of $(x,\ y)$ and of $(z,\ w)$, the equation is

$$x^2 z + y^2 w = 0,$$

which may be considered as the canonical form of the equation. It may be noticed that the Hessian of the form is x^2y^2.

29. We may of course establish the theory of the surface from the equation $x^2z + y^2w = 0$; the equation is satisfied by $x = \lambda y$, $w = -\lambda^2 z$, which are the equations of a line meeting the line $(x = 0,\ y = 0)$ (1) and the line $(z = 0,\ w = 0)$ (1′). The generating line meets also any plane section of the surface; in fact, if the equation of the plane of the section be $\alpha x+\beta y+\gamma z+\delta w = 0$, then we have at once

$$x : y : z : w = \delta\lambda^3 - \gamma\lambda : \delta\lambda^2 - \gamma : \alpha\lambda + \beta : -\alpha\lambda^3 - \beta\lambda^2$$

for the coordinates of the point of intersection.

30. The form of the equation shows that there are on the line 1 two points, viz. the points $(x = 0,\ y = 0,\ z = 0)$ and $(x = 0,\ y = 0,\ w = 0)$, through each of which there passes a pair of coincident generating lines: calling these A and B, then, if the coincident lines through A meet the line 1′ in C, and the coincident lines through B meet the line 1′ in D, it is easy to see that $x = 0$, $y = 0$, $z = 0$, and $w = 0$ will denote the equations of the planes BAC, BAD, BCD, and ACD respectively.

31. We obtain also the following construction: take a cubic curve having a node, and from any point K on the curve draw to the curve the tangents Kp, Kq; through the points of contact draw at pleasure the lines pAC and qBD; through the node draw a line meeting these two lines in the points A, B respectively, this will be the line 1; and through the point K a line meeting the same two lines in the points C and D respectively, this will be the line 1′; and, the equations $x = 0$, $y = 0$, $z = 0$, $w = 0$ denoting as above, the equation of the surface will be $x^2z+y^2w = 0$.

The points A and B are cuspidal points on the nodal line; any section of the scroll by a plane through one of these points is a cubic curve having at the point in question a cusp.

32. It is to be noticed however that the cuspidal points are not of necessity real; if for x, y we write $x + iy$, $x - iy$, and in like manner $z + iw$, $z - iw$ for z, w, then the equation takes the form

$$(x^2 - y^2)\, z - 2xy\, w = 0,$$

which is a cubic scroll $S(1,\ 1,\ 3)$ with the cuspidal points imaginary.

In the last-mentioned case the nodal line is throughout its whole length crunodal; in the case first considered, where the equation is $x^2z+y^2w = 0$, the nodal line is for that part of its length for which z, w have opposite signs, crunodal; and for the remainder of its length, or where z, w have the same sign, acnodal. There are two different forms, according as the line is for the portion intermediate between the cuspidal points crunodal and for the extramediate portions acnodal, or as it is for the intermediate portion acnodal and for the extramediate portions crunodal.

Cubic Scroll $S(\overline{1,\ 1},\ 3)$.

33. Starting from the equation

$$(\lambda x + \mu y)\,(yw - xz) + (* \mid x,\ y)^3 = 0,$$

then putting $w - \mu z$ for w and λz for z, this may be written

$$(\lambda x + \mu y)\,\big(yw - z\,(\lambda x + \mu y)\big) + (* \mid x,\ y)^3 = 0,$$

or, what is the same thing,

$$\lambda'\,(yw - xz) + (* \mid x,\ y)^3 = 0;$$

and then, if $(* \mid x,\, y)^3 = (\alpha,\, \beta,\, \gamma,\, \delta \mid x,\, y)^3$, this may be written

$$\lambda' \left\{ y\,(w + \beta x + \gamma y) - x\,(z - \alpha x) \right\} + \delta y^3 = 0;$$

or changing the values of w and z, we have

$$\lambda\,(yw - xz) + y^3 = 0$$

for the equation of the scroll $S(\overline{1,\, 1},\, 3)$ [3].

34. The Hessian of the form is x^4, and it thus appears that the plane $x = 0$ is a determinate plane through the double line. But $y = 0$ is not a determinate plane; in fact, if for y we write $y + \lambda x$, the equation is

$$-x^2 z + xw\,(y + \lambda x) + (y + \lambda x)^3 = 0,$$

that is

$$-x^2\,(z - \lambda w - 3\lambda^2 y - \lambda^3 x) + xy\,(w + 3\lambda x) + y^3 = 0,$$

which, changing z and w, is still of the form $x\,(yw - xz) + y^3 = 0$.

The planes $z = 0$, $w = 0$ will alter with the plane $y = 0$, but they are not determined even when the plane $y = 0$ is determined; in fact we may, without altering the equation, change w, z into $w + \theta y$, $z + \theta x$ respectively.

35. In the equation $x\,(yw - xz) + y^3 = 0$, writing $y = \lambda x$, we find for the equations of a generating line, $y = \lambda x$, $z = \lambda w + \lambda^3 x$. Considering the section by the plane $\alpha x + \beta y + \gamma z + \delta w = 0$, we have

$$x : y : z : w = -\gamma\lambda^2 - \delta : -\gamma\lambda^3 - \delta\lambda : -\delta\lambda^4 + \beta\lambda^2 + \alpha\lambda : \gamma\lambda^4 + \beta\lambda + \alpha$$

for the coordinates of the point where the generating line meets the section.

The generating line meets the nodal line at the intersection of the nodal line by the plane $z = \lambda w$; that is, the points $z = \lambda w$ on the nodal line correspond to the planes $y = \lambda x$ through the nodal line. In particular the point $w = 0$ on the nodal line corresponds to the plane $x = 0$ through the nodal line: the point $\gamma z + \delta w = 0$ on the nodal line (that is, the point where this line is met by the plane $\alpha x + \beta y + \gamma z + \delta w = 0$) corresponds to the plane $\gamma x + \delta y = 0$ through the nodal line; the intersections of the plane $\alpha x + \beta y + \gamma z + \delta w = 0$ by this plane $\gamma x + \delta y = 0$, and by the plane $x = 0$, are the tangents of the section at the node.

Quartic Scrolls, Article Nos. 36 to 50.

36. We may consider, first, the quartic scrolls $S(1,\, 1,\, 4)$. The section is a quartic curve having an α-tuple point and a β-tuple point, where $\alpha + \beta = 4$; that is, we have $\alpha = 2$, $\beta = 2$, a quartic with two nodes (double points), or else $\alpha = 3$, $\beta = 1$, a quartic with a triple point. But the case $\alpha = 2$, $\beta = 2$ gives rise to two species: viz., in general the quartic has only the two double points, and we have then a scroll with two nodal (2-tuple) directrix lines, and without any nodal generator; the section may however have a third double point, and the scroll has then a nodal (double) generator. For the case $\alpha = 3$, $\beta = 1$, the section admits of no further singularity, and we have a quartic scroll with a triple directrix line and a single directrix line.

37. Next for the quartic scroll $S(\overline{1,\, 1},\, 4)$. The section is here a quartic curve with an $\alpha\,(+\beta)$ tuple point, where $\alpha + \beta = 4$; that is, $\alpha = 2$, $\beta = 2$, or else $\alpha = 3$, $\beta = 1$. In the former case the section has a $2\,(+2)$ tuple point, that is, a double point where the two branches have a common tangent—otherwise, two coincident double points: say the curve has a tacnode; the

[3] It is somewhat more convenient to change the sign of z, and take $x\,(yw + xz) + y^3 = 0$ as the canonical form.

line of approach is the tangent at the tacnode. We have here a scroll with a twofold double line; there are however two cases: viz., in general the section has, besides the tacnode, no other double point; that is, the scroll has no nodal generator: the section may however have a third double point, and the scroll has then a nodal (double) generator. In the case $\alpha = 3$, $\beta = 1$ the section has a triple point, and the line of approach is the tangent at one of the branches at the triple point; the scroll has a twofold, say a $3\,(+1)$ tuple directrix line: as the section admits of no further singularity, this is the only case. The foregoing enumeration gives three species of quartic scrolls $S(1,\,1,\,4)$, and three species of quartic scrolls $S(\overline{1,\,1},\,4)$, together six species, viz. these are as follows:

Quartic Scroll, First Species, $S(1_2,\,1_2,\,4)$, with two double directrix lines, and without a nodal generator.

38. Taking $(x = 0,\ y = 0)$ and $(z = 0,\ w = 0)$ for the equations of the two directrix lines respectively, the equation of the scroll is

$$(*\mid x,\,y)^2\,(z,\,w)^2 = 0.$$

Quartic Scroll, Second Species, $S'(1_2,\,1_2,\,4)$, with two double directrix lines and with a double generator.

39. This is in fact a specialized form of the first species, the difference being that there is a nodal (double) generator. Supposing as before that the equations of the directrix lines are $(x = 0,\ y = 0)$ and $(z = 0,\ w = 0)$ respectively; let the equations of the nodal generator be $(x + y = 0,\ z + w = 0)$; then, observing that for the first species the equation may be written $(*\mid x,\,y)^2\,(z,\,z+w)^2 = 0$, it is clear that if the terms in z^2 and $z\,(z+w)$ are divisible by $(x+y)^2$ and $(x+y)$ respectively, the surface will have as a new double line the line $(x+y = 0,\ z+w = 0)$, which will be a double generator; and we thus arrive at the equation of the second species of quartic scrolls, viz. this is

$$\big((x+y)^2,\,(x+y)(x,\,y),\,(x,\,y)^2 \mid z,\,z+w\big)^2 = 0.$$

Quartic Scroll, Third Species, $S(1_3,\,1,\,4)$, with a triple directrix line and a single directrix line.

40. Taking $(x = 0,\ y = 0)$ for the equations of the triple directrix line, and $(z = 0,\ w = 0)$ for the equations of the single directrix line, the equation is

$$(*\mid x,\,y)^3\,(z,\,w) = 0.$$

Quartic Scroll, Fourth Species, $S(\overline{1_2,\,1_2},\,4)$, with a twofold $(2\,(+2)$ tuple$)$ directrix line, and without a nodal generator.

41. Taking $(x = 0,\ y = 0)$ for the equations of the directrix line, $z = 0$ for that of a plane section of the scroll, $y = 0$ for the equation of a plane through the tangent at the tacnode of the section, and supposing (see *ante*, No. 22) that the plane through the directrix line and the corresponding point on this line are respectively given by the equations $x = \lambda y$ and $z = \lambda w$, the equation of the scroll is

$$(yw - xz)^2 + (yw - xz)\,(x,\,y)^2 + (x,\,y)^4 = 0.$$

Quartic Scroll, Fifth Species, $S'(\overline{1_2,\, 1_2},\, 4)$*, with a twofold* $(2\,(+2)$ *tuple) generating line, and with a double generator.*

42. Let the equations of the double generator be $x + y = 0$, $z + w = 0$; then the line in question must be a double line on the surface represented by the last-mentioned equation, and this will be the case if only the second and third terms contain the factors $(x + y)$ and $(x + y)^2$ respectively. The equation for the fifth species consequently is

$$(yw - xz)^2 + 2\,(yw - xz)\,(x + y)\,(x,\, y) + (x + y)^2\,(x,\, y)^2 = 0.$$

Quartic Scroll, Sixth Species, $S(\overline{1_3,\, 1},\, 4)$*, with a twofold* $(3\,(+1)$ *tuple) generating line.*

43. Taking $(x = 0,\, y = 0)$ for the equations of the directrix line, $z = 0$ for the equation of a plane section, and assuming that the plane $y = 0$ passes through the tangent which is the line of approach, and that the plane through the directrix line and the corresponding point on this line are respectively given by the equations $x = \lambda y$ and $z = \lambda w$, the equation of the scroll is

$$(yw - xz)\,(x,\, y)^2 + (x,\, y)^4 = 0.$$

I refrain on the present occasion from a more particular discussion of the foregoing six species of quartic scrolls. I establish two other species, as follows:

Quartic Scroll, Seventh Species, $S(1,\, 2,\, 2)$*, with nodal directrix line, and nodal directrix conic which meet, and with a simple directrix conic which meets the nodal conic in two points.*

44. We see, *à priori*, that the scroll generated as above will be of the order 4, that is, a quartic scroll. In fact using the formula (*ante*, No. 5),

$$\text{Order} = 2mnp - \alpha m - \beta n - \gamma p,$$

we have here

$$\begin{aligned} \text{Nodal conic,}\quad & m = 2, \quad \alpha = 0, \\ \text{Simple conic,}\quad & n = 2, \quad \beta = 1, \\ \text{Line,}\quad & p = 1, \quad \gamma = 2, \end{aligned}$$

and hence

$$\text{Order} = 8 - 2 - 2,\ = 4.$$

45. Take $(x = 0,\, y = 0)$ for the equations of the directrix line, $z = 0$ for the equation of the plane of the simple conic, $w = 0$ for that of the plane of the nodal conic; since the conics intersect in two points, they lie on a quadric surface, say the surface $U = 0$; the equations of the simple conic thus are $z = 0$, $U = 0$; those of the nodal conic are $w = 0$, $U = 0$. The directrix line $x = 0$, $y = 0$ meets the nodal conic; that is, U must vanish identically for $x = 0$, $y = 0$, $w = 0$; and this will be the case if only the term in z^2 is wanting; that is, we must have

$$U = (a,\, b,\, 0,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid x,\, y,\, z,\, w)^2.$$

But we may in the first instance omit the condition in question, and write

$$U = (a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid x,\, y,\, z,\, w)^2;$$

this would lead to a sextic instead of a quartic scroll.

46. The equations of a generating line (since it meets the directrix line $x = 0$, $y = 0$) may be taken to be

$$x = \alpha y, \quad z = \beta\left(y - \frac{w}{\theta}\right);$$

the condition in order to the intersection of the generating line with the nodal conic is at once found to be

$$a\alpha^2 + 2h\alpha + b + 2\beta\,(f + g\alpha) + c\beta^2 = 0,$$

and that for its intersection with the simple conic

$$a\alpha^2 + 2h\alpha + b + 2\theta\,(m + l\alpha) + d\theta^2 = 0;$$

and writing the equations of the generating line in the form

$$\frac{y}{x} = \frac{1}{\alpha}, \quad \frac{\beta y - z}{\beta} = \frac{w}{\theta},$$

the elimination of α, β, θ from these four equations gives the required equation of the scroll. Writing for a moment

$$F = g\alpha + f,$$
$$M = h\alpha + m,$$

we find

$$c\beta^2 + 2F\beta + \Theta = 0,$$
$$(\Theta y^2 + 2Myw + dw^2)\,\beta^2 - 2\,(\Theta yz + Mwz)\,\beta + \Theta z^2 = 0;$$

or, introducing at this place the condition $c = 0$, the first equation gives β linearly, and we thence obtain

$$\Theta\,(\Theta y^2 + 2Myw + dw^2) + 4F\,(\Theta yz + Mwz) + 4F^2 z^2 = 0,$$

or, what is the same thing,

$$(\Theta y + 2Fz)^2 + 4Mw\,(\Theta y + 2Fz) + 4dw^2 = 0;$$

whence, observing that we have

$$\Theta = \frac{a\alpha^2 + 2h\alpha\, y + by^2}{y^2}, \quad F = \frac{g\alpha + fy}{y}, \quad M = \frac{h\alpha + my}{y},$$

the equation of the scroll is

$$\begin{aligned}&(ax^2 + 2hxy + by^2 + 2gzx + 2fyz)^2\\ &\quad + 2\,(ax^2 + 2hxy + by^2 + 2gzx + 2fyz)\,(lx + my)\,w\\ &\quad + (ax^2 + 2hxy + by^2)\,dw^2 = 0.\end{aligned}$$

We see from the equation that the surface contains the line ($x = 0$, $y = 0$) as a double line, the conic

$$w = 0, \quad ax^2 + 2hxy + by^2 + 2gzx + 2fyz = 0$$

as a double curve, also the conic

$$z = 0, \quad ax^2 + 2hxy + by^2 + 2lxw + 2myw + dw^2 = 0$$

as a simple curve on the surface, the complete intersection by the plane $z = 0$ being in fact the last-mentioned conic, and the pair of lines

$$z = 0, \quad ax^2 + 2hxy + by^2 = 0.$$

Quartic Scroll, Eighth Species, $S(1,\, 3^2)$, with a directrix line, and a directrix skew cubic met twice by each generating line.

47. We see, *à priori*, that the scroll is of the order 4, that is, a quartic scroll; in fact for the quartic scroll $S(1,\, m^2)$ the order is $= [m]^2 + M$ (first memoir, p. 457 [*ante* p. 172]), and we have here $m = 3$, $M = h - \frac{1}{2}[m]^2 = 1 - 3 = -2$; that is, order $= 6 - 2, = 4$.

48. The equations of the cubic curve may be taken to be

$$\begin{vmatrix} x, & y, & z \\ y, & z, & w \end{vmatrix} = 0,$$

or, what is the same thing,

$$xz - y^2 = 0, \quad xw - yz = 0, \quad yw - z^2 = 0;$$

those of the directrix line may be represented by

$$\alpha x + \beta y + \gamma z + \delta w = 0,$$

$$\alpha' x + \beta' y + \gamma' z + \delta' w = 0;$$

or, what is the same thing, if

$$\begin{aligned} \beta\gamma' - \beta'\gamma &= \mathbf{a}, \quad & \alpha\delta' - \alpha'\delta &= \mathbf{f}, \\ \gamma\alpha' - \gamma'\alpha &= \mathbf{b}, \quad & \beta\delta' - \beta'\delta &= \mathbf{g}, \\ \alpha\beta' - \alpha'\beta &= \mathbf{c}, \quad & \gamma\delta' - \gamma'\delta &= \mathbf{h}, \end{aligned}$$

(and therefore identically $\mathbf{af} + \mathbf{bg} + \mathbf{ch} = 0$), the line is defined by means of its "six coordinates" $(\mathbf{a},\, \mathbf{b},\, \mathbf{c},\, \mathbf{f},\, \mathbf{g},\, \mathbf{h})$.

49. The equations of the cubic curve are satisfied by writing therein

$$x : y : z : w = 1 : t : t^2 : t^3,$$

and therefore the coordinates of any two points on the curve may be represented by $(1,\, \theta,\, \theta^2,\, \theta^3)$ and $(1,\, \phi,\, \phi^2,\, \phi^3)$; hence, if x, y, z, w are the coordinates of a point in the line joining the last mentioned two points, we have

$$x : y : z : w = l + m : l\theta + m\phi : l\theta^2 + m\phi^2 : l\theta^3 + m\phi^3,$$

which equations, treating therein l, m as indeterminate parameters, give the equations of the line in question. And putting moreover

$$p = yw - z^2, \quad q = yz - xw, \quad r = xz - y^2,$$

we have identically

$$p : q : r = \theta\phi : -(\theta + \phi) : 1.$$

50. In order that the line in question may meet the directrix line, we must have

$$l\,(\alpha + \beta\theta + \gamma\theta^2 + \delta\theta^3) + m\,(\alpha + \beta\phi + \gamma\phi^2 + \delta\phi^3) = 0,$$

$$l\,(\alpha' + \beta'\theta + \gamma'\theta^2 + \delta'\theta^3) + m\,(\alpha' + \beta'\phi + \gamma'\phi^2 + \delta'\phi^3) = 0;$$

that is, eliminating l and m, we must have

$$\begin{vmatrix} \alpha + \beta\theta + \gamma\theta^2 + \delta\theta^3, & \alpha + \beta\phi + \gamma\phi^2 + \delta\phi^3 \\ \alpha' + \beta'\theta + \gamma'\theta^2 + \delta'\theta^3, & \alpha' + \beta'\phi + \gamma'\phi^2 + \delta'\phi^3 \end{vmatrix} = 0,$$

or, developing,

$$(\alpha\beta' - \alpha'\beta)(\phi - \theta) + (\alpha\gamma' - \alpha'\gamma)(\phi^2 - \theta^2) + (\alpha\delta' - \alpha'\delta)(\phi^3 - \theta^3)$$
$$+(\beta\gamma' - \beta'\gamma)(\theta\phi^2 - \theta^2\phi) + (\beta\delta' - \beta'\delta)(\theta\phi^3 - \theta^3\phi) + (\gamma\delta' - \gamma'\delta)(\theta^2\phi^3 - \theta^3\phi^2) = 0;$$

the several terms in $(\theta,\, \phi)$, each divided by $\phi - \theta$, give respectively

$$1, \quad \phi + \theta, \quad (\phi + \theta)^2 - \phi\theta, \quad \theta\phi, \quad \theta\phi\,(\phi + \theta), \quad \theta^2\phi^2,$$

which are equal to

$$(r^2, \quad -qr, \quad q^2 - pr, \quad pr, \quad -pq, \quad p^2);$$

hence replacing also $\alpha\beta' - \alpha'\beta$, &c. by their values $\mathbf{c}$, &c., we find

$$(\mathbf{c},\, -\mathbf{b},\, \mathbf{f},\, \mathbf{a},\, \mathbf{g},\, \mathbf{h} \mid r^2,\, -qr,\, q^2 - pr,\, pr,\, -pq,\, p^2) = 0,$$

or, what is the same thing,

$$(\mathbf{h},\, \mathbf{f},\, \mathbf{c},\, \mathbf{b},\, \mathbf{a} - \mathbf{f},\, -\mathbf{g} \mid p,\, q,\, r)^2 = 0,$$

where the coefficients $(\mathbf{a},\, \mathbf{b},\, \mathbf{c},\, \mathbf{f},\, \mathbf{g},\, \mathbf{h})$ satisfy the relation $\mathbf{af} + \mathbf{bg} + \mathbf{ch} = 0$; $p,\, q,\, r$ stand respectively for $yw - z^2$, $yz - xw$, $xz - y^2$.

Writing for greater convenience

$$(\mathbf{h},\, \mathbf{f},\, \mathbf{c},\, \mathbf{b},\, \mathbf{a} - \mathbf{f},\, -\mathbf{g}) = (\mathfrak{a},\, \mathfrak{b},\, \mathfrak{c},\, 2\mathfrak{f},\, 2\mathfrak{g},\, 2\mathfrak{h}),$$

or, what is the same thing,

$$(\mathbf{a},\, \mathbf{b},\, \mathbf{c},\, \mathbf{f},\, \mathbf{g},\, \mathbf{h}) = (\mathfrak{b} + 2\mathfrak{g},\, 2\mathfrak{f},\, \mathfrak{c},\, \mathfrak{b},\, -2\mathfrak{h},\, \mathfrak{a}),$$

then we have

$$\mathbf{af} + \mathbf{bg} + \mathbf{ch} = \mathfrak{ac} + \mathfrak{b}^2 + 2\mathfrak{bg} - 4\mathfrak{fh} = 0;$$

and hence finally we have for the equation of the scroll $S(1,\, 3^2)$,

$$(\mathfrak{a},\, \mathfrak{b},\, \mathfrak{c},\, \mathfrak{f},\, \mathfrak{g},\, \mathfrak{h} \mid yw - z^2,\, yz - xw,\, xz - y^2)^2 = 0,$$

where the coefficients satisfy the relation

$$\mathfrak{ac} + \mathfrak{b}^2 + 2\mathfrak{bg} - 4\mathfrak{fh} = 0.$$

The equations of the directrix cubic are of course

$$yw - z^2 = 0, \quad yz - xw = 0, \quad xz - y^2 = 0;$$

and the directrix line is given by its six coordinates,

$$(\mathfrak{b} + 2\mathfrak{g},\, 2\mathfrak{f},\, \mathfrak{c},\, \mathfrak{b},\, -2\mathfrak{h},\, \mathfrak{a}).$$

On the general Theory of Scrolls, Article Nos. 51 to 53.

51. I annex in conclusion the following considerations on the general theory of scrolls. Consider a scroll of the nth order; the intersection by an arbitrary plane, say the plane $w = 0$, is a curve of the nth order $(* \mid x,\ y,\ z)^n = 0$; any point $(x,\ y,\ z,\ 0)$ where $(x,\ y,\ z)$ satisfy the foregoing equation, is the foot of a generating line; and we may imagine this generating line determined by means of the coordinates $(X,\ Y,\ Z,\ W)$, given functions of $(x,\ y,\ z)$ of a point on the line. This being so, the "six coordinates," say $(p,\ q,\ r,\ s,\ t,\ u)$, of the line are

$$\begin{vmatrix} X, & Y, & Z, & W \\ x, & y, & z, & 0 \end{vmatrix},$$

viz.

$$\begin{aligned} p &= Yz - Zy, & s &= -Wx, \\ q &= Zx - Xz, & t &= -Wy, \\ r &= Xy - Yx, & u &= -Wz; \end{aligned}$$

or, writing for greater convenience $-v$ in the place of W, the six coordinates of the line are $p,\ q,\ r,\ vx,\ vy,\ vz$, where $p,\ q,\ r$ are functions of $(x,\ y,\ z)$, connected by the relation $px+qy+rz = 0$; and v is also a function of $(x,\ y,\ z)$.

52. Consider the intersection of the surface by an arbitrary line, the six coordinates whereof are $(A,\ B,\ C,\ F,\ G,\ H)$; then for the generating lines which meet this line we have

$$v\,(Ax + By + Cz) + Fp + Gq + Hr = 0,$$

and this equation, together with the equation $(* \mid x,\ y,\ z)^n = 0$, determines $(x,\ y,\ z)$, the coordinates of the foot of a generating line which meets the arbitrary line $(A,\ B,\ C,\ F,\ G,\ H)$. Since the order of the scroll is $= n$, the number of such generating lines should be $= n$, that is, there should be n relevant intersections of the two curves,

$$v\,(Ax + By + Cz) + Fp + Gq + Hr = 0,$$

$$(* \mid x,\ y,\ z)^n = 0;$$

but if $(p,\ q,\ r,\ vx,\ vy,\ vz)$ are each of the order k, the number of actual intersections is $= kn$, which is too many by $(k-1)\,n$.

53. Suppose that the curves

$$p = 0, \quad q = 0, \quad r = 0, \quad vx = 0, \quad vy = 0, \quad vz = 0,$$

or say the curves

$$p = 0, \quad q = 0, \quad r = 0, \quad v = 0$$

have in common θ intersections, and let these be points of the multiplicities $a_1,\ a_2,\ a_3, \ldots\ a_\theta$ on the curve $(* \mid x,\ y,\ z)^n = 0$ (viz. according as the curve does not pass through any one of the intersections in question, or passes once, twice, &c. through such intersection, we have for that intersection $a_i = 0,\ 1,\ 2$, &c., as the case may be, and so for the other intersections); then the kn points of intersection include the $a_1 + a_2 + \ldots + a_\theta$, or say the Σa intersections; but these, being independent of the line $(A,\ B,\ C,\ F,\ G,\ H)$ under consideration, are irrelevant points, and the number of relevant points of intersection is $kn - \Sigma a$; that is, if we have $\Sigma a = (k-1)\,n$, then the scroll in question, viz. the scroll generated by a line which meets the plane $w = 0$ in the curve $(* \mid x,\ y,\ z)^n = 0$, and which has for its six coordinates $(p,\ q,\ r,\ vx,\ vy,\ vz)$, will be a scroll of the nth order.

341. On the Sextactic Points of a Plane Curve

[From the Philosophical Transactions of the Royal Society of London, vol. CLV. (for the year 1865), pp. 545–578. Received November 5,—Read December 22, 1864.]

It is, in my memoir "On the Conic of Five-pointic Contact at any point of a Plane Curve," Phil. Trans. vol. CXLIX. (1859), pp. 371–400, [261], remarked that as in a plane curve there are certain singular points, viz. the points of inflexion, where three consecutive points lie in a line, so there are singular points where six consecutive points of the curve lie in a conic; and such a singular point is there termed a "sextactic point." The memoir in question (here cited as "former memoir") contains the theory of the sextactic points of a cubic curve; but it is only recently that I have succeeded in establishing the theory for a curve of the order m. The result arrived at is that the number of sextactic points is $= m\,(12m-27)$, the points in question being the intersections of the curve m with a curve of the order $12m-27$, the equation of which is

$$\begin{aligned}&(12m^2-54m+57)\,H\cdot \text{Jac.}(U,\,H,\,\Omega_{\mathfrak{H}})\\ &\quad+(m-2)(12m-27)\,H\cdot\text{Jac.}(U,\,H,\,\Omega_U)\\ &\quad+40\,(m-2)^2\cdot\text{Jac.}(U,\,H,\,\Psi)=0,\end{aligned}$$

where $U=0$ is the equation of the given curve of the order m, H is the Hessian or determinant formed with the second differential coefficients $(a,\,b,\,c,\,f,\,g,\,h)$ of U, and, $(\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H})$ being the inverse coefficients ($\mathfrak{A}=bc-f^2$, &c.), then

$$\Omega=(\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H}\mid\partial_x,\,\partial_y,\,\partial_z)^2,$$

and Jac. denotes the Jacobian or functional determinant, viz.

$$\text{Jac.}(U,\,H,\,\Psi)=\begin{vmatrix}\partial_xU, & \partial_yU, & \partial_zU\\ \partial_xH, & \partial_yH, & \partial_zH\\ \partial_x\Psi, & \partial_y\Psi, & \partial_z\Psi\end{vmatrix},$$

and Jac.$(U,\,H,\,\Omega)$ would of course denote the like derivative of $(U,\,H,\,\Omega)$; the subscripts $(\mathfrak{H},\,U)$ of Ω denote restrictions in regard to the differentiation of this function, viz. treating Ω as a function of U and H,

$$\Omega=(\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H}\mid a',\,b',\,c',\,f',\,2g',\,2h')$$

if $(a',\,b',\,c',\,f',\,g',\,h')$ are the second differential coefficients of H, then we have

$$\begin{aligned}\partial_x\Omega=(\partial_x\mathfrak{A},\,\ldots\mid a',\,\ldots)\quad(=\partial_x\Omega_{\mathfrak{H}})\\ +(\mathfrak{A},\,\ldots\mid\partial_xa',\,\ldots)\quad(=\partial_x\Omega_U);\end{aligned}$$

viz. in $\partial_x\Omega_{\mathfrak{H}}$ we consider as exempt from differentiation $(a',\,b',\,c',\,f',\,g',\,h')$ which depend upon H, and in $\partial_x\Omega_U$ we consider as exempt from differentiation $(\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H})$ which depend upon U. We have similarly

$$\partial_y\Omega=\partial_y\Omega_{\mathfrak{H}}+\partial_y\Omega_U,\quad\partial_z\Omega=\partial_z\Omega_{\mathfrak{H}}+\partial_z\Omega_U;$$

and in like manner

$$\text{Jac.}(U,\,H,\,\Omega)=\text{Jac.}(U,\,H,\,\Omega_{\mathfrak{H}})+\text{Jac.}(U,\,H,\,\Omega_U),$$

which explains the signification of the notations Jac.$(U,\,H,\,\Omega_{\mathfrak{H}})$, Jac.$(U,\,H,\,\Omega_U)$.

The condition for a sextactic point is in the first instance obtained in a form involving the arbitrary coefficients $(\lambda,\,\mu,\,\nu)$; viz. we have an equation of the order 5 in $(\lambda,\,\mu,\,\nu)$ and of the order $12m - 22$ in the coordinates $(x,\, y,\, z)$. But writing $\vartheta = \lambda x + \mu y + \nu z$, by successive transformations we throw out the factors ϑ^5, ϑ^4, ϑ^3, ϑ^2, ϑ, thus arriving at a result independent of $(\lambda,\, \mu,\, \nu)$; viz. this is the before-mentioned equation of the order $12m - 27$. The difficulty of the investigation consists in obtaining the transformations by means of which the equation in its original form is thus divested of these irrelevant factors.

Articles Nos. 1 to 6.—Investigation of the Condition for a Sextactic Point.

1. Following the course of investigation in my former memoir, I take $(X,\, Y,\, Z)$ as current coordinates, and I write

$$T = (* \mid X,\, Y,\, Z)^m = 0$$

for the equation of the given curve; $(x,\, y,\, z)$ are the coordinates of a particular point on the given curve, viz. the sextactic point; and U, $= (* \mid x,\, y,\, z)^m$, is what T becomes when $(x,\, y,\, z)$ are written in place of $(X,\, Y,\, Z)$: we have thus $U = 0$ as a condition satisfied by the coordinates of the point in question.

2. Writing for shortness

$$DU = (X\partial_x + Y\partial_y + Z\partial_z)\, U,$$

and taking $\Pi = aX + bY + cZ = 0$ for the equation of an arbitrary line, the equation

$$D^2U - \Pi\, DU = 0$$

is that of a conic having an ordinary (two-pointic) contact with the curve at the point $(x,\, y,\, z)$; and the coefficients of Π are in the former memoir determined so that the contact may be a five-pointic one; the value obtained for Π is

$$\Pi = \tfrac{1}{2}\, \frac{1}{H}\, DH + \Lambda\, DU,$$

where

$$\Lambda = \frac{1}{9H}\, (-3\Omega H + 4\Psi).$$

3. This result was obtained by considering the coordinates of a point of the curve as functions of a single arbitrary parameter, and taking

$$x + dx + \tfrac{1}{2}d^2x + \tfrac{1}{6}d^3x + \tfrac{1}{24}d^4x, \quad y + \&c., \quad z + \&c.$$

for the coordinates of a point consecutive to $(x,\, y,\, z)$; for the present purpose we must go a step further, and write for the coordinates

$$\begin{gathered} x + dx + \tfrac{1}{2}d^2x + \tfrac{1}{6}d^3x + \tfrac{1}{24}d^4x + \tfrac{1}{120}d^5x, \\ y + dy + \tfrac{1}{2}d^2y + \tfrac{1}{6}d^3y + \tfrac{1}{24}d^4y + \tfrac{1}{120}d^5y, \\ z + dz + \tfrac{1}{2}d^2z + \tfrac{1}{6}d^3z + \tfrac{1}{24}d^4z + \tfrac{1}{120}d^5z. \end{gathered}$$

4. Hence if

$$\partial_1 = dx\,\partial_x + dy\,\partial_y + dz\,\partial_z, \quad \partial_2 = d^2x\,\partial_x + d^2y\,\partial_y + d^2z\,\partial_z, \quad \&c.,$$

$$U = 0,$$

we have, in addition to the equations

$$\partial_1 U = 0,$$

$$(\partial_1^2 + \partial_2)\, U = 0,$$

$$(\partial_1^3 + 3\partial_1\partial_2 + \partial_3)\,U = 0,$$
$$(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2 + \partial_4)\,U = 0,$$

of my former memoir, the new equation

$$(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2 + 5\partial_1\partial_4 + 10\partial_2\partial_3 + \partial_5)\,U = 0,$$

and in addition to the equations, $(P = ax + by + cz)$,

$$\begin{aligned}
&-\tfrac{1}{2}\left[(m-1)\,\partial_1^3 + 3\,(m-2)\,\partial_1\partial_2\right]U + P\cdot\tfrac{1}{6}\,(\partial_1^4 + 3\partial_1\partial_3)\,U + \partial_1 P\cdot\tfrac{1}{2}\partial_1^2 U = 0,\\
&-\tfrac{1}{6}\left[(m-1)(\partial_1^4 + 6\partial_1^2\partial_2) + (m-2)(4\partial_1\partial_3 + 3\partial_2^2)\right]U\\
&\quad + P\cdot\tfrac{1}{24}\,(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2)\,U + \partial_1 P\cdot\tfrac{1}{6}\,(\partial_1^3 + 3\partial_1\partial_2)\,U + \partial_2 P\cdot\tfrac{1}{2}\partial_1^2 U = 0,
\end{aligned}$$

giving in the first instance

$$P = 2\,(m-2),$$
$$\partial_1 P = \frac{1}{2}\,\frac{(\partial_1^4 + 6\partial_1^2\partial_2)\,U}{\partial_1^2 U} - \frac{1}{H}\cdot\frac{(\partial_1^3 + 3\partial_1\partial_2)\,U}{2},$$

and leading ultimately to the before-mentioned value of Π, we have the new equation

$$\begin{aligned}
&-\tfrac{1}{24}\left[(m-1)(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2) + (m-2)(5\partial_1\partial_4 + 10\partial_2\partial_3)\right]U\\
&\quad + P\cdot\tfrac{1}{120}\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2 + 5\partial_1\partial_4 + 10\partial_2\partial_3)\,U\\
&\quad + \partial_1 P\cdot\tfrac{1}{24}\,(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2)\,U\\
&\quad + \tfrac{1}{2}\partial_2 P\cdot\tfrac{1}{6}\,(\partial_1^3 + 3\partial_1\partial_2)\,U\\
&\quad + \tfrac{1}{6}\partial_3 P\cdot\tfrac{1}{2}\partial_1^2 U = 0.
\end{aligned}$$

5. This may be written in the form

$$\begin{aligned}
&2\left[(m-1)(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2) + (m-2)(5\partial_1\partial_4 + 10\partial_2\partial_3)\right]U\\
&\quad + P\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2 + 5\partial_1\partial_4 + 10\partial_2\partial_3)\,U\\
&\quad + 5\partial_1 P\,(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2)\,U\\
&\quad + 10\partial_2 P\,(\partial_1^3 + 3\partial_1\partial_2)\,U\\
&\quad + 10\partial_3 P\,(\partial_1^2 U) = 0;
\end{aligned}$$

or putting for P its value, $= 2\,(m-2)$, the equation becomes

$$\begin{aligned}
&2\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2)\,U\\
&\quad + 5\partial_1 P\,(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2)\,U\\
&\quad + 10\partial_2 P\,(\partial_1^3 + 3\partial_1\partial_2)\,U\\
&\quad + 10\partial_3 P\cdot\partial_1^2 U = 0;
\end{aligned}$$

or as this may also be written,

$$2\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2)\,U + 5\partial_1 P\cdot\partial_4 U + 10\partial_2 P\cdot\partial_3 U + 10\partial_3 P\cdot\partial_2 U = 0.$$

6. But the equation

$$\Pi = \tfrac{1}{2}\,\frac{1}{H}\,DH + \Lambda\,DU,$$

which is an identity in regard to (X, Y, Z), gives

$$\partial_1 P = \tfrac{1}{2}\,\frac{1}{H}\,\partial_1 H,$$

$$\partial_2 P = \tfrac{1}{2}\,\frac{1}{H}\,\partial_2 H + \Lambda\,\partial_2 U,$$

$$\partial_3 P = \tfrac{1}{2}\,\frac{1}{H}\,\partial_3 H + \Lambda\,\partial_3 U;$$

and substituting these values, the foregoing equation becomes

$$\begin{aligned} &2\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2)\,U \\ &\qquad + (5\partial_1 U\,\partial_4 H + 10\partial_2 U\,\partial_3 H + 10\partial_3 U\,\partial_2 H)\,\frac{1}{H} + \Lambda\cdot 20\,\partial_2 U\,\partial_3 U = 0; \end{aligned}$$

or putting for Λ its value, $= \frac{1}{9H}\,(-3\Omega H + 4\Psi)$, and multiplying by $9H^2$, this is

$$\begin{aligned} &9H^2\,(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2)\,U \\ &\quad + 15H\,(\partial_1 U\,\partial_4 H + 2\partial_2 U\,\partial_3 H + 2\partial_3 U\,\partial_2 H) \\ &\quad + \tfrac{1}{H}\,(-3\Omega H + 4\Psi)\cdot 10\,\partial_2 U\,\partial_3 U = 0, \end{aligned}$$

which is, in its original or unreduced form, the condition for a sextactic point.

Article Nos. 7 and 8.—Notations and Remarks.

7. Writing, as in my former memoir, A, B, C for the first differential coefficients of U, we have $B\nu - C\mu$, $C\lambda - A\nu$, $A\mu - B\lambda$ for the values of dx, dy, dz, and instead of the symbol D used in my former memoir, I use indifferently the original symbol ∂_1, or write instead thereof ∂, to denote the resulting value

$$\partial_1\,(=\partial) = (B\nu - C\mu)\,\partial_x + (C\lambda - A\nu)\,\partial_y + (A\mu - B\lambda)\,\partial_z,$$

and I remark here that for any function whatever Ω, we have

$$\partial\Omega = \begin{vmatrix} A, & B, & C \\ \lambda, & \mu, & \nu \\ \partial_x\Omega, & \partial_y\Omega, & \partial_z\Omega \end{vmatrix} = \text{Jac.}(U,\,\vartheta,\,\Omega),$$

where $\vartheta = \lambda x + \mu y + \nu z$. I write, as in the former memoir,

$$\Phi = (\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H}\mid\lambda,\,\mu,\,\nu)^2;$$

And also

$$\nabla = (\mathfrak{A},\,\mathfrak{B},\,\mathfrak{C},\,\mathfrak{F},\,\mathfrak{G},\,\mathfrak{H}\mid\lambda,\,\mu,\,\nu\mid\partial_x,\,\partial_y,\,\partial_z),$$

which new symbol ∇ serves to express the functions Π, $\square$ occurring in the former memoir; viz. we have $\Pi = 2\nabla\Phi$, $\square = 2\nabla H$, so that the symbols Π, $\square$ are not any longer required.

8. I remark that the symbols ∂, ∇ are each of them a linear function of $(\partial_x,\,\partial_y,\,\partial_z)$, with coefficients which are functions of the variables $(x,\,y,\,z)$, and this being so, that for any function Π whatever, we have

$$\partial\,(\nabla\Pi) = (\partial.\nabla)\,\Pi + \partial\nabla\Pi,$$

viz. in $\partial\,(\nabla\Pi)$ we operate with ∇ on Π, thereby obtaining $\nabla\Pi$, and then with ∂ on $\nabla\Pi$; in $(\partial.\nabla)\,\Pi$ we operate with ∂ upon ∇ in so far as ∇ is a function of $(x,\,y,\,z)$, thus obtaining a new operating symbol $\partial.\nabla$, a linear function of $(\partial_x,\,\partial_y,\,\partial_z)$, and then operate with $\partial.\nabla$ upon Π; and lastly, in $\partial\nabla\Pi$, we simply multiply together ∂ and ∇, thus obtaining a new operating symbol $\partial\nabla$ of the form $(\partial_x,\,\partial_y,\,\partial_z)^2$, and then operate therewith on Π; it is clear that, as regards the last-mentioned mode of combination, the symbols ∂ and ∇ are convertible, or $\partial\nabla = \nabla\partial$, that is, $\partial\nabla\Pi = \nabla\partial\Pi$.

It is to be observed throughout the memoir that the point (.) is used (as above in $\partial.\nabla$) when an operation is performed upon a symbol of operation as operand; the mere apposition of two or more symbols of operation (as above in $\partial\nabla$) denotes that the symbols of operation are simply multiplied together; and when $\partial\nabla$ is followed by a letter Π denoting not a symbol of operation, but a mere function of the coordinates, that is in an expression such as $\partial\nabla\Pi$, the resulting operation $\partial\nabla$ is performed upon Π as operand; if instead of the single letter Π we have a compound symbol such as HU or $H\,\nabla\Phi$, so that the expression is ∂HU, $\partial H\,\nabla\Phi$, $\partial\nabla HU$ or $\partial\nabla H\,\nabla\Phi$, then it is to be understood that it is merely the immediately following function H which is operated upon by ∂ or $\partial\nabla$; in the few instances where any ambiguity might arise a special explanation is given.

Article Nos. 9 to 11.—First transformation.

9. We have, assuming always $U = 0$, the following formulae (see *post*, Article Nos. 31 to 33):

$$\begin{aligned}(\partial_1^5 + 10\partial_1^3\partial_2 + 10\partial_1^2\partial_3 + 15\partial_1\partial_2^2)\,U = {} & \frac{1}{(m-1)^4}\left\{(27m^3 - 96m + 81)\,H\partial\Phi + (17m^2 - 56m + 51)\,\Phi\partial H\right\} \\ & + \frac{1}{(m-1)^3}\left\{(-14m - 22)\,(\partial.\nabla)\,H - (10m - 18)\,\partial\nabla H\right\} \\ & + \frac{1}{(m-1)^2}\left\{9\Pi\right\},\end{aligned}$$

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The Collected Mathematical Papers of Arthur Cayley

Volume V

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Arthur Cayley

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341. On the Sextactic Points of a Plane Curve *(continued)*

[Continuation from p. 226, in the calculation of the condition for a sextactic point, near the end of Art. 9 and into Art. 10.]

... [the terms of the foregoing expansion combine, and using the various intermediate substitutions of the foregoing articles] we have

$$\partial_x U\,\partial_x H + 2\,\partial_y U\,\partial_y H + 2\,\partial_z U\,\partial_z H$$

expressible as a linear combination of $H\,d^2\Omega$, $H^2\partial\Phi$, and $\Phi\,\partial H$ (with rational coefficients in m).

10. And by means of these the condition becomes

$$0 \;=\; \tfrac{1}{(m-1)^7}\big\{(153m^2-594m+549)H\partial^2\Omega+(-102m^2+396m+366)\Phi\,\partial H\big\}$$
$$+\;\tfrac{1}{(m-1)^6}\big\{(-96m+168)H(\partial.\nabla)H+(-90m+162)H\partial\nabla H+(120m-240)\partial H\nabla H\big\}$$
$$+\;\tfrac{1}{(m-1)^5}\big(9H^2\partial\Omega-45H\Omega\partial H+40\Phi\partial H\big),$$

this being, as already remarked, of the degree 5 in the arbitrary coefficients (λ,μ,ν), and of the order $12m-22$ in the coordinates (x,y,z).

11. But throwing out the factor $\frac{1}{(m-1)^5}$ and observing that in the first line the quadratic functions of m are each a numerical multiple of $51m^2-198m+183$, the condition becomes

$$0 \;=\; (51m^2-198m+183)\,H\,(3H\partial^2\Omega-2\Phi\,\partial H)$$

$$+ \tfrac{1}{m-1}\big\{(-96m+168)H(\partial.\nabla)H + (-90m+162)H\partial\nabla H + (120m-240)\partial H\nabla H\big\}$$
$$+ \tfrac{1}{(m-1)^2}\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big).$$

Article Nos. 12 and 13. – Second transformation.

12. We effect this by means of the formula

$$(m-2)(3H\partial^2\Omega - 2\Phi\,\partial H) \;=\; \tfrac{1}{H}\ \text{Jac.}(U,\Phi,H), \qquad (J)\ (^1)$$

for, substituting this value of $(3H\partial^2\Omega - 2\Phi\,\partial H)$, the equation becomes divisible by H; and dividing out accordingly, the condition becomes

$$\frac{51m^2-198m+183}{m-2}\ H\ \text{Jac.}(U,\Phi,H)$$
$$+ (-96m+168)H^2(\partial.\nabla)H + (-90m+162)H^2\partial\nabla H + (120m-240)H\partial H\nabla H$$
$$+ \tfrac{1}{m-1}\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0.$$

1

13. We have (see post. Article Nos. 36 to 40)

$$\text{Jac.}(U,\Phi,H) \;=\; -(\partial.\nabla)H,$$

and introducing also $\partial\nabla H$ in place of $\nabla\partial H$ by means of the formula [as in former memoir], the condition becomes

$$\big(51m^2-198m+183-(6m-6)\big)H^2(\partial.\nabla)H$$
$$+ (-90m+162)H^2\partial\nabla H + 120(m-2)H\partial H\nabla H$$
$$+ \tfrac{1}{m-1}\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0,$$

or, as this may be written,

$$(45m^2-180m+171)H^2(\partial.\nabla)H + (-90m+162)(m-2)H^2(\partial\nabla H) + 120(m-2)^2H\partial H\nabla H$$
$$+ (m-2)\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0.$$

Article Nos. 14 to 17. – Third transformation.

14. We have the following formulæ,

$$\tfrac{1}{H}\ \text{Jac.}(U,\nabla H,H) = -(5m-11)\partial H\,\nabla H + (3m-6)H\partial(\nabla H) = 0, \quad (J)$$
$$\tfrac{1}{H}\ \text{Jac.}(U,\nabla,H)H = -(2m-4)\partial H\,\nabla H + (3m-6)H(\partial.\nabla)H = 0, \quad (J)$$

in the latter of which, treating ∇ as a function of the coordinates, we first form the symbol Jac.(U,∇,H), and then operating therewith on H, we have Jac.$(U,\nabla,H)H$; these give

$$H\partial(\nabla H) \;=\; \frac{5m-11}{3m-6}\,\partial H\,\nabla H - \tfrac{1}{3(m-2)}\ \text{Jac.}(U,\nabla H,H),$$
$$H(\partial.\nabla)H \;=\; \frac{2m-4}{3m-6}\,\partial H\,\nabla H - \tfrac{1}{3(m-2)}\ \text{Jac.}(U,\nabla,H)H;$$

[1] $(^1)$ Here and elsewhere (J) refers to the Jacobian Formula, see post. Article Nos. 34 and 35.

and substituting these values, the resulting coefficient of $H\partial H\nabla H$ is

$$(45m^2 - 180m + 171) + 120 + \tfrac{(-90m+162)(5m-11)}{3} \;=\; 0.$$

15. Hence the condition will contain the factor H, and throwing this out, and also the constant factor $\frac{2}{m-2}$, it becomes

$$(-15m^2 + 60m - 57)\, H \text{ Jac.}(U, \nabla, H)H \;+\; (30m - 54)(m-2)\, H \text{ Jac.}(U, \nabla H, H)$$
$$+(m-2)^2\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0.$$

16. We have

$$\partial_x(\nabla H) \;=\; \partial_x.\nabla H + \partial_x \nabla H,$$

viz. in $(\partial_x.\nabla)H$, treating ∇ as a function of (x, y, z) we operate upon it with ∂_x to obtain the new symbol $\partial_x.\nabla$, and with this we operate on H; in $\partial_x\nabla H$ we simply multiply together the symbols ∂_x and ∇, giving a new symbol of the form $(\partial^2, \partial_x\partial_y, \partial_x\partial_z)$ which then operates on H. We have the like values of $\partial_y(\nabla H)$ and $\partial_z(\nabla H)$; and thence also

$$\text{Jac.}(U, \nabla H, H) \;=\; \text{Jac.}(U, \nabla, H)H + \text{Jac.}(U, \nabla H, H),$$

viz. in the determinant Jac.(U, ∇, H) the second line corresponding to ∇ is $\partial_x.\nabla$, $\partial_y.\nabla$, $\partial_z.\nabla$ (∇ being the operand); and the Jacobian thus obtained is a symbol which operates on H giving Jac.$(U, \nabla, H)H$; and in the determinant Jac.$(U, \nabla H, H)$ the second line is $\partial_x\nabla H$, $\partial_y\nabla H$, $\partial_z\nabla H$ (∇ being simply multiplied by $\partial_x, \partial_y, \partial_z$ respectively).

17. Substituting, the condition becomes

$$(-15m^2 + 60m - 57)H \text{ Jac.}(U, \nabla, H)H$$
$$+\ (30m - 54)(m-2)\big\{H \text{ Jac.}(U, \nabla, H)H + \text{Jac.}(U, \nabla H, H)\big\}$$
$$+\ (m-2)^2\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0,$$

or, what is the same thing,

$$(15m^2 - 54m + 51)\, H \text{ Jac.}(U, \nabla, H)H \;+\; (30m - 54)(m-2)\, H \text{ Jac.}(U, \nabla H, H)$$
$$+(m-2)^2\big(9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H\big) \;=\; 0.$$

Article Nos. 18 to 27. – Fourth transformation, and final form of the condition for a Sextactic Point.

18. I write

$$(5m - 12)\,\Omega\,\partial H - (3m - 6)\,H\,\partial\Omega \;=\; \tfrac{1}{H}\text{ Jac.}(U, \Omega, H), \quad (J)$$

$$\Omega\partial H + H\partial\Omega \;=\; \partial(\Omega H);$$

and, introducing for convenience the new symbol Ψ,

$$-5\Psi H + H\partial\Omega \;=\; W, \qquad = 0,$$

so that we have

$$\begin{vmatrix} 5m-12 & -(3m-6) & \frac{1}{H}\text{Jac.}(U, \Omega, H) \\ 1 & 1 & \partial(\Omega H) \\ -5 & 1 & W \end{vmatrix}$$

or, what is the same thing,

$$(8m-18)W + 6\partial\,\text{Jac.}(U,\Omega,H) + (10m-18)\partial(\Omega H) \;=\; 0,$$

$$W \;=\; \Omega\partial H - 5\Omega\partial H \;=\; -\tfrac{3}{4m-9}\,\Phi\;\text{Jac.}(U,\Omega,H) - \tfrac{5m-9}{4m-9}\,\partial(\Omega H).$$

19. We have also (*J*)

$$(8m-18)\Psi\partial H - (3m-6)H\partial\Psi - \tfrac{1}{H}\,\text{Jac.}(U,\Psi,H) \;=\; 0,$$

that is

$$\Psi H^2 \;=\; \tfrac{3m-6}{8m-18}\,H\partial\Psi + \tfrac{1}{(8m-18)H}\;\text{Jac.}(U,\Psi,H),$$

and thence

$$\begin{aligned}\Phi HW + W\partial H \;&=\; 9H^2\partial\Omega - 45H\Omega\partial H + 40\Phi\partial H,\\ &= -\tfrac{9(m-2)}{4m-9}\,H\partial(\Omega H) + \tfrac{60(m-2)}{4m-9}\,\Phi\partial H\\ &\quad + \tfrac{1}{4m-9}\big\{-27H\,\text{Jac.}(U,\Omega,H) + 40\,\text{Jac.}(U,\Psi,H)\big\}.\end{aligned}$$

20. The condition thus becomes

$$\begin{aligned}&(15m^2-54m+51)(4m-9)\,H\;\text{Jac.}(U,\nabla,H)H\\ &+ 6(5m-9)(m-2)(4m-9)\,H\;\text{Jac.}(U,\nabla H,H)\\ &+ 3(m-2)\big\{-3(5m-9)(m-2)\partial(\Omega H) + 20(m-2)^2\Phi\partial H\big\}\\ &+ (m-2)^2\big\{-27H\,\text{Jac.}(U,\Omega,H) + 40\,\text{Jac.}(U,\Psi,H)\big\} \;=\; 0,\end{aligned}$$

which for shortness I represent by

$$3H\Pi + (m-2)^2\big\{-27H\,\text{Jac.}(U,\Omega,H) + 40\,\text{Jac.}(U,\Psi,H)\big\} \;=\; 0,$$

so that we have

$$\begin{aligned}\Pi \;=\; &(5m^2-18m+17)(4m-9)\;\text{Jac.}(U,\nabla,H)H\\ &+ 2(5m-9)(m-2)(4m-9)\;\text{Jac.}(U,\nabla H,H)\\ &+ (m-2)\big\{-3(5m-9)(m-2)\partial(\Omega H) + 20(m-2)^2\Phi\partial H\big\}.\end{aligned}$$

21. Write

$$\Psi_1 \;=\; (\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}' \mid A,B,C)^2,$$

where (A,B,C) are as before the differential coefficients of U, and (a',b',c',f',g',h') being the second differential coefficients of H, $(\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}')$ are the inverse coefficients, viz., $\mathfrak{A}' = b'c' - f'^2$, etc. We have

$$-(m-1)^2\,\partial\Psi_1 \;=\; (3m-6)(3m-7)\partial(\Omega H) - (3m-7)^2\,\partial\Psi$$

(see post, Nos. 41 to 46), that is

$$(3m-6)\,\partial(\Omega H) \;=\; (3m-7)\,\partial\Psi - \tfrac{(m-1)^2}{3m-7}\,\partial\Psi_1,$$

and thence

$$\begin{aligned}\Pi \;=\; &(5m^2-18m+17)(4m-9)\;\text{Jac.}(U,\nabla,H)H\\ &+ 2(5m-9)(m-2)(4m-9)\;\text{Jac.}(U,\nabla H,H)\\ &+ (m-2)^2\Big\{(5m^2-18m+17)\partial\Psi + \tfrac{(m-1)^2(5m-9)}{3m-7}\,\partial\Psi_1\Big\}.\end{aligned}$$

22. Now

$$\Psi \;=\; (\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H} \mid A',B',C')^2 \;=\; (\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}' \mid A,B,C)^2,$$

and writing for shortness

$$E_\Psi \;=\; (\partial\mathfrak{A},\ldots \mid A',B',C')^2, \qquad F_\Psi \;=\; (\mathfrak{A},\ldots \mid A',B',C' \mid \partial A',\partial B',\partial C'),$$

$$E_{\Psi_1} \;=\; (\partial\mathfrak{A}',\ldots \mid A,B,C)^2, \qquad F_{\Psi_1} \;=\; (\mathfrak{A}',\ldots \mid A,B,C \mid \partial A,\partial B,\partial C),$$

(we might, in a notation above explained, write $E_\Psi = \partial_N\Psi$, $F_\Psi = \partial_{N'}\Psi$, and in like manner $E_{\Psi_1} = \partial_{N',\Psi}$, $F_{\Psi_1} = \partial_{N,\Psi}$), then we have

$$\partial\Psi \;=\; E_\Psi + 2F_\Psi, \qquad \partial\Psi_1 \;=\; E_{\Psi_1} + 2F_{\Psi_1}.$$

We have moreover

$$\text{Jac.}(U,\nabla H,H) \;=\; -\tfrac{m-1}{3m-7}\,F_{\Psi_1}, \qquad \text{post, Nos. 47 to 50.}$$

$$\text{Jac.}(U,\nabla,H)H \;=\; -E_{\Psi_1}, \qquad \text{post, Nos. 51 to 53.}$$

23. The just-mentioned formulæ give

$$\begin{aligned}\Pi \;=\; &-(5m^2-18m+17)(4m-9)\,E_{\Psi_1}\\ &-2(5m-9)(m-2)(4m-9)\tfrac{m-1}{3m-7}\,F_{\Psi_1}\\ &+(m-2)(5m^2-18m+17)(E_\Psi+2F_\Psi)\\ &+\tfrac{(5m-9)(m-1)^2(m-2)}{3m-7}\,(E_{\Psi_1}+2F_{\Psi_1}),\end{aligned}$$

which simplifies to

$$\begin{aligned}\Pi \;=\; &-(3m-7)(5m^2-18m+17)\,E_{\Psi_1}\\ &+2(m-2)(5m^2-18m+17)\,F_\Psi\\ &+(5m-9)(m-1)^2(m-2)/(3m-7)\,F_{\Psi_1}\\ &-2(m-1)(m-2)(3m-8)(5m-9)/(3m-7)\,F_{\Psi_1},\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}(3m-7)\Pi \;=\; &-(5m^2-18m+17)\big\{-2(m-1)(m-2)F_{\Psi_1}+(3m-7)^2E_{\Psi_1}\big\}\\ &-(5m-9)(m-2)\big\{(m-1)(3m-8)F_{\Psi_1}+(3m-7)(3m-8)F_\Psi-(m-1)^2E_{\Psi_1}\big\}\\ &+(25m^2-103m+106)(m-2)\big\{-(m-1)F_{\Psi_1}+(3m-7)F_\Psi\big\}.\end{aligned}$$

24. But recollecting that

$$\Omega \;=\; (\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H} \mid \partial_x,\partial_y,\partial_z)^2 H$$

and putting

$$= (\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H} \mid a',b',c',2f',2g',2h'),$$

$$E_\Omega \;=\; (\partial\mathfrak{A},\ldots \mid a',\ldots)\;\;(=\partial_N\Omega),$$

$$F_\Omega \;=\; (\mathfrak{A},\ldots \mid \partial a',\ldots)\;\;(=\partial_{N'}\Omega),$$

we have, post, Nos. 41 to 46,

$$-2(m-1)(m-2)F_{\Psi_1}+(3m-7)^2E_{\Psi_1} \;=\; (3m-6)(3m-7)HE\Omega,$$

$$(m-1)(3m-8)F_{\Psi_1} + (3m-7)(3m-8)F_\Psi - (m-1)^2 HE_{\Psi_1} \;=\; (3m-6)(3m-7)HF\Omega,$$
$$-(m-1)F_{\Psi_1} + (3m-7)F_\Psi \;=\; (3m-7)\,\Omega\,\partial H,$$

and the foregoing equation becomes

$$\begin{aligned}(3m-7)\Pi \;=\; & -(5m^2-18m+17)(3m-6)(3m-7)\,HE\Omega \\ & -(5m-9)(m-2)(3m-6)(3m-7)\,HF\Omega \\ & +(m-2)(25m^2-103m-106)(3m-7)\,\Omega\,\partial H.\end{aligned}$$

25. But we have

$$\tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_U) - (3m-6)HE\Omega + (2m-4)\Omega\partial H \;=\; 0, \quad (J)$$

that is

$$\tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_H) - (3m-6)HF\Omega + (3m-6)\Omega\partial H \;=\; 0, \quad (J)$$
$$3(m-2)HE\Omega \;=\; 2(m-2)\Omega\partial H + \tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_U),$$
$$3(m-2)HF\Omega \;=\; (3m-8)\Omega\partial H + \tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_H),$$

and we thus obtain

$$\begin{aligned}\Pi \;=\; & -(5m^2-18m+17)\big(2(m-2)\Omega\partial H + \tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_U)\big) \\ & -(5m-9)(m-2)\big((3m-8)\Omega\partial H + \tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_H)\big) \\ & +(25m^2-103m+106)(m-2)\,\Omega\,\partial H,\end{aligned}$$

where the coefficient of $(m-2)\Omega\partial H$ is

$$-(10m^2-36m+34) - (5m-9)(3m-8) + (25m^2-103m+106) \;=\; 0.$$

Hence

$$\Pi \;=\; -(5m^2-18m+17)\tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_U) - (5m-9)(m-2)\tfrac{1}{H}\,\mathrm{Jac.}(U,H,\Omega_H).$$

26. Substituting this in the equation

$$3H\Pi + (m-2)^2\big\{-27H\,\mathrm{Jac.}(U,H,\Omega) + 40\,\mathrm{Jac.}(U,H,\Psi)\big\} \;=\; 0,$$

the result contains the factor H, and, throwing this out, the condition is

$$\begin{aligned}& 3H\big\{-(5m^2-18m+17)\,\mathrm{Jac.}(U,H,\Omega_U) - (5m-9)(m-2)\,\mathrm{Jac.}(U,H,\Omega_H)\big\} \\ & +(m-2)^2\big\{27H\,\mathrm{Jac.}(U,H,\Omega) - 40\,\mathrm{Jac.}(U,H,\Psi)\big\} \;=\; 0,\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}& -(15m^2-54m+51)H\,\mathrm{Jac.}(U,H,\Omega_U) - 3(5m-9)(m-2)H\,\mathrm{Jac.}(U,H,\Omega_H) \\ & +27(m-2)^2\big\{H\,\mathrm{Jac.}(U,H,\Omega_U) + H\,\mathrm{Jac.}(U,H,\Omega_H)\big\} \\ & -40(m-2)^2\,\mathrm{Jac.}(U,H,\Psi) \;=\; 0.\end{aligned}$$

27. Hence the condition finally is

$$(12m^2-54m+57)\,H\ \mathrm{Jac.}(U,H,\Omega_U) + (m-2)(12m-27)\,H\ \mathrm{Jac.}(U,H,\Omega_H)$$

$$-40(m-2)^2 \text{ Jac.}(U, H, \Psi) \;=\; 0,$$

or, as this may also be written,

$$-3(m-1)\,H \text{ Jac.}(U, H, \Omega_U) + (m-2)(12m-27)\,H \text{ Jac.}(U, H, \Omega)$$
$$-40(m-2)^2 \text{ Jac.}(U, H, \Psi) \;=\; 0,$$

viz. the sextactic points are the intersections of the curve m with the curve represented by this equation; and observing that U, H, $H\Omega$ and Ψ are of the orders m, $3m-6$, $8m-18$ respectively, the order of the curve is as above mentioned $= 12m-27$.

Article Nos. 28 to 30. – Application to a Cubic.

28. I have in my former memoir, No. 30, shown that for a cubic curve

$$\Omega \;=\; (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \;=\; -2S\cdot U \;=\; 0;$$

this implies $\text{Jac.}(U, H, \Omega) = 0$, and hence if one of the two Jacobians, $\text{Jac.}(U, H, \Omega_U)$, $\text{Jac.}(U, H, \Omega_H)$ vanish, the other will also vanish. Now, using the canonical form, we have

$$\Omega \;=\; (\mathfrak{A}, \ldots \mid a', \ldots)$$
$$= \bigl(yz - l^2x^2,\; zx - l^2y^2,\; xy - l^2z^2,\; l^2yz - lx^2,\; l^2zx - ly^2,\; l^2xy - lz^2\bigr)$$

acting on $(-3l^2, 0, 0, (1+2l^3), 0, 0, \ldots)$, the development of which in fact gives the last-mentioned result. But applying this formula to the calculation of $\text{Jac.}(U, H, \Omega_U)$, then disregarding numerical factors, we have

$$\partial_x\Omega_U \;=\; (yz - l^2x^2, \ldots \mid l^2yz - lx^2, \ldots \mid -3l^2, 0, 0, (1+2l^3), 0, 0)$$
$$= -3l^2(yz - l^2x^2) + (1+2l^3)(l^2yz - lx^2)$$
$$= (-l + l^4)(x^3 + 2l\,yz) \;=\; S\,\partial_x U;$$

and in like manner $\partial_y\Omega_U = S\,\partial_y U$, $\partial_z\Omega_U = S\,\partial_z U$, and therefore

$$\text{Jac.}(U, H, \Omega_U) \;=\; S \text{ Jac.}(U, H, U) \;=\; 0,$$

whence also $\text{Jac.}(U, H, \Omega_H) = 0$. And the condition for a sextactic point assumes the more simple form,

$$\text{Jac.}(U, H, \Psi) \;=\; 0.$$

29. Now (former memoir, No. 32) we have

$$\Psi \;=\; (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2$$
$$= (1+8l^3)^2(y^2z^2 + z^2x^2 + x^2y^2)$$
$$+\, (-9l^2)^2(x^6 + y^6 + z^6)$$
$$+\, (-2l - 5l^4 - 20l^7)(x^3 + y^3 + z^3)\,xyz$$
$$+\, (-15l^2 - 78l^5 + 12l^8)\,x^2y^2z^2,$$

or, observing that $x^3 + y^3 + z^3$ and xyz, and therefore the last three lines of the expression of Ψ, are functions of $U\,(= x^3 + y^3 + z^3 + 6l\,xyz)$ and $H\,(= -l^2(x^3 + y^3 + z^3) + (1+2l^3)xyz)$, and consequently give rise to the term $= 0$ in $\text{Jac.}(U, H, \Psi)$, we may write

$$\Psi \;=\; (1+8l^3)^2\,(y^2z^2 + z^2x^2 + x^2y^2).$$

30. We have then, disregarding a constant factor,

$$\begin{aligned}
\text{Jac.}(U, H, \Psi) &= \text{Jac.}\big(x^3+y^3+z^3,\ xyz,\ y^2z^2+z^2x^2+x^2y^2\big),\\
&= \begin{vmatrix} x^2 & y^2 & z^2 \\ yz & zx & xy \\ x(y^2+z^2) & y(z^2+x^2) & z(x^2+y^2) \end{vmatrix}\\
&= x^3(y^4-z^4)+y^3(z^4-x^4)+z^3(x^4-y^4)\\
&= (y^2-z^2)(z^2-x^2)(x^2-y^2)(x+y+z),
\end{aligned}$$

so that the sextactic points are the intersections of the curve

$$(y^2-z^2)(z^2-x^2)(x^2-y^2)(x+y+z) \;=\; 0$$

with the curve

$$U \;=\; x^3+y^3+z^3+6l\,xyz \;=\; 0.$$

Article Nos. 31 to 33. – Proof of identities for the first transformation.

31. Calculation of $(\partial_1^4+10\partial_1^2\partial_2+10\partial_1\partial_3+15\partial_2\partial_2^2)U$.

Writing ∂ in place of ∂_1, we have (former memoir, No. 20)

$$\big(\partial_2+\tfrac{1}{m-1}\,\partial^2\big)U \;=\; \tfrac{1}{(m-1)^2}\big\{-\Phi+\tfrac{3m-6}{m-1}H\partial^2\big\}.$$

But

$$\partial_3{}^m \;=\; \tfrac{1}{m-1}\big(\,[\text{former memoir, Nos. 21 and 22}]\,\big),$$

and thence

$$\partial_2 \;=\; \tfrac{-6(m-1)-(3m-7)}{m-1}\,\Phi-\tfrac{6m+\ldots}{(m-1)^2}H+\tfrac{m-1}{1}\big(\ldots\big),$$

$$(\partial_1^4+6\partial_1^2\partial_2)U \;=\; \tfrac{1}{(m-1)^2}\big[\tfrac{1}{(m-1)^2}(18m^2-66m+60)\Phi^2+\ldots\big],$$

whence operating on each side with ∂_1, $=\partial$, we have

$$\begin{aligned}
(\partial_1^4+10\partial_1^2\partial_2+6\partial_1\partial_3+12\partial_2\partial_3)U \;=\; &-\tfrac{1}{(m-1)^2}(18m^2-66m+60)(H\partial\Phi+\Phi\partial H)\\
&+\tfrac{1}{(m-1)^3}(-10m+18)\big((\partial.\nabla)H+\partial\nabla H\big)+\tfrac{1}{(m-1)^4}\ \ldots\ \partial\Omega.
\end{aligned}$$

We have besides (see Appendix, Nos. 69 to 74),

$$\partial_2^2U \;=\; \tfrac{1}{(m-1)^4}\big(\Phi^2H-\Phi^2\partial H+\ldots\big),$$

and thence

$$(4\partial_1\partial_3+3\partial_2^2)U \;=\; \tfrac{1}{(m-1)^4}\big\{(33m-21)\Phi^2\partial H+(-9m+9)\Phi\partial H\,\partial\Phi\big\},$$

and adding this to the foregoing expression for $(\partial_1^4+10\partial_1^2\partial_2+6\partial_1\partial_3+12\partial_2\partial_3)U$, we have

$$\begin{aligned}
&(\partial_1^4+10\partial_1^2\partial_2+10\partial_1\partial_3+15\partial_2\partial_2^2)U\\
&= \tfrac{1}{(m-1)^4}\big\{(27m^2-96m+81)H\partial\Phi+(17m^2-56m+51)\Phi\partial H\big\}\\
&\quad+\tfrac{1}{(m-1)^3}\big\{(-14m+22)(\partial.\nabla)H+(-10m+18)\partial\nabla H\big\}\\
&\quad+\tfrac{1}{(m-1)^2}\,\Phi\partial\Omega.
\end{aligned}$$

32. Calculation of $\partial_x U\,\partial_x H + 2\partial_y U\,\partial_y H + 2\partial_z U\,\partial_z H$.
We have

$$\partial_x U \;=\; \tfrac{1}{(m-1)^2}\,\partial^2 H, \qquad \partial_y U \;=\; \tfrac{1}{m-1}\,\partial H, \qquad \partial_z U \;=\; \partial H,$$

$$\partial_z H \;=\; \partial H, \qquad \partial_y H \;=\; \partial_y H,$$

$$\partial_x H \;=\; \tfrac{1}{(m-1)^2}\big(-(3m-6)H\partial^2 H + \Phi\,\partial H + \tfrac{m-1}{1}(\partial.\nabla)H\big),$$

for which values see Appendix, No. 58. And hence the expression sought for is

$$\tfrac{1}{(m-1)^4}\Big\{H\big((-3m+6)H\partial^2 H - \Phi\partial H + (m-1)(\partial.\nabla)H\big)$$

$$+\,2(m-1)\,\partial H\,\partial_x H + 2H\big((-3m+6)H\partial^2 H - \Phi\partial H + (m-1)(\partial.\nabla)H\big)\Big\},$$

which reduces (using former memoir, Nos. 21 and 25) to

$$\partial_x U\,\partial_x H + 2\partial_y U\,\partial_y H + 2\partial_z U\,\partial_z H$$

$$= \tfrac{1}{(m-1)^4}\big[(-6m^2+18m-12)H^2\partial^2 H + (-17m^2+60m-55)H^2\Phi\partial H\big]$$

$$+\;\tfrac{1}{(m-1)^3}\big[(2m-2)H(\partial.\nabla)H + (8m-16)\partial H\nabla H\big]$$

$$+\;\tfrac{1}{(m-1)^2}\,\Omega\,\partial H.$$

33. Calculation of $\partial_y U\,\partial_y U$.
This is

$$\tfrac{1}{(m-1)^4}\,H\partial H.$$

Article Nos. 34 and 35. – The Jacobian Formula.

34. Identically, if P, Q, R, S be functions of the degrees p, q, r, s respectively, we have

$$\begin{vmatrix} pP & qQ & rR & sS \\ \partial_x P & \partial_x Q & \partial_x R & \partial_x S \\ \partial_y P & \partial_y Q & \partial_y R & \partial_y S \\ \partial_z P & \partial_z Q & \partial_z R & \partial_z S \end{vmatrix} \;=\; 0,$$

or, what is the same thing,

$$pP\ \text{Jac.}(Q,R,S) - qQ\ \text{Jac.}(R,S,P) + rR\ \text{Jac.}(S,P,Q) - sS\ \text{Jac.}(P,Q,R) \;=\; 0.$$

Hence in particular if $P = U$, and assuming $U = 0$, we have

$$-qQ\ \text{Jac.}(R,S,U) + rR\ \text{Jac.}(S,U,Q) - sS\ \text{Jac.}(U,Q,R) \;=\; 0.$$

If moreover $Q = \frac{1}{H}$, and therefore $S = 1$, we have

$$-\tfrac{1}{H}\,\text{Jac.}(R,S,U) + rR\ \text{Jac.}(S,U,\tfrac{1}{H}) - sS\ \text{Jac.}(U,\tfrac{1}{H},R) \;=\; 0;$$

or, as this may also be written,

$$-\tfrac{1}{H}\,\text{Jac.}(U,R,S) + rR\,\partial S - sS\,\partial R \;=\; 0,$$

that is

$$-\tfrac{1}{H}\,\text{Jac.}(U,R,S) + rR\,\partial S - sS\,\partial R \;=\; 0.$$

35. Particular cases are

$$\begin{aligned}
(2m-4)\Phi\partial H-(3m-6)H\partial\Phi &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Phi,H), \quad \text{ante, No. 12},\\
(5m-11)\nabla H\partial H-(3m-6)H\partial(\nabla H) &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\nabla H,H),\\
(2m-4)\nabla:\partial H-(3m-6)H\partial.\nabla &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\nabla,H),\\
(5m-12)\Omega\partial H-(3m-6)H\partial\Omega &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega,H), \quad \text{Art. 18},\\
(8m-18)\Psi\partial H-(3m-6)H\partial\Psi &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Psi,H), \quad \text{Art. 25},\\
(2m-4)\Omega\partial H-(3m-6)HE\Omega &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega_U,H),\\
(3m-8)\Omega\partial H-(3m-6)HF\Omega &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega_H,H),
\end{aligned}$$

where it is to be observed that in the third of these formulæ I have, in accordance with the notation before employed, written $\partial.\nabla$ to denote the result of the operation ∂ performed on ∇ as operand. I have also written $\nabla:\partial H$ to show that the operation ∇ is not to be performed on the following ∂H as an operand, but that it remains as an unperformed operation. As regards the last two equations, it is to be remarked that the demonstration in the last preceding number depends merely on the homogeneity of the functions, and the orders of these functions: in the former of the two formulæ, the differentiation of Ω is performed upon Ω in regard to the coordinates (x,y,z) in so far only as they enter through U, and Ω is therefore to be regarded as a function of the order $2m-4$; in the latter of the two formulæ, the differentiation is to be performed in regard to the coordinates in so far only as they enter through H, and Ω is therefore to be regarded as a function of the order $3m-8$. The two formulæ might also be written

$$\begin{aligned}
(2m-4)\Omega\partial H-(3m-6)H\partial\Omega_U &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega_U,H),\\
(3m-8)\Omega\partial H-(3m-6)H\partial\Omega_H &= \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega_H,H);
\end{aligned}$$

and it may be noticed that, adding these together, we obtain the foregoing formula,

$$(5m-12)\Omega\partial H-(3m-6)H\partial\Omega = \tfrac{1}{H}\,\mathrm{Jac.}(U,\Omega,H).$$

Article Nos. 36 to 40. – Proof of equation $(\partial.\nabla)H=\mathrm{Jac.}(U,H,\Phi)$, used in the second transformation.

36. We have

$$\begin{aligned}
\nabla &= (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid \partial_x,\partial_y,\partial_z)\\
&= (\mathfrak{A}\partial_x+\mathfrak{H}\partial_y+\mathfrak{G}\partial_z,\ \mathfrak{H}\partial_x+\mathfrak{B}\partial_y+\mathfrak{F}\partial_z,\ \mathfrak{G}\partial_x+\mathfrak{F}\partial_y+\mathfrak{C}\partial_z \mid \lambda,\mu,\nu).
\end{aligned}$$

Also

$$\begin{aligned}
\partial &= (B\nu-C\mu)\partial_x+(C\lambda-A\nu)\partial_y+(A\mu-B\lambda)\partial_z\\
&= \lambda P+\mu Q+\nu R,
\end{aligned}$$

if for a moment $P,Q,R=C\partial_y-B\partial_z,\ A\partial_z-C\partial_x,\ B\partial_x-A\partial_y$. Hence

$$\partial.\nabla = (P\lambda+Q\mu+R\nu)\cdot(\mathfrak{A}\partial_x+\mathfrak{H}\partial_y+\mathfrak{G}\partial_z,\ \mathfrak{H}\partial_x+\mathfrak{B}\partial_y+\mathfrak{F}\partial_z,\ \mathfrak{G}\partial_x+\mathfrak{F}\partial_y+\mathfrak{C}\partial_z \mid \lambda,\mu,\nu),$$

viz. coefficient of λ^2

$$= P\mathfrak{A}\partial_x+P\mathfrak{H}\partial_y+P\mathfrak{G}\partial_z,$$

and so for the other terms; whence also in $(\partial.\nabla)H$ the coefficients of λ^2, etc. are

$$(P\mathfrak{A}\partial_x+P\mathfrak{H}\partial_y+P\mathfrak{G}\partial_z)H,\ \text{etc.}$$

37. Again, in Jac.(U, H, Φ), where $\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2$, the coefficients of λ^2, etc. are Jac.$(U, H, \mathfrak{A})$, etc.; and hence the assumed equation

$$(\partial.\nabla)H \;=\; \text{Jac.}(U, H, \Phi),$$

in regard to the term in λ^2, is

$$(P\mathfrak{A}\partial_x + P\mathfrak{H}\partial_y + P\mathfrak{G}\partial_z)H \;=\; \text{Jac.}(U, H, \mathfrak{A}),$$

$$\text{Jac.}(U, H, \mathfrak{A}) \;=\; \begin{vmatrix} A & B & C \\ \partial_x H & \partial_y H & \partial_z H \\ \partial_x \mathfrak{A} & \partial_y \mathfrak{A} & \partial_z \mathfrak{A} \end{vmatrix},$$

so that the equation is

$$\begin{aligned} &\big[\partial_x H(C\partial_y - B\partial_z) + \partial_y H(A\partial_z - C\partial_x) + \partial_z H(B\partial_x - A\partial_y)\big]\mathfrak{A} \\ &= (\partial_x H \cdot P + \partial_y H \cdot Q + \partial_z H \cdot R)\mathfrak{A} \\ &= P\mathfrak{A}\partial_x H + Q\mathfrak{A}\partial_y H + R\mathfrak{A}\partial_z H, \end{aligned}$$

or, as this may be written,

$$\begin{aligned} &\big[(B\partial_x - C\partial_y)\mathfrak{H} - (C\partial_x - A\partial_z)\mathfrak{A}\big]\partial_y H \\ &+ \big[(B\partial_z - C\partial_y)\mathfrak{G} - (A\partial_y - B\partial_x)\mathfrak{A}\big]\partial_z H \;=\; 0. \end{aligned}$$

38. The coefficient of $\partial_y H$ is, in virtue of the identity, post, No. 40, $\partial_x \mathfrak{A} + \partial_y \mathfrak{H} + \partial_z \mathfrak{G} = 0$,

$$= A\partial_x \mathfrak{A} + B\partial_x \mathfrak{H} + C\partial_x \mathfrak{G};$$

and in like manner the coefficient of $\partial_z H$ is

$$= -(A\partial_y \mathfrak{A} + B\partial_y \mathfrak{H} + C\partial_y \mathfrak{G}),$$

so that the equation is

$$(A\partial_x \mathfrak{A} + B\partial_x \mathfrak{H} + C\partial_x \mathfrak{G})\partial_y H - (A\partial_y \mathfrak{A} + B\partial_y \mathfrak{H} + C\partial_y \mathfrak{G})\partial_z H \;=\; 0.$$

39. But we have

$$\begin{aligned} \mathfrak{A}a + \mathfrak{H}h + \mathfrak{G}g \;&=\; H, \\ \mathfrak{A}h + \mathfrak{H}b + \mathfrak{G}f \;&=\; 0, \\ \mathfrak{A}g + \mathfrak{H}f + \mathfrak{G}c \;&=\; 0, \end{aligned}$$

or multiplying by x, y, z and adding, $= xH$; whence also

$$(m-1)(\mathfrak{A}A + \mathfrak{H}B + \mathfrak{G}C) \;=\; xH,$$

that is

$$(m-1)(\mathfrak{A}\partial_x A + \mathfrak{H}\partial_x B + \mathfrak{G}\partial_x C) \;=\; x\partial_x H,$$

and in like manner

$$\begin{aligned} (m-1)(\mathfrak{A}\partial_y A + \mathfrak{H}\partial_y B + \mathfrak{G}\partial_y C) \;&=\; x\partial_y H, \\ (m-1)(\mathfrak{A}\partial_z A + \mathfrak{H}\partial_z B + \mathfrak{G}\partial_z C) \;&=\; x\partial_z H, \end{aligned}$$

whence the equation in question. The terms in λ^2 are thus shown to be equal, and it might in a similar manner be shown that the terms in $\mu\nu$ are equal; the other terms will then be equal, and we have therefore

$$(\partial.\nabla)H \;=\; \text{Jac.}(U, H, \Phi).$$

40. The identity

$$\partial_x\mathfrak{A} + \partial_y\mathfrak{H} + \partial_z\mathfrak{G} \;=\; 0$$

assumed in the course of the foregoing proof is easily proved. We have in fact

$$\partial_x\mathfrak{A} + \partial_y\mathfrak{H} + \partial_z\mathfrak{G} \;=\; \partial_x(bc - f^2) + \partial_y(fg - ch) + \partial_z(fh - bg)$$

$$= b(\partial_x c - \partial_z f) + c(\partial_x b - \partial_y h) + f(-2\partial_x f + \partial_y g + \partial_z h) + g(\partial_y f - \partial_z b) + h(-\partial_y c + \partial_z f),$$

where the coefficients of b, c, f, g, h separately vanish: we have of course the system

$$\partial_x\mathfrak{A} + \partial_y\mathfrak{H} + \partial_z\mathfrak{G} \;=\; 0,$$

$$\partial_x\mathfrak{H} + \partial_y\mathfrak{B} + \partial_z\mathfrak{F} \;=\; 0,$$

$$\partial_x\mathfrak{G} + \partial_y\mathfrak{F} + \partial_z\mathfrak{C} \;=\; 0.$$

Article Nos. 41 to 46. – Proof of identities for the fourth transformation.

41. Consider the coefficients (a, b, c, f, g, h) and the inverse set $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$, and the coefficients (a', b', c', f', g', h'), and the inverse set $(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}')$; then we have identically

$$(a, \ldots \mid x, y, z)^2\,(\mathfrak{A}', \ldots \mid \alpha, \ldots) - (\mathfrak{A}', \ldots \mid ax + hy + gz, \ldots)^2$$

$$= (a', \ldots \mid x, y, z)^2\,(\mathfrak{A}, \ldots \mid a', \ldots) - (\mathfrak{A}, \ldots \mid a'x + h'y + g'z, \ldots)^2,$$

where $(\mathfrak{A}', \ldots \mid a, \ldots)$ and $(\mathfrak{A}, \ldots \mid a', \ldots)$ stand for

$$(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}' \mid a, b, c, 2f, 2g, 2h)$$

and

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid a', b', c', 2f', 2g', 2h')$$

respectively.

42. Taking (a, b, c, f, g, h), the second differential coefficients of a function U of the order m, and in like manner (a', b', c', f', g', h'), the second differential coefficients of a function U' of the order m', we have

$$m(m-1)U \cdot (\mathfrak{A}', \ldots \mid \partial_x, \partial_y, \partial_z)^2 U - (m-1)^2(\mathfrak{A}', \ldots \mid \partial_x U, \partial_y U, \partial_z U)^2$$

$$= m'(m'-1)U' \cdot (\mathfrak{A}, \ldots \mid \partial_x, \partial_y, \partial_z)^2 U' - (m'-1)^2(\mathfrak{A}, \ldots \mid \partial_x U', \partial_y U', \partial_z U')^2;$$

and in particular if U' be the Hessian of U, then $m' = 3m - 6$.

43. Hence writing

$$\Omega \;=\; (\mathfrak{A}, \ldots \mid \partial_x, \partial_y, \partial_z)^2 H, \qquad \Psi \;=\; (\mathfrak{A}, \ldots \mid \partial_x H, \partial_y H, \partial_z H)^2,$$

we have

$$\Omega_1 \;=\; (\mathfrak{A}', \ldots \mid \partial_x, \partial_y, \partial_z)^2 U, \qquad \Psi_1 \;=\; (\mathfrak{A}', \ldots \mid \partial_x U, \partial_y U, \partial_z U)^2;$$

$$m(m-1)U\Omega_1 - (m-1)^2\Psi_1 \;=\; (3m-6)(3m-7)H\Omega - (3m-7)^2\Psi;$$

or if $U = 0$, then

$$-(m-1)^2\Psi_1 \;=\; (3m-6)(3m-7)H\Omega - (3m-7)^2\Psi;$$

whence also

$$-(m-1)^2\,\partial\Psi_1 \;=\; (3m-6)(3m-7)(H\partial\Omega + \Omega\partial H) - (3m-7)^2\,\partial\Psi,$$

which is the formula, ante No. 21.

44. Recurring to the original formula, since this is an actual identity, we may operate on it with the differential symbol ∂ on the three assumptions:

1. (a, b, c, f, g, h), $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are alone variable.
2. (a', b', c', f', g', h'), $(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}')$ are alone variable.
3. (x, y, z) are alone variable.

We thus obtain

$$(\partial a, \ldots \mid x, y, z)^2(\mathfrak{A}', \ldots \mid a, \ldots) \;=\; (a', \ldots \mid x, y, z)^2(\partial\mathfrak{A}', \ldots \mid a, \ldots) - 2(\mathfrak{A}', \ldots \mid ax+hy+gz, \ldots \mid x\partial a+y\partial h+z\partial g$$

$$(a, \ldots \mid x, y, z)^2(\partial\mathfrak{A}', \ldots \mid a, \ldots) - (\partial\mathfrak{A}', \ldots \mid ax+hy+gz, \ldots)^2 \;=\; (\partial a', \ldots \mid x, y, z)^2(\mathfrak{A}, \ldots \mid a', \ldots) + (a', \ldots \mid x, y,$$

$$2(a, \ldots \mid x, y, z \mid \partial x, \partial y, \partial z)(\mathfrak{A}', \ldots \mid a, \ldots) \;=\; 2(a', \ldots \mid x, y, z \mid \partial x, \partial y, \partial z)(\mathfrak{A}, \ldots \mid a', \ldots) - 2(\mathfrak{A}', \ldots \mid ax+hy+g$$

45. If in these equations respectively we suppose as before that (a, b, c, f, g, h) are the second differential coefficients of a function U of the order m, and (a', b', c', f', g', h') the second differential coefficients of a function U' of the order m'; and that (A, B, C), (A', B', C') are the first differential coefficients of these functions respectively, then after some easy reductions we have

$$\begin{aligned}(m-1)(m-2)\partial U(\mathfrak{A}', \ldots \mid a, \ldots) \;=\;& m'(m'-1)U'(\partial\mathfrak{A}, \ldots \mid a', \ldots) - (m'-1)^2(\partial\mathfrak{A}, \ldots \mid A', B', C')^2\\ &+\; m(m-1)U(\mathfrak{A}', \ldots \mid \partial a, \ldots) - 2(m-1)(m-2)(\mathfrak{A}', \ldots \mid A, B, C \mid \partial A, \partial B,\end{aligned}$$

$$\begin{aligned}m(m-1)U(\partial\mathfrak{A}', \ldots \mid a, \ldots) - (m-1)^2(\partial\mathfrak{A}', \ldots \mid A, B, C)^2 \;=\;& (m'-1)(m'-2)\partial U'(\mathfrak{A}, \ldots \mid a', \ldots)\\ &+\; m'(m'-1)U'(\mathfrak{A}, \ldots \mid \partial a', \ldots),\end{aligned}$$

$$\begin{aligned}2(m-1)\partial U(\mathfrak{A}', \ldots \mid a, \ldots) - 2(m-1)(m-2)(\mathfrak{A}, \ldots \mid A', B', C' \mid \partial A', \partial B', \partial C') \;=\;& 2(m'-1)\partial U'(\mathfrak{A}, \ldots \mid a', \ldots\\ &-\; 2(m-1)(\mathfrak{A}', \ldots \mid A, B, C \mid\end{aligned}$$

equations which may be verified by remarking that their sum is

$$\begin{aligned}&m(m-1)\big\{\partial U(\mathfrak{A}', \ldots \mid a, \ldots) + U[(\partial\mathfrak{A}', \ldots \mid a, \ldots) + (\mathfrak{A}', \ldots \mid \partial a, \ldots)]\big\}\\ &-(m-1)^2\big\{(\partial\mathfrak{A}', \ldots \mid A, B, C)^2 + (\mathfrak{A}', \ldots \mid A, B, C \mid \partial A, \partial B, \partial C)\big\} \;=\; m'(m'-1) \text{ \&c.},\end{aligned}$$

viz., this is the derivative with ∂ of the equation

$$m(m-1)U(\mathfrak{A}', \ldots \mid a, \ldots) - (m-1)^2(\mathfrak{A}', \ldots \mid A, B, C)^2 \;=\; m'(m'-1) \text{ \&c.}, \qquad \partial U = 0.$$

46. Taking now $U' = H$, and therefore $m' = 3m-6$; putting also $U = 0$, and writing as before

$$E_\Psi = (\partial\mathfrak{A}, \ldots \mid A', B', C')^2, \quad F_\Psi = (\mathfrak{A}, \ldots \mid A', B', C' \mid \partial A', \partial B', \partial C'),$$

$$E_{\Psi_1} = (\partial\mathfrak{A}', \ldots \mid A, B, C)^2, \quad F_{\Psi_1} = (\mathfrak{A}', \ldots \mid A, B, C \mid \partial A, \partial B, \partial C),$$

$$E_\Omega = (\partial\mathfrak{A},\ldots \mid a',\ldots),\quad F_\Omega = (\mathfrak{A},\ldots \mid \partial a',\ldots),$$

then the three equations are

$$\begin{aligned}
-2(m-1)(m-2)F_{\Psi_1} &= (3m-6)(3m-7)HE\Omega - (3m-7)^2E_\Psi,\\
-(m-1)^2E_{\Psi_1} &= (3m-7)(3m-8)\Omega\partial H + (3m-6)(3m-7)HF\Omega - 2(3m-7)(3m-8)F_\Psi,\\
-2(m-1)F_{\Psi_1} &= 2(3m-7)\Omega\partial H - 2(3m-7)F_\Psi,
\end{aligned}$$

whence, adding, we have

$$\begin{aligned}
-(m-1)^2(E_{\Psi_1}+2F_{\Psi_1}) &= -(3m-7)^2(E_\Psi+2F_\Psi)\\
&\quad + (3m-6)(3m-7)\big\{\Omega\partial H + H(E\Omega+F\Omega)\big\},
\end{aligned}$$

(that is

$$-(m-1)^2\,\partial\Psi_1 = -(3m-7)^2\,\partial\Psi + (3m-6)(3m-7)\partial.\Omega H,$$

which is right). And by linearly combining the three equations, we deduce

$$\begin{aligned}
(3m-6)(3m-7)HE\Omega &= -2(m-1)(m-2)F_{\Psi_1} + (3m-7)^2E_\Psi,\\
(3m-7)\Omega\partial H &= -(m-1)F_{\Psi_1} + (3m-7)F_\Psi,\\
(3m-6)(3m-7)HF\Omega &= (m-1)(3m-8)F_{\Psi_1} + (3m-7)(3m-8)F_\Psi - (m-1)^2E_{\Psi_1},
\end{aligned}$$

which are the formulæ, ante No. 24.

Article Nos. 47 to 50. – Proof of an identity used in the fourth transformation, viz.,

$$\text{Jac.}(U,\nabla H,H) = -\tfrac{m-1}{3m-7}\,F_{\Psi_1},$$

or say

$$\text{Jac.}(U,H,\nabla H) = \tfrac{m-1}{3m-7}\,(\mathfrak{A}',\ldots \mid A,B,C \mid \partial A,\partial B,\partial C).$$

47. We have

$$\begin{aligned}
\nabla &= (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid \partial_x,\partial_y,\partial_z)\\
&= \big((\mathfrak{A},\mathfrak{H},\mathfrak{G} \mid \lambda,\mu,\nu),\ (\mathfrak{H},\mathfrak{B},\mathfrak{F} \mid \lambda,\mu,\nu),\ (\mathfrak{G},\mathfrak{F},\mathfrak{C} \mid \lambda,\mu,\nu) \mid \partial_x,\partial_y,\partial_z\big);
\end{aligned}$$

or, attending to the effect of the bar as denoting the exemption of the $(\mathfrak{A},\ldots)$ from differentiation,

$$\begin{aligned}
\text{Jac.}(U,H,\nabla H) &= (\mathfrak{A},\mathfrak{H},\mathfrak{G} \mid \lambda,\mu,\nu)\ \text{Jac.}(U,H,\partial_x H)\\
&\quad + (\mathfrak{H},\mathfrak{B},\mathfrak{F} \mid \lambda,\mu,\nu)\ \text{Jac.}(U,H,\partial_y H)\\
&\quad + (\mathfrak{G},\mathfrak{F},\mathfrak{C} \mid \lambda,\mu,\nu)\ \text{Jac.}(U,H,\partial_z H).
\end{aligned}$$

48. Now

$$\text{Jac.}(U,H,\partial_x H) = -\tfrac{1}{m-2}\ \text{Jac.}(U,\ x\partial_x H + y\partial_y H + z\partial_z H,\ \partial_x H),$$

and the last-mentioned Jacobian is

$$\begin{aligned}
&= \partial_x H\ \text{Jac.}(U,x,\partial_x H) + \partial_y H\ \text{Jac.}(U,y,\partial_x H) + \partial_z H\ \text{Jac.}(U,z,\partial_x H)\\
&\quad + y\ \text{Jac.}(U,\partial_y H,\partial_x H) + z\ \text{Jac.}(U,\partial_z H,\partial_x H),
\end{aligned}$$

where the second line is

$$= -y\ \text{Jac.}(U,\partial_x H,\partial_y H) + z\ \text{Jac.}(U,\partial_x H,\partial_z H),$$

or, writing (A', B', C') = first differential coefficients of H, this is the second differential coefficients of H and (a', b', c', f', g', h') for those, in

$$\begin{vmatrix} A & B & C \\ a' & h' & g' \\ h' & b' & f' \end{vmatrix} - y \ldots + z \ldots \;=\; -y(\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C) + z(\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C).$$

The first line, reduced by the formulæ

$$(3m-7)(A', B', C') \;=\; (a'x + h'y + g'z,\ h'x + b'y + f'z,\ g'x + f'y + c'z),$$

is

$$= \tfrac{1}{3m-7}\big\{-y(\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C) + z(\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C)\big\}.$$

Hence we have

$$\begin{aligned} \text{Jac.}(U, H, \partial_x H) \;&=\; \tfrac{1}{3m-7}\big(1 + \tfrac{3m-7}{m-2}\big)\big\{-y(\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C) + z(\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C)\big\} \\ &= \tfrac{1}{m-2}\big\{-y(\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C) + z(\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C)\big\}; \end{aligned}$$

and in like manner

$$\begin{aligned} \text{Jac.}(U, H, \partial_y H) \;&=\; \tfrac{1}{m-2}\big\{-z(\mathfrak{A}', \mathfrak{H}', \mathfrak{G}' \mid A, B, C) + x(\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C)\big\}, \\ \text{Jac.}(U, H, \partial_z H) \;&=\; \tfrac{1}{m-2}\big\{-x(\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C) + y(\mathfrak{A}', \mathfrak{H}', \mathfrak{G}' \mid A, B, C)\big\}. \end{aligned}$$

49. We thence have

$$\text{Jac.}(U, H, \nabla H) \;=\; \tfrac{1}{3m-7}\begin{vmatrix} (\mathfrak{A}', \mathfrak{H}', \mathfrak{G}' \mid A, B, C) & (\mathfrak{H}', \mathfrak{B}', \mathfrak{F}' \mid A, B, C) & (\mathfrak{G}', \mathfrak{F}', \mathfrak{C}' \mid A, B, C) \\ \lambda & \mu & \nu \\ x & y & z \end{vmatrix},$$

or multiplying the two sides by

$$H \;=\; \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix},$$

the right-hand side is

$$\tfrac{1}{3m-7}\begin{vmatrix} H\lambda & H\mu & H\nu \\ X & Y & Z \\ (m-1)A & (m-1)B & (m-1)C \end{vmatrix}$$

which is

$$= H \cdot \tfrac{m-1}{3m-7}\begin{vmatrix} \lambda & \mu & \nu \\ X & Y & Z \\ A & B & C \end{vmatrix},$$

if for a moment

$$\begin{aligned} X &= (\mathfrak{A}', \ldots \mid A, B, C \mid a, h, g), \\ Y &= (\mathfrak{A}', \ldots \mid A, B, C \mid h, b, f), \\ Z &= (\mathfrak{A}', \ldots \mid A, B, C \mid g, f, c). \end{aligned}$$

50. Hence observing that these equations may be written

$$X = (\mathfrak{A}', \ldots \mid A, B, C \mid \partial_x A, \partial_x B, \partial_x C),$$

$$Y = (\mathfrak{A}', \ldots \mid A, B, C \mid \partial_y A, \partial_y B, \partial_y C),$$

$$Z = (\mathfrak{A}', \ldots \mid A, B, C \mid \partial_z A, \partial_z B, \partial_z C),$$

and that we have

$$\partial \;=\; \begin{vmatrix} \lambda & \mu & \nu \\ \partial_x & \partial_y & \partial_z \\ A & B & C \end{vmatrix},$$

we obtain for H Jac.$(U, H, \nabla H)$ the value

$$= H \cdot \tfrac{m-1}{3m-7} (\mathfrak{A}', \ldots \mid A, B, C \mid \partial A, \partial B, \partial C),$$

or throwing out the factor H, we have the required result.

Article Nos. 51 to 53. – Proof of identity used in the fourth transformation, viz.

$$\text{Jac.}(U, \nabla, H)H \;=\; -E_{\Psi_1}, \quad \text{or say } \text{Jac.}(U, H, \nabla)H \;=\; (\partial\mathfrak{A}', \ldots \mid A, B, C)^2.$$

51. We have

$$\nabla \;=\; (\mathfrak{A}, \ldots \mid \lambda, \mu, \nu \mid \partial_x, \partial_y, \partial_z),$$

and thence

$$(\partial_x.\nabla)H \;=\; (\partial_x\mathfrak{A}, \partial_x\mathfrak{H}, \partial_x\mathfrak{G} \mid \lambda, \mu, \nu \mid x, y, z),$$

with the like values for $(\partial_y.\nabla)H$ and $(\partial_z.\nabla)H$. And then

$$\text{Jac.}(U, H, \nabla)H \;=\; \begin{vmatrix} A & B & C \\ A' & B' & C' \\ (\partial_x.\nabla)H & (\partial_y.\nabla)H & (\partial_z.\nabla)H \end{vmatrix},$$

in which the coefficient of λ^2 is

$$= (C\partial_y - B\partial_z)(\mathfrak{A}, \mathfrak{H}, \mathfrak{G} \mid \lambda, \mu, \nu);$$

or putting for shortness

$$(C\partial_y - B\partial_z, A\partial_z - C\partial_x, B\partial_x - A\partial_y) \;=\; (P, Q, R),$$

the coefficient is $\partial = (P\lambda + Q\mu + R\nu)$, and thence

$$\text{coefficient } \lambda^2 - \partial\mathfrak{A} \;=\; (P\mathfrak{A}, P\mathfrak{H}, P\mathfrak{G} \mid \lambda, \mu, \nu) - (P\mathfrak{A}, Q\mathfrak{A}, R\mathfrak{A} \mid \lambda, \mu, \nu)$$

which is

$$\begin{aligned} &= \mu\big\{(C\partial_y - B\partial_z)\mathfrak{H} - (A\partial_z - C\partial_x)\mathfrak{A}\big\} \\ &+ \nu\big\{(C\partial_y - B\partial_z)\mathfrak{G} - (B\partial_x - A\partial_y)\mathfrak{A}\big\}, \end{aligned}$$

where coefficient of μ is

$$= -A\partial_z\mathfrak{A} - B\partial_y\mathfrak{H} + C(\partial_x\mathfrak{A} + \partial_y\mathfrak{H}) \;=\; -(A\partial_z\mathfrak{A} + B\partial_z\mathfrak{H} + C\partial_z\mathfrak{G}) \;=\; -\tfrac{1}{m-1}\, x\partial_z H,$$

and coefficient of ν is

$$= +(A\partial_y\mathfrak{A} + B\partial_y\mathfrak{H} + C\partial_y\mathfrak{G}) \;=\; \tfrac{1}{m-1}\, x\partial_y H,$$

so that

$$\text{coefficient } \lambda^2 - \partial\mathfrak{A} \;=\; -\tfrac{1}{m-1}\, x(\mu\partial_z H - \nu\partial_y H).$$

53. By forming in a similar manner the coefficients of the other terms, it appears that

$$\text{Jac.}(U, H, \nabla)H - (\partial\mathfrak{A}, \ldots \mid A', B', C')^2$$

$$\propto \begin{vmatrix} \lambda & \mu & \nu \\ \partial_x H & \partial_y H & \partial_z H \\ A' & B' & C' \end{vmatrix} (A'x + B'y + C'z),$$

or since the determinant

$$\begin{vmatrix} \lambda & \mu & \nu \\ \partial_x H & \partial_y H & \partial_z H \\ A' & B' & C' \end{vmatrix} = 0,$$

we have the required equation,

$$\text{Jac.}(U, H, \nabla)H = (\partial\mathfrak{A}, \dots \mid A', B', C')^2.$$

This completes the series of formulæ used in the transformations of the condition for the sextactic point.

Appendix, Nos. 54 to 74.

For the sake of exhibiting in their proper connexion some of the formulæ employed in the foregoing first transformation of the condition for a sextactic point, I have investigated them in the present Appendix, which however is numbered continuously with the memoir.

54. The investigations of my former memoir and the present memoir have reference to the operations

$$\begin{aligned}\partial_1 &= dx\,\partial_x + dy\,\partial_y + dz\,\partial_z,\\ \partial_2 &= d^2x\,\partial_x + d^2y\,\partial_y + d^2z\,\partial_z,\\ \partial_3 &= d^3x\,\partial_x + d^3y\,\partial_y + d^3z\,\partial_z,\end{aligned}$$

where, if (A, B, C) are arbitrary constants, then we have $\partial_1 = \partial$, the first differential coefficients of a function $U' = (* \mid x, y, z)^{m'}$, so that putting $dx = B\nu - C\mu$, $dy = C\lambda - A\nu$, $dz = A\mu - B\lambda$;

$$\partial = (B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z = \begin{vmatrix} A & B & C \\ \lambda & \mu & \nu \\ \partial_x & \partial_y & \partial_z \end{vmatrix},$$

we have $\partial_1 = \partial$. The foregoing expressions of (dx, dy, dz) determine of course the values of (d^2x, d^2y, d^2z), (d^3x, d^3y, d^3z), etc., and it is throughout assumed that these values are substituted in the symbols ∂_2, ∂_3, etc., so that $\partial_1 = \partial$, and ∂_2, ∂_3, etc. denote each of them an operator such as $X\partial_x + Y\partial_y + Z\partial_z$, where (X, Y, Z) are functions of the coordinates; such operator, in so far as it is a function of the coordinates, may therefore be made an operand, and be operated upon by itself or any other like operator.

55. Taking (a, b, c, f, g, h) for the second differential coefficients of U, $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ for the inverse coefficients, and H for the Hessian, I write also

$$\Phi = (\mathfrak{A}, \dots \mid \lambda, \mu, \nu)^2.$$

$$\Pi = \lambda x + \mu y + \nu z,$$

$$\Psi = (\mathfrak{A}, \dots \mid \partial_x H, \partial_y H, \partial_z H)^2,$$

$$\nabla = (\mathfrak{A}, \dots \mid \mu\partial_z - \nu\partial_y,\ \nu\partial_x - \lambda\partial_z,\ \lambda\partial_y - \mu\partial_x),$$

and I notice that we have

$$\nabla\Pi^2 = 2\Phi, \quad \nabla\Pi U = \tfrac{1}{m-1}H, \quad \Omega U = SH,$$

the last of which is proved, post No. 65; the others are found without any difficulty.

56. I form the Table [of substitution rules and coefficients to follow; consult facsimile p. 248 for the explicit Table; symbolic entries involve powers of $\frac{1}{m-1}$ multiplied by combinations of Φ, Ψ, and H].

And assuming $U = 0$, we have $\partial U = 0$, and

$$(\partial_2 H)^2 \;=\; (SH)^2 \;=\; \tfrac{1}{(m-1)^2}\big(\Phi^2\partial^2 H - 2\Phi\Psi\partial H + \Psi^2\big),$$

which are for the most part given in my former memoir; the expressions for $\partial_3 U$, $\partial_4 U$, which are not explicitly given, follow at once from the equations

$$(\partial_1^3 + \partial_2)U \;=\; 0, \qquad (\partial_1^4 + 2\partial_2\partial_3 + \partial_4)U \;=\; 0;$$

those for $\partial_2\partial_3 U$, $\partial_3^2 U$, and $\partial_4 U$ are new, but when the expressions for $\partial_3\partial_2 U$ and $\partial_3 U$ are known, that for $\partial_4 U$ is at once found from the equation

$$(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2 + \partial_4)U \;=\; 0.$$

57. Before going further, I remark that we have identically

$$(a,\dots \mid x,y,z)^2(a,\dots \mid \mu y - \nu\beta,\ \nu\alpha - \lambda y,\ \lambda\beta - \mu\alpha)^2$$

$$\propto \begin{vmatrix} ax+hy+gz & hx+by+fz & gx+fy+cz \\ \alpha & \beta & \gamma \\ \lambda & \mu & \nu \end{vmatrix}$$

$$=\; (\mathfrak{A},\dots \mid \lambda p - \alpha\eta,\ \mu p - \beta\eta,\ \nu p - \gamma\eta)^2,$$

(if for shortness $p = \alpha x + \beta y + \gamma z$, $\eta = \lambda x + \mu y + \nu z$)

$$= p^2(\mathfrak{A},\dots \mid \lambda,\mu,\nu)^2 - 2p\eta(\mathfrak{A},\dots \mid \lambda,\mu,\nu \mid \alpha,\beta,\gamma) + \eta^2(\mathfrak{A},\dots \mid \alpha,\beta,\gamma)^2.$$

58. If in this equation we take (a,b,c,f,g,h) to be the second differential coefficients of U, and write also $(\alpha,\beta,\gamma) = (\partial_x,\partial_y,\partial_z)$, the equation becomes

$$m(m-1)U\Gamma - (m-1)^2\Omega \;=\; \Phi(x\partial_x + y\partial_y + z\partial_z)^2 - 2\Pi(x\partial_x + y\partial_y + z\partial_z)\nabla$$

which is a general equation for the transformation of $\partial_1^2 (= \partial^2)$.

59. If with the two sides of this equation we operate on U, we obtain

$$m(m-1)U\Gamma U - (m-1)^2\,\partial^2 U \;=\; m(m-1)\Phi U - 2(m-1)\Pi\nabla U,$$

and substituting the values we find the before-mentioned expression of ${\partial_1}^2 U$.

60. Operating with the two sides of the same equation on a function H of the order m', we find

$$m(m-1)U\Gamma H - (m-1)^2\,\partial^2 H \;=\; m'(m'-1)\Phi H - 2(m'-1)\Pi\nabla H + \Phi\Omega H;$$

and in particular if H is the Hessian, then writing $m' = 3m-6$, and putting $U = 0$, we find the before-mentioned expression for $\partial^2 H$.

61. But we may also from the general identical equation deduce the expression for $(\partial H)^2$. In fact taking H a function of the degree m' and writing

$$(\alpha,\beta,\gamma) \;=\; (\partial_x H, \partial_y H, \partial_z H),$$

we have

$$m(m-1)U(\mathfrak{A},\ldots \mid \mu\partial_z H - \nu\partial_y H,\ \nu\partial_x H - \lambda\partial_z H,\ \lambda\partial_y H - \mu\partial_x H)^2 - (m-1)^2(\partial H)^2$$
$$= \ m'^2\Phi H^2 - 2m'\Pi H\nabla H + \Pi^2(\mathfrak{A},\ldots \mid \partial_x H, \partial_y H, \partial_z H)^2;$$

and if H be the Hessian, then writing $m' = 3m-6$ and putting also $U = 0$, we find the before-mentioned expression for $(\partial H)^2$.

62. Proof of equation $\partial \ = \ \frac{1}{m-1}\big(x\partial_x + y\partial_y + z\partial_z\big) + \frac{1}{m-1}\Pi\nabla$.

We have

$$\partial \ = \ \tfrac{1}{m-1}\big(\lambda(C\partial_y - B\partial_z) + \mu(A\partial_z - C\partial_x) + \nu(B\partial_x - A\partial_y)\big),$$

where

$$= \lambda(C'\partial_y - B'\partial_z) + \mu(A'\partial_z - C'\partial_x) + \nu(B'\partial_x - A'\partial_y),$$
$$A' \ = \ \partial A \ = \ a(B\nu - C\mu) + h(C\lambda - A\nu) + g(A\mu - B\lambda)$$
$$= \lambda(hC - gB) + \mu(gA - aC) + \nu(aB - hA),$$

with the like values for B' and C'. Substituting the values

$$(m-1)(A, B, C) \ = \ (ax + hy + gz,\ hx + by + fz,\ gx + fy + cz),$$

we have

$$(m-1)A' \ = \ \lambda(\mathfrak{H}y - \mathfrak{G}z) + \mu(\mathfrak{G}y - \mathfrak{H}z) + \nu(\mathfrak{H}y - \mathfrak{G}z);$$

and similarly

$$(m-1)B' \ = \ \lambda(\mathfrak{H}z - \mathfrak{G}x) + \mu(\mathfrak{F}z - \mathfrak{G}x) + \nu(\mathfrak{C}z - \mathfrak{G}x),$$
$$(m-1)C' \ = \ \lambda(\mathfrak{H}x - \mathfrak{B}y) + \mu(\mathfrak{F}x - \mathfrak{B}y) + \nu(\mathfrak{C}x - \mathfrak{F}y),$$

and then

$$(m-1)(C'\partial_y - B'\partial_z) \ = \ \lambda\big\{(\mathfrak{B}x - \mathfrak{A}y)\partial_y - (\mathfrak{H}z - \mathfrak{G}x)\partial_z\big\}$$
$$+ \ \mu\big\{(\mathfrak{F}x - \mathfrak{H}y)\partial_y - (\mathfrak{B}z - \mathfrak{F}x)\partial_z\big\}$$
$$+ \ \nu\{\ldots\}$$
$$= x(\mathfrak{A}, \mathfrak{H}, \mathfrak{G} \mid \partial_x, \partial_y, \partial_z \mid \lambda, \mu, \nu) - \mathfrak{A}(x\partial_x + y\partial_y + z\partial_z);$$

and so

$$(m-1)(A'\partial_z - C'\partial_x) \ = \ y(\mathfrak{H}, \mathfrak{B}, \mathfrak{F} \mid \lambda, \mu, \nu \mid \partial_x, \partial_y, \partial_z) - \mathfrak{H}(x\partial_x + y\partial_y + z\partial_z),$$
$$(m-1)(B'\partial_x - A'\partial_y) \ = \ z(\mathfrak{G}, \mathfrak{F}, \mathfrak{C} \mid \lambda, \mu, \nu \mid \partial_x, \partial_y, \partial_z) - \mathfrak{G}(x\partial_x + y\partial_y + z\partial_z);$$
$$(m-1)\partial \ = \ (\lambda x + \mu y + \nu z)\nabla - (\mathfrak{A},\ldots \mid x, \mu, \nu)(x\partial_x + y\partial_y + z\partial_z),$$
$$= \Pi\nabla - \Phi(x\partial_x + y\partial_y + z\partial_z);$$

or finally,

$$\partial \ = \ -\tfrac{\Phi}{m-1}(x\partial_x + y\partial_y + z\partial_z) + \tfrac{1}{m-1}\Pi\nabla.$$

63. This leads to the expression for $\partial^2 U$; we have

$$\partial^2 \ = \ \tfrac{1}{(m-1)^2}\big[\Phi^2(x\partial_x + y\partial_y + z\partial_z)^2 - 2\Phi\Pi\nabla(x\partial_x + y\partial_y + z\partial_z) + \Pi^2\nabla^2\big],$$

and operating herewith on U, we find

$$\partial^2 U \ = \ \tfrac{1}{(m-1)^2}\big[m(m-1)\Phi^2 U - 2(m-1)\Phi\Pi\nabla U + \Pi^2\nabla^2 U\big];$$

or since $\nabla U = \frac{1}{m-1}H$, this is

$$\partial^2 U \;=\; \tfrac{1}{(m-1)^2}\big[m(m-1)\Phi^2 U - 2\Phi\Pi H + \Pi^2\nabla^2 U\big].$$

64. We have $\partial_4 U = 0$, and thence

$$(\partial_1^4 + \partial_4)U \;=\; 0,$$

that is

$$\partial_4 U \;=\; -\partial_1^4 U - \partial_1\partial_3 U;$$

or substituting the values of $\partial_1\partial_3 U$ and $\partial_1^4 U$, we find the value of $\partial_4 U$ as given in the Table. And then from the equation

$$(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2 + \partial_4)U \;=\; 0,$$

we find the value of $\partial_4 U$, and the proof of the expressions in the Table is thus completed.

65. Proof of equation $\nabla.\partial = 0$.

We have

$$\begin{aligned}\nabla.\partial \;&=\; \nabla.\big((B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z\big)\\ &= \nabla.\big(\lambda(\mu\partial_z - \nu\partial_y) + \mu(\nu\partial_x - \lambda\partial_z) + \nu(\lambda\partial_y - \mu\partial_x)\big)\end{aligned}$$

and then

$$\begin{aligned}\nabla A \;&=\; (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid a,h,g) \;=\; H\lambda,\\ \nabla B \;&=\; (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid h,b,f) \;=\; H\mu,\\ \nabla C \;&=\; (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid g,f,c) \;=\; H\nu;\end{aligned}$$

or substituting these values, we have the equation in question.

66. Proof of the expression for ∂_2.

We have, operating on the two sides respectively with $\partial_1 = \partial$, we have

$$\begin{aligned}\partial_2 \;&=\; \tfrac{1}{m-1}\big\{\partial(\Phi(x\partial_x + y\partial_y + z\partial_z)) + \Phi\partial.(x\partial_x + y\partial_y + z\partial_z)\big\}\\ &= \tfrac{1}{m-1}\big\{\partial\Phi\cdot\nabla + \Phi\partial\nabla\big\}.\end{aligned}$$

67. Proof of expression for $\partial_2 H$.

Operating with ∂_2 upon H, we have at once

$$\partial_2 H \;=\; -\tfrac{m'-1}{m-1}H\partial^2 + \tfrac{1}{m-1}\Phi\partial H + \tfrac{1}{m-1}(\partial.\nabla)H.$$

The remainder of the present Appendix is preliminary, or relating to the investigation of the expressions for $\partial_2\partial_3 U$ and $\partial_3\partial_2 U$, used ante, No. 31.

68. Proof of equation $\nabla^2.\partial U \;=\; \Psi\,\partial H - H\,\partial\Phi$.

We have identically

$$(\mathfrak{A},\ldots \mid \lambda,\mu,\nu)^2(\mathfrak{A},\ldots \mid \partial_x,\partial_y,\partial_z)^2 - (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid \partial_x,\partial_y,\partial_z)^2$$

that is

$$= (ax - \&c.)(\mathfrak{A},\ldots \mid \mu\partial_z - \nu\partial_y,\ \nu\partial_x - \lambda\partial_z,\ \lambda\partial_y - \mu\partial_x)^2;$$

$$\Phi\Omega - \nabla^2 \;=\; H\Gamma;$$

and then multiplying by ∂ and with the result operating on U, we find

$$\Phi\Omega U - \nabla^2.\partial U \;=\; H\partial\Gamma U.$$

Now

$$\Omega U \;=\; (\mathfrak{A},\ldots \mid \partial_x,\partial_y,\partial_z)^2 U \;=\; (\mathfrak{A},\ldots \mid a,b,c,2f,2g,2h),$$

and thence

$$\Omega.\partial U \;=\; (\mathfrak{A},\ldots \mid \partial a,\partial b,\partial c,2\partial f,2\partial g,2\partial h);$$

and observing that

$$H \;=\; \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}, \qquad \partial H \;=\; \begin{vmatrix} \partial a & h & g \\ \partial h & b & f \\ \partial g & f & c \end{vmatrix} + \begin{vmatrix} a & \partial h & g \\ h & \partial b & f \\ g & \partial f & c \end{vmatrix} + \begin{vmatrix} a & h & \partial g \\ h & b & \partial f \\ g & f & \partial c \end{vmatrix}$$

$$= (\mathfrak{A},\mathfrak{H},\mathfrak{G} \mid \partial a,\partial h,\partial g) + (\mathfrak{H},\mathfrak{B},\mathfrak{F} \mid \partial h,\partial b,\partial f) + (\mathfrak{G},\mathfrak{F},\mathfrak{C} \mid \partial g,\partial f,\partial c),$$
$$= (\mathfrak{A},\ldots \mid \partial a,\partial b,\partial c,2\partial f,2\partial g,2\partial h),$$

we see that

$$\Omega\,\partial U \;=\; \partial H.$$

Moreover

$$(\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid \partial_x,\partial_y,\partial_z)^2.\partial U$$
$$= a(b\rho^2 + c\mu^2 - 2f\mu\nu) + b(c\lambda^2 + a\nu^2 - 2g\nu\lambda) + c(a\mu^2 + b\lambda^2 - 2h\lambda\mu)$$
$$+\, 2f(-f\lambda^2 + g\nu^2 + h\lambda\nu - c\mu\nu) + 2g(\ldots) + 2h(\ldots)$$
$$= \nabla^2 H = (\mathfrak{A},\ldots \mid \lambda,\mu,\nu \mid \partial_x H,\partial_y H,\partial_z H)$$
$$= (\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H} \mid \lambda,\mu,\nu)\cdot\partial H.$$

Hence the equation

$$\Phi U\partial U - \nabla^2\partial U \;=\; H\Gamma\partial U$$

becomes

$$\nabla^2\partial U \;=\; \Psi\,\partial H - H\,\partial\Phi.$$

69. Proof of equation $\partial\partial_1^2 U \;=\; \frac{1}{(m-1)^2}(\Psi\partial H - H\partial\Phi)$.

We have

$$\partial^2 \;=\; \tfrac{1}{(m-1)^2}\big(\Phi^2(x\partial_x + y\partial_y + z\partial_z)^2 - 2\Phi\Pi(x\partial_x + y\partial_y + z\partial_z)\nabla + \Pi^2\nabla^2\big),$$

and thence multiplying by $\partial_1 = \partial$ and with the result operating upon U, we find

$$((m-1)^2\,\partial_1^2 U) \;=\; \tfrac{1}{(m-1)^4}\big(\Phi^2 HU - 2\Phi\Pi H + \Pi^2\nabla^2 U\big).$$

But $\partial U = 0$, and thence also $\nabla(\partial U) = 0$, that is $(m-1)(\nabla.\partial)U + \nabla\partial U = 0$; moreover $\nabla.\partial = 0$, and therefore $(\nabla.\partial)U = 0$, whence also $\nabla\partial U = 0$. Therefore

$$\partial\partial_1^2 U \;=\; \tfrac{1}{(m-1)^2}\,\Pi^2\nabla^2.\partial U;$$

or substituting for $\nabla^2.\partial U$ its value $= \Psi\partial H - H\partial\Phi$, we have the required expression for $\partial\partial_1^2 U$.

70. Proof of equation [for $\partial_1\partial_2 U$].

We have

$$\partial_2 \;=\; \tfrac{1}{m-1}\Phi(x\partial_x + y\partial_y + z\partial_z) + \tfrac{1}{m-1}\nabla,$$

and thence multiplying by $\partial_1 = \partial$, and operating on U, [reduction by similar manipulations as above, using $\nabla\partial U = 0$ and the explicit expressions in Article No. 31].

To reduce $(\partial.\nabla)\partial^2 U$, we have

$$\partial(\nabla\partial^2 U) \;=\; \nabla\partial^3 U + (\partial.\nabla)\partial^2 U,$$

$$= \nabla\partial^3 U + \big((\partial.\nabla)\partial^2 + \nabla(\partial.\partial^2)\big)U,$$
$$= \nabla\partial^3 U + (\partial.\nabla)\partial^2 U + 2\nabla\partial\partial_2 U,$$

and since

$$\partial \;=\; -\tfrac{\Phi}{m-1}(x\partial_x + y\partial_y + z\partial_z) + \tfrac{1}{m-1}\Pi\nabla,$$

multiplying by $\nabla\partial$, and with the result operating on U, we obtain

$$\nabla\partial.\partial U \;=\; \tfrac{m-1}{m-1}\,\nabla.\partial U\cdot(\ldots),$$

or since $\nabla\partial U = 0$, this is

$$(\partial.\nabla)\partial^2 U \;=\; \partial(\nabla\partial^2 U) - \nabla\partial^3 U - \tfrac{2}{m-1}\nabla^2\partial U.$$

Hence

$$\partial.\nabla\partial^2 U \;=\; \partial(\nabla\partial^2 U) - \nabla\partial^3 U - \tfrac{2}{m-1}\nabla^2\partial U.$$

Substituting this value of $(\partial.\nabla)\partial^2 U$, we find

$$\partial_1\partial_2 U \;=\; -\tfrac{1}{m-1}\Phi\partial^2 U - \tfrac{1}{m-1}\Phi\partial^2 U + \big(-2\nabla^2\partial U\big),$$

the three lines whereof are to be separately further reduced.

71. For the first line we have

$$\partial^2 \;=\; \tfrac{1}{(m-1)^2}\big(\Phi^2 - \Phi\Pi\nabla + (m-1)\Pi^2\Omega\big),$$

and hence

$$\text{first line of } \partial_1\partial_2 U \;=\; \tfrac{1}{(m-1)^3}\big((m-2)m\Phi^2 + \Phi\partial H\big).$$

72. For the second line, we have

$$\nabla(\partial^2 U) \;=\; \nabla\partial^2 U + 2(\nabla.\partial)\partial U$$

or writing

$$= \nabla\partial^2 U, \quad \text{since } \nabla.\partial = 0, \text{ and therefore } (\nabla.\partial)\partial U = 0;$$
$$\nabla\partial^2 U \;=\; \nabla(\partial^2 U) \;=\; \nabla\big(\tfrac{\Phi}{m-1}\Phi - \tfrac{1}{m-1}\Phi\nabla\big).$$

By computation,

$$\nabla\partial^2 U \;=\; \tfrac{m'}{(m-1)^2}(\Phi\nabla\Phi + \Phi\nabla\Psi) - \tfrac{1}{(m-1)^2}(\Psi\nabla H + 2\Phi\nabla\Psi);$$
$$U = 0,\ \nabla U = \tfrac{1}{m-1}H,\ \nabla\partial U = \Phi,$$

this is

$$\nabla\partial^2 U \;=\; \tfrac{1}{(m-1)^2}\big(m\Phi\nabla\Phi - \Psi\nabla H\big),$$

whence also

$$\nabla\partial^3 U \;=\; \tfrac{1}{(m-1)^2}(m\Phi\nabla\Phi - 2\Phi\partial H) - \tfrac{2}{(m-1)^2}\Pi\nabla\nabla\partial^2 U,$$
$$\nabla\partial^3 U \;=\; \nabla(\partial^3 U) \;=\; \tfrac{1}{(m-1)^3}\big[(\text{combinations of } \Phi, \Psi, H, \partial H, \nabla H, \text{\&c.})\big].$$

Similarly

$$\nabla\partial^4 U \;=\; \nabla(\partial^4 U) \;=\; \tfrac{1}{(m-1)^4}\big[\ldots\big],$$
$$= \tfrac{m'}{(m-1)^3}(\Phi\nabla H + \Phi\nabla\Psi) - \tfrac{1}{(m-1)^4}(\Phi^2\nabla H + 2\Phi\Psi\partial H + 2\Phi^2\nabla\Psi);$$

or putting

$$U = 0,\ \nabla U = \tfrac{1}{m-1}H,\ \nabla\partial U = \Phi,$$

and observing also that $\nabla(\partial H) = \nabla\partial H + (\nabla.\partial)H$ is equal to $\nabla\partial H$, that is to $\partial\nabla H$, we obtain

$$\nabla\partial^2 U \;=\; -\tfrac{1}{(m-1)^2}(m\Phi\partial\Phi - 2\Phi\partial H) - \tfrac{1}{(m-1)^2}\partial\nabla H;$$

and then from the above value of $\partial(\nabla\partial^2 U)$, we find

$$\partial(\nabla\partial^2 U) - \nabla\partial^3 U \;=\; \tfrac{1}{(m-1)^2}\big(-2\Psi\partial\Phi + m\Phi\partial H\big) + \tfrac{1}{(m-1)^3}\big(-(\partial.\nabla)H\big);$$

or observing that the term multiplied by $\frac{1}{(m-1)^3}$ is $= -(\partial.\nabla)H$, we find

$$\text{second line of } \partial_1\partial_2 U \;=\; \tfrac{1}{(m-1)^3}\big(-2\Psi\partial\Phi + m\Phi\partial H\big) + \tfrac{1}{(m-1)^4}\big(-(\partial.\nabla)H\big).$$

73. For the third line, substituting for $\nabla^2\partial U$ its value $= \Psi\partial H - H\partial\Phi$, we have

$$\text{third line of } \partial_1\partial_2 U \;=\; -\tfrac{2}{(m-1)^2}(\Psi\partial H - H\partial\Phi).$$

74. Hence, uniting the three lines, we have

$$\tfrac{1}{(m-1)^3}\big((m-2)m\Phi^2 + \Phi\partial H\big) \;+\; \tfrac{1}{(m-1)^3}\big(-2\Psi\partial\Phi + m\Phi\partial H\big) \;+\; \tfrac{1}{(m-1)^4}\big(-(\partial.\nabla)H\big)$$
$$+\; \tfrac{1}{(m-1)^2}\big((2m-2)H\partial\Phi + (-2m+2)\Psi\partial H\big),$$

and, reducing, we have the above-mentioned value of $\partial_1\partial_2 U$.

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342. On the Conics which Pass through Three Given Points and Touch a Given Line.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vi. (1864), pp. 24–30.]

Consider the system of conics which pass through three given points and touch a given line; if among these we select the conics which touch an assumed line, it is easy to show analytically that there are four such conics, all real or else all imaginary; viz. the three points form a triangle, and if the two lines cut the three sides produced or cut the same two sides and the third side produced, then the conics are all real; but in every other case they are all imaginary. The latter part of the theorem may also be seen geometrically; in fact, if a triangle is inscribed in a conic, say first in an ellipse, or in a parabola, or in one branch of a hyperbola, then all the tangents of the conic (and therefore any two tangents whatever) cut the three sides produced, but if the triangle is inscribed in the two branches of a hyperbola (that is, two vertices on one branch and the remaining vertex on the other branch), then all the tangents of the conic (and therefore any two tangents whatever) cut the same two sides and the third side produced: and thus the only real conics are those which cut the three sides produced, or else the same two sides and the third side produced.

The analytical proof referred to is as follows: taking ($x = 0$, $y = 0$, $z = 0$) for the equations of the sides of the triangle, the equation of a conic through the three points is

$$\tfrac{f}{x} + \tfrac{g}{y} + \tfrac{h}{z} \;=\; 0,$$

or, what is the same thing,

$$2fyz + 2gzx + 2hxy \;=\; 0,$$

that is

$$(0,\, 0,\, 0,\, f,\, g,\, h \mid x,\, y,\, z)^2 \;=\; 0.$$

The inverse coefficients are

$$(-f^2,\, -g^2,\, -h^2,\, gh,\, hf,\, fg),$$

and hence the condition in order that the conic may touch the line $\alpha x + \beta y + \gamma z = 0$ is

$$(f^2,\, g^2,\, h^2,\, -gh,\, -hf,\, -fg \mid \alpha,\, \beta,\, \gamma)^2 \;=\; 0,$$

or, what is the same thing,

$$\sqrt{\alpha f} + \sqrt{\beta g} + \sqrt{\gamma h} \;=\; 0.$$

Similarly the condition in order that the conic may touch the line $lx + my + nz = 0$ is

$$\sqrt{lf} + \sqrt{mg} + \sqrt{nh} \;=\; 0.$$

Hence if the conic touch the two lines, we have

$$\sqrt{lf} : \sqrt{mg} : \sqrt{nh} \;=\; \sqrt{\beta n} - \sqrt{\gamma m} : \sqrt{\gamma l} - \sqrt{\alpha n} : \sqrt{\alpha m} - \sqrt{\beta l},$$

or, what is the same thing,

$$f : g : h \;=\; \beta n + \gamma m - 2\sqrt{\beta\gamma mn} : \gamma l + \alpha n - 2\sqrt{\gamma\alpha nl} : \alpha m + \beta l - 2\sqrt{\alpha\beta lm},$$

which, since the radicals must be so taken that the product may be $= \alpha\beta\gamma lmn$, gives in all four conics: and these will be all real if the signs of (l, m, n) are the same with, or opposite to those of (α, β, γ) respectively; which proves the theorem.

In particular since infinity is a line meeting the three sides produced; if the given line meet the three sides produced, the system will contain four real parabolas; but, if the given line meets two sides and a side produced, there is not any real parabola. In the latter case, as is obvious geometrically, the conics of the system are all hyperbolas.

Any side of the triangle, and the line joining the opposite vertex with the point of intersection of the side and given line, form a pair of lines passing through the three points and meeting on the given line; such pair of lines is a conic of the system; and we have thus three pairs of lines, each pair a conic of the system.

We may by what precedes form some idea of the nature of the system of conics which pass through the three given points and touch the given line. In fact writing at the point of contact the letters H, P, E, L according as the conic is a hyperbola, parabola, ellipse, or pair of lines, then if the given line cut the three sides produced, we have as in fig. 1.

[Fig. 1: schematic showing labels H, H, L along the given line, indicating sequence of conic types.]

Whereas, if the given line cuts two sides and a side produced, we have more simply as in fig. 2.

[Fig. 2: simpler diagram with labels along the given line.]

But to gain a more precise knowledge, it is proper to consider the curve which is the locus of the centres of the conics of the system.

Such locus which, as will presently be seen, is a curve of the fourth order, must, it is clear, pass through the points of intersection (L_1, L_2, L_3 in figs. 1 and 2 respectively and p, q, r in fig. 3

presently referred to) of the sides with the given line; and it is not difficult to show geometrically that it touches, at these points, the sides of the triangle. It may be shown also that the curve has three nodes (double points), viz. the middle point of each side of the triangle is a node of the curve. In fact if upon any side as base we apply an equal and opposite triangle so as to form with the given triangle a parallelogram, then any conic through the four vertices of the parallelogram will have for its centre the central point of the parallelogram; that is, the middle point of the side in question. But we may through the four vertices describe two conics, each of them touching the given line; that is the middle point of the side is the centre of two different conics of the system, and it is therefore a node upon the curve of centres. And moreover the node will be a crunode or an acnode (i.e. a double point with two real branches, or else a conjugate or isolated point) according as the conics are real or imaginary: and it is easy to see that if the given line does not cut the parallelogram, or if it cuts two opposite sides, the conics will be both real; but if it cuts two adjacent sides the conics will be both imaginary; that is, in the former case we have a crunode, and in the latter an acnode. Through each node may be drawn two tangents to the curve; and it is a known property of curves of the fourth order that the six points of contact lie on a conic; one of the tangents through the node is however the side whereon the node lies, and the points of contact of the three sides lie on a line, viz. the given line: hence the last mentioned conic is composed of the given line, and another line; that is, the three points of contact of the other tangents through the three nodes lie on this other line.

It is proper to add that the points at infinity of the curve of centres are the centres of the four parabolas; that is, there will be four infinite branches, if the parabolas are real, viz. if the given line cuts the three sides produced; but no infinite branch if the parabolas are imaginary, viz. if the given line cut two sides and a side produced.

The triangle and the three triangles applied to the three sides form together a triangle similar to the original triangle but of double the linear magnitude, and the form of the curve of centres depends as has been shown on the position of the given line in regard to the triangle and the double triangle. The cases to be considered are tolerably numerous, but it is easy from the foregoing considerations, to determine in any particular case what is the form of the curve of centres; for facility of selection I select a form without infinite branches, see fig. 3, in which the given line cuts the two sides CA, CB, and the third side AB produced; it is moreover to be observed

[Fig. 3: triangle ABC with the given line cutting CA, CB and producing across AB; middle points Q, P, R marked; the curve of centres is drawn through p, q, r with two crunodes at Q, P and an acnode at R.]

that as the figure is drawn the given line cuts the two sides CA, CB below their middle points Q and P respectively. By what precedes it appears that the middle points Q, P of these two sides CA, CB are each of them crunodes, but that the middle point R of the remaining side AB is an acnode. And this being so the general form of the curve is at once perceived to be that shown by fig. 3.

It is very interesting to trace the corresponding positions of the point of contact on the given line, and of the centre on the curve of centres. When the point of contact is at ∞, the centre is at I; as the point of contact moves from ∞ to q, the centre moves from I to q, and at q the two coincide; as the point of contact moves from q to a point Q_2, the centre moves from q to Q (along the branch Q_2); as the point of contact moves from Q_2 to a point P_1, the centre moves from Q to P (along the branch $Q_2\,1\,P$); as the point of contact moves from P_1 to p, the centre moves from P to p (along the branch P_1) and at p the point of contact and the centre again coincide; as the point of contact moves from p to r, the centre moves from p to r and at r they again coincide; as the point of contact moves from r to a point P_2 the centre moves from r to P (along the branch $2P$); as the point of contact moves from P_2 to a point Q_1, the centre moves from P to Q (along the branch $P_2\,1\,Q$) and finally as the point of contact moves from Q_1 to ∞, the centre moves from Q (along the branch Q_1) to I, thus completing the circuit.

The equation of the curve of centres was given in the late Mr Hearn's "*Researches on Curves of the Second Order*, &c. London, 1846," viz. if $x = 0,\ y = 0,\ z = 0$ be the equations of the sides of the triangle formed by the given points; $x + y + z = 0$ the equation of the line infinity, and $\alpha x + \beta y + \gamma z = 0$ the equation of the given line, then the equation of the curve of centres is

$$\sqrt{\alpha x(-x+y+z)} + \sqrt{\beta y(x-y+z)} + \sqrt{\gamma z(x+y-z)} \;=\; 0,$$

or more generally if $x + y + z = 0$ be the equation of an assumed line, then this equation is that of the locus of the pole of the assumed line in regard to the conics passing through the given points and touching the given line, see my paper "Note on a Family of Curves of the Fourth Order," *Cambridge and Dublin Mathematical Journal*, t. V. (1850), pp. 148–152, [85], where I have noticed the above mentioned property, that the conic through the points of contact of the tangents through the nodes breaks up into a pair of lines. It is I think worth while to show how the equation is obtained. The equation of a conic through the given points and touching the given line is

$$(0,\,0,\,0,\,f,\,g,\,h \mid x,\,y,\,z)^2 \;=\; 0$$

with the condition $\sqrt{\alpha f} + \sqrt{\beta g} + \sqrt{\gamma h} = 0$, and this being so, the coordinates of the pole in relation thereto, of the assumed line $x + y + z = 0$, are

$$x : y : z \;=\; (-f+g+h)f : (f-g+h)g : (f+g-h)h.$$

We have thence

$$-x+y+z \ \text{ proportional to } \ -(-f+g+h)f + (f-g+h)g + (f+g-h)h,$$

that is, to $f^2 - (g-h)^2$, which is $= (f-g+h)(f+g-h)$, and combining with this the equation

$$\alpha x \ \text{ proportional to } \ (-f+g+h)f,$$

we obtain

$$\alpha x(-x+y+z) \ \text{ proportional to } \ \alpha f,$$

that is

$$\alpha x(-x+y+z) : \beta y(x-y+z) : \gamma z(x+y-z) \;=\; \alpha f : \beta g : \gamma h,$$

so that from the equation $\sqrt{\alpha f} + \sqrt{\beta g} + \sqrt{\gamma h} = 0$, we have at once the foregoing equation

$$\sqrt{\alpha x(-x+y+z)} + \sqrt{\beta y(x-y+z)} + \sqrt{\gamma z(x+y-z)} \;=\; 0.$$

The rationalised form is

$$(1,\,1,\,1,\,-1,\,-1,\,-1 \mid \alpha x(-x+y+z),\ \beta y(x-y+z),\ \gamma z(x+y-z))^2 \;=\; 0,$$

which shows what has been all along assumed, that the curve is of the fourth order. This equation may be transformed into

$$\begin{gathered}
x^2(\alpha^2x^2 + \beta^2y^2 + \gamma^2z^2 - 2\beta\gamma yz + 2\gamma\alpha zx + 2\alpha\beta xy) \\
+\, y^2(\alpha^2x^2 + \beta^2y^2 + \gamma^2z^2 + 2\beta\gamma yz - 2\gamma\alpha zx + 2\alpha\beta xy) \\
+\, z^2(\alpha^2x^2 + \beta^2y^2 + \gamma^2z^2 + 2\beta\gamma yz + 2\gamma\alpha zx - 2\alpha\beta xy) \\
-\, 2yz(\alpha x + \beta y + \gamma z)(-\alpha x + \beta y + \gamma z) \\
-\, 2zx(\alpha x + \beta y + \gamma z)(\alpha x - \beta y + \gamma z)
\end{gathered}$$

$$- 2xy(\alpha x + \beta y + \gamma z)(\alpha x + \beta y - \gamma z) \;=\; 0;$$

if with this equation we combine the equation $\alpha x + \beta y + \gamma z = 0$, we find at the points of intersection with the given line

$$x^2 \cdot 2\beta\gamma yz + y^2 \cdot 2\gamma\alpha zx + z^2 \cdot 2\alpha\beta xy \;=\; 0,$$

that is

$$xyz(\beta\gamma x + \gamma\alpha y + \alpha\beta z) \;=\; 0,$$

so that the points in question are the intersections of the given line $\alpha x + \beta y + \gamma z = 0$, with the lines $x = 0$, $y = 0$, $z = 0$, $\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 0$. The point $\alpha x + \beta y + \gamma z = 0$, $\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 0$ corresponds to the conic which touches the given line at its intersection with the assumed line $x + y + z = 0$, the pole in relation to this conic is obviously a point on the given line. The point in question, if $x + y + z = 0$ denote the line infinity, is the point I of fig. 3.

It may be proper to mention a far less symmetrical form of the equation of the conic, but which has the advantage of putting in evidence the point of contact; viz. the equation is expressed in terms of the parameter α denoting the distance of the point of contact from a given point in the base line, and which is therefore very convenient for tracing the changes of form of the conic. Assuming as before that the base line cuts the sides produced, then (see fig. 4) if of the three points 1 denote

[Fig. 4: schematic with axes x and the base line $y = 0$, points labelled 1 farthest, 2 closest to base line.]

that which is furthest from, and 2 that which is nearest to the base line, and if the base line be taken as the axis of x, and 23 as the axis of y; the equation of the base line is $y = 0$, and the equations of the sides 23, 31, 12 are $x = 0$, $\frac{x}{a} + \frac{y}{b} = 1$, $\frac{x}{a'} + \frac{y}{b'} = 1$, where a, b, a', b', $a' - a$, $b' - b$, $a'b - ab'$ are all positive, so that, by choosing the axes as above, we avoid the consideration of the several cases corresponding to different signs of these quantities. And this being so, if $x = \alpha$ is the coordinate of the point of contact, the equation of the conic is

$$(A,\, B,\, C,\, F,\, G,\, H \mid x,\, y,\, 1)^2 \;=\; 0,$$

where

$$A \;=\; 2bb'(a' - a),$$
$$B \;=\; 2\alpha^2(\alpha' - c),$$

and these give

$$C \;=\; -2\alpha aa'b(b' - b)\alpha,$$
$$F \;=\; -2\alpha bb'(a' - a),$$
$$G \;=\; -2abb'(a' - a),$$
$$H \;=\; \big\{(\alpha^2 + aa')(b' - b) - 2\alpha(ab' - a'b)\big\},$$
$$AB - H^2 \;=\; -\big[\big((a-\alpha)\sqrt{b'} + (a-a')\sqrt{b}\big)^2 - (a'-a)(a'b-ab')\big] \times \big[\big((a-\alpha)\sqrt{b'} - (a-a')\sqrt{b}\big)^2 - (a'-a)(a'b-ab')\big],$$
$$BC - F^2 \;=\; -a^4(a' - a)^2(b' - b)^2,$$
$$CA - G^2 \;=\; 0,$$
$$GH - AF \;=\; -2bb'(a' - a)(b' - b)\alpha(a - \alpha)(\alpha - a'),$$
$$HF - BG \;=\; -(a' - a)(b' - b)\alpha^2\big\{(b' + b)(\alpha^2 + aa') - 2\alpha(ab' + a'b)\big\},$$
$$FG - CH \;=\; -2bb'(a' - a)(b' - b)\alpha^2(a - \alpha)(\alpha - a').$$

The condition that the conic may be a parabola is $AB - H^2 = 0$, which gives, as it should do, four real values of α.

2, Stone Buildings, W.C.

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343. On the Cusp of the Second Kind or Nodecusp.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vi. (1864), pp. 74–75.]

The so-called cusp of the second kind or ramphoid cusp, is not an ordinary singularity of plane curves, but it is a singularity of a higher order. It is however particularly considered in Plücker's *Theorie der Analytischen Curven*, 1839; and it is there, not only in the analytical discussion of the singularities of plane curves, but in the author's theory of the generation of a curve, considered as described and enveloped by a point moving along a line which at the same time rotates round the point; when the motion along the line vanishes, we have a cusp; when the motion round the point vanishes, we have an inflexion; when the two motions vanish together, we have a cusp of the second kind, which thus presents itself as a singularity uniting the characters of a cusp or stationary point, and an inflexion or stationary tangent: (I remark in passing that in this explanation it is not clear what is the independent variable wherewith the motions are compared). But there is another point of view from which the singularity in question may be considered, viz. it may be regarded as a singularity arising from the union and amalgamation of a cusp, and a double point or node; in fact, in the figure, which represents a curve having a cusp and also a node, we have only to imagine the node approaching nearer and nearer to and ultimately coinciding with the cusp, and it will be at once seen that the point will become a cusp of the second kind; or as it might properly, with reference to this generation of it, be termed, a "nodecusp." It is to be noticed that in the point-theory of curves, there is between the cusp and the nodecusp the intermediate singularity of the tacnode, which arises from the union and amalgamation of two nodes, and possesses the character of a cusp.

I return to the nodecusp; taking the point in question as the origin, and the tangent for the axis of x, the equation will be a specialised form of the equation

$$\tfrac{1}{2}y^2 + \tfrac{1}{6}(a,\, b,\, c,\, d \mid x,\, y)^3 + \tfrac{1}{24}(a',\, b',\, c',\, d',\, e' \mid x,\, y)^4 + \&c. \;=\; 0,$$

which belongs to the case of a cusp, viz. (see Plücker, p. 165) the conditions satisfied by the special form are $a = 0,\ a' = 3b^2$, or the equation is

$$\tfrac{1}{2}(y + \tfrac{1}{2}bx^2)^2 + \tfrac{1}{6}y^2(3cx + dy) + \tfrac{1}{24}y(4b'x^3 + 6c'x^2y + 4d'xy^2 + e'y^3) + \&c. \;=\; 0,$$

which is most easily verified, by observing that (this being so) the expansion of y in terms of x will be of the form

$$y \;=\; -\tfrac{1}{2}bx^2 + Ax^3 + \&c.$$

It is now to be shown how the foregoing conditions $a = 0,\ a' = 3b^2$, are obtained by assuming that the curve has, besides the cusp, a node which ultimately coincides with the cusp. Let $(\alpha,\, \beta)$ be the coordinates of the node; we must have

$$\tfrac{1}{2}\beta^2 + \tfrac{1}{6}(a,\, b,\, c,\, d \mid \alpha,\, \beta)^3 + \tfrac{1}{24}(a',\, b',\, c',\, d',\, e' \mid \alpha,\, \beta)^4 + \&c. \;=\; 0,$$

$$\tfrac{1}{2}(a,\, b,\, c \mid \alpha,\, \beta)^2 + \tfrac{1}{6}(a',\, b',\, c',\, d' \mid \alpha,\, \beta)^3 + \&c. \;=\; 0,$$

$$\beta + \tfrac{1}{2}(b,\, c,\, d \mid \alpha,\, \beta)^2 + \tfrac{1}{6}(b',\, c',\, d',\, e' \mid \alpha,\, \beta)^3 + \&c. \;=\; 0.$$

Assume $\beta = m\alpha^2$, and then let α vanish; the equations become in the first instance

$$\tfrac{1}{2}\alpha\alpha^2 + (\tfrac{1}{6}m^2 + \tfrac{1}{2}bm + \tfrac{1}{6}a)\alpha^3 + \&c. \;=\; 0,$$

$$\tfrac{1}{2}a\alpha^2 + \&c. \;=\; 0,$$

$$(m + \tfrac{1}{2}b)\alpha^2 + \&c. \;=\; 0,$$

the second and third equation give $a = 0,\ m + \frac{1}{2}b = 0$, and the first equation then gives

$$\tfrac{1}{2}m^2 + \tfrac{1}{2}bm + \tfrac{1}{6}a' \;=\; 0;$$

or substituting for m its value $= -\frac{1}{2}b$, this is $a' = 3b^2$, or the required conditions are $a = 0,\ a' = 3b^2$, ut sup. The single condition $a = 0$ corresponds to the case of the cusp at a node.

2, Stone Buildings, W.C., September 17, 1862.

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344. On Certain Developable Surfaces.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vi. (1864), pp. 108–126.[2]]

If $U = 0$ be the equation of a developable surface, or say a developable, then the Hessian HU vanishes, not identically, but only by virtue of the equation $U = 0$ of the surface; that is, HU contains U as a factor, or we may write $HU = U \cdot PU$; the function PU, which for the developable replaces as it were the Hessian HU, is termed the *Prohessian*; and (since if r be the order of U the order of HU is $4r - 8$) we have $3r - 8$ for the order of the Prohessian. If $r = 4$, the order of the Prohessian is also 4, and in fact, as is known, the Prohessian is in this case $= U$. The Prohessian is considered, but not in much detail, in Dr Salmon's *Geometry of Three Dimensions*, (1862), pp. 338 and 426 [Ed. 4 (1882), p. 408]: the theorem given in the latter place is almost all that is known on the subject. I call to mind that the tangent plane along a generating line of the developable meets the developable in this line taken 2 times, and in a curve of the order $r - 2$; the line touches the curve at the point of contact, or say the *ineunt*, on the edge of regression, and besides meets it in $r - 4$ points. The ineunt taken 3 times, and the $r - 4$ points form a linear system of the order $r - 1$, and the Hessian of this system (considered as a curve of one dimension, or binary quantic) is a linear system of $2r - 6$ points; viz. it is composed of the ineunt taken 4 times, and of $2r - 10$ other points. This being so, the theorem is that the generating line meets the Prohessian in the ineunt taken 6 times, in the $r - 4$ points, and in the $2r - 10$ points $(6 + r - 4 + 2r - 10 = 3r - 8)$; it is assumed that r is at least $= 5$.

The developables which first present themselves are those which are the envelopes of a plane

$$(a,\, b,\, \ldots \mid t,\, 1)^n \;=\; 0,$$

where t is an arbitrary parameter, and the coefficients $(a,\, b,\, \ldots)$ are linear functions of the coordinates; the equation of the developable is

$$\text{Disct.}(a,\, b,\, \ldots \mid t,\, 1)^n \;=\; 0,$$

[2]Presented to the Royal Society and read 27 Nov., 1862, but withdrawn by permission of the Council.

the discriminant being taken in regard to the parameter t. Such developable is in general of the order $2n-2$, but if the second coefficient b is $=0$, or, more generally, if it is a mere numerical multiple of a, then a will divide out from the equation, and we have a developable of the order $2n-3$: the like property of course exists in regard to the last but one, and the last, of the coefficients of the function. We thus obtain developables of the orders 4, 5, and 6, sufficiently simple to allow of the actual calculation of their Prohessians, and the chief object of the present Memoir is to exhibit these Prohessians; but the Memoir contains some other researches in relation to the developables in question.

Quartic Developable, Nos. 1 to 6.

1. I consider first the developable of the fourth order

$$U \;=\; a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

derived from the cubic function $(a,\, b,\, c,\, d \mid t,\, 1)^3$, and which is in fact the general quartic developable.

2. Taking $(a,\, b,\, c,\, d)$ as coordinates and omitting common numerical factors, the first derived functions are

$$\begin{gathered} ad^2 - 3bcd + 2c^3, \\ -3acd + 6b^2d - 3bc^2, \\ -3abd + 6ac^2 - 3b^2c, \\ a^2d - 3abc + 2b^3, \end{gathered}$$

(quantities which, if $(X,Y,Z,W \mid t,1)^3$ denote the cubicovariant of $(a,b,c,d \mid t,1)^3$, are equal to $(-W,\, 3Z,\, -3Y,\, X)$ respectively). And the second derived functions are

$$\begin{array}{cccc} d^2 & -3cd & -3bd+6c^2 & 2ad-2bc \\ -3cd & 12bd-3c^2 & -3ad-6bc & -3ac+6b^2 \\ -3bd+6c^2 & -3ad-6bc & 12ac-3b^2 & -3ab \\ 2ad-3bc & -3ac+6b^2 & -3ab & a^2 \end{array}$$

3. Representing these by

$$\begin{pmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & P \end{pmatrix},$$

and expressing the determinant in the partially developed form

$$= (AM-LH)(FN-GM) + (AN-LG)(FM-BN) + (AP-L^2)(BC-F^2)$$

$$+\,(HN-GM)^2 + (HP-LM)(FG-CH) + (GP-LN)(HF-BG),$$

and proceeding to the calculation, we find a tabular expansion of seven component products $AM-LH$, $FN-GM$, $AN-LG$, $FM-BN$, $AP-L^2$, $BC-F^2$, etc., each producing several terms in the monomials $a^pb^qc^rd^s$ with explicit integer coefficients. *[See facsimile pp. 268–269 for the full numerical tableau.]*

4. Hence, forming the six parts and collecting, we find the Hessian (the first column of the developed tableau). This is in fact $=U^2$, and hence the Prohessian is

$$PU \;=\; U \;=\; a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

viz. for the quartic developable the Prohessian coincides with U, as expected.

5. To complete the theory it is proper to calculate the inverse coefficients

$$\begin{array}{c}\mathfrak{A},\ \mathfrak{H},\ \mathfrak{G},\ \mathfrak{L},\\ \mathfrak{H},\ \mathfrak{B},\ \mathfrak{F},\ \mathfrak{M},\\ \mathfrak{G},\ \mathfrak{F},\ \mathfrak{C},\ \mathfrak{N},\\ \mathfrak{L},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{P}.\end{array}$$

We have for example

$$\mathfrak{P} \ = \ ABC - AF^2 - BG^2 - CH^2 + 2FGH,$$

which is found to be

$$= \ 3(ad^2 - 3bcd + 2c^3)^2 - 4d^2(a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2),$$

that is $= 3W^2 - 4d^2U$; and calculating in like manner the other coefficients, the system is found to be

$$\begin{array}{cccc}
3X^2 - 4a^2U, & 3XY - 4abU, & 3XZ + (2ac - 6b^2)U, & 3XW + (5ad - 9bc)U,\\
3YX - 4abU, & 3Y^2 - 4acU, & 3YZ - (7ad + 3bc)U, & 3YW + (2bd - 6c^2)U,\\
3ZX + (2ac - 6b^2)U, & 3ZY - (7ad + 3bc)U, & 3Z^2 - 4bdU, & 3ZW - 4cdU,\\
3WX + (5ad - 9bc)U, & 3WY + (2bd - 6c^2)U, & 3WZ - 4cdU, & 3W^2 - 4d^2U
\end{array}$$

6. Let $(\lambda,\ \mu,\ \nu,\ \rho)$ be any arbitrary multipliers, and write

$$\begin{array}{rcl}(\mathfrak{X},\ \mathfrak{Y},\ \mathfrak{Z},\ \mathfrak{W}) & = & (\mathfrak{A},\ \mathfrak{H},\ \mathfrak{G},\ \mathfrak{L} \mid \lambda,\ \mu,\ \nu,\ \rho),\\ & & (\mathfrak{H},\ \mathfrak{B},\ \mathfrak{F},\ \mathfrak{M}),\\ & & (\mathfrak{G},\ \mathfrak{F},\ \mathfrak{C},\ \mathfrak{N}),\\ & & (\mathfrak{L},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{P}),\end{array}$$

then if $\theta = 3(\lambda X + \mu Y + \nu Z + \rho W)$, we have

$$(\mathfrak{X},\ \mathfrak{Y},\ \mathfrak{Z},\ \mathfrak{W}) \ = \ (\theta X + \alpha U,\ \theta Y + \beta U,\ \theta Z + \gamma U,\ \theta W + \delta U).$$

The functions $(X,\ Y,\ Z,\ W \mid t,\ 1)^3$ are the cubicovariant of $(a,\ b,\ c,\ d \mid t,\ 1)^3$, respectively, and if we also write for a moment the functions $u + \theta v$, $U = a^2d^2 - \&c. = \mathit{ff}(a, b, c, d)$, then

$$\mathit{ff}(a + \theta X,\ b + \theta Y,\ c + \theta Z,\ d + \theta W) \ = \ \text{Disct.}(u + \theta v) \ = \ (1 - \theta^2 U)\, U,$$

by a formula given in my "Fifth Memoir on Quantics," *Phil. Trans.*, t. CXLVIII. (1858), see p. 442 [156]; the function on the left-hand side thus contains U as a factor, and it at once follows that the function

$$U(a + \mathfrak{X},\ b + \mathfrak{Y},\ c + \mathfrak{Z},\ d + \mathfrak{W});$$

viz., the function obtained from U by writing therein $(a + \mathfrak{X},\ b + \mathfrak{Y},\ c + \mathfrak{Z},\ d + \mathfrak{W})$ in the place of $(a,\ b,\ c,\ d)$ respectively, contains U as a factor, and therefore vanishes if $U = 0$; that is $a + \mathfrak{X},\ b + \mathfrak{Y},\ c + \mathfrak{Z},\ d + \mathfrak{W}$, are the coordinates of a point on the surface $U = 0$; they are in fact the coordinates of a point on the generating line through $(a,\ b,\ c,\ d)$; this is a theorem which applies to any developable whatever, as appears by the following considerations.

Remarks on the General Theory of Developables, Nos. 7 to 9.

7. In general for any surface whatever, taking a point on the surface, the successive polars of this point (the last of them being the tangent plane) all touch at this point; and not only so, but the tangents to the two branches of the curve in which the surface itself (or any of its polars down to the quadric polar) is intersected by the ultimate polar or tangent plane, are respectively coincident. Suppose that for any point on the surface, the quadric polar becomes a cone: the vertex of this cone is not the point itself; hence the tangent plane at the point touches the cone along a generating line; that is the tangents to the curve of intersection with the surface, or with any of its polars, coincide with the generating line of the cone—and the curve of intersection of the tangent plane with the surface, or any of its polars, at the point of contact (instead of, as in general, a node) has a cusp. In particular the curve of intersection with the surface has at the point of contact a cusp. The condition that the quadric polar may be a cone is $HU = 0$, and when this differential equation is satisfied in virtue of the equation $U = 0$ (that is, when we have identically $HU = U \cdot PU$), the surface is a developable. Now all that is proved in the first instance by the equation $HU = 0$ is that every point of the surface has the above mentioned property; viz., that the tangent plane at the point cuts the surface in a curve having a cusp at the point in question.

8. What really happens in the case of a developable is more than this; viz. the curve of intersection is made up of the generating line taken twice, and of a curve of an order less by 2 than the order of the surface. Let $(x,\ y,\ z,\ w)$ be the coordinates of the point on the developable, $U = 0$ the equation of the developable, $(A,\ B,\ C,\ P,\ F,\ Q,\ H,\ L,\ M,\ N)$ the second derived functions of U, $(\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{F},\ \mathfrak{G},\ \mathfrak{H},\ \mathfrak{L},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{P})$ the inverse system, K the determinant formed with the second derived functions, so that we have $K = HU = 0$. The coordinates of the vertex of the cone are given by the equations

$$\begin{aligned}\dot{x} : \dot{y} : \dot{z} : \dot{w} \ &= \ \mathfrak{A} : \mathfrak{H} : \mathfrak{G} : \mathfrak{L}\\ &= \mathfrak{H} : \mathfrak{B} : \mathfrak{F} : \mathfrak{M}\\ &= \mathfrak{G} : \mathfrak{F} : \mathfrak{C} : \mathfrak{N}\\ &= \mathfrak{L} : \mathfrak{M} : \mathfrak{N} : \mathfrak{P};\end{aligned}$$

these several sets of ratios being equivalent to each other in virtue of the equation $K = 0$. Hence $(\lambda,\ \mu,\ \nu,\ \rho)$ being arbitrary multipliers, if we write

$$\begin{aligned}(\mathfrak{X},\ \mathfrak{Y},\ \mathfrak{Z},\ \mathfrak{W}) \ = \ (&\mathfrak{A},\ \mathfrak{H},\ \mathfrak{G},\ \mathfrak{L} \mid \lambda,\ \mu,\ \nu,\ \rho),\\ &\mathfrak{H},\ \mathfrak{B},\ \mathfrak{F},\ \mathfrak{M}),\\ &\mathfrak{G},\ \mathfrak{F},\ \mathfrak{C},\ \mathfrak{N}),\\ &\mathfrak{L},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{P}),\end{aligned}$$

the coordinates of the vertex of the cone will be as $\mathfrak{X} : \mathfrak{Y} : \mathfrak{Z} : \mathfrak{W}$, and hence observing that the absolute magnitudes of these quantities are arbitrary, $x + \mathfrak{X} : y + \mathfrak{Y} : z + \mathfrak{Z} : w + \mathfrak{W}$ will represent the coordinates of any point on the line joining the point $(x,\ y,\ z,\ w)$ with the vertex of the cone, that is, the generating line through the point $(x,\ y,\ z,\ w)$; which is the theorem in question, the coordinates being in the present investigation denoted by $(x,\ y,\ z,\ w)$ instead of the $(a,\ b,\ c,\ d)$ of the example.

9. Reverting to the developable $U = a^2d^2 - \&c. = 0$, the results previously obtained show that the coordinates of the vertex of the cone which is the quadric polar of the point $(a,\ b,\ c,\ d)$ are as $X : Y : Z : W$ (these quantities denoting as above the coefficients of the cubicovariant), and thence also that the coordinates of any point on the generating line will be as $a + \theta X : b + \theta Y : c + \theta Z : d + \theta W$, where θ is arbitrary.

Special Quintic Developable, Nos. 10 to 25.

10. We have, secondly, the developable of the fifth order

$$U \;=\; a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 \;=\; 0,$$

derived from the quartic function $(a,\, 2b,\, 3c,\, 0,\, -27e \mid t,\, 1)^4$, or, what is the same thing, $at^4 + 8bt^3 + 18ct^2 - 27e = 0$, where it will be observed that, as well in the quartic function as in the equation of the developable, the sum of the numerical coefficients is = zero; it was on this account that the foregoing form of the quartic function was selected in preference to the form $(a,\, b,\, c,\, 0,\, e \mid t,\, 1)^4$. The last mentioned form has for its discriminant

$$(ae + 3c^2)^3 - 27(ace - b^2e - c^3)^2 \;=\; e(a^3e^2 - 18a^2c^2e + 54ab^2ce + 81ac^4 - 27b^4e - 54b^2c^3),$$

and writing therein in place of $(a,\, b,\, c,\, e)$, the values $(a,\, 2b,\, 3c,\, -27e)$, the second factor divided by 729 gives the foregoing expression for U, belonging to the form

$$(a,\, 2b,\, 3c,\, 0,\, -27e \mid t,\, 1)^4.$$

11. Taking $(a,\, b,\, c,\, e)$ as coordinates, and omitting common numerical factors, the first derived functions of U are

$$3a^2e^2 + 12ac^2e - 24b^2ce + 9c^4,$$
$$-48abce + 64b^3e - 16bc^3,$$
$$12a^2ce - 24ab^2c + 36ac^3 - 24b^2c^2,$$
$$2a^3e + 6a^2c^2 - 24ab^2c + 16b^4,$$

and the second derived functions are

$$\begin{array}{cccc}
3(ae^2 + 2c^2e) & -24bce & 6(2ace - 2b^2e + 3c^3) & 3(a^2e + 2ac^2 - 4b^2c) \\
-24bce & 8(-3ace + 12b^2e - c^3) & 24(-abe - bc^2) & 8(-3abc + 4b^3) \\
6(2ace - 2b^2e + 3c^3) & 24(-abe - bc^2) & 6(a^2e + 9ac^2 - 4b^2c) & 6(a^2c - 2ab^2) \\
3(a^2e + 2ac^2 - 4b^2c) & 8(-3abc + 4b^3) & 6(a^2c - 2ab^2) & a^3
\end{array}$$

12. Representing these by

$$\begin{pmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & P \end{pmatrix},$$

and expressing the determinant in the partially developed form

$$P\,(ABC - AF^2 - BG^2 - CH^2 + 2FGH)$$
$$-\,L^2(BC - F^2) - M^2(CA - G^2) - N^2(AB - H^2)$$
$$-\,2MN(GH - AF) - 2NL(HF - BG) - 2LM(FG - CH),$$

then, proceeding to the calculation, we have a tabular expansion of the component products, each producing several terms in the monomials with explicit integer coefficients. *[See facsimile pp. 272–274 for the full numerical tableau, with detailed expansion of A^2, AH, AG, AL, H^2, HG, HL, M^2, JK, JL, $2MJ$, $2LM$, etc., and the resulting expression of HU as a tableau of monomials and integer coefficients.]*

13. It is now easy to form the seven component terms of the determinant, and thence the determinant itself; each of the component terms divides by 576, and omitting this factor, the sum of the seven terms divides by 2; the result is the explicit polynomial expansion of HU. *[See facsimile p. 273 for the full tableau.]*

14. This divides as it should do by

$$U \;=\; a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3,$$

and the quotient, which is the Prohessian, is found to be

$$PU \;=\; -24a^2e^3 + 3a^2c^2e + 10a^2b^2c^2e + 32ac^4 + 8ab^2ce - 24ac^3 + 32b^2e + 32b^4c^3, \;\; \text{etc.}$$

[Full polynomial coefficients appear in the facsimile expansion p. 274.]

15. But before discussing the Prohessian, I will further consider the developable itself. Regarding it as derived from the equation $(a,\, 2b,\, 3c,\, 0,\, -27e \mid t,\, 1)^4 = 0$, we have, observing that

$$eU \;=\; (27)^2\big[(ae-c^2)^3 + (-3ace + 4b^2e - c^3)^2\big] \;=\; 0,$$
$$-3ace + 4b^2e - c^3 \;=\; c(ae-c^2) - 4e(ac-b^2),$$

it appears that the equations of the cuspidal curve or edge of regression of the developable are $ae-c^2 = 0,\; ac-b^2 = 0$ (so that the cuspidal curve is a curve of the fourth order, the intersection of two quadric surfaces, or say a quadri-quadric curve). This is perhaps better seen by writing the equation of the developable in the form

$$U \;=\; a(ae-c^2)^2 - 8c(ae-c^2)(ac-b^2) + 16e(ac-b^2)^2 \;=\; 0,$$

or what is the same thing

$$U \;=\; (a,\, c,\, e \mid ae-c^2,\; 4ac-4b^2)^2 \;=\; 0,$$

where the discriminant of the quadric function is $= ae-c^2$, which vanishes for the curve $(ae-c^2 = 0,\; ac-b^2 = 0)$.

16. Another form of the equation is

$$U \;=\; a(ae+3c^2)^2 - 8b^2(3ace - 2b^2e + c^3) \;=\; 0,$$

which shows that the conic $ae + 3c^2 = 0,\; b = 0$, is a nodal curve on the developable.

And, again, another form is

$$U \;=\; c^3(9ac - 8b^2) + e(a^3e + 6a^2c^2 - 24ab^2c + 16b^4) \;=\; 0,$$

which shows that the conic $9ac - 8b^2 = 0,\; e = 0$, is a simple line on the developable.

17. In my paper "On the Developable Surfaces which arise from Two Surfaces of the Second Order," *Camb. and Dub. Math. Jour.*, t. v. (1850), pp. 46–57, [84], I considered first the developable having for its edge of regression the intersection of two quadric surfaces; in the general case the developable is of the order 8; but if the two surfaces have an ordinary contact it is of the order 6; and if they have a singular contact (as there explained) it is of the order 5. And in the last mentioned case, if the equations of the two quadric surfaces are taken to be $x^2 - 2wz = 0$, $y^2 - 2zx = 0$, then the equation of the developable was found to be

$$4z^2w^2 + 12z^2x^2 + 9zx^4 - 24zxy^2w - 4x^3y^2 + 8y^4w \;=\; 0,$$

which putting therein $z = a,\; y = 2b,\; x = 2c,\; w = 2e$, becomes

$$a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 \;=\; 0,$$

which is the before mentioned developable $U = 0$; the two equations $x^2 - 2wz = 0$, $y^2 - 2zx = 0$ become by the same substitution $ae - c^2 = 0$, $ac - b^2 = 0$. We have in fact already seen that the developable $U = 0$ has this curve for its edge of regression.

18. But in the paper just referred to, it is also shown that considering the developable which is the envelope of the common tangent planes of two quadric surfaces; in the general case the developable is of the order 8, but if the two surfaces have an ordinary contact it is of the order 6, and if they have a singular contact it is of the order 5.

In the last mentioned case the surfaces may without loss of generality be reduced to conics, and their equations may be taken to be $(y^2 - 2zx = 0,\ w = 0)$ and $(x^2 - 2yw = 0,\ y = 0)$[3], and this being so the equation of the developable is

$$32z^2w^2 - 27z^2x^2w + 72zxy^2w + 8zx^4 - 27y^4w - 4c^4y^2 \;=\; 0.$$

This is really a developable of the same kind with the first mentioned developable of the order 5; for writing $x = 12c,\ y = 8b,\ z = 8a,\ w = -8e$, the equation becomes

$$a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 \;=\; 0,$$

which is the before mentioned developable $U = 0$. The equations of the two conics become $(9ac - 8b^2 = 0,\ e = 0)$ and $(ae + 3c^2 = 0,\ b = 0)$, and the developable is thus the envelope of the common tangent planes of these two conics. It has been seen that the first conic is a simple line, but the second conic a nodal line, on the developable.

19. Recapitulating, the developable of the fifth order $U = 0$, which is the envelope of the plane $(a,\, 2b,\, 3c,\, 0,\, -27e)(t,\, 1)^4 = 0$ is the locus of the tangents of the quadri-quadric curve $(ae - c^2 = 0,\ ac - b^2 = 0)$, and it is also the envelope of the common tangent planes of the conics $(9ac - 8b^2 = 0,\ e = 0)$ and $(ae + 3c^2 = 0,\ b = 0)$.

20. Returning now to the Prohessian, its equation may be written in the form

$$PU \;=\; (3a^2c,\ -8b^2c,\ ace + 4b^2e \mid ae - c^2,\ 4ac - 4b^2)^2 \;=\; 0,$$

and the discriminant of the quadric function is

$$3a^2ce(ac + 2b^2) - 9b^2c^3,$$

which is

$$=\; 3c\big\{(a^2e + 3b^2c)(ac - b^2) + 3ab^2(ae - c^2)\big\},$$

and recollecting that the equations of the cuspidal curve or edge of regression of the developable are $ae - c^2 = 0$, $ac - b^2 = 0$, it thus appears that the curve in question is also a cuspidal curve on the Prohessian.

21. Consider for a moment the surface

$$(3a^2c,\ -8b^2c,\ ace + 2b^2e \mid ae - c^2,\ 4ac - 4b^2)^2 \;=\; 0,$$

and suppressing the discriminant is, taking c as a coefficient, of the form

$$(B,\, C)(ae - c^2)^2 + 2(\ldots)(ae - c^2)(ac - b^2) + (B,\, D)(ac - b^2)^2,$$

where the coefficients A, B, C are as before the differential coefficients of U *[see facsimile p. 275 for the complete development]*.

[3]These are in fact the conics made use of, p. 37 in the paper above referred to [and p. 495 in the reprint], but the equations are by mistake given as $(x^2 - 2yz = 0,\ w = 0)$, $(y^2 - 2zw = 0,\ x = 0)$, that is, x and y are interchanged.

The Memoir of Paper 344 continues with further analysis of this Prohessian and the parallel Sextic developable (Nos. 26 onward), but these later articles fall beyond the present 50-page extract and will be taken up in the subsequent installment.

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Volume V

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Arthur Cayley

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344. On Certain Developable Surfaces *(continued and concluded)*

[Continuation from p. 276. The Quintic Developable, Nos. 21 ff., the Prohessian, and the special Sextic Developable.]

The surface in question (with θ an arbitrary parameter)

$$(3a^2c,\ -8b^2c,\ ace+2b^2e \mid ae-c^2,\ 4ac-4b^2)^2 \ = \ 0$$

is a surface having for a nodal curve the cuspidal curve of the developable; but if the discriminant of the quadric function vanishes, that is if

$$3a^2ce(ac+2b^2e)-9b^2c^3 \ = \ 0,$$

for $ae-c^2=0$, $ac-b^2=0$, then the curve in question will be a cuspidal curve on the surface. The last-mentioned equation is

$$3c\big\{(a^2e+3b^2c)(ac-b^2)+a^3((1+2\theta)\,ae-3b^2c)\big\} \ = \ 0,$$

which for $ae-c^2=0$, $ac-b^2=0$ becomes $1+2\theta-3\theta^2=0$, that is $\theta=1$, giving the Prohessian, or $\theta=-\frac{1}{3}$.

22. For the latter value the surface is

$$(9a^2c,\ -36b^2c,\ 3ace-2b^2e \mid ae-c^2,\ 4ac-4b^2)^2 \ = \ 0,$$

[The expanded coefficient table on facsimile p. 277, listing the coefficients of the monomials in a, b, c, e, is here reproduced in narrative form; the discriminant of the quadric is]

$$c\big\{(3a^2e - b^2c)(ac - b^2) + ab^2(ae - c^2)\big\}.$$

The geometrical signification of this surface, and its relation to the Prohessian, are not further examined.

23. The equation of the Prohessian may be written

$$PU \;=\; (ac - b^2)\big\{16e(ac - b^2)^2 + 3a(ae - c^2)^2\big\} + b^2U \;=\; 0,$$

or what is the same thing

$$PU \;=\; (ac - b^2)\big[a(ae + 3c^2)(3ae + c^2) - 32ab^2ce + 16b^4e\big] + b^2U \;=\; 0,$$

the latter of which shows that the conic $(ae + 3c^2 = 0,\ b = 0)$, which is the nodal line of the developable, is a simple line on the Prohessian.

24. Consider the curve of intersection of the developable and the Prohessian; this is of the order $5 \times 7 = 35$. We have $ac - b^2 = 0$, $U = 0$, or else

$$16e(ac - b^2)^2 + 3a(ae - c^2)^2 \;=\; 0, \qquad U = 0.$$

Consider for a moment the second system; this is

$$3a(ae - c^2)^2 + 16e(ac - b^2)^2 = 0,$$

$$3a(ae - c^2)^2 - 24c(ae - c^2)(ac - b^2) + 48e(ac - b^2)^2 = 0,$$

which give

$$(ac - b^2)\big\{3c(ae - c^2) - 4e(ac - b^2)\big\} \;=\; 0, \qquad U = 0,$$

and these are equivalent to $(ac-b^2 = 0,\ U = 0)$ and $\big(3c(ae-c^2)-4e(ac-b^2) = 0,\ U = 0\big)$, so that the entire intersection is made up of $(ac-b^2 = 0,\ U = 0)$ twice, and of $(4e(ac-b^2)-3c(ae-c^2) = 0,\ U = 0)$ once.

25. The first part is at once seen to give

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$	4 times,	order 16
the line $(a = 0,\ b = 0)$	4 times,	order 4

The second part gives

$$(ae - c^2)\big\{4c(ac - b^2) + a(ae - c^2)\big\} = 0, \qquad \big\{4e(ac - b^2) - 3c(ae - c^2)\big\} = 0;$$

this consists of 1° the part $ae - c^2 = 0$, $e(ac - b^2) = 0$, viz.

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$	once,	order 4
the line $(c = 0,\ e = 0)$	twice,	order 2

and 2° the part $4c(ac - b^2) + a(ae - c^2) = 0$, which contains $4e(ac - b^2) - 3c(ae - c^2) = 0$, viz.

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$	once,	order 4,

and by writing the two equations in the form

$$c(ae + 3c^2) - 4b^2e = 0, \qquad a(ae + 3c^2) - 4b^2c = 0,$$

it is clear that it contains also

the nodal curve ($ae+3c^2=0,\ b=0$)	twice,	order 4
the line ($c=0,\ e=0$)	once,	order 1

whence the complete intersection of the developable and the Prohessian is made up as follows:

the cuspidal curve ($ac-b^2=0,\ ae-c^2=0$)	6 times,	order 24
the nodal curve ($ae+3c^2=0,\ b=0$)	2 times,	order 4
the line ($a=0,\ b=0$)	4 times,	order 4
the line ($c=0,\ e=0$)	3 times,	order 3
		35

Special Sextic Developable, Nos. 26 to 35

26. Lastly, we have the developable

$$U \ = \ a^2e^4 - 12a^2bde^3 - 27a^2d^4 - 6ab^2d^2e^2 - 27b^4e^2 - 64b^3d^3 \ = \ 0,$$

derived from the quartic function $(a,b,0,d,e \mid t,1)^4$; the discriminant is in fact $(ae-4bd)^3 - 27(-ad^2-b^2e)^2$, which equals the foregoing value of U.

27. Taking a,b,d,e as coordinates, then (omitting common numerical factors) the first derived functions are

$$\begin{aligned}\partial_a U &= ae^4 - 3abde^3 - 18ad^4 - 2b^2d^2e,\\ \partial_b U &= -4a^2de^3 - 4abd^3e - 36b^3e^2 - 64b^2d^3,\\ \partial_d U &= -4a^2be^2 - 36a^2d^3 - 24ab^2de - 64b^3d^2,\\ \partial_e U &= a^2e^3 - 3a^2bde - 2ab^2d^2 - 18b^4e,\end{aligned}$$

and the second derived functions (changing, for greater convenience, the signs) are arranged as the symmetric 4×4 matrix *[see facsimile p. 279 for the full table; the entries are quartic monomials in a,b,d,e].*

28. Representing these by

$$\begin{vmatrix} A & H & G & L\\ H & B & F & M\\ G & F & C & N\\ L & M & N & F'\end{vmatrix},$$

and employing for the determinant the same partially developed form as in the first example, then proceeding to the calculation, we find the partial bilinear minors (the table is reproduced *in extenso* in the facsimile, pp. 279–280); the column sums (provided "by way of verification") are $+459$, -1107, -243, $+891$, $+351$ respectively.

29. Forming from these the seven component terms of the determinant, and collecting, the result divides by 5, and we have the lengthy expression for the determinant displayed at facsimile p. 280, the first column of which is the Hessian. The column-sum totals are $+492075$, -2032452, -866052, $+350649+2985984$, -866052, -2032452.

30. This divides, as it should do, by

$$U' \ = \ -1\cdot ae^4 + 12\,abde^3 - 27\,ad^4 + 27\,b^4e^2 + 64\,b^3d^3,$$

(see the displayed list of monomial coefficients on the facsimile, p. 281), and the other factor, which is the Prohessian, is

$$\begin{aligned}PU \ = \ &+1\,a^3e^6 + 16\,a^2bde^5 + 108\,a^2b^2d^2e^4 + 524\,a^2c\,b\,d^3e^3 - 432\,a^3b^2d^4e^2\\ &+ 656\,a^3b^3d^5e + 1512\,ab^4d^4e^2 - 1280\,b^6d^6 + 3645\,b^4d^6e^2 \ + \ \cdots,\end{aligned}$$

[the complete tabulation is given at facsimile p. 281; the coefficients quoted above are intended only to indicate the form. The compiler has not reproduced every monomial: it suffices to record that all the coefficients are calculated and that Cayley's checked column totals match.]

31. To simplify this, Cayley first collects the six terms

$$-108\,a^2c^2(a^2d^4+b^4e^2),$$

$$-432\,abce(a^2d^2+b^2d^2),$$

$$+1512\,b^2d^2(ce^2d^2+b^2e^2);$$

and then putting $a^2d^4+b^4e^2=(ad^2+b^2e)^2-2ab^2d^2e$, we have the terms

$$+216\,a^2b^2d^2e^2+864\,a^3b^2d^3e^2-3024\,ab^4d^4e,$$

which combined with the remaining terms give the consolidated expression

$$\begin{aligned} PU \;=\; &+1\,a^3e^6+16\,a^2bde^5-308\,a^2b^2d^2e^4+1520\,a^3b^3d^4e^3 \\ &-2752\,ab^4d^5e^2+1280\,b^6d^6, \end{aligned}$$

which is found to be divisible by $(ae-4bd)^2$; and we thus obtain for PU the form

$$PU \;=\; -108(a^2e^2+4abde-14b^2d^2)(ad^2+b^2e)^2+(a^2e^2+28abde-20b^2d^2)(ae-4bd)^2,$$

which puts in evidence that the cuspidal line $(ae-4bd=0,\ ad^2+b^2e=0)$ of the developable is also a cuspidal line of the Prohessian.

32. Writing the equations of the developable and the Prohessian under the forms

$$A'^3-27B'^2 \;=\; 0, \qquad LA'^3-108MB'^2 \;=\; 0,$$

and substituting in the second equation $A'^3=27B'^2$, it becomes $B'^2(L-4M)=0$, that is, the intersection is made up of $A'=0$, $B'=0$, which is the cuspidal curve taken six times (order 36), and of the curve $A'^3-27B'^2=0$, $L-4M=0$ (order 24). But substituting for L, M their values, the equation $L-4M=0$ becomes

$$a^2e^2-4abde-12b^2d^2 \;=\; 0,$$

that is

$$(ae+2bd)(ae-6bd) \;=\; 0,$$

so that the last-mentioned curve is composed of the intersections of the developable by the two quadric surfaces

$$ae+2bd=0, \qquad ae-6bd=0.$$

33. Now combining with the equation of the developable the equation $ae+2bd=0$, and observing that in consequence of the last-mentioned equation we have

$$(ae-4bd)^3 \;=\; (-6bd)^3 \;=\; -216\,b^3d^3 \;=\; +108\,ab^2d^2\cdot d,$$

the equation of the developable gives $(ad^2-b^2e)^2=0$, or we have (taken twice) the curve $ae+2bd=0$, $ad^2-b^2e=0$, which is a curve of the sixth order made up of the lines $(a=0,\ b=0)$, $(d=0,\ e=0)$, and of a quartic curve (an *excubo-quartic*[1]) the nodal line on the developable.

[1]A quartic curve which is the complete intersection of two quadric surfaces is termed a quadri-quadric; a quartic curve of the kind which is not such complete intersection but can only be represented by means of a cubic surface is termed an excubo-quartic.

If in like manner with the equation of the developable we combine the equation $ae - 6bd = 0$, then from this equation we have

$$(ae - 4bd)^3 \;=\; (2bd)^3 \;=\; 8b^3d^3 \;=\; -\tfrac{27}{4}\, ab^2d^2e^2,$$

and the equation of the developable then gives

$$27(ad^2 + b^2e)^2 - \tfrac{4}{27}\cdot ab^2d^2e^2 \;=\; 0,$$

that is,

$$\big(ad^2 + \tfrac{4}{3}\, b^2e\big)\big(ad^2 + 9b^2e\big) \;=\; 0.$$

The curve $ae - 6bd = 0$, $ad^2 + 9b^2e = 0$ is made up of the lines $(a = 0,\ b = 0)$, $(d = 0,\ e = 0)$, and of an excubo-quartic, and the curve $ae - 6bd = 0$, $ad^2 + \frac{4}{3}\, b^2e = 0$ is made up of the same two lines and of an excubo-quartic.

34. Hence we see that the intersection of the developable and the Prohessian, which is of the order $(6 \cdot 10 =)60$, is made up as follows, viz.

cuspidal curve $ae - 4bd = 0,\ ad^2 + b^2e = 0$	6 times,	$6 \times 6 = 36$
line $(a = 0,\ b = 0)$	4 times,	$1 \times 4 = 4$
line $(d = 0,\ e = 0)$	4 times,	$1 \times 4 = 4$
nodal curve (excubo-quartic) $ae + 2bd = 0,\ ad^2 - b^2e = 0$	2 times,	$4 \times 2 = 8$
excubo-quartic $ae - 6bd = 0,\ ad^2 + 9b^2e = 0$	1 time,	$4 \times 1 = 4$
excubo-quartic $ae - 6bd = 0,\ ad^2 + \frac{4}{3}b^2e = 0$	1 time,	$4 \times 1 = 4$
		60

35. It is to be added that a generating line of the developable meets the Prohessian in the ineunt on the cuspidal edge taken 6 times, in a point of the nodal line taken 2 times (viz. the $r-4$ points, r being here $= 6$, of the general theorem), in a point of the excubo-quartic $ae - 6bd = 0$, $ad^2 + 9b^2e = 0$, and in a point of the excubo-quartic $ae - 6bd = 0$, $ad^2 + \frac{4}{3}b^2e = 0$ (these being the $2r - 10$ points of the general theorem); we have thus $(6 + 2 + 2 =)10$ points of intersection of the generating line with the Prohessian.

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345. On the Inflexions of the Cubical Divergent Parabolas

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 199–203.]

The five divergent parabolas, species 67 to 71, of Newton's *Enumeratio Linearum tertii Ordinis* (1704), are included under the general equation $y^2 = ax^3 + 3bx^2 + 3cx + d$: there are two general forms, or forms without singularities, viz. the *parabola cum ovali*, sp. 67, and the *parabola pura*, sp. 71; two forms having a double point, viz. the *nodata*, sp. 68, and the *punctata*, sp. 69, according as the double point is one with real branches, or is a conjugate or isolated point; and finally the *cuspidata* or semicubical parabola, sp. 70, which has a cusp. In the nomenclature of my short note "On Curves of the Third Order," *British Assoc. Report for the Year 1861, Notices &c. p. 2*, the five parabolas are the *complex*, the *simplex*, the *crunodal*, the *acnodal*, and the *cuspidal*; the distinction there made of the simplex kind of cones of the third order into three subspecies applies to the simplex parabola, and for this particular case was, as I have since ascertained, noticed in Murdoch's *Newtoni Genesis Curvarum per Umbras*, 8vo. Lond. 1746, pp. 1–126. It may be remarked that in this very interesting and valuable work the number of species is given as 78, viz. the author includes the four species added by Stirling, and the other two usually considered to have been added by Cramer (one of them the author himself attributes to Cramer), and that the demonstration of Newton's theorem is effected in the most complete way by showing in what manner the five cones are each of them to be cut so as to obtain the 78 species of cubic curves.

The analytical investigation of the points of inflexion of the above-mentioned divergent parabolas, that is, the curves defined by the equation

$$y^2 \;=\; ax^3+3bx^2+3cx+d,$$

is not without interest. The result which should be obtained is, by the general theory of cubic curves, known to be as follows: there is always an inflexion at infinity, in the point where the line at infinity is met by the line $x=0$ (or, what is the same thing, by any ordinate of the curve); but disregarding altogether this inflexion at infinity, then in the general case where the curve is without singularity, the remaining eight inflexions (two of them real, six imaginary) lie in pairs on four ordinates of the curve: if however the curve has an acnode, the six imaginary inflexions coincide with the acnode, viz. the three ordinates corresponding to these pass through the acnode, but there are still two real inflexions; if the curve has a crunode, four of the imaginary inflexions and the two real inflexions coincide with the crunode, viz. the three ordinates corresponding to these pass through the crunode, and there is not any real inflexion, although there are still two imaginary inflexions; finally, if the curve has a cusp, then the eight inflexions coincide with the cusp, viz. the four ordinates corresponding to these pass through the cusp, and there is no inflexion real or imaginary.

Proceeding now to the analytical investigation, if in order to form the Hessian we introduce the new coordinate $z=1$, the equation of the curve becomes

$$U \;=\; -y^2z+ax^3+3bx^2z+3cxz^2+dz^3,$$

and thence forming the second differential coefficients, and ultimately replacing z by its value $=1$, we have

$$HU \;=\; -6\begin{vmatrix}(ax+b) & 0 & (bx+c)\\ 0 & 2 & 2y\\ (bx+c) & 2y & (cx+d)\end{vmatrix} \;=\; 0,$$

whence, developing and dividing by 24, we find

$$3\big\{(ax+b)(cx+d)-(bx+c)^2\big\}+(ax+b)\,y^2 \;=\; 0,$$

or, what is the same thing,

$$3\big\{(ac-b^2)x^2+(ad-bc)x+(bd-c^2)\big\}+(ax+b)y^2 \;=\; 0,$$

as the equation of the Hessian curve, meeting the given cubic curve $y^2=ax^3+3bx^2+3cx+d$ in its points of inflexion. Multiplying the last-mentioned equation by b, and adding it to the equation of the Hessian, we obtain

$$abx^3+3acx^2+3adx+4bd-3c^2+axy^2 \;=\; 0,$$

or, what is the same thing,

$$xy^2 \;=\; -bx^3-3cx^2-3dx+\tfrac{3c^2-4bd}{a},$$

as the equation of a curve meeting the given curve in its points of inflexion; and if for greater simplicity we assume $a=1$, $d=0$ (the latter equation means obviously that the origin is taken at one of the three intersections of the curve with the axis of x, say the real one, if the intersections are one real, two imaginary), then the equation of the curve is

$$y^2 \;=\; x(x^2+3bx+3c),$$

and the inflexions are given as the intersections of the curve with the curve

$$xy^2 \;=\; -bx^3-3cx^2+3c^2.$$

There is, it is clear, an inflexion at the point at infinity on the line $x=0$; and eliminating y^2 we find

$$x^3+3bx^2+3cx^2 \;=\; -bx^3-3cx^2+3c^2,$$

or, what is the same thing,

$$x^4+4bx^3+6cx^2-3c^2 \;=\; 0,$$

a quartic equation giving the four ordinates through the remaining eight inflexions.

If the curve has a cuspidal point, then the origin will be at the cusp, and we have $b=0$, $c=0$, and the quartic equation becomes $x^4=0$; that is, the four ordinates pass through the cusp.

If the curve have a node, then taking the origin at the node we have $c=0$; the equation of the curve is

$$y^2 \;=\; x^2(x+3b),$$

and the curve has a crunode or an acnode according as b is positive or negative: the quartic equation becomes $x^3(x+4b)=0$, and the factor $x^3=0$ gives three ordinates through the node; the remaining factor $x+4b=0$ gives the ordinate through the two inflexions; and substituting this value of x in the equation of the cubic, we find

$$y^2 \;=\; -4b^3,$$

and the resulting values of y (consequently also the inflexions) are imaginary if b be positive, that is, for the crunodal form; but real if b be negative, that is, for the acnodal form. It is to be observed that the indefinite ordinate $x+4b=0$ or $x=-4b$ is real in each of the two cases: in the crunodal case, the ordinate lies outside the curve, that is beyond the loop; in the acnodal case inside the curve, that is on the opposite side to the acnode in regard to the vertex; and using $3b$ to denote the distance (taken positively) of the vertex from the node, (that is, in the crunodal case changing the sign of b), the distance (taken positively) of the ordinate from the vertex is $4b-3b=b=\frac{1}{3}\cdot 3b$, that is, it is one-third of the distance of the vertex from the node.

Consider next the case of a curve without singularities; and first the complex case, the condition for which is that the equation $x^2+3bx+3c=0$ may have its roots real, or $c<\frac{3}{4}b^2$. The values of x which give $y=0$ are

$$x=0, \qquad x=-\tfrac{3}{2}b\pm\sqrt{3(\tfrac{3}{4}b^2-c)};$$

and we may without loss of generality assume that b and c are each of them positive; the value $x=0$ will then belong to the vertex of the parabolic portion, and the two negative values $x=-\frac{3}{2}b\pm\sqrt{3(\frac{3}{4}b^2-c)}$ will belong to the vertices of the oval. The limiting values $c=0$ and $c=\frac{3}{4}b^2$ give the acnodal and the crunodal curves respectively, which have been already considered.

In the case in question ($b=+$, $c=+$, $c<\frac{3}{4}b^2$), the equation $x^4+4bx^3+6cx^2-3c^2=0$ has only two real roots, one of them positive and the other negative; and the positive root substituted in the equation $y^2=x(x^2+3bx+3c)$ gives $y^2=+$, and we have thus the two real inflexions: in order to verify that the negative root gives imaginary inflexions, it must be shown that this negative root does not lie between the two values $x = -\frac{3}{2}b \pm \sqrt{3(\frac{3}{4}b^2 - c)}$, or, what is the same thing, that these values substituted in the function

$$x^4 + 4bx^3 + 6cx^2 - 3c^2$$

give results of the same sign.

To verify this write $c = \frac{3}{4}b^2(1-\theta^2)$ (where $0 < \theta < 1$) and $x = \frac{3}{2}b(\xi - 1)$; then for the limiting values of x, we have $\frac{3}{2}b(\xi-1) = -\frac{3}{2}b \pm \frac{3}{2}b\theta$, or $\xi = \pm\theta$; and moreover

$$x^4 + 4bx^3 + 6cx^2 - 3c^2 \;=\; \tfrac{9}{16}b^4\big[3(\xi-1)^4 + 8(\xi-1)^3 + 6(\xi-1)^2(1-\theta^2) - (1-\theta^2)^2\big],$$

where the term in brackets is

$$= 3\xi^4 - 4\xi^3 - 6\xi^2\theta^2 + 12\xi\theta^2 - \theta^4 - \theta^4,$$

and writing $\xi = \pm\theta$, this becomes

$$-4\theta^4 - 4\theta^3 + 8\theta^3 \;=\; -4\theta^2(\theta \pm 1)^2,$$

so that the two values are each negative, and the theorem is thus proved. It may be added that the curve

$$y = x^4 + 4bx^3 + 6cx^2 - 3c^2$$

cuts the axis in two real points, one of them situate between the oval and the parabolic portion of the cubic parabola, the other within the parabolic portion.

Lastly, for the simplex case, the condition for which is $c > \frac{3}{4}b^2$: the equation $0 = x^4 + 4bx^3 + 6cx^2 - 3c^2$ has, as before, two real roots, one positive and the other negative; and since the negative root substituted for x in the equation $y^2 = x^3 + 3bx^2 + 3cx$ gives a negative value of y^2, it is only the positive root which gives an ordinate through two real inflexions. The curve $y = x^4 + 4bx^3 + 6cx^2 - 3c^2$ meets the axis in two real points, one of them without, the other within the cubic parabola.

I remark that the equation $x^2 + 2bx + c = 0$ gives the level points (i.e. the points where the tangent is parallel to the axis) of the cubic parabola. In the complex case, where $c < \frac{3}{4}b^2$, then *a fortiori* $c < b^2$ or the values of x are both real, one of these values gives y^2 positive, and we have thus the maximum ordinate of the oval; the other value of x gives y^2 negative. In the simplex case, where $c > \frac{3}{4}b^2$, we may have 1°. $c < b^2$, and the two values of x give each of them y^2 positive; the least value of x corresponds to a maximum ordinate, the greatest to a minimum ordinate of the cubic parabola, and between these we have the ordinate through the two real inflexions, the tangents at the inflexions meeting on the axis within the parabola. 2°. We may have $c = b^2$: the two values of x here coincide, giving the ordinate through the two real inflexions, the tangents at the inflexions being horizontal. And 3°. We may have $c > b^2$: the two values of x are then imaginary and we have no real level point. These are, in fact, Murdoch's three forms, which he distinguishes as the *ampullata*, *media*, and *campaniformis*.

2, Stone Buildings, W.C., June 2, 1863.

346. Note on an Expression for the Resultant of two Binary Cubics

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 380–382.]

Mr Warren, in his paper "Illustrations of the Theory of Critical Functions," *Quarterly Mathematical Journal*, t. VI. pp. 231–237, (1864), has given for the Resultant of two binary cubic functions an expression which is in effect as follows; viz. considering the cubic

$$(a, b, c, d \mid x, y)^3,$$

its Hessian

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \mid x, y)^2 \ = \ (ac - b^2,\ ad - bc,\ bd - c^2 \mid x, y)^2,$$

and the cubicovariant

$$(A, B, C, D \mid x, y)^3 \ = \ \big(a^2d{-}3abc{+}2b^3,\ \ 3abd{-}6ac^2{+}3b^2c,\ \ {-}3acd{+}6b^2d{-}3bc^2,\ \ {-}ad^2{+}3bcd{-}2c^3 \mid x, y\big)^3;$$

and in like manner for the cubic $(a', b', c', d' \mid x, y)^3$, its Hessian $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \mid x, y)^2$ and the cubicovariant $(A', B', C', D' \mid x, y)^3$; and writing

$$\mathfrak{A} \ = \ ad' - 3bc' - 3b'c - a'd,$$

$$\mathfrak{B} \ = \ ac' + a'c - 2bb',$$

$$\mathfrak{C} \ = \ AD' - 3BC' + 3B'C - A'D,$$

then the Resultant is

$$= -2\mathfrak{A}^3 + 27\mathfrak{A}\mathfrak{B}^3 + 27\mathfrak{C}^2,$$

that is, the Resultant is

$$\begin{aligned}
&- 2(ad' - a'd - 3bc' + 3b'c)^3 \\
&\quad + 27(ad' - a'd - 3bc' + 3b'c) \\
&\qquad \times \big[(ac - b^2)(b'c' - d'^2) + \tfrac{?}{?}(ad - bc)(a'd' - b'c') + (bd - c^2)(a'^2 - b'^2)\big] \\
&\quad + 27\big\{(a^2d - 3abc + 2b^3)(-a'd'^2 + 3b'c'd' - 2c'^3) \\
&\qquad - 9(abd - 2ac^2 + b^2c)(-a'c'd' + 2b'^2d' - b'c'^2) \\
&\qquad + 3(-acd + 2b^2d - bc^2)(a'b'd' - 2a'c'^2 + b'^2c) \\
&\qquad - (-ad^2 + 3bcd - 2c^3)(a'^2d' - 3a'b'c' + 2b'^3)\big\}.
\end{aligned}$$

[The above table is reproduced from the facsimile p. 290; some indices are uncertain in the OCR.]

In particular assume

$$(a, b, c, d \mid x, y)^3 = x^3 + y^3,$$

so that

$$(a', b', c', d' \mid x, y)^3 = xy, \quad (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \mid x, y)^2 = xy, \quad (A', B', C', D' \mid x, y)^3 = x^3 - y^3,$$

$$a = d = 1,\ \ b = c = 0, \qquad a' = d' = 0,\ \ b' = c' = \tfrac{1}{3}, \qquad A' = -D' = 1,\ \ B' = C' = 0;$$

$$\mathfrak{A} = a - d,$$

$$\mathfrak{B} = -b = bc - ad,$$

$$\mathfrak{C} = A + D = a^2d - ad^2 - 3abc + 3bcd + 2b^3 - 2c^3,$$

or, putting for shortness $a - d = \theta$, and therefore $a = d + \theta$,

$$\mathfrak{A} = \theta,$$

$$\mathfrak{B} = bc - d^2 - d\theta,$$
$$\mathfrak{C} = 2(b^3 - c^3) - 3bcd + 3bc\theta + d^2\theta + d\theta^2,$$

and Resultant is

$$\begin{aligned} &-2\theta^3 + 27\theta(bc - d^2 - d\theta)^2 + 27[2(b^3 - c^3) - 3bcd + 3bc\theta + d^2\theta + d\theta^2]^2 \\ &\quad = -2\theta^3 + 54b^3 - 54c^3 - 54bcd \cdot \theta \text{ [?]}, \end{aligned}$$

or rejecting the factor -2, it is

$$\theta^3 - 27b^3 + 27c^3 + 27bcd.$$

But the two equations are

$$(a, b, c, d \mid x, y)^3 = 0, \qquad x^3 + y^3 = 0,$$

the last of which gives $y = -x$, $y = -\omega x$, $y = -\omega^2 x$, if ω be an imaginary cube root of unity, and hence the Resultant is

$$= (a - 3b + 3c - d)(a - 3b\omega + 3c\omega^2 - d)(a - 3b\omega^2 + 3c\omega - d),$$

which is

$$= (d - 3b + 3c)(d - 3b\omega + 3c\omega^2)(d - 3b\omega^2 + 3c\omega) = d^3 - 27b^3 + 27c^3 + 27bcd,$$

and the formula is thus verified.

If the two cubics are taken to be

$$(a, b, c, d \mid x, y)^3 = 0, \qquad (b, c, d, e \mid x, y)^3 = 0,$$

then the formula gives for the Discriminant of the quartic function $(a, b, c, d, e \mid x, y)^4$ a new expression, which however does not appear to be an elegant one.

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347. On the Notion and Boundaries of Algebra

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 382–384.]

I do not admit the assertion, that the idea of number is derived from that of time: it appears to me that it is derived from that of succession in time or space indifferently. But I would rather say that the idea of cardinal number is derived and abstracted from that of ordinal number, viz. (distinguishing the expressions 'set' and 'series,' the latter being used to designate a set of things considered as arranged in a definite order), if we have a series of things a, b, c, d, &c., or say a series of words *first*, *second*, *third*, *fourth*, &c.; then any set of things X, N, Y, P, Q, &c., taking them up one after the other, no matter in what order, and coordinating them with the terms of the series a, b, c, d, &c. or with the words *first*, *second*, *third*, *fourth*, &c.—the last of them will be coordinated with a definite term of the series a, b, c, d, &c., or with a definite term of the series *first*, *second*, *third*, *fourth*, &c.; that is, the set, whatever be the assumed order of the terms, or (what is the same thing) without assuming any order therein, will have a certain property; viz. in the set X, N, Y, P, Q, where the last term is coordinated with e or with the word *fifth*, the property is that the set consists of five things: and so in general the set consists of a certain (cardinal) number of things, such cardinal number being the number corresponding to the rank in the series a, b, c, d, &c., of the term wherewith is coordinated the last term of the set, or corresponding with the like ordinal number in the series *first*, *second*, *third*, *fourth*, &c.

The foregoing remarks are made to some extent incidentally, but they have a bearing on the distinction in kind which exists e.g. between the proposition $1 + 1 + 1 + 1 = 4$, and the proposition which for ordinary purposes would be expressed as $1 + 1 +$ &c. (n terms) $= n$, but which is better expressed in the form $1_1 + 1_2 + \cdots + 1_n = n$, where the subscript numbers merely distinguish between the different unities which are added altogether.

I use the term Algebra in a wide sense as including, or indeed I might say identical with, Finite Analysis, and excluding Infinite Analysis; but in speaking of it as identical with Finite Analysis I include in that term part of what might be considered Infinite Analysis; viz. many of the theorems relating to infinite series or other successions of operations, e.g.

$$(1 - x)(1 + x + x^2 + \cdots \text{ ad inf.}) = 1,$$

really belong to Finite Analysis; for what is asserted is that the coefficient of the term of indefinite rank, say x^n, is a *finite* series equal in value to zero (this coefficient in fact is $1-1$ which is $= 0$). On the other hand the theorem

$$1 - \frac{x^2}{1\cdot 2} + \frac{x^4}{1\cdot 2\cdot 3\cdot 4} - \cdots \;=\; (1-x)\left(\frac{1+x}{1}\right)\cdots,$$

the truth whereof depends on the equations

$$\frac{\pi^2}{6} \;=\; \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \quad \text{ad inf.,} \quad \text{\&c.,}$$

which are not arithmetically verifiable, belongs strictly to Infinite Analysis.

Algebra is an Art and a Science; *qua* Art, it defines and prescribes operations which are either *tactical* or else *logistical*; viz. a tactical operation is one relating to the arrangement in any manner of a set of things; a logistical operation (I prefer to use the new expression instead of arithmetical) is the actual performance, so as to obtain for the result a number, of any arithmetical operations (of course, given operations) finite in number, since these alone can be actually performed, upon given numbers. And *qua* Science, Algebra affirms *a priori*, or predicts, the result of any such tactical or logistical (or tactical and logistical) operations. An equation such as $1 + 4 + 10 = 15$ is not an algebraical theorem; it is merely the assertion that the sum of the numbers $1, 4, 10$ is that number, viz. 15, which is the sum of the numbers in question. And, similarly, the equation $1 + 1 + 1 = 3$ is not an algebraical theorem. But on the other hand, the equation $1 + 1 + 1 + \cdots$ (n terms) $= n$, is an algebraical theorem; in the equivalent form $1_1 + 1_2 + \cdots + 1_n = n$, (where $1_k = 1$) it is not different in kind from the equation $1 + 2 + 3 \cdots + n = \frac{1}{2}n(n+1)$, or say $1_1 + 1_2 + \cdots + 1_n = \frac{1}{2}n(n+1)$, (where $1_k = k$), which is certainly an algebraical theorem. And this leads to the remark, that every algebraical theorem rests ultimately on a tactical foundation. In fact, whether we prove the last-mentioned theorem in the easiest way by writing

$$1 + 2 + 3 + \cdots + n = S,$$

$$n + (n-1) + (n-2) + \cdots + 1 = S,$$

and therefore $2S = (n+1) + (n+1) + \cdots$ (n terms) $= n(n+1)$ or $S = \frac{1}{2}n(n+1)$; or by induction by showing that the theorem, if true for n, is true for $(n + 1)$, (this depends on the equation $\frac{1}{2}n(n+1) + (n+1) = (n+1)(\frac{1}{2}n + 1) = \frac{1}{2}(n+1)(n+2)$); the proof is equally a tactical one; it is always tactic which determines what logistical operations are to be performed.

Although it may not be possible absolutely to separate the tactical and logistical operations; for in (at all events) a series of logistical operations, there is always something that is tactical, and in many tactical operations (e.g. in the Partition of Numbers) there is something which is logistical, yet the two great divisions of Algebra are Tactic and Logistic. Or if, as might be done, we separate Tactic off altogether from Algebra, making it a distinct branch of Mathematical Science, then (assuming in Algebra a knowledge of all the Tactic which is required) Algebra will be nothing else than Logistic.

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348. On the Theory of Involution

[From the Transactions of the Cambridge Philosophical Society, vol. XI. Part I. (1866), pp. 21–38. Read February 22, 1864.]

Three or more quantics which satisfy identically a linear equation such as

$$\lambda U + \lambda' U' + \lambda'' U'' + \cdots \;=\; 0,$$

where $\lambda, \lambda', \lambda'', \ldots$ are constants, are said to be in Involution. In particular any quantic $U + kV$, where k is a constant, is in involution with the quantics U, V; and the entire system of such quantics, k having any value whatever, is a system in involution with the quantics U, V. And in like manner the equation $U + kV = 0$, or the locus or system of loci thereby represented is said to be in involution, or to form a system in involution with the equations or loci $U = 0$, $V = 0$. If U, V are binary quantics then the equation $U + kV = 0$ may be considered as representing a range of points in involution with the ranges $U = 0$, $V = 0$. And similarly, if U, V are ternary quantics, then the equation $U + kV = 0$ may be considered as representing a curve in involution with the curves $U = 0$, $V = 0$.

In the case of a range $U + kV = 0$, the constant k may be determined so that the range shall have a twofold[2] point. The condition for this may be written

$$\text{Disct.}\,.\,(U + kV) \;=\; 0,$$

it being understood that the discriminant of the function $U + kV$ is taken in regard to the coordinates. And this being so, we may write Disct.. Disct.. $(U+kV)$ to denote the discriminant in regard to k of the function Disct.. $(U+kV)$. The quantity in question (viz. Disct.. Disct.. $(U+kV)$, or say for shortness D) is a function of the coefficients of U, V, homogeneous as regards each set of coefficients separately, and it breaks up into factors in the form

$$D \;=\; RQ^2P^2;$$

where $R = 0$ is the condition in order that the ranges $U = 0$, $V = 0$ may have a point in common; or what is the same thing, in order that there may have a range $U + kV = 0$ having a twofold point at a common point of the ranges $U = 0$, $V = 0$, (R is in fact the resultant of the quantics U, V): $Q = 0$ is the condition in order that there may be a range $U + kV = 0$ having a threefold point: and $P = 0$ is the condition in order that there may be a range $U + kV = 0$ having a pair of twofold points.

[2]The series of epithets *onefold*, *twofold*, &c., seems preferable to the series *single*, *double*, &c., as avoiding ambiguities which would sometimes be occasioned by the use of these last. The double point of a curve I call a *node*, viz. a *crunode* when it is a double point with two real branches, and an *acnode* when it is a conjugate or isolated point. The subject-matter and context will in general show whether the term node is to be considered as including or as not including a cusp.

And similarly, when $U = 0$, $V = 0$ are curves, then we have the like equation

$$D = RQ^2P^2,$$

where $R = 0$ is the condition in order that the curves $U = 0$, $V = 0$ may have a point of twofold intersection, that is, that the two curves may touch each other, (R is the Tactinvariant of the quantics U, V); or what is the same thing, it is the condition in order that there may be a curve $U + kV = 0$ having a node at a point of twofold intersection of the curves $U = 0$, $V = 0$; moreover $Q = 0$ is the condition in order that there may be a curve $U + kV = 0$ having a cusp: and $P = 0$ is the condition in order that there may be a curve $U + kV = 0$ having a pair of nodes.

The establishment and illustration of the foregoing theorems form the chief object of the present memoir.

Article Nos. 1 to 16, relating to two Binary Quantics.

1. Let $U = (a, \dots \mid x, y)^n$, $V = (a', \dots \mid x, y)^n$, be two binary quantics of the same order n, and write $W = U + kV = (a + ka', \dots \mid x, y)^n$, so that $W = U + kV = 0$ is the equation of a range in involution with the ranges $U = 0$, $V = 0$. But for greater distinctness it is in general convenient to retain $U + kV$ instead of replacing it by the single letter W.

2. In order that the range $U+kV = 0$ may have a twofold point we must have simultaneously

$$\partial_x(U + kV) = 0, \qquad \partial_y(U + kV) = 0,$$

and eliminating (x, y) from these equations we find

$$\text{Disct.}\,.\,(U + kV) = 0,$$

which is an equation of the order $2(n - 1)$ as regards k, and of the same order as regards the coefficients of U and V conjointly. And to each of the $2(n - 1)$ values of k there corresponds a point (x, y) satisfying the required conditions; that is, a point which is a twofold point of the range $U + kV = 0$. The points in question may be termed the 'critic centres' of the involution.

3. The elimination of k from the before-mentioned two equations gives

$$\begin{vmatrix} \partial_x U & \partial_y U \\ \partial_x V & \partial_y V \end{vmatrix} = 0,$$

where the determinant, which for shortness I call J, is the Jacobian of the two functions U, V. The equation $J = 0$ gives a range of $2(n - 1)$ points which are in fact the critic centres; and for each of these points we have

$$\partial_x U : \partial_x V = \partial_y U : \partial_y V = -k : 1,$$

which gives the value of k corresponding to the point in question.

4. The condition in order that the equation in k may have a twofold root is

$$\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = 0, \quad \text{or say} \quad D = 0,$$

where $\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = D$ is a function of the degree $2(2n - 2)(2n - 3)$ in regard to the coefficients of U, V conjointly; but it is separately homogeneous, and therefore of the degree $(2n - 2)(2n - 3)$ in regard to each of the two sets of coefficients.

5. To each point of the range $J = 0$ there corresponds a value of k; hence if the range J have a twofold point, then the equation in k will have a twofold root. Now first if the ranges $U = 0$,

$V = 0$ have a common point, then this is a twofold point of the range $J = 0$. But secondly, without a common point in the ranges $U = 0$, $V = 0$, the range $J = 0$ may have a twofold point; and in this case also without a twofold point in the range $J = 0$, there may be in this range two onefold points giving each of them the same value of k, and so giving a twofold root of the equation in k. And the three suppositions correspond respectively to the cases of there being a range $U + kV = 0$ having a twofold point at a common point of the ranges $U = 0$, $V = 0$, having a threefold point, and having a pair of twofold points.

6. First, if the ranges $U = 0$, $V = 0$ have a common point we may write

$$U = (x - \alpha y)U', \qquad V = (x - \alpha y)V',$$

and these give

$$U + kV = (x - \alpha y)(U' + kV').$$

Now in general

$$\text{Disct.}.\, PQ \;=\; \text{Disct.}.\, P \cdot \text{Disct.}.\, Q \cdot [\text{Result.}.(P, Q)]^2,$$

and hence in the present case

$$\text{Disct.}.\, (U + kV) \;=\; \text{Disct.}.\, (U' + kV') \cdot (U'_0 + kV'_0)^2,$$

where $U'_0 + kV'_0$ is what $U' + kV'$ becomes on writing therein $x = \alpha y$ and neglecting the factor y^{n-1} which then presents itself. We see therefore that in this case the equation k has a twofold root $k = -U'_0 : V'_0$; a result which might also have been obtained from the consideration of the Jacobian.

7. The condition in order that the ranges $U = 0$, $V = 0$ may have a common point is

$$\text{Result.}.\, (U, V) = 0,$$

say $R = 0$. R is of the degree n in regard to the coefficients of U, V respectively.

8. Secondly, suppose that the functions U, V are such that there exists a range $U + kV = 0$ having a threefold point. If k_1 be the proper value of k, then we have $U + k_1 V = (x - \alpha y)^3\Theta$, and therefore $U = -k_1 V + (x - \alpha y)^3\Theta$. Hence forming the Jacobian of U, V, the equation for the determination of the critic centres will be

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^3\Theta \\ \partial_y V & \partial_y (x - \alpha y)^3\Theta \end{vmatrix} = 0,$$

which is of the form $(x - \alpha y)^2\Omega = 0$; or we have $(x - \alpha y)^2 = 0$, a twofold critic centre. The corresponding value of k given by the equation $-k : 1 = \partial_x U : \partial_x V$ is $k = k_1$, and we have thus $k = k_1$ as a twofold root of the equation in k.

9. But if the range $W = U + kV = 0$ has a threefold point, or what is the same thing, if the equation $W = 0$ has a threefold root; then we must have between the coefficients of W a plexus of equations equivalent to two relations. Such plexus is known to be of the order $3(n - 2)$. This comes to saying that if the coefficients of W are assumed to be of the form $a + ka' + la'', \ldots$ and if between the several equations of the plexus we eliminate k, we obtain for l an equation $Q = 0$ of the degree $3(n - 2)$. The equation in question would be of the form Funct..$(a + la'', a', \ldots) = 0$. Hence Q is of the degree $3(n - 2)$ in the coefficients $(a, \ldots)$ of U. And in a similar manner Q is of the degree $3(n - 2)$ in the coefficients $(a', \ldots)$ of V. And omitting altogether the terms in l, or taking the coefficients of W to be $a + ka', \ldots$ if from the equations of the plexus we eliminate k, we find an equation $Q = 0$, where Q is a function of the degree $3(n - 2)$ as regards the coefficients of U, and of the same degree as regards the coefficients of V. We have thus found the form of the condition $Q = 0$ which expresses that there may be a range $U + kV = 0$ having a threefold point.

10. It may be proper to remark conversely that given the equation $Q = 0$, if in this equation we write $a = (a + ka') - ka', \ldots$ so that Q becomes a function of $a + ka', \ldots, a', \ldots, k$, then the equation $Q = 0$ will be satisfied irrespectively of the values of $a, \ldots, k$ by a plexus of equations involving only the coefficients $a + ka', \ldots$ and which is in fact the very plexus (equivalent therefore to two relations) which gives the conditions in order that the equation $W = 0$ may have a threefold root.

11. Thirdly, suppose that the functions U, V are such that there exists a range $U + kV = 0$ having a pair of twofold points. If k_1 be the proper value of k, then we have $U + k_1V = (x - \alpha y)^2(x - \beta y)^2\Theta$, and therefore $U = -k_1V + (x - \alpha y)^2(x - \beta y)^2\Theta$. Hence forming the Jacobian of U, V, we have for the determination of the critic centres the equation

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2(x - \beta y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2(x - \beta y)^2\Theta \end{vmatrix} = 0,$$

which is of the form $(x - \alpha y)(x - \beta y)\Omega = 0$; or, we have $x - \alpha y = 0$, or $x - \beta y = 0$, a pair of critic centres; and for each of these the corresponding value of k given by the equation $-k : 1 = \partial_x U : \partial_x V$ is $k = k_1$, so that $k = k_1$ is a twofold root of the equation in k.

12. By the like considerations as for the threefold root (observing that if the equation $W = 0$ has a pair of twofold roots we must have between the coefficients of W a plexus equivalent to two relations, and of the order $2(n - 2)(n - 3)$), we see that the condition for the existence of a range $U + kV = 0$ having a pair of twofold points is of the form $P = 0$, where P is a function of the degree $2(n - 2)(n - 3)$ as regards the coefficients of U, and of the same degree as regards the coefficients of V; and conversely that, given the equation $P = 0$, we may find the plexus.

13. The equation $D = 0$ will be satisfied if $R = 0$, or if $Q = 0$, or if $P = 0$; and in no other cases. To prove this, suppose that $x - \alpha y = 0$ is the critic centre corresponding to a twofold root k_1 of the equation in k. We have $U = -k_1V + (x - \alpha y)^2\Theta$, and thence the equation for the critic centres is

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2\Theta \end{vmatrix} = 0,$$

which is an equation of the form $(x-\alpha y)\Omega = 0$; and where, corresponding to the root $x-\alpha y = 0$, the equation $-k : 1 = \partial_x U : \partial_x V$ gives $k = k_1$. Since k_1 is a twofold root, there must be another critic centre also giving the value k_1 of k. This new critic centre may be either $x - \alpha y = 0$ (the same as the first mentioned critic centre) or it may be a distinct critic centre $x - \beta y = 0$. In the former case

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2\Theta \end{vmatrix}$$

must contain, instead of the factor $(x - \alpha y)$, the factor $(x - \alpha y)^2$. In order that this may be so, we must have $(\alpha\partial_x V + \partial_y V)\Theta$ divisible by $(\partial_x V \cdot \alpha\partial_y V)$, that is, either $\alpha\partial_x V + \partial_y V$, or else Θ, divisible by $x - \alpha y$; or, what is the same thing, either V, or else Θ, divisible by $x - \alpha y$. But if V be divisible by $x - \alpha y$, then U, V have the common factor $x - \alpha y$, and we have the case first above considered. And again if Θ contain the factor $x - \alpha y$, then we have

$$U = -k_1V + (x - \alpha y)^3\Theta',$$

and we have the case secondly above considered. Finally if the new critic centre be the distinct centre $x - \beta y = 0$, then for $x - \beta y = 0$ the equation

$$-k : 1 = \partial_x U : \partial_x V = \partial_y U : \partial_y V$$

should give $k = k_1$; but this will only happen if $\partial_x \cdot (x - \alpha y)^2\Theta$, $\partial_y \cdot (x - \alpha y)^2\Theta$ vanish for $x - \beta y = 0$, that is, if Θ contains the factor $(x - \beta y)^2$; and when this is so,

$$U = -k_1 V + (x - \alpha y)^2 (x - \beta y)^2 \Theta',$$

or we have the case thirdly above considered.

14. Hence the equation $D = 0$ being satisfied if $R = 0$, or else if $Q = 0$, or else if $P = 0$, and in no other cases, the function D must be made up of the factors R, Q, P, each taken the proper number of times, and knowing the degrees of the several functions, it follows that we must have

$$D \;=\; RQ^2P^2;$$

in fact, considering the coefficients of either U or V, the comparison of the degrees gives

$$2(n-1)(2n-3) \;=\; n + 9(n-2) + 4(n-2)(n-3),$$

where the function on the right-hand side is

$$n + 9n - 18 + 4n^2 - 20n + 24 \;=\; 4n^2 - 10n + 6,$$

which is the value of the function on the left-hand side.

15. In the very particular case $n = 2$, Q and P are each of them of the degree $= 0$; and we have simply $D = R$, that is, the resultant of the two quadric functions

$$U = (a, b, c \mid x, y)^2, \qquad V = (a', b', c' \mid x, y)^2,$$

is

$$= \text{Disct.}.(U + kV) \;=\; \text{Disct.}.\big((ac - b^2,\;\; ac' + a'c - 2bb',\;\; a'c' - b'^2) \mid 1, k\big)^2,$$

which is Prof. Boole's ancient theorem referred to in my Fifth Memoir on Quantics[3], but which is now first exhibited in connexion with the general theory to which it belongs.

16. It may be noticed that the condition for a twofold critic centre, or (what is the same thing) a twofold factor of the Jacobian (which condition is of the degree $2(2n-3)$ in regard to the coefficients of U or V) is $RQ = 0$; and that we in fact have

$$2(2n-3) \;=\; n + 3(n-2).$$

This remark is due to Dr Salmon.

Article Nos. 17 to 42, relating to two Ternary Quantics.

17. Suppose now that $U = (a, \ldots \mid x, y, z)^n$, $V = (a', \ldots \mid x, y, z)^n$ are two ternary quantics of the same order n, and write $W = U + kV = (a + ka', \ldots \mid x, y, z)^n$, so that

$$W = U + kV = 0$$

is the equation of a curve in involution with the curves $U = 0$, $V = 0$. But for greater distinctness it is in general proper to retain $U + kV$ in place of W.

18. In order that the curve $U + kV = 0$ may have a node, we must have simultaneously

$$\partial_x(U + kV) = 0, \qquad \partial_y(U + kV) = 0, \qquad \partial_z(U + kV) = 0,$$

[3] *Phil. Trans.* vol. CXLVIII. (1858), pp. 415–427, [156].

and eliminating (x, y, z) from these equations we have

$$\text{Disct..}\,(U + kV) \;=\; 0,$$

which is an equation of the degree $3(n-1)^2$ as regards k, and of the same order as regards the coefficients of U, V conjointly.

19. To each of the $3(n-1)^2$ values of k there corresponds a point satisfying the conditions in question, and which is therefore a node of the corresponding nodal curve $U + kV = 0$; the points in question are the critic centres of the involution.

20. The critic centres may be differently obtained as follows; viz. if from the three equations we eliminate k, we find

$$\begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

a plexus of three curves, each of them of the order $2(n-1)$; any two of the three curves intersect in $4(n-1)^2$ points; but $(n-1)^2$ of these do not lie on the third curve; the remaining $3(n-1)^2$ of them lie on all three of the curves, and they are the critic centres of the involution.

21. More generally the critic centres lie on any curve whatever of the form

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

and any such curve, viz. any curve of the order $2(n-1)$ passing through the $3(n-1)^2$ critic centres, may be termed a *diacritic* curve.

22. For any one of the critic centres we have

$$\partial_x U : \partial_y U : \partial_z U = \partial_x V : \partial_y V : \partial_z V = -k : 1,$$

which gives the value of k corresponding to the point in question.

23. The condition in order that the equation in k may have a twofold root is

$$\text{Disct.. Disct..}\,(U + kV) = 0, \quad \text{or say} \quad D = 0,$$

where Disct.. Disct.. $(U + kV) = D$ is a function of the degree $2 \cdot 3(n-1)^2\{3(n-1)^2 - 1\}$ in regard to the coefficients of U, V conjointly; but it is separately homogeneous, and therefore of the degree $3(n-1)^2\{3(n-1)^2 - 1\}$ in regard to each set of coefficients.

24. To each of the critic centres there corresponds a value of k. Hence if two of the critic centres coincide, or say if there is a twofold critic centre, the equation in k will have a twofold root. Now first if the curves $U = 0$, $V = 0$ touch each other (have a point of contact or twofold intersection), then the diacritic curves will all touch (have a point of twofold intersection) at the point in question, which is therefore a twofold critic centre. It may be remarked in passing that the diacritic curves do not at the twofold critic centre touch the curves $U = 0$, $V = 0$. But secondly, the diacritic curves may touch at a point which is not a point of contact of the curves $U = 0$, $V = 0$. Such a point is a twofold critic centre. In each of these two cases the equation in k has a twofold root. Moreover, in the first case the curve $U + kV = 0$ corresponding to the twofold root has a node at the point of contact of the two curves $U = 0$, $V = 0$; in the second case the curve $U + kV = 0$ corresponding to the twofold root has the twofold centre (not a mere node but) a cusp. And thirdly, without any twofold critic centre, two distinct critic centres may give by the equations

$$\partial_x U : \partial_y U : \partial_z U = \partial_x V : \partial_y V : \partial_z V = -k : 1$$

the same value of k, and then the curve $U + kV = 0$ corresponding to such value of k is a curve having a node at each of the critic centres in question, that is, it has two nodes.

25. First, if the curves $U = 0$, $V = 0$ touch each other, then, (x, y, z) being the coordinates of the point of contact, we have

$$\partial_x(U + k_1V) = 0, \quad \partial_y(U + k_1V) = 0, \quad \partial_z(U + k_1V) = 0,$$

where k_1 denotes the value given by the equations

$$\partial_x U : \partial_y U : \partial_z U = \partial_x V : \partial_y V : \partial_z V = -k_1 : 1$$

belonging to the point of contact. It at once follows that every diacritic curve passes through the point in question. But it is somewhat more difficult to show that the diacritic curves touch at this point.

26. I represent for shortness the first and second differential coefficients of U by (L, M, N), (a, b, c, f, g, h), and similarly those of V by (L', M', N'), (a', b', c', f', g', h'), these values all belonging to the point of contact: we have therefore

$$L + k_1L' = 0, \quad M + k_1M' = 0, \quad N + k_1N' = 0.$$

The equation of the diacritic curve is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L & M & N \\ L' & M' & N' \end{vmatrix} = 0;$$

to find the tangent we must operate on the left-hand side with $X\partial_x + Y\partial_y + Z\partial_z$, where X, Y, Z are current coordinates. Calling the foregoing symbol D, this gives, after development and substitution *[see facsimile pp. 303–304 for full computations]*, the tangent equation

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L + \theta L' & M + \theta M' & N + \theta N' \\ DL + k_1DL' & DM + k_1DM' & DN + k_1DN' \end{vmatrix} = 0,$$

or, substituting for D its value, the equation of the tangent is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L + \theta L' & M + \theta M' & N + \theta N' \\ (a + ka', \ldots \mid X, Y, Z) & (h + kh', \ldots \mid X, Y, Z) & (g + kg', \ldots \mid X, Y, Z) \end{vmatrix} = 0.$$

27. Now if the diacritics touch, this equation should be independent of α, β, γ. Putting for shortness $a + k_1a' = a$, &c., and also taking as we may do $\theta = 0$, the parts multiplied by α, β, γ respectively are

$$M(gX + fY + cZ) - N(hX + bY + fZ),$$

$$N(aX + hY + gZ) - L(gX + fY + cZ),$$

$$L(hX + bY + fZ) - M(aX + hY + gZ),$$

and we ought therefore to have

$$Mg - Nh : Mf - Nb : Mc - Nf = Na - Lg : Nh - Lf : Ng - Lc = Lh - Ma : Lb - Mh : Lf - Mg.$$

These equations are in fact satisfied; for take any one of them, for example, the equation

$$\frac{Mg - Nh}{Na - Lg} = \frac{Mf - Nb}{Nh - Lf};$$

this is

$$MN(gh-af)-N^2(h^2-ab)-LM(fg-fg)+LN(fh-bg) \;=\; 0,$$

or, omitting the term in LM, and throwing out the factor N,

$$L(hf-bg)+M(gh-af)+N(ab-h^2) \;=\; 0.$$

But the equations $L+kL'=0$, &c., give

$$ax+hy+gz=0, \qquad hx+by+fz=0, \qquad gx+fy+cz=0,$$

that is

$$x:y:z \;=\; bc-f^2:fg-ch:hf-bg \;=\; fg-ch:ca-g^2:gh-af \;=\; hf-bg:gh-af:ab-h^2,$$

so that the above written equation is $Lx+My+Nz=0$, which is true in virtue of the equation $U=0$; and similarly for all the other equations which were to be verified.

28. It is to be noticed that the determination of the tangent of the diacritics depends only on the second differential coefficients (a,b,c,f,g,h), (a',b',c',f',g',h') of U, V. The tangent in question will be the same if instead of the curves $U=0$, $V=0$ we have the conics passing through the point of contact of the two curves, and whose tangents are coincident with those of the two curves $U=0$, $V=0$ respectively; that is, the conics touch at the point in question. They consequently intersect in two more points; the chord of intersection or line joining the last-mentioned two points meets the common tangent in a point; the polars of this point in regard to the two conics respectively pass through the point of contact, and moreover they are one and the same line; this line is the required tangent of the diacritics. The proof will be given, *post*, No. 41.

29. Let $R=0$ be the condition in order that the two curves $U=0$, $V=0$ may touch each other, or say let R be the Tactinvariant of U, V. When the curves U, V are of the degrees m, n respectively, then R is of the degrees $n(2m+n-3)$, $m(m+2n-3)$ in regard to the coefficients of U, V respectively. Hence in the present case where U, V are each of the degree n, R is of the degree $3n(n-1)$ in regard to each set of coefficients.

30. Secondly, if the functions U, V are such that there exists a curve $U+kV=0$ (say the curve $U+k_1V=0$) which has a cusp, then it is to be shown that $k=k_1$ is a twofold root of the equation in k; and to do this it has to be shown that the cusp is a twofold critic centre; or that the diacritic curves touch at the cusp: it may be added that the cuspidal tangent is the common tangent of the diacritic curves. Now the cuspidal curve being $U+k_1V=0$, then at the cusp the first derived functions $L+k_1L'$, $M+k_1M'$, $N+k_1N'$ vanish identically; and moreover the second derived functions $a+k_1a'$, … are such that (X,Y,Z) being any magnitudes whatever, $(a+k_1a',\ldots \mid X,Y,Z)^2$ is a perfect square, $=(\lambda X+\mu Y+\nu Z)^2$ suppose. Now X,Y,Z being current coordinates, and D denoting the operation $D=X\partial_x+Y\partial_y+Z\partial_z$, the equation of the tangent to the diacritic curve (by an investigation similar to that for this same tangent in the case first above considered) is found to be

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L+\theta L' & M+\theta M' & N+\theta N' \\ (a+k_1a',\ldots \mid X,Y,Z) & (h+k_1h',\ldots \mid X,Y,Z) & (g+k_1g',\ldots \mid X,Y,Z) \end{vmatrix} = 0,$$

and we have

$$(a+k_1a',\ldots \mid X,Y,Z) = \tfrac{1}{2}\partial_X(a+k_1a',\ldots \mid X,Y,Z)^2 = \tfrac{1}{2}\partial_X(\lambda X+\mu Y+\nu Z)^2 = \lambda(\lambda X+\mu Y+\nu Z),$$

and similarly the values of $(h + k_1h', \dots \mid X, Y, Z)$ and $(g + k_1g', \dots \mid X, Y, Z)$ are $= \mu(\lambda X + \mu Y + \nu Z)$ and $= \nu(\lambda X + \mu Y + \nu Z)$ respectively. Hence the equation of the tangent to the diacritic curve is

$$\lambda X + \mu Y + \nu Z = 0,$$

that is, the tangent being independent of the values of (α, β, γ) is the same for all the diacritic curves, and is the tangent at the cusp of the cuspidal curve $U + k_1V = 0$.

31. The conditions in order that the curve $W = U + kV = 0$ may have a cusp are given by a plexus equivalent to three relations between the coefficients $a + ka', \dots$ of W, and using for a moment β to denote the degree of the plexus or system, then eliminating k between the equations of the plexus we find between the coefficients $a, \dots$ of U and the coefficients $a', \dots$ of V an equation $Q = 0$ of the degree β in regard to the two sets of coefficients respectively. Conversely, given the equation $Q = 0$, we may find the plexus between the coefficients $a+ka', \dots$ of W. The value of β, as will be shown *post*, Annex, is

$$= 12(n - 1)(n - 2).$$

32. Thirdly, when the functions U, V are such that there exists a curve $U+kV = 0$ (suppose the curve $U + k_1V = 0$) which has a pair of nodes; each of these nodes is a critic centre, and (by means of the equation $-k : 1 = \partial_xU : \partial_xV$) gives the value k_1 of k, that is, k_1 is a twofold root of the equation in k.

33. The conditions in order that the curve $W = U + kV = 0$ may have a pair of nodes are given by a plexus of the degree α; then the coefficients being $a + ka', \dots$ if we eliminate k between the equations of the plexus, we find between the coefficients $a, \dots$ of U and $a', \dots$ of V an equation $P = 0$ of the degree α in the two sets of coefficients respectively. And conversely, given the equation $P = 0$, we may find the plexus between the coefficients $a + ka', \dots$ of W. I have not succeeded in finding directly the value of α, but only derive it from the equation $D = RQ^2P^2$, which if α had been found independently, would have been verified by means of such value of α; the value is

$$\alpha \;=\; \tfrac{1}{2} \cdot 3(n - 1)(n - 2)(3n^2 - 3n - 11).$$

34. The equation $D = 0$ is satisfied if $R = 0$, or if $Q = 0$, or if $P = 0$, and it may be seen that it is not satisfied in any other case. Hence D is made up of the factors P, Q, R, and I assume that its form is the same as in the case of a binary quantic, that is, that we have

$$D \;=\; RQ^2P^2.$$

35. Comparing the degrees of the two sides we have

$$3(n - 1)^2(3n^2 - 6n + 2) \;=\; 3n(n - 1) + 36(n - 1)(n - 2) + 2(n - 1)(n - 2)(3n^2 - 3n - 11);$$

or, what is the same thing,

$$(n - 1)(3n^2 - 6n + 2) \;=\; n + 12(n - 2) + (n - 2)(3n^2 - 3n - 11),$$

which is true, but, as just remarked, this equation itself was used to find the value

$$\alpha \;=\; \tfrac{1}{2} \cdot 3(n - 1)(n - 2)(3n^2 - 3n - 11).$$

36. Recapitulating, the equation in k will have a twofold root

1°. if $R = 0$, that is, if the curves $U = 0$, $V = 0$ touch each other, and in this case there is a twofold critic centre at the point of contact;

2°. if $Q = 0$, that is, if there be a curve $U + kV = 0$ having a cusp, and in this case the cusp is a twofold critic centre;

3°. if $P = 0$, that is, if there is a curve $U + kV = 0$ having a pair of nodes.

37. The three cases may be geometrically illustrated by supposing that the curves $U = 0$, $V = 0$ are in the first instance nearly, but not exactly, in the several relations in question.

First, if the curves $U = 0$, $V = 0$ are about to touch each other, that is, if there are two points of intersection about to coincide with each other. There are here two critic centres in the neighbourhood of the two points of intersection, and which, when the two points of intersection become a point of contact, coincide each with the point of contact.

Secondly, when the curves are such that there are two critic centres which become ultimately a twofold centre.

Thirdly, when the curves are such that there are two critic centres which remaining distinct from each other belong ultimately to the same critic curve.

38. The curves 1 and 2 in the left-hand figures respectively represent the nodal curves corresponding to slightly different values of k, which in the right-hand figures respectively give the curve corresponding to a twofold value of k. In the first pair of figures the curves $U = 0$ and $V = 0$, about to touch in the left-hand figure, touching in the right-hand figure, are shown by dotted lines. It will be observed that in the second case in the left-hand figure the two nodes which give rise to a cusp are the one of them an acnode and the other a crunode; this is in fact the only mode of drawing the figure so that a cusp shall present itself. The transition of form is one of ordinary occurrence in cubic curves and in curves of a higher order; thus if $y^2 = (x - a)(x - b)(x - c)$, where $a < b < c$, then if $a = b$, we have an acnodal curve, if $b = c$ a crunodal curve, and if $a = b = c$ a cuspidal curve.

39. In the case of two conics, $n = 2$. We have here simply $D = R$, where $R = 0$ is the condition in order that the two conics may touch each other. The nodal curves are of course the three pairs of lines passing through the points of intersection of the two conics, and the nodes of these curves, or critic centres, are the centres of the quadrangle formed by the four points in question; or, what is the same thing, they are the system of conjugate points common to the two conics, viz. the points which are such that the polar of one of them taken with respect to either of the conics is the line joining the other two of them. The diacritics are any conics passing through the three points.

40. If the two conics are

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0, \qquad (a', b', c', f', g', h' \mid x, y, z)^2 = 0,$$

and if the determinant formed with $a + ka'$, &c., is denoted by $(K, \Theta, \Theta', K' \mid 1, k)^3$, so that

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

$$\Theta = a'(bc - f^2) + \text{\&c.},$$

$$\Theta' = a(b'c' - f'^2) + \text{\&c.},$$

$$K' = a'b'c' - a'f'^2 - b'g'^2 - c'h'^2 + 2f'g'h',$$

then the before-mentioned equation $D = R = 0$, which gives the condition that the conics may touch, is

$$\text{Disct.}\,.\,(K, \Theta, \Theta', K' \mid 1, k)^3 \;=\; 0,$$

where the left-hand side is of the order 6 in the coefficients of the two conics respectively: this is a known formula.

41. If the equation in k have a twofold root, the two conics will touch: two of the critic centres will then coincide at the point of contact, or this point is a twofold critic centre: the remaining or onefold critic centre is the intersection of the common tangent and of the line joining the two points of intersection of the conics. In virtue of the general property, the first-mentioned two centres must be considered as lying on the line which is the polar of the onefold critic centre in regard to either of the conics. The diacritics pass through the critic centres, that is, they pass through the onefold centre, and touch the polar in question at the point of contact of the two conics, or twofold critic centre; this is in fact the property mentioned *ante*, No. 28.

42. The equation

$$\text{Disct.. Disct..}(U + kV) \;=\; 0,$$

as applied to a conic and a circle leads at once to the equation of the curve parallel to a given conic; such parallel curve is in fact the envelope of the circles of a given constant radius which touch the given conic. This method is in effect due to Dr Salmon, who applied the corresponding theorem *in solido* to the determination of the surface parallel to an ellipsoid.

Annex, referred to No. 31. Investigation of the order of the plexus or system for the existence of a Cusp.

Considering for a moment the curve

$$U = (* \mid x, y, z)^n = 0, \quad \text{let } (L, M, N),\ (a, b, c, f, g, h)$$

be the first and second differential coefficients of U; (A, B, C, F, G, H) the inverse system, viz. $A = bc - f^2$, &c. At a cusp we have

$$L = 0,\ M = 0,\ N = 0,\ A = 0,\ B = 0,\ C = 0,\ F = 0,\ G = 0,\ H = 0,$$

a system of equations which is contained in the system $L = 0$, $M = 0$, $N = 0$, $A = 0$. But this system contains besides the cusp system the irrelevant system $L = 0$, $M = 0$, $N = 0$, $x^2 = 0$. In fact the equations $L = 0$, $M = 0$, $N = 0$ give

$$ax + hy + gz = 0, \quad hx + by + fz = 0, \quad gx + fy + cz = 0,$$

and thence

$$x^2 : y^2 : z^2 : yz : zx : xy = A : B : C : F : G : H.$$

Hence the equation $A = 0$, if $x^2 = 0$, implies the entire system

$$A = 0,\ B = 0,\ C = 0,\ F = 0,\ G = 0,\ H = 0.$$

But if $L = 0$, $M = 0$, $N = 0$, $x^2 = 0$, then these equations give $A = 0$ (and also $H = 0$, $G = 0$), but they do not give the remaining equations $B = 0$, $C = 0$, $F = 0$. Or the same thing may be shown in a less symmetrical form, but more clearly thus; we have identically

$$-cx(ax+hy+gz)+(hx-fz)(hx+by+fz)+hz(gx+fy+cz)-(bc-f^2)z^2+(ab-h^2)x^2 = 0,$$

whence the equations $L = 0$, $M = 0$, $N = 0$, $A = 0$ give $Cx^2 = 0$, that is, $C = 0$, or $x^2 = 0$. But the equations $L = 0$, $M = 0$, $N = 0$, $A = 0$, $C = 0$ give (as it is easy to show) the entire system $A = 0$, $B = 0$, $C = 0$, $F = 0$, $G = 0$, $H = 0$. That is, the system

$$L = 0,\ M = 0,\ N = 0,\ A = 0$$

is made up of the cusp system, and of the system ($L = 0,\ M = 0,\ N = 0,\ A = 0,\ x^2 = 0$); or since $A = 0$ is a consequence of the other equations, the second system is

$$(L = 0,\ M = 0,\ N = 0,\ x^2 = 0).$$

Consider now the curve $\lambda U + \mu U' + \nu U'' = 0$, which will have a cusp if the ratios $\lambda : \mu : \nu$ are properly determined. And to each set of values of $\lambda : \mu : \nu$ there corresponds a set of values (x, y, z), the coordinates of a cusp of the curve; so that the number of such sets, that is, the number of points each whereof is the cusp of a corresponding curve $\lambda U + \mu V + \nu W = 0$ is precisely equal to the number of sets of values of $\lambda : \mu : \nu$; or it is equal to the order of the system of conditions for the existence of a cusp.

Denoting as before the first and second differential coefficients of U by $L, M, N, a, b, c, f, g, h$, and those of U', U'' in a corresponding manner, and taking for the cusp the system before represented by $L = 0$, $M = 0$, $N = 0$, $A = 0$, we have

$$\lambda L + \mu L' + \nu L'' = 0,$$

$$\lambda M + \mu M' + \nu M'' = 0,$$

$$\lambda N + \mu N' + \nu N'' = 0,$$

$$(\lambda b + \mu b' + \nu b'')(\lambda c + \mu c' + \nu c'') - (\lambda f + \mu f' + \nu f'')^2 = 0,$$

which last equation, to denote that it is of the second order in regard to the differential coefficients a, b, &c., a', &c., I represent by

$$((a, b, \ldots) \mid \lambda, \mu, \nu)^2 = 0.$$

But this system of four equations contains not only the cusp system, but the system made of the three linear equations and the equation $x^2 = 0$. Eliminating λ, μ, ν, the last-mentioned system is

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0, \qquad x^2 = 0,$$

where the first equation is that of a curve of the order $3(n - 1)$. And the two equations give together $6(n - 1)$ points, viz. the points of intersection of the curve by the line $x = 0$, each reckoned as a twofold point.

Returning to the first-mentioned system, this may be replaced by

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0, \qquad ((a, b, \ldots) \mid L'M'' - L''M',\ L''M - LM'',\ LM' - L'M)^2 = 0,$$

which are curves of the orders $3(n - 1)$ and $6n - 8$ respectively. But each of these curves passes through the $3(n - 1)^2$ points given by the equations

$$L = 0,\ M = 0,\ N = 0;$$

and these points are moreover nodes on the curve of the order $6n-8$; hence the points in question reckon as $6(n - 1)^2$ intersections of the two curves. The number of the remaining intersections is

$$3(n - 1)(6n - 8) - 6(n - 1)^2 \ = \ 6(n - 1)\big(3n - 4 - (n - 1)\big) \ = \ 6(n - 1)(2n - 3),$$

but among these are included the $6(n-1)$ intersections of the curve of the order $3(n-1)$ by the twofold line $x^2=0$; or, subtracting these, the number of the remaining points is

$$6(n-1)(2n-3-1) \;=\; 12(n-1)(n-2);$$

which number is consequently the order of the cusp system.

It may be remarked that considering the entire series of equations at first denoted by $(L=0,\ M=0,\ N=0)$, $(A=0,\ B=0,\ C=0,\ F=0,\ G=0,\ H=0)$, the elimination of λ,μ,ν from the three linear equations gives as before

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0,$$

which is a curve of the order $3(n-1)$: and the eliminating of $\lambda^2,\mu^2,\nu^2,\ \mu\nu,\nu\lambda,\lambda\mu$ from the six quadric equations gives

$$\left|bc-f^2,\ \ b'c'-f'^2,\ \ b''c''-f''^2,\ \ b'c''+b''c'-2f'f'',\ \ b''c+bc''-2ff'',\ \ bc'+b'c-2ff'\right|=0,$$

which is a curve of the order $12(n-2)$; the two curves would intersect in $36(n-1)(n-2)$ points, but as this is precisely three times the number $12(n-1)(n-2)$, I infer that these are in fact the $12(n-1)(n-2)$ points three times repeated, that is, that each of these is a point of threefold intersection of the two curves.

Cambridge, 7th November, 1863.

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349. On a Case of the Involution of Cubic Curves *(begins)*

[From the Transactions of the Cambridge Philosophical Society, vol. XI. Part I. (1866), pp. 39–80.—Read 22 February, 1864.]

The present memoir relates to the involution

$$xyz+k(x+y+z)^2(\lambda x+\mu y+\nu z) \;=\; 0,$$

viz. treating x,y,z as coordinates, and k as a variable parameter, this equation represents the series of cubic curves passing through the intersections of the two cubics

$$xyz=0, \qquad (x+y+z)^2(\lambda x+\mu y+\nu z)=0;$$

or, what is the same thing, the line $x+y+z=0$ meets any cubic of the series in three points the tangents at which are $x=0$, $y=0$, $z=0$, and these tangents again meet the cubic in three points lying on the line $\lambda x+\mu y+\nu z=0$; so that in the language which I have used elsewhere, the lines $x+y+z=0$, $\lambda x+\mu y+\nu z=0$ are in regard to the cubic a *primary* and a *satellite* line respectively. The investigation (which is a development of two short papers already published in the *Philosophical Magazine*)[4] was undertaken in order to applying it to the explanation and discussion of Plücker's Classification of Curves of the Third Order; but such application will properly be made in a separate memoir, *On the Classification of Cubic Curves*, and it has also

[4] *On the Cubic Centres of a Line with respect to Three Lines and a Line.—Phil. Mag.* t. XX. pp. 418–423 (1860), [257]. *Ditto, Second Paper*, t. XXII. pp. 433–436 (1861), [315].

appeared to me convenient to give therein the discussion of the geometrical forms of certain loci which present themselves in the present memoir.

I remark that the involution intended to be here considered is a case of the more general one $U + kV = 0$, where $U = 0$, $V = 0$ are any two cubic curves whatever. It appears from my memoir *On the Theory of Involution*, [348], that the equation $\text{Disct.}\,.\,(U + kV) = 0$, which determines the critic values of k is in the general case of the order 12; the special case is however in the present memoir treated irrespectively of the general one, and the equation for the critic values of k is found to be of the order 3; this of course means that the equation of the order 12 breaks up into two equations of the orders 9 and 3 respectively, but I have not attempted to show how the decomposition and reduction arise. Moreover, in the general case the equation $\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = 0$, which is the condition for the existence of a twofold critic value, presents itself in the form $RQ^2P^2 = 0$, where $R = 0$ is the condition that the two cubics $(U = 0, V = 0)$ shall touch each other; $Q = 0$ the condition that there shall be in the involution $U + kV = 0$ a curve having (not a mere node but) a cusp; and $P = 0$ the condition that there shall be a curve having two nodes, or (what is the same thing) breaking up into a line and conic. But in the special case, which, as already noticed, is here considered irrespectively of the general one, the equation $\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = 0$, for the existence of a twofold critic value presents itself in the reduced form $Q = 0$, giving the condition that corresponding to the twofold critic value there shall be a curve having (not a mere node but) a cusp.

Article Nos. 1 to 18, Explanations, Definitions, and Results.

1. I consider the involution

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \;=\; 0,$$

where $x = 0$, $y = 0$, $z = 0$, $x + y + z = 0$ may be considered as representing any four lines no three of which meet in a point, and $\lambda x + \mu y + \nu z = 0$ as representing any fifth line whatever: k is a variable parameter. The lines $x + y + z = 0$, $\lambda x + \mu y + \nu z = 0$, are a primary line and a satellite line of any cubic of the series, viz. the tangents $x = 0$, $y = 0$, $z = 0$ at the points of intersection with the primary line $x + y + z = 0$ meet the cubic in three points lying on the satellite line $\lambda x + \mu y + \nu z = 0$.

2. A *critic value* of k is a value for which the corresponding curve

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \;=\; 0$$

has a node; and such node, or say rather the site of such node, is a *critic centre*.

3. The critic values of k are in effect determined by a cubic equation, and the coordinates of the critic centre are then given rationally in terms of k; there are consequently three critic values of k; and the same number of critic centres, and of nodal curves: it is however found to be convenient to express as well the critic value of k, as the coordinates of the critic centre, rationally in terms of an auxiliary parameter θ, which is given by a cubic equation.

4. The cubic equation in k (or what is the same thing, that in θ) may have a twofold root (pair of equal roots); or, say rather, it may have a twofold root and a one-with-twofold root: corresponding to the twofold value of k we have a twofold critic centre, which is not a mere node but a cusp on the cubic, or instead of a merely nodal cubic we have a cuspidal cubic; and corresponding to the one-with-twofold value of k we have a one-with-twofold critic centre, being of course a mere node on the nodal cubic.

5. In the case in question of a twofold and one-with-twofold value of k, the line $\lambda x+\mu y+\nu z = 0$, or say the satellite line, envelopes a curve which might be termed the twofold and one-with-twofold envelope, but which is spoken of simply as the *envelope*.

The locus of the twofold centre is a curve which is called the *twofold centre locus.*

The locus of the one-with-twofold centre is a curve which is called the *one-with-twofold centre locus.*

These definitions premised, the following results may be stated:

6. The equation in θ may be represented in the three equivalent forms

$$\frac{1}{\theta+\lambda}+\frac{1}{\theta+\mu}+\frac{1}{\theta+\nu}-1=0, \qquad \frac{1}{\theta}=\frac{\theta}{\theta+\lambda}+\frac{\theta}{\theta+\mu}+\frac{\theta}{\theta+\nu},$$

$$\theta^3-\theta(\mu\nu+\nu\lambda+\lambda\mu)-2\lambda\mu\nu=0.$$

7. The critic value of k and the coordinates of the critic centre are then given by the equations

$$k=\frac{-\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)},$$

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

8. The condition for a twofold and one-with-twofold value of k is

$$\sqrt{\lambda}+\sqrt{\mu}+\sqrt{\nu}=0,$$

or, what is the same thing,

$$(\mu\nu+\nu\lambda+\lambda\mu)^2-27\,\lambda^2\mu^2\nu^2=0,$$

which equations may either of them be considered as the line-equation of the envelope. The equation in the coordinates (x,y,z), or point-equation of the envelope is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}=0,$$

or, in its rationalised form,

$$x^4+y^4+z^4-4(yz^3+y^3z+zx^3+z^3x+xy^3+x^3y)+6(y^2z^2+z^2x^2+x^2y^2)-124(x^2yz+xy^2z+xyz^2)=0.$$

9. The equation of the twofold centre locus is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}=0,$$

or, in its rationalised form,

$$x^2+y^2+z^2-2yz-2zx-2xy=0;$$

the curve is therefore a conic, and it may be spoken of as the *twofold centre conic.*

10. The equation of the one-with-twofold centre locus is

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0;$$

the curve is therefore a cubic, and it may be spoken of as the *one-with-twofold centre cubic.*

11. The before-mentioned equation $\lambda^{-1/2}+\mu^{-1/2}+\nu^{-1/2}=0$ is satisfied by

$$\lambda:\mu:\nu=\alpha^{-2}:\beta^{-2}:\gamma^{-2},$$

where $\alpha+\beta+\gamma=0$, and it is very convenient to introduce these quantities α,β,γ into the formulae.

12. The equation of the satellite line giving a twofold and one-with-twofold centre is

$$\frac{x}{\alpha^2} + \frac{y}{\beta^2} + \frac{z}{\gamma^2} = 0,$$

the coordinates of the point of contact with the envelope are $x : y : z = \alpha^4 : \beta^4 : \gamma^4$.

The equation in θ gives $\theta_1 = \theta_2 = -\frac{1}{\alpha\beta\gamma}$ for the values corresponding to the twofold centre; and $\theta_3 = \frac{2}{\alpha\beta\gamma}$ for the value corresponding to the one-with-twofold centre.

The coordinates of the twofold centre, or cusp, are $x : y : z = \alpha^3 : \beta^3 : \gamma^3$.

The coordinates of the one-with-twofold centre, or node, are

$$x : y : z = \alpha^2(\beta - \gamma) : \beta^2(\gamma - \alpha) : \gamma^2(\alpha - \beta).$$

The equation of the tangent at the cusp is

$$(\beta - \gamma)\frac{x}{\alpha} + (\gamma - \alpha)\frac{y}{\beta} + (\alpha - \beta)\frac{z}{\gamma} = 0.$$

The equation of the line joining the cusp and the node, which line is also one of the tangents at the node, is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0.$$

The equation of the other tangent at the node is

$$(2\beta\gamma + \alpha^2)\frac{x}{\alpha} + (2\gamma\alpha + \beta^2)\frac{y}{\beta} + (2\alpha\beta + \gamma^2)\frac{z}{\gamma} = 0.$$

13. Considering the critic centre corresponding to a root θ of the cubic equation, the equation of the line joining the other two critic centres is

$$\frac{\lambda x}{\theta + \lambda} + \frac{\mu y}{\theta + \mu} + \frac{\nu z}{\theta + \nu} = 0,$$

which is the polar of the critic centre in regard to the twofold centre conic. The critic centres are consequently conjugate poles in regard to the twofold centre conic.

14. The equation of the tangents at the critic centre considered as a node of the corresponding cubic curve is

$$\left(\theta + 4\lambda,\ \theta + 4\mu,\ \theta + 4\nu,\ -\theta - \tfrac{2\mu\nu}{\theta},\ -\theta - \tfrac{2\nu\lambda}{\theta},\ -\theta - \tfrac{2\lambda\mu}{\theta}\right)(x, y, z)^2 = 0.$$

15. The last-mentioned formulae lead to some which involve the three critic centres viz. if $X = 0$, $Y = 0$, $Z = 0$ are the equations of the sides of the triangle formed by the critic centres, then the equations of the tangents at the three critic centres respectively are of the form

$$BY^2 + CZ^2 = 0, \qquad AX^2 + CZ^2 = 0, \qquad AX^2 + BY^2 = 0,$$

so that the tangents in question are in fact the tangents from the three nodes respectively to the conic

$$AX^2 + BY^2 + CZ^2 = 0 :$$

the three nodes or critic centres being thus conjugate poles in regard to the conic; this is called "the three-centre conic."

16. The equation of a nodal cubic is also expressible in a simple form in terms of the new coordinates X, Y, Z. In the formulae for these transformations, and indeed throughout the memoir, the three roots of the equation in θ are represented by $\theta_1, \theta_2, \theta_3$, and I write also

$$l_1 = \theta_2 - \theta_3, \qquad l_2 = \theta_3 - \theta_1, \qquad l_3 = \theta_1 - \theta_2,$$

$$e_i = (\theta_i + \lambda)(\theta_i + \mu)(\theta_i + \nu) \quad (i = 1, 2, 3).$$

17. If $\lambda a + \mu b + \nu c = 0$, that is, if (a, b, c) are the coordinates of a point on the line $\lambda x + \mu y + \nu z = 0$, then the critic centres lie on the cubic

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} - \frac{2(a+b+c)}{x+y+z} = 0,$$

or, what is the same thing, this curve is the locus of the critic centres corresponding to the several lines $\lambda x + \mu y + \nu z = 0$ through the point (a, b, c).

18. In particular, taking in succession for the point (a, b, c) the point of intersection of the line $\lambda x + \mu y + \nu z = 0$ with the lines $x = 0$, $y = 0$, $z = 0$, $x + y + z = 0$, the critic centres lie on the conics

$$\frac{\mu}{y} + \frac{\nu}{z} - \frac{2(\mu+\nu)}{x+y+z} = 0,$$
$$\frac{\nu}{z} + \frac{\lambda}{x} - \frac{2(\nu+\lambda)}{x+y+z} = 0,$$
$$\frac{\lambda}{x} + \frac{\mu}{y} - \frac{2(\lambda+\mu)}{x+y+z} = 0,$$
$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

which are useful for the construction of the critic centres for a given line $\lambda x + \mu y + \nu z = 0$. The last of the four conics passes through the point $(1, 1, 1)$, which is the harmonic of the line $x + y + z = 0$ in regard to the triangle formed by the lines $x = 0$, $y = 0$, $z = 0$; and I call it the *harmonic conic.*

Article Nos. 19 to 21. General Formulae for the Critic Centres.

19. I consider the involution

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) = 0.$$

Writing the equation in the form

$$k(\lambda x + \mu y + \nu z) = \frac{-xyz}{(x+y+z)^2},$$

and differentiating with regard to x, y, z respectively, we obtain

$$\begin{aligned} -k\lambda(x+y+z)^3 &= yz(-x+y+z), \\ -k\mu(x+y+z)^3 &= zx(\ \ x-y+z), \\ -k\nu(x+y+z)^3 &= xy(\ \ x+y-z), \end{aligned}$$

which determine the coordinate ratios $x : y : z$ of the node or critic centre; and the corresponding value of k.

20. Writing the equations under the form

$$\frac{-k(x+y+z)^3}{xyz} = \frac{-x+y+z}{\lambda x} = \frac{x-y+z}{\mu y} = \frac{x+y-z}{\nu z} = \frac{2}{\theta},$$

where θ is an auxiliary parameter to be determined, we find

$$x\Big(1 + \tfrac{2\lambda}{\theta}\Big) + y + z = 2x\Big(1 + \tfrac{\lambda}{\theta}\Big),$$

that is $x+y+z=2x\big(1+\frac{\lambda}{\theta}\big)$, and consequently

$$x+y+z \;=\; 2x\Big(1+\tfrac{\lambda}{\theta}\Big) \;=\; 2y\Big(1+\tfrac{\mu}{\theta}\Big) \;=\; 2z\Big(1+\tfrac{\nu}{\theta}\Big),$$

and we have thence an equation which may also be written in the form

$$\frac{1}{\theta+\lambda}+\frac{1}{\theta+\mu}+\frac{1}{\theta+\nu}-\frac{1}{\theta} \;=\; 0,$$

that is,

$$\theta^3-\theta(\mu\nu+\nu\lambda+\lambda\mu)-2\lambda\mu\nu \;=\; 0;$$

and we then have

$$k \;=\; -\frac{\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)}.$$

21. We see that θ is determined by a cubic equation, and that the ratios $x:y:z$ and the parameter k are rational functions of θ. There are thus three nodes or critic centres, and the like number of nodal curves and of critic values of k.

The form secondly obtained for the equation in θ shows that we may write

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z \;=\; \frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

Article Nos. 22 to 32, relating to a Twofold and a One-with-Twofold Centre.

22. If k has a twofold and a one-with-twofold value, then θ will have also a twofold and a one-with-twofold value; and conversely. The equation in θ will have a twofold and a one-with-twofold root if

$$(\mu\nu+\nu\lambda+\lambda\mu)^3-27\,\lambda^2\mu^2\nu^2=0;$$

or, what is the same thing, if

$$\mu\nu+\nu\lambda+\lambda\mu-3(\lambda\mu\nu)^{2/3} \;=\; 0,$$

or finally if

$$\sqrt{\lambda}+\sqrt{\mu}+\sqrt{\nu} \;=\; 0,$$

so that the condition is satisfied if $\lambda=\alpha^{-2}$, $\mu=\beta^{-2}$, $\nu=\gamma^{-2}$ where $\alpha+\beta+\gamma=0$. In fact with these values the equation in θ becomes

$$(\alpha\beta\gamma\,\theta)^3-3\alpha\beta\gamma\,\theta-2=0,$$

that is

$$(\alpha\beta\gamma\,\theta+1)^2(\alpha\beta\gamma\,\theta-2)=0,$$

so that the twofold value is $\theta_1=\theta_2=-\frac{1}{\alpha\beta\gamma}$; and the one-with-twofold value is $\theta_3=\frac{2}{\alpha\beta\gamma}$.

23. It is throughout assumed that the quantities α,β,γ satisfy the condition $\alpha+\beta+\gamma=0$. The result just obtained shows that the line

$$\frac{x}{\alpha^2}+\frac{y}{\beta^2}+\frac{z}{\gamma^2}=0$$

is a twofold and one-with-twofold satellite line. From this equation, considering α,β,γ as variable parameters satisfying the condition $\alpha+\beta+\gamma=0$, we find at once the equation of the curve enveloped by the line in question, which curve is called simply the envelope—viz. the coordinates of the point of contact are found to be $x : y : z = \alpha^4 : \beta^4 : \gamma^4$, and thence the equation of the envelope is $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$, or rationalising, it is

$$x^4+y^4+z^4-4(yz^3+y^3z+zx^3+z^3x+xy^3+x^3y)+6(y^2z^2+z^2x^2+x^2y^2)-124(x^2yz+y^2zx+z^2xy) = 0.$$

The before-mentioned equation $\lambda^{-1/2} + \mu^{-1/2} + \nu^{-1/2} = 0$, or

$$(\mu\nu + \nu\lambda + \lambda\mu)^3 - 27\,\lambda^2\mu^2\nu^2 = 0,$$

may be considered as the tangential equation; the envelope is thus of the order 4, and the class 6.

24. It is easy to show that the curve has three nodes the coordinates whereof are $(-4, 1, 1)$, $(1, -4, 1)$, $(1, 1, -4)$; and this being known, the equation may be transformed so as to put the nodes in evidence. I effect the transformation synthetically as follows, viz. writing $x+y+z = \sigma$, $yz + zx + xy = q$, $xyz = r$, the equation of the curve is

$$(\sigma^4 - 2q\sigma^2 + 2q^2 + 4r\sigma) - 4(q\sigma^2 - 2q^2 - r\sigma) + 6(q^2 - 2r\sigma) - 124(r\sigma) = 0,$$

viz. it is

$$\sigma^4 - 6q\sigma^2 + 16q^2 - 128r\sigma = 0,$$

which is $(\sigma^2 + 4q)^2 - 16\sigma(3\sigma^2 + 4q\sigma + 8r) = 0$, or, putting for a moment $l = -\frac{16}{5}$, and therefore $5(l - 2) = -16$, it is

$$(\sigma^2 + 4q)^2 + (l - 2)\,5\sigma(3\sigma^2 + 4q\sigma + 8r) = 0.$$

Now writing $x' = \sigma + 2x = 3x + y + z$, $y' = \sigma + 2y = x + 3y + z$, $z' = \sigma + 2z = x + y + 3z$, we find

$$y'z' + z'x' + x'y' = 7\sigma^2 + 4q, \qquad x' + y' + z' = 5\sigma, \qquad x'y'z' = 3\sigma^3 + 4q\sigma + 8r,$$

so that the equation is

$$(y'z' + z'x' + x'y')^2 + (l - 2)\,x'y'z'(x' + y' + z') = 0,$$

that is

$$y'^2z'^2 + z'^2x'^2 + x'^2y'^2 + l\,x'y'z'(x' + y' + z') = 0;$$

or, putting for l its value, the equation is

$$5(y'^2z'^2 + z'^2x'^2 + x'^2y'^2) - 16\,x'y'z'(x' + y' + z') = 0;$$

or, as this may also be written,

$$(5,\ 5,\ 5,\ -3,\ -3,\ -3 \mid y'z',\ z'x',\ x'y')^2 = 0;$$

a form which shows that the curve has three nodes at the angles of the triangle $x' = 0$, $y' = 0$, $z' = 0$.

25. It is easy to see that the curve is touched by the lines $x = 0$, $y = 0$, $z = 0$ at their intersections with the lines $y - z = 0$, $z - x = 0$, $x - y = 0$ respectively, or (what is the same thing) in the points $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ respectively. It may be added that the line $y - z = 0$ meets the curve in the node $(-4, 1, 1)$, being of course a point of twofold intersection, in the point $(0, 1, 1)$ on the line $x = 0$, and besides in the point $(16, 1, 1)$: and the like for the lines $z - x = 0$ and $x - y = 0$.

26. It may be noticed that although any line passing through one of the nodes is in a sense a tangent to the envelope, yet that it is not a proper tangent and does not give rise to a twofold centre. It is in fact shown (*post*, Nos. 73 and 74) that the critic centres for a line $\lambda x+\mu y+\nu z=0$ passing through the point $(-4,1,1)$ are three points lying, one of them on the line $y+z=0$, and the other two on the conic $x(x+y+z)-6yz=0$.

27. Assume that θ has its twofold value $=-\frac{1}{\alpha\beta\gamma}$; the equations

$$x:y:z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu},$$

substituting therein for λ,μ,ν the values $\alpha^{-2},\beta^{-2},\gamma^{-2}$, give for the coordinates of the twofold centre

$$x:y:z=\frac{1}{\beta\gamma-\alpha^2}:\frac{1}{\gamma\alpha-\beta^2}:\frac{1}{\alpha\beta-\gamma^2};$$

but in virtue of the relation $\alpha+\beta+\gamma=0$ we have

$$\beta\gamma-\alpha^2=\beta\gamma+\gamma\alpha+\alpha\beta=\gamma\alpha-\beta^2=\alpha\beta-\gamma^2;$$

or the values are $x:y:z=\alpha^3:\beta^3:\gamma^3$. Hence also we have as the equation of the locus of the twofold centre, $\sqrt{x}+\sqrt{y}+\sqrt{z}=0$, or, what is the same thing,

$$x^2+y^2+z^2-2yz-2zx-2xy=0,$$

which is a conic touching the lines $x=0$, $y=0$, $z=0$ at their intersections with the lines $y-z=0$, $z-x=0$, $x-y=0$ respectively, or, what is the same thing, in the points $(0,1,1)$, $(1,0,1)$, $(1,1,0)$ respectively.

28. Similarly, if θ has its one-with-twofold value $=\frac{2}{\alpha\beta\gamma}$, the equations

$$x:y:z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu},$$

substituting therein for λ,μ,ν the values $\alpha^{-2},\beta^{-2},\gamma^{-2}$, give for the coordinates of the one-with-twofold centre

$$x:y:z=\frac{1}{2\alpha^2+\beta\gamma}:\frac{1}{2\beta^2+\gamma\alpha}:\frac{1}{2\gamma^2+\alpha\beta};$$

but in virtue of $\alpha+\beta+\gamma=0$ we have

$$2\alpha^2+\beta\gamma=\alpha^2-\alpha(\beta+\gamma)+\beta\gamma=(\alpha-\beta)(\alpha-\gamma)=-(\gamma-\alpha)(\alpha-\beta),$$

and similarly

$$2\beta^2+\gamma\alpha=-(\alpha-\beta)(\beta-\gamma),\qquad 2\gamma^2+\alpha\beta=-(\beta-\gamma)(\gamma-\alpha);$$

and thence these values are

$$x:y:z=\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta),$$

for the coordinates of the one-with-twofold centre.

We thence deduce

$$y+z=\beta^2\gamma-\beta^3+\gamma^2\alpha-\gamma^3\beta=(\beta\gamma-\alpha\beta-\alpha\gamma)(\beta-\gamma)=(\beta\gamma+\alpha^2)(\beta-\gamma),$$

and consequently $-x+y+z=\beta\gamma(\beta-\gamma)$, that is

$$x:y:z:-x+y+z:x-y+z:x+y-z=\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta):\beta\gamma(\beta-\gamma):\gamma\alpha(\gamma-\alpha):\alpha\beta(\alpha-\beta),$$

and these give

$$-(-x+y+z)(x-y+z)(x+y-z)+xyz=0,$$

or, what is the same thing,

$$x^3 + y^3 + z^3 - (yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y) + 3xyz = 0,$$

as the equation of the locus of the one-with-twofold centre, which locus is thus a cubic curve.

29. The equation

$$x^3 + y^3 + z^3 - (yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y) + 3xyz = 0$$

of the one-with-twofold centre locus may be transformed as follows, viz. writing for a moment $x + y + z = -w$, we have

$$\begin{aligned}(9x + 4w)(9y + 4w)(9z + 4w) - w^3 &= 729\,xyz + 324\,w(yz + zx + xy) - 144w^2 + 64w^3 - w^3,\\ &= 81\{9xyz + 4w(yz + zx + xy) - w^3\},\\ &= 81\big[9xyz - 12xyz - 4(yz^2 + \text{\&c.}) + (x^3 + y^3 + z^3) - (3yz^2 + \text{\&c.}) + 6xyz\big],\\ &= 81\{x^3 + y^3 + z^3 - (yz^2 + \text{\&c.}) + 3xyz\},\end{aligned}$$

so that the equation may be written

$$(9x + 4w)(9y + 4w)(9z + 4w) - w^3 = 0,$$

or, what is the same thing,

$$(5x - 4y - 4z)(-4x + 5y - 4z)(-4x - 4y + 5z) + (x + y + z)^3 = 0,$$

which shows that the intersections of the line $x+y+z = 0$ with the sides $x = 0$, $y = 0$, $z = 0$ of the triangle are inflexions on the curve; and that the tangents at these points are respectively $5x - 4y - 4z = 0$, $-4x + 5y - 4z = 0$, $-4x - 4y + 5z = 0$.

30. The curve passes through the point $(1, 1, 1)$, which is the harmonic of $x + y + z = 0$ in regard to the triangle; and this point is moreover a node on the curve; in fact if the equation be represented by $W = 0$, then we have

$$\partial_x W = 3x^2 - 2x(y + z) - y^2 + 3yz - z^2,$$

$= 0$ for the point in question; and similarly $\partial_y W = 0$, and $\partial_z W = 0$.

31. The equation for the twofold centre conic may also be obtained as follows; viz. the equations

$$\frac{-x + y + z}{\lambda x} = \frac{x - y + z}{\mu y} = \frac{x + y - z}{\nu z} = \frac{2}{\theta},$$

give

$$(-x + y + z)(x - y + z)(x + y - z) = \tfrac{8}{\theta^3}\, xyz\, \lambda\mu\nu,$$

which substituting for θ the twofold value $= -\frac{1}{\alpha\beta\gamma}$, gives

$$(-x + y + z)(x - y + z)(x + y - z) + 8xyz = 0,$$

an equation which may be written

$$(x + y + z)^2(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) = 0,$$

which is the former result affected by the extraneous factor $x + y + z$.

If, instead, we substitute for θ the onefold value $= \frac{2}{\alpha\beta\gamma}$, we find

$$-(-x + y + z)(x - y + z)(x + y - z) + xyz = 0,$$

or, what is the same thing,

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0,$$

which is the one-with-twofold centre cubic.

32. Recollecting that

$$k=-\frac{\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)},$$

we deduce for the twofold value of k

$$k_1=k_2=-\frac{2\alpha^2\beta^2\gamma^2}{(\beta\gamma+\gamma\alpha+\alpha\beta)^3},$$

and for the one-with-twofold value

$$k_3=\frac{4\alpha\beta\gamma}{(2\alpha^2+\beta\gamma)(2\beta^2+\gamma\alpha)(2\gamma^2+\alpha\beta)}=-\frac{4\alpha\beta\gamma}{(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2}.$$

Article Nos. 33 to 38, relating to the Tangents at a Node or Critic Centre.

33. I proceed to investigate the equation of the tangents at the node of the curve

$$xyz+k(x+y+z)^2(\lambda x+\mu y+\nu z)=0;$$

it will be recollected that if x,y,z are the coordinates of the node, then we have

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

Representing for a moment the equation of the curve by $U=0$, then the second derived functions of U are

$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\lambda,$$

$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\mu,$$

$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\nu,$$

$$x+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\mu+\nu),$$

$$y+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\nu+\lambda),$$

$$z+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\lambda+\mu),$$

or calling these (a,b,c,f,g,h) respectively, and substituting the values $x=\frac{1}{\theta+\lambda}$, &c., we find

$$a=2k+\frac{4k\lambda}{\theta}=\frac{2k}{\theta}(\theta+4\lambda),$$

with the like values for b, c; and

$$f=\frac{1}{\theta+\lambda}+\frac{2k}{\theta}(\theta+2(\mu+\nu)),$$

where the term in $(\ldots)$ is, after computation,

$$=-2\theta-2\mu-2\nu-\tfrac{2\mu\nu}{\theta}+\theta+2\mu+2\nu=-\theta-\tfrac{2\mu\nu}{\theta};$$

$$\text{i.e. } f=\frac{2k}{\theta}\left(-\theta-\tfrac{2\mu\nu}{\theta}\right),$$

with the like values for g and h; or omitting the common factor $\frac{2k}{\theta}$, we have

$$(a, b, c, f, g, h) = \bigl(\theta + 4\lambda,\ \theta + 4\mu,\ \theta + 4\nu,\ -\theta - \tfrac{2\mu\nu}{\theta},\ -\theta - \tfrac{2\nu\lambda}{\theta},\ -\theta - \tfrac{2\lambda\mu}{\theta}\bigr),$$

and thence, taking now x, y, z as current coordinates, the equation of the tangents at the node is

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0.$$

34. Substituting for λ, μ, ν the values $\alpha^{-2}, \beta^{-2}, \gamma^{-2}$, and for θ the twofold value $-\frac{1}{\alpha\beta\gamma}$, the equation of the tangents at the twofold centre becomes

$$\bigl(\beta^2\gamma^2(4\beta\gamma - \alpha^2),\ \ldots,\ \alpha^2\beta\gamma(\beta\gamma + 2\alpha^2),\ \ldots\bigr)(x, y, z)^2 = 0,$$

which is at once reduced to

$$\bigl(\beta^2\gamma^2(\beta - \gamma)^2,\ \ldots,\ \alpha^2\beta\gamma(\gamma - \alpha)(\alpha - \beta),\ \ldots\bigr)(x, y, z)^2 = 0,$$

or, what is the same thing,

$$\bigl[\beta\gamma(\beta - \gamma)x + \gamma\alpha(\gamma - \alpha)y + \alpha\beta(\alpha - \beta)z\bigr]^2 = 0,$$

which shows that the twofold centre is a cusp, and that the tangent is

$$\beta\gamma(\beta - \gamma)x + \gamma\alpha(\gamma - \alpha)y + \alpha\beta(\alpha - \beta)z = 0,$$

or, what is the same thing,

$$(\beta - \gamma)\frac{x}{\alpha} + (\gamma - \alpha)\frac{y}{\beta} + (\alpha - \beta)\frac{z}{\gamma} = 0.$$

35. Writing in like manner $\lambda, \mu, \nu = \alpha^{-2}, \beta^{-2}, \gamma^{-2}$ and θ for the one-with-twofold value $= \frac{2}{\alpha\beta\gamma}$, we find for the equation of the tangents at the one-with-twofold centre

$$\bigl(2\beta^2\gamma^2(2\beta\gamma + \alpha^2),\ \ldots,\ -\alpha^2\beta\gamma(4\beta\gamma + \alpha^2),\ \ldots\bigr)(x, y, z)^2 = 0,$$

which may be reduced to

$$\bigl(2\beta^2\gamma^2(2\beta\gamma + \alpha^2),\ \ldots,\ \alpha^2\beta\gamma(2\gamma\alpha + \beta^2)(2\alpha\beta + \gamma^2),\ \ldots\bigr)(x, y, z)^2 = 0,$$

or, what is the same thing,

$$(\beta\gamma x + \gamma\alpha y + \alpha\beta z)\bigl[(2\beta\gamma + \alpha^2)\beta\gamma x + (2\gamma\alpha + \beta^2)\gamma\alpha y + (2\alpha\beta + \gamma^2)\alpha\beta z\bigr] = 0.$$

Hence at the one-with-twofold centre the equation of one of the tangents is

$$(2\beta\gamma + \alpha^2)\beta\gamma x + (2\gamma\alpha + \beta^2)\gamma\alpha y + (2\alpha\beta + \gamma^2)\alpha\beta z = 0,$$

or, as this may otherwise be written,

$$(2\beta\gamma + \alpha^2)\frac{x}{\alpha} + (2\gamma\alpha + \beta^2)\frac{y}{\beta} + (2\alpha\beta + \gamma^2)\frac{z}{\gamma} = 0.$$

The memoir of Paper 349 continues from Art. 36 with the discussion of the harmonic-conic transformation, the three-centre conic, formulae for X, Y, Z, and the general formulae for arbitrary critic centres. The remainder of the paper, comprising Arts. 36–80 (facsimile pp. 326–354), falls beyond the present 50-page extract and will be taken up in the subsequent installment.

End of excerpt: PDF pp. 301–350 of Cayley, Coll. Math. Papers, Vol. V.

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Arthur Cayley

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349. On a Case of the Involution of Cubic Curves *(continued and concluded)*

[Continuation from p. 326. Arts. 36–87, treating the equation of the second tangent at a one-with-twofold centre, the alternative form of the nodal-tangent equation in terms of the node-coordinates, the triangle of critic centres, the harmonic-conic transformation, the three-centre conic, the transformation of the equation of the cubic, the harmonic conic and harmonoconic, and miscellaneous investigations.]

36. The equation of the other tangent at a one-with-twofold centre is

$$\beta^2\gamma\, x \;+\; \gamma^2\alpha\, y \;+\; \alpha^2\beta\, z \;=\; 0,$$

or, what is the same thing,

$$\frac{x}{\alpha} \;+\; \frac{y}{\beta} \;+\; \frac{z}{\gamma} \;=\; 0.$$

This is in fact equivalent to

$$\begin{vmatrix} x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha^2(\beta-\gamma) & \beta^2(\gamma-\alpha) & \gamma^2(\alpha-\beta) \end{vmatrix} \;=\; 0\,;$$

for, developing the determinant, we find

$$x\,\beta^2\gamma^2(2\alpha-\beta-\gamma) \;+\; y\,\gamma^2\alpha^2(2\beta-\gamma-\alpha) \;+\; z\,\alpha^2\beta^2(2\gamma-\alpha-\beta) \;=\; 0,$$

or, what is the same thing (using $\alpha+\beta+\gamma=0$),

$$\beta^2\gamma^2\, x \;+\; \gamma^2\alpha^2\, y \;+\; \alpha^2\beta^2\, z \;=\; 0,$$

hence the line

$$\alpha^2\beta^2\gamma^2\Big(\tfrac{x}{\alpha^2}+\tfrac{y}{\beta^2}+\tfrac{z}{\gamma^2}\Big) \;=\; 0\,,$$

which is one of the tangents at the one-with-twofold centre, is also the line joining this point with the twofold centre.

37. The equation of the tangents at a critic centre or node may be obtained in a different form, involving, instead of the parameter θ, the coordinates (x, y, z) of the node. We have

$$(\theta+\lambda)\,x = \theta\big(-x+y+z\big),$$

or, what is the same thing,

$$\lambda x = -\,\theta\,(x+y+z) + 2\theta\,x \;=\; -\theta\,(-x+y+z),$$

and similarly

$$\mu y = -\,\theta\,(\,x-y+z), \qquad \nu z = -\,\theta\,(\,x+y-z);$$

thence also

$$(\theta+4\lambda)\,x = \theta\,(\,-x+2y+2z),$$

$$(\theta^2+4\mu\nu)\,yz = \theta^2\left\{yz + \tfrac{1}{4}(x-y+z)(x+y-z)\right\} = \tfrac{1}{4}\theta^2\big(x^2-y^2-z^2+2yz\big).$$

From which we obtain

$$a:b:c:f:g:h$$

proportional to

$$2yz(-x+2y+2z),\;\; 2zx(2x-y+z),\;\; 2xy(2x+2y-z),$$

$$-x(-x^2-y^2-z^2+4yz),\;\; -y(-x^2+y^2-z^2+4zx),\;\; -z(-x^2-y^2+z^2+4xy),$$

which are the required new forms.

38. We have

$$bc-f^2 \;=\; 4xyz\,(-x+y+z)(2x+2y-z) \;-\; x^2\big(-x^2-y^2-z^2+4yz\big)^2,$$

$$=\; (x+y+z)^2\,\big(x^2+y^2+z^2-2yz-2zx-2xy\big),$$

which is $=0$ if $x+y+z=0$, or if $x^2-2x(y+z)+(y-z)^2=0$. In the former case, viz. if $x+y+z=0$, we find $a=b=c=f=g=h=-6xyz$, and therefore

$$(a,b,c,f,g,h \mid x',y',z')^2 \;=\; -6xyz\,(x'+y'+z')^2,$$

but this corresponds merely to the value $k=\infty$, for which the cubic is

$$(x+y+z)^2(\lambda x+\mu y+\nu z) \;=\; 0,$$

which is not a proper cuspidal curve. In the latter case, or where

$$x^2+y^2+z^2-2yz-2zx-2xy \;=\; 0,$$

or, what is the same thing, $\sqrt{x}+\sqrt{y}+\sqrt{z}=0$, we have a proper cuspidal curve.

Article Nos. 39 to 43, relating to the Triangle of the Critic Centres.

39. The equation

$$\frac{\lambda x}{\theta_1+\lambda} \;+\; \frac{\mu y}{\theta_1+\mu} \;+\; \frac{\nu z}{\theta_1+\nu} \;=\; 0$$

is satisfied by substituting therein

$$x : y : z \;=\; \frac{1}{\theta_2+\lambda} - \frac{1}{\theta_3+\lambda} \;:\; \frac{1}{\theta_2+\mu} - \frac{1}{\theta_3+\mu} \;:\; \frac{1}{\theta_2+\nu} - \frac{1}{\theta_3+\nu};$$

in fact, for these values the equation becomes

$$\frac{\lambda}{(\theta_1+\lambda)(\theta_2+\lambda)} - \frac{\lambda}{(\theta_1+\lambda)(\theta_3+\lambda)} \;+\; \cdots \;=\; 0,$$

or as this may be written

$$\frac{1}{\theta_1-\theta_3}\left(\frac{1}{\theta_1+\lambda} - \frac{1}{\theta_3+\lambda}\right) - \frac{1}{\theta_1-\theta_2}\left(\frac{1}{\theta_1+\lambda} - \frac{1}{\theta_2+\lambda}\right) \;+\; \cdots \;=\; 0,$$

that is, $1-1=0$; and similarly for the second set of values. Hence the equation in question is that of the line joining the critic centres corresponding to the roots θ_2 and θ_3. Hence

$$\frac{\lambda x}{\theta_1+\lambda} + \frac{\mu y}{\theta_1+\mu} + \frac{\nu z}{\theta_1+\nu} \;=\; 0,$$

$$\frac{\lambda x}{\theta_2+\lambda} + \frac{\mu y}{\theta_2+\mu} + \frac{\nu z}{\theta_2+\nu} \;=\; 0,$$

$$\frac{\lambda x}{\theta_3+\lambda} + \frac{\mu y}{\theta_3+\mu} + \frac{\nu z}{\theta_3+\nu} \;=\; 0,$$

are the equations of the sides of the triangle formed by the three critic centres.

40. It is to be remarked that the line

$$\frac{\lambda x}{\theta_1+\lambda} + \frac{\mu y}{\theta_1+\mu} + \frac{\nu z}{\theta_1+\nu} \;=\; 0$$

is the polar of the critic centre $\left(\frac{1}{\theta_1+\lambda}, \frac{1}{\theta_1+\mu}, \frac{1}{\theta_1+\nu}\right)$ in regard to the twofold-centre conic

$$x^2+y^2+z^2-2yz-2zx-2xy \;=\; 0\,;$$

in fact, forming the equation of the polar in question, this is

$$\left(-\tfrac{1}{\theta_1+\lambda} + \tfrac{1}{\theta_1+\mu} + \tfrac{1}{\theta_1+\nu}\right)x + \cdots \;=\; 0,$$

but from the equation in θ,

$$\frac{1}{\theta_1+\lambda} - \frac{1}{\theta_1+\mu} - \frac{1}{\theta_1+\nu} \;=\; \frac{2}{\theta_1} - \frac{2\lambda}{\theta_1(\theta_1+\lambda)},$$

and the like for the coefficients of y and z; this proves the theorem, and it thus appears that the critic centres are conjugate poles in regard to the twofold-centre conic.

Article Nos. 41 to 50. Transformation of the Equation of the Nodal Tangents; the Three-Centre Conic.

41. Writing as above,

$$l_1=\theta_2-\theta_3, \quad l_2=\theta_3-\theta_1, \quad l_3=\theta_1-\theta_2,$$

$$\mathfrak{S}_1=(\theta_1+\lambda)(\theta_1+\mu)(\theta_1+\nu), \quad \mathfrak{S}_2=(\theta_2+\lambda)(\theta_2+\mu)(\theta_2+\nu), \quad \mathfrak{S}_3=(\theta_3+\lambda)(\theta_3+\mu)(\theta_3+\nu),$$

I put for greater convenience

$$X = -\frac{\theta_1}{l_1}\left(\frac{\lambda x}{\theta_1+\lambda} + \frac{\mu y}{\theta_1+\mu} + \frac{\nu z}{\theta_1+\nu}\right),$$

$$Y = -\frac{\theta_2}{l_2}\left(\frac{\lambda x}{\theta_2+\lambda} + \frac{\mu y}{\theta_2+\mu} + \frac{\nu z}{\theta_2+\nu}\right),$$

$$Z = -\frac{\theta_3}{l_3}\left(\frac{\lambda x}{\theta_3+\lambda} + \frac{\mu y}{\theta_3+\mu} + \frac{\nu z}{\theta_3+\nu}\right),$$

so that $X = 0$, $Y = 0$, $Z = 0$ are the equations of the sides of the triangle formed by the critic centres.

42. Then X, Y, Z may, if we please, be considered as new coordinates replacing the original coordinates x, y, z; the relation between the two sets being given by the equations last written down; the values of x, y, z in terms of X, Y, Z are given by the converse system

$$x = \frac{X}{\theta_1+\lambda}+\frac{Y}{\theta_2+\lambda}+\frac{Z}{\theta_3+\lambda}, \quad y = \frac{X}{\theta_1+\mu}+\frac{Y}{\theta_2+\mu}+\frac{Z}{\theta_3+\mu}, \quad z = \frac{X}{\theta_1+\nu}+\frac{Y}{\theta_2+\nu}+\frac{Z}{\theta_3+\nu}.$$

43. To exhibit the identity of the two systems, I start from the last-mentioned one; this gives, on substitution,

$$\frac{\lambda x}{\theta_1+\lambda} = X\left(\frac{1}{\theta_1+\lambda} - \frac{1}{\theta_2+\lambda} - \frac{1}{\theta_3+\lambda}\right) + \cdots,$$

where the coefficient of X reduces, after a short calculation, to

$$\frac{(\mu-\nu)(\nu-\lambda)(\lambda-\mu)(\theta_1-\theta_2)(\theta_2-\theta_3)(\theta_3-\theta_1)}{(\theta_1+\lambda)(\theta_1+\mu)(\theta_1+\nu)\,(\theta_2+\lambda)(\theta_2+\mu)(\theta_2+\nu)\,(\theta_3+\lambda)(\theta_3+\mu)(\theta_3+\nu)},$$

or, what is the same thing,

$$\frac{-\,(\mu-\nu)(\nu-\lambda)(\lambda-\mu)\,l_1l_2l_3}{\mathfrak{S}_1\mathfrak{S}_2\mathfrak{S}_3}.$$

The first side is a linear function of x, y, z which vanishes for

$$x:y:z = \frac{1}{\theta_2+\lambda} : \frac{1}{\theta_2+\mu} : \frac{1}{\theta_2+\nu},$$

and for

$$x:y:z = \frac{1}{\theta_3+\lambda} : \frac{1}{\theta_3+\mu} : \frac{1}{\theta_3+\nu},$$

and hence it is of the form

$$K\left(\frac{\lambda x}{\theta_1+\lambda} + \frac{\mu y}{\theta_1+\mu} + \frac{\nu z}{\theta_1+\nu}\right),$$

and by comparing coefficients of x we have

$$\frac{K\lambda}{\theta_1+\lambda} = \frac{1}{(\theta_2+\mu)(\theta_2+\nu)} - \frac{1}{(\theta_3+\mu)(\theta_3+\nu)} = \frac{(\theta_3-\theta_2)(\theta_2+\theta_3+\mu+\nu)}{(\theta_2+\mu)(\theta_2+\nu)(\theta_3+\mu)(\theta_3+\nu)};$$

that is

$$K = \frac{-\,l_1(\mu-\nu)(\theta_1+\lambda)}{(\theta_2+\mu)(\theta_2+\nu)(\theta_3+\mu)(\theta_3+\nu)\lambda},$$

and it is easy to see that

$$K = \frac{-\,\theta_1}{l_1},$$

so that, using

$$(\theta_1+\lambda)(\theta_2+\lambda)(\theta_3+\lambda) \;=\; -\lambda\,(\nu-\lambda)(\lambda-\mu),$$

the equation becomes

$$X \;=\; -\frac{\theta_1}{l_1}\left(\frac{\lambda x}{\theta_1+\lambda}+\frac{\mu y}{\theta_1+\mu}+\frac{\nu z}{\theta_1+\nu}\right),$$

which is right; and similarly for the values of Y and Z.

44. The equation of the tangents at the node corresponding to the root θ_1 is

$$\Big(\theta_1+4\lambda,\ \theta_1+4\mu,\ \theta_1+4\nu,\ -\theta_1-\tfrac{2\mu\nu}{\theta_1+\lambda},\ -\theta_1-\tfrac{2\nu\lambda}{\theta_1+\mu},\ -\theta_1-\tfrac{2\lambda\mu}{\theta_1+\nu}\Big)(x,y,z)^2 \;=\; 0;$$

and substituting for x, y, z their values in terms of X, Y, Z, it appears in the first place that the coefficients of X^2, XY, XZ, YZ all of them vanish.

45. In fact,

$$\text{coeff.}\,X^2 \;=\; (\theta_1+4\lambda)\,\tfrac{1}{(\theta_1+\lambda)^2}+\cdots-2\big(\theta_1+\tfrac{2\mu\nu}{\theta_1+\lambda}\big)\,\tfrac{1}{(\theta_1+\mu)(\theta_1+\nu)}-\cdots.$$

The first term is

$$\sum\frac{\theta_1+4\lambda}{(\theta_1+\lambda)^2} \;=\; \sum\frac{1}{\theta_1+\lambda}\;+\;3\sum\frac{\lambda}{(\theta_1+\lambda)^2} \;=\; \frac{2}{\theta_1}+\frac{3}{\theta_1^2}\big\{3\theta_1^2-(\mu\nu+\nu\lambda+\lambda\mu)\big\},$$

where the value of $\sum\frac{\lambda}{(\theta_1+\lambda)^2}$ is most readily found from the identical equation

$$\frac{\lambda}{\theta_1+\lambda}+\frac{\mu}{\theta_1+\mu}+\frac{\nu}{\theta_1+\nu} \;=\; \frac{\theta^3-\theta\,(\mu\nu+\nu\lambda+\lambda\mu)-\lambda\mu\nu}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)}$$

by differentiating and then writing $\theta=\theta_1$. The second term is

$$-\frac{2}{\theta_1^2}\big\{3\theta_1^2+(\lambda+\mu+\nu)\theta_1+2(\mu\nu+\nu\lambda+\lambda\mu)+\tfrac{2\lambda\mu\nu}{\theta_1}\big\}.$$

Whole is $\frac{2}{\theta_1^2}$ multiplied into

$$\theta_1^2+3\big\{3\theta_1^2-(\mu\nu+\nu\lambda+\lambda\mu)\big\}-2\big\{3\theta_1^2+(\lambda+\mu+\nu)\theta_1+2(\mu\nu+\nu\lambda+\lambda\mu)+\tfrac{2\lambda\mu\nu}{\theta_1}\big\}$$
$$=\;\tfrac{2}{\theta_1^2}\big\{\theta_1^3-(\mu\nu+\nu\lambda+\lambda\mu)\theta_1-2\lambda\mu\nu\big\},$$

which is $=0$.

46. We have next

$$\text{coeff.}\,XY \;=\; \big(\theta_1+4\lambda,\ldots,-\theta_1-\tfrac{2\mu\nu}{\theta_1+\lambda},\ldots\big)\Big(\tfrac{1}{\theta_1+\lambda},\ \tfrac{1}{\theta_1+\mu},\ \tfrac{1}{\theta_1+\nu}\Big)\Big(\tfrac{1}{\theta_2+\lambda},\ \tfrac{1}{\theta_2+\mu},\ \tfrac{1}{\theta_2+\nu}\Big).$$

This expands as

$$\sum\frac{\theta_1+4\lambda}{(\theta_1+\lambda)(\theta_2+\lambda)}-\sum\big(\theta_1+\tfrac{2\mu\nu}{\theta_1+\lambda}\big)\Big\{\tfrac{1}{(\theta_1+\mu)(\theta_2+\nu)}+\tfrac{1}{(\theta_1+\nu)(\theta_2+\mu)}\Big\}.$$

First term reduces to

$$\sum\frac{\theta_1+4\lambda}{(\theta_1+\lambda)(\theta_2+\lambda)} \;=\; \sum\frac{1}{\theta_2-\theta_1}\left(\frac{1}{\theta_1+\lambda}-\frac{1}{\theta_2+\lambda}\right)+3\sum\frac{\lambda}{(\theta_1+\lambda)(\theta_2+\lambda)},$$

and after using

$$\sum\frac{\lambda}{(\theta_1+\lambda)(\theta_2+\lambda)} \;=\; \frac{\lambda\mu\nu}{\theta_1\theta_2}\cdot\frac{1}{\theta_2-\theta_1}\left(\frac{2}{\theta_1}-\frac{2}{\theta_2}\right)=\frac{2}{\theta_1\theta_2},$$

one finds that the first term equals $\frac{2}{\theta_1\theta_2}$.

The second term, written out at full length and collecting the terms which contain $\frac{1}{\theta_1+\lambda}$, becomes

$$-\frac{1}{\theta_1\theta_2}\sum\frac{2\mu\nu}{(\theta_1+\lambda)}-\frac{1}{\theta_1\theta_2}\sum\frac{1}{\theta_1-\theta_2}\left(\frac{2\lambda^2}{\theta_1+\lambda}-\frac{2\lambda^2}{\theta_2+\lambda}\right).$$

Observing that

$$\frac{2\mu\nu}{\theta_1+\lambda} \;=\; 3\theta_1+(\lambda+\mu+\nu)-6(\theta_1+\lambda)-2\lambda \;=\; -3\theta_1+\lambda+\mu+\nu,$$

together with the analogous expression for $\frac{2\nu\lambda}{\theta_2+\lambda}$, the second part reduces to $-\frac{3}{\theta_1\theta_2}$.

47. Hence the whole second term is $-\frac{2}{\theta_1\theta_2}$, and combining the two terms we have

$$\text{coeff.}\, XY \;=\; \tfrac{2}{\theta_1\theta_2}-\tfrac{2}{\theta_1\theta_2} \;=\; 0.$$

In the same manner precisely it appears that

$$\text{coeff.}\, XZ \;=\; 0.$$

48. Next,

$$\text{coeff.}\, YZ \;=\; (\theta_1+4\lambda,\ldots,-\theta_1-\tfrac{2\mu\nu}{\theta_1+\lambda},\ldots)\left(\tfrac{1}{\theta_2+\lambda},\tfrac{1}{\theta_2+\mu},\tfrac{1}{\theta_2+\nu}\right)\left(\tfrac{1}{\theta_3+\lambda},\tfrac{1}{\theta_3+\mu},\tfrac{1}{\theta_3+\nu}\right).$$

The first term is

$$\sum\frac{\theta_1+4\lambda}{(\theta_2+\lambda)(\theta_3+\lambda)} \;=\; \sum\frac{1}{\theta_3-\theta_2}\left(\frac{1}{\theta_2+\lambda}-\frac{1}{\theta_3+\lambda}\right)+3\sum\frac{\lambda}{(\theta_2+\lambda)(\theta_3+\lambda)},$$

which reduces to

$$\frac{2}{\theta_2\theta_3}\cdot\frac{\theta_1\, l_1}{\theta_2-\theta_3}\left(\frac{1}{\theta_2}-\frac{1}{\theta_3}\right)\;-\;\frac{l_1}{\theta_2\theta_3}\;=\;-\frac{l_1}{\theta_2\theta_3}.$$

The second term, written out at full length and rearranged, is easily seen to be

$$-\frac{l_1}{\theta_2\theta_3}\left\{\sum\frac{2\mu\nu}{\theta_1+\lambda}\cdot\frac{1}{\theta_2-\theta_3}\left(\frac{1}{\theta_2+\mu}+\frac{1}{\theta_3+\mu}\right)\;-\;\cdots\right\},$$

so that the whole second term reduces to $-\frac{l_1}{\theta_2\theta_3}$; whence combining the two we have

$$\text{coeff.}\, YZ \;=\; \tfrac{l_1}{\theta_2\theta_3}-\tfrac{l_1}{\theta_2\theta_3} \;=\; 0.$$

49. We have now to find the coefficients of Y^2 and Z^2.

$$\text{coeff.}\, Y^2 \;=\; \left(\theta_1+4\lambda,\ldots,-\theta_1-\tfrac{2\mu\nu}{\theta_1+\lambda},\ldots\right)\left(\tfrac{1}{\theta_2+\lambda},\tfrac{1}{\theta_2+\mu},\tfrac{1}{\theta_2+\nu}\right)^2,$$

and observing that the terms of this expression differ from those of coeff. X^2 only by having θ_2 in place of θ_1, so that the corresponding part vanishes, the only residue arises from the coefficients involving θ_1 explicitly. After collection,

$$\text{coeff.}\, Y^2 \;=\; (\theta_1-\theta_2)\left\{\sum\frac{1}{(\theta_2+\lambda)^2}-2\sum\frac{1}{(\theta_2+\mu)(\theta_2+\nu)}\right\}+\frac{2}{\theta_1}\left(2(\lambda+\mu+\nu)+\tfrac{2\mu\nu}{\theta_1+\lambda}+\tfrac{2\nu\lambda}{\theta_1+\mu}+\tfrac{2\lambda\mu}{\theta_1+\nu}\right),$$

and substituting for $\theta_1-\theta_2$ its value l_3, one obtains

$$\text{coeff.}\, Y^2 \;=\; \frac{l_3}{\theta_2^2}\left\{-3\theta_2^2+(\mu\nu+\nu\lambda+\lambda\mu)(\theta_2+2\theta_1)+6\lambda\mu\nu\right\}.$$

But $\theta_1+\theta_2+\theta_3=0$, whence $\theta_2+2\theta_1=\theta_1-\theta_3=l_2$; and $\mu\nu+\nu\lambda+\lambda\mu=-(\theta_1\theta_2+\theta_2\theta_3+\theta_3\theta_1)$, $\lambda\mu\nu=-\theta_1\theta_2\theta_3$; so

$$\text{coeff.}\,Y^2 \;=\; \frac{l_3}{\theta_2^2}\{-3\theta_2^2-(\theta_1\theta_2+\theta_2\theta_3+\theta_3\theta_1)l_2-6\theta_1\theta_2\theta_3\}\,,$$

that is

$$\text{coeff.}\,Y^2 \;=\; -\,\frac{l_3\,l_2}{\theta_2}\,,$$

and, by merely interchanging θ_2 and θ_3,

$$\text{coeff.}\,Z^2 \;=\; -\,\frac{l_2\,l_3}{\theta_3}.$$

50. Hence the equation of the tangents is

$$-\,\frac{l_3\,l_2}{\theta_2}\,Y^2 \;-\; \frac{l_2\,l_3}{\theta_3}\,Z^2 \;=\; 0,$$

or, what is the same thing,

$$\frac{Y^2}{\theta_2} \;+\; \frac{Z^2}{\theta_3} \;=\; 0,$$

or putting

$$A=\frac{l_1^2}{\theta_1},\quad B=\frac{l_2^2}{\theta_2},\quad C=\frac{l_3^2}{\theta_3},$$

the equation of the tangents at the node corresponding to θ_1 is $BY^2+CZ^2=0$. Hence the equations of the tangents at the three nodes respectively are

$$BY^2+CZ^2 \;=\; 0,$$

$$AX^2+CZ^2 \;=\; 0,$$

$$AX^2+BY^2 \;=\; 0;$$

that is, the nodes or critic centres are conjugate poles in regard to a conic

$$AX^2+BY^2+CZ^2 \;=\; 0,$$

which is the *three-centre conic*; and the tangents at each node are the tangents from such node to the conic in question.

Article Nos. 51 and 52. Special Case of the Three-Centre Conic.

51. Write for a moment

$$X' \;=\; \frac{\lambda x}{\theta_1+\lambda}+\frac{\mu y}{\theta_1+\mu}+\frac{\nu z}{\theta_1+\nu},$$

$$Y' \;=\; \frac{\lambda x}{\theta_2+\lambda}+\frac{\mu y}{\theta_2+\mu}+\frac{\nu z}{\theta_2+\nu},$$

$$Z' \;=\; \frac{\lambda x}{\theta_3+\lambda}+\frac{\mu y}{\theta_3+\mu}+\frac{\nu z}{\theta_3+\nu},$$

so that $X=-\frac{\theta_1}{l_1}X'$, etc.; the equation of the three-centre conic in terms of (X',Y',Z') is

$$\frac{l_1^2}{\theta_1}\,\frac{\theta_1^2}{l_1^2}\,X'^2+\frac{l_2^2}{\theta_2}\,\frac{\theta_2^2}{l_2^2}\,Y'^2+\frac{l_3^2}{\theta_3}\,\frac{\theta_3^2}{l_3^2}\,Z'^2 \;=\; 0,$$

say

$$A'X'^2 + B'Y'^2 + C'Z'^2 \;=\; 0.$$

When $\theta_3 = \infty$, we have $C' = \infty$, $A' = -B'$, $X' = Y'$; writing the equation in the form

$$(A' + B')X'^2 + B'(Y'^2 - X'^2) + C'Z'^2 \;=\; 0,$$

and observing that in the limit $Y'^2 - X'^2 = 2X'(Y' - X')$, we see that the equation will assume the form

$$(\theta_1\theta_2 - \theta_1^2)X'\Sigma - \frac{l_1}{\theta_1 + 2\theta_1}\theta_2 Z'^2 \;=\; 0,$$

where Σ is a finite function; $X' = 0$ is the line joining the twofold centre and the one-with-twofold centre, $\Sigma = 0$ is the other tangent at the one-with-twofold centre, $Z' = 0$ the tangent at the twofold centre or cusp; the form $X'\Sigma + Z'^2 = 0$ shows that the three-centre conic reduces itself to a pair of points, viz. the twofold centre or cusp, and the point where the tangent at the cusp is met by the other tangent (that is the tangent not passing through the cusp) at the one-with-twofold centre.

52. To verify the value of Σ I proceed as follows:

$$A' + B' \;=\; \frac{1}{\theta_1 - \theta_2}\{\theta_1(\theta_2 - \theta_3) + \theta_2(\theta_3 - \theta_1)\} \;=\; (\theta_1 - \theta_2)(\lambda + \mu + \nu + 3\theta_1);$$

$$\theta_1^3 - \theta_2^3 + (\lambda + \mu + \nu)(\theta_1^2 - \theta_2^2) + (\mu\nu + \nu\lambda + \lambda\mu)(\theta_1 - \theta_2) \;=\; (\theta_1 - \theta_2)\{6\theta_1^2 + 2(\lambda + \mu + \nu)\theta_1\}.$$

Moreover

$$Y'^2 - X'^2 \;=\; 2X'\Big\{\lambda x\Big(\tfrac{1}{\theta_1+\lambda} \cdot \tfrac{1}{\theta_2+\lambda}\Big) + \mu y\Big(\tfrac{1}{\theta_1+\mu} \cdot \tfrac{1}{\theta_2+\mu}\Big) + \nu z\Big(\tfrac{1}{\theta_1+\nu} \cdot \tfrac{1}{\theta_2+\nu}\Big)\Big\},$$

and hence

$$B'(Y'^2 - X'^2) \;=\; -\frac{2\theta_1}{l_1}X'\,\Sigma',$$

in which we have only now to substitute $(\lambda, \mu, \nu) = (\alpha^{-2}, \beta^{-2}, \gamma^{-2})$ and $\theta_1 = -\frac{1}{\alpha\beta\gamma}$. We have

$$\theta_1 + \lambda \;=\; \tfrac{\beta\gamma}{\alpha^2\beta^2\gamma^2}\,k_1, \quad \theta_1 + \mu \;=\; \tfrac{\gamma\alpha}{\alpha^2\beta^2\gamma^2}\,k_2, \quad \theta_1 + \nu \;=\; \tfrac{\alpha\beta}{\alpha^2\beta^2\gamma^2}\,k_3,$$

where $k = -3(\beta\gamma - \alpha^2) = -3(\beta\gamma + \gamma\alpha + \alpha\beta)$; and then observing that the equation $\Sigma = 0$ becomes

$$2(\beta\gamma + \gamma\alpha + \alpha\beta)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) + 3(\alpha x + \beta y + \gamma z) \;=\; 0,$$

or, what is the same thing,

$$(2\beta\gamma + \alpha^2)\frac{x}{\alpha} + (2\gamma\alpha + \beta^2)\frac{y}{\beta} + (2\alpha\beta + \gamma^2)\frac{z}{\gamma} \;=\; 0,$$

which agrees with a former result.

Article Nos. 53 to 55. Transformation of the Equation of the Cubic.

53. Let it be required to express the cubic

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \;=\; 0$$

in terms of the coordinates X, Y, Z. We have

$$\lambda x + \mu y + \nu z \;=\; X + Y + Z,$$

and the equation therefore is

$$\Pi\left(\frac{X}{\theta_1+\lambda}+\frac{Y}{\theta_2+\lambda}+\frac{Z}{\theta_3+\lambda}\right)+k\left(\frac{X}{\theta_1}+\frac{Y}{\theta_2}+\frac{Z}{\theta_3}\right)^2(X+Y+Z) \;=\; 0,$$

where Π denotes the product of the three factors obtained by writing λ, μ, ν successively in the place of λ.

For one of the nodal cubics we have

$$k \;=\; -\tfrac{1}{4}\theta_1,$$

and the equation multiplied by θ_1 is

$$\theta_1\Pi(\cdots)-\tfrac{1}{4}\theta_1^2\big(\tfrac{X}{\theta_1}+\tfrac{Y}{\theta_2}+\tfrac{Z}{\theta_3}\big)^2(X+Y+Z) \;=\; 0,$$

which it is clear *à priori* must be of the form

$$\big(\cdots+X\big)^2(\cdots)(X+Y+Z) \;=\; 0,$$

and there is in fact no difficulty in verifying that the coefficients of X^3, X^2Y, X^2Z, XYZ all of them vanish. To find K, comparing the coefficients of XY^2 we have

$$K \;=\; 2(\theta_1+\lambda)(\theta_2+\mu)(\theta_2+\nu)-\tfrac{1}{4}\theta_1(\theta_1+2\theta_2)\theta_2\,;$$

that is

$$K \;=\; \tfrac{1}{4}(\theta_1-\theta_2)\,l_2^2 \;+\; \tfrac{1}{4}\,\Sigma\,(\theta_1+\theta_3)\,\theta_2\,,$$

and similarly

$$K \;=\; \tfrac{1}{4}\big\{\theta_1^2\theta_2^2-6\theta_1\theta_2(\mu\nu+\nu\lambda+\lambda\mu)-(\theta_1+\theta_2)^2\theta_3^2\big\} \;=\; \tfrac{1}{4}\big\{\theta_1^2\theta_2^2+6\theta_1\theta_2(\theta_1\theta_2+\theta_2\theta_3+\theta_3\theta_1)-(\theta_1+\theta_2)^2\theta_3^2\big\},$$

so that we have

$$K \;=\; \tfrac{1}{4}\theta_3^2\,l_3^2,$$

that is

$$K \;=\; \tfrac{1}{4}\theta_3\,l_3^2,$$

and the equation of the nodal cubic is

$$-\tfrac{1}{4}\Pi\,X\left(\frac{Y^2}{\theta_2^2\,l_2^2}+\frac{Z^2}{\theta_3^2\,l_3^2}\right)+\tfrac{1}{4}\theta_1\Pi\left(\frac{Y}{\theta_2\,l_2}-\frac{Z}{\theta_3\,l_3}\right)^2\cdot\Sigma \;=\; 0.$$

54. To complete the reduction we have

$$\text{coeff.}\,Y^2 \;=\; \tfrac{1}{4}\cdot\tfrac{\theta_1\theta_3^2\,l_3^2}{\theta_2^2}\cdot\big(\theta_1\theta_2^2\,l_2^2/(\theta_2^2\,l_2^2)\big),$$

$$\text{coeff.}\,YZ \;=\; \tfrac{\theta_1}{2}\cdot\tfrac{l_3l_2}{\theta_2\theta_3}\cdot\big(\theta_2\theta_3^2l_3^2-\theta_2^2\theta_3l_2^2\big)\,,$$

so that substituting for $\sqrt{\theta_1-\theta_2\theta_3}$ its value $-\frac{1}{2}\theta_1 l_1^2$, the terms in Y^2 and YZ are

$$\cdots$$

and in like manner the terms in YZ^2 and Z^3 are

$$\cdots\,.$$

55. So that the terms in $(Y,Z)^3$ are

$$-\tfrac{1}{4}\theta_1^2\, l_1^2\Big\{\tfrac{Y^3}{\theta_2 l_2}\Big(\tfrac{1}{\theta_2}-\tfrac{1}{\theta_3}\,l_2^2\Big)\;-\;\tfrac{Z^3}{\theta_3 l_3}\Big(\tfrac{1}{\theta_3}-\tfrac{1}{\theta_2}\,l_3^2\Big)\Big\}\,,$$

and the equation, omitting the factor $-\frac{1}{4}\theta_1^2 l_1^2$, is

$$\frac{Y}{\theta_2^2 l_2^3}-\frac{Z}{\theta_3^2 l_3^3}\;=\;0.$$

But the term in [...] is

$$\frac{Y}{\theta_2^2 l_2^3}-\frac{Z}{\theta_3^2 l_3^3}\;=\;\frac{1}{l_1^2\theta_2^2\theta_3^2}\left(\frac{l_3 Y}{\theta_2 l_2}-\frac{l_2 Z}{\theta_3 l_3}\right),$$

and the equation of the nodal cubic is finally

$$\frac{Y}{\theta_2^2 l_2^3}-\frac{Z}{\theta_3^2 l_3^3}\;=\;0.$$

The lines $Y=0$, $Z=0$, $\frac{Y}{\theta_2^2 l_2^2}-\frac{Z}{\theta_3^2 l_3^2}=0$ each pass through the node and meet the cubic in a third point; the three points of intersection lie in the line $l_1X+\theta_1(Y-Z)=0$.

Article Nos. 56 to 66. The Cubic Locus, Harmoconics and Harmonic Conic.

56. Suppose that the line $\lambda x+\mu y+\nu z=0$ passes through a given point (a,b,c), then we have

$$\lambda a+\mu b+\nu c\;=\;0;$$

and observing that $\theta+\lambda$, $\theta+\mu$, $\theta+\nu$, θ are proportional to

$$\tfrac{1}{x},\quad \tfrac{1}{y},\quad \tfrac{1}{z},\quad \tfrac{2}{x+y+z}\,,\quad \text{respectively,}$$

we find

$$\frac{a}{x}+\frac{b}{y}+\frac{c}{z}\;=\;\frac{2(a+b+c)}{x+y+z}\,,$$

the equation of a cubic curve, the locus of the critic centres corresponding to the several lines $\lambda x+\mu y+\nu z=0$ which pass through the point (a,b,c). The cubic curve passes, it is clear, through the six points which are the angles of the quadrilateral

$$x=0,\ y=0,\ z=0,\ x+y+z=0.$$

57. If we take for (a,b,c) the coordinates of the point of intersection of the line $\lambda x+\mu y+\nu z=0$ with any one of the lines $x=0$, $y=0$, $z=0$, $x+y+z=0$, then in each case the cubic breaks up into the same line and a conic, viz. we have the conics

$$\frac{\nu}{y}-\frac{\mu}{z}-\frac{2(\nu-\mu)}{x+y+z}\;=\;0,$$

$$\frac{\lambda}{z}-\frac{\nu}{x}-\frac{2(\lambda-\nu)}{x+y+z}\;=\;0,$$

$$\frac{\mu}{x}-\frac{\lambda}{y}-\frac{2(\mu-\lambda)}{x+y+z}\;=\;0,$$

$$\frac{\lambda}{x}+\frac{\mu}{y}+\frac{\nu}{z}\;=\;0,$$

and it is to be noticed that in each case the critic centres all of them lie on the conic. In fact, since the point $(0,\nu,-\mu)$ is an arbitrary point on the line $x=0$, a line $\lambda x+\mu y+\nu z=0$ passing through the point in question is an absolutely arbitrary line, and the corresponding critic centres therefore do not lie on the line $x = 0$; that is, they lie on the conic

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu-\mu)}{x+y+z} \;=\; 0,$$

and it may also be remarked that the elimination of λ, θ from the system

$$\theta+\mu \;:\; \theta+\nu \;:\; \theta \;=\; \tfrac{1}{y} \;:\; \tfrac{1}{z} \;:\; \tfrac{2}{x+y+z}\,,$$

gives the last-mentioned equation, unencumbered by the factor $x = 0$.

We have thus four conics, each of them passing through the three critic centres which correspond to the line $\lambda x + \mu y + \nu z = 0$; as to the signification of the first three of these conics, I remark as follows.

58. The 'harmoconic' of a point A as to the line T in respect of the conic Ω may be defined as follows; viz. considering the pencil of lines through A, the locus of the fourth harmonic of the point in which a line of the pencil meets T, in regard to the two points in which the same line meets the conic Ω, is a conic which is the harmoconic in question. (In particular, if the line T pass through the point A the harmoconic breaks up into the line T and into the polar of A.) The conic Ω may of course be a pair of lines.

Consider any three lines x, y, z, a line S, and the line T; then the harmoconics being all as to the same line T, we have the theorem

- Harmoconic of intersection of x, S in regard to pair of lines y, z,
- Ditto of y, S in regard to pair of lines z, x,
- Ditto of z, S in regard to pair of lines x, y,

all pass through the same three points.

And taking $x = 0$, $y = 0$, $z = 0$ for the equations of the lines x, y, z; $\lambda x + \mu y + \nu z = 0$ for the equation of the line S; and $x + y + z = 0$ for the equation of the line T, the harmoconics just spoken of are the above-mentioned three conics respectively.

59. In fact, considering the harmoconic of intersection of x, S in regard to the pair y, z; and taking x', y', z' as the coordinates of a point P of the harmoconic, then the equation of the line AP is

$$\det\begin{pmatrix} x & y & z \\ x' & y' & z' \\ 0 & \nu & -\mu \end{pmatrix} \;=\; 0,$$

that is

$$x(\mu y' + \nu z') - x'(\mu y + \nu z) \;=\; 0,$$

and at the point of intersection with the line T or $x + y + z = 0$, we have

$$(y+z)(\mu y' + \nu z') + x'(\mu y + \nu z) \;=\; 0,$$

or, what is the same thing,

$$y(\mu x' + \mu y' + \nu z') + z(\nu x' + \mu y' + \nu z') \;=\; 0,$$

which is the line through the last-mentioned point and the point $(y = 0, z = 0)$.

The line from the point A to the point $(y = 0, z = 0)$ is

$$y\,z' - z\,y' \;=\; 0.$$

60. By the definition of the harmoconic, the last-mentioned two lines are harmonics in regard to the lines $y = 0$, $z = 0$; that is, we have for the equation of the harmoconic in question

$$-y'(\mu x' + \mu y' + \nu z') + z'(\nu x' + \mu y' + \nu z') \;=\; 0,$$

this equation may also be written

$$(yz' - zy')(\mu x' + y' + z') - 2(\nu - \mu)y'z' \;=\; 0,$$

or, what is the same thing,

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu-\mu)}{x+y+z} \;=\; 0,$$

hence writing x, y, z in place of x', y', z', we see that this harmoconic is in fact the first of the above-mentioned three conics.

61. The fourth conic through the critic centres is the conic

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} \;=\; 0,$$

which it will be observed passes through the vertices of the triangle $x = 0$, $y = 0$, $z = 0$, and also through the point $(1, 1, 1)$, which is the harmonic of the line $x + y + z = 0$ in regard to the triangle: I call it the 'harmonic conic.' Representing the equation by

$$2fyz + 2gzx + 2hxy \;=\; 0,$$

we have $f = \mu - \nu$, $g = \nu - \lambda$, $h = \lambda - \mu$, and therefore $f + g + h = 0$.

62. It is easy to show that the coordinates of the pole of the line $x+y+z = 0$ in regard to the harmonic conic are $x : y : z = f^2 : g^2 : h^2$; these values satisfy the condition $\sqrt{x}+\sqrt{y}+\sqrt{z} = 0$, that is, the pole in question lies on the twofold-centre conic.

63. The equation of the tangents to the harmonic conic at its intersection with the line $x + y + z = 0$ (which tangents meet of course in the last-mentioned pole, that is in a point of the twofold-centre conic) is found to be

$$2fgh(x + y + z)^2 + \Omega\,(2fyz + 2gzx + 2hxy) \;=\; 0;$$

if for shortness

$$\Omega \;=\; f^2 + g^2 + h^2 - 2gh - 2hf - 2fg,$$

or, what is the same thing,

$$\Omega \;=\; -4(gh + hf + fg) \;=\; 2(f^2 + g^2 + h^2).$$

64. We have identically

$$-6fgh\bigl(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy\bigr) + (hz)(ghx + hfy + fgz)$$

$$=\; 2fgh(x + y + z)^2 + \Omega(2fyz + 2gzx + 2hxy) - 8(fx + gy + hz)(ghx + hfy + fgz),$$

so that the tangents in question meet the twofold-centre conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy \;=\; 0$$

at its intersections with the lines $fx+gy+hz = 0$ and $ghx+hfy+fgz = 0$: the latter of these is in fact the tangent of the conic at the point (f^2, g^2, h^2) of intersection of the two tangents. Hence the two tangents meet at the point (f^2, g^2, h^2) of the twofold-centre conic and they besides meet the conic at its points of intersection with the line $fx + gy + hz = 0$.

65. The line $\lambda x + \mu y + \nu z = 0$ may be expressed in the form,

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} + \tfrac{1}{3}(x + y + z) \;=\; 0,$$

(where, *ut supra*, $\alpha + \beta + \gamma = 0$). The corresponding values of f, g, h are

$$f \;=\; \tfrac{1}{\beta} - \tfrac{1}{\gamma}, \quad g \;=\; \tfrac{1}{\gamma} - \tfrac{1}{\alpha}, \quad h \;=\; \tfrac{1}{\alpha} - \tfrac{1}{\beta},$$

or, what is the same thing,

$$f : g : h \;=\; \alpha^2(\beta - \gamma) \;:\; \beta^2(\gamma - \alpha) \;:\; \gamma^2(\alpha - \beta),$$

or, again,

$$f : g : h \;=\; \alpha(\beta^2 - \gamma^2) \;:\; \beta(\gamma^2 - \alpha^2) \;:\; \gamma(\alpha^2 - \beta^2).$$

The equation $fx + gy + hz = 0$ may be written

$$\det\begin{pmatrix} x & y & z \\ \frac{1}{\alpha^2} & \frac{1}{\beta^2} & \frac{1}{\gamma^2} \\ 1 & 1 & 1 \end{pmatrix} \;=\; 0,$$

that is, the line in question is the line joining the harmonic point $(1, 1, 1)$ with the inverse of the point $(\alpha^2, \beta^2, \gamma^2)$, which is (*ante*, No. 27) the point of contact of the satellite line with the envelope.

66. The harmonic conic passes through the vertices of the triangle $x = 0$, $y = 0$, $z = 0$, through the harmonic point $(1, 1, 1)$, and through the critic centres. Hence if one of the critic centres be given, the harmonic conic passes through five given points and is thus completely determined. But a critic centre being given, the line joining the other two critic centres is the polar of the given centre in regard to the twofold-centre conic (*ante*, No. 40), and it is thus completely determined; and the other two critic centres are of course the intersections of this line with the harmonic conic.

Article Nos. 67 to 87. Miscellaneous Investigations.

67. I demonstrate by means of the last-mentioned formulae a theorem already in effect demonstrated by the investigation which led to the three-centre conic, viz. that the tangents at a node or critic centre, and the lines drawn to the other two critic centres, form a harmonic pencil.

In fact the tangents at the node or critic centre are given by the equation

$$\big(\theta + 4\lambda,\; \ldots,\; -\theta - \tfrac{2\mu\nu}{\theta+\lambda},\; \ldots\big)(x, y, z)^2 \;=\; 0;$$

the other two critic centres are given as the intersection of the line

$$\frac{\lambda x}{\theta + \lambda} + \frac{\mu y}{\theta + \mu} + \frac{\nu z}{\theta + \nu} \;=\; 0$$

with the conic

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} \;=\; 0;$$

the theorem will be true if the pair of tangents and the last-mentioned conic are cut harmonically by the last-mentioned line.

Now in general the condition in order that the line $\xi x+\eta y+\zeta z=0$ may cut harmonically the conics $(a,b,c,f,g,h \mid x,y,z)^2$ and $(a',b',c',f',g',h')(x,y,z)^2=0$ is

$$\big(bc'+b'c-2ff',\,\ldots,\,gh'+g'h-af'-a'f,\,\ldots \mid \xi,\eta,\zeta\big)^2 \;=\; 0,$$

and if $a'=b'=c'=0$, then the condition is

$$\big(-2ff',\,\ldots,\,gh'+g'h-af',\,\ldots \mid \xi,\eta,\zeta\big)^2 \;=\; 0.$$

68. In the present case the equations of the two conics may be written

$$\big(\theta+4\lambda,\,\ldots,\,-\theta-\tfrac{2\mu\nu}{\theta+\lambda},\,\ldots\big)(x,y,z)^2 \;=\; 0,$$

$$\big(0,\,\ldots,\,\mu-\nu,\,\ldots\big)(x,y,z)^2 \;=\; 0,$$

and we have

$$-2ff' \;=\; -2(\mu-\nu)\big(\theta+\tfrac{2\mu\nu}{\theta+\lambda}\big),$$

$$\begin{aligned} gh'+g'h-af' \;&=\; -\big(\theta+\tfrac{2\nu\lambda}{\theta+\mu}\big)(\lambda-\mu)-\big(\theta+\tfrac{2\lambda\mu}{\theta+\nu}\big)(\nu-\lambda)-(\mu-\nu)(\theta+4\lambda)\\ &=\; -\theta(\lambda-\mu+\mu-\nu+\nu-\lambda)-\tfrac{2}{\theta}(\nu\lambda^2+\nu^2\lambda-\mu\lambda^2-\mu^2\nu)+4\lambda(\mu-\nu)\\ &=\; (\mu-\nu)(\theta-4\lambda), \end{aligned}$$

and the condition is

$$\Big\{-2(\mu-\nu)\Big(\theta+\tfrac{2\mu\nu}{\theta+\lambda}\Big),\,\ldots,\,(\mu-\nu)(\theta-4\lambda),\,\ldots \mid \tfrac{\lambda}{\theta+\lambda},\,\tfrac{\mu}{\theta+\mu},\,\tfrac{\nu}{\theta+\nu}\Big\}^2 \;=\; 0.$$

Writing this in the form

$$\cdots$$

$$-2\lambda\mu\nu\Big(\theta+\tfrac{2\mu\nu}{\theta+\lambda}\Big)\,\tfrac{1}{(\theta+\mu)(\theta+\nu)}(\mu-\nu)+\cdots \;=\; 0,$$

then observing that

$$\tfrac{\theta+\mu+\nu}{\theta+\lambda} \;=\; \tfrac{2}{\theta}\,(\lambda+\theta),$$

the first part is

$$\frac{2\lambda\mu\nu(\mu-\nu)(\nu-\lambda)(\lambda-\mu)}{\theta^2(\theta+\lambda)(\theta+\mu)(\theta+\nu)}\,(\theta+\lambda),$$

and observing that the sum is

$$\sum 2\lambda(\mu-\nu)(\theta^2+2\mu\nu)-\lambda\mu\nu\sum\frac{2(\mu-\nu)(\theta+2\mu\nu)}{\theta+\lambda} \;=\; \frac{2}{\theta}\lambda\mu\nu(\mu-\nu)(\nu-\lambda)(\lambda-\mu),$$

the first part is

$$-\,\frac{2\lambda\mu\nu(\mu-\nu)(\nu-\lambda)(\lambda-\mu)}{\theta^2}\,\frac{1}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)}.$$

The second part is

$$\frac{2\lambda\mu\nu}{\theta(\theta+\lambda)(\theta+\mu)(\theta+\nu)}\sum(\mu-\nu)(\lambda^2-\lambda\theta-2\theta^2),$$

in which the sum is

$$\sum(\mu-\nu)(\lambda^2-\lambda\theta-2\theta^2) \;=\; 2\lambda^2(\mu-\nu) \;=\; -(\mu-\nu)(\nu-\lambda)(\lambda-\mu),$$

so that the second part is

$$-\,\frac{2\lambda\mu\nu(\mu-\nu)(\nu-\lambda)(\lambda-\mu)}{\theta^2(\theta+\lambda)(\theta+\mu)(\theta+\nu)}\,,$$

and the sum of the two parts is $= 0$, which proves the theorem.

70. Let x_1, y_1, z_1 be the coordinates of a critic centre, then the equation of the polar in regard to the twofold-centre conic is

$$(-x_1 + y_1 + z_1)x + (x_1 - y_1 + z_1)y + (x_1 + y_1 - z_1)z \;=\; 0,$$

and the equation of the conic through the five points is

$$\frac{x_1(y_1 - z_1)}{x} + \frac{y_1(z_1 - x_1)}{y} + \frac{z_1(x_1 - y_1)}{z} \;=\; 0,$$

and these equations together determine the remaining two critic centres.

71. I remark in passing that the equation of the one-with-twofold centre locus may also be obtained by means of the equations

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} \;=\; 0,$$

$$(-x_1 + y_1 + z_1)x + (x_1 - y_1 + z_1)y + (x_1 + y_1 - z_1)z \;=\; 0,$$

which determine the remaining two critic centres corresponding to a given critic centre (x_1, y_1, z_1); in fact, in order that the centre (x_1, y_1, z_1) may be accompanied by a twofold centre the line must touch the conic; and the analytical condition, substituting therein (x, y, z) in the place of (x_1, y_1, z_1), is found to be

$$xyz\{x^3 + y^3 + z^3\} - \big(yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y\big) + 3xyz \;=\; 0;$$

the three lines $xyz = 0$ are not properly part of the locus, but their appearance may be accounted for without difficulty.

72. Assume that the line $\lambda x + \mu y + \nu z = 0$ passes successively through the points

$$(x = 0,\ y - z = 0),\ (y = 0,\ z - x = 0),\ (z = 0,\ x - y = 0),$$

or, what is the same thing, the points $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$: then (*ante*, No. 56) the critic centres are in all these cases respectively on the conics

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu - \mu)}{x + y + z} \;=\; 0,$$

$$\frac{\lambda}{z} - \frac{\nu}{x} - \frac{2(\lambda - \nu)}{x + y + z} \;=\; 0,$$

$$\frac{\mu}{x} - \frac{\lambda}{y} - \frac{2(\mu - \lambda)}{x + y + z} \;=\; 0,$$

or, as these may be written,

$$-zy + x(y + z) \;=\; 0,$$

$$-xy + y(z + x) \;=\; 0,$$

$$-yy + z(x + y) \;=\; 0,$$

the first of which is a conic touching the lines $x = 0$, $y + z = 0$ at the points of intersection with the line $y - z = 0$; and similarly for the other two conics.

73. Suppose that the line $\lambda x + \mu y + \nu z = 0$ passes through the point $(4, -1, -1)$, or let $4\lambda - \mu - \nu = 0$; we have $(\alpha, \beta, \gamma) = (4, -1, -1)$; and the critic centres lie on the curve

$$\frac{4}{x} - \frac{1}{y} - \frac{1}{z} - \frac{4}{x + y + z} \;=\; 0,$$

that is

$$\frac{4(y+z)}{x(x+y+z)} - \frac{y+z}{yz} \;=\; 0,$$

or, as this may be written,

$$(y+z)\big\{x(x+y+z) - 4yz\big\} \;=\; 0,$$

so that the cubic locus breaks up into the line $y+z=0$ and into the conic

$$x(x+y+z) - 4yz \;=\; 0.$$

74. I say that the critic centres lie, one of them on the line, and the other two on the conic. In fact, putting $\lambda = \frac{1}{4}(\mu+\nu)$ the equation in θ is

$$\theta^3 - \theta\,\mu\nu - 2\lambda\mu\nu \;=\; 0,$$

that is

$$\big(\theta + \tfrac{1}{2}(\mu+\nu)\big)\big\{\theta^2 - \tfrac{1}{2}(\mu+\nu)\theta - \mu\nu\big\} \;=\; 0,$$

and we have

$$x : y : z \;=\; \frac{1}{\theta+\lambda} \;:\; \frac{1}{\theta+\mu} \;:\; \frac{1}{\theta+\nu}.$$

75. Hence if $\theta + \frac{1}{2}(\mu+\nu) = 0$, we obtain

$$x : y : z \;=\; \frac{1}{\frac{1}{4}(\mu+\nu) + \frac{1}{2}(\mu+\nu) + \mu} \;:\; \cdots\,,$$

whence also

$$\begin{aligned}(\mu+\nu)x + (\mu-\nu)y \;&=\; 0,\\ (\mu+\nu)x - (\mu-\nu)z \;&=\; 0,\\ y+z \;&=\; 0,\end{aligned}$$

so that the corresponding critic centre lies on the line $y+z=0$; the last-mentioned equations, restoring the value 4λ in place of $\mu+\nu$, may also be written

$$\begin{aligned}4\lambda x + (\mu-\nu)y \;&=\; 0,\\ 4\lambda x - (\mu-\nu)z \;&=\; 0,\\ y+z \;&=\; 0.\end{aligned}$$

76. If on the other hand

$$\theta^2 - \tfrac{1}{2}(\mu+\nu)\theta - \mu\nu \;=\; 0,$$

or, as this equation may be written,

$$5\theta^2 - (\theta+2\mu)(\theta+2\nu) \;=\; 0,$$

then observing that in general, in virtue of the equation

$$\frac{1}{\theta+\lambda} + \frac{1}{\theta+\mu} + \frac{1}{\theta+\nu} \;=\; \frac{2}{\theta},$$

we have

$$\frac{1}{\theta+\mu} + \frac{1}{\theta+\nu} \;=\; \frac{\theta+2\mu}{\theta(\theta+\mu)},$$

and consequently

$$y : x + z \;=\; \theta : \theta + 2\mu, \qquad z : x + y \;=\; \theta : \theta + 2\nu,$$

the foregoing equation

$$5\theta^2 - (\theta + 2\mu)(\theta + 2\nu) \;=\; 0$$

gives

$$5yz - (x + z)(x + y) \;=\; 0,$$

that is

$$x(x + y + z) - 4yz \;=\; 0;$$

or the critic centres corresponding to the two values of θ lie on the conic. The line joining them is the polar of the point $(4, -1, -1)$ in regard to the twofold-centre conic; the equation therefore is

$$(\mu - \nu)x - (3\mu + \nu)y + (\mu + 3\nu)z \;=\; 0.$$

77. Starting with a critic centre on the line $y + z = 0$, the other two critic centres lie on the conic $x(x + y + z) - 4yz = 0$, and they are the intersections of the conic by the polar of the first centre in regard to the twofold-centre conic.

78. Starting with a critic centre on the conic $x(x + y + z) - 4yz = 0$, the other two critic centres lie one on the conic, and the other on the line $y + z = 0$; viz. the polar of the first centre in regard to the twofold-centre conic meets the line in one point, and the conic in two points; of these one is the harmonic of the point on the line in regard to the twofold-centre conic; this point on the conic, and the point on the line, are the other two centres.

79. The point $(4, -1, -1)$ is of course one of a system of three points; viz. these are $(4, -1, -1)$, $(-1, 4, -1)$, $(-1, -1, 4)$; and the corresponding loci of the critic centres are

$$(y + z)\big\{x(x + y + z) - 4yz\big\} \;=\; 0,$$
$$(z + x)\big\{y(x + y + z) - 4zx\big\} \;=\; 0,$$
$$(x + y)\big\{z(x + y + z) - 4xy\big\} \;=\; 0,$$

the three points in question are (*ante*, No. 24) shown to be nodes of the twofold-centre envelope.

80. The line $5x + y + z = 0$ is the line through the points $(-1, 4, -1)$, $(-1, -1, 4)$, and as such the corresponding critic centres lie

- one on the line $z + x = 0$, two on the conic $x(x + y + z) - 4yz = 0$,
- one on the line $x + y = 0$, two on the conic $y(x + y + z) - 4zx = 0$.

The two lines meet in the point $(1, -1, -1)$.

The two conics meet in the points $(1, 0, 0)$, $(2, 3, 3)$; and touch at the point $(0, 1, -1)$, the common tangent being $5x + y + z = 0$: this appears by writing the equations of the two conics in the forms

$$(y - z)(5x + y + z) + (y + z)(-3x + y + z) \;=\; 0,$$
$$-(y - z)(5x + y + z) + (y + z)(-3x + y + z) \;=\; 0,$$

for we have then the four points of intersection put in evidence; viz. these are

$$y - z = 0,\ y + z = 0,\ \text{that is } (1, 0, 0),$$
$$y - z = 0,\ -3x + y + z = 0,\ (2, 3, 3),$$
$$5x + y + z = 0,\ y + z = 0,\ (0, 1, -1),$$
$$5x + y + z = 0,\ -3x + y + z = 0,\ (0, 1, -1).$$

The point of intersection $(1, 0, 0)$, which is an angle of the triangle, is not a critic centre; the three critic centres are the other point of intersection $(2, 3, 3)$; the point of contact $(0, 1, -1)$; and the point of intersection $(1, -1, -1)$ of the two lines.

81. To obtain in a different manner the last-mentioned result it may be remarked that for the line $3x + y + z = 0$, for which $(\lambda, \mu, \nu) = (3, 1, 1)$, the equation in θ is

$$\theta^3 - 7\theta - 6 \;=\; (\theta+1)(\theta+2)(\theta-3) \;=\; 0,$$

so that the values of $\theta + \lambda$, $\theta + \mu$, $\theta + \nu$ are

$$\begin{aligned} \text{for }\ \theta &= -1, \quad 2,\ 0,\ 0,\\ \theta &= -2, \quad 1,\ -1,\ -1,\\ \theta &= 3, \quad 6,\ 4,\ 4, \end{aligned}$$

and the corresponding values of $x : y : z$ are

$$\begin{aligned} \theta = -1 :&\quad \tfrac{1}{2} : \infty : \infty, \ \text{ that is } (0, 1, -1),\\ \theta = -2 :&\quad 1 : -1 : -1, \ \text{ that is } (1, -1, -1),\\ \theta = 3 :&\quad \tfrac{1}{6} : \tfrac{1}{4} : \tfrac{1}{4}, \ \text{ that is } (2, 3, 3), \end{aligned}$$

which points are therefore the critic centres for the line $3x + y + z = 0$.

The last-mentioned line, it is clear, is one of the system of three lines

$$3x + y + z = 0, \quad x + 3y + z = 0, \quad x + y + 3z = 0.$$

82. If $\lambda = 0$, that is if the line $\lambda x + \mu y + \nu z = 0$ pass through an angle $y = 0$, $z = 0$ of the triangle; then reverting to the original equations

$$\frac{-x+y+z}{\lambda x} \;=\; \frac{x-y+z}{\mu y} \;=\; \frac{x+y-z}{\nu z},$$

these give $(y = 0, z = 0)$ or else $(-x + y + z = 0,\ \nu z^2 - \mu y^2 = 0)$, that is, one of the three critic centres is the angle $(y = 0, z = 0)$ of the triangle; and the other two are the intersections of the line $-x + y + z = 0$ with the pair of lines $\nu z^2 - \mu y^2 = 0$.

It should be remarked that, given the critic centre $y = y_1 = 0$, $z = z_1 = 0$, the remaining two centres cannot be determined as the intersection of the polar $-x+y+z = 0$ with the conic

$$\frac{x_1(y_1 - z_1)}{x} + \frac{y_1(z_1 - x_1)}{y} + \frac{z_1(x_1 - y_1)}{z} \;=\; 0,$$

inasmuch as the equation of this conic becomes the identity $0 = 0$.

83. The critic centres for the case in question, $\lambda = 0$, may also be determined by means of the equation of the cubic through the three centres; in fact, since $\lambda = 0$, the equation $\lambda\alpha + \mu\beta + \nu\gamma = 0$ becomes $\mu\beta + \nu\gamma = 0$, that is $\beta : \gamma = \nu : -\mu$; and the equation of the cubic therefore is

$$\nu\left(\frac{1}{y} - \frac{1}{x+y+z}\right) - \mu\left(\frac{1}{z} - \frac{1}{x+y+z}\right) \;=\; 0,$$

and since the ratio $\alpha : \beta$ is arbitrary we have the two equations

$$\frac{1}{x} - \frac{2}{x+y+z} \;=\; 0, \qquad \frac{\nu}{y} - \frac{\mu}{z} \;=\; 0,$$

which resolve themselves into the above-mentioned two equations, $-x+y+z = 0$, $\nu z^2 - \mu y^2 = 0$.

84. Consider a critic centre the coordinates of which are $(0, y_1, z_1)$, that is, which is an arbitrary point on the side $x = 0$ of the triangle: it is to be remarked that there is not any position of the line $\lambda x + \mu y + \nu z = 0$, which properly gives rise to such a critic centre. For writing $x_1 = 0$ the equations

$$\frac{-x_1 + y_1 + z_1}{\lambda x_1} = \frac{x_1 - y_1 + z_1}{\mu y_1} = \frac{x_1 + y_1 - z_1}{\nu z_1}$$

give $\mu = 0$, $\nu = 0$, that is, the line $\lambda x + \mu y + \nu z = 0$ is found to be $x = 0$; but in this case the cubic is $x\bigl(yz + k(x + y + z)^2\bigr) = 0$, which irrespectively of the value of k has nodes at the points $x = 0$, $yz + k(y + z)^2 = 0$, and which only for the value $k = 0$ acquires a third node at the point $y = 0$, $z = 0$: the case is a singular and exceptional one.

85. If notwithstanding we assume a critic centre at the point $(0, y_1, z_1)$, then the other two critic centres are by the general theorem given as the intersection of the line

$$(y_1 + z_1)x - (y_1 - z_1)(y - z) = 0$$

with the conic (pair of lines) $x(y - z) = 0$, that is, we have a twofold centre $x = 0$, $y - z = 0$, or what is the same thing a twofold centre $(0, 1, 1)$.

86. If a critic centre lie on the line $y - z = 0$, then of the other two critic centres, one lies on this same line and the other is the point $x = 0$, $x + y + z = 0$, or say the point $(0, 1, -1)$. And in this case the line $\lambda x + \mu y + \nu z = 0$ passes through the last-mentioned point; that is, we have $\mu = \nu$. Conversely, starting from the equation $\mu = \nu$, so that the line $\lambda x + \mu y + \nu z = 0$ is $\lambda x + \mu(y + z) = 0$, a line through the intersection of the lines $x = 0$, $x + y + z = 0$, the equation in θ is

$$(\theta + \mu)(\theta^2 - \mu\theta - 2\lambda\mu) = 0,$$

where the factor $\theta + \mu = 0$ corresponds to the critic centre $x = 0$, $x + y + z = 0$, or $(0, 1, -1)$ (it will presently be shown that this is so), and the quadric equation $\theta^2 - \mu\theta - 2\lambda\mu = 0$ corresponds to two critic centres on the line $y - z = 0$. We have

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \mu},$$

and thence $y - z = 0$; and $\theta(x - y) = -\lambda x + \mu y$, which substituted in the equation $\theta^2 - \mu\theta - 2\lambda\mu = 0$ gives

$$(\lambda x - \mu y)\bigl\{(\lambda + \mu)x - 2\mu y\bigr\} - 2\lambda\mu(x - y)^2 = 0,$$

and the two critic centres are given as the intersections of this conic by the line $y - z = 0$.

87. Consider for a moment the case $\nu = \mu + \varepsilon$, where ε is ultimately $= 0$. The equation in θ is

$$\frac{1}{\theta + \lambda} + \frac{1}{\theta + \mu} + \frac{1}{\theta + \mu + \varepsilon} = \frac{2}{\theta};$$

then if a root is $\theta = -\mu + A\varepsilon$, we have

$$\frac{1}{A\varepsilon + \lambda - \mu} + \frac{1}{A\varepsilon} + \frac{1}{(A + 1)\varepsilon} = \frac{2}{A\varepsilon - \mu};$$

so that, ε being indefinitely small, we have

$$\frac{1}{A} + \frac{1}{A + 1} = 0, \quad \text{that is } 2A + 1 = 0 \text{ or } A = -\tfrac{1}{2},$$

and then

$$\theta = -\mu - \tfrac{1}{2}\varepsilon, \quad \theta + \lambda = \lambda - \mu - \tfrac{1}{2}\varepsilon, \quad \theta + \mu = -\tfrac{1}{2}\varepsilon, \quad \theta + \nu = +\tfrac{1}{2}\varepsilon,$$

which gives

$$x:y:z \;=\; \frac{1}{\theta+\lambda} \;:\; \frac{1}{\theta+\mu} \;:\; \frac{1}{\theta+\nu} \;=\; \frac{1}{\lambda-\mu-\frac{1}{2}\varepsilon} \;:\; -\frac{2}{\varepsilon} \;:\; \frac{2}{\varepsilon},$$

or, ε being indefinitely small, $x:y:z = 0:1:-1$, so that the factor $\theta+\mu=0$ corresponds, as mentioned above, to the critic centre $(0,1,-1)$.

[End of Paper 349. The paper concludes with Article 87 above, completing the lengthy analytical treatment of the involution of cubic curves of the form $xyz + k(x+y+z)^2(\lambda x+\mu y+\nu z) = 0$.]

350. On the Classification of Cubic Curves

[From the Transactions of the Cambridge Philosophical Society, vol. xi. Part i. (1866), pp. 81–128. Read April 18, 1864.]

The notion of a curve of a given order may be considered as arising from Descartes' invention of the method of coordinates; and one of the earliest applications of the method was made by Sir Isaac Newton in the *Enumeratio linearum tertii Ordinis* (1706), a work worthy of its author, and which opened a new field of geometrical science. The classification is according to the nature of the infinite branches; there are fourteen genera containing together seventy-two species, but four species were added by Stirling in his *Lineae tertii Ordinis Newtonianae; sive Illustratio dx.* (1717), and two more by Murdoch or Cramer,[1] making in all seventy-eight species. A new classification was made by Plücker in his *System der Analytischen Geometrie*, 1835; this is likewise according to the nature of the infinite branches, but after his six head divisions, and some subordinate divisions thereof, Plücker establishes the divisions called *Groups*, which have nothing analogous to them in the Newtonian theory; there are sixty-one groups, and the total number of species is 219.

The present Memoir contains an exposition of the foregoing classifications, and of the principles on which they are founded, in so far as relates to the superior divisions of the two classifications; and in particular I develope more completely than was done by Plücker the theory of the division into groups. I do not however consider otherwise than very slightly the ultimate division into species.

The above-mentioned work of Newton contains, under the heading "Genesis Curvarum per Umbras," the remarkable theorem that the curves of the third order may all of them be considered as the shadows of the five Divergent Parabolas; I reserve for a separate Memoir the whole series of considerations to which this theorem gives rise.

I commence by establishing the theory of the classification of cubic curves according to the nature of their infinite branches, in what appears to me the scientifically correct manner as follows.

The Seven Head Divisions, Article Nos. 1 to 4.

1. A line in general, and therefore the line Infinity, meets a cubic curve in three points, and these may be

- Three onefold points,
- A twofold point and a onefold (or, as it may also be termed, a one-with-twofold) point,
- A threefold point.

2. But in the second case the line Infinity

[1] The two additional species are, I believe, first mentioned in Murdoch's *Genesis Curvarum per Umbras* (1746), but one of them is there ascribed to Cramer.

- may be a proper tangent to the curve,
- may pass through a node,
- may pass through a cusp;

and in the third case the line Infinity

- may touch the curve at an inflexion,
- may at a node touch one of the two branches,
- may touch the curve at a cusp.

3. The first case, the three divisions of the second case, and the three divisions of the third case, give in all seven divisions, which, as will appear in the sequel, fall in with Newton's classification, and can be named in his language, viz.

Three onefold points,	The Hyperbolas.
A onefold and a twofold point;	
Infinity a proper tangent.	The Parabolic Hyperbolas.
Do. through a node.	The Central Hyperbolisms.
Do. through a cusp.	The Parabolic Hyperbolisms.
A threefold point;	
Infinity a tangent at an inflexion.	The Divergent Parabolas.
Do. Do. at a node, to one branch.	The Trident Curve.
Do. Do. at a cusp.	The Cubical Parabola.

4. As regards the signification of these terms, it may be remarked that the Hyperbolas have hyperbolic branches, the Parabolic Hyperbolas, hyperbolic and parabolic branches; where by a hyperbolic branch is meant one having an asymptote, and by a parabolic branch one not having an asymptote. The hyperbolism of any curve is the curve derived from it by altering the ordinate in the ratio of the abscissa to any given line

$$\big(y : y' = x : a, \text{ or say } y' = \tfrac{ay}{x}\big);$$

the expression Central Hyperbolism is used to include Newton's hyperbolisms of the hyperbola and ellipse; and the expression Parabolic Hyperbolism to denote his hyperbolism of the parabola. The Divergent Parabolas are curves the branches of which ultimately diverge from each other as in the semicubical parabola $y^2 = x^3$. The names Trident Curve and Cubical Parabola are in fact not generic but specific; it so happens that the genera to which they respectively belong contain each only a single species. The names for the several kinds of curves are not scientifically devised ones, but it is convenient to have them such as they are.

The foregoing seven divisions, uniting in one the Central Hyperbolisms and the Parabolic Hyperbolisms, are the six head divisions of Plücker.

Asymptotes, &c. Equations for the Seven Head Divisions. Article Nos. 5 to 22.

5. For a Hyperbola there is at each of the points at infinity a tangent, which is an asymptote; and the hyperbola has thus three asymptotes.

6. For a Parabolic Hyperbola there is at the onefold point at infinity a tangent, which is an asymptote. There may be described a conic having with the curve at the twofold point at infinity a five-pointic intersection.[2] Such conic, as having the line infinity for a tangent, is a parabola, and it may be termed the asymptotic parabola: the Parabolic Hyperbola has thus an asymptote and an asymptotic parabola.

[2] I have elsewhere spoken of the conic of five-pointic contact: the expression five-pointic intersection is more accurate.

7. For a Central Hyperbolism there is at the onefold point at infinity a tangent which is an asymptote, and which for distinction may be called the onefold asymptote; and at the node or twofold point at infinity there is a pair of tangents which are the parallel asymptotes.

8. For a Parabolic Hyperbolism there is at the onefold point at infinity a tangent which is an asymptote, and which may be called the onefold asymptote; and at the cusp or twofold point at infinity a twofold tangent which is an asymptote, and which may be called the twofold asymptote.

9. For a Divergent Parabola there is not any asymptote or asymptotic conic; but we may consider an asymptotic cubic, viz. this will be a semicubical parabola ($y^2 = x^3$), which is in fact one of the divergent parabolas, the cuspidal divergent parabola, and which may be in general so determined as to have at the inflexion or threefold point at infinity a seven-pointic intersection. For the asymptotic cubic, in order that it may have a cusp, must satisfy two conditions; it may therefore be made to satisfy seven more conditions, or to have a seven-pointic intersection; and then the original curve having an inflexion or threefold point at infinity, the asymptotic cubic will *ipso facto* have the same point as an inflexion, or threefold point at infinity, and be thus a cuspidal divergent parabola.

10. For the Trident Curve, we have at the node or threefold point at infinity, viz. to the branch which is not touched by the line infinity, a tangent which is an asymptote: this cuts at the node the other branch of the curve, and it is therefore an asymptote of three-pointic intersection. We may describe a conic having at the node a five-pointic intersection with the other branch of the curve; such conic as touching the line infinity is a parabola, and it may be called the asymptotic parabola; since the parabola cuts at the node the first-mentioned branch of the curve, viz. the branch not touched by the line infinity, the parabola is in fact a parabola of six-pointic intersection. The Trident Curve has thus an asymptote and an asymptotic parabola of six-pointic intersection.

11. For the Cubical Parabola there is not any asymptote or asymptotic conic: the curve *quâ* curve having a cusp (viz. the cusp or threefold point at infinity) has a single inflexion; and the line joining the cusp with the inflexion, regarded as a threefold line, has with the curve a six-pointic intersection at infinity, and may be considered as an asymptotic cubic.

12. We have in every case a cubic curve $V = 0$ having with the original curve an intersection at infinity which is at least six-pointic, and which I call the asymptotic aggregate: viz. the asymptotic aggregate is

- For the Hyperbolas; the three asymptotes, intersection six-pointic.
- For the Parabolic Hyperbolas; the asymptote and the asymptotic parabola, intersection seven-pointic.
- For the Central Hyperbolisms; the onefold asymptote and the parallel asymptotes, intersection eight-pointic.
- For the Parabolic Hyperbolisms; the onefold asymptote and the twofold asymptote regarded as a twofold line; intersection eight-pointic.
- For the Divergent Parabolas; the asymptotic semicubical parabola, intersection seven-pointic.
- For the Trident Curve; the asymptote and the asymptotic parabola, intersection nine-pointic.
- For the Cubical Parabola; the line joining the cusp at infinity with the inflexion, regarded as a threefold line, intersection six-pointic.

13. I have said that the intersection at infinity is at least six-pointic; but more than this, the intersection at any onefold point at infinity is at least two-pointic; at a twofold point at infinity it is at least four-pointic; and at a threefold point at infinity it is at least six-pointic.

It follows that the intersections at infinity of the cubic and the asymptotic aggregate include the six intersections of the cubic by the line infinity considered as a twofold line; and hence the remaining three intersections of the cubic and the asymptotic aggregate must lie in a line $s = 0$ (Plücker's line S), which I call the *satellite line*. And writing $z = 0$ for the equation of the line infinity, the equation of the cubic is of the form $U = V + \mu z^2 s = 0$. It is to be observed moreover, that when, as for the Hyperbolas and the Cubical Parabola, the intersection at infinity is six-pointic, the line $s = 0$ is an arbitrary line; when as for the Parabolic Hyperbolas, and the Divergent Parabolas, the intersection is seven-pointic, the line $s = 0$ meets the cubic in a given point at infinity, viz. the twofold or the threefold point at infinity; and when as for the Central Hyperbolisms and the Parabolic Hyperbolisms the intersection is eight-pointic, the line $s = 0$ has with the cubic a given twofold intersection at infinity; this however merely implies that the line $s = 0$ passes through the node or cusp at infinity, and so imposes only one condition on the line $s = 0$. Finally, when as in the Trident Curve the intersection is nine-pointic, the line $s = 0$ has with the curve a given threefold intersection at infinity; that is, it coincides with the line infinity, $z = 0$.

14. The preceding considerations in regard to the asymptotic aggregate $V = 0$, lead very directly to the best analytical form of the function V, and therefore to that of the equation $U = V + \mu z^2 s = 0$, of the cubic.

15. For the Hyperbolas; the equations of the asymptotes being $p = 0$, $q = 0$, $r = 0$, we have $V = pqr = 0$ for the asymptotic aggregate; the satellite line is arbitrary, and hence

$$\text{Equation of the Hyperbolas is} \quad pqr + \mu z^2 s \;=\; 0.$$

16. For the Parabolic Hyperbolas. Imagine parallel to the asymptote a line $q = 0$ touching the asymptotic parabola; and let the line joining the point of contact with the twofold point at infinity have for its equation $q = 0$; the equation of the asymptote is

$$p + \kappa z \;=\; 0,$$

that of the asymptotic parabola is $q^2 + \lambda pz = 0$, and hence the equation of the asymptotic aggregate is $(p+\kappa z)(q^2+\lambda pz) = 0$; the satellite line passes through the twofold point at infinity, or its equation is $q + \sigma z = 0$; hence

$$\text{Equation of the Parabolic Hyperbolas is} \quad (p + \kappa z)(q^2 + \lambda pz) + \mu z^2(q + \sigma z) \;=\; 0.$$

17. For the Central Hyperbolisms; the equation of the onefold asymptote is taken to be $p = 0$, and that of the parallel asymptotes to be $q^2 + \kappa z^2 = 0$; hence the equation of the asymptotic aggregate is $p(q^2 + \kappa z^2) = 0$, the satellite line passes through the twofold point at infinity, its equation is $q + \sigma z = 0$; hence

$$\text{Equation of the Central Hyperbolisms is} \quad p(q^2 + \kappa z^2) + \mu z^2(q + \sigma z) \;=\; 0.$$

18. For the Parabolic Hyperbolisms: the only difference is that instead of the parallel asymptotes $q^2 + \kappa z^2 = 0$ we have the twofold asymptote $q^2 = 0$; hence

$$\text{Equation of the Parabolic Hyperbolisms is} \quad pq^2 + \mu z^2(q + \sigma z) \;=\; 0.$$

19. For the Divergent Parabolas: the asymptotic aggregate is a semicubical parabola; let $q = 0$ be the equation of the cuspidal tangent, $p = 0$ the equation of the line joining the cusp with the inflexion at infinity, then the equation is $p^3+\lambda q^2 z=0$. The satellite line passes through the threefold point at infinity, its equation is $p+\sigma z=0$; hence

$$\text{Equation of the Divergent Parabolas is}\quad p^3+\lambda q^2 z+\mu z^2(p+\sigma z) \;=\; 0.$$

20. For the Trident Curve: let $p=0$ be the equation of the asymptote, $q=0$ that of the tangent to the asymptotic parabola at the point not at infinity where it is met by the asymptote, then the equation of the parabola is $p^2+\lambda qz=0$, and that of the asymptotic aggregate is $p(p^2+\lambda qz)=0$; the satellite line is the line infinity, $z=0$; hence

$$\text{Equation of the Trident Curve is}\quad p(p^2+\lambda qz)+\mu z^3 \;=\; 0.$$

21. For the Cubical Parabola: let $p=0$ be the equation of the line joining the inflexion with the cusp at infinity, then the asymptotic aggregate is this line taken as a threefold line, or the equation is $p^3=0$; the satellite line is arbitrary; hence

$$\text{Equation of the Cubical Parabola is}\quad p^3+\mu z^2 s \;=\; 0.$$

22. It is convenient to notice here that for the Hyperbolas the line $s=0$ is determined as follows, viz. the line infinity meets the curve in three points, and the tangents at these points (the asymptotes) again meet the curve in three points lying in a line which is the line in question; in other words, the line $s=0$ is (in the sense in which I have elsewhere used the term) the satellite line of infinity. For the other kinds of cubic curves, the line $s=0$ is not, in the sense just referred to, the satellite line of infinity; but in the present Memoir I shall in every case call the line $s=0$, the satellite line.

The Thirteen Divisions. Article Nos. 23 to 33.

23. The characters of the foregoing seven divisions are irrespective of reality; and before going further it may be remarked, that as to the Hyperbolas and the Parabolic Hyperbolas a subdivision also irrespective of reality may be made as follows.

24. For a Hyperbola, the three asymptotes may not meet in a point, or they may meet in a point. For shortness I say that in the former case we have a Hyperbola A, in the latter case a Hyperbola B. I consider more particularly (post. No. 41) the special case of a Hyperbola B.

25. For a Parabolic Hyperbola, the asymptote may meet the asymptotic parabola in two onefold points; or in a twofold point.

26. I come now to the divisions which depend on reality: it is assumed that the curve is real.

27. For the Hyperbola the three points at infinity may be all real or else one real, two imaginary. In the former case, the asymptotes are all real, and we have the *redundant hyperbola*; in the latter case the real point at infinity gives rise to a real asymptote, the imaginary points to imaginary asymptotes: we have in this case the *defective hyperbola*. It is to be noticed that the imaginary asymptotes meet in a real point, called the *asymptote-point*; and that such point, if we regard it as an indefinitely small ellipse given as to the position and ratio of its axes, determines the imaginary asymptotes. Combining the division with the A, B, we have four subdivisions of the Hyperbola.

28. For a Hyperbola A redundant the three asymptotes form a triangle, and for a Hyperbola B redundant they meet in a point. For a Hyperbola A defective, the asymptote-point does not lie on the real asymptote; for a Hyperbola B defective it does lie on the real asymptote.

29. For a Parabolic Hyperbola: the onefold point and the twofold point at infinity are of necessity real, as are also the asymptote and the asymptotic parabola. If the asymptote meets the asymptotic parabola in two onefold points, these may be both real or both imaginary; if it meets it in a twofold point, this is real. We have thus three subdivisions of the Parabolic Hyperbola. For the Central Hyperbolism, the onefold point, and the node or twofold point at infinity, are both real; the asymptote is also real. But the node may be a crunode or an acnode; that is, the tangents at the node, or parallel asymptotes, may be both real, or both imaginary: we have thus two subdivisions, viz. the Hyperbolism of the hyperbola, and the Hyperbolism of the ellipse.

30. For the Parabolic Hyperbolism, the onefold point and the cusp or twofold point at infinity, and also the onefold asymptote and the twofold asymptote are all real.

31. For the Divergent Parabola, the inflexion or threefold point at infinity is real.

32. For the Trident Curve the node or threefold point at infinity is real, and inasmuch as one of the tangents is the line infinity, the node is a crunode, and the other tangent, or asymptote of the curve, is also real.

33. For the Cubical Parabola, the cusp or threefold point at infinity and the tangent at this point are each real.

Reckoning the hyperbolas as 4, the parabolic hyperbolas as 3, the central hyperbolisms as 2, and the parabolic hyperbolisms, the divergent parabolas, the trident curve, and the cubical parabola, each as 1, we have in all 13 divisions.

The Notion of a Group. Article No. 34.

34. I remark that the characters as well of the 7 divisions as of the 13 divisions have exclusive reference to the form of the asymptotic aggregate $V = 0$; we have an ulterior division depending on the relation of the satellite line to the asymptotic aggregate, and which I regard as the proper origin of Plücker's Groups: viz. for a given form of the asymptotic aggregate $V = 0$, and corresponding to each characteristically distinct position in relation thereto of the satellite line $s = 0$, we have a *Group*. The determination of the characteristically distinct positions of the satellite line cannot be completely effected *à priori*; for instance, in the case of the Hyperbolas A redundant, the distinctions which immediately present themselves are that the satellite line cuts the three sides produced, or two sides and the third side produced, of the triangle formed by the asymptotes, or passes through an angle of the triangle, &c.; but these are not all the distinctions which have to be made; to determine them, taking the satellite line as given, we discuss the series of curves represented by the equation $V + \mu z^2 s = 0$; for instance (and it is on this that the discussion chiefly turns), we see that the parameter μ may be so determined that the curve shall have a node, but the reality or non-reality of the roots of the equation in μ, and therefore the existence of a real nodal curve or curves, will depend on the position of the satellite line $s = 0$; and it is thus only by the discussion of the group that we arrive at an enumeration of the different groups.

Osculating Asymptotes and other Specialities. Article Nos. 35 to 41.

35. But Plücker nevertheless, prior to the establishment of his groups, introduces certain intermediate divisions as to osculating asymptotes, &c., which have really reference to the position of the satellite line; an osculating asymptote gives rise to a 'diameter,' and the diameter is a distinctive character in the Newtonian genera; to explain how all this is, I proceed as follows.

36. The parallel asymptotes of a Central Hyperbolism, the twofold asymptote of a Parabolic Hyperbolism and the asymptote of the Trident Curve are singular asymptotes, that is, each of them touches the curve at a node or a cusp, and is thus an asymptote of three-pointic intersection. Excluding these, and using the term asymptote to denote a non-singular asymptote, an asymptote is in general an ordinary tangent or asymptote of two-pointic intersection; if, however, the point of contact is an inflexion, then the asymptote is an asymptote of three-pointic intersection, or osculating asymptote. In particular for the Hyperbolas, the asymptotes may be all ordinary, or they may be two ordinary and one osculating, or all three osculating; but they cannot be only two of them osculating; for the line through two inflexions meets a cubic curve in a third point which is also an inflexion; that is, if two asymptotes are osculating, the third is also an osculating asymptote. The foregoing remarks apply as well to the defective as the redundant Hyperbolas; it is to be noticed, however, as regards the defective Hyperbolas that the osculating asymptote, when there is only one, is necessarily the real asymptote, and consequently that the cases are – asymptotes ordinary; the real asymptote alone osculating; three osculating asymptotes. For the Parabolic Hyperbolas the asymptote, and for the Central Hyperbolisms and the Parabolic Hyperbolisms the onefold asymptote, may be ordinary or osculating.

37. The distinction of ordinary and osculating asymptotes has reference to the position of the satellite line; viz. for the Hyperbolas, when there is a single osculating asymptote, the satellite line passes through the point at infinity of the osculating asymptote, or what is the same thing, the satellite line is parallel to the osculating asymptote; and when there are three osculating asymptotes, the satellite line coincides with the line infinity. And, conversely, when the satellite line is parallel to an asymptote such asymptote is an osculating one, and when the satellite line is at infinity the three asymptotes are osculating. For the Parabolic Hyperbolas the asymptote, and for the Hyperbolisms the onefold asymptote, is an osculating asymptote when the satellite line is at infinity; and conversely.

38. There is in regard to the Divergent Parabolas a distinction which may be mentioned here; viz. the satellite line may disappear altogether ($\mu = 0$), and the curve thus coincide with the asymptotic semicubical parabola. Or, what is the general case, the satellite line may be distinct from the line infinity, – and it may cut in two real points, touch, or cut in two imaginary points the asymptotic semicubical parabola; or the satellite line may coincide with the line infinity, the asymptotic semicubical parabola being in this case of nine-pointic intersection.

39. The term "diameter" is used by Newton in the *Enumeratio* in two different senses; viz. for any given direction of the ordinates there exists a right line or "diameter," such that measuring the ordinates from this line the sum $y + y' + y''$ of the three ordinates is $= 0$. Such diameter is in fact the second or line polar in regard to the cubic of an arbitrary point on the line infinity. But the term diameter is afterwards and will be here used to denote a diameter *absolute dictum*, viz. for a direction of the ordinates parallel to a non-singular asymptote there may exist a right line or "diameter" such that the ordinates measured from this point are equal and opposite to each other, or what is the same thing, such that the sum $y + y'$ of the two ordinates is $= 0$; this implies that the asymptote is an osculating asymptote. In fact, the first or conic polar of any inflexion of the cubic breaks up into a pair of lines, one of which is the tangent at the inflexion, the other of them, the 'polar' of the inflexion, a line which cuts harmonically the chords through the inflexion, and which when the inflexion is at infinity becomes a diameter. The remarks previously made as to osculating asymptotes apply therefore to diameters, viz. the Hyperbolas may have no diameter, a single diameter, or three diameters, &c.

40. Newton speaks also of the "centre" of a cubic curve; viz. there may be a point on the curve such that for any line through this point the two radius vectors are equal and opposite to each other, or that the sum $r + r'$ of the two radius vectors is $= 0$. The centre is in fact a point of inflexion which has for its polar the line infinity. The curves which may have a centre are the Hyperbolas (redundant or defective), the Central Hyperbolisms (of the hyperbola or ellipse) and the Cubical Parabola. For the hyperbolas, the three asymptotes and the satellite line must meet in a point of the curve, which point is then the centre; for the central hyperbolisms the onefold asymptote and the satellite line must meet in a point of the curve, which point is then a centre; and for the cubical parabola no condition is required, but the inflexion is a centre. I remark here, in passing, that the notion of a centre as just explained has no place in Plücker's Classification, and that the two Newtonian species 58 and 59 (hyperbolisms of the hyperbola) and the two Newtonian species 61 and 62 (hyperbolisms of the ellipse) which differ, the two of a pair from each other, according as there is no centre or a single centre, form each pair a single species with Plücker; viz. they are 198 and 201 respectively.

41. It has been already remarked that the three asymptotes of a Hyperbola may meet in a point. As to this it is to be noticed that from any point we may draw six tangents to a cubic, the points of contact lie on a conic, the conic polar of the point; if, however the point lie on the Hessian of the cubic, then the conic breaks up into a pair of lines, each of which is a tangent to the Pippian; the two lines meet in a point of the Hessian, which point forms with the first mentioned point a pair of conjugate poles of the cubic.[3]

Conversely, any tangent of the Pippian meets the cubic in three points, the tangents at which meet in a point of the Hessian; and from this point we may draw to the cubic three other tangents the points of contact of which lie on a line which is also a tangent of the Pippian, and the two tangents of the Pippian meet in a point of the Hessian; the two points of the Hessian being conjugate poles of the cubic. In particular, if the line infinity is a tangent of the Pippian, then the three asymptotes meet in a point of the Hessian, and the three tangents from this point to the cubic touch the cubic in three points lying on a line which is a tangent of the Pippian, and which meets the line infinity in a point forming with the first mentioned point a pair of conjugate poles of the cubic.

I proceed now to explain the classification of Newton so far as relates to the division into genera, and the classification of Plücker so far as relates to the divisions immediately superior to the groups.

Newton's Classification. Article Nos. 42 to 46.

42. Newton establishes in the first instance the following four cases; viz. the equation of a cubic curve is one of the forms

$$\begin{aligned}
\text{I.}\quad & xy^2+ey \;=\; ax^3+bx^2+cx+d,\\
\text{II.}\quad & xy \;=\; ax^3+bx^2+cx+d,\\
\text{III.}\quad & y^2 \;=\; ax^3+bx^2+cx+d,\\
\text{IV.}\quad & y \;=\; ax^3+bx^2+cx+d.
\end{aligned}$$

It is not, I think, necessary to reproduce here the very interesting reasoning by means of which this most important step in the classification was effected.

43. Starting from the four cases, Newton obtains his 14 genera, viz. Case I gives 11 genera, and Cases II, III, IV give each a single genus. But these genera group themselves as follows, viz. 1, 2, 3, 4, 5, 6 are Hyperbolas; 7 and 8, Parabolic Hyperbolas; 9 and 10, Central Hyperbolisms; 11, Parabolic Hyperbolisms; 12, the Trident Curve; 13, the Divergent Parabolas; and 14, the Cubical Parabola. And the equations are as follows:

$$\begin{aligned}
\text{the Hyperbolas}\quad & xy^2+ey \;=\; ax^3+bx^2+cx+d,\\
\text{the Parabolic Hyperbolas}\quad & xy^2+ey \;=\; bx^2+cx+d,\\
\text{the Central Hyperbolisms}\quad & xy^2+ey \;=\; cx+d,\\
\text{the Parabolic Hyperbolisms}\quad & xy^2+ey \;=\; d,\\
\text{the Divergent Parabolas}\quad & y^2 \;=\; ax^3+bx^2+cx+d,\\
\text{the Trident Curve}\quad & xy \;=\; ax^3+bx^2+cx+d,\\
\text{the Cubical Parabola}\quad & y \;=\; ax^3+bx^2+cx+d,
\end{aligned}$$

where it is to be understood that the highest expressed power on the right-hand side of each equation does not vanish.

[3]See as to this theory my Memoir on *Curves of the Third Order*, Phil. Trans. p. 147 (1856), [145].

44. In these equations the axes $x = 0$, $y = 0$ are not for the most part lines precisely determined in relation to the curve, but it is easy to see as well analytically as geometrically how by a proper transformation of the equations they may be brought into forms such as those previously obtained, in which the several lines $p = 0$, etc., stand in a determinate relation to the curve. Thus, taking the equation of the Cubical Parabola, this may be written $y = a\big(x + \frac{b}{3a}\big)^3 + c'x + d'$; or, what is the same thing, $y' = ax'^3$. Or geometrically, we see that $x = 0$ is a line completely determined as to its direction, it is in fact a line through the cusp at infinity; but that $y = 0$ is an arbitrary line in regard to the curve; taking for $y = 0$ the tangent at the inflexion, and for $x = 0$ the line from the inflexion to the cusp at infinity, then the curve must pass through the point $(x = 0, y = 0)$, and $y = 0$ must give a threefold value of x; the equation thus is $y = ax^3$. And so in other cases.

45. In the division into genera, Newton distinguishes the Hyperbolas into the redundant and defective, and the redundant hyperbolas into those for which the asymptotes form a triangle, and those for which the asymptotes meet in a point. The redundant hyperbolas with asymptotes forming a triangle are distinguished according as they have no diameter, a single diameter, or three diameters. The like distinction might have been, but is not, made as to the redundant hyperbolas with asymptotes meeting in a point; these are in fact included in a single genus; but the distinction presents itself in the species of that genus. As to the defective hyperbolas, Newton attends only to the real asymptote; and the only distinction is according as they have no diameter or a real diameter. The Parabolic Hyperbolas are in like manner divided according as they have no diameter or a diameter. The Central Hyperbolisms, according as the parallel asymptotes are real or imaginary, are the hyperbolisms of the hyperbola or of the ellipse. The hyperbolisms of the hyperbola form a single genus. Each of the Hyperbolisms might have been distinguished according as there is no diameter or a single diameter; this distinction appears in the species. The Trident Curve, the Divergent Parabolas, and the Cubical Parabola, form each a single genus.

46. We have thus the following Table of the Newtonian genera: I show in it the species in each genus, retaining Newton's numbers, and distinguishing by the numbers $10'$, $13'$, $22'$, $22''$ the four species added by Stirling, and by $56'$ and $56''$ the two species added by Murdoch or Cramer: I show also the division of genus 4, according to the number of diameters; and I also show the five species of curves having a centre.

Table of the Newtonian Genera.

1. Redundant Hyperbolas with asymptotes forming a triangle, and without a diameter.
Sp. 1, 2, 3, 4, 5, 6, 7, 8, 9.

2. Redundant Hyperbolas with asymptotes forming a triangle, and with a single diameter.
Sp. 10, $10'$, 11, 12, 13, $13'$, 14, 15, 16, 17, 18, 19, 20, 21.

3. Redundant Hyperbolas with asymptotes forming a triangle, and with three diameters.
Sp. 22, $22'$, $22''$, 23.

4. Redundant Hyperbola with asymptotes meeting in a point.
Without a diameter, Sp. 24, 25, 26, 27. With one diameter, Sp. 28, 29, 30, 31. With three diameters, Sp. 32. Sp. 27 has a centre.

5. Defective Hyperbolas without a diameter.
Sp. 33, 34, 35, 36, 37, 38. Sp. 38 has a centre.

6. Defective Hyperbolas with a diameter.
Sp. 39, 40, 41, 42, 43, 44, 45.

7. Parabolic Hyperbolas without a diameter.
Sp. 46, 47, 48, 49, 50, 51, 52.

8. Parabolic Hyperbolas with a diameter.
Sp. 53, 54, 55, 56, $56'$, $56''$.

9. Hyperbolisms of the hyperbola.
Without a diameter, Sp. 57, 58, 59. With a diameter, Sp. 60. Sp. 59 has a centre.

10. Hyperbolisms of the ellipse.
Without a diameter, Sp. 61, 62. With a diameter, Sp. 63. Sp. 62 has a centre.

11. Hyperbolisms of the parabola.
Without a diameter, Sp. 64. With a diameter, Sp. 65.

12. Trident Curve, Sp. 66.

13. Divergent Parabolas. Sp. 67, 68, 69, 70, 71.

14. Cubical Parabola. Sp. 72.

Plücker's Classification. Article Nos. 47 to 49.

47. Plücker in the first instance obtains by analytical considerations his six head divisions, corresponding to the seven divisions in the present Memoir, the Central Hyperbolisms and the Parabolic Hyperbolisms forming with him a single division. The equations are obtained in the form already mentioned, the only difference being that he writes $z = 1$; his six head divisions with their equations thus are

$$\begin{array}{rl}
\text{Hyperbolas} & pqr + \mu s \;=\; 0, \\
\text{Parabolic Hyperbolas} & (p+\kappa)(q^2+\lambda p) + \mu(q+\sigma) \;=\; 0, \\
\text{Hyperbolisms} & p(q^2+\kappa) + \mu(q+\sigma) \;=\; 0, \\
\text{Divergent Parabolas} & p^3 + \lambda q^2 + \mu(p+\sigma) \;=\; 0, \\
\text{Trident Curve} & p(p^2+\lambda q) + \mu \;=\; 0, \\
\text{Cubical Parabola} & p^3 + \mu s \;=\; 0.
\end{array}$$

48. He then divides the Hyperbolas into the redundant and defective. The redundant hyperbolas are then divided as they have no osculating asymptote, one osculating asymptote, or three osculating asymptotes; and each of these according as the asymptotes form a triangle or meet in a point. As regards the defective hyperbolas he attends to the imaginary asymptotes, represented by means of their real point of intersection, the "asymptote-point," and the division is thus similar to that of the redundant hyperbolas, viz. the defective hyperbolas are distinguished according as they have no osculating asymptote, a real osculating asymptote, or three osculating asymptotes; and each of these according as the asymptotes form a triangle or meet in a point; that is, according as the asymptote-point does not, or does, lie on the real asymptote.

The Parabolic Hyperbolas are distinguished according as the asymptote is ordinary and the asymptotic parabola one of five-pointic intersection; or, as the asymptote is osculating and the parabola one of six-pointic intersection; and each of these according as the asymptote cuts, touches, or does not cut, the parabola. The Hyperbolisms are distinguished into those of the hyperbola, ellipse, and parabola, and each of these according as the asymptote is ordinary or osculating. The Divergent Parabolas are distinguished in the manner already mentioned; viz. according as the curve is the semicubical parabola; or, as there is a satellite line not at infinity, and an asymptotic semicubical parabola of seven-pointic intersection, and which is cut, touched, or not cut, by the satellite line; or, as the satellite line is at infinity, and the semicubical parabola is of nine-pointic intersection. The Trident Curve and the Cubical Parabola are not divided.

49. I annex the following Table showing the Groups included in each division: for shortness I use the before-mentioned symbols A, B to denote that the asymptotes form a triangle or meet in a point respectively.

Table of the Plückerian Divisions.

Hyperbolas:

Redundant;
No osculating asymptote,
A I, II, III, IV, V, VI
B VII, VIII
One osculating asymptote,
A IX, X, XI, XII, XIII, XIV
B XV
Three osculating asymptotes,
A XVI
B XVII
Defective;
No osculating asymptote,
A XVIII, XIX, XX, XXI, XXII, XXIII
B XXIV, XXV, XXVI, XXVII
Real osculating asymptote,
A XXVIII, XXIX, XXX, XXXI, XXXII, XXXIII
B XXXIV
Three osculating asymptotes,
A XXXV
B XXXVI

Parabolic Hyperbolas:

Ordinary asymptote and five-pointic parabola.
asymptote cuts parabola XXXVII, XXXVIII, XXXIX, XL, XLI
does not cut „ XLII
touches „ XLIII, XLIV

Osculating asymptote and six-pointic parabola.
asymptote cuts parabola XLV
does not cut „ XLVI
touches „ XLVII

Hyperbolisms:

Of hyperbola;
asymptote ordinary XLVIII, XLIX
„ osculating L
Of ellipse;
asymptote ordinary LI
„ osculating LII
Of parabola;
asymptote ordinary LIII
„ osculating LIV

Divergent Parabolas:

Semicubical Parabola LV
Asymptotic ditto, seven-pointic.
satellite line cuts semic. parab. LVI
does not cut „ LVII
touches (i.e. passes through the vertex of) LVIII
Asymptotic ditto, nine-pointic LIX

Trident Curve LX
Cubical Parabola LXI

As to the Theory of Groups. Article No. 50.

50. It has been already remarked that the division into groups depends on the different positions of the satellite line in regard to the asymptotic aggregate, and that the investigation relates in a great measure to the discussion of the values of the parameter μ in the equation $V + \mu z^2 s = 0$, which are such as to give rise to a nodal curve. It is to be noticed that except for the Hyperbolas and for the Cubical Parabola, the satellite $s = 0$ is either a line passing through a given point at infinity (determinate, that is, as regards its direction), or in the case of the Trident Curve it is the given line infinity; there is at most only a single series of positions to be considered, and the theory is a short and easy one; and for the Cubical Parabola, although the satellite line $s = 0$ is here an arbitrary line, yet on account of the cusp at infinity, there is not any critic value of μ, or in fact any distinction of cases. The only other case where the satellite line $s = 0$ is an arbitrary line, admitting therefore of a double series of positions, is that of the Hyperbolas; and the division into groups constitutes an extensive and interesting theory, which is insufficiently discussed by Plücker; and it was with a view to the development of this theory that my Memoir, *On a Case of the Involution of two Cubic Curves*, (*ante*, pp. 39 to 81), referred to in the sequel as Memoir on Involution, [349], was written. I remark that of the three curves there established as material to the theory, and which are further spoken of in the sequel of the present memoir, viz. the envelope, the twofold-centre locus, and the one-with-twofold-centre locus, Plücker considers only the twofold-centre locus. I proceed to apply the results of that Memoir to the present theory.

As to the Groups of the Hyperbolas A. Article Nos. 51 to 53.

51. The assumed form of equation was $pqr + \mu z^2 s = 0$, but using now

$$x = 0, \quad y = 0, \quad z = 0$$

(instead of $p = 0$, $q = 0$, $r = 0$) for the equations of the asymptotes, we may imagine the implicit constants so determined that the line infinity (before represented by $z = 0$) shall have for its equation $x + y + z = 0$; writing moreover $\lambda x + \mu y + \nu z = 0$ for the satellite line $s = 0$, and k in the place of μ, the equation becomes

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \ = \ 0,$$

which is the form considered in the Memoir just referred to.

52. It is there shown that for an arbitrary position of the satellite line, the parameter k, or, what is the same thing, the auxiliary parameter θ, may be determined by a cubic equation in such manner that the curve shall have a node; the node, or rather the site of the node is termed a *critic centre*; and there are consequently three critic centres (all real or else one real, two imaginary). If however the satellite line touches a certain curve called the envelope, then two of the critic centres unite together, forming a *twofold centre* which is (not a mere node but) a cusp on the corresponding cubic curve; the other critic centre is termed a *one-with-twofold centre*; and the loci of the twofold centre and of the one-with-twofold centre respectively are determinate curves, the former a conic, the latter a cubic. The critic centres corresponding to the entire series of satellite lines which pass through a certain fixed point lie on a cubic, which when the fixed point lies on the line $x + y + z = 0$, here the line infinity, degenerates into a conic called the "Harmonic Conic," and when one of the critic centres is known the other two are determined as the intersections of the harmonic conic by the polar of the given critic centre in regard to the twofold-centre conic.

53. For establishing the theory of the groups of the hyperbolas A, it is necessary to consider the geometrical forms of the several curves which have been just referred to; viz. this is to be done, on the assumption always that the line $x + y + z = 0$ is the line infinity, for the Redundant Hyperbolas taking the lines $x = 0$, $y = 0$, $z = 0$ to be real lines and for the Defective Hyperbolas, taking them to be one real and the other two imaginary. The formulae of the memoir above referred to are in their actual form adapted to the former case, but they can of course be transformed so as to adapt them to the latter case. I proceed to examine the two cases separately.

The hyperbolas A Redundant (See fig. 1). Article Nos. 54 to 71.

54. Everything is symmetrical with respect to the three asymptotes, and to fix the ideas and without any real loss of generality we may consider the asymptotes as forming an equilateral triangle. Taking the perpendicular distance of a vertex from the opposite side as unity, the absolute magnitudes of the coordinates may be fixed by assuming $x+y+z=1$; x, y, z will then denote the perpendicular distances of a point from the three sides respectively. If the coordinates of a point are proportional to α, β, γ, then the absolute magnitudes are of course

$$\frac{\alpha}{\alpha+\beta+\gamma},\quad \frac{\beta}{\alpha+\beta+\gamma},\quad \frac{\gamma}{\alpha+\beta+\gamma};$$

the point may be spoken of indifferently as the point (α,β,γ) or the point

$$\left(\tfrac{\alpha}{\alpha+\beta+\gamma},\ \tfrac{\beta}{\alpha+\beta+\gamma},\ \tfrac{\gamma}{\alpha+\beta+\gamma}\right),$$

and it is sometimes convenient to use both of the two notations; thus the Harmonic point (that is, the point the harmonic of infinity in regard to the triangle) is the point $(1,1,1)$ or $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$. The lines $y-z=0$, $z-x=0$, $x-y=0$, which are the lines joining the harmonic point with the three vertices respectively, are in the case of the equilateral triangle the perpendiculars from the vertices on the three sides respectively, and they may be spoken of simply as the perpendiculars, but it is to be borne in mind that the former is the proper construction of these lines.

55. The equation of the envelope is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}\ =\ 0,$$

or, what is the same thing,

$$x^4+y^4+z^4-4(yz^3+y^3z+zx^3+z^3x+xy^3+x^3y)+6(y^2z^2+z^2x^2+x^2y^2)-124(x^2yz+xy^2z+xyz^2)\ =\ 0.$$

The curve consists, as shown in the figure, of a trigonoid branch inscribed in the triangle and of three acnodes outside the triangle.

56. The side $x=0$ touches the curve in the point $(0,1,1)$ or $(0,\frac{1}{2},\frac{1}{2})$, which is its intersection with the perpendicular $y-z=0$; the side $x=0$ has with the curve at the point in question a four-pointic intersection. The last-mentioned line $y-z=0$ meets the curve in the point $(-4,1,1)$ or $(2,-\frac{1}{2},-\frac{1}{2})$, which is one of the acnodes, and therefore a point of twofold intersection; then again in the point $(16,1,1)$ or $(\frac{8}{9},\frac{1}{18},\frac{1}{18})$ which may be considered as a vertex of the trigonoid branch, and finally in the before-mentioned point $(0,1,1)$ or $(0,\frac{1}{2},\frac{1}{2})$, which is the point of contact with the side $x=0$.

57. The equation of the twofold-centre locus is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}\ =\ 0,$$

or in a rational form

$$x^2+y^2+z^2-2yz-2zx-2xy\ =\ 0,$$

which is, in the case of an equilateral triangle, a circle inscribed in the triangle and touching the sides at their midpoints respectively. The circle is shown in the figure.

58. The equation of the one-with-twofold-centre locus is

$$-(-x+y+z)(x-y+z)(x+y-z)+xyz\ =\ 0,$$

or, what is the same thing,

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+8xyz \;=\; 0.$$

It is a cubic having the harmonic point $(1,1,1)$ or $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$ for an acnode, touching the sides of the triangle externally at their midpoints respectively, and having the three asymptotes

$$5x-4y-4z=0, \quad -4x+5y-4z=0, \quad -4x-4y+5z=0,$$

or, what is the same thing, $x=\frac{4}{5}$, $y=\frac{4}{5}$, $z=\frac{4}{5}$; the form of the curve is shown in the figure.

59. The equation of the harmonic conic corresponding to the satellite line $\lambda x+\mu y+\nu z=0$ is

$$\frac{\lambda}{x}+\frac{\mu}{y}+\frac{\nu}{z} \;=\; 0,$$

or say

$$2fyz+2gzx+2hxy \;=\; 0,$$

where $f+g+h=0$. It is a hyperbola passing through the angles of the triangle, and through the harmonic point $(1,1,1)$ or $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$. Observing that the points in question are the intersections of the two rectangular hyperbolas (pairs of lines) $x(y-z)=0$, $y(z-x)=0$, it follows that the harmonic conic is a rectangular hyperbola.

60. The coordinates of the centre are f^2, g^2, h^2 and the centre is consequently a point on the circle which is the twofold-centre locus.

The asymptotes of course meet in the centre, and they again meet the circle in two points which are the intersections of the circle with the line $fx+gy+hz=0$.

61. The Harmonic Conic is the same for the satellite lines which have a given direction, and we may to determine it take a satellite line which touches the envelope. If the constants α, β, γ satisfy the condition $\alpha+\beta+\gamma=0$; then the equation of a satellite line tangent to the envelope is $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0$; the coordinates of the point of contact with the envelope are as $\alpha^4:\beta^4:\gamma^4$; the coordinates of the twofold centre as $\alpha^2:\beta^2:\gamma^2$; the coordinates of the one with twofold centre as

$$\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta).$$

The values of f, g, h are as $\alpha^2(\beta^2-\gamma^2):\beta^2(\gamma^2-\alpha^2):\gamma^2(\alpha^2-\beta^2)$, or what is the same thing, as $\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta)$, or as $\alpha(\beta^2-\gamma^2):\beta(\gamma^2-\alpha^2):\gamma(\alpha^2-\beta^2)$: the last-mentioned values show that the line $fx+gy+hz=0$ passes through the harmonic point $(1,1,1)$ or $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$, and also through the point $\left(\frac{1}{\alpha^2},\frac{1}{\beta^2},\frac{1}{\gamma^2}\right)$, the inverse of the point $(\alpha^2,\beta^2,\gamma^2)$ which is the twofold centre.

62. On account of the symmetry of the figure in regard to the three asymptotes, it is sufficient to construct the harmonic conic for a direction of the satellite line inclined to the base at an angle not $>30°$, and this is what is accordingly done: it may however be remarked that for the limiting inclination $=0°$, that is, when the satellite line is parallel to the base, the harmonic conic becomes a pair of right lines, the base and perpendicular; but that for the other limiting inclination $=30°$, that is, when the satellite line is perpendicular to one of the legs of the triangle, the harmonic conic is still a proper hyperbola, and is situate symmetrically in regard to the leg in question; the two limiting cases will be readily understood by means of the general case shown in the figure.

63. Imagine now the satellite line moving parallel to itself through the series of positions $ABCMDEA'$; to simplify the figure these are not delineated in their proper positions (but they are merely indicated according to their order of succession), and it is to be understood that they have the following positions, viz.

- A, at infinity;[4]
- B, through the vertex B,
- C, touching the envelope,
- M, through the vertex M_1,
- D, through the vertex D_1,
- E, through node X of the envelope,
- A', at infinity;

then the corresponding positions of the critic centres are, on one branch of the hyperbola: A_1 at infinity; CM_1, a one-with-twofold centre; C_1, a one-with-twofold centre; AE_1; A_1 at infinity. On the other branch: A_2, at infinity; A_3, the harmonic point; AB_1; C_{12}, a twofold centre; M_1, M_2 are imaginary; C'_{12}, a twofold centre; D_1, D_2; E_1, E_2; A'_2, at infinity; A'_3, the harmonic point.

64. For the further explanation of the figure it is to be observed that B_1, B_2 lie on the line joining the midpoints of two sides; and in like manner D_1, D_2 on the line joining the midpoints of two sides; (the imaginary points M_1, M_2 are in like manner on the line joining the midpoints of two sides): these relations depend on the theorem, No. 81, of the Memoir on Involution, viz. that for the satellite lines which pass through a vertex $(1, 0, 0)$ of the triangle, one of the critic centres is the vertex $(1, 0, 0)$, and the other two critic centres are points on the line $-x+y+z = 0$, or, what is the same thing, $x = \frac{1}{2}$.

65. Again, the point E_3 is on the line ($x = 1$) through the vertex D, parallel to the base, and the points E_1, E_2 are on the hyperbola (indicated by a dotted line in the figure) $(y + \frac{1}{4})(z + \frac{1}{4}) = \frac{1}{16}$; this depends on the theorem Nos. 73 and 74 of the Memoir on Involution, viz. the critic centres corresponding to the satellite lines through the point

$$(-4, 1, 1) \text{ or } (2, -\tfrac{1}{2}, -\tfrac{1}{2})$$

lie one of them on the line $y + z = 0$, and the other two on the conic $x(x + y + z) - 4yz = 0$; reducing by the condition $x + y + z = 1$, these equations become respectively $x = 1$, and

$$(y + \tfrac{1}{4})(z + \tfrac{1}{4}) - \tfrac{1}{16} \ = \ 0.$$

66. The foregoing positions of the satellite line, and the critic centres, as exhibited in the figure, were selected partly for facility of delineation; I wished however to examine the effect of the passage of the satellite line through a node of the envelope; and it appears that such passage does not give rise to any marked peculiarity in regard to the critic centres. The selected positions are sufficient to indicate the circumstances of the critic centres as the satellite line passes from the position A at infinity continuously to the position A' at infinity; in particular we see that as the line passes from A to C, or from C' to A', there are three real centres; but that as the line passes from C to C' there is only one real centre.

67. The case of the satellite line parallel to the asymptote $x = 0$, is included (as already mentioned) as a limiting case in the foregoing one; the harmonic conic is here the pair of lines $x(y - z) = 0$: and we have two critic centres on the line $y - z = 0$, (the perpendicular), and the third (not properly a critic centre) at infinity on the asymptote $x = 0$; in fact, starting with a critic centre on the line $y - z = 0$, the polar of the centre in regard to the twofold-centre conic or circle is a line parallel to the asymptote $x = 0$, and which therefore meets the harmonic conic $x(y - z) = 0$ in a second centre on the line $y - z = 0$, and in the point at infinity on the line $x = 0$. But the analytical theory of the case is peculiar and may be specially considered.

[4]Strictly speaking a line at infinity is the line infinity, and as such has no definite direction; but we may of course consider a line which moves parallel to itself in opposite senses as having for its limit the line infinity.

68. Writing $\mu = \nu$, the equation of the satellite line is $\lambda x + \mu(y+z) = 0$, or putting $x+y+z=0$ (sic, $x+y+z=1$) this is $x = \frac{\mu}{\mu-\lambda}$. The equation in θ (see Memoir on Involution, No. 20) becomes

$$(\theta+\mu)(\theta^2 - \theta\mu - 2\lambda\mu) \;=\; 0;$$

or, disregarding the factor $\theta+\mu=0$, which corresponds to the centre at infinity, the equation is

$$\theta^2 - \theta\mu - 2\lambda\mu \;=\; 0,$$

which is a quadratic equation, giving therefore two values of θ, and the corresponding critic centres lie on the perpendicular $y-z=0$, the x coordinate being given by the equation

$$\theta \;=\; \frac{\lambda x}{\frac{1}{2}-x}.$$

We have therefore conversely

$$\lambda x^2 - \mu x\left(\tfrac{1}{2}-x\right) - 2\mu\left(\tfrac{1}{2}-x\right)^2 \;=\; 0,$$

or, what is the same thing,

$$(\lambda-\mu)x^2 + \tfrac{5}{2}\mu x - \tfrac{1}{2}\mu \;=\; 0,$$

so that putting for shortness $\frac{\mu}{\mu-\lambda} = \varpi$ (where ϖ denotes the distance of the satellite line from the asymptote $x=0$), then the equation which determines the distance of the critic centres from the asymptote is

$$x^2 - \tfrac{5}{2}\varpi x + \tfrac{1}{2}\varpi \;=\; 0,$$

or we have

$$x \;=\; \tfrac{5}{4}\varpi \pm \tfrac{1}{4}\sqrt{25\varpi^2 - 8\varpi}.$$

The condition for a twofold centre is ($\varpi = 0$, which may be disregarded, or else) $\varpi = \frac{8}{25}$, or, what is the same thing, $8\lambda + \mu = 0$.

69. If x_1, x_2 are the coordinates of the two centres, we have

$$2x_1^2 = \varpi(3x_1 - 1),$$

$$2x_2^2 = \varpi(3x_2 - 1),$$

or, reducing,

$$\frac{x_1}{x_2} \;=\; \frac{3x_1-1}{3x_2-1},$$

$$x_1 + x_2 - 3x_1x_2 \;=\; 0,$$

a relation connecting the two values x_1, x_2; this equation however only expresses the known relation that the two centres are harmonics of each other in regard to the twofold-centre conic or circle.

70. The foregoing examination of the form of the envelope shows very readily what are the positions of the satellite line which give rise to Plücker's groups for the Hyperbolas A Redundant. We have in fact first,

Hyperbolas A Redundant, no osculating asymptote.

The satellite line is not parallel to a side of the triangle; and the different positions give the following six of Plücker's groups, viz.

I. Satellite line cuts three sides produced.

II. „ passes through a vertex and cuts opposite side produced.

III. „ passes through a vertex and cuts opposite side.

IV. „ cuts two sides and a side produced, but does not cut the envelope.

V. „ touches the envelope.

VI. „ cuts the envelope.

Next,

Hyperbolas A Redundant, one osculating asymptote.

The satellite line is parallel to the osculating asymptote, say to the base of the triangle; and the different positions give the following six of Plücker's groups, viz.

IX. Satellite line above the vertex.

X. „ through the vertex.

XI. „ below the vertex, but not cutting envelope.

XII. „ touches envelope.

XIII. „ cuts envelope.

XIV. „ lies below the base.

And finally,

Hyperbolas A Redundant, three osculating asymptotes.

The position of the satellite line is here completely determined, giving one of Plücker's groups, viz.

XVI. Satellite line at infinity.

71. It may be remarked that in this enumeration no account is taken of the nodes of the envelope: the enumeration was in fact made by Plücker by considerations relating to the critic centres, but without arriving at or making use of the envelope at all: if account were taken of the nodes of the envelope several of the foregoing groups would have to be subdivided according to the different positions of the satellite line in regard to these nodes: but the effect produced by the passage of the satellite line through a node of the envelope is so slight, that I am inclined to think that the enumeration may be properly effected in the foregoing manner, without any account being taken of these nodes.

The Hyperbolas A Defective (See fig. 2). Article Nos. 72 to 101.

72. If in the formulae for the Hyperbolas A Redundant we write

$$\tfrac{1}{2}(x+yi) \text{ for } x, \qquad \tfrac{1}{2}(x-yi) \text{ for } y, \qquad \lambda-\mu i \text{ for } \lambda, \qquad \lambda+\mu i \text{ for } \mu,$$

then the equation of the satellite line is

$$\tfrac{1}{2}(\lambda-\mu i)(x+yi)+\tfrac{1}{2}(\lambda+\mu i)(x-yi)+\nu z \;=\; 0,$$

which is $\lambda x+\mu y+\nu z=0$ as before; the equation of the line infinity is $x+z=0$, and the equation of the curve is

$$\tfrac{1}{4}(x^2+y^2)z+k(x+z)^2(\lambda x+\mu y+\nu z) \;=\; 0.$$

[The treatment of the Hyperbolas A Defective continues from Art. 73 with the equation of the envelope reduced to a quartic in x and y, the discussion of the nodes of the envelope, the twofold-centre conic in the defective case, the one-with-twofold-centre cubic, the harmonic conic for the defective hyperbola, and the analogous enumeration of Plücker's groups XVIII–XXIII for the

Defective Hyperbolas A without an osculating asymptote, then groups XXVIII–XXXIII for the cases of a real osculating asymptote, and finally group XXXV for the three osculating asymptotes. The Article Nos. 73–101 carry through this analysis in close parallel with the redundant case treated above, and they continue past the present 50-page extract. The remainder of Paper 350, covering the Hyperbolas B, the Parabolic Hyperbolas, the various Hyperbolisms, the Divergent Parabolas, the Trident Curve, and the Cubical Parabola, falls beyond PDF p. 400 and will be taken up in the subsequent installment.]

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